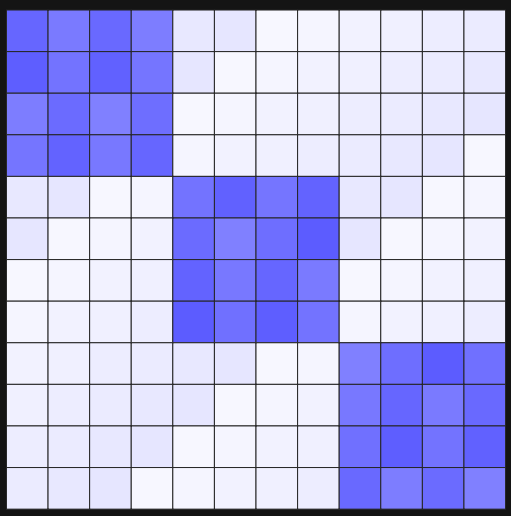
STRUCTURAL ECONOMICS

The General Case



Werner L. Hernani-Limarino

$$W^{CE} = \tfrac{1}{N}\mathbf{1}\mathbf{1}' \quad \longrightarrow \quad W = W^{CE} + \Delta(W)$$

STRUCTURAL ECONOMICS

The General Case

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DRAFT — November 10, 2026

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For my daughter — the park I care about most.

"O subdesenvolvimento não constitui uma etapa necessária do processo de formação das economias capitalistas modernas. É, em si, um processo particular, resultante da penetração de empresas capitalistas modernas em estruturas arcaicas."

— Celso Furtado, *Desenvolvimento e Subdesenvolvimento* (1961)

"The competitive benchmark is the special case $W = \frac{1}{N}\mathbf{1}\mathbf{1}'$*.*
Structural economics studies the general case."

— This book

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Notation

Vectors are bold and are column vectors; $'$ is the transpose and $\mathbf{1}$ the vector of ones. N is the number of nodes (agents, sectors, groups, outlets or countries) of the matrix in question. Measures of concentration (N_{eff}, $\text{Var}(\boldsymbol{\pi})$, $G(\boldsymbol{\pi})$) are comparable across matrices only on a common node set, or after normalizing by N.

Symbol	Meaning (where defined)
Weight matrices	
W, W_{ij}	A row-stochastic interaction matrix ($W \geq 0$, $W\mathbf{1} = \mathbf{1}$); W_{ij} is the weight agent i places on agent j (Chapter 1, Section 1.7).
K, D	A raw nonnegative intensity matrix and its row-sum diagonal, $D = \text{diag}(K\mathbf{1})$; then $W = D^{-1}K$.
W^U	Utility network: who cares about whom; enters $V_i = U_i + \gamma_U \sum_j W^U_{ij} U_j$ (Definition 5.1). The benchmark has $W^U = 0$ (no externalities).
W^P	Production weight matrix, $W^P = D^{-1}A$: input-cost shares, row-stochastic (Definition 5.4). Never used for multipliers or linkages.
W^I	Information network: who listens to whom; drives DeGroot learning (Definition 5.6).
α_k	Weights of the composite $W = \sum_k \alpha_k W^k$, defined only when all layers live on the same node set (Definition 5.10).
A, a_{ij}	Technical coefficients: units of good j per unit of good i; balance $\mathbf{y} = A'\mathbf{y} + \mathbf{d}$ (Definition 1.12).
L	Leontief inverse $(I - A')^{-1}$, requiring $\rho(A) < 1$ (Theorem 1.14). Multipliers and backward/forward linkages come from L. *Also:* L is the number of goods in Chapter 1 and, with a subscript, a welfare loss (below).
$\chi(A)$	Cross-block mass: share of intermediate requirements that cross blocks (Definition 3.5).
$\mathbf{s}$	Wealth (expenditure) shares (Definition 1.17).

Symbol	Meaning (where defined)
W^{mkt}	Market interaction matrix $\mathbf{1}\mathbf{s}'$: anonymous, rank one, $\Gamma = 1$, $\boldsymbol{\pi} = \mathbf{s}$ (Definition 1.17).
W^{CE}	Competitive benchmark $\frac{1}{N}\mathbf{1}\mathbf{1}'$, the equal-endowment case of W^{mkt} (Definition 1.18).
$\Delta(W)$	Departure from the benchmark, $W - W^{\text{CE}}$, summarized by the two independent coordinates d_π and $1 - \Gamma$ (Section 1.7).
Power and integration	
$\boldsymbol{\pi}$	Power: the left Perron vector, $\boldsymbol{\pi}'W = \boldsymbol{\pi}'$, $\boldsymbol{\pi} > 0$, $\sum_i \pi_i = 1$ (Definitions 1.20 and 6.4). Uniform iff W is doubly stochastic (Proposition 6.8). Superscripts ($\boldsymbol{\pi}^P$, $\boldsymbol{\pi}^I$, $\boldsymbol{\pi}^U$) name the layer.
N_{eff}	Effective number of agents, $1/\sum_i \pi_i^2 \in [1, N]$ (Definition 6.15).
$\text{Var}(\boldsymbol{\pi})$	Variance of power, $\frac{1}{N}\sum_i(\pi_i - \frac{1}{N})^2 = \frac{1}{N}(\frac{1}{N_{\text{eff}}} - \frac{1}{N}) \leq \frac{N-1}{N^2}$ (Definition 6.15).
$G(\boldsymbol{\pi})$	Power Gini: the Gini coefficient of the entries of $\boldsymbol{\pi}$ (Definition 6.15).
d_π	Power distance, $\frac{1}{2}\|\boldsymbol{\pi} - \frac{1}{N}\mathbf{1}\|_1 \in [0, 1 - \frac{1}{N}]$ (Section 1.7).
λ_2	Eigenvalue of W of second-largest modulus.
Γ	Spectral gap $1 - \lvert\lambda_2(W)\rvert$: integration (mixing speed), never concentration (Definition 6.14). Independent of $\boldsymbol{\pi}$ (Proposition 6.16).
τ_1	Dobrushin coefficient $\frac{1}{2}\max_{i,j}\sum_k \lvert W_{ik} - W_{jk}\rvert$; $\lvert\lambda_2\rvert \leq \tau_1$ and τ_1 is submultiplicative (Proposition 10.5; Chapter 23).
Z	Fundamental matrix $(I - W + \mathbf{1}\boldsymbol{\pi}')^{-1}$; $d\boldsymbol{\pi}'/dt = \boldsymbol{\pi}'EZ$ for a perturbation E with $E\mathbf{1} = 0$ (Lemma 6.19).
$\mathbf{b}^{(0)}$, b^*	Initial beliefs and the DeGroot consensus $b^* = (\boldsymbol{\pi}^I)'\mathbf{b}^{(0)}$ (Theorem 8.1).
σ_B	Between-block share of inequality (Definition 17.3).
Allocation and welfare	
X, $\mathbf{x}$	Allocation matrix ($N \times L$) and allocation vector (Definition 1.1).
$\boldsymbol{\omega}$, ω_i	Endowments. In Chapter 10 ω_i is also the endowment term of the bargaining rule. *Also:* $\omega_m, \omega_r, \omega_n, \omega_d$ are the source weights of the composite W^I (Chapter 21).
Ω	Total resources (one good) or total joint output to be divided. In Chapter 1 also the $N \times L$ endowment matrix.
U, κ	Private utility; in examples $U(x) = x - \frac{\kappa}{2}x^2$, with every allocation below satiation, $\kappa x_i < 1$ (Chapter 1).
d_i, $\bar{d}$	Disagreement payoffs (outside options); $\bar{d}$ when they are equal.

Symbol	Meaning (where defined)
Φ	Bargaining surplus over disagreement, $\Phi = \Omega - N\bar{d}$ (not $1/\kappa$). Allocation rule $x_i = \Omega/N + (\pi_i - \frac{1}{N})\Phi$ (Proposition 7.3), with $\boldsymbol{\pi}$ as Nash weights by assumption. *Also:* conductance in the Cheeger inequality (Chapter 25).
c_j	Externality centrality: how much others care about j, the column sum of W^U (Definition 2.2).
γ_U, γ_P	Intensity of utility externalities and of production spillovers (Chapter 5); γ in Chapter 2.
V_i, $\mathcal{S}$	Interdependent utility and social welfare $\mathcal{S}(\mathbf{x}) = \sum_i V_i(\mathbf{x})$ (Section 7.4).
$\mathcal{W}$	Welfare totals, e.g. $\mathcal{W}^{FB}$ (first best) and $\mathcal{W}^0$ (endowments) (Proposition 1.5). *Also:* the set of row-stochastic matrices (Chapter 15), and $\mathcal{W}_\varepsilon$, those with all entries $\geq \varepsilon$ (Chapter 13).
L_{power}	Distributional (utilitarian) loss from unequal power, $\frac{\kappa N}{2}\Phi^2 \operatorname{Var}(\boldsymbol{\pi})$; never a Pareto inefficiency (Theorem 7.8).
$L_{\text{externality}}$	Loss of equal division relative to the social optimum when the c_j differ; to second order $\propto \gamma^2 \operatorname{Var}(c_j)$ (Theorem 7.8; Proposition 2.9).
$L_{\text{interaction}}$	Interaction term; its leading part $\propto -\operatorname{Cov}(c_j, \pi_j)$ offsets the others when the powerful are the most cared about (Theorem 7.8).
L_{block}	Between-block welfare loss (Proposition 2.11).
$L_{\text{formation}}$	Additional loss from formed preferences, used in Chapter 18 (Definition 18.15); not one of the three terms above.
Preference formation	
$\boldsymbol{\theta}$	Preference vector (aspirations, patience, ...) (Definition 9.1).
ρ, μ	Intergenerational persistence and weight of the network signal in the formation equation (Definition 9.7).
θ°	Anchor of the formation equation ($\bar{\theta}$ in Chapter 9), distinct from the cross-sectional mean $\bar{\theta}_t$ (Definition 10.1).
r_j, r^*	Observed returns and the reference return.
h, ϕ	Bargaining map $x_i = h_i(\boldsymbol{\pi}, \boldsymbol{\theta}, \boldsymbol{\omega}) = \omega_i + \pi_i\Phi + \phi(\theta_i - \bar{\theta})$ and its preference weight ϕ (Definition 10.1). *Also:* $\phi(\mathbf{a}_i, \mathbf{a}_j)$ is the assortative mixing function of Chapter 18.
T	Formation map $\boldsymbol{\theta}_{t+1} = T(\boldsymbol{\theta}_t)$ (equation (10.4)); $T(\cdot\,; W)$ in Chapter 13. *Also:* a horizon in generations, as in $\Psi_i(\Delta W, T)$.
J, C, P	Formation Jacobian $J = \rho I + \mu\phi W^I C P$, with return Jacobian $C = \operatorname{diag}(\boldsymbol{\beta}) + \operatorname{diag}(\boldsymbol{\gamma})W^P$ and centering matrix $P = I - \frac{1}{N}\mathbf{1}\mathbf{1}'$ (Proposition 10.2).
$\mathcal{M}$	Formation multiplier $\mu/(1-\rho)$ (Definition 10.3).

Symbol	Meaning (where defined)
E_i, F_i	Enforcement and formation effects of a stated perturbation ΔW (Definition 9.13).
Ψ	Formation-to-enforcement ratio $\Psi_i(\Delta W, T) = F_i / E_i$ of a perturbation, not of the economy; $\bar{\Psi}$ is its aggregate (Definition 9.13). $(1 + \Psi)$ is the formation amplifier of Chapter 14.
$\hat{\Psi}_{\text{var}}$	Cohort–year variance ratio: a descriptive statistic, not Ψ (Definition 11.4).
Network evolution, attributes, time and complexity	
$\mathcal{E}$	Network evolution map $W_{t+1} = \mathcal{E}(W_t; \mathbf{x}_t^*)$ (Definition 13.6); $\mathcal{N}$ is its clip-and-renormalize step (Definition 13.5). Written $\mathcal{E}$, not $\mathcal{T}$, to keep it apart from the formation map T.
$\mathcal{J}$	Jacobian of the co-evolutionary system in $(W, \boldsymbol{\theta})$ (Chapter 13, Appendix 13B).
λ_k	Assortative mixing (openness) parameter for attribute k: cross-group edges are scaled by $\lambda_k \in [0, 1]$ (Definition 18.2; e.g. λ_g for gender). Not an eigenvalue.
ι	Intersectional interaction $\phi - \prod_k \phi_k$ (equation (18.9)).
v_Γ, SCI	Structural velocity and Structural Change Index (Definitions 23.12 and 23.17).
M, ECI	Country $\times$ product specialization matrix and the Economic Complexity Index, computed from M, not from W (Definitions 16.5 and 16.7).
$\mathcal{D}$, τ_c	Dense subgraph of the product space and the truncation index (Definitions 25.1 and 25.3).

Preface

The Preface (9 pp.) is maintained separately and is not yet part of this source tree.

Visual Prologue: Seeing the Matrix

The Visual Prologue (9 pp.) is maintained separately and is not yet part of this source tree.

Part I

The Matrix Reformulation

Chapter 1

Economics in Matrix Notation

The Standard Model, Rewritten So the Structure Becomes Visible

> *"La realidad está destruyendo en la América Latina aquel pretérito esquema de la división internacional del trabajo que, después de haber adquirido gran vigor en el siglo XIX, seguía prevaleciendo doctrinariamente hasta muy avanzado el presente."*
>
> ["Reality is destroying in Latin America that outdated scheme of the international division of labor which, after acquiring great vigor in the nineteenth century, continued to prevail doctrinally until well into the present."]
>
> — Raúl Prebisch, *El desarrollo económico de la América Latina* (1949)

> *"The purpose of abstracting is not to be vague, but to create a new semantic level in which one can be absolutely precise."*
>
> — Edsger W. Dijkstra, *The Humble Programmer* (1972)

This chapter does one thing: it rewrites the economics you already know in a notation that reveals the economics you have not been taught.

Every result here is standard. If you have taken intermediate microeconomics, you know the consumer's problem, the producer's problem, and the idea of market equilibrium. If you have taken a first course in linear algebra, you know what a matrix is and how to multiply them. This chapter puts the two together.

The payoff comes at the end. In Section 1.6, we will name an object that is present in every microeconomics textbook but that no microeconomics textbook names: the *interaction matrix* — the mathematical description of how agents in the economy relate to each other. The standard competitive model implies a very specific interaction matrix: anonymous, rank one, with each agent weighted by her

wealth share. Structural economics is what happens when you replace that matrix with the one the world actually has — non-anonymous, with utility interdependence, and with a distribution of power that the network itself produces.

But first, the standard model.

1.1 Who Gets What: The Allocation Matrix

Economics studies how scarce resources are distributed among people. Let us write this down in the simplest possible way.

1.1.1 A Small Economy

Suppose there are 3 people and 2 goods — call them Food and Clothing. Each person consumes some amount of each good. We can organize this information in a table:

	Food	Clothing
Ana	4	2
Boris	3	5
Carmen	1	3

Ana eats 4 units of Food and wears 2 units of Clothing. Boris eats 3, wears 5. Carmen eats 1, wears 3. This table is the entire economy. Every question economics asks — Is this allocation efficient? Fair? Sustainable? Better or worse than some alternative? — is a question about the numbers in this table.

Now write the table as a matrix:

$$X = \begin{pmatrix} 4 & 2 \\ 3 & 5 \\ 1 & 3 \end{pmatrix}. \tag{1.1}$$

Each *row* is a person. Each *column* is a good. The entry $X_{i\ell}$ tells us how much of good ℓ person i consumes.

Intuition: Reading the matrix two ways

Read the matrix by rows and you see individuals: Ana's bundle is $(4, 2)$, Boris's is $(3, 5)$, Carmen's is $(1, 3)$. Read it by columns and you see the distribution of each good: Food is split $(4, 3, 1)$ across three people, Clothing is split $(2, 5, 3)$. The row view is how microeconomics usually thinks — one person at a time. The column view is how a planner or a sociologist thinks — one resource at a time, distributed across a population. This book uses both.

Definition 1.1 (Allocation Matrix)**.** *For an economy with N agents and L goods, the **allocation matrix** is the $N \times L$ matrix*

$$X = \begin{pmatrix} x_1' \\ x_2' \\ \vdots \\ x_N' \end{pmatrix}, \tag{1.2}$$

where $x_i = (x_{i1}, x_{i2}, \ldots, x_{iL})' \in \mathbb{R}^L_+$ is agent i's consumption vector. Row i is person i's bundle. Column ℓ is the distribution of good ℓ across agents.

How much Food exists in total? Sum the first column: $4 + 3 + 1 = 8$. In matrix notation: $\mathbf{1}'Xe_1 = 8$, where $\mathbf{1} = (1, 1, 1)'$ is the vector of ones and $e_1 = (1, 0)'$ picks out the first column.

More generally, the total endowment of each good is the vector $\mathbf{1}'X$: a $1 \times L$ row vector whose ℓ-th entry is the total amount of good ℓ consumed.

1.1.2 Endowments: Where People Start

Before any trade or production happens, each person starts with some stuff. Ana might have 6 units of Food and 1 of Clothing (she is a farmer). Boris might have 1 of Food and 6 of Clothing (he is a tailor). Carmen might have 1 of each.

$$\Omega = \begin{pmatrix} 6 & 1 \\ 1 & 6 \\ 1 & 3 \end{pmatrix}. \tag{1.3}$$

The endowment matrix Ω has the same shape as X: $N \times L$. Entry $\omega_{i\ell}$ is how much of good ℓ person i starts with. The total supply of each good is $\mathbf{1}'\Omega$: here, $(8, 10)$ — 8 units of Food and 10 of Clothing exist in total.

Definition 1.2 (Feasibility)**.** *An allocation X is **feasible** if nobody consumes more of any good than exists in total:*

$$\mathbf{1}'X \leq \mathbf{1}'\Omega \qquad (\textit{component-wise for each good}). \tag{1.4}$$

With equality (no waste), this is $\mathbf{1}'X = \mathbf{1}'\Omega$: total consumption equals total endowment.

Check: in our example, $\mathbf{1}'X = (8, 10)$ and $\mathbf{1}'\Omega = (8, 10)$. The allocation is feasible.

1.2 The Consumer's Problem

1.2.1 What People Want

Each person has preferences over bundles. Ana might prefer more Food, Boris might prefer more Clothing, Carmen might want a balanced mix. We represent preferences by a utility function $U_i(x_i)$: a number that is higher when person i is happier with her bundle x_i.

Each person chooses the bundle that makes her happiest, subject to what she can afford. If prices are $p = (p_F, p_C)' = (1, 2)'$ (Food costs 1, Clothing costs 2), then Ana's budget is $p'\omega_{\text{Ana}} = 1 \cdot 6 + 2 \cdot 1 = 8$ dollars. She can buy any bundle (x_F, x_C) with $x_F + 2x_C \leq 8$.

Definition 1.3 (The Consumer's Problem)**.** *Agent i solves:*

$$\max_{x_i \geq 0} \; U_i(x_i) \quad \textit{subject to} \quad p'x_i \leq p'\omega_i. \tag{1.5}$$

*The solution $x_i^*(p, w_i)$ is agent i's **Marshallian demand**, where $w_i = p'\omega_i$ is her wealth.*

For the entire economy, stack the N problems. Define the wealth vector $\mathbf{w} = \Omega p$: an $N \times 1$ vector whose i-th entry is $w_i = p'\omega_i$. The budget constraints become:

$$Xp \leq \Omega p = \mathbf{w}. \tag{1.6}$$

Intuition: What this equation hides

Look at equation (1.5) carefully. Agent i's utility depends only on x_i — her *own* consumption. Nobody else's consumption appears. Ana does not care what Boris eats. Boris does not care what Carmen wears. Each person optimizes in isolation, connected to others only through the price vector p.

This is an assumption, not a fact. In the real world, people care intensely about what others consume, earn, and do. The keeping-up-with-the-Joneses effect, peer pressure, status competition, network externalities — all of these are cases where U_i depends on x_j for $j \neq i$. The standard model assumes all of these away.

We will make this assumption visible and then relax it.

1.2.2 A Worked Example: Cobb-Douglas

Suppose all three agents have the same preferences:

$$U_i(x_{i1}, x_{i2}) = x_{i1}^{1/2} \, x_{i2}^{1/2}. \tag{1.7}$$

This is the Cobb-Douglas utility function with equal weights on both goods. The Marshallian demands (derived by setting up the Lagrangian and solving the first-order conditions — see any intermediate micro text) are:

$$x_{i1}^* = \frac{w_i}{2p_1}, \qquad x_{i2}^* = \frac{w_i}{2p_2}. \tag{1.8}$$

Each person spends half her income on each good. Let $p = (1, 1)'$ (both goods cost 1). Then:

$$w_{\text{Ana}} = 1 \cdot 6 + 1 \cdot 1 = 7, \tag{1.9}$$

$$w_{\text{Boris}} = 1 \cdot 1 + 1 \cdot 6 = 7, \tag{1.10}$$

$$w_{\text{Carmen}} = 1 \cdot 1 + 1 \cdot 3 = 4. \tag{1.11}$$

The demands are:

$$X^* = \begin{pmatrix} 3.5 & 3.5 \\ 3.5 & 3.5 \\ 2.0 & 2.0 \end{pmatrix}. \tag{1.12}$$

Intuition: Two things to notice

First, Ana and Boris have *identical demands* even though they start with very different endowments (Ana has mostly Food, Boris has mostly Clothing). The market — through trade at prices p — erases the difference. This is the power of markets: they allow people to trade away from their endowments toward their preferred bundles.

Second, Carmen consumes less of *both* goods. Not because she has different preferences (all three have the same U_i), but because she is poorer. Her endowment has lower total value. The allocation matrix X^* reflects not just preferences but wealth — and wealth reflects endowments, which reflect history.

This is the first hint of the structural question: where do the endowments come from?

Now write the demand in matrix form. From (1.8), each person's bundle is $x_i^* = \frac{w_i}{2}(1/p_1, 1/p_2)'$. Stack them:

$$X^* = \frac{1}{2}\mathbf{w} \cdot (1/p_1,\ 1/p_2) = \frac{1}{2}\begin{pmatrix} 7 \\ 7 \\ 4 \end{pmatrix}\begin{pmatrix} 1 & 1 \end{pmatrix}. \tag{1.13}$$

The allocation matrix is the *outer product* of the wealth vector $\mathbf{w}$ and the price-dependent expenditure-share vector $(1/2p_1, 1/2p_2)$. Every row is a scalar multiple of the same vector. People differ only in *how much* they consume, not in *what* they consume.

Key Structural Insight

In the Cobb-Douglas model, the allocation matrix X^* is a **rank-1 matrix**: it can be written as a column vector times a row vector. Rank 1 means all rows are proportional. This is the mathematical signature of an economy where people differ only in wealth, not in preferences or social position.

When we introduce social structure in Chapter 5, the allocation matrix will have higher rank. Different agents will consume different *kinds* of bundles, not just different *amounts* of the same bundle. The rank of X is a measure of structural differentiation.

1.2.3 Market Clearing: Does Everything Add Up?

Check feasibility: $\mathbf{1}'X^* = (3.5 + 3.5 + 2.0,\ 3.5 + 3.5 + 2.0) = (9, 9)$. But total endowment is $\mathbf{1}'\Omega = (8, 10)$. Food demand exceeds supply by 1; Clothing demand falls short by 1. The allocation is not feasible at these prices.

This means $p = (1, 1)$ is not an equilibrium price. The correct price must adjust to equate supply and demand. We need to find p^* such that $\mathbf{1}'X^*(p^*) = \mathbf{1}'\Omega$.

Setting up the market-clearing condition for Food (normalizing $p_1 = 1$):

$$\frac{w_1}{2} + \frac{w_2}{2} + \frac{w_3}{2} = 8 \quad \Rightarrow \quad \frac{(6 + p_2) + (1 + 6p_2) + (1 + 3p_2)}{2} = 8. \tag{1.14}$$

Solving: $\frac{8+10p_2}{2} = 8$, so $8 + 10p_2 = 16$, giving $p_2^* = 0.8$.

At $p^* = (1, 0.8)$: $w_{\text{Ana}} = 6.8$, $w_{\text{Boris}} = 5.8$, $w_{\text{Carmen}} = 3.4$. The equilibrium allocation is:

$$X^* = \begin{pmatrix} 3.40 & 4.25 \\ 2.90 & 3.63 \\ 1.70 & 2.13 \end{pmatrix}. \tag{1.15}$$

Verify: column sums are $(8.0, 10.0)$. Markets clear. This is the **Walrasian equilibrium**.

1.3 The One-Good Economy: Welfare and Power

The multi-good case demonstrates trade. But to see *power* clearly, we need the simplest possible setting: one good, N agents, and the question of who gets how much.

1.3.1 Setup

N agents share a total endowment $\Omega > 0$ of a single good. Person i starts with $\omega_i \geq 0$ and $\sum_i \omega_i = \Omega$. Preferences are quadratic:

$$U_i(x_i) = x_i - \frac{\kappa}{2}x_i^2, \qquad \kappa > 0. \tag{1.16}$$

This utility function is concave: the first unit is worth approximately 1, each additional unit is worth less, and the marginal utility hits zero at $x_i = 1/\kappa$. The parameter κ controls how fast the returns diminish. Think of it as how quickly a person gets "full."

Example 1.4 (Three People, One Good). *Let $N = 3$, $\kappa = 0.5$, $\Omega = 3$. Endowments: $\omega_1 = 1.8$, $\omega_2 = 0.6$, $\omega_3 = 0.6$. Satiation is at $1/\kappa = 2$, so every allocation considered below satisfies $\kappa x_i < 1$: more is always better.*

Person 1 is rich (starts with 1.8 out of 3). Persons 2 and 3 are poor (start with 0.6 each).

What is the best possible allocation? *A utilitarian planner maximizes total welfare $\sum_i U_i(x_i)$ subject to $\sum_i x_i = 3$. The first-order conditions require equal marginal utility: $1 - \kappa x_i = 1 - \kappa x_j$, so $x_i = x_j$ for all i, j. Equal division: $x_i^{FB} = 1$ for all i.*

Total welfare at the first-best: $\mathcal{W}^{FB} = 3 \times (1 - 0.25) = 2.25$.

What happens without a planner? *With one good there is nothing to trade, so each person consumes her endowment: $x_1 = 1.8$, $x_2 = x_3 = 0.6$.*

Welfare: $\mathcal{W}^0 = (1.8 - 0.81) + 2(0.6 - 0.09) = 0.99 + 1.02 = 2.01$.

The rich person's last units are worth little: her marginal utility is $1 - 0.5 \times 1.8 = 0.1$, while each poor person's is $1 - 0.5 \times 0.6 = 0.7$. Moving a unit from person 1 to person 2 would raise total utility by roughly 0.6.

The welfare loss: *$L = \mathcal{W}^{FB} - \mathcal{W}^0 = 2.25 - 2.01 = 0.24$, about 11% of first-best welfare. This is a utilitarian (distributional) loss: the endowment allocation is Pareto efficient, since any reallocation makes someone worse off.*

Proposition 1.5 (Welfare Loss from Inequality). *In the one-good economy with quadratic utility (1.16), the welfare loss from endowment inequality is:*

$$L = \frac{\kappa N}{2} \cdot Var(\omega_i), \tag{1.17}$$

where $Var(\omega_i) = \frac{1}{N}\sum_i \omega_i^2 - \left(\frac{\Omega}{N}\right)^2$ is the variance of endowments. The loss is zero if and only if all endowments are equal.

Intuition: What this says in words

The cost of inequality is proportional to three things: how curved preferences are (κ), how many people there are (N), and how unequally the endowment is distributed ($\text{Var}(\omega_i)$). More curvature means the rich waste more of what they have (diminishing returns bite harder). More people means more total waste. More inequality means more mismatch between who has stuff and who can use it.

This is the simplest possible welfare cost of inequality. In the structural model, the question becomes: where does $\text{Var}(\omega_i)$ come from? The answer will be: from the weight matrix.

Proof. First-best welfare: $\mathcal{W}^{FB} = N \cdot U(\Omega/N) = N[\Omega/N - \frac{\kappa}{2}(\Omega/N)^2] = \Omega - \frac{\kappa\Omega^2}{2N}$.

Endowment welfare: $\mathcal{W}^0 = \sum_i U_i(\omega_i) = \sum_i [\omega_i - \frac{\kappa}{2}\omega_i^2] = \Omega - \frac{\kappa}{2}\sum_i \omega_i^2$.

The difference:

$$L = \mathcal{W}^{FB} - \mathcal{W}^0 = \frac{\kappa}{2}\left(\sum_i \omega_i^2 - \frac{\Omega^2}{N}\right) = \frac{\kappa}{2} \cdot N \cdot \mathrm{Var}(\omega_i). \tag{1.18}$$

The last step uses the identity $\sum_i \omega_i^2 - \Omega^2/N = N \cdot \mathrm{Var}(\omega_i)$. Since $\kappa N/2 > 0$, $L = 0$ if and only if $\mathrm{Var}(\omega_i) = 0$, that is, if and only if every $\omega_i = \Omega/N$. (The planner's first-best is equal division because $\sum_i U(x_i)$ is strictly concave and symmetric, so its unique maximizer on $\sum_i x_i = \Omega$ is the symmetric point.) □

Example 1.6 (Verifying the Formula)**.** *In Example 1.4: $Var(\omega) = \frac{1}{3}(3.24 + 0.36 + 0.36) - 1^2 = 1.32 - 1 = 0.32$. So $L = \frac{0.5 \cdot 3}{2} \cdot 0.32 = 0.24$. Matches.*

1.3.2 Bargaining: When Power Enters

In a one-good exchange economy there is nothing to bargain over: the endowment allocation is already Pareto efficient. Bargaining needs a *surplus*. So change the story. Agent 1 is a landowner; agents 2 and 3 are laborers. Working together they produce a harvest of Ω; if they fail to agree, each falls back on a disagreement payoff d_i (what the landowner can grow alone, what the laborers earn elsewhere), with $\sum_i d_i < \Omega$. The joint surplus $\Omega - \sum_i d_i$ is what bargaining power divides. Payoffs are in units of the good, so utility is transferable for the purpose of bargaining; we use the concave U in (1.16) only to *evaluate* welfare.

We model the division through the **generalized (asymmetric) Nash bargaining solution**: the allocation maximizes a weighted product of gains over disagreement.

Definition 1.7 (Bargaining Power)**.** *The **bargaining weight vector** is $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_N)'$ with $\lambda_i > 0$ and $\sum_i \lambda_i = 1$. Agent i's weight λ_i represents her share of total bargaining power.*

Definition 1.8 (Nash Bargaining Solution)**.** *The **Nash bargaining allocation** maximizes:*

$$\max_{\mathbf{x}} \sum_{i=1}^{N} \lambda_i \ln(x_i - d_i) \quad \text{subject to} \quad \sum_i x_i = \Omega, \quad x_i > d_i, \tag{1.19}$$

*where d_i is agent i's **disagreement payoff** — what she gets if bargaining fails.*

Proposition 1.9 (Nash Division of the Surplus)**.** *Let $\lambda_i > 0$ for all i and $\Omega > \sum_j d_j$. Then (1.19) has the unique solution*

$$x_i = d_i + \lambda_i\Big(\Omega - \sum_j d_j\Big). \tag{1.20}$$

Every agent receives her disagreement payoff plus a share λ_i of the joint surplus. The allocation exhausts the harvest, so it is Pareto efficient for every $\boldsymbol{\lambda}$: in units of the good, and in the utilities U of (1.16) whenever $\kappa x_i \leq 1$ for every i.

Proof. This is the asymmetric Nash solution (Kalai, 1977) for transferable utility. The feasible set $\{x : \sum_i x_i = \Omega,\ x_i > d_i\}$ is nonempty because $\Omega > \sum_j d_j$, and the objective is strictly concave on it and tends to $-\infty$ at its boundary, so a maximizer exists, is interior and is unique. The Lagrange condition is $\lambda_i/(x_i - d_i) = \mu$, so $x_i - d_i = \lambda_i/\mu$. Summing over i and using $\sum_i \lambda_i = 1$ gives $1/\mu = \Omega - \sum_j d_j$. Efficiency: let $\mathbf{x}' \neq \mathbf{x}$ be any other allocation with $\sum_i x'_i \leq \Omega$. Then $x'_j < x_j$ for some j. In units of the good j is worse off. In utility, U is strictly increasing on $(-\infty, 1/\kappa]$, so $\kappa x_j \leq 1$ gives $U(x'_j) < U(x_j)$. Hence no feasible reallocation makes someone better off without making someone worse off. □

Intuition: What bargaining power does

The weight λ_i acts like a "voting share" in the allocation decision. An agent with high λ_i pulls the allocation toward her preferred outcome. In the extreme, if $\lambda_1 = 1$ and all others are zero, agent 1 gets everything she wants (subject to feasibility). If $\lambda_i = 1/N$ for all i, bargaining is symmetric and the allocation is "fair" in a specific sense.

The critical question for this book: where does $\boldsymbol{\lambda}$ come from? The symmetric Nash solution sets $\lambda_i = 1/N$; competitive markets involve no bargaining at all. In this book we will *postulate* that $\boldsymbol{\lambda}$ equals the Perron vector $\boldsymbol{\pi}$ of a weight matrix W, so that bargaining power comes from position in the network of economic relationships. That is a modelling assumption, not a theorem (Section 1.7); Chapters 5–7 develop it.

Example 1.10 (Bargaining with Asymmetric Power)**.** *Same preferences as Example 1.4 ($N = 3$, $\kappa = 0.5$). Cooperation produces a harvest $\Omega = 3$; disagreement payoffs are $\mathbf{d} = (0.6, 0.3, 0.3)$, so the joint surplus is $\Omega - \sum_j d_j = 1.8$. Let $\boldsymbol{\lambda} = (0.6, 0.2, 0.2)$ — agent 1 has three times the bargaining power of each other agent.*

By Proposition 1.9: $x_1^B = 0.6 + 0.6 \times 1.8 = 1.68$ and $x_2^B = x_3^B = 0.3 + 0.2 \times 1.8 = 0.66$.

Compare the allocations:

	Agent 1	*Agent 2*	*Agent 3*	*Welfare* $\sum_i U_i$
Disagreement $\mathbf{d}$	*0.60*	*0.30*	*0.30*	*1.065*
First-best (planner)	*1.00*	*1.00*	*1.00*	*2.250*
Symmetric Nash ($\lambda_i = 1/3$)	*1.20*	*0.90*	*0.90*	*2.235*
Asymmetric Nash ($\lambda_1 = 0.6$)	*1.68*	*0.66*	*0.66*	*2.077*

The powerful agent gets more than the equal-division first-best and is better off for it ($U_1 = 0.974$ *versus* 0.75)*. The weak agents get less. Everyone is better off than at disagreement, and the allocation is Pareto efficient.*

The utilitarian loss relative to the first-best is $2.250 - 2.077 = 0.173$*. Of this,* 0.015 *would remain even with symmetric bargaining (it comes from the unequal disagreement payoffs); the remaining* 0.158 *is due to the asymmetry of power. The planner's first-best would be reached by bargaining weights* $\lambda_i = (1 - d_i)/1.8 = (0.22, 0.39, 0.39)$ *— the weak would need more power, not equal power, to offset their worse outside options.*

Key Structural Insight

In the one-good economy, there are two sources of distributional loss. The first is *endowment inequality*: some people start with too much, others with too little (or, in bargaining, with better outside options). The second is *power inequality*: bargaining power determines who captures the surplus. Neither is a Pareto inefficiency — asymmetric Nash bargaining is efficient — but both lower utilitarian welfare when utility is concave. Competitive markets sidestep the second source because price-takers do not bargain. This book studies economies in which allocation runs through relationships, so that power matters.

1.4 Production: The Leontief System

We now move from consumption to production. The production side introduces the first genuinely structural matrix in standard economics: the *input-output matrix*.

1.4.1 Indirect Requirements

Consider a simple economy with three sectors: Farming, Milling, and Baking. To produce bread, you need flour. To produce flour, you need wheat. To produce wheat, you need…bread? Well, you need to feed the farmers.

This creates a web of dependencies:

- Baking needs Milling (for flour).
- Milling needs Farming (for wheat).
- Farming needs Baking (for food for the farmers).

Now suppose the country wants to *export* 100 loaves of bread. How much wheat must be grown? Not just enough for 100 loaves — also enough for the flour that those loaves require, and enough to feed the farmers who grow the wheat for the flour, and enough for the millers who mill the wheat that feeds the farmers who…

The cascade of indirect requirements goes on forever. Each round is smaller than the last (you don't need as much farmer-bread as you need consumer-bread), so the total converges. But computing the total requires summing an infinite series.

This is exactly what the Leontief inverse does.

1.4.2 The Input-Output Matrix

Example 1.11 (A Three-Sector Economy)**.** *Let us assign numbers. To produce \$1 of output:*

- *Farming uses \$0.10 of its own output (seeds), \$0.05 from Milling, and \$0.10 from Baking.*
- *Milling uses \$0.20 from Farming, \$0.05 of its own output, and \$0.05 from Baking.*
- *Baking uses \$0.05 from Farming, \$0.20 from Milling, and \$0.10 of its own output.*

We organize this as a matrix where entry a_{ij} is the amount of good j needed to produce one unit of good i:

$$A = \begin{pmatrix} 0.10 & 0.05 & 0.10 \\ 0.20 & 0.05 & 0.05 \\ 0.05 & 0.20 & 0.10 \end{pmatrix}. \tag{1.21}$$

Row i tells you what sector i needs from each sector. Column j tells you who uses sector j's output.

Definition 1.12 (Technical Coefficients Matrix)**.** *The **input-output matrix** A is the $N \times N$ matrix where $a_{ij} \geq 0$ is the amount of good j required to produce one unit of good i.*

The material balance for each sector says: what is produced (y_i) must equal what is used as inputs by all sectors ($\sum_j a_{ji} y_j$, the total demand for i's output as intermediate inputs) plus what is consumed by final users (d_i, household consumption, exports, government):

$$y_i = \sum_{j=1}^{N} a_{ji} y_j + d_i. \tag{1.22}$$

Wait — read that carefully. The sum $\sum_j a_{ji} y_j$ uses a_{ji}, not a_{ij}. This is because sector j uses a_{ji} units of good i to produce y_j units. The total intermediate demand for good i is the sum over all sectors of how much each sector uses.

In matrix form, stacking all N equations:

$$\mathbf{y} = A'\mathbf{y} + \mathbf{d} \qquad \Leftrightarrow \qquad (I - A')\mathbf{y} = \mathbf{d}. \tag{1.23}$$

(Some textbooks define A with rows and columns transposed; we follow the convention where a_{ij} = "sector i needs good j," so the material balance uses A'. The reader should always check the convention.)

1.4.3 Why Infinite Rounds? The Cascade

Before solving the system, let us see what it means.

Example 1.13 (The Cascade for 100 Loaves)**.** *Suppose final demand is* $\mathbf{d} = (0, 0, 100)'$*: we want 100 units of Baking output and nothing else.*

Round 0 (direct): *We need 100 units of Baking. Output needed:* $(0, 0, 100)$.

Round 1 (first indirect): *To produce 100 units of Baking, we need* $100 \times (0.05, 0.20, 0.10)$*, that is,* $(5, 20, 10)$ *units of inputs. So Farming must produce 5 more, Milling 20 more, Baking 10 more. Total after round 1:* $(5, 20, 110)$.

Round 2: *The 5 extra Farming output needs inputs* $5 \times (0.10, 0.05, 0.10)$*, that is,* $(0.5, 0.25, 0.5)$. *The 20 extra Milling needs* $20 \times (0.20, 0.05, 0.05)$*, that is,* $(4, 1, 1)$. *The 10 extra Baking needs* $10 \times (0.05, 0.20, 0.10)$*, that is,* $(0.5, 2, 1)$. *Round 2 total:* $(5, 3.25, 2.5)$.

Each round is smaller. By round 5, the additions are tiny. The total converges to the solution of $(I - A')\mathbf{y} = \mathbf{d}$.

Intuition: Why the series converges

Each round of indirect effects is smaller because every sector needs less than one dollar of inputs per dollar of output: the total input requirement of sector i (the row sum of A, equivalently the column sum of A') is 0.25, 0.30 and 0.35 here. Small individual entries $a_{ij} < 1$ would not be enough; it is the row sums that matter. Because the row sums are below one, the cascade "leaks" value at each round — some of each round's output becomes final product rather than being recycled as input. The leak ensures convergence.

Technically, convergence requires that the **spectral radius** of A — the largest eigenvalue in magnitude — is less than 1. We explain eigenvalues in Appendix 1.9. For now: if the economy doesn't require more than a dollar of inputs to produce a dollar of output, the cascade converges.

1.4.4 The Leontief Inverse: Solving the System

Theorem 1.14 (The Leontief Inverse)**.** *If $\rho(A) < 1$ (the economy is **productive** — see Appendix 1.9 for what this means), then:*

$$\mathbf{y} = (I - A')^{-1}\mathbf{d}. \tag{1.24}$$

*The matrix $L = (I - A')^{-1}$ is the **Leontief inverse**. Its entry ℓ_{ij} gives the total output of sector i required to deliver one unit of good j to final demand, counting all direct and indirect effects.*

Intuition: What the Leontief inverse says in words

If you want one more loaf of bread (one more unit of final demand for Baking), how much extra wheat must be grown? Not just the direct input — also the wheat to feed the millers, the wheat to feed the farmers who grow the extra wheat, and so on. The Leontief inverse answers this question exactly: $\ell_{\text{Farming, Baking}}$ is the total Farming output needed for one unit of Baking final demand, including all indirect rounds.

The proof uses the *Neumann series*, which is the matrix version of the geometric series. We develop this in Appendix 1.9. Here we state the key formula:

$$(I - A')^{-1} = I + A' + (A')^2 + (A')^3 + \cdots = \sum_{k=0}^{\infty} (A')^k. \tag{1.25}$$

Proof. Since A' and A have the same eigenvalues, $\rho(A') = \rho(A) < 1$. By Gelfand's

formula (Horn and Johnson, 2012, §5.6) there are $C > 0$ and $r < 1$ with $\|(A')^k\| \leq Cr^k$ for all k (Appendix 1.9), so the partial sums $S_K = \sum_{k=0}^{K}(A')^k$ converge absolutely to a matrix S. Telescoping, $(I - A')S_K = S_K(I - A') = I - (A')^{K+1} \to I$, so $I - A'$ is invertible with $(I - A')^{-1} = S$. The material balance (1.23) therefore has the unique solution $\mathbf{y} = L\mathbf{d}$, and $\partial y_i / \partial d_j = \ell_{ij}$. Finally $L = \sum_k (A')^k \geq 0$ entrywise because $A \geq 0$, with $L \geq I$; the term $(A')^k$ collects the k-th round of indirect requirements. □

Round 0 is I (direct demand). Round 1 is A' (first-order indirect demand). Round 2 is $(A')^2$ (second-order). The Leontief inverse sums all rounds.

Example 1.15 (Computing the Leontief Inverse)**.** *For the matrix in Example 1.11, computing $(I - A')^{-1}$ numerically (do this with Python — see Exercise 1.7):*

$$L = (I - A')^{-1} \approx \begin{pmatrix} 1.14 & 0.25 & 0.12 \\ 0.09 & 1.08 & 0.25 \\ 0.13 & 0.09 & 1.14 \end{pmatrix}. \tag{1.26}$$

Reading column 3 (Baking): to deliver 1 unit of bread to final demand requires 1.14 units of Baking output (the 1 unit of bread plus 0.14 units used as input by Baking itself and by the sectors that supply it), 0.25 units of Milling output, and 0.12 units of Farming output. For the 100 loaves of Example 1.13, total output is $L\mathbf{d} = (11.78, 24.57, 113.79)'$.

*The **output multiplier** of each sector is the column sum of L:*

Sector	*Output multiplier*
Farming	$1.138 + 0.088 + 0.131 = 1.357$
Milling	$0.246 + 1.084 + 0.088 = 1.417$
Baking	$0.118 + 0.246 + 1.138 = 1.501$

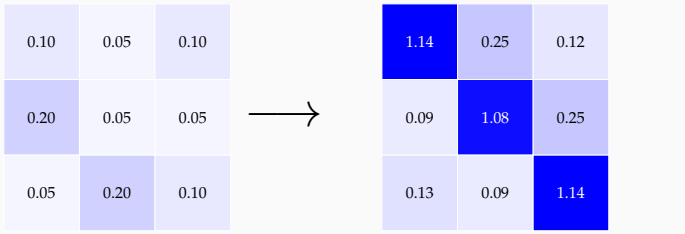
Baking has the highest multiplier: stimulating bread demand generates the most total economic activity, because bread production is deeply embedded in the input chain (it needs flour, which needs wheat, which needs food). Farming has the lowest: it has fewer backward linkages.

Key Structural Insight

The Leontief inverse is the first **structural multiplier** in this book. It shows that the effect of a change in one sector propagates through the entire economy, amplified by the network of inter-sectoral dependencies. The structure of A — which sectors depend on which — determines the pattern and magnitude of propagation. Two economies with identical technologies but different A matrices will respond differently to the same demand shock. The matrix A *is* the production structure.

In Chapter 16, we will apply this to the global economy, with countries as "sectors" and international trade as the input-output matrix. Whether the center-periphery structure that Prebisch described shows up in the column sums (backward linkages) of the global A is a hypothesis that chapter tests.

Visualization: The Leontief Inverse as a Heatmap



A **(inputs)** $L = (I - A')^{-1}$ **(total effects)**

Both panels use one colour scale, from 0 (white) to 1.14 (the largest entry of L). The input matrix A of Example 1.11 (left) is dim: weak direct dependencies. The Leontief inverse $L = (I - A')^{-1}$ (right; Example 1.15) is brighter — indirect effects add to direct ones. Its diagonal is brightest because each unit of final demand requires at least one unit of own output. Bright off-diagonal cells mean demand propagates across sectors; dark ones (a block-diagonal A) mean it stays localized.

1.5 General Equilibrium: Markets Clear

1.5.1 The Walrasian Equilibrium

We now combine consumption and production. The economy has N agents, L goods, prices p, and an allocation X.

Definition 1.16 (Walrasian Equilibrium)**.** *A* ***Walrasian equilibrium*** *is a price vector p^* and an allocation X^* such that:*

1. *Every agent optimizes: x_i^* solves (1.5) at prices p^*.*

2. *Markets clear:* $\mathbf{1}'X^* = \mathbf{1}'\Omega$ *(total demand equals total supply for each good).*

The equilibrium p^* is a *fixed point*: it is the price at which the demands generated by p^* are exactly consistent with p^*. Under standard conditions (continuous preferences, positive endowments), such a price exists — this is the content of the Arrow-Debreu existence theorem, proved using Brouwer's fixed-point theorem.

We will not reproduce the proof here (see Mas-Colell et al., 1995, Chapter 17). The textbook proof is written for agents who interact only through p, but existence survives other channels: Debreu's (1952) social-equilibrium theorem already covers consumption externalities. What externalities destroy is *efficiency*, not existence (Chapter 2). Chapter 7 returns to existence for the bargaining equilibrium used in this book.

1.6 The Implicit Weight Matrix

We are now ready for the chapter's punchline.

1.6.1 How Do Agents Interact in the Standard Model?

Consider the economy we have developed: N agents, each maximizing $U_i(x_i)$ subject to $p'x_i \leq w_i$, with market-clearing prices p^*. How does agent i affect agent j?

The answer: only through p^*. Agent i's demand for goods affects aggregate demand, which affects the equilibrium price, which affects j's budget set and optimal choice. The interaction is:

(a) **Anonymous**: the effect does not depend on who i is — only on i's demand.

(b) **Wealth-weighted**: in the aggregate, agent i's demand counts in proportion to her expenditure. In the Cobb-Douglas economy, aggregate demand for good ℓ is $\sum_i w_i/(2p_\ell)$, so agent i's weight is her **wealth share** $s_i = w_i/\sum_j w_j$. At the equilibrium of Section 1.2, $\mathbf{s} = (6.8, 5.8, 3.4)/16 = (0.425, 0.363, 0.213)$ — not $(1/3, 1/3, 1/3)$.

(c) **Indirect**: i does not enter j's utility function. The effect passes through the market.

(d) **Price-taking**: no single agent's choice moves p by a non-negligible amount; everyone takes the aggregate as given.

These four properties describe a very specific pattern of interaction. We can write it as a matrix.

1.6.2 Naming the Matrix

Definition 1.17 (The Market Interaction Matrix)**.** *Let* $\mathbf{s}$ *be the vector of wealth (expenditure) shares,* $s_i = w_i / \sum_j w_j$*. The* ***market interaction matrix*** *is*

$$W^{mkt} = \mathbf{1}\mathbf{s}'. \tag{1.27}$$

Every row is the same vector $\mathbf{s}'$*: each agent interacts with the market aggregate, in which agent* j *carries weight* s_j*. Applied to any vector,* $W^{mkt}\mathbf{x} = (\mathbf{s}'\mathbf{x})\mathbf{1}$ *replaces every entry by the wealth-weighted mean.*

Definition 1.18 (The Competitive Benchmark)**.** *With equal endowments,* $\mathbf{s} = \frac{1}{N}\mathbf{1}$ *and the market interaction matrix becomes the* ***competitive benchmark***

$$W^{CE} = \frac{1}{N}\mathbf{1}\mathbf{1}' = \frac{1}{N}\begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{pmatrix}. \tag{1.28}$$

Every entry is $1/N$*: each agent affects every other agent equally.*

W^{mkt} is what the Walrasian model implies; W^{CE} is its equal-endowment special case, and the benchmark this book measures departures from. It has remarkable properties, which we now catalog.

Proposition 1.19 (Properties of the Competitive Interaction Matrix)**.** $W^{CE} = \frac{1}{N}\mathbf{1}\mathbf{1}'$ *satisfies:*

1. ***Doubly stochastic****: every row sums to 1, and every column sums to 1. Nobody gives or receives more total influence than anyone else.*
2. ***Symmetric****:* $W^{CE} = (W^{CE})'$*. The effect of* i *on* j *equals the effect of* j *on* i*.*
3. ***Rank 1****: all rows are identical. The matrix can be written as the outer product of two vectors:* $W^{CE} = \frac{1}{N}\mathbf{1} \cdot \mathbf{1}'$*. This means* W^{CE} *contains no information about individual positions — every agent looks the same.*
4. ***Equal power****: its Perron vector (Section 1.7) is uniform,* $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$*.*
5. ***Unit spectral gap****:* W^{CE} *has eigenvalue 1 (with eigenvector* $\mathbf{1}$*) and eigenvalue 0 (with multiplicity* $N-1$*), so* $\Gamma = 1 - |\lambda_2| = 1$*. (See Appendix 1.9 for eigenvalues.)*

Properties (3) and (5) are shared by every rank-one row-stochastic matrix, including $W^{mkt} = \mathbf{1}\mathbf{s}'$ *for any* $\mathbf{s}$*. What singles out* W^{CE} *is (4): it is the only rank-one row-stochastic matrix with uniform* $\boldsymbol{\pi}$*.*

Proof. (1) $\sum_j W^{CE}_{ij} = \sum_j 1/N = 1$. Similarly for column sums.

(2) $(W^{CE})' = (1/N \cdot \mathbf{1}\mathbf{1}')' = 1/N \cdot \mathbf{1}\mathbf{1}' = W^{CE}$.

(3) Every row is $(1/N, 1/N, \ldots, 1/N)$, so all rows are proportional (in fact, identical). Rank $= 1$.

(4) $\frac{1}{N}\mathbf{1}'W^{CE} = \frac{1}{N^2}(\mathbf{1}'\mathbf{1})\mathbf{1}' = \frac{1}{N}\mathbf{1}'$, so $\frac{1}{N}\mathbf{1}$ is a stationary vector; it is the unique one because $W^{CE} > 0$ is irreducible (Perron–Frobenius, Chapter 6, Theorem 6.6). Uniqueness of W^{CE} among rank-one row-stochastic matrices with uniform $\boldsymbol{\pi}$: Section 1.7.

(5) $W^{CE}\mathbf{1} = \frac{1}{N}\mathbf{1}(\mathbf{1}'\mathbf{1}) = \frac{1}{N}\mathbf{1} \cdot N = \mathbf{1}$, so $\mathbf{1}$ is an eigenvector with eigenvalue 1. For any $\mathbf{x}$ perpendicular to $\mathbf{1}$ (i.e., $\mathbf{1}'\mathbf{x} = 0$): $W^{CE}\mathbf{x} = \frac{1}{N}\mathbf{1}(\mathbf{1}'\mathbf{x}) = 0$. So eigenvalue 0 has multiplicity $N - 1$. □

Intuition: What W^{CE} does to a vector

Apply W^{CE} to any vector $\mathbf{x}$:

$$W^{CE}\mathbf{x} = \frac{1}{N}\mathbf{1}(\mathbf{1}'\mathbf{x}) = \bar{x} \cdot \mathbf{1}, \tag{1.29}$$

where $\bar{x} = \frac{1}{N}\sum_i x_i$ is the mean.

The competitive benchmark **replaces every entry with the mean**. It destroys all information about individual positions and keeps only the aggregate. The market matrix W^{mkt} does the same with the wealth-weighted mean $\mathbf{s}'\mathbf{x}$: individual identities are washed out by the price mechanism, and only the aggregate (total demand, total supply) determines the outcome. Wealth, however, is not washed out.

1.6.3 The Autarky Matrix

The other extreme is no interaction at all:

$$W^A = I = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}. \tag{1.30}$$

Each agent interacts only with herself. No trade, no externalities, no social contact. Robinson Crusoe, N times. As a weight matrix, I is reducible: every probability vector is stationary, so power $\boldsymbol{\pi}$ is not defined, and $\Gamma = 0$.

1.6.4 The Space of All Possible Structures

Between $W = I$ (autarky) and $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$ (anonymous competitive market) lies a vast space of possible interaction structures. Most real economies live somewhere in this space:

	Autarky $W = I$	**General** W	**Competitive** $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$
Interaction	None	Structured	Anonymous
Power $\boldsymbol{\pi}$	Undefined (reducible)	$\boldsymbol{\pi}(W)$, computed	$\mathbf{s}$ ($= \frac{1}{N}\mathbf{1}$ if equal endowments)
Utility interdependence	None	$W^U \neq 0$ possible	$W^U = 0$
Welfare theorems	Fail (no gains from trade)	FWT fails if $W^U \neq 0$	Hold, for any $\mathbf{s}$

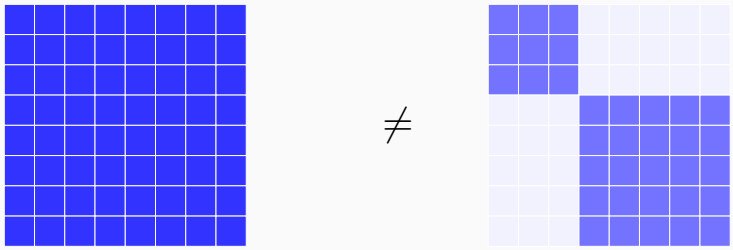
Key Structural Insight

The program of this book is now visible. The benchmark competitive model lives at the right edge of this table. Its interaction is anonymous and rank one, $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$, with no utility interdependence; with equal endowments it is W^{CE}. The First Welfare Theorem holds for *any* wealth distribution $\mathbf{s}$; what it needs, in our language, is $W^U = 0$ together with complete markets and price-taking. The model does not assume equal power — it assumes that power does not enter, because nobody bargains.

Structural economics replaces anonymous interaction with a general W: non-anonymous, possibly fragmented (slow integration, small Γ), and with a power distribution $\boldsymbol{\pi}$ that the network produces. Power becomes endogenous (Chapter 6). Preferences become endogenous to the network (Chapter 10). Externalities break efficiency (Chapter 2); unequal power is distributional, and the book asks how it is produced and changed. The question shifts from "is the equilibrium efficient?" to "what is the structure, who does it benefit, and can it be changed?"

The standard model is not wrong. It is a special case. This book is the general case.

Visualization: W^{CE} vs. the real world



$W^{CE} = \frac{1}{N}\mathbf{1}\mathbf{1}'$
$\Gamma = 1$, $N_{\text{eff}} = 8$

Stylized dual economy [ILLUSTRATIVE]
$\Gamma \approx 0.16$, $N_{\text{eff}} \approx 7.9$

Left: the competitive benchmark (Prologue pattern 2, "Market"). Right: a stylized dual economy (pattern 6, "Dual Economy"; cell weights 70, 55 and 5, rows normalized), not an estimate for any country. Its two blocks are weakly coupled, so it integrates slowly ($\Gamma \approx 0.16$), yet power is almost uniform ($N_{\text{eff}} \approx 7.9$ of 8). Weak integration and power concentration are different properties (Section 1.7); the book studies both.

1.7 Conventions Used Throughout the Book

The rest of the book uses the objects introduced above in a fixed way. This section states the conventions once; later chapters cite it rather than restating them.

1.7.1 Matrices and Normalization

Interaction matrices are row-stochastic. An **interaction (weight) matrix** is a nonnegative $N \times N$ matrix W with $W\mathbf{1} = \mathbf{1}$; W_{ij} is the weight agent i places on agent j. This covers the information network W^I, the utility network W^U, composites, and any generic W whose Perron vector is read as power. A raw nonnegative matrix K (counts, flows, overlaps) is turned into a weight matrix by row normalization, $W = D^{-1}K$ with $D = \text{diag}(K\mathbf{1})$; we say so whenever we do it.

Production uses two different objects.

- The **technical-coefficient matrix** A of Definition 1.12: a_{ij} is the amount of good j needed per unit of good i, so row i is sector i's recipe. The balance is $\mathbf{y} = A'\mathbf{y} + \mathbf{d}$ and the Leontief inverse is $L = (I - A')^{-1}$, which requires $\rho(A) < 1$. Output multipliers and backward/forward linkages are computed from A (or a social-accounting coefficient matrix), never from a row-normalized matrix.

- The **production weight matrix** $W^P := D^{-1}A$ with $D = \operatorname{diag}(A\mathbf{1})$: row i gives the input-cost shares of sector i. It is row-stochastic, and it is the input-share matrix of Acemoglu et al. (2012a). It is the object whose Perron vector and spectral gap we compute.

The reason for the separation: for *any* row-stochastic W and $0 \leq \gamma < 1$, $(I - \gamma W)\mathbf{1} = (1-\gamma)\mathbf{1}$, hence $(I - \gamma W)^{-1}\mathbf{1} = \frac{1}{1-\gamma}\mathbf{1}$. All row-sum "linkages" of a row-normalized matrix coincide and carry no information.

Composites. A composite $W = \sum_k \alpha_k W^k$ requires $\alpha_k \geq 0$, $\sum_k \alpha_k = 1$, and every layer W^k row-stochastic on the *same* node set. Layers defined on different node sets (sectors, media outlets, individuals) do not form a composite.

1.7.2 Power: the Perron Vector

Definition 1.20 (Perron Vector)**.** *For an irreducible row-stochastic W, the **Perron vector** $\boldsymbol{\pi}$ is the unique vector with $\boldsymbol{\pi}'W = \boldsymbol{\pi}'$, $\boldsymbol{\pi} > 0$ and $\sum_i \pi_i = 1$. (Existence and uniqueness: Perron–Frobenius, Chapter 6.)*

Each chapter states which of three readings of $\boldsymbol{\pi}$ is in force:

1. **Information (DeGroot).** If beliefs update as $\mathbf{b}^{(t+1)} = W\mathbf{b}^{(t)}$, the consensus is $b^* = \boldsymbol{\pi}'\mathbf{b}^{(0)}$: π_j is exactly agent j's weight in the long-run consensus. If W is also aperiodic, $W^t \to \mathbf{1}\boldsymbol{\pi}'$ and the beliefs converge to it.

2. **Production.** $\boldsymbol{\pi}' = \lim_{\gamma\to 1} \frac{1-\gamma}{N}\mathbf{1}'(I - \gamma W^P)^{-1}$, the limit of the normalized influence vector. For $\gamma < 1$, the influence vector of Acemoglu et al. (2012a), $\frac{\alpha}{N}\mathbf{1}'(I - (1-\alpha)W^P)^{-1}$, is a Katz/Leontief object, not the Perron vector.

3. **Bargaining.** Using $\boldsymbol{\pi}$ as the Nash weights $\boldsymbol{\lambda}$ of Definition 1.7 is a **modelling assumption**, not a result. In strategic models of bargaining on networks (Manea, 2011; Abreu and Manea, 2012a,b), payoffs are generally *not* eigenvector centralities.

For the market matrix, $\boldsymbol{\pi}(W^{\text{mkt}}) = \mathbf{s}$: since $\mathbf{s}'\mathbf{1}\mathbf{s}' = \mathbf{s}'$, wealth shares are the market's "power" vector.

Measuring concentration. Unequal power means a concentrated $\boldsymbol{\pi}$. We measure it by:

- the **effective number of agents** $N_{\text{eff}} = 1/\sum_i \pi_i^2 \in [1, N]$;

- the **variance of power** $\mathrm{Var}(\pi) = \frac{1}{N}\sum_i(\pi_i - \frac{1}{N})^2 = \frac{1}{N}\big(\frac{1}{N_{\text{eff}}} - \frac{1}{N}\big)$, which uses $\sum_i(\pi_i - \frac{1}{N})^2 = \sum_i \pi_i^2 - \frac{1}{N}$. Its maximum is $(N-1)/N^2$, attained at $\boldsymbol{\pi} = e_1$: 0.1875 for $N = 4$, 0.0475 for $N = 20$;

- the **power distance** $d_\pi = \frac{1}{2}\|\boldsymbol{\pi} - \frac{1}{N}\mathbf{1}\|_1 \in [0, 1 - \frac{1}{N}]$, and the power Gini.

In the Cobb-Douglas equilibrium of Section 1.2, $N_{\text{eff}}(\mathbf{s}) = 2.80$ of 3.

Proposition 1.21 (When Power Is Uniform)**.** *For an irreducible row-stochastic W, $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$ if and only if W is doubly stochastic.*

Proof. If $\mathbf{1}'W = \mathbf{1}'$, then $\frac{1}{N}\mathbf{1}'$ is a stationary probability vector, and by uniqueness of the stationary vector of an irreducible stochastic matrix (Perron–Frobenius; Horn and Johnson 2012, Thm. 8.4.4) it equals $\boldsymbol{\pi}'$. Conversely, $\frac{1}{N}\mathbf{1}'W = \frac{1}{N}\mathbf{1}'$ says every column sums to one. □

For a nonnegative W that is *not* row-stochastic, the condition is "all column sums equal", which is why we avoid non-stochastic W when $\boldsymbol{\pi}$ is read as power.

1.7.3 The Spectral Gap Measures Integration

Definition 1.22 (Spectral Gap)**.** *For a row-stochastic W with eigenvalues ordered $1 = |\lambda_1| \geq |\lambda_2| \geq \cdots$, the **spectral gap** is $\Gamma(W) = 1 - |\lambda_2(W)|$.*

Γ measures **integration**: deviations from consensus (or from the stationary distribution) decay like $|\lambda_2|^t$, with half-life $\ln 2/(-\ln|\lambda_2|)$. $\Gamma = 0$ when W is reducible (several unit eigenvalues, as for $W = I$) *or* periodic (an eigenvalue of modulus one other than 1, as for a bipartite star or an even ring without self-loops); the lazy version $\frac{1}{2}(I + W)$ removes periodicity. For reversible W with conductance Φ_c, Cheeger's inequality $\Phi_c^2/2 \leq 1 - \lambda_2 \leq 2\Phi_c$ (Levin et al., 2009, Thm. 13.14) says that weak cross-block coupling implies a small gap. The converse fails: a small gap can come from periodicity rather than fragmentation.

Γ **does not measure power.** For every $\boldsymbol{\pi}$ in the open simplex and every $\gamma \in (0, 1]$, the matrix $W = \gamma\,\mathbf{1}\boldsymbol{\pi}' + (1-\gamma)I$ is irreducible, aperiodic and row-stochastic, with $\boldsymbol{\pi}(W) = \boldsymbol{\pi}$ (because $\boldsymbol{\pi}'W = \gamma\boldsymbol{\pi}' + (1-\gamma)\boldsymbol{\pi}'$) and spectrum $\{1, 1-\gamma, \ldots, 1-\gamma\}$, so $\Gamma(W) = \gamma$. Any degree of concentration is compatible with any gap (Chapter 6, Proposition 6.16). With $N = 10$ and $\pi_1 = 0.91$, $N_{\text{eff}} = 1.21$ for every γ, while Γ ranges over $(0, 1]$. The stylized dual economy in the visualization above makes the opposite point: small Γ with almost uniform power.

1.7.4 Distance from the Benchmark

We write $\Delta(W) := W - W^{CE}$ for the departure from the benchmark and summarize it by **two** coordinates, which the example above shows to be independent: the power distance d_π (or N_{eff}/N) and the integration deficit $1 - \Gamma$.

Two facts about the benchmark. Every rank-one row-stochastic matrix has the form $\mathbf{1}\boldsymbol{\pi}'$ (if $W = \mathbf{u}\mathbf{v}'$ and $W\mathbf{1} = \mathbf{1}$, then $\mathbf{u} \propto \mathbf{1}$), and its Perron vector is $\boldsymbol{\pi}$. Hence W^{CE} is the *unique* rank-one row-stochastic matrix with uniform power. No scalar norm of $\Delta(W)$ determines welfare: outcomes depend on W *and* on primitives (curvature, endowments, the game being played) that are not functions of $\Delta(W)$.

1.7.5 The Competitive Benchmark and the Welfare Theorems

The statement used throughout the book is:

- Walrasian price formation is anonymous. Aggregate demand weights agents by wealth shares $\mathbf{s}$, so the implied interaction matrix is $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$: rank one, $\Gamma = 1$, $\boldsymbol{\pi} = \mathbf{s}$. The benchmark $W^{CE} = \frac{1}{N}\mathbf{1}\mathbf{1}'$ is the equal-endowment special case.

- The First Welfare Theorem holds for *any* $\mathbf{s}$. It requires local nonsatiation, complete markets, price-taking and no externalities — in this book's language, $W^U = 0$.

- Utility interdependence ($W^U \neq 0$) and missing markets break efficiency. Non-anonymous interaction with unequal $\boldsymbol{\pi}$ is **distributional**: weighted Nash bargaining is Pareto efficient (Proposition 1.9), and the associated loss is a utilitarian loss, not a Pareto inefficiency.

- Autarky ($W = I$) is not a case in which the welfare theorems "trivially hold": the gains from trade go unrealized.

The textbook benchmark is thus anonymous interaction with no externalities — in its equal-endowment form, $W = W^{CE}$. Structural economics studies non-anonymous W (its structure: blocks, integration Γ) and the distribution of power $\boldsymbol{\pi}$, and how both are produced and changed.

1.7.6 The Allocation Rule and the Calibration of Utility

Proposition 1.23 (The Allocation Rule)**.** *In the transferable-utility bargaining problem of Definition 1.8, let the weights be $\boldsymbol{\lambda} = \boldsymbol{\pi}$, the Perron vector of an irreducible row-stochastic W (so $\pi_i > 0$), and let all disagreement payoffs be equal, $d_i = \bar{d}$. Define the*

bargaining surplus $\Phi := \Omega - N\bar{d}$ *and assume* $\Phi > 0$*. Then*

$$x_i = \frac{\Omega}{N} + \Big(\pi_i - \frac{1}{N}\Big)\Phi. \tag{1.31}$$

If utility is $U(x) = x - \frac{\kappa}{2}x^2$*, the utilitarian loss relative to equal division is exactly*

$$L_{power} := \sum_i \Big[U\big(\tfrac{\Omega}{N}\big) - U(x_i)\Big] = \frac{\kappa N}{2}\,\Phi^2\,\mathrm{Var}(\pi). \tag{1.32}$$

Proof. The hypotheses of Proposition 1.9 hold ($\pi_i > 0$, $\Omega - \sum_j d_j = \Phi > 0$), so by (1.20), $x_i = \bar{d} + \pi_i\Phi = \frac{\Omega}{N} + (\pi_i - \frac{1}{N})\Phi$, since $\bar{d} = \frac{\Omega}{N} - \frac{\Phi}{N}$. Write $m = \Omega/N$ and $z_i = x_i - m$, so $\sum_i z_i = 0$. Because U is quadratic, $U(m + z_i) = U(m) + U'(m)z_i - \frac{\kappa}{2}z_i^2$ exactly. Summing, the linear terms cancel and $L_{\text{power}} = \frac{\kappa}{2}\sum_i z_i^2 = \frac{\kappa}{2}\Phi^2\sum_i(\pi_i - \frac{1}{N})^2 = \frac{\kappa N}{2}\Phi^2\,\mathrm{Var}(\pi)$. □

Chapter 7 derives the same rule in the network setting (Proposition 7.3). Two cautions. First, Φ is the *surplus over disagreement*, not a curvature parameter. In a one-good exchange economy with endowments as the disagreement point there is nothing to bargain over and $\Phi = 0$. Second, every allocation used in an example must satisfy $\kappa x_i < 1$, so that utility is increasing. Past the satiation point $1/\kappa$, "getting more" makes an agent worse off and the welfare comparisons become meaningless. For example, with $\Omega = 3$, $\bar{d} = 0.3$ (so $\Phi = 2.1$), $\boldsymbol{\pi} = (0.6, 0.2, 0.2)$ and $\kappa = 0.5$: $\mathbf{x} = (1.56, 0.72, 0.72)$, $\mathrm{Var}(\pi) = 0.0356$, and $L_{\text{power}} = 0.75 \times 4.41 \times 0.0356 = 0.118$.

1.8 Summary

This chapter established five results, all of them standard economics expressed in matrix notation:

1. The **allocation matrix** X organizes who gets what. Read by rows: individual bundles. Read by columns: distribution of each good. The rank of X measures structural differentiation.

2. The **consumer's problem** determines each row of X independently under the standard model. Agent i's utility depends only on x_i. This is a structural assumption: the weight agent i places on agent j's consumption is zero.

3. The **welfare loss from inequality** in the one-good economy is $L = \frac{\kappa N}{2}\mathrm{Var}(\omega_i)$. With Nash bargaining over a joint surplus, $x_i = d_i + \lambda_i(\Omega - \sum_j d_j)$: the allo-

cation depends on who has power, not just on who has outside options. The resulting loss is distributional, not a Pareto inefficiency.

4. The **Leontief inverse** $(I - A')^{-1}$ is the first structural matrix in standard economics. It encodes all direct and indirect inter-sectoral dependencies. Its properties (column sums, spectral radius) characterize the production structure.

5. The **implicit interaction matrix** of the competitive model is $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$: anonymous, rank one, $\Gamma = 1$, with power equal to wealth shares; its equal-endowment case is the benchmark $W^{CE} = \frac{1}{N}\mathbf{1}\mathbf{1}'$. The First Welfare Theorem holds for any $\mathbf{s}$ provided $W^{U} = 0$. Power concentration (N_{eff}, $\text{Var}(\pi)$) and integration (Γ) are independent coordinates of a general W (Section 1.7).

No new economics was introduced. But the notation reveals an object — the weight matrix W — that was always present but never named. The next chapters will show what changes when W is not anonymous.

1.9 Exercises

1.1 Reading the allocation matrix. For $N = 4$ agents and $L = 3$ goods (Food, Clothing, Housing), the allocation matrix is:

$$X = \begin{pmatrix} 5 & 3 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 5 \\ 4 & 2 & 2 \end{pmatrix}. \tag{1.33}$$

(a) What is agent 3's consumption bundle? What is agent 3 likely "good at" (what might she produce)?

(b) What is the total consumption of Housing? Which agent consumes the most Housing?

(c) Is this allocation feasible if total endowments are $\mathbf{1}'\Omega = (12, 10, 10)$?

(d) Compute the rank of X. Is this a "structurally differentiated" economy in the sense discussed above?

1.2 Cobb-Douglas equilibrium. Three agents with $U_i = x_{i1}^{1/3} x_{i2}^{2/3}$ and endowments $\omega_1 = (4, 1)'$, $\omega_2 = (1, 3)'$, $\omega_3 = (3, 2)'$.

(a) Derive the Marshallian demand functions.

(b) Find the equilibrium price p_2^* (normalize $p_1 = 1$) by solving the market-clearing condition.

(c) Compute the equilibrium allocation X^* and verify feasibility.

1.3 Welfare loss computation. For $N = 5$, $\kappa = 0.1$, $\Omega = 5$ (check that $\kappa\omega_i < 1$ in every case):

(a) Compute L for $\boldsymbol{\omega} = (1, 1, 1, 1, 1)$. (Answer: 0.)

(b) Compute L for $\boldsymbol{\omega} = (3, 1, 0.5, 0.25, 0.25)$.

(c) Compute L for $\boldsymbol{\omega} = (5, 0, 0, 0, 0)$ (one person has everything).

(d) Plot L as a function of the Gini coefficient of $\boldsymbol{\omega}$.

1.4 Bargaining with different power. Two agents ($N = 2$, $\kappa = 0.5$) produce $\Omega = 2$ together; their disagreement payoffs are $\mathbf{d} = (0.6, 0.2)$. Use Proposition 1.9.

(a) Find the first-best (utilitarian) allocation and welfare.

(b) Find the bargaining allocation for $\boldsymbol{\lambda} = (0.5, 0.5)$ (equal power). Compare to the first-best.

(c) Find the bargaining allocation for $\boldsymbol{\lambda} = (0.8, 0.2)$ (asymmetric power). Who benefits?

(d) For what $\boldsymbol{\lambda}$ does the bargaining allocation equal the first-best?

1.5 The cascade in action. For the input-output matrix:

$$A = \begin{pmatrix} 0.2 & 0.1 \\ 0.3 & 0.1 \end{pmatrix}. \tag{1.34}$$

(a) Compute the first five terms of the Neumann series: I, A', $(A')^2$, $(A')^3$, $(A')^4$. (Use a calculator or Python.)

(b) Sum them: $\sum_{k=0}^{4} (A')^k$. Compare with the exact $(I - A')^{-1}$. How close are they?

(c) What is the spectral radius $\rho(A)$? (See Appendix 1.9.) Verify $\rho(A) < 1$.

(d) Compute output multipliers (column sums of $(I - A')^{-1}$). Which sector has stronger backward linkages?

1.6 Properties of W^{CE}.

(a) Show that W^{CE} is idempotent: $(W^{CE})^2 = W^{CE}$. (Hint: compute $\frac{1}{N}\mathbf{1}\mathbf{1}' \cdot \frac{1}{N}\mathbf{1}\mathbf{1}'$.)

(b) Show that for any vector $\mathbf{x}$, $W^{CE}\mathbf{x} = \bar{x}\mathbf{1}$. Interpret: the competitive market replaces individual variation with the average.

(c) Prove: W^{CE} has eigenvalue 1 with eigenvector $\mathbf{1}$, and eigenvalue 0 with multiplicity $N - 1$. Find the eigenvectors for eigenvalue 0 when $N = 3$.

(d) For $N = 3$, consider the matrix $W = \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 2 & 0.5 & 0.5 \\ 0 & 1.5 & 1.5 \end{pmatrix}$. Is this doubly stochastic? Compute $W\mathbf{x}$ for $\mathbf{x} = (3, 6, 9)'$ and compare with $W^{CE}\mathbf{x}$. What is different?

1.7 Bolivia's input-output table (data exercise). Download the Bolivian input-output or supply-use tables from INE (Bolivia is not a separate economy in the OECD ICIO tables; multi-region alternatives that include it are Eora and CEPAL's regional input-output matrices). Aggregate to 5–10 sectors.

(a) Construct A in this book's convention: a_{ij} = purchases by sector i from sector j divided by sector i's gross output. (Published tables usually record sales from row i to column j and divide each *column* by gross output; that matrix is A' in our notation.)

(b) Compute the Leontief inverse $(I - A')^{-1}$.

(c) Compute output multipliers for: Mining/Gas, Agriculture, Manufacturing, Services.

(d) Which sector has the highest multiplier? Lowest? Relate to Pinto's structural heterogeneity: is Bolivia's economy "structurally connected" or "block-diagonal"?

(e) *Challenge*: Is the gas sector connected to the rest of the economy, or is it an enclave?

1.8 From story to matrix. Doña María sells api morado on Calle Comercio in La Paz. To produce api, she buys corn from a wholesaler in El Alto, sugar from a shop on Calle Max Paredes, cinnamon from a market distributor, and charcoal from a supplier in the altiplano. The wholesaler buys from farmers in the Yungas.

(a) Define 5 agents: María, Wholesaler, Sugar Shop, Cinnamon Distributor, Charcoal Supplier. Draw the directed dependency graph.

(b) Write the 5×5 adjacency matrix A ($A_{ij} = 1$ if agent i uses agent j's output as input, 0 otherwise).

(c) Is A symmetric? What does the asymmetry mean economically?

(d) Suppose the Wholesaler disappears. Set row and column 2 of A to zero. Which other agents are directly affected? Which are indirectly affected (via the Leontief cascade)?

(e) Compare this adjacency matrix to W^{CE} for $N = 5$. What structural information does A contain that W^{CE} destroys?

1.9 From sectors to countries. Suppose the "agents" in the Leontief system are 3 countries:

$$A = \begin{pmatrix} 0.05 & 0.30 & 0.30 \\ 0.01 & 0.05 & 0.05 \\ 0.01 & 0.05 & 0.05 \end{pmatrix}. \tag{1.35}$$

Column 1 shows who needs Country 1's output: $0.05, 0.01, 0.01$ — it is barely used by others. Row 1 shows that Country 1 needs a lot from Countries 2 and 3 (0.30 each).

(a) Compute the output multipliers (column sums of $(I - A')^{-1}$). Which country has the strongest backward linkages?

(b) If final demand rises by 1 unit in Country 2 only, how much total output is generated in each country?

(c) In Prebisch's account the periphery exports primary goods to the center and imports manufactures *from* it. Does this A have that pattern, or its reverse? Modify A so that Countries 2 and 3 depend on Country 1's output, and recompute (a).

(d) Argue informally that the center-periphery structure is a property of the asymmetry of A.

Mathematical Appendices

Appendix 1A: Matrices as Economic Objects

This appendix covers the matrix concepts used in Chapter 1, starting from scratch.

What is a matrix? A matrix is a rectangular table of numbers. An $N \times L$ matrix has N rows and L columns. In this book, matrices represent *relationships*: the entry in row i, column j describes the relationship between agent (or sector) i and agent (or sector) j.

Matrix	What it describes
Allocation X ($N \times L$)	How much of each good each person consumes
Endowment Ω ($N \times L$)	How much of each good each person starts with
Input-output A ($N \times N$)	How much of each sector's output each sector needs
Interaction W ($N \times N$)	How much each agent affects each other agent

Special vectors. Two vectors appear constantly:

- $\mathbf{1} = (1, 1, \ldots, 1)'$: the vector of all ones. $\mathbf{1}'x = \sum_i x_i$ sums the entries of x.
- e_k: the k-th standard basis vector (1 in position k, 0 elsewhere). $e_k'x = x_k$ picks out the k-th entry.

Matrix multiplication as "flow through a system." If A is $N \times N$ and $\mathbf{x}$ is $N \times 1$, then $A\mathbf{x}$ is $N \times 1$ with $(A\mathbf{x})_i = \sum_j a_{ij}x_j$. Think of x_j as the "amount of stuff at node j" and a_{ij} as "how much of node j's stuff flows to node i." The product $A\mathbf{x}$ is the result of flowing everything through the system once.

Flow through twice: $A(A\mathbf{x}) = A^2\mathbf{x}$. Flow k times: $A^k\mathbf{x}$. The Leontief inverse $(I - A)^{-1} = I + A + A^2 + \cdots$ sums the result of flowing through all possible numbers of times.

The outer product $\mathbf{1}\mathbf{1}'$. This is an $N \times N$ matrix of all ones:

$$\mathbf{1}\mathbf{1}' = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & \cdots & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{pmatrix}. \tag{1.36}$$

Dividing by N gives $W^{CE} = \frac{1}{N}\mathbf{1}\mathbf{1}'$, where every entry is $1/N$. This matrix has a beautiful property: $W^{CE}\mathbf{x} = \bar{x}\mathbf{1}$ — it replaces every entry of $\mathbf{x}$ with the mean. It is the mathematical representation of "anonymous, symmetric interaction."

The identity matrix I. The $N \times N$ identity matrix has 1 on the diagonal and 0 elsewhere. $I\mathbf{x} = \mathbf{x}$ for any $\mathbf{x}$: it does nothing. It represents "no interaction" — each agent affects only herself.

Row sums and column sums. For an $N \times N$ matrix M:

- Row sums: $M\mathbf{1}$ is an $N \times 1$ vector whose i-th entry is $\sum_j m_{ij}$ (total outflow from i).
- Column sums: $\mathbf{1}'M$ is a $1 \times N$ vector whose j-th entry is $\sum_i m_{ij}$ (total inflow to j).

A matrix is **row-stochastic** if all row sums equal 1: $M\mathbf{1} = \mathbf{1}$. It is **column-stochastic** if all column sums equal 1: $\mathbf{1}'M = \mathbf{1}'$. It is **doubly stochastic** if both. The competitive interaction matrix W^{CE} is doubly stochastic.

Appendix 1B: Eigenvalues and the Spectral Radius

The key idea. An eigenvalue is a number that tells you how a matrix *stretches* things. If $A\mathbf{v} = \lambda\mathbf{v}$ for some nonzero vector $\mathbf{v}$, then λ is an eigenvalue and $\mathbf{v}$ is the corresponding eigenvector. Applying A to $\mathbf{v}$ just rescales it by λ — the direction stays the same, only the magnitude changes.

Why economists care. Eigenvalues tell you whether a system explodes, converges, or oscillates:

- $|\lambda| < 1$: applying A repeatedly shrinks the component along $\mathbf{v}$. The system converges.
- $|\lambda| = 1$: the component stays the same. Steady state.
- $|\lambda| > 1$: the component grows without bound. Explosion.

The spectral radius. The **spectral radius** of A is $\rho(A) = \max_i |\lambda_i|$ — the largest eigenvalue in magnitude. It determines the long-run behavior of the system:

- $\rho(A) < 1$: $A^k \to 0$ as $k \to \infty$. The cascade of indirect effects dies out. The Leontief inverse exists.

- $\rho(A) = 1$: the system is on the boundary. Convergence is borderline.
- $\rho(A) > 1$: A^k explodes. The system requires more inputs than it produces. The economy is "unproductive."

For the input-output matrix, $\rho(A) < 1$ means the economy is productive: it produces more output than it uses as input.

Computing eigenvalues. For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the eigenvalues solve $\det(A - \lambda I) = 0$:

$$\lambda^2 - (a+d)\lambda + (ad - bc) = 0. \tag{1.37}$$

The solutions are $\lambda = \frac{(a+d) \pm \sqrt{(a+d)^2 - 4(ad-bc)}}{2}$. For larger matrices, use numerical software.

Example. For $A = \begin{pmatrix} 0.2 & 0.1 \\ 0.3 & 0.1 \end{pmatrix}$ (from Exercise 1.5): trace $= 0.3$, determinant $= 0.02 - 0.03 = -0.01$. Eigenvalues: $\lambda = \frac{0.3 \pm \sqrt{0.09 + 0.04}}{2} = \frac{0.3 \pm 0.36}{2}$. So $\lambda_1 = 0.33$ and $\lambda_2 = -0.03$. Spectral radius: $\rho(A) = 0.33 < 1$. The economy is productive.

The spectral gap. For a row-stochastic W the largest eigenvalue is $\lambda_1 = 1$, and the **spectral gap** is $\Gamma = 1 - |\lambda_2|$, where λ_2 is the eigenvalue of second-largest modulus (Definition 1.22). The spectral gap measures *integration* — how fast the system forgets its starting point:

- Large Γ: deviations from the long-run state die out quickly (like $|\lambda_2|^t$). Every rank-one matrix $\mathbf{1}\boldsymbol{\pi}'$, including W^{CE}, has $\Gamma = 1$.
- Small Γ: slow convergence. This happens when the network is split into weakly coupled blocks, or when it is periodic (a bipartite star or an even ring without self-loops has $\Gamma = 0$).

The spectral gap says nothing about how concentrated power is: Section 1.7 shows that every $\boldsymbol{\pi}$ is compatible with every Γ. Chapter 6 develops these connections formally. For now: eigenvalues measure how a system amplifies or dampens, and the spectral gap measures how fast it integrates.

Appendix 1C: The Neumann Series

Start with scalars. You know the geometric series: if $|r| < 1$, then

$$1 + r + r^2 + r^3 + \cdots = \frac{1}{1-r}. \tag{1.38}$$

Rearranging: $(1-r)(1+r+r^2+\cdots)=1$, so $1+r+r^2+\cdots$ is the *inverse* of $(1-r)$.

Now matrices. Replace the scalar r with a matrix A and the number 1 with the identity matrix I:

$$I+A+A^2+A^3+\cdots=(I-A)^{-1}, \tag{1.39}$$

provided the series converges. The condition for convergence is $\rho(A)<1$ (the spectral radius is less than 1) — the matrix analog of $|r|<1$.

Why it converges. The entries of A^k are *not* bounded by $\rho(A)^k$ in general (for $A=\begin{pmatrix}0.5 & 0.9\\ 0 & 0.5\end{pmatrix}$, $\rho=0.5$ but $a_{12}=0.9$). What is true is Gelfand's formula, $\|A^k\|^{1/k}\to\rho(A)$: for any $\varepsilon>0$ there is a constant C with $\|A^k\|\le C(\rho(A)+\varepsilon)^k$. Since $\rho(A)<1$, choose ε with $\rho(A)+\varepsilon<1$; the powers decay geometrically and $A^k\to 0$ as $k\to\infty$. The partial sums $\sum_{k=0}^{K}A^k$ converge to a finite matrix, which is $(I-A)^{-1}$.

Verification. Multiply a partial sum by $(I-A)$:

$$(I-A)(I+A+\cdots+A^K)=(I+A+\cdots+A^K)-(A+A^2+\cdots+A^{K+1}) \tag{1.40}$$

$$=I-A^{K+1}. \tag{1.41}$$

The sum telescopes. Since $A^{K+1}\to 0$, letting $K\to\infty$ gives $(I-A)\sum_{k=0}^{\infty}A^k=I$ (and likewise with the factors in the other order). This confirms that $\sum_{k=0}^{\infty}A^k$ is indeed the inverse of $(I-A)$.

Economic interpretation. In the Leontief system:

- I: direct demand (round 0). Each unit of final demand requires one unit of direct output.
- A': first-round indirect demand. Producing the direct output requires A' units of intermediate inputs.
- $(A')^2$: second-round indirect demand. Producing the first-round inputs requires further inputs.
- $(A')^k$: k-th round. The cascade continues.

- $(I - A')^{-1}$: the total of all rounds. The complete direct + indirect requirement.

A numerical example. For $A' = \begin{pmatrix} 0.2 & 0.3 \\ 0.1 & 0.1 \end{pmatrix}$:

	$(1,1)$	$(1,2)$	$(2,1)$	$(2,2)$
I	1.000	0.000	0.000	1.000
$I + A'$	1.200	0.300	0.100	1.100
$I + A' + (A')^2$	1.270	0.390	0.130	1.140
$I + A' + \cdots + (A')^3$	1.293	0.420	0.140	1.153
$I + A' + \cdots + (A')^4$	1.301	0.430	0.143	1.157
$(I - A')^{-1}$ (exact)	1.304	0.435	0.145	1.159

By round 4, every entry of the partial sum is within about 1.1% of the exact inverse. The cascade converges fast when $\rho(A)$ is small.

Structuralist Case Study: Prebisch's 1949 Manifesto and the Weight Matrix

In 1949, Raúl Prebisch — an Argentine economist who had served as head of his country's central bank — published what Albert Hirschman later called the "Latin American Manifesto": *El desarrollo económico de la América Latina y algunos de sus principales problemas*. The essay served as the introduction to CEPAL's first *Estudio económico de América Latina* and laid the intellectual foundation for half a century of structuralist economics.

Prebisch's argument has three steps, each of which corresponds to a property of the weight matrix developed in this chapter.

Step 1: The global economy has a structure. Prebisch rejected the presumption that the gains from trade and technical progress are spread evenly across trading partners. He argued that the world economy is divided into an industrialized "center" and a primary-commodity-exporting "periphery," connected by trade linkages that are fundamentally *asymmetric*: the periphery exports raw materials with low value-added and imports manufactures with high value-added. In the notation of this chapter, the hypothesis is that the global input-output matrix A_{global} has a core-periphery topology: the center's column sums in the Leontief inverse $(I - A')^{-1}$ are large (strong backward linkages), and the periphery's are small. Interaction is not anonymous; it is structured (Section 1.6).

Step 2: The structure generates asymmetric outcomes. Prebisch argued that the fruits of technical progress are distributed unequally: center countries retain

the productivity gains from industrialization (through higher wages negotiated by strong unions), while periphery countries transfer theirs to the center through deteriorating terms of trade. His mechanism ran through prices — asymmetric wage and price behavior over the trade cycle and different income elasticities of demand for primary goods and manufactures — and no object in this chapter captures it directly. What this chapter does supply is the distributional logic: when a joint surplus is divided by bargaining, the side with the larger weight captures more of it (Proposition 1.9), and the cost is a distributional loss of the form (1.32). Where the weights come from — the Perron vector $\boldsymbol{\pi}$ of a global weight matrix — is the question of Chapter 6.

Step 3: The doctrine that defends the structure is produced by the structure. Prebisch opened his essay by declaring that "reality is destroying that outdated scheme of the international division of labor." The "scheme" he attacked was classical trade theory, which claimed that free trade was optimal for all participants. His point: the theory itself was produced at the center and propagated to the periphery through academic training, policy advice, and institutional pressure. In the language of Chapter 5: the global information network W^I has a center-periphery topology — the periphery absorbs economic doctrine from the center (the Perron eigenvector of W^I assigns dominant weight to center economists), and the resulting "consensus" reflects center interests. Chapter 8 builds the DeGroot model in which one can ask whether a later policy consensus, such as the Washington Consensus, is the long-run belief of such a W^I; this chapter offers no evidence that it is.

Each step corresponds to a formal object in this book:

Prebisch's argument	**Formal object**	**Chapter**
The global economy has a structure	Non-anonymous W; asymmetric column sums of $(I - A')^{-1}$ (a testable hypothesis)	1, 16
The gains of progress are unevenly shared	Surplus divided by weights $\boldsymbol{\lambda} \neq \frac{1}{N}\mathbf{1}$; distributional loss (1.32)	1, 6, 7
The structure produces the doctrine	W^I has center-periphery topology	5, 8
Terms of trade deteriorate	A price mechanism, not a Leontief quantity object	16

Prebisch wrote all of this in prose. This chapter supplies the first row and the logic of the second; the equations for the rest are the content of Chapters 5–8 and 16. The structuralists built the cathedral in words. This book provides the engineering drawings. But the blueprint was always theirs.

Further Reading

Standard microeconomics: Mas-Colell et al. (1995), Chapters 1–3 (consumer theory), 5 (production), 15–17 (general equilibrium).

The Leontief system: Leontief (1986), *Input-Output Economics*. Miller and Blair (2009), *Input-Output Analysis*, 2nd edition. Sargent and Stachurski (2024), *Economic Networks*, Chapters 1–4.

Network perspective on production: Acemoglu et al. (2012a), "The Network Origins of Aggregate Fluctuations."

Linear algebra for economists: Strang (2016), *Introduction to Linear Algebra*, 5th edition. Simon and Blume (1994), *Mathematics for Economists*. Sargent and Stachurski's QuantEcon lectures (`quantecon.org`).

The structuralist tradition: Prebisch (1949), *El desarrollo económico de la América Latina y algunos de sus principales problemas*, CEPAL (the full text, in Spanish, is available at `repositorio.cepal.org`). Hirschman (1958), *The Strategy of Economic Development*, Chapters 6–7. Taylor (1983), *Structuralist Macroeconomics*, Chapter 1. Bielschowsky (2009), "Sixty years of ECLAC: structuralism and neo-structuralism."

Answers to the Prologue Challenge

A: Center–Periphery (Pattern 7). Persons 1–3 form the "center" (bright upper-left). The lower-left rectangle is brighter than the upper-right: dependence runs one way.

B: Hub / Star (Pattern 4). One blazing column at position 4 — not position 1. You had to read the columns.

C: Chain / Ring (Pattern 3). Bright band along the diagonal, wrapping at corners.

D: Triple Economy (Pattern 6, extended). Three bright squares: persons 1–3, 4–6, 7–8.

E: Market / Complete (Pattern 2). Uniform brightness. $W = \frac{1}{N}\mathbf{1}\mathbf{1}'$.

F: Hierarchy (Pattern 5). Brightness gradient from upper-left to lower-right.

Chapter 2

Welfare Theorems as Matrix Properties

What Breaks Efficiency, What Is Distributional — and Where It Breaks

> *"O subdesenvolvimento não é uma etapa necessária do processo de formação das economias capitalistas modernas. É um processo particular, resultante da penetração de empresas capitalistas modernas em estruturas arcaicas."*
>
> ["Underdevelopment is not a necessary stage in the formation of modern capitalist economies. It is a particular process, resulting from the penetration of modern capitalist enterprises into archaic structures."]
> — Celso Furtado, *Desenvolvimento e Subdesenvolvimento* (1961)

Chapter 1 showed that Walrasian interaction is anonymous: the implied weight matrix is $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$, and with equal endowments the benchmark $W^{CE} = \frac{1}{N}\mathbf{1}\mathbf{1}'$. This chapter asks: what happens to the welfare theorems when interaction is not anonymous?

The First Welfare Theorem says: the competitive equilibrium is Pareto efficient. The Second says: any efficient allocation can be achieved by suitable transfers. Both are beautiful results. Neither requires equal power or a particular weight matrix; what the First requires, in this book's language, is the absence of utility interdependence, $W^U = 0$ (together with complete markets and price-taking).

This chapter separates three structural departures and says which of them is an efficiency problem. Unpriced externalities ($W^U \neq 0$) break efficiency; with $W_{ii} = 0$, they leave the utilitarian benchmark intact exactly when W^U is doubly stochastic. Asymmetric power ($\boldsymbol{\pi} \neq \frac{1}{N}\mathbf{1}$) is distributional: bargaining remains efficient, but the surplus tilts toward the powerful, with an exact loss $\frac{\kappa N}{2}\Phi^2 \operatorname{Var}(\pi)$.

Block disconnection leaves gains from integration unrealized. The spectral gap measures integration, not power, and it does not bound the power loss.

2.1 The First Welfare Theorem

2.1.1 The Standard Statement and Proof

Theorem 2.1 (First Welfare Theorem)**.** *In an exchange economy, suppose: (i) every agent maximizes utility given prices, (ii) markets clear, (iii) preferences are locally nonsatiated, and (iv) each agent's utility U_i depends only on her own bundle x_i (no externalities, $W^U = 0$). Then the competitive equilibrium allocation is Pareto efficient (Mas-Colell et al., 1995, Prop. 16.C.1).*

Proof. Let (p^*, X^*) be a competitive equilibrium. Suppose it is not Pareto efficient: there exists a feasible allocation $\hat{X}$ with $U_i(\hat{x}_i) \geq U_i(x_i^*)$ for all i and $U_j(\hat{x}_j) > U_j(x_j^*)$ for some j.

For agent j: since x_j^* maximizes U_j on the budget set and $\hat{x}_j$ is strictly preferred, $\hat{x}_j$ must not be affordable: $p^{*\prime}\hat{x}_j > p^{*\prime}\omega_j$.

For all other agents $i \neq j$: by local nonsatiation, $U_i(\hat{x}_i) \geq U_i(x_i^*)$ implies $p^{*\prime}\hat{x}_i \geq p^{*\prime}\omega_i$. Indeed, if $p^{*\prime}\hat{x}_i < p^{*\prime}\omega_i$, every bundle close enough to $\hat{x}_i$ is still affordable, and local nonsatiation gives one such bundle x' with $U_i(x') > U_i(\hat{x}_i) \geq U_i(x_i^*)$, contradicting the optimality of x_i^*. (Both steps use (iv): "$\hat{x}_i$ is at least as good for i" is a statement about $\hat{x}_i$ alone.)

Summing over all agents:

$$\sum_{i=1}^{N} p^{*\prime}\hat{x}_i > \sum_{i=1}^{N} p^{*\prime}\omega_i. \tag{2.1}$$

But feasibility requires $\sum_i \hat{x}_{i\ell} = \sum_i \omega_{i\ell}$ for all goods ℓ. Multiplying by p_ℓ^* and summing: $\sum_i p^{*\prime}\hat{x}_i = \sum_i p^{*\prime}\omega_i$. Contradiction. □

Intuition: Where does structure enter the proof?

The proof uses two features of the competitive mechanism: every agent faces the *same* price vector p^* and takes it as given, and each agent's utility depends only on her own bundle. The price aggregates individual demands anonymously — the identity of the agent does not affect the price she faces. Agent i's weight in that aggregate is her wealth share s_i, not her "position" in any network.

In weight-matrix language the interaction is $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$ (Chapter 1, Section 1.6), and the proof goes through for *every* $\mathbf{s}$: unequal wealth does not break it. What would break it is hypothesis (iv), which the textbook statement often leaves implicit. If U_i depended on x_j (an externality, $W^U \neq 0$), "$\hat{x}_i$ is at least as good" would no longer translate into a statement about i's own budget. If agents did not take prices as given (market power), x_i^* would not maximize U_i on the budget set.

2.1.2 Externalities in Matrix Form

To see what externalities do in the simplest setting, consider the one-good economy from Chapter 1 with utility externalities. With one good there is no market, so we compare two *utilitarian* benchmarks rather than an equilibrium and an optimum. Agent i's utility now includes the effect of others:

$$V_i(\mathbf{x}) = U_i(x_i) + \gamma \sum_{j \neq i} W_{ij} U_j(x_j), \tag{2.2}$$

where $W_{ij} \geq 0$ is the weight i places on j's utility and $\gamma \geq 0$ is the externality intensity.

A social planner maximizes total social welfare $\sum_i V_i(\mathbf{x})$ subject to $\sum_i x_i = \Omega$. Expanding:

$$\sum_{i=1}^{N} V_i = \sum_{i=1}^{N} U_i(x_i) + \gamma \sum_{i=1}^{N} \sum_{j \neq i} W_{ij} U_j(x_j). \tag{2.3}$$

Rearranging the double sum (collect terms by U_j): the coefficient on $U_j(x_j)$ is

$$\alpha_j = 1 + \gamma \sum_{i \neq j} W_{ij}. \tag{2.4}$$

The term $\sum_{i \neq j} W_{ij}$ is the column sum of W at column j (excluding the diagonal). It measures how much agent j is *cared about* by others — her total incoming externality weight.

Definition 2.2 (Externality Centrality)**.** *Agent j's **externality centrality** is the total weight others place on j's utility, $c_j = \sum_{i \neq j} W_{ij}$. The planner's effective weight on j is $\alpha_j = 1 + \gamma c_j$.*

Proposition 2.3 (Externalities and the Utilitarian Benchmark)**.** *In the one-good economy with externalities (2.2), let all U_j be strictly concave and differentiable, and consider interior allocations with $U_j' > 0$. The allocation that maximizes $\sum_j U_j(x_j)$ (the "externality-blind" benchmark, which ignores W) also maximizes social welfare $\sum_i V_i = \sum_j \alpha_j U_j(x_j)$, for every total Ω, if and only if*

$$\alpha_j = \alpha_k \quad \text{for all } j,k \qquad \Longleftrightarrow \qquad c_j = c_k \quad \text{for all } j,k. \tag{2.5}$$

That is: all agents must have equal externality centrality.

Proof. The blind optimum x^U is characterized by $U_j'(x_j^U) = \mu$ for all j; the social optimum by $\alpha_j U_j'(x_j) = \mu'$. If all α_j equal α, the two conditions coincide (with $\mu' = \alpha\mu$), and since both objectives are strictly concave ($\alpha_j \geq 1 > 0$) on the convex feasible set, each has at most one maximizer, so the solutions coincide. If $\alpha_j \neq \alpha_k$ for some j,k, then at x^U we have $\alpha_j U_j'(x_j^U) \neq \alpha_k U_k'(x_k^U)$ because $U_j' = U_k' = \mu > 0$; moving a small amount from the agent with the lower α to the one with the higher α raises $\sum_j \alpha_j U_j$, so x^U is not socially optimal. □

Remark 2.4. This is a statement about utilitarian optima, not about Pareto efficiency or competitive equilibrium. With one good there is no market, and with weak altruism every interior allocation can be Pareto efficient with respect to the V_i. In a multi-good economy with consumption externalities ($W^U \neq 0$), the competitive equilibrium is in general not Pareto efficient; the proposition isolates the one-good analogue of that failure.

Proposition 2.5 (The Doubly Stochastic Condition)**.** *Let W be row-stochastic ($\sum_j W_{ij} = 1$ for all i) with zero diagonal ($W_{ii} = 0$, which is natural in (2.2) because the diagonal never enters). Then the condition $c_j = c_k$ for all j,k is equivalent to: W **is doubly stochastic** ($\sum_i W_{ij} = 1$ for all j).*

Proof. With $W_{jj} = 0$, $c_j = \sum_i W_{ij}$ is the j-th column sum. Row-stochasticity gives $\sum_j c_j = \sum_i \sum_j W_{ij} = N$. If all c_j are equal, each equals 1, so W is doubly stochastic. Conversely, if W is doubly stochastic, every $c_j = 1$. □

The zero-diagonal hypothesis matters: $W = \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.5 & 0.4 \\ 0.4 & 0.4 & 0.2 \end{pmatrix}$ is row-stochastic with $c = (0.5, 0.5, 0.5)$, but its column sums are $(1.3, 1.0, 0.7)$.

Intuition: What doubly stochastic means here

Row-stochastic means each agent distributes her total attention/dependency across others, summing to 1. This is a normalization — it says your total outgoing weight is fixed.

Column-stochastic means each agent receives the same total weight from others. Nobody is "more cared about" than anyone else. Nobody generates more externality.

Doubly stochastic means both: the giving is normalized AND the receiving is equal. This is the condition under which externalities do not tilt the social optimum away from the externality-blind one.

When column sums are unequal, the planner wants to give more to the agents with high column sums (others care about them, so their consumption counts several times). An allocation rule that ignores externalities gives everyone the same marginal utility. It under-provides to the agents others care about and over-provides to those nobody cares about.

Two special cases satisfy the condition trivially:

Corollary 2.6 (Two Sufficient Conditions)**.** *The blind and social optima coincide if:*

1. $\gamma = 0$ *(no externalities). Then* $\alpha_j = 1$ *for all* j *regardless of* W*.*

2. $W = \frac{1}{N}\mathbf{1}\mathbf{1}'$ *(competitive benchmark). Then* $c_j = (N-1)/N$ *for all* j*.*

Proof. Under the hypotheses of Proposition 2.3, the two optima coincide if all $\alpha_j = 1 + \gamma c_j$ are equal. (1) With $\gamma = 0$, $\alpha_j = 1$. (2) With $W_{ij} = 1/N$, $c_j = \sum_{i \neq j} 1/N = (N-1)/N$ for every j. (In case (2) the diagonal of W is not zero, so Proposition 2.5 does not apply; the conclusion comes directly from the equal c_j.) □

2.2 Three Failures

We distinguish three structural departures from the benchmark. Only the second is an efficiency failure; the first is distributional, and the third is about integration.

2.2.1 Failure 1: Asymmetric Power

For a row-stochastic irreducible W, the Perron vector $\boldsymbol{\pi}$ departs from $\frac{1}{N}\mathbf{1}$ exactly when W is not doubly stochastic (Chapter 1, Proposition 1.21). Chapter 6 develops $\boldsymbol{\pi}$ as power. Here we adopt the book's bargaining assumption: a joint surplus

is divided by asymmetric Nash bargaining with weights $\boldsymbol{\pi}$, from equal disagreement payoffs $\bar{d}$. The allocation then tilts toward the powerful (Chapter 1, Proposition 1.23; Chapter 7, Proposition 7.3):

$$x_i^* = \frac{\Omega}{N} + \Big(\pi_i - \frac{1}{N}\Big)\Phi, \qquad \Phi = \Omega - N\bar{d}. \tag{2.6}$$

Proposition 2.7 (Welfare Loss from Power Asymmetry)**.** *With $U(x) = x - \frac{\kappa}{2}x^2$ and allocation (2.6), the utilitarian loss relative to equal division is exactly*

$$L_{power} = \frac{\kappa N}{2}\,\Phi^2\,\mathrm{Var}(\pi_i), \tag{2.7}$$

where $\mathrm{Var}(\pi_i) = \frac{1}{N}\sum_i(\pi_i - 1/N)^2$.

Proof. Chapter 1, Proposition 1.23: with $z_i = x_i - \Omega/N$ and $\sum_i z_i = 0$, the linear terms of the exact quadratic expansion cancel, leaving $\frac{\kappa}{2}\sum_i z_i^2$. □

The loss is proportional to the variance of the Perron vector. When $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$ (equal power), $\mathrm{Var}(\pi_i) = 0$ and the loss vanishes. It is a *distributional* loss: Nash bargaining is Pareto efficient for every $\boldsymbol{\pi}$, so no allocation makes everyone better off.

Example 2.8 (Power Asymmetry with $N = 4$)**.** *Four agents, $\Omega = 20$, $\kappa = 0.1$ (satiation at 10, so $\kappa x_i < 1$ throughout), equal disagreement payoffs $\bar{d} = 4.5$, hence surplus $\Phi = 2$. Compare three hypothetical power vectors:*

Structure	π_1	π_2	π_3	π_4	$\mathrm{Var}(\pi_i) \times 10^3$
Competitive ($\frac{1}{4}\mathbf{1}\mathbf{1}'$)	*0.25*	*0.25*	*0.25*	*0.25*	*0*
Mild asymmetry	*0.30*	*0.28*	*0.22*	*0.20*	*1.70*
Strong asymmetry (star)	*0.55*	*0.15*	*0.15*	*0.15*	*30.0*

Bargaining allocations (from (2.6)):

	x_1^*	x_2^*	x_3^*	x_4^*	L_{power}
Competitive	*5.00*	*5.00*	*5.00*	*5.00*	*0*
Mild	*5.10*	*5.06*	*4.94*	*4.90*	*0.0014*
Star	*5.60*	*4.80*	*4.80*	*4.80*	*0.0240*

The star produces a loss 17.6 times larger than the mildly asymmetric case — the ratio of the two variances, since Φ and κ are the same. The powerful agent ($\pi_1 = 0.55$) captures 12% more than equal division despite identical outside options and preferences. This is inequality produced entirely by structure (given the bargaining assumption).

2.2.2 Failure 2: Unpriced Externalities

Even if $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$, externalities that nobody prices generate a genuine efficiency problem.

The social optimum equates $\alpha_j U_j'(x_j)$ across agents. The externality-blind allocation (equal division, with identical U) equates $U_j'(x_j)$. When α_j varies (unequal externality centrality), these differ.

Proposition 2.9 (Welfare Loss from Unpriced Externalities). *Let $U(x) = x - \frac{\kappa}{2}x^2$, $m = \Omega/N$ with $1 - \kappa m > 0$, and $\alpha_j = 1 + \gamma c_j$, where $c_j = \sum_{i \neq j} W^U_{ij}$. Let $H = N/\sum_j \alpha_j^{-1}$ be the harmonic mean of the α_j. If the social optimum is interior with $\kappa x_j < 1$, the loss of equal division, measured in social welfare $\sum_i V_i$, is exactly*

$$L_{externality} = \frac{(1-\kappa m)^2}{2\kappa} \sum_j \frac{(\alpha_j - H)^2}{\alpha_j} \;=\; \frac{N(1-\kappa m)^2}{2\kappa}\, \gamma^2 \,\mathrm{Var}(c_j) + O(\gamma^3), \quad (2.8)$$

where $\mathrm{Var}(c_j) = \frac{1}{N}\sum_j (c_j - \bar{c})^2$. *The social optimum is* $x_j = m + \frac{(1-\kappa m)(\alpha_j - H)}{\kappa \alpha_j}$.

Proof. Write $x_j = m + z_j$ with $\sum_j z_j = 0$. Exactly, $\sum_j \alpha_j U(m + z_j) = \sum_j \alpha_j U(m) + U'(m)\sum_j \alpha_j z_j - \frac{\kappa}{2}\sum_j \alpha_j z_j^2$. Maximizing over z subject to $\sum_j z_j = 0$ gives $z_j = (U'(m)\alpha_j - \nu)/(\kappa\alpha_j)$, and the constraint gives $\nu = U'(m)H$. Let $Q = \sum_j (U'(m)\alpha_j - \nu)^2/\alpha_j = U'(m)^2 \sum_j (\alpha_j - H)^2/\alpha_j$. Since $\alpha_j z_j = U'(m)(\alpha_j - H)/\kappa$ and $\sum_j H/\alpha_j = N$, the identity $\sum_j (\alpha_j - H)^2/\alpha_j = \sum_j (\alpha_j - H) - H\sum_j (1 - H/\alpha_j) = \sum_j (\alpha_j - H)$ gives $U'(m)\sum_j \alpha_j z_j = Q/\kappa$ and $\frac{\kappa}{2}\sum_j \alpha_j z_j^2 = Q/(2\kappa)$, so the gain over $z = 0$ is $Q/(2\kappa)$. The objective is strictly concave ($\alpha_j > 0$), so this stationary point is the maximum; the hypothesis that it is interior with $\kappa x_j < 1$ makes it the social optimum of the constrained problem. For the expansion, $\sum_j \alpha_j^{-1} = N - \gamma N \bar{c} + O(\gamma^2)$ gives $H = 1 + \gamma\bar{c} + O(\gamma^2)$, so $\alpha_j - H = \gamma(c_j - \bar{c}) + O(\gamma^2)$; with $\alpha_j = 1 + O(\gamma)$, $\sum_j (\alpha_j - H)^2/\alpha_j = \gamma^2 N \,\mathrm{Var}(c_j) + O(\gamma^3)$. □

This loss exists even with equal power. To second order it is proportional to the *variance of column sums* in W^U — how unevenly the caring is distributed. If everybody is cared about equally (c_j constant), the loss is zero even with large γ. It vanishes at satiation ($\kappa m = 1$), which is one more reason to calibrate below it.

Example 2.10 (The Unpriced Bond). $N = 3$, $\gamma = 0.5$, $\kappa = 0.5$, $\Omega = 3$ *(so $m = 1$ and $1 - \kappa m = 0.5$).*

Case A: *Symmetric externalities.* $W^U = \begin{pmatrix} 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \\ 0.5 & 0.5 & 0 \end{pmatrix}$. *Column sums:* $(1, 1, 1)$. *Equal.* $L_{externality} = 0$.

Case B: *Asymmetric.* $W^U = \begin{pmatrix} 0 & 0.6 & 0 \\ 0.5 & 0 & 0 \\ 0.0 & 0.5 & 0 \end{pmatrix}$. *Column sums:* $(0.5, 1.1, 0)$. *Agent 2 is cared about by both others; Agent 3 is cared about by nobody.* $\mathrm{Var}(c_j) = 0.202$ *and* $\boldsymbol{\alpha} = (1.25, 1.55, 1.00)$. *The social optimum is* $\mathbf{x}^V = (1.02, 1.21, 0.77)$*, and the exact loss of equal division is* $L_{externality} = 0.030$. *(The second-order approximation gives* 0.038*;* $\gamma = 0.5$ *is not small.)*

The planner gives more to Agent 2 (whose consumption others value). An allocation rule that ignores externalities equalizes marginal utilities, under-providing for Agent 2 and over-providing for Agent 3.

2.2.3 Failure 3: Block Disconnection

When W is block-diagonal with k blocks, the economy fragments into independent components. Allocation may equalize *within* each block, but gains from cross-block reallocation are unrealized.

Proposition 2.11 (Block-Diagonal Welfare Loss)**.** *Let W have k blocks with sizes $N_1, \ldots, N_k$ and mean endowments $\bar{\omega}_1, \ldots, \bar{\omega}_k$, with no cross-block weight. Suppose resources are equalized within each block but do not cross blocks, and $U(x) = x - \frac{\kappa}{2}x^2$. The utilitarian loss relative to full equal division is:*

$$L_{block} = \frac{\kappa}{2} \sum_{b=1}^{k} N_b (\bar{\omega}_b - \bar{\omega})^2. \tag{2.9}$$

This is N times the ***between-group variance*** *of endowments, weighted by $\kappa/2$.*

Proof. Each block b solves its own welfare problem independently, achieving within-block equal division at $\bar{\omega}_b$. The first-best (full-economy equal division) would give everyone $\bar{\omega}$. The difference in welfare:

$$L = \sum_b N_b \left[U(\bar{\omega}) - U(\bar{\omega}_b) \right] = \sum_b N_b \left[(\bar{\omega} - \bar{\omega}_b) - \frac{\kappa}{2}(\bar{\omega}^2 - \bar{\omega}_b^2) \right] \tag{2.10}$$

$$= -\frac{\kappa}{2} \sum_b N_b (\bar{\omega}^2 - \bar{\omega}_b^2) = \frac{\kappa}{2} \sum_b N_b (\bar{\omega}_b - \bar{\omega})^2, \tag{2.11}$$

using $\sum_b N_b \bar{\omega}_b = N\bar{\omega}$ (block means average to the population mean) twice: it removes the linear terms, and it gives $\sum_b N_b \bar{\omega}_b^2 - N\bar{\omega}^2 = \sum_b N_b (\bar{\omega}_b - \bar{\omega})^2$. □

Intuition: Between-group inequality as a structural loss

The block-diagonal welfare loss equals the between-group variance of endowments. If all blocks have the same average ($\bar{\omega}_b = \bar{\omega}$), the loss is zero even with total disconnection — there is nothing to redistribute across blocks. But if blocks differ in average endowment (one is rich, another poor), disconnection prevents the equalization that would improve welfare.

This is the structural diagnosis: the welfare cost of fragmentation is not fragmentation per se, but fragmentation *in the presence of between-group inequality*. An economy with two equally poor sectors loses nothing from disconnection. An economy with one gas sector and one subsistence sector loses enormously.

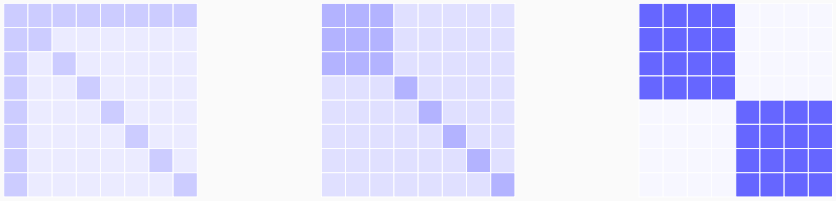
Example 2.12 (Two Blocks with Different Means)**.** $N = 8$, $\kappa = 0.1$ *(satiation at 10). Block 1 (agents 1–3): gas/formal sector,* $\bar{\omega}_1 = 8$*. Block 2 (agents 4–8): traditional sector,* $\bar{\omega}_2 = 3.2$*. Overall mean:* $\bar{\omega} = \frac{3\times 8+5\times 3.2}{8} = 5$*.*

Between-group variance: $\frac{1}{8}[3(8-5)^2 + 5(3.2-5)^2] = \frac{1}{8}[27 + 16.2] = 5.4$*.*

Block-diagonal welfare loss: $L_{block} = \frac{0.1}{2} \times 8 \times 5.4 = 2.16$*.*

Now connect the blocks: increase cross-block weights to 0.15 (structural policy). If the connection lets resources partially equalize across blocks, the welfare gain is bounded above by $L_{block} = 2.16$ *— the maximum achievable improvement from structural integration. How much of it is realized depends on the allocation mechanism, not on W alone.*

Visualization: Three failures, three heatmaps



Asymmetric Power
$\pi_1 = 0.44$, $N_{\text{eff}} = 4.2$
(yet $\Gamma = 0.89$)

Clustered Externalities
$\text{Var}(c_j) = 0.060$

Block Disconnection
$\Gamma = 0.095$, $N_{\text{eff}} = 8$

Three departures from W^{CE} (heatmap weights, rows normalized; diagnostics by `code/ch02.py`). Only the middle one, unpriced externalities, is an efficiency failure. The left is distributional: power is concentrated, although the spectral gap is large. The right leaves gains from integration unrealized: the gap is small, although power is uniform. The real world typically has all three simultaneously.

2.3 The Second Welfare Theorem

2.3.1 The Standard Statement

Theorem 2.13 (Second Welfare Theorem)**.** *In an exchange economy with no externalities ($W^U = 0$), let every agent's preferences be continuous, convex and strongly monotone. Then any Pareto-efficient allocation x^* with $x_i^* \gg 0$ for every i can be supported as a competitive equilibrium after suitable lump-sum transfers of endowments: there is a price vector $p \geq 0$, $p \neq 0$, such that each x_i^* maximizes U_i on $\{x_i : p'x_i \leq p'x_i^*\}$.*

Proof sketch. This is Mas-Colell et al. (1995, Props. 16.D.1 and 16.D.2). The set of aggregate bundles $\sum_i x_i$ with $x_i \succ_i x_i^*$ for every i is convex (convex preferences) and, by efficiency, does not contain the aggregate endowment. The separating hyperplane theorem gives $p \neq 0$ with $p' \sum_i x_i \geq p' \sum_i x_i^*$ on that set; monotonicity gives $p \geq 0$, and local nonsatiation turns this into: $x_i \succ_i x_i^*$ implies $p'x_i \geq p'x_i^*$ (a price quasi-equilibrium with transfers $p'x_i^* - p'\omega_i$). Because $x_i^* \gg 0$ and $p \geq 0$, $p \neq 0$, each agent has a strictly cheaper bundle, and continuity upgrades "$\succ_i$ implies $\geq$" to "$\succ_i$ implies $>$", which is utility maximization on the budget set. □

The standard interpretation: equity and efficiency are separable. A benevolent planner can achieve any desired distribution by (1) computing the required endowment transfers, (2) executing them, and (3) letting the market handle the allocation. The market handles efficiency; the planner handles equity.

This separation is the intellectual foundation of "growth first, redistribute later": get the prices right (efficiency), then transfer income (equity).

2.3.2 Transfers and the Power Term

The SWT assumes that after the transfer, allocation runs through an anonymous mechanism. When allocation runs through bargaining with weights $\boldsymbol{\pi}(W)$, transfers still work — but they act on a different object from the one power acts on.

Proposition 2.14 (Transfers Do Not Remove the Power Term)**.** *Let the surplus be divided by asymmetric Nash bargaining with weights $\boldsymbol{\pi}$ from disagreement payoffs $\mathbf{d}$, so that $x_i = d_i + \pi_i \Phi$ with $\Phi = \Omega - \sum_j d_j$ (Chapter 1, Proposition 1.9). Let $\boldsymbol{\tau}$ be a transfer with $\sum_i \tau_i = 0$.*

1. *A transfer that changes disagreement payoffs, $\mathbf{d}' = \mathbf{d} + \boldsymbol{\tau}$, changes the allocation one-for-one: $\mathbf{x}' = \mathbf{x} + \boldsymbol{\tau}$. There is no clawback.*

2. *The power term $\pi_i \Phi$ is unchanged, because Φ depends only on $\sum_j d_j$ and $\boldsymbol{\pi}$ depends only on W.*

3. *Equal division is reached if and only if* $\tau_i = \frac{\Omega}{N} - d_i - \pi_i\Phi$. *With equal disagreement payoffs this is* $\tau_i = (\frac{1}{N} - \pi_i)\Phi$, *a gross transfer volume* $\frac{1}{2}\sum_i |\tau_i| = \Phi\, d_\pi$, *where* $d_\pi = \frac{1}{2}\|\boldsymbol{\pi} - \frac{1}{N}\mathbf{1}\|_1$.

4. *If the surplus is re-bargained every period from disagreement payoffs fixed by structural position, a transfer of consumption made after bargaining does not carry over: offsetting power costs* $\Phi\, d_\pi$ *per period. A change in* W *that makes it doubly stochastic (*$\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$*) reaches equal division with no transfers.*

Proof. (1)–(2): substitute $\mathbf{d}'$ into $x_i = d_i + \pi_i(\Omega - \sum_j d_j)$ and use $\sum_j \tau_j = 0$. (3): solve $d_i + \tau_i + \pi_i\Phi = \Omega/N$; with $d_i = \bar{d}$, $\Omega/N - \bar{d} = \Phi/N$. The positive parts of $\boldsymbol{\tau}$ sum to $\Phi\sum_i(\frac{1}{N} - \pi_i)^+ = \Phi\, d_\pi$. (4): each period's allocation is $d_i + \pi_i\Phi$ again; uniform $\boldsymbol{\pi}$ makes it Ω/N by (3) with $\boldsymbol{\tau} = 0$. □

Example 2.15 (Transfer vs. Structural Policy)**.** [ILLUSTRATIVE] $N = 4$, $\Omega = 20$, $\kappa = 0.1$, *equal disagreement payoffs* $\bar{d} = 3$, *so* $\Phi = 8$. *Agents 1–2 form a gas/government block, agents 3–4 a traditional block:*

$$W = \begin{pmatrix} 0.49 & 0.49 & 0.01 & 0.01 \\ 0.49 & 0.49 & 0.01 & 0.01 \\ 0.02 & 0.02 & 0.48 & 0.48 \\ 0.02 & 0.02 & 0.48 & 0.48 \end{pmatrix}.$$

The traditional block depends on the gas block twice as much (0.02 per entry) as the gas block depends on it (0.01). The Perron vector is $\boldsymbol{\pi} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{6}, \frac{1}{6})$, *so* $N_{eff} = 3.6$; *the spectral gap is* $\Gamma = 0.06$.

No policy. $\mathbf{x}^* = (5.67, 5.67, 4.33, 4.33)$; *the gas block gets 13% more than equal division.* $L_{power} = 0.089$.

Policy A — Transfers. *The transfer* $\boldsymbol{\tau} = (-0.67, -0.67, 0.67, 0.67)$ *restores equal division, at a volume of* $\Phi d_\pi = 1.33$ *per bargaining round. A one-off transfer of 2 units of disagreement payoff (say, land) from Agent 2 to Agent 3 sticks one-for-one:* $\mathbf{x}' = (5.67, 3.67, 6.33, 4.33)$. *Agent 3 gains exactly 2.00, but the power term is untouched.*

Policy B1 — Integration only. *Multiply all cross-block entries by five (0.05 and 0.10). The spectral gap rises from 0.06 to 0.30, but* $\boldsymbol{\pi}$ *does not move: block masses are set by the ratio of the cross-block flows (*$2 \times 0.02 : 2 \times 0.01$*), not by their level. The allocation is unchanged.*

Policy B2 — Rebalancing. *The gas block sources more from the traditional block: its cross entries rise from 0.01 to 0.015 (the traditional block's stay at 0.02). Now* $\boldsymbol{\pi} = (0.286, 0.286, 0.214, 0.214)$, $N_{eff} = 3.92$, *and* $\mathbf{x}' = (5.29, 5.29, 4.71, 4.71)$. *The recurring transfer needed to finish the job falls from 1.33 to 0.57 per round.*

	x_1	x_2	x_3	x_4	*Max gap*	Γ	*Transfer/round*
No policy	*5.67*	*5.67*	*4.33*	*4.33*	*1.33*	*0.06*	*1.33*
A: transfers each round	*5.00*	*5.00*	*5.00*	*5.00*	*0*	*0.06*	*1.33*
B1: integration only	*5.67*	*5.67*	*4.33*	*4.33*	*1.33*	*0.30*	*1.33*
B2: rebalancing	*5.29*	*5.29*	*4.71*	*4.71*	*0.57*	*0.07*	*0.57*

Transfers can always reach equal division, but they must keep offsetting a term that is a property of W. Integration alone does nothing for the distribution. A structural change that alters the direction of dependence reduces the power term itself.

Key Structural Insight

The Second Welfare Theorem is a theorem about $\boldsymbol{\omega}$: it says you can achieve any efficient allocation by choosing the right endowments, and it assumes the mechanism that converts endowments into allocations is anonymous. When allocation runs through bargaining with weights $\boldsymbol{\pi}(W)$, transfers still move the allocation one-for-one, but they act on $\boldsymbol{\omega}$ (or $\mathbf{d}$) while the power term $(\pi_i - \frac{1}{N})\Phi$ is a property of W. Offsetting it costs Φd_π every time the surplus is re-divided; the associated loss is proportional to $\mathrm{Var}(\pi_i)$.
Three policy levers exist: (1) transfers change $\boldsymbol{\omega}$, (2) structural policy changes W, (3) formation interventions change W^I and eventually preferences. The SWT assumes only lever (1) is needed. When allocation is not anonymous, lever (2) can substitute for a recurring use of lever (1) — provided it changes $\boldsymbol{\pi}$, not merely Γ.

2.4 The Welfare Loss Decomposition

Combining the three failures, the total welfare loss from structural departures:

Theorem 2.16 (Structural Welfare Loss)**.** *Let allocation follow (2.6) with $m = \Omega/N$, evaluate welfare by $\sum_i V_i = \sum_j \alpha_j U(x_j)$ with $U(x) = x - \frac{\kappa}{2}x^2$ and $\alpha_j = 1 + \gamma c_j$, and assume the social optimum is interior as in Proposition 2.9. The loss of the bargaining allocation relative to the social optimum is exactly*

$$L(W, \gamma) = L_{externality} + L_{power} + L_{interaction}, \tag{2.12}$$

with $L_{externality}$ given by (2.8), $L_{power} = \frac{\kappa N}{2}\Phi^2 \mathrm{Var}(\pi_i)$ as in (2.7), and (as in Chapter 7,

Theorem 7.8, with $\gamma = \gamma_U$)

$$L_{interaction} = \frac{\kappa\gamma\Phi^2}{2}\sum_j c_j\Big(\pi_j - \frac{1}{N}\Big)^2 - \gamma\Phi N(1-\kappa m)\,\mathrm{Cov}(\pi_j, c_j). \tag{2.13}$$

Proof. Let $z_j = (\pi_j - \frac{1}{N})\Phi$, so $\sum_j z_j = 0$. Exactly, $\sum_j \alpha_j U(m+z_j) = \sum_j \alpha_j U(m) + (1-\kappa m)\sum_j \alpha_j z_j - \frac{\kappa}{2}\sum_j \alpha_j z_j^2$. Since $\sum_j z_j = 0$, $\sum_j \alpha_j z_j = \gamma\sum_j c_j z_j = \gamma\Phi N\,\mathrm{Cov}(\pi_j, c_j)$, and $\sum_j \alpha_j z_j^2 = \Phi^2\sum_j(\pi_j - \frac{1}{N})^2 + \gamma\Phi^2\sum_j c_j(\pi_j - \frac{1}{N})^2$. The social optimum exceeds $\sum_j \alpha_j U(m)$ by $L_{\text{externality}}$ (Proposition 2.9). Subtract. □

The leading interaction term is *minus* $\gamma\Phi N(1-\kappa m)\,\mathrm{Cov}(\pi_j, c_j)$. When the powerful are also the agents others care about (Cov > 0), the bargaining tilt toward them is partly what a planner who counts that caring would choose, so the interaction *offsets* the other two terms. The distortions compound when Cov < 0: power held by agents nobody cares about.

Example 2.17 (Decomposition for Two Structures)**.** [ILLUSTRATIVE] $N = 4$, $\gamma = 0.3$, $\kappa = 0.1$, $\Omega = 20$, $\Phi = 8$. *The power vectors come from the production and information layers; the externality centralities c_j are the column sums of a zero-diagonal, row-stochastic W^U (so they sum to N).*

Connected economy: $\boldsymbol{\pi} = (0.26, 0.25, 0.25, 0.24)$ *and* $\mathbf{c} = (1.02, 1.01, 0.99, 0.98)$*:*

$$L_{externality} = 0.00009, \quad L_{power} = 0.00064, \quad L_{interaction} = -0.00029, \quad L = 0.00044.$$

Enclave economy: $\boldsymbol{\pi} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{6}, \frac{1}{6})$ *(Example 2.15) and* $\mathbf{c} = (1.24, 1.30, 0.72, 0.74)$*:*

$$L_{externality} = 0.0254, \quad L_{power} = 0.0889, \quad L_{interaction} = -0.0813, \quad L = 0.0329.$$

The enclave's loss is about 75 times the connected economy's. But the interaction term is large and negative: the gas-government elite are both powerful and cared about by each other, so a social welfare function that counts their mutual caring endorses much of the tilt toward them. Whether such caring should count in social welfare is a normative choice, not a matter of arithmetic (Chapter 12).

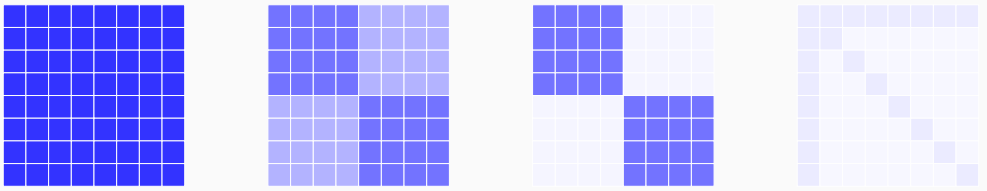
2.5 What the Spectral Gap Measures

It is tempting to look for a single number that bounds the power loss. The spectral gap $\Gamma = 1 - |\lambda_2(W)|$ is not that number. It measures integration — how fast the network mixes (Chapter 1, Section 1.7) — and it places no restriction on $\boldsymbol{\pi}$. For every $\boldsymbol{\pi}$ in the open simplex and every $\gamma \in (0, 1]$, $W = \gamma\mathbf{1}\boldsymbol{\pi}' + (1-\gamma)I$ has Perron vector $\boldsymbol{\pi}$ and $\Gamma = \gamma$ (Chapter 6, Proposition 6.16). With $N = 10$ and

$\pi_1 = 0.91, \Gamma = 1$ coexists with $N_{\text{eff}} = 1.21$, so no bound of the form $\text{Var}(\pi) \leq f(\Gamma)$ with $f(1) = 0$ can hold. Conversely, a doubly stochastic block matrix has uniform power and a gap near zero.

What the gap *does* diagnose is the third failure. Weakly coupled blocks integrate slowly (small Γ, by Cheeger's inequality for reversible W), and that is the situation in which gains from cross-block reallocation (Proposition 2.11) go unrealized. Power concentration is read off $\boldsymbol{\pi}$ itself: N_{eff}, $\text{Var}(\pi)$, d_π.

Visualization: Four structures: integration and power are different



Competitive
$\Gamma = 1.0$, $N_{\text{eff}} = 8$
$L = 0$

Weakly Dual
$\Gamma = 0.67$, $N_{\text{eff}} = 7.98$
$L = 0.001$

Enclave
$\Gamma = 0.13$, $N_{\text{eff}} = 7.95$
$L = 0.002$

Monopoly
$\Gamma = 0.36$, $N_{\text{eff}} = 3.87$
$L = 0.43$

[ILLUSTRATIVE] Stylized matrices, not estimates for any country. Same economy, same preferences: $N = 8$, $\Omega = 40$, $\kappa = 0.1$, $\Phi = 8$. Only W changes (heatmap weights, rows normalized). L is the power loss (2.7). The enclave has the smallest gap but almost no power loss; the monopoly has a larger gap and by far the largest loss. Concentration (N_{eff}), not the gap, predicts L_{power}.

2.6 Summary

1. The **First Welfare Theorem** holds for any wealth distribution; what it needs is $W^U = 0$. With utility externalities and $W_{ii} = 0$, the externality-blind allocation is socially optimal exactly when W^U is doubly stochastic (equal externality centrality).

2. Three structural departures differ in kind: **unpriced externalities** ($\text{Var}(c_j) > 0$) are an efficiency failure; **asymmetric power** ($\text{Var}(\pi_i) > 0$) is distributional, with exact loss $L_{\text{power}} = \frac{\kappa N}{2}\Phi^2 \text{Var}(\pi_i)$; **block disconnection** leaves gains from integration unrealized, with loss equal to the between-group term.

3. The **Second Welfare Theorem** survives as arithmetic: transfers move the bargaining allocation one-for-one. But the power term $(\pi_i - \frac{1}{N})\Phi$ is a property of W, and offsetting it by transfers costs Φd_π every time the surplus is re-divided.

4. The **welfare loss** decomposes exactly as $L = L_{\text{externality}} + L_{\text{power}} + L_{\text{interaction}}$. The leading interaction term is $-\gamma\Phi N(1-\kappa m)\,\mathrm{Cov}(\pi_j, c_j)$: it offsets the others when the powerful are also the most cared about.

5. **The spectral gap measures integration, not power.** It does not bound L_{power}; raising Γ without changing the direction of dependence leaves $\boldsymbol{\pi}$, and the allocation, unchanged.

2.7 Exercises

2.1 Doubly stochastic check. For each matrix, determine if it is doubly stochastic. If not, find the Perron eigenvector $\boldsymbol{\pi}$ and compute $\mathrm{Var}(\pi_i)$:

(a) $W = \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

(b) $W = \begin{pmatrix} 0.6 & 0.3 & 0.1 \\ 0.1 & 0.6 & 0.3 \\ 0.3 & 0.1 & 0.6 \end{pmatrix}$

(c) $W = \begin{pmatrix} 0.8 & 0.2 \\ 0.5 & 0.5 \end{pmatrix}$

(d) $W = \begin{pmatrix} 0.4 & 0.3 & 0.3 \\ 0.3 & 0.4 & 0.3 \\ 0.3 & 0.3 & 0.4 \end{pmatrix}$

2.2 Is there a clawback? Using Example 2.15: let τ be a transfer of disagreement payoff from Agent 2 to Agent 3.

(a) Compute the bargaining allocation for $\tau = 0.5, 1, 2, 3$ and verify that Agent 3 receives exactly τ (Proposition 2.14).

(b) Now let the gas block's cross-block entry be $b \in \{0.01, 0.0125, 0.015, 0.02\}$, with the traditional block's fixed at 0.02. For each b, compute $\boldsymbol{\pi}$, the allocation, and the per-round transfer volume Φd_π needed to reach equal division. Plot it against b.

(c) Repeat (b) after multiplying all cross-block entries by 5. Which of your answers change, and why?

2.3 Welfare decomposition. For $N = 3$, $\gamma = 0.5$, $\kappa = 0.5$, $\Omega = 3$, $\Phi = 1$, taking $W^U = W$ for the externality weights:

(a) Let $W = \frac{1}{3}\mathbf{1}\mathbf{1}'$. Verify $L_{\text{power}} = L_{\text{externality}} = L_{\text{interaction}} = 0$.

(b) Let $W = \begin{pmatrix} 0.5 & 0.4 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.3 & 0.6 \end{pmatrix}$. Compute $\boldsymbol{\pi}$, $\text{Var}(\pi_i)$, the off-diagonal column sums c_j, $\text{Var}(c_j)$, and $\text{Cov}(\pi_i, c_i)$. Then compute the three terms of Theorem 2.16.

(c) Is $L_{\text{interaction}}$ positive or negative? What does the sign mean?

2.4 Spectral gap across topologies. For $N = 8$, construct: (a) W^{CE}, (b) a ring, (c) two disconnected blocks of 4, (d) a star. For each, compute Γ and, where $\boldsymbol{\pi}$ is unique, N_{eff} and $\text{Var}(\pi_i)$. Explain why the ring and the star without self-loops have $\Gamma = 0$ although they are connected, and why the two blocks have no unique $\boldsymbol{\pi}$. Repeat with the lazy versions $\frac{1}{2}(I + W)$. Do Γ and N_{eff} move together? Use Python.

2.5 Block disconnection with data. Using the Bolivian input-output or supply-use tables from INE (Exercise 1.7), aggregate to 2 sectors: Hydrocarbons + Mining vs. Everything Else. Combine value added with sectoral employment (from INE's employment surveys) and compute the between-group variance of value added per worker. How large is L_{block} relative to the within-group component?

2.6 El Alto vs. Zona Sur. [ILLUSTRATIVE] Suppose El Alto and Zona Sur are two blocks of an 8×8 weight matrix: 5 agents in El Alto and 3 in Zona Sur. Each agent puts weight ε on every agent of the other block and spreads the rest evenly within her own block. Mean consumption: $\bar{\omega}_{\text{Alto}} = 800$ Bs/month, $\bar{\omega}_{\text{Sur}} = 4500$ Bs/month (illustrative figures, not survey estimates).

(a) Compute the between-group welfare loss L_{block} with $\kappa = 0.0001$, and check that $\kappa\bar{\omega}_b < 1$ in both blocks.

(b) Compute Γ for $\varepsilon = 0.02$ and $\varepsilon = 0.15$. Explain why Γ measures how fast the blocks integrate but does not change L_{block}, which depends on whether resources actually cross blocks.

(c) What real-world intervention would increase ε?

2.7 Furtado's thesis formalized. Furtado claimed that underdevelopment is not a stage but a stable structural condition. Formalize: let W_t evolve according to $W_{t+1} = f(W_t, \mathbf{x}^*(W_t))$ (the structure at time $t+1$ depends on

the current structure and the allocation it produces). A stable underdevelopment equilibrium is a stable fixed point W^* with weak cross-block coupling and $L_{\text{block}}(W^*) > 0$. (The case study gives a one-parameter example.)

(a) Construct a simple 2×2 example where the block-diagonal W^* is a fixed point: the allocation produced by W^* reinforces the cross-block disconnection. (Hint: if the traditional sector has low income, it cannot invest in cross-block linkages.)

(b) Under what conditions is the fixed point stable (small perturbations return to W^*)?

(c) What does "breaking out of underdevelopment" mean in this framework?

Mathematical Appendices

Appendix 2A: Doubly Stochastic Matrices

Definition. An $N \times N$ nonnegative matrix W is **row-stochastic** if $W\mathbf{1} = \mathbf{1}$, **column-stochastic** if $\mathbf{1}'W = \mathbf{1}'$, and **doubly stochastic** if both.

The Birkhoff-von Neumann theorem. The set of $N \times N$ doubly stochastic matrices is the convex hull of the $N!$ permutation matrices. That is: every doubly stochastic matrix can be written as $W = \sum_k \alpha_k P_k$ where P_k are permutation matrices, $\alpha_k \geq 0$, and $\sum_k \alpha_k = 1$. This means doubly stochastic matrices are "averages" of perfect matchings.

Key property. If W is doubly stochastic and irreducible, the left Perron eigenvector is $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$.

Proof. $\boldsymbol{\pi}'W = \frac{1}{N}\mathbf{1}'W = \frac{1}{N}\mathbf{1}' = \boldsymbol{\pi}'$. By uniqueness of the Perron eigenvector (Perron-Frobenius, Chapter 6), this is the only left eigenvector with eigenvalue 1. □

Measuring departure from double stochasticity. A useful scalar is the column-sum deviation

$$\delta_{\text{col}}(W) = \sqrt{\sum_{j=1}^{N}\left(\sum_{i=1}^{N} W_{ij} - 1\right)^2} = \|\mathbf{1}'W - \mathbf{1}'\|_2. \tag{2.14}$$

$\delta_{\text{col}} = 0$ iff W is column-stochastic. For row-stochastic irreducible W: $\delta_{\text{col}} = 0$ iff W is doubly stochastic iff $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$. (The book reserves $\Delta(W) = W - W^{CE}$ for the full departure from the benchmark, summarized by d_π and $1 - \Gamma$; Chapter 1, Section 1.7.)

Nearby doubly stochastic matrices. Given a positive (or, more generally, totally supported) W, Sinkhorn's algorithm — alternately normalize rows and columns until convergence — converges to the doubly stochastic matrix $D_1 W D_2$ with positive diagonal D_1, D_2. That scaling is the doubly stochastic matrix closest to W in relative entropy (Kullback–Leibler), not in Frobenius norm; the Frobenius-nearest doubly stochastic matrix $\arg\min_{D \in \mathcal{DS}} \|W - D\|_F$ is a quadratic program with a different solution in general.

Appendix 2B: Variance Decomposition

Total variance. For $\omega_1, \ldots, \omega_N$ with mean $\bar{\omega}$:

$$\text{Var}(\omega) = \frac{1}{N}\sum_{i=1}^{N}(\omega_i - \bar{\omega})^2. \tag{2.15}$$

Group decomposition. Partition agents into k groups with sizes $N_1, \ldots, N_k$ and group means $\bar{\omega}_1, \ldots, \bar{\omega}_k$:

$$\text{Var}(\omega) = \underbrace{\sum_{b=1}^{k}\frac{N_b}{N}(\bar{\omega}_b - \bar{\omega})^2}_{\text{Between-group}} + \underbrace{\sum_{b=1}^{k}\frac{N_b}{N}\text{Var}_b(\omega)}_{\text{Within-group}}. \tag{2.16}$$

The between-group component is the structural welfare loss from block disconnection (Proposition 2.11). The within-group component is the loss addressable by within-block redistribution.

Connection to $\text{Var}(\pi_i)$. The Perron eigenvector variance plays the same role for *power* inequality that the endowment variance plays for *wealth* inequality. Both generate welfare losses proportional to their variance. Endowment inequality can be addressed once and for all by transfers; power inequality can be offset by transfers only by repeating them every time the surplus is re-divided (Proposition 2.14), or removed by changing W.

Structuralist Case Study: Furtado's Paraíba and the Fixed Point of Underdevelopment

Celso Furtado was born in 1920 in Pombal, Paraíba — deep in the Sertão, the semi-arid interior of northeastern Brazil. The Northeast was Brazil's original economic center (sugar, then cotton), but by the twentieth century it had become the country's poorest region while São Paulo industrialized. Furtado did not need to theorize structural heterogeneity. He grew up inside the dim lower-right block.

His 1961 book *Desenvolvimento e Subdesenvolvimento* advanced a thesis that maps directly onto the welfare-theoretic framework of this chapter. The argument has three parts.

Part 1: Underdevelopment is not a stage. The modernization view (Rostow's five "stages of growth": traditional society, preconditions for take-off, take-off, drive to maturity, high mass consumption) held that every economy passes through the same sequence and that the underdeveloped economies are simply

earlier in the queue. Furtado rejected this. His claim: underdevelopment is not a phase that precedes development but a *specific structural condition* produced by the same historical process that produced development elsewhere. In the language of this chapter: the weight matrix W of the peripheral economy is not a transient state on the way to $W^{CE} = \frac{1}{N}\mathbf{1}\mathbf{1}'$. It is a *fixed point* — a stable equilibrium of the structural dynamical system. The losses of this chapter do not "eventually" disappear; they persist, because the matrix that produces them reproduces itself.

A one-parameter example [ILLUSTRATIVE] makes this precise. Let ε_t be the cross-block coupling of a two-block W_t. Linkages are built out of the traditional block's income, and that income rises with the linkages, with a threshold: $\varepsilon_{t+1} = F(\varepsilon_t) = \bar{\varepsilon}\,\varepsilon_t^2/(\varepsilon_t^2 + h^2)$. With $\bar{\varepsilon} = 0.3$ and $h = 0.1$, F has three fixed points: $\varepsilon = 0$ (stable, $F'(0) = 0$), $\varepsilon = 0.038$ (unstable, $F' = 1.75$), and $\varepsilon = 0.262$ (stable, $F' = 0.25$). The block-diagonal economy is a stable equilibrium, not a stage. Starting from $\varepsilon_0 = 0.03$ the coupling decays to zero; starting from 0.05 it converges to 0.262. Small, transfer-sized perturbations die out; only a structural change large enough to cross the unstable threshold escapes. (Exercise 2.7 asks for the general version.)

Part 2: Modern enterprises penetrate archaic structures without transforming them. Furtado's key mechanism: when capitalist firms enter a peripheral economy, they do not replace the pre-existing structure. They graft onto it. The modern enterprise and the archaic structure coexist within the same country, connected by a few asymmetric linkages (labor recruitment, raw material extraction) but otherwise operating independently. This is the block-diagonal W from Proposition 2.11: two blocks with weak cross-block entries. The modern block has high productivity and high $\bar{\omega}$; the archaic block has low productivity and low $\bar{\omega}$. The between-group variance is enormous. The welfare loss from disconnection is $L_{\text{block}} = \frac{\kappa}{2}\sum_b N_b(\bar{\omega}_b - \bar{\omega})^2$. In Paraíba, this was not an abstraction.

Part 3: The structure reproduces the conditions for its own persistence. The modern sector's profits are repatriated or spent on imported consumption goods (the center-periphery leakage). The archaic sector's surplus is too small to finance the investments that would create cross-block linkages. Each of these mechanisms is a feedback loop: the allocation produced by W reinforces W. Proposition 2.14 is the static half of the argument — transfers move allocations but leave the power term of W in place. The fixed-point example adds the dynamic half: the current allocation makes the future W no less block-diagonal than the present one.

Furtado's argument	Formal object	This chapter
Underdevelopment is not a stage	W^* is a stable fixed point	Case study example; Ex. 2.7
Modern grafted onto archaic	Block-diagonal W with weak cross-links	Prop. 2.11
Between-region inequality	$L_{\text{block}} \propto \sum_b N_b(\bar{\omega}_b - \bar{\omega})^2$	Eq. (2.9)
Structure reproduces itself	$\varepsilon_{t+1} = F(\varepsilon_t)$ with stable $\varepsilon^* = 0$	Case study example
Transfers don't transform	Transfers move $\mathbf{x}$ one-for-one but leave the power term $(\pi_i - \frac{1}{N})\Phi$	Prop. 2.14

In 1959, Furtado was appointed head of SUDENE (Superintendência do Desenvolvimento do Nordeste) — an agency created specifically to change W^P_{Brazil} by building cross-block linkages between the Northeast and the industrializing South. He was removed after the 1964 military coup. The coup did not change the welfare theorems. It changed who had the power to change W.

Further Reading

Welfare theorems: Mas-Colell et al. (1995), *Microeconomic Theory*, Chapter 16. Arrow (1951), "An Extension of the Basic Theorems of Classical Welfare Economics." Debreu (1959), *Theory of Value*, Chapter 6.

Structural failures of efficiency: Greenwald and Stiglitz (1986), "Externalities in Economies with Imperfect Information and Incomplete Markets." Acemoglu (2003), "Why Not a Political Coase Theorem?"

The structuralist critique: Furtado (1961), *Desenvolvimento e Subdesenvolvimento*, Rio de Janeiro: Fundo de Cultura (the key text, available through Centro Celso Furtado: `centrocelsofurtado.org.br`). Furtado (1959), *Formação Econômica do Brasil* — the historical companion, tracing the evolution of Brazil's W^P from colonial plantation to ISI. Taylor (1983), *Structuralist Macroeconomics*, Chapter 1.

Sinkhorn's algorithm: Sinkhorn (1964), "A Relationship Between Arbitrary Positive Matrices and Doubly Stochastic Matrices." Used in Appendix 2A for the doubly stochastic scaling $D_1 W D_2$ of a positive matrix.

Chapter 3

Growth and Distribution in Matrix Form

Why Growth Makes Some Countries Rich and Others More Unequal

> *"Los enormes beneficios que derivan de la creciente productividad no han llegado a la periferia en medida comparable a la que han logrado disfrutar las poblaciones de los grandes países industriales. De ahí las diferencias tan acentuadas en los niveles de vida de las masas de éstos y de aquéllos y las notorias discrepancias entre sus respectivas fuerzas de capitalización."*
>
> ["The enormous benefits derived from increasing productivity have not reached the periphery in a measure comparable to that enjoyed by the peoples of the great industrial countries. Hence the great differences in standards of living, and the manifest discrepancies in their respective capacities for capital accumulation."]
>
> — Raúl Prebisch, *El desarrollo económico de la América Latina* (1949)

Prebisch saw it without the math: growth in the periphery does not work the same way as growth in the center, and the difference is structural. The production structure of the center—diversified, interconnected, industrialized—propagates productivity gains internally, raising wages and living standards broadly. The production structure of the periphery—concentrated in primary commodities, weakly linked internally, dependent on imported technology—leaks productivity gains outward through deteriorating terms of trade, leaving internal inequality unchanged or worse.

Chapter 2 separated efficiency from distribution and showed that power is a property of W. This chapter turns to growth and asks how the production network shapes who gains from it.

We develop growth theory in matrix form, starting from the standard Solow model (one sector, one country, growth converges to a steady state) and progressively introducing the structural elements that make growth in the periphery different from growth in the center. The key mathematical object is the production network A—the input-output matrix from Chapter 1—now embedded in a dynamic growth model where the structure of A determines not just who produces what today but how productivity gains propagate across sectors over time.

The central result: whether growth reaches the traditional sector depends on the *cross-block entries* of A and hence on the off-diagonal entries of its Leontief inverse. When the sectors are well connected (large cross-block input shares), demand growth in one sector propagates to the others. When A is nearly block-diagonal, growth stays in the modern sector. Pinto's structural heterogeneity is a property of the coupling between blocks — measurable by the cross-block mass of A or, for the row-normalized $W^P = D^{-1}A$, by its spectral gap, which here measures integration and nothing else.

3.1 The Standard Story: One Sector, One Country

3.1.1 Solow in One Paragraph

The Solow growth model says: a country accumulates capital, capital produces output, output is split between consumption and saving, and saving becomes next period's capital. The economy converges to a steady state where capital per worker is constant and output grows only through technological progress.

The model has one sector, one representative agent, and one production function: $Y_t = K_t^{\alpha}(Z_t L_t)^{1-\alpha}$, where K is capital, L is labor, Z is technology (we reserve A for the input-output matrix), and $\alpha \in (0, 1)$ is the capital share. The savings rate s is exogenous. Capital accumulates as $K_{t+1} = sY_t + (1-\delta)K_t$.

In per-effective-worker terms ($k = K/(ZL)$), the dynamics reduce to one equation:

$$k_{t+1} = \frac{sk_t^{\alpha} + (1-\delta)k_t}{(1+g)(1+n)}, \tag{3.1}$$

where g is the growth rate of technology and n is population growth. The steady state k^* satisfies $sf(k^*) = (\delta + g + n + gn)k^* \approx (\delta + g + n)k^*$.

Intuition: What Solow gets right and what it misses

Solow gets convergence right: poor countries with low k have high marginal product of capital, invest at the same rate s, and grow faster until they catch up. This predicts convergence—and conditional convergence (controlling for s, δ, n) is one of the most robust findings in empirical growth.

What Solow misses is everything structural. There is one sector, so there is no structural heterogeneity. There is one savings rate, so there is no differential accumulation by group. There is one production function, so there are no inter-sectoral linkages, no Leontief multipliers, no question of whether growth in mining propagates to agriculture. The country is a point, not a structure. Prebisch would say: this describes Sweden, not Bolivia.

3.1.2 Solow with Two Groups: The OLG Version

To introduce structure, we move from the Solow model to a model of overlapping generations (OLG), in which savings decisions are made by *different groups* with *different parameters*.

Example 3.1 (A Two-Group Economy)**.** *Consider an economy with two groups: M (men) and F (women), each comprising half the population. Each agent lives two periods: young (work, earn, save) and old (consume savings).*

A young agent of group $g \in \{M, F\}$ has utility over young and old consumption:

$$U^g = \ln(c^{g,y}) + \theta^g \ln(c^{g,o}), \tag{3.2}$$

*where $\theta^g > 0$ is the **patience parameter**—how much group g values future consumption relative to present consumption.*

The budget constraints are $c^{g,y} + s^g = w_t$ (young: consume or save out of wage income) and $c^{g,o} = (1 + r_{t+1})s^g$ (old: consume return on savings). The optimal savings rate is:

$$\sigma(\theta^g) = \frac{\theta^g}{1 + \theta^g}. \tag{3.3}$$

A group with higher θ (more patient) saves a larger fraction of income. If $\theta^M = \theta^F$ (same patience), both groups save the same share and the economy behaves like standard Solow. If $\theta^M > \theta^F$ — in this book's reading, because the structure has produced groups that value future consumption differently (Chapter 10), not because of anything innate — then men save a larger share of the same wage and consume more in old age. With no bequests the gap does not compound across generations: each cohort starts again from the wage. What changes is aggregate saving, and with it the economy's capital stock.

Aggregate capital next period is:

$$K_{t+1} = \frac{1}{2}\left[\sigma(\theta_t^M) + \sigma(\theta_t^F)\right](1-\alpha)K_t^\alpha. \tag{3.4}$$

This is the Solow model with a twist: the aggregate savings rate depends on the *composition* of the population. If the two groups have different patience parameters, the savings rate—and therefore the growth path—depends on who has what θ.

Intuition: Where does θ come from?

In the standard OLG model, θ^g is given. Men and women are either equally patient or not, and the model does not ask why. In the formation model (Chapter 10), θ^g is produced by the developmental environment (what Chapter 10 calls the "park"): it determines how much each group values future consumption, investment, and career building. Women raised in a normative environment where female economic participation is invisible or discouraged will have lower θ^F—not because of biology but because the signal structure of that environment made future consumption appear less attainable or less valuable.

For now, θ^g is a parameter. Later, it becomes an output of W^I.

3.2 Multi-Sector Growth: The Production Network

Now we introduce multiple sectors. This is where the Leontief system from Chapter 1 becomes dynamic.

3.2.1 Gas and the Traditional Economy

Stylize Bolivia as two economies. The first is the gas economy: capital-intensive, high-productivity, connected to global commodity markets, a large share of export revenue in boom years and a small share of employment. The second is the traditional economy (potatoes, llamas, small-scale agriculture, informal commerce): labor-intensive, low-productivity, producing mainly for domestic markets, and employing most of the population. The data bear out the contrast. Fuels were 54.5% of Bolivia's merchandise exports in 2013, at the height of the boom, 32.1% in 2019 and 18.9% in 2024 (WDI series TX.VAL.FUEL.ZS.UN; World Bank, 2026b). Mining and quarrying, hydrocarbons included, employed 1.3% of workers in 2019, against 27.4% in agriculture and 17.1% in commerce (ILOSTAT, employment by ISIC section; International Labour Organization, 2026). Chapter 17 gives

the full sectoral picture from INE's supply-use tables.

A dollar of investment in the gas sector produces output that is sold internationally, generates fiscal revenue, and feeds back into the gas sector (maintenance, exploration, expansion). Very little of it flows to traditional producers. A dollar of investment in traditional agriculture feeds local consumption, local transport, local markets — but generates little fiscal revenue and few export earnings.

This is structural heterogeneity. And the question is: if the gas sector grows at 5% per year for a decade, does the traditional economy grow?

The answer depends on the structure of A — the input-output matrix. If the gas sector buys inputs from the domestic economy (backward linkages) and sells intermediate products to domestic manufacturers (forward linkages), then gas growth propagates. If the gas sector imports its inputs and exports its output, with no domestic linkages, then gas growth is an enclave: it enriches the gas sector and the treasury without touching the traditional economy.

Prebisch Connection: The enclave economy

This is exactly Prebisch's center-periphery argument applied within a country. The gas sector is the "center" of the Bolivian economy—high productivity, strong international linkages, favorable terms of trade (when commodity prices are high). The traditional sector is the "periphery"—low productivity, weak linkages, dependent on domestic demand. Pinto's contribution was to see that this center-periphery structure exists *inside* peripheral countries, not just between countries. Bolivia is peripheral in the global economy *and* internally heterogeneous. The gas sector's relationship to the traditional sector replicates, at national scale, the center's relationship to the periphery at global scale.

One qualification is forced by the data. "Center" here means productivity and export earnings, not a central position in the *domestic* production network. In INE's 2019 supply-use table the hydrocarbons sector sits at the periphery of the domestic W^P: its Perron weight is $\pi = 0.039$, well below the uniform share $1/11 \approx 0.091$. It exports 70% of its output, buys only 0.225 of domestic intermediate inputs per unit of output (manufacturing: 0.558), and pays 6% of its value added as compensation of employees (INE supply-use tables, base 2017; author's calculations, `data/compute/compute_bolivia_io.py`; Instituto Nacional de Estadística (INE), Bolivia, 2026). That is Prebisch's and Pinto's enclave, measured by $\boldsymbol{\pi}$ and by export orientation. Its rents reach the rest of the economy through the fiscal system (the direct hydrocarbons tax, IDH), not through production linkages.

3.2.2 The Multi-Sector Solow Model

Consider N sectors, each with its own capital stock K_{it}, labor force L_i, and production function. We use a reduced-form demand-pull model. Sector i's output is

$$Y_{it} = K_{it}^{\alpha_i} L_i^{1-\alpha_i} + \gamma \sum_{j=1}^{N} a_{ji} Y_{jt}, \tag{3.5}$$

where A is the technical-coefficient matrix of Chapter 1 (a_{ji} = input i per unit of output j) and $\gamma \in (0, 1]$ scales the strength of the linkage. The second term is the demand for sector i's output coming from the sectors that use it as an input, exactly as in the material balance $\mathbf{y} = A'\mathbf{y} + \mathbf{d}$ of Chapter 1.

In matrix form, stacking all sectors:

$$\mathbf{Y}_t = \mathbf{f}(\mathbf{K}_t, \mathbf{L}) + \gamma A' \mathbf{Y}_t, \tag{3.6}$$

where $\mathbf{f}$ is the vector of Cobb-Douglas production functions, playing the role of the final-demand impulse. Solving for $\mathbf{Y}_t$ (as in Chapter 1):

$$\mathbf{Y}_t = (I - \gamma A')^{-1} \mathbf{f}(\mathbf{K}_t, \mathbf{L}). \tag{3.7}$$

This is the Leontief system made dynamic: at each period t, output is determined by capital (which accumulates) filtered through the production network (which amplifies through the Leontief inverse). It requires $\gamma\rho(A) < 1$. Two cautions about the reduced form. It transmits impulses the way the quantity system transmits final demand: *upstream*, from users to their suppliers. A genuine cost-reducing productivity shock travels the other way, *downstream* to the users of the cheaper good, through the price dual $\mathbf{p} = (I - A)^{-1}\mathbf{v}$ ($\mathbf{v}$ = value added per unit); network models of productivity shocks (Acemoglu et al., 2012a) use that dual. And we treat $\mathbf{Y}$ as the base for saving below; with gross output one would first net out intermediate use.

Capital accumulates sector by sector:

$$K_{i,t+1} = s_i Y_{it} + (1 - \delta_i) K_{it}, \tag{3.8}$$

where s_i is sector i's savings (reinvestment) rate and δ_i is depreciation. In matrix form:

$$\mathbf{K}_{t+1} = S\mathbf{Y}_t + (I - \Delta)\mathbf{K}_t, \tag{3.9}$$

where $S = \text{diag}(s_1, \ldots, s_N)$ and $\Delta = \text{diag}(\delta_1, \ldots, \delta_N)$.

Substituting the output equation:

$$\mathbf{K}_{t+1} = S(I - \gamma A')^{-1} \mathbf{f}(\mathbf{K}_t, \mathbf{L}) + (I - \Delta)\mathbf{K}_t. \tag{3.10}$$

Key Structural Insight

Equation (3.10) is the multi-sector growth equation. It says: next period's capital depends on this period's capital, transformed by two structural objects:

- $(I - \gamma A')^{-1}$: the Leontief inverse, which amplifies each sector's output through inter-sectoral linkages.
- S: the savings matrix, which determines how much of each sector's output is reinvested.

The *structure* of growth—which sectors grow, how fast, and whether growth spreads across sectors—is governed by the interaction of these two matrices.

3.2.3 The Steady State

At the steady state $\mathbf{K}^*$, capital is constant: $\mathbf{K}^* = S(I - \gamma A')^{-1}\mathbf{f}(\mathbf{K}^*, \mathbf{L}) + (I - \Delta)\mathbf{K}^*$, or equivalently

$$\Delta\mathbf{K}^* = S(I - \gamma A')^{-1}\mathbf{f}(\mathbf{K}^*, \mathbf{L}). \tag{3.11}$$

Finding $\mathbf{K}^*$ is a fixed-point problem; with $\alpha_i < 1$, diminishing returns deliver a positive solution under standard conditions. The linear case ($\alpha = 1$, $f_i = K_i$) is different: the system becomes $(\Delta - S(I - \gamma A')^{-1})\mathbf{K}^* = \mathbf{0}$, which has a positive solution only if $\Delta - S(I - \gamma A')^{-1}$ is singular. Otherwise capital grows or shrinks at the rate given by the Perron root of $(I - \Delta) + S(I - \gamma A')^{-1}$ — a multi-sector AK model with endogenous growth. With $\alpha < 1$, the transition path is governed by the eigenvalues of the *growth mapping* $G = (I - \Delta) + S(I - \gamma A')^{-1}F'(\mathbf{K}^*)$, where F' is the Jacobian of the production function.

3.3 When Does Growth Reduce Inequality? The Kuznets Question

3.3.1 The Standard Answer

Simon Kuznets (1955) proposed that inequality first rises and then falls with development — the "inverted-U" hypothesis. The mechanism: early industrialization moves workers out of a low-income, relatively equal agricultural sector into a higher-income, more unequal industrial sector. While the shift is under way, the gap between the sectors raises overall inequality; once most workers have moved, that between-sector component shrinks and overall inequality falls.

The inverted-U has been one of the most tested hypotheses in economics. The evidence is mixed at best: cross-country data show no robust inverted-U, and the within-country trajectories vary enormously. Some countries (South Korea, Taiwan) grew rapidly and reduced inequality. Others (Brazil in the 1960s–70s) grew and saw inequality rise. Bolivia is a warning against easy stories. The Gini of per-capita household income fell from 0.600 in 2005 to 0.426 in 2019 (INE, Encuestas de Hogares; the World Bank's Poverty and Inequality Platform gives 0.585 and 0.416, World Bank, 2026a), and extreme poverty fell from 38.2% in 2005 to 15.2% in 2018 (INE, poverty lines of 2005–2018). Over the same years the gas sector remained an enclave in input-output terms (Section 3.2), and the integration of the domestic production network barely moved: $\Gamma(W^P)$ was 0.518 in 2006 and 0.484 in 2013 (12 sectors, input shares; INE input-output matrices, base 1990; author's calculations, `data/compute/compute_bolivia_io.py`; Instituto Nacional de Estadística (INE), Bolivia, 2026). Large equity gains came without any rise in structural integration. Any structural account has to confront that fact, and the transfer discussion in Section 3.5 does.

The matrix framework offers one part of the explanation: propagation depends on A.

3.3.2 The Structural Answer: It Depends on A

Example 3.2 (Two Economies, Same Technology, Different A)**.** *Consider two economies, each with $N = 2$ sectors: Modern (M) and Traditional (T). Both have the same technology:*

$$Y_M = K_M^{0.4} L_M^{0.6}, \qquad Y_T = K_T^{0.3} L_T^{0.7}. \tag{3.12}$$

Modern is more capital-intensive ($\alpha_M = 0.4 > \alpha_T = 0.3$). Both sectors have savings rate $s = 0.2$ and depreciation $\delta = 0.05$.

Economy A (Connected): *Strong inter-sectoral linkages (rows: Modern, Traditional; Chapter 1 convention).*

$$A_A = \begin{pmatrix} 0.10 & 0.25 \\ 0.20 & 0.10 \end{pmatrix}. \tag{3.13}$$

Modern uses Traditional inputs ($a_{MT} = 0.25$): factories buy agricultural raw materials. Traditional uses Modern inputs ($a_{TM} = 0.20$): farmers buy manufactured fertilizer and equipment.

Economy B (Disconnected): *Weak inter-sectoral linkages.*

$$A_B = \begin{pmatrix} 0.10 & 0.02 \\ 0.02 & 0.10 \end{pmatrix}. \tag{3.14}$$

The sectors barely interact. Modern imports its inputs and exports its output. Traditional produces for local consumption.

Now give both economies an impulse to the Modern sector: its productivity Z_M doubles, raising f_M.

Example 3.3 (Continued: The Propagation)**.** *The Leontief inverse for Economy A:*

$$L_A = (I - \gamma A_A')^{-1} \approx \begin{pmatrix} 1.18 & 0.26 \\ 0.33 & 1.18 \end{pmatrix} \quad (\text{with } \gamma = 1). \tag{3.15}$$

A unit impulse to Modern ($\Delta f_M = 1$, column 1) raises output by 1.18 *in Modern (direct plus own-sector feedback) and* 0.33 *in Traditional (cross-sector spillover). The Traditional sector benefits because it sells inputs to Modern ($a_{MT} = 0.25$).*

The Leontief inverse for Economy B:

$$L_B = (I - \gamma A_B')^{-1} \approx \begin{pmatrix} 1.11 & 0.02 \\ 0.02 & 1.11 \end{pmatrix}. \tag{3.16}$$

The same impulse to Modern generates 1.11 *in Modern and only* 0.02 *in Traditional. The spillover is negligible. Growth in Modern stays in Modern.*

Over the transition to the new steady state, Economy A shows both sectors growing (Traditional pulled up by its linkages to Modern). Economy B shows Modern growing and Traditional nearly unchanged, so the gap between the sectors widens with growth (Exercise 3.6 asks you to simulate both).

Proposition 3.4 (A Bound on Cross-Sector Propagation)**.** *Consider the multi-sector growth model (3.10) with technical coefficients A. An impulse to f_j raises the output of sector i by the (i, j)-th entry ℓ_{ij} of the Leontief inverse $(I - \gamma A')^{-1}$. If $\|\cdot\|$ is the spectral norm (or any submultiplicative matrix norm with $|M_{ij}| \le \|M\|$ and $\|A'\| = \|A\|$) and $\gamma\|A\| < 1$, then for every $i \neq j$*

$$\ell_{ij} \le \frac{\gamma\|A\|}{1 - \gamma\|A\|}, \tag{3.17}$$

and so is the average of the off-diagonal entries. Propagation is small when the linkages are weak in norm.

Intuition: What this says in words

Whether growth in one sector helps another sector depends on how connected they are in the production network. If the Leontief inverse has large off-diagonal entries (sectors are strongly linked), growth propagates. If the off-diagonal entries are small (sectors are weakly linked), growth concentrates. The overall strength of the linkages, measured by a matrix norm, bounds the strength of inter-sectoral transmission.

Proof. The Leontief inverse is $(I - \gamma A')^{-1} = \sum_{k=0}^{\infty} \gamma^k (A')^k$, convergent because $\gamma \|A'\| < 1$. The off-diagonal entries are $\ell_{ij} = \sum_{k=1}^{\infty} \gamma^k (A'^k)_{ij}$ (the $k = 0$ term is I, contributing only to the diagonal). Each $(A'^k)_{ij}$ sums over paths of length k from j to i, and $|(A'^k)_{ij}| \leq \|A'^k\| \leq \|A'\|^k$. Summing the geometric series gives the bound. □

The spectral radius cannot replace the norm. Entries of powers are not bounded by ρ^k: $\begin{pmatrix} 0.5 & 0.9 \\ 0 & 0.5 \end{pmatrix}$ has $\rho = 0.5$ but an entry of 0.9. For the productive matrix $\begin{pmatrix} 0.85 & 0.88 & 0.05 \\ 0 & 0.86 & 0.72 \\ 0 & 0 & 0.88 \end{pmatrix}$ ($\rho = 0.88$), the average off-diagonal entry of $(I - A')^{-1}$ is 56.5, far above $\|A\|_2/(1-\rho) = 12.7$. Because $\rho \leq \|A\|$, the norm bound is the safe one; for the examples of this chapter it gives 0.27 (Bolivia) and 0.49 (connected Bolivia).

3.4 Structural Heterogeneity: The Block-Diagonal Economy

3.4.1 Pinto's Diagnosis

Aníbal Pinto observed in the 1960s that Latin American economies were not uniformly "underdeveloped." They contained a modern sector with productivity levels comparable to the center, coexisting with a traditional sector with productivity levels comparable to the 19th century. The gap between sectors—within the same country—was often larger than the gap between the modern sector and the developed world.

This was not a transitional phase. It was a stable configuration. Growth did not close the gap; it widened it. The modern sector absorbed capital and technology, grew rapidly, and became more productive. The traditional sector absorbed labor, grew slowly or not at all, and stagnated. The structural disconnection between them prevented the propagation that would have equalized productivity.

We now formalize this.

Definition 3.5 (Structural Heterogeneity)**.** *An economy with technical coefficients A has **structural heterogeneity of degree** k if, after ordering sectors, A is approximately block-diagonal with k blocks:*

$$A \approx \begin{pmatrix} A_{11} & \varepsilon_{12} \\ \varepsilon_{21} & A_{22} \end{pmatrix}, \tag{3.18}$$

*where A_{11} and A_{22} are the within-block linkage matrices and $\varepsilon_{12}, \varepsilon_{21}$ are the cross-block linkages, with $\|\varepsilon_{12}\|, \|\varepsilon_{21}\| \ll \|A_{11}\|, \|A_{22}\|$. We measure the degree of coupling by the **cross-block mass** $\chi(A) = \sum_{\text{cross-block}} a_{ij} / \sum_{i,j} a_{ij}$, the share of all intermediate input requirements that cross block boundaries.*

Degree $k = 1$: the economy is structurally integrated (one block). Degree $k = 2$: the economy has a dual structure (Pinto's case). Degree $k > 2$: multiple disconnected sectors.

Example 3.6 (A Stylized Degree-2 Economy: Gas and Tradition)**.** [ILLUSTRATIVE] *A dual economy of the kind Pinto described, with Bolivian sector names, can be stylized as two blocks. This is a stylized economy, not an estimate of Bolivia's structure; the end of this section compares it with the measured one.*

***Block 1 (Modern):** Gas extraction, mining, large-scale agriculture (soybeans), financial services. High capital intensity, high productivity, strong linkages to global markets, weak linkages to the domestic traditional sector.*

***Block 2 (Traditional):** Small-scale agriculture (potatoes, llamas), informal commerce, small-scale manufacturing, domestic services. Labor-intensive, low productivity, strong internal linkages (farmers sell to local markets, local markets sell to local consumers), negligible linkages to the modern sector.*

The stylized coefficient matrix (rows: Modern, Traditional), with numbers chosen for illustration rather than estimated:

$$A_{dual} = \begin{pmatrix} 0.15 & 0.03 \\ 0.02 & 0.20 \end{pmatrix}. \tag{3.19}$$

Within the modern sector, linkages are moderate (0.15): gas companies buy services from financial firms, mining needs transport. Within the traditional sector, linkages are strong (0.20): the farmer sells to the market seller who buys from the small manufacturer. But cross-sector linkages are tiny: the modern sector buys 0.03 of traditional inputs per unit of output (it imports its equipment rather than buying from Bolivian workshops), and the traditional sector buys 0.02 of modern inputs. The cross-block mass is $\chi = 0.05/0.40 = 0.125$.

The Leontief inverse $L = (I - A')^{-1}$*:*

$$L \approx \begin{pmatrix} 1.18 & 0.03 \\ 0.04 & 1.25 \end{pmatrix}. \tag{3.20}$$

A unit impulse to the gas sector (column 1) generates 1.18 units of output in the modern sector and 0.04 in the traditional sector. Almost none of the growth propagates. The enclave multiplier is 1.22; of this, 96.4% stays in the modern sector.

Compare with a hypothetical connected version of the same economy, where both cross-sector coefficients are 0.15 *(*$\chi = 0.46$*):*

$$A_{connected} = \begin{pmatrix} 0.15 & 0.15 \\ 0.15 & 0.20 \end{pmatrix}, \quad L_{connected} \approx \begin{pmatrix} 1.22 & 0.23 \\ 0.23 & 1.29 \end{pmatrix}. \tag{3.21}$$

(A symmetric A has a symmetric inverse.) Now the same gas impulse generates 0.23 units in the traditional sector — about five times more propagation. The multiplier rises to 1.44, with 16% reaching the traditional sector.

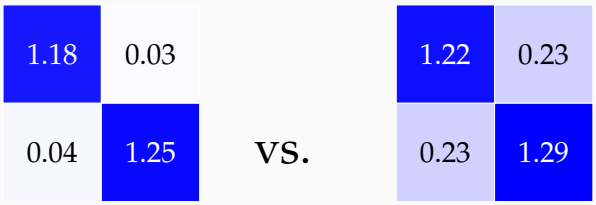
Prebisch Connection: The multiplier gap is the development gap

The difference between the two economies above is not technology, not savings rates, not human capital, not institutions in the Acemoglu-Robinson sense. It is the *structure of inter-sectoral linkages* — the cross-block entries of A. Both economies have the same production functions, the same capital, the same labor. They differ only in how connected the sectors are. And this difference produces, over time, radically different development trajectories.

This is a reinterpretation of Prebisch in the linkage language of Hirschman, not Prebisch's own mechanism. Prebisch explained the deterioration of the periphery's terms of trade by asymmetric wage and price behavior over the trade cycle and by different income elasticities of demand for primary goods and manufactures. The linkage argument adds a quantity side: when the center grows, its interconnected production structure amplifies the growth internally; when the periphery grows, its disconnected structure leaks the growth externally (through commodity export revenues that purchase imported manufactures) without amplifying it internally.

The multiplier gap — the difference in off-diagonal entries of the Leontief inverse between a connected and a disconnected economy — is one measurable part of the development gap. It is computable. It is a property of A. And it can be changed.

Visualization: Two possible Leontief inverses of a stylized dual economy



$(I - A')^{-1}_{\text{enclave}}$
Off-diagonal ≈ 0. Growth stays.

$(I - A')^{-1}_{\text{connected}}$
Off-diagonal bright. Growth propagates.

Same economy, same production functions, same capital, same labor; one colour scale for both panels. The only difference is the cross-sector coefficients of A (0.03 and 0.02 vs. 0.15). Left: the Leontief inverse is nearly diagonal — a gas impulse generates 0.04 in the traditional sector. Right: the same impulse generates 0.23 — about five times more. The dark off-diagonal squares on the left are the trickle-down problem in two pixels.

3.4.2 The Spectral Signature of Heterogeneity

Coupling between blocks has a spectral signature, but it lives in the row-normalized *production weight matrix* $W^P = D^{-1}A$ (input-cost shares; Chapter 1, Section 1.7), not in A itself. The spectral radius and second eigenvalue of A mostly track the overall level of the linkages: scaling A by c scales $\rho(A) - |\lambda_2(A)|$ by c, and the fully disconnected $A = \text{diag}(0.6, 0.05)$ has $\rho - |\lambda_2| = 0.55$, larger than either version of the stylized economy. W^P removes the scale.

Proposition 3.7 (Coupling, Integration and Block Weights)**.** *Let every row of A be nonzero, so that $W^P = D^{-1}A$ is row-stochastic.*

1. ***Exactly decoupled blocks.*** *If A is block-diagonal with k irreducible blocks, W^P has eigenvalue 1 with multiplicity k; for $k \geq 2$, $\Gamma(W^P) = 0$ and the Perron vector is not unique.*

2. ***Two sectors.*** *If $W^P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ with $a, b > 0$ and $a + b \leq 1$, its eigenvalues are 1 and $1 - a - b$, so $\Gamma(W^P) = a + b$: the integration speed is exactly the total cross-sector input share. Its Perron vector is $\boldsymbol{\pi} = (b, a)/(a + b)$.*

3. ***Reversible W^P, any number of sectors.*** *Let W^P be irreducible and reversible ($\pi_i W^P_{ij} = \pi_j W^P_{ji}$), let λ_2 be its second-largest (real) eigenvalue, and let $\Phi_c = \min_{S:\, 0<\pi(S)\leq 1/2} \sum_{i\in S, j\notin S} \pi_i W^P_{ij} / \pi(S)$ be its conductance. Then $\Phi_c^2/2 \leq 1 - \lambda_2 \leq 2\Phi_c$ (Cheeger), and hence $\Gamma(W^P) \leq 2\Phi_c$. Weak cross-block coupling (small Φ_c) therefore implies slow integration.*

In (2) the block weights depend only on the ratio a/b; multiplying both cross shares by the same factor (keeping $a + b \leq 1$) raises Γ and leaves $\boldsymbol{\pi}$ unchanged.

Proof. (1) Each diagonal block W^P_{bb} is row-stochastic and irreducible, so by Perron–Frobenius (Chapter 6, Theorem 6.6) it has the eigenvalue 1 exactly once; the characteristic polynomial of a block-diagonal matrix is the product of those of its blocks, so 1 has multiplicity exactly k, and $|\lambda_2| = 1$, $\Gamma = 0$ when $k \geq 2$. The stationary vector of each block, padded with zeros, is a stationary vector of W^P; for $k \geq 2$ these are linearly independent, so the stationary vector is not unique. (2) $W^P\mathbf{1} = \mathbf{1}$ gives the eigenvalue 1; the trace is $2 - a - b$, so the other eigenvalue is $1 - a - b \in [0, 1)$, and $\Gamma = 1 - |1 - a - b| = a + b$. Also $(b, a)W^P = (b(1 - a) + ab,\ ab + a(1 - b)) = (b, a)$, and the stationary vector is unique because $a, b > 0$ make W^P irreducible. Scaling (a, b) to (ta, tb) leaves $(b, a)/(a + b)$ unchanged and multiplies Γ by t. (3) The two inequalities are Cheeger's inequality for reversible chains (Levin et al., 2009, Thm. 13.14). Reversibility makes W^P similar to a symmetric matrix, so its eigenvalues are real, and $\Gamma(W^P) = 1 - \max_{k\geq 2} |\lambda_k| \leq 1 - \lambda_2 \leq 2\Phi_c$. □

More generally, in a nearly completely decomposable W^P (Simon and Ando, 1961) the Perron weights of the blocks are set, to first order in the coupling, by the balance of cross-block flows rather than by their level; for two sectors this is exactly part (2).

For the stylized dual economy, $W^P = \begin{pmatrix} 0.833 & 0.167 \\ 0.091 & 0.909 \end{pmatrix}$, so $\Gamma = 0.26$. For its connected version, $W^P = \begin{pmatrix} 0.50 & 0.50 \\ 0.43 & 0.57 \end{pmatrix}$ and $\Gamma = 0.93$. The production Perron vector of the enclave is $(0.35, 0.65)$: in the production reading, the traditional sector carries the larger weight, because the modern sector relies on it for a larger share of its inputs (0.167) than it relies on the modern sector (0.091). Integration and power are different questions.

Intuition: What this says in words

A nearly block-diagonal economy integrates slowly: an impulse that starts in one block takes a long time to reach the other, so productivity differences across blocks persist. The spectral gap of the row-normalized W^P measures exactly that speed, and for two sectors it equals the total cross-sector input share.

What the gap does *not* measure is who benefits. Strengthening every cross-block linkage in proportion speeds up integration without changing the block weights; changing the *direction* of dependence changes the weights. Pinto's structural heterogeneity is a statement about coupling — measured by the cross-block mass $\chi(A)$ and, in the row-normalized network, by $\Gamma(W^P)$ — not about power.

What the data say. On the integration coordinate the stylized economy is wrong about Bolivia. INE's 2019 supply-use table, aggregated to the book's 12 sectors (domestic flows, input shares), gives $\Gamma(W^P) = 0.467$, not 0.26, with $N_{\text{eff}} = 6.48$ of 11 sectors. On the ISIC-based aggregation used for cross-country comparison, Bolivia's Γ is 0.475, inside the range of the comparators in the OECD ICIO tables for the same year: USA 0.463, Korea 0.511, Germany 0.544, Chile 0.557, China 0.582, Mexico 0.649 (INE supply-use tables, base 2017, Instituto Nacional de Estadística (INE), Bolivia, 2026; OECD ICIO 2023, OECD, 2023; author's calculations, `data/compute/compute_bolivia_io.py` and `compute_oecd_io.py`). A single Γ also depends on aggregation and normalization: Bolivia's 2019 value ranges from 0.33 to 0.69 across input and sales shares, with and without the diagonal, at 12 and 39 sectors. Bolivia's dualism is real, but aggregate Γ does not detect it, and it does not separate Bolivia's development path from Korea's. The enclave shows up instead in where the gas sector sits in the network and where its output goes (Section 3.2). Chapter 17 gives the full table.

3.5 Does Growth Trickle Down? A Formal Answer

3.5.1 The Trickle-Down Hypothesis in Matrix Form

The "trickle-down" hypothesis says: growth in the modern sector benefits the traditional sector through market linkages—increased demand for traditional products, higher wages that spread through labor mobility, technology transfer. In our framework, trickle-down is the claim that the off-diagonal entries of the Leontief

inverse are large enough that modern-sector growth significantly raises traditional-sector output.

Definition 3.8 (The Trickle-Down Condition)**.** *Growth in sector j (the modern sector)* ***trickles down*** *to sector i (the traditional sector) if*

$$\ell_{ij} = [(I - \gamma A')^{-1}]_{ij} > \tau, \tag{3.22}$$

where $\tau > 0$ is a threshold representing "economically meaningful" propagation. Since $\ell_{ij} = \sum_{k \geq 1} \gamma^k (A'^k)_{ij}$, the condition aggregates all chains of production linkages from j to i; no single chain needs to be strong.

Intuition: When does trickle-down fail?

Trickle-down fails when ℓ_{ij} is small—when the Leontief inverse has small off-diagonal entries. This happens in three cases:

1. **Weak direct linkages**: a_{ji} is small (the modern sector buys few inputs from the traditional sector).

2. **Weak indirect linkages**: there is no chain of intermediate sectors connecting j to i with sufficient strength. All paths through the network from j to i pass through weak links.

3. **The enclave**: the modern sector is connected to the *global* economy but disconnected from the *domestic* economy. Its linkages point outward (imports, exports) rather than inward (domestic input purchases, domestic sales).

In the stylization of Example 3.6, Bolivia's gas sector fails on all three counts: it buys few inputs from the domestic traditional sector (case 1), there are few intermediate sectors linking gas to traditional agriculture (case 2), and its linkages run to export pipelines and international commodity markets rather than to Bolivian workshops (case 3). The 2019 supply-use table bears out cases 1 and 3 for the hydrocarbons sector: it buys 0.225 of domestic intermediate inputs per unit of output, against 0.558 for manufacturing, and exports 70% of its output (Section 3.2). Chapter 17 examines the rest.

3.5.2 The Transfer Perversity Result

Here is where the formation model connects to the growth model. Suppose the government, observing that gas growth does not trickle down, intervenes with a

fiscal transfer: tax the modern sector and redistribute to the traditional sector. (Bolivia has done something of this kind since the mid-2000s, with transfers financed in part by gas revenue, such as the Bono Juancito Pinto, introduced in 2006, and the Renta Dignidad, introduced in 2008.)

The multi-sector model above has no labor mobility, no wage premium and no sector choice, so it cannot derive what such a transfer does to structural transformation. What follows is therefore a conjecture, stated so that a model with those ingredients (a Lewis or Harris–Todaro block) could test it.

Conjecture 3.9 (Transfer Perversity Under Structural Heterogeneity)**.** *In a block-diagonal economy with structural heterogeneity, an unconditional transfer τ from the modern sector to the traditional sector:*

1. ***Raises traditional-sector income*** *(direct effect: $+\tau$).*

2. ***Raises traditional-sector capital*** *(through higher savings: $\Delta K_T = s_T \tau$).*

3. ***Lowers the modern-sector wage premium*** *(by raising traditional-sector productivity relative to the gap that motivates sector-switching).*

4. ***Weakens the signal for modern-sector investment*** *(the return differential between sectors narrows).*

5. ***Makes the traditional sector more attractive relative to the modern sector****, reducing the incentive for structural transformation.*

The net effect can be perverse: the transfer improves welfare within the traditional sector while making it less likely that the economy escapes structural heterogeneity. Items (1)–(2) follow from the accounting of this chapter; items (3)–(5) require the model with sector choice.

The conjecture concerns structural transformation, not inequality. Bolivia's measured inequality fell during the period of these transfers (Section 3.3); the conjecture would be consistent with that, and with persistent heterogeneity at the same time.

Intuition: Why transfers can entrench the trap

The transfer does not change A. It does not add linkages between sectors. It does not make the gas sector buy inputs from Bolivian workshops. It does not make traditional farmers invest in modern-sector skills. It takes money from the modern sector and gives it to the traditional sector, which spends it on traditional-sector goods (because that is what its information environment has trained its population to want; Chapter 10), which raises traditional-sector income, which makes the traditional sector more comfortable, which reduces the already-weak incentive to invest in the structural transformation that would connect the two sectors.

This is not an argument against transfers. Transfers reduce poverty and they should be done. It is an argument that transfers are not structural policy. Structural policy changes A — it builds the linkages, the roads, the value chains, the input-output connections that make growth propagate. Transfers change $\boldsymbol{\omega}$ (who has what). Structural policy changes A (who is connected to whom).

3.6 The Three Scales

We can now see the three-scale structure that Chapter 16 will develop fully:

Global scale. Countries are nodes. A_{global} is the international input-output matrix. Prebisch's hypothesis, which Chapter 16 tests, is that the center captures the gains from trade and the periphery transfers productivity gains outward.

National scale. Sectors are nodes. A_{national} is the domestic input-output matrix. If block-diagonal, the economy has structural heterogeneity. Modern-sector growth stays in the modern sector. Traditional-sector stagnation is self-reproducing. This is Pinto.

Individual scale. People are nodes. W^I is the information/signal network that shapes aspirations and preferences. If W^I channels traditional-sector people's attention toward traditional-sector signals, their aspirations are calibrated to the traditional sector even when the modern sector is accessible. This is the formation model (Chapter 10).

Key Structural Insight

The same mathematical framework — coupling between blocks of a network — operates at all three scales. At the global and national scales, cross-block entries of A determine how far growth in one block propagates to the others; the spectral gap of the row-normalized W^P says how fast it does so. At the individual scale, the cross-block weights of W^I determine whether normative change reaches the formation-constrained population, and $\Gamma(W^I)$ how fast. None of these gaps says who holds power; that is read off π (Chapter 6). At each scale, the answer to "does growth trickle?" is: it depends on the off-diagonal entries of the Leontief inverse. And those entries are a function of A, which is the structure.

Prebisch saw this at the global scale. Pinto saw it at the national scale. Neither had the matrix. Now we do.

3.7 Summary

This chapter established five results:

1. The **standard Solow model** assumes one sector and one savings rate. It predicts convergence. The real world has multiple sectors with heterogeneous productivity and differential savings, and convergence is conditional on the structure of inter-sectoral linkages.

2. The **multi-sector growth model** (3.10) embeds the Leontief system in a dynamic framework. Growth propagation depends on the technical coefficients A through the Leontief inverse $(I - \gamma A')^{-1}$; the reduced form transmits demand-side impulses upstream, while cost shocks travel downstream through the price dual.

3. The **Kuznets question**—does growth reduce inequality?—has a structural part of its answer: it depends on A. Connected economies (large cross-block entries, large off-diagonal Leontief entries) propagate growth across sectors. Disconnected economies (nearly block-diagonal A) concentrate growth in the modern sector. Measured inequality also depends on transfers and labor markets, as Bolivia shows: its Gini fell from 0.600 (2005) to 0.426 (2019) while $\Gamma(W^P)$ stayed roughly flat.

4. **Structural heterogeneity** (Pinto) is a nearly block-diagonal A with weak cross-block linkages, measured by the cross-block mass $\chi(A)$. In the row-normalized $W^P = D^{-1}A$ it shows up as slow integration: for two sectors,

$\Gamma(W^P)$ equals the total cross-sector input share. The gap measures integration, not who holds power. The stylized dual economy is a degree-2 case; Bolivia's measured 12-sector $\Gamma(W^P)$ (0.467 in 2019) is not small, so its enclave shows up in π and export orientation, not in Γ.

5. **Transfers do not change** A. They alleviate poverty within the traditional sector without building the inter-sectoral linkages that would make growth propagate. Structural policy changes the matrix. Transfer policy changes the endowment vector.

3.8 Exercises

3.1 Solow with two groups. For the OLG model with $\alpha = 0.3$, $\theta^M = 2$, $\theta^F = 0.5$:

(a) Compute the savings rates $\sigma(\theta^M)$ and $\sigma(\theta^F)$.

(b) Compute the aggregate savings rate $\bar{s} = \frac{1}{2}[\sigma(\theta^M) + \sigma(\theta^F)]$.

(c) Find the steady-state capital K^* as a function of $\bar{s}$.

(d) How much lower is K^* compared to an economy where both groups have $\theta = 2$? This gap is the macroeconomic cost of the formation mechanism.

3.2 Propagation in a 3-sector economy. For:

$$A = \begin{pmatrix} 0.10 & 0.20 & 0.05 \\ 0.15 & 0.10 & 0.15 \\ 0.05 & 0.10 & 0.10 \end{pmatrix}. \tag{3.23}$$

(a) Compute the Leontief inverse $(I - A')^{-1}$.

(b) Sector 2 receives a unit impulse ($\Delta f_2 = 1$). How much does each sector's output change?

(c) Which sector has the highest output multiplier? Why?

(d) Compute $\rho(A)$. Then row-normalize to $W^P = D^{-1}A$ and compute its spectral gap. Is this economy "connected" or "heterogeneous"?

3.3 The enclave. Modify the 3-sector economy: set $a_{31} = a_{32} = 0.01$ and $a_{13} = a_{23} = 0.01$ (sector 3 is an enclave).

(a) Recompute the Leontief inverse. How much does a shock to sector 3 propagate to sectors 1 and 2?

(b) Compare with the connected case. By what factor does propagation decline?

(c) Compute the spectral gap of the new W^P. Has it changed? Has the cross-block mass?

(d) Relate to Bolivia's gas sector.

3.4 Transfer vs. structural policy. Using the stylized dual economy (Example 3.6):

(a) Suppose the government transfers 10% of modern-sector output to the traditional sector. What happens to traditional-sector income? To the cross-sector productivity gap?

(b) Now suppose the government spends the same 10% building linkages: the cross-sector coefficients of A (0.03 and 0.02) increase to 0.10. Recompute the Leontief inverse. What happens to the cross-sector propagation?

(c) Compare the two policies after 10 periods. Which one closes the productivity gap?

3.5 Prebisch in the global I-O table. Download a multi-region input-output table that includes Bolivia (Bolivia is not a separate economy in the OECD ICIO tables; Eora and CEPAL's regional input-output matrices include it). Aggregate to 5 countries or regions — for example USA, Germany, China, Bolivia, Nigeria — and build the country-level coefficient matrix A and its row normalization W^P.

(a) Compute column sums of W^P (how much the others rely on each country's output). Which countries have the highest column sums?

(b) Compute the Leontief inverse $(I - A')^{-1}$. Which countries have the highest output multipliers?

(c) A productivity shock hits Bolivia (agricultural sector). How much propagates to the USA vs. to Nigeria?

(d) Is the global economy "connected" or "block-diagonal"? Compute the cross-block mass and the spectral gap of W^P.

(e) *Challenge*: Compute the left Perron eigenvector of the global W^P. Which countries have the most weight in the production reading of $\boldsymbol{\pi}$ (Chapter 1, Section 1.7)? Is this consistent with the center-periphery hypothesis?

3.6 Kuznets revisited. Simulate the multi-sector growth model for $T = 50$ periods with $N = 2$ sectors, starting from $K_M = K_T = 1$, with a productivity advantage in the modern sector ($Z_M = 2, Z_T = 1$):

(a) Case A: A connected (off-diagonal = 0.15). Plot the ratio Y_M/Y_T over time.

(b) Case B: A disconnected (off-diagonal = 0.02). Plot the same ratio.

(c) The ratio Y_M/Y_T is not an inequality index. Assign each sector's output to its workers and compute the between-sector variance of log output per worker over time. Does it trace an inverted U in either case?

(d) How does the long-run ratio Y_M/Y_T depend on the cross-sector linkage strength? With diminishing returns in each sector, is there any linkage strength at which the ratio diverges, or does it always converge?

3.7 The formation connection (preview of Chapter 10). In the two-group OLG model, suppose θ^F is not fixed but evolves according to:

$$\theta^F_{t+1} = 0.8\theta^F_t + 0.2\bar{\theta} + \mu(\lambda_t r_t - r^*), \tag{3.24}$$

where r_t is the return on savings (take $r_t = 0.1$ for all t) and $\lambda_t < 1$ is the signal discount (women perceive lower returns).

(a) Starting from $\theta^F_0 = 0.5$ and $\theta^M = 2$ (fixed), simulate for 20 generations with $\lambda = 0.7$, $\mu = 0.1$, $\bar{\theta} = 2$, $r^* = 0.1$.

(b) Plot the trajectory of θ^F_t and the gender gap $\theta^M - \theta^F_t$.

(c) At what generation does the gap halve?

(d) Now set $\lambda = 1$ (signal distortion removed) at $t = 10$. How many additional generations until the gap halves from its $t = 10$ value? Compare with the persistence results of Chapter 10.

Mathematical Appendices

Appendix 3A: Block Matrices and Their Eigenvalues

Block matrices. A **block matrix** is a matrix partitioned into rectangular submatrices. A 2×2 block matrix:

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \tag{3.25}$$

where A is $n_1 \times n_1$, D is $n_2 \times n_2$, B is $n_1 \times n_2$, and C is $n_2 \times n_1$. The total matrix M is $(n_1 + n_2) \times (n_1 + n_2)$.

Block-diagonal matrices. When $B = 0$ and $C = 0$, the matrix is **block-diagonal**: $M = \text{diag}(A, D)$. The eigenvalues of a block-diagonal matrix are simply the union of the eigenvalues of each block. If A has eigenvalues $\{\alpha_1, \ldots, \alpha_{n_1}\}$ and D has eigenvalues $\{\delta_1, \ldots, \delta_{n_2}\}$, then M has eigenvalues $\{\alpha_1, \ldots, \alpha_{n_1}, \delta_1, \ldots, \delta_{n_2}\}$.

Why this matters. A block-diagonal coefficient matrix means the economy's sectors don't interact across blocks. Each block has its own dynamics and its own eigenvalues. For the row-normalized W^P, each block is itself row-stochastic and contributes an eigenvalue 1, so a block-diagonal W^P has $\Gamma = 0$: two separate "modes" that never mix (Proposition 3.7).

Near-block-diagonal matrices. When B and C are small but nonzero (weak cross-block linkages), the eigenvalues are perturbed from the block-diagonal case. The Bauer-Fike theorem (Horn and Johnson, 2012, Thm. 6.3.2) states: if $\text{diag}(A, D) = V\Lambda V^{-1}$ is diagonalizable, λ is an eigenvalue of M and λ_0 is the nearest eigenvalue of $\text{diag}(A, D)$, then $|\lambda - \lambda_0| \leq \kappa(V)\|E\|_2$, where $\kappa(V) = \|V\|_2\|V^{-1}\|_2$ is the condition number of the eigenvector matrix and $E = M - \text{diag}(A, D)$ is the off-diagonal perturbation (the blocks B and C).

In plain language: small cross-block linkages produce small changes in eigenvalues (when the eigenvector matrix is well conditioned). The qualitative structure (two slowly mixing modes, small spectral gap of W^P) is preserved under small perturbation. This is why structural heterogeneity is robust: adding a few weak linkages between the gas sector and the traditional sector does not fundamentally change the spectral structure. The economy remains approximately block-diagonal until the cross-block linkages are comparable in strength to the within-block linkages.

Example. For the stylized dual-economy coefficient matrix of Example 3.6:

$$A = \begin{pmatrix} 0.15 & 0.03 \\ 0.02 & 0.20 \end{pmatrix}. \tag{3.26}$$

Block-diagonal part: $\operatorname{diag}(0.15, 0.20)$ with eigenvalues 0.15 and 0.20. Full matrix: the characteristic equation is $\lambda^2 - 0.35\lambda + 0.0294 = 0$, with discriminant $0.1225 - 0.1176 = 0.0049$, so $\lambda_1 = 0.21$ and $\lambda_2 = 0.14$. The perturbation moves each eigenvalue by 0.01, well within $\|E\|$. The spectral radius $\rho(A) = 0.21 < 1$ confirms that the economy is productive.

For the connected case ($A_{\text{connected}}$ from Example 3.6), $\lambda_1 = 0.327$ and $\lambda_2 = 0.023$.

The difference $\rho(A) - |\lambda_2(A)|$ (0.07 versus 0.30) is not a measure of integration: it scales with the level of A, and the fully disconnected $\operatorname{diag}(0.6, 0.05)$ would score 0.55. Integration is measured on the row-normalized matrices: $\Gamma(W^P) = 0.26$ for the enclave and 0.93 for the connected case (Proposition 3.7).

Appendix 3B: Convergence in Multi-Sector Growth Models

The convergence question. In the one-sector Solow model, the economy converges to a unique steady state at rate $\beta = (1-\alpha)(\delta + g + n)$. In the multi-sector model, convergence depends on the eigenvalues of the *Jacobian of the growth mapping* evaluated at the steady state.

The growth mapping. Define $G(\mathbf{K}) = S(I - \gamma A')^{-1}\mathbf{f}(\mathbf{K}) + (I - \Delta)\mathbf{K}$ from equation (3.10). The Jacobian at steady state $\mathbf{K}^*$ is:

$$J = DG(\mathbf{K}^*) = S(I - \gamma A')^{-1}F'(\mathbf{K}^*) + (I - \Delta), \tag{3.27}$$

where $F'(\mathbf{K}^*) = \operatorname{diag}(\alpha_1 f_1/K_1^*, \ldots, \alpha_N f_N/K_N^*)$ is the diagonal matrix of marginal products.

Convergence condition. The economy converges to $\mathbf{K}^*$ from all nearby initial conditions if all eigenvalues of J have modulus less than 1, $\rho(J) < 1$; if $\rho(J) > 1$ the steady state is unstable (the linearization theorem for maps; the case $\rho(J) = 1$ is not decided by J). The *speed* of convergence is $-\ln \rho(J)$: faster when $\rho(J)$ is further below 1.

How A affects convergence speed. The Leontief inverse $(I - \gamma A')^{-1}$ amplifies the marginal products $F'(\mathbf{K}^*)$, which makes the eigenvalues of J larger (closer to 1). Stronger inter-sectoral linkages (larger $\gamma\rho(A)$) mean stronger amplification, which means *slower* convergence in levels but stronger propagation of shocks across sectors.

The tradeoff: a well-connected economy ($\gamma\rho(A)$ close to 1) propagates growth broadly but converges slowly to the steady state. A disconnected economy (small $\gamma\rho(A)$) converges fast *within* each block but transmits little *between* blocks.

The two-speed result. In a block-diagonal economy, the Jacobian J is approximately block-diagonal, and convergence proceeds at two speeds:

- **Within-block convergence**: governed by the eigenvalues of the individual blocks. Fast if each block has moderate linkages.
- **Between-block transmission**: governed by the cross-block entries of J, which are small (proportional to ε). An impulse to one block reaches the other only weakly.

Nothing in this model pulls the two blocks' productivity levels toward each other: each sector converges to its own steady state, determined by its own parameters. This is the "two-speed" pattern that Pinto observed: the modern sector grows rapidly to its (high) steady state; the traditional sector grows slowly to its (low) steady state; and the gap between the steady states is closed only by cross-block linkages strong enough to transmit the modern sector's growth, which here are small.

In the formation model (Chapter 10), the same two-speed pattern appears for preferences: enforcement effects converge fast (within-generation timescale), formation effects converge slow (between-generation timescale, governed by ρ). The mathematical structure is identical: block-diagonal dynamics with weak cross-block coupling.

Structuralist Case Study: Pinto's Structural Heterogeneity and Weak Coupling

Aníbal Pinto Santa Cruz (1919–1996) was a Chilean economist who spent most of his career at CEPAL. In 1970, he published a short essay in *El Trimestre Económico* — "Naturaleza e implicaciones de la 'heterogeneidad estructural' de la América Latina" — that reframed the structuralist research program. His argument was that the center-periphery framework, while correct about international asymmetries, had missed an equally important asymmetry *within* peripheral economies.

The problem was not only the asymmetry between center and periphery but its reproduction inside the periphery.

Pinto's thesis has three parts, each of which maps to a result in this chapter.

Part 1: The peripheral economy is not one economy but three. Pinto identified a tripartite structure within Latin American countries: a "primitive" stratum (with productivity levels comparable to colonial times), a "modern pole" (with productivity levels comparable to the industrialized countries), and an "intermediate" stratum (near the national average). The productivity ratio between the modern pole and the primitive stratum was very large (the figure of 20:1 often attributed to the essay needs checking against the text).

In this chapter's notation: the domestic coefficient matrix A has at least two blocks (we simplified to two for tractability), with *weak cross-block entries* — a small cross-block mass $\chi(A)$. The tripartite structure is a nearly block-diagonal matrix with three blocks of very different productivity. The spectral gap $\Gamma(W^P)$ of the row-normalized matrix is small because the coupling between blocks is weak: impulses mix slowly across them. Whether measured production networks show this at the level of aggregate sectors is a separate, empirical question; for Bolivia at 12 sectors they do not (Section 3.4).

Part 2: Import-substitution industrialization added new layers of heterogeneity without eliminating old ones. The ISI strategy, which CEPAL had championed, successfully built a modern industrial sector in several Latin American countries (Brazil, Mexico, Argentina). But Pinto observed that this modern sector coexisted with the pre-existing traditional sector rather than absorbing it. The economy became more complex — more blocks in the matrix — without becoming more *connected*. The off-diagonal entries of W^P remained small. He called this "modernización excluyente" — exclusionary modernization.

This is the dynamic result from Section 3.5: new blocks can be added to a block-diagonal matrix (new sectors created through ISI) without increasing the cross-block mass or the spectral gap of W^P, if the new blocks connect only to themselves and to the global economy rather than to the pre-existing traditional sector. The Leontief inverse gains new rows and columns, but the off-diagonal entries between the new modern block and the old traditional block remain dark. The heatmap gets bigger without getting brighter.

Part 3: The structural policy implication. Pinto's prescription was not more ISI (adding more blocks) but structural integration: building the cross-block linkages that would allow productivity gains in the modern sector to propagate to the traditional sector. In the language of this chapter: the policy goal is to increase the cross-block entries of A — to make the heatmap brighter off the diagonal —

which raises the off-diagonal Leontief entries and speeds up integration (larger $\Gamma(W^P)$). Whether it also reduces the between-group welfare loss of Chapter 2 (Proposition 2.11) depends on whether the linkages actually narrow the gap in block means; and whether it changes who holds power depends on the direction of the new dependence, not on Γ.

Pinto's argument	**Formal object**	**This chapter**
Tripartite structure	Nearly block-diagonal A with 2–3 blocks	Sec. 3.4
Large productivity gap between strata	Different block means; between-group loss	Chapter 2, Prop. 2.11
ISI adds blocks, not links	New blocks with dark cross-entries	Eq. (3.18)
Exclusionary modernization	Growth stays in bright blocks	$\ell_{ij} \approx 0$ for $i \neq j$
Weak coupling	Small $\chi(A)$; small $\Gamma(W^P)$ ($= w_{12} + w_{21}$ for two sectors)	Prop. 3.7
Structural integration	Increase off-diagonal entries	Example 3.6: $0.03 \to 0.15$

Pinto's essay is short. Its mathematical content is a 2×2 nearly block-diagonal matrix, its Leontief inverse, and the coupling between its blocks. He did not have the notation. He had the diagnosis.

Further Reading

Growth theory: Acemoglu (2009), *Introduction to Modern Economic Growth*, chapters on Solow, OLG, multi-sector models. Barro and Sala-i Martin (2004), *Economic Growth*, for the empirical convergence literature.

Multi-sector growth with networks: Acemoglu et al. (2012a), "Network Origins of Aggregate Fluctuations." Baqaee and Farhi (2019), "The Macroeconomic Impact of Microeconomic Shocks." Liu and Tsyvinski (2020), "Dynamical Structure and Spectral Properties of Input-Output Networks."

Structural heterogeneity: Pinto (1970), "Naturaleza e implicaciones de la 'heterogeneidad estructural'." This is the source. Read it. CEPAL (2010), *Time for Equality: Closing Gaps, Opening Trails*, for the modern institutional statement.

The CEPAL tradition: Prebisch (1950), start here. Furtado (1964), *Development*

and Underdevelopment. Ocampo et al. (2009), *Growth and Policy in Developing Countries: A Structuralist Approach*. Bielschowsky (2009), "Sixty Years of ECLAC."

Bolivia: Morales and Sachs (1990), "Bolivia's Economic Crisis" for the structural adjustment context. Weisbrot et al. (2009), "Bolivia: The Economy During the Morales Administration" for the resource-transfer period. The National Statistics Institute (INE) input-output tables for the data.

Chapter 4

What the Standard Model Assumes

Six Properties of One Matrix

> *"El subdesarrollo no es un 'momento' ni una 'etapa' en la evolución de una sociedad aislada y autónoma, sino parte del proceso histórico global de desarrollo del capitalismo."*
>
> ["Underdevelopment is not a 'moment' or a 'stage' in the evolution of an isolated and autonomous society, but part of the global historical process of capitalist development."]
> — Osvaldo Sunkel and Pedro Paz, *El subdesarrollo latinoamericano y la teoría del desarrollo* (1970)

The first three chapters rewrote the standard model in matrix notation. Chapter 1 showed that anonymous, price-mediated interaction implies $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$, and named its equal-endowment case $W^{CE} = \frac{1}{N}\mathbf{1}\mathbf{1}'$. Chapter 2 showed which departures from it break efficiency ($W^{U} \neq 0$) and which are distributional (unequal $\boldsymbol{\pi}$). Chapter 3 showed that growth propagation depends on the production structure A.

This chapter pulls the threads together. It catalogs, in one place, the assumptions from which the benchmark model abstracts — six properties, most of them properties of one matrix — and for each assumption states: (a) the formal condition, (b) when it approximately holds, (c) when it fails, (d) what the world looks like when it does, and (e) which chapter in this book develops the general case.

This is the hinge. Part I ends here. Part II begins with W as an object to be studied rather than assumed.

4.1 The Six Assumptions

Recall from Chapter 1 that the competitive benchmark $W^{CE} = \frac{1}{N}\mathbf{1}\mathbf{1}'$ has five properties: doubly stochastic, symmetric, rank 1, equal power, unit spectral gap. The benchmark Arrow–Debreu model abstracts from several kinds of structure; the six assumptions below make that abstraction explicit. The sixth is not a property of any matrix but of the model that uses it: preferences are exogenous.

Definition 4.1 (The Six Assumptions of the Standard Model)**.** *The benchmark competitive model abstracts from the following ("assumes" in what follows means "is built so that"):*

A1. No utility externalities. *$W^U = 0$ (or equivalently $\gamma_U = 0$). Agent i's utility depends only on x_i.*

A2. Exogenous production structure. *The technical coefficients A (and hence $W^P = D^{-1}A$) are taken as given from input-output data: a technological fact, not a social choice.*

A3. No information network. *Beliefs are not formed through social channels: there is no W^I along which one agent's beliefs move another's.*

A4. Anonymous market interaction. *The price mechanism aggregates demands anonymously, weighting agents by wealth shares: $W^{mkt} = \mathbf{1}\mathbf{s}'$ (Chapter 1, Definition 1.17).*

A5. Equal power. *The left Perron vector of the composite interaction matrix is $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$. No agent has more structural influence than any other. In the benchmark this holds only with equal endowments, since $\boldsymbol{\pi}(W^{mkt}) = \mathbf{s}$.*

A6. Exogenous preferences. *θ_i (the preference parameter governing patience, risk aversion, time discount, labor supply) is fixed. It does not evolve in response to the network.*

We now examine each assumption in detail.

4.2 A1: No Utility Externalities ($W^U = 0$)

4.2.1 The Formal Condition

Agent i maximizes $U_i(x_i)$, not $V_i(\mathbf{x}) = U_i(x_i) + \gamma_U \sum_j W^U_{ij} U_j(x_j)$. Nobody's consumption enters anyone else's utility function. No altruism, no envy, no status competition, no compadrazgo bonds, no peer effects.

4.2.2 When It Approximately Holds

Large anonymous markets for standardized goods. The buyer of a barrel of oil on the NYMEX does not care who produced it. The seller of wheat at the Chicago Board of Trade does not know who will eat it. In these settings, the externality weights W_{ij}^U are effectively zero because the transacting parties have no social relationship.

4.2.3 When It Fails

Small communities. In a village or a tight-knit neighborhood, people know each other. The api vendor cares about her comadre's income. The shopkeeper cares about the schoolteacher's welfare. Utility externalities are large, local, and asymmetric.

Status competition. Veblen (1899), Duesenberry (1949), Frank (1985): consumption generates externalities through social comparison. A neighbor's new car reduces your satisfaction with your old one. The externality is negative and proportional to the visibility of consumption.

Network effects. Adoption of a technology by agent j raises the value of the technology for agent i (telephones, social media, payment systems). $W_{ij}^U > 0$ for complementary users.

Proposition 4.2 (Welfare Cost of Ignoring W^U)**.** *In the one-good quadratic economy of Chapter 2, under the hypotheses of Proposition 2.9 ($1 - \kappa m > 0$ and an interior social optimum with $\kappa x_j < 1$), when $W^U \neq 0$ and $\gamma_U > 0$, the externality-blind allocation (which ignores W^U) generates a social-welfare loss of*

$$L_{A1} = \gamma_U^2 \cdot \mathrm{Var}(c_j^U) \cdot K_{ext} + O(\gamma_U^3), \qquad K_{ext} = \frac{N(1-\kappa m)^2}{2\kappa}, \tag{4.1}$$

where $c_j^U = \sum_{i \neq j} W_{ij}^U$ is agent j's externality centrality in the utility network and $m = \Omega/N$ (Chapter 2, Proposition 2.9, which also gives the exact formula).

Proof. With identical U, the externality-blind allocation is equal division $x_j = m$. Proposition 2.9 with $\gamma = \gamma_U$ and $c_j = c_j^U$ gives its loss as $\frac{N(1-\kappa m)^2}{2\kappa}\gamma_U^2 \,\mathrm{Var}(c_j^U) + O(\gamma_U^3) = \gamma_U^2 \,\mathrm{Var}(c_j^U) K_{\mathrm{ext}} + O(\gamma_U^3)$. □

Relaxed in: Chapters 5 (the utility network defined), 7 (bargaining equilibrium with externalities), 12 (welfare under endogenous preferences).

4.3 A2: Exogenous Production Structure (W^P Given)

4.3.1 The Formal Condition

The technical coefficients matrix A is treated as a technological parameter: determined by engineering, not by economic choices. The Leontief system takes A as data and solves for output given final demand. This is a feature of the fixed-coefficient benchmark, not of general-equilibrium theory as such: with Cobb-Douglas or CES technologies, input shares respond to prices, and recent work makes the network itself endogenous (Oberfield, 2018; Acemoglu and Azar, 2020). What such work rarely does is treat A as an object of development policy.

4.3.2 When It Approximately Holds

Short-run analysis. In a given year, a steel plant requires a specific amount of iron ore, coal, and electricity per ton of steel. These technical coefficients change slowly. For short-run comparative statics, treating A as fixed is reasonable.

4.3.3 When It Fails

ISI as structural policy. The entire history of import-substitution industrialization is an attempt to *change* A — to build domestic production of goods previously imported. Tavares (1972) documented how each external bottleneck provoked a structural response that altered the domestic input-output matrix. A is a policy variable, not a constant.

Endogenous specialization. Countries choose what to produce. The product space (Hidalgo et al., 2007) shows that diversification follows path-dependent trajectories shaped by the existing productive structure: a country that produces wine can more easily diversify into grapes, bottles, and corks than into semiconductors. The production network evolves.

FDI and global value chains. When a multinational opens an assembly plant, it inserts new rows and columns into the domestic A, connecting the domestic economy to the global production network. The entries of A reflect corporate decisions, not just technology.

Proposition 4.3 (Endogenous W^P Changes the Growth Trajectory)**.** *In the multi-sector growth model of Chapter 3, write* $L_\gamma = (I - \gamma A')^{-1}$ *and* $F' = diag(\partial f_i/\partial K_i)$ *at the steady state* $\mathbf{K}^*$*. If* $\Delta - SL_\gamma F'$ *is invertible, a small change* ΔA *in the technical*

coefficients shifts the steady-state capital vector by

$$\Delta \mathbf{K}^* \approx \left(\Delta - SL_\gamma F'\right)^{-1} S\, L_\gamma\, (\gamma\, \Delta A')\, L_\gamma\, \mathbf{f}(\mathbf{K}^*). \tag{4.2}$$

The effect is amplified by the Leontief inverse twice — once for the direct impact, once for the propagation through the existing network — and then by the steady-state feedback $(\Delta - SL_\gamma F')^{-1}$*. A small change in* A *can have a large effect on* $\mathbf{K}^*$ *if the Leontief inverse has large off-diagonal entries.*

Proof. Assume $\gamma\rho(A) < 1$, so that L_γ exists and depends smoothly on A, and that $\mathbf{f}$ is continuously differentiable near $\mathbf{K}^* \gg 0$. The steady state solves $H(\mathbf{K}, A) := \Delta\mathbf{K} - SL_\gamma(A)\mathbf{f}(\mathbf{K}) = 0$, and $D_{\mathbf{K}}H = \Delta - SL_\gamma F'$ is invertible by hypothesis, so by the implicit function theorem the steady state is locally a C^1 function $\mathbf{K}^*(A)$. Differentiate: $\Delta\, d\mathbf{K} = S\,(dL_\gamma)\,\mathbf{f} + SL_\gamma F' d\mathbf{K}$, and $dL_\gamma = L_\gamma(\gamma\, dA')L_\gamma$ (differentiate $L_\gamma(I - \gamma A') = I$). Solve for $d\mathbf{K}$. (For the two-sector economy of Chapter 3, Example 3.2, raising a_{12} by 0.01 moves $\mathbf{K}^*$ by $(0.089, 0.269)$ by this formula and by $(0.089, 0.270)$ by direct computation; `code/ch04.py`.) □

Relaxed in: Chapters 3 (production network in growth), 13 (how networks evolve), 15 (operator design — policy as deliberate change of W^P).

4.4 A3: No Information Network (W^I Absent)

4.4.1 The Formal Condition

The benchmark Arrow–Debreu model abstracts from belief formation through social networks: agents know the relevant prices and probabilities, and there is no mechanism through which one agent's beliefs influence another's — no W^I. Mainstream economics relaxes much of this, with asymmetric information and with Bayesian learning in social networks (Acemoglu et al., 2011). What it rarely does is make the weights W^I themselves the object of study.

4.4.2 When It Approximately Holds

Financial markets with mandatory disclosure, standardized goods with transparent pricing, and repeated interactions where agents learn the true model. In these settings, heterogeneous beliefs converge to the truth through Bayesian updating, making W^I unimportant.

4.4.3 When It Fails

DeGroot-style learning. Agents in many real-world settings do not observe the data-generating process. They observe their neighbors' behavior, their social media feeds, their peers' outcomes. There is experimental evidence that, in some settings, people update closer to a weighted average of neighbors' beliefs than to the Bayesian benchmark (Chandrasekhar et al., 2020). Under such averaging the long-run consensus weights are the Perron vector of W^I (DeGroot, 1974): the agents with the most epistemic power determine the consensus, whether or not their initial beliefs are correct. Golub and Jackson (2010) characterize when the consensus is nevertheless correct in large networks (the "wisdom of crowds").

Media concentration. When a few outlets control information flow, W^I is star-shaped: everyone listens to the hub. The consensus converges to the hub's beliefs. State media, monopoly platforms, and algorithmic feeds all produce this topology.

Aspiration formation. The girl in Potosí does not observe the "true" return to education. She observes the returns *in her network* — her mother, her school friends, her cousin in Buenos Aires, her TikTok feed. Her perceived returns are W^I-weighted averages of the returns experienced by her reference group. This shapes her aspirations, her investment in education, and ultimately her economic outcomes.

Proposition 4.4 (Epistemic Power and Belief Convergence)**.** *Under DeGroot learning on an information network W^I (row-stochastic, irreducible, aperiodic):*

$$\lim_{t\to\infty} \mathbf{b}^{(t)} = (\boldsymbol{\pi}^I)'\mathbf{b}^{(0)} \cdot \mathbf{1}, \tag{4.3}$$

where $\boldsymbol{\pi}^I$ is the left Perron eigenvector of W^I. The long-run consensus is a weighted average of initial beliefs, with weights given by epistemic power π_j^I. Agents with $\pi_j^I \gg 1/N$ dominate the consensus.

Proof. $\mathbf{b}^{(t)} = (W^I)^t\mathbf{b}^{(0)}$. For a primitive (irreducible and aperiodic) stochastic matrix, $(W^I)^t \to \mathbf{1}(\boldsymbol{\pi}^I)'$ by Perron–Frobenius (Chapter 6, Theorem 6.6; Horn and Johnson 2012, Thm. 8.5.1). Directly: write $W = W^I$, $\boldsymbol{\pi} = \boldsymbol{\pi}^I$ and $P = \mathbf{1}\boldsymbol{\pi}'$. Since $W\mathbf{1} = \mathbf{1}$, $\boldsymbol{\pi}'W = \boldsymbol{\pi}'$ and $\boldsymbol{\pi}'\mathbf{1} = 1$, we have $WP = PW = P^2 = P$, hence $(W - P)^t = W^t - P$ for $t \geq 1$. The spectrum of $W - P$ is that of W with the simple eigenvalue 1 replaced by 0, and primitivity puts every other eigenvalue of W strictly inside the unit circle, so $\rho(W - P) = |\lambda_2(W)| < 1$ and $(W - P)^t \to 0$ (Chapter 1, Appendix 1.9). □

Relaxed in: Chapters 5 (the information network defined), 8 (belief manipula-

tion and epistemic power), 9–12 (the formation model, where W^I shapes preferences).

4.5 A4: Anonymous Market Interaction ($W^{\mathbf{mkt}} = \mathbf{1s}'$)

4.5.1 The Formal Condition

The competitive price mechanism aggregates demands anonymously. Each agent's weight in aggregate demand is her expenditure share s_i ($1/N$ with equal budgets). Nobody has market power. The interaction is $W^{\text{mkt}} = \mathbf{1s}'$, which is W^{CE} under equal endowments (Chapter 1, Section 1.6).

4.5.2 When It Approximately Holds

Markets with many small agents, homogeneous goods, free entry and exit, and complete information. The textbook examples: wheat markets, foreign exchange (at the retail level), standardized commodity markets with many participants.

4.5.3 When It Fails

Monopsony and monopoly. A single buyer (monopsony employer in a company town) or single seller (a state monopoly in domestic gas distribution) sets prices rather than taking them. The interaction matrix has a star topology: $W^{\text{mkt}}_{ij} \approx 0$ for all j except the hub.

Platform economies. Amazon, Uber, and Airbnb are hubs through which millions of small agents must transact. The platform sets the terms of trade. W^{mkt} is star-shaped with the platform at the center.

Bargaining. In thin markets (few participants, differentiated goods, relationship-specific investments), prices are determined by bilateral bargaining rather than anonymous market clearing. The allocation depends on relative bargaining power, which is a function of the network position (Chapter 7).

Proposition 4.5 (Distributional Loss from Market Power)**.** *When interaction is not anonymous and the surplus is divided by asymmetric Nash bargaining with weights $\boldsymbol{\pi}(W)$ from equal disagreement payoffs (the book's bargaining assumption; Chapter 7, Proposition 7.3), allocations are*

$$x_i^* = \frac{\Omega}{N} + \left(\pi_i(W) - \frac{1}{N}\right)\Phi, \tag{4.4}$$

where Φ is the bargaining surplus, with quadratic-utility loss $L_{A4} = \frac{\kappa N}{2}\Phi^2 \operatorname{Var}(\pi_i)$

(Chapter 2, Proposition 2.7). We assume W irreducible, $\Phi > 0$ and $\kappa x_i^* \leq 1$ for all i. The loss is distributional: the bargaining outcome is Pareto efficient.

Proof. The allocation and the loss are Chapter 1, Proposition 1.23, with $\boldsymbol{\lambda} = \boldsymbol{\pi}(W)$. Pareto efficiency is Proposition 1.9, which applies because $\pi_i > 0$ (irreducibility), $\Phi > 0$ and $\kappa x_i^* \leq 1$. □

Relaxed in: Chapters 6 (centrality and power from W), 7 (the bargaining equilibrium).

4.6 A5: Equal Power ($\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$)

4.6.1 The Formal Condition

The left Perron vector of the composite weight matrix $W = \alpha_U W^U + \alpha_P W^P + \alpha_I W^I + \alpha_M W^{\text{mkt}}$ — all layers row-stochastic on the same set of N agents, $\alpha_k \geq 0$, $\sum_k \alpha_k = 1$, with absent layers given weight zero — is uniform: $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$. No agent has more structural influence than any other. For irreducible W this holds if and only if W is doubly stochastic (Chapter 1, Proposition 1.21).

4.6.2 When It Approximately Holds

Economies where interactions are dense, symmetric, and relatively uniform. Small cooperatives, well-functioning democracies with distributed power, competitive markets with many participants of similar size.

4.6.3 When It Fails

Wherever the composite fails to be doubly stochastic. That happens when W^U has clustered externalities, W^I is star-shaped, or W^{mkt} is concentrated — but it also happens with A1–A4 intact: with unequal endowments, $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$ has $\boldsymbol{\pi} = \mathbf{s} \neq \frac{1}{N}\mathbf{1}$. A5 is therefore a separate assumption, and in the benchmark it amounts to equal endowments. (A block-diagonal W^P by itself need not concentrate power: with balanced cross-block flows, $\boldsymbol{\pi}$ can be uniform; Chapter 3, Proposition 3.7.)

Proposition 4.6 (Power Decomposition)**.** *Let the composite $W = \alpha_U W^U + \alpha_P W^P + \alpha_I W^I$ (layers row-stochastic on the same node set, $\alpha_k \geq 0$, $\sum_k \alpha_k = 1$; a market layer adds a fourth term in the same way) be irreducible. Its Perron vector satisfies:*

$$\pi_i = \alpha_U \underbrace{\sum_j \pi_j W^U_{ji}}_{\text{social power}} + \alpha_P \underbrace{\sum_j \pi_j W^P_{ji}}_{\text{production power}} + \alpha_I \underbrace{\sum_j \pi_j W^I_{ji}}_{\text{epistemic power}} . \tag{4.5}$$

An agent's total power is the weighted sum of power directed toward her through all three channels: how much powerful agents care about her (W^U), depend on her output (W^P), and listen to her signals (W^I).

Proof. W is row-stochastic (Appendix 4.12) and irreducible, so $\boldsymbol{\pi}$ exists and is unique (Chapter 6, Theorem 6.6). Write the i-th entry of $\boldsymbol{\pi}' = \boldsymbol{\pi}'W$ and expand $W_{ji} = \alpha_U W^U_{ji} + \alpha_P W^P_{ji} + \alpha_I W^I_{ji}$. Each underbraced term is computed with the composite $\boldsymbol{\pi}$, not with the layer's own Perron vector. □

Intuition: Power is recursive

The equation is self-referential: π_i depends on π_j for all j who point toward i, and π_j depends on π_k for all k who point toward j, and so on. Power comes from being valued, depended upon, and listened to *by other powerful agents*. This recursion is the defining feature of the Perron eigenvector and the mathematical content of "privilege reproduces itself."

Relaxed in: Chapters 5 (three networks), 6 (Perron-Frobenius), 7 (bargaining allocation).

4.7 A6: Exogenous Preferences (θ_i Fixed)

4.7.1 The Formal Condition

The preference parameter θ_i — patience, risk aversion, time discount, aspiration level, labor supply elasticity — is a primitive of the model. It does not change over time, does not respond to the network, and is not influenced by the allocations that the model produces.

This is close to the Stigler and Becker (1977) position, "de gustibus non est disputandum": tastes are stable over time and similar across people, and apparent differences in tastes are to be explained by prices, incomes and accumulated consumption capital rather than by changing preferences. The economist's job is to derive behavior from stable preferences, not to explain the preferences themselves.

4.7.2 When It Approximately Holds

Short-run analysis with stable social environments. If nothing in the agent's information network changes, and no significant shocks alter her reference group or her perceived returns, then treating θ_i as fixed over a budget period or a business cycle is harmless.

4.7.3 When It Fails

Intergenerational change. Preferences visibly evolve across generations: attitudes toward gender roles, savings behavior, educational aspirations, labor force participation norms. These are not random mutations — they correlate with changes in the information environment. The girl in Potosí with a phone has different aspirations than her mother, not because of a random preference shock but because W^I changed.

Peer effects. Sacerdote (2001), Duflo and Saez (2003): randomly assigned roommates and coworkers' information influence academic outcomes and retirement-savings decisions. Choices are contagious — they travel along W^I; whether the underlying preferences travel too is what Part III asks.

Advertising and media. W^I is deliberately shaped by actors who benefit from specific preference configurations. Advertising can shift preferences as well as inform; to the extent it does, it changes θ.

Definition 4.7 (The Formation Equation (Preview))**.** *When θ_i evolves according to the information network, we write:*

$$\theta_{i,t+1} = \rho\theta_{i,t} + (1-\rho)\bar{\theta} + \mu\left(\sum_j W^I_{ij} r_{j,t} - r^*\right), \tag{4.6}$$

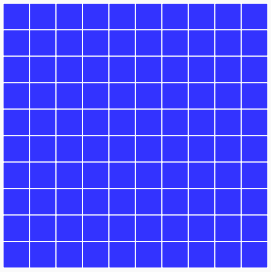
where $\rho \in (0,1)$ is persistence, $\bar{\theta}$ is a long-run anchor, $r_{j,t}$ is the return observed from agent j, and r^* is the reference return. The term $\sum_j W^I_{ij} r_{j,t}$ is the W^I-weighted average of observed returns: what you see depends on who you watch.

This equation is developed in full in Chapter 10. Its key implication: *the structure produces the preferences that justify the structure.* If W^I is block-diagonal (the girl in Potosí observes only traditional-sector outcomes), she forms low aspirations, which lead to low investment, which reproduces the traditional-sector allocation, which keeps W^I block-diagonal. The fixed point is Furtado's "autonomous process of underdevelopment."

Relaxed in: Chapters 9 (where preferences come from), 10 (the formation model), 11 (the Ψ ratio), 12 (welfare under endogenous preferences).

4.8 The Assumptions Together

Visualization: Six assumptions, one heatmap



$W^{CE} = \frac{1}{N}\mathbf{1}\mathbf{1}'$

$\longleftarrow$

A1: $W^U = 0$ (no off-diagonal in utility)
A2: W^P exogenous (technology, not choice)
A3: W^I absent (no belief channel)
A4: Identical rows (anonymous aggregation)
A5: $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$ (equal power)
A6: θ_i fixed (preferences exogenous)

The uniform-blue heatmap is the visual signature of the equal-endowment benchmark, with all six assumptions holding. With unequal endowments the anonymous benchmark is $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$: identical rows, columns of different brightness. Any other departure — a dark off-diagonal region, a block structure, a bright column not explained by wealth — violates at least one assumption.

The six assumptions are not independent. They form a rough hierarchy:

Remark 4.8 (The Hierarchy of Assumptions).

1. A1–A3 are about the **existence** of the three networks. The benchmark model abstracts from W^U, takes W^P as exogenous, and has no W^I.

2. A4 is about the **anonymity** of the market mechanism. Given that W^U and W^I are absent, the remaining interaction (through prices) is anonymous and wealth-weighted: $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$.

3. A5 is **not** a consequence of A1–A4. If the only interaction is through the anonymous market, then $\boldsymbol{\pi} = \mathbf{s}$, which is uniform only with equal endowments. A5 adds equal endowments; and the First Welfare Theorem holds without it (Chapter 1, Section 1.7).

4. A6 is the **deepest** assumption: it says the structure cannot change preferences. Without A6, even a correct diagnosis of the other five becomes dynamically unstable — the preferences used to evaluate welfare are themselves produced by the network.

Key Structural Insight

The six assumptions can be summarized in one sentence: *the benchmark model abstracts from social structure, so that structure cannot matter for economic outcomes.*

$W^U = 0$: social bonds don't affect utility. W^P exogenous: production structure is engineering, not social choice. W^I absent: information doesn't flow through social channels. $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$: the market treats everyone anonymously, weighting only by wealth. $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$: with equal endowments, nobody has structural power. θ fixed: the structure doesn't shape who you become.

Structural economics is the systematic relaxation of this sentence.

4.9 The Structural Program

Part II of this book introduces the three networks (W^U, W^P, W^I) as distinct objects, derives power from the Perron eigenvector, and computes the bargaining equilibrium — relaxing A1, A3, A4, and A5.

Part III relaxes A6 — the formation model, where W^I shapes preferences.

Part IV studies dynamics — how the networks evolve and reproduce, relaxing A2 (production structure becomes endogenous).

Parts V–VII apply the framework to three scales (global, national, individual), develop measurement tools, and analyze structural change.

Assumption	Benchmark model	Structural economics	Chapters
A1 (W^U)	$= 0$	General W^U	5, 7, 12
A2 (W^P)	Exogenous	Endogenous, policy variable	3, 13, 15
A3 (W^I)	Absent	Drives beliefs and aspirations	5, 8, 9–12
A4 (Market)	$\mathbf{1}\mathbf{s}'$	Non-anonymous, concentrated	6, 7
A5 (Power)	$\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$	$\boldsymbol{\pi}(W)$, computed	6, 7, 16–19
A6 (Preferences)	Fixed θ	$\theta_{t+1} = f(W^I, \theta_t)$	9–12

Each row of this table is a chapter (or group of chapters). Each cell is a research program. The table is the book.

4.10 How Far Is the Real World from W^{CE}?

A natural question: are we tilting at windmills? How badly do the assumptions fail in practice?

For each assumption we can define a scalar measure of departure that could be estimated from data. Table 4.1 lists them, with orders of magnitude that are *illustrative* placeholders, not estimates; Parts V and VI of the book estimate them.

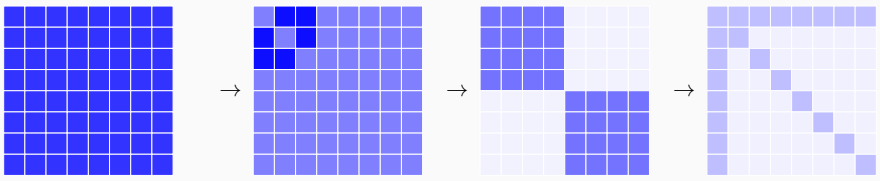
	Departure measure	Data source	Illustrative range
A1	$\gamma_U \lVert W^U \rVert_F$ (externality intensity)	Social surveys	0.1–0.5
A2	$\lVert \Delta A \rVert / \lVert A \rVert$ (I-O change per decade)	I-O time series	5–20%
A3	$N_{\text{eff}}(\boldsymbol{\pi}^I)/N$ (epistemic concentration)	Media and network data	0.2–0.8
A4	$d_\pi(W^{\text{mkt}})$ (power distance of market shares)	Market shares	0.1–0.5
A5	$N_{\text{eff}}(\boldsymbol{\pi})/N$ (power concentration)	Composite W	0.3–0.9
A6	Ψ ratio (formation vs. enforcement; Definition 9.13, estimation in Chapter 11)	Cohort-year data	0.5–4.0

Table 4.1: Departure measures for the six assumptions. [ILLUSTRATIVE] The ranges are placeholders chosen to be internally consistent ($N_{\text{eff}}/N \in [1/N, 1]$, $d_\pi \in [0, 1 - 1/N]$), not estimates from the data sources listed.

Whether the departures are large is an empirical question that the later parts of the book take up for Bolivia and for a cross-country sample. Variances of $\boldsymbol{\pi}$ are hard to compare across samples of different size — their maximum is $(N-1)/N^2$ — which is why the table uses N_{eff}/N.

The standard model is a useful benchmark — like Euclidean geometry in a curved universe. The curvature is measurable, and the book's claim, to be tested, is that in development economics it is large.

Visualization: From W^{CE} to the real world: five departures



Progressive departure from the competitive benchmark (heatmap weights, rows normalized; diagnostics from `code/ch04.py`). Each step adds a structural feature that the benchmark model abstracts from. The block structure slows integration (Γ falls) without concentrating power; the hub concentrates power (N_{eff} falls) while the gap stays high ($\Gamma = 0.855$). The book's claim is that the real world is on the right.

4.11 Summary

1. The benchmark competitive model abstracts from structure in six ways, five of them expressible as properties of weight matrices: $W^U = 0$, W^P exogenous, no W^I, anonymous markets $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$, $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$, and θ fixed.

2. The assumptions form a rough hierarchy: A1–A3 concern the existence of three networks; A4 asserts anonymity of the remaining interaction; A5 does *not* follow from A1–A4 — it adds equal endowments, since the anonymous market gives $\boldsymbol{\pi} = \mathbf{s}$; A6 freezes the structure by fixing preferences.

3. Each assumption has a measurable departure: externality intensity, I-O change rates, epistemic and overall power concentration (N_{eff}/N), the power distance of market shares, and the Ψ ratio. How large they are in developing economies is an empirical question for Parts V–VI.

4. Structural economics is the systematic relaxation of these six assumptions: Parts II–VII develop general W^U, endogenous W^P, general W^I, asymmetric power, and endogenous preferences.

5. The standard model is not wrong. It is the special case of anonymous interaction ($W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$, or W^{CE} with equal endowments) with $W^U = 0$ and θ fixed. This book is the general case.

4.12 Exercises

4.1 Classifying failures. For each scenario, identify which of A1–A6 is violated and which is the most important:

(a) A landlord and three tenant farmers negotiate sharecropping rates in a village with no outside labor market.

(b) A social media platform shows users algorithmically selected content that shifts purchasing behavior.

(c) Bolivia's gas revenues are redistributed through transfers, but the gas sector remains an enclave.

(d) A girl in Potosí with a smartphone develops aspirations her mother never had.

(e) Two ethnic groups in the same city trade goods but do not intermarry, share schools, or attend the same churches.

4.2 Measuring departure. For the 3×3 row-stochastic matrix

$$W = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.1 & 0.7 \end{pmatrix}:$$

(a) Compute column sums. Is W doubly stochastic?

(b) Compute the column-sum deviation $\delta_{\text{col}}(W) = \|\mathbf{1}'W - \mathbf{1}'\|_2$ (Chapter 2, Appendix 2A).

(c) Compute the Perron eigenvector $\boldsymbol{\pi}$ and $\text{Var}(\pi_i)$.

(d) If this were Bolivia's composite W (with $N = 3$ sectors), which assumption is most severely violated?

4.3 The hierarchy in action. Construct a 4×4 economy where:

(a) A1 fails but A4 holds (utility externalities exist, but the market interaction is still symmetric). Is $\boldsymbol{\pi}$ uniform? (Hint: the composite W includes W^U.)

(b) A4 fails but A1 holds (market power exists, no utility externalities). Is $\boldsymbol{\pi}$ uniform?

(c) Both A1 and A4 fail simultaneously. Compute $\boldsymbol{\pi}$. Is the welfare loss larger than the sum of the individual losses? (This tests whether the interaction term $L_{\text{interaction}}$ is positive.)

4.4 Bolivia's six assumptions. Start from the stylized two-sector dual economy of Chapter 3 (W^P with Gas and Traditional; Example 3.6), then check it against the measured values in Chapter 3, Section 3.4:

(a) Assess each of A1–A6 for Bolivia. Which hold approximately? Which fail clearly?

(b) For each failing assumption, estimate the departure measure from publicly available data (INE, CEPAL).

(c) Rank the six departures from largest to smallest. Which structural feature of Bolivia is the most severe deviation from the competitive benchmark?

4.5 Sunkel's five processes. Sunkel (1971) argued, in substance, that development, underdevelopment, dependency, marginalization, and spatial inequality are manifestations of a single process of transnational integration and national disintegration.

4.6 A6 is the deepest. Consider two economists:

(a) Economist A relaxes A1–A5 but keeps A6 (preferences are fixed). She finds the optimal policy: a transfer T^* that maximizes welfare given the structural power $\boldsymbol{\pi}(W)$.

(b) Economist B relaxes all six, including A6. She finds that the policy changes W^I, which changes θ, which changes the welfare function itself.

(c) Show that Economist A's policy can be welfare-*reducing* under Economist B's model. (Hint: the transfer may reinforce the information structure that produces low aspirations.)

(d) What does this imply about the "growth first, redistribute later" consensus?

4.7 The spectral gap of the assumptions. For $N = 8$, construct a sequence of matrices $W^{(k)}$ for $k = 0, 1, \ldots, 5$ where $W^{(0)} = \frac{1}{8}\mathbf{1}\mathbf{1}'$ and each successive matrix adds one structural feature (a cluster in W^U, a block in W^P, a hub in W^I, etc.). Compute $\Gamma^{(k)}$, $N_{\text{eff}}^{(k)}$ and $L_{\text{power}}^{(k)}$ for each (use $\Omega = 8$, $\kappa = 0.5$, $\Phi = 1$, and check $\kappa x_i < 1$). Plot all three as functions of k. Do Γ and N_{eff} move together? Does the welfare loss grow linearly, convexly, or concavely in the number of structural features?

Mathematical Appendix

Appendix 4A: Composite Matrices and Their Perron Eigenvectors

The composite. Given three nonnegative matrices W^U, W^P, W^I (each $N \times N$) and weights $\alpha_U + \alpha_P + \alpha_I = 1$ with $\alpha_k \geq 0$:

$$W = \alpha_U W^U + \alpha_P W^P + \alpha_I W^I. \tag{4.7}$$

If each component is row-stochastic, so is W: $W\mathbf{1} = \alpha_U W^U\mathbf{1} + \alpha_P W^P\mathbf{1} + \alpha_I W^I\mathbf{1} = \alpha_U\mathbf{1} + \alpha_P\mathbf{1} + \alpha_I\mathbf{1} = \mathbf{1}$.

Perron eigenvector of the composite. In general, $\boldsymbol{\pi}(W)$ is NOT a convex combination of $\boldsymbol{\pi}(W^U)$, $\boldsymbol{\pi}(W^P)$, $\boldsymbol{\pi}(W^I)$. The eigenvectors of a sum of matrices are not the sums of the eigenvectors. This is a nonlinearity that makes the composite power vector genuinely different from any single-network power vector.

Special cases. If all three components have the same eigenvector ($\boldsymbol{\pi}^U = \boldsymbol{\pi}^P = \boldsymbol{\pi}^I = \boldsymbol{\pi}$), then $\boldsymbol{\pi}(W) = \boldsymbol{\pi}$ — the composite inherits the common power structure. This is the "aligned networks" case from Chapter 5: all three networks agree on who is powerful. In this case, power is reinforced across channels.

If the components have different eigenvectors, the composite $\boldsymbol{\pi}(W)$ is a nonlinear blend that depends on the α_k weights. Changing α_P (the importance of production linkages) relative to α_I (the importance of information) moves $\boldsymbol{\pi}$ along a curve, not a straight line. As long as the composite is irreducible, its Perron root is simple and $\boldsymbol{\pi}$ is an analytic — in particular continuous — function of the α_k: there are no jumps. (For the block layer and the hub layer of the visualization above, $\boldsymbol{\pi}$ never moves by more than 0.0007 when α_P changes by 0.001.) The change can nevertheless be fast when the composite is close to reducible, because the sensitivity of $\boldsymbol{\pi}$ grows as $1 - \lambda_2$ shrinks (Chapter 6, Lemma 6.19). Genuine discontinuities arise only at the reducible boundary, where $\boldsymbol{\pi}$ ceases to be unique.

Computing $\boldsymbol{\pi}(W)$. Form $W = \alpha_U W^U + \alpha_P W^P + \alpha_I W^I$, then apply power iteration (Appendix 6A).

Structuralist Case Study: Sunkel and Paz's Systematic Critique

In 1970, Osvaldo Sunkel and Pedro Paz published *El subdesarrollo latinoamericano y la teoría del desarrollo* through ILPES/Siglo XXI — a 385-page treatise that remains

the most comprehensive systematic critique of orthodox development theory from the structuralist perspective. Where Prebisch (1949) was a manifesto and Furtado (1961) was a theoretical essay, Sunkel and Paz was a textbook: methodical, exhaustive, and organized as a point-by-point examination of what mainstream economics assumes and why those assumptions fail in the periphery.

Their argument maps directly onto the six assumptions of this chapter.

Part 1: Mainstream theory assumes a closed, self-contained economy. Sunkel and Paz began by noting that the neoclassical model analyzes an economy in isolation — abstracting from its insertion in the global system. They argued that this is not an innocuous simplification but a fundamental distortion: the peripheral economy's structure is shaped by its relationship to the center. In our notation: the standard model treats W^P as a domestic object, ignoring the global production network in which the domestic matrix is embedded. The domestic W^P is not independent of the global W^P — the off-diagonal entries linking the domestic economy to the world are among the most important entries in the matrix.

Part 2: One process, several manifestations. The book's central thesis was that development and underdevelopment are two faces of a single historical process, not successive stages. Sunkel carried the thesis further in a 1971 essay, "Capitalismo transnacional y desintegración nacional en América Latina": development, underdevelopment, dependency, marginalization, and spatial inequality are not separate phenomena but manifestations of one process — the simultaneous transnational integration and national disintegration produced by global capitalism. In matrix terms: the global weight matrix W_{global} is being rewired in a way that increases cross-country connectivity among elite nodes (transnational integration) while decreasing within-country connectivity between elite and non-elite nodes (national disintegration). The five "processes" are five observables of one matrix transformation.

Part 3: The theory is part of the structure. Sunkel and Paz argued that the theories of development taught at the center and exported to the periphery — above all the theories of stages — were not neutral descriptions of the periphery but generalizations from the center's own history, a theme Quijano would later radicalize. The extension to the formal models is this book's, not theirs. The theory of comparative advantage, the Heckscher-Ohlin model and the welfare theorems abstract from the structures catalogued in this chapter. Teaching them in La Paz without noting what they abstract from is an exercise of epistemic power: it naturalizes the abstractions, and so lends legitimacy to the structure that the theorems leave out.

Sunkel & Paz's argument	This chapter's assumption	Formal object
Economy not self-contained	A2: W^P treated as domestic	Global embedding of W^P
Five processes, one process	A1–A5 fail simultaneously	$W \neq \frac{1}{N}\mathbf{1}\mathbf{1}'$, all channels
National disintegration	A2: domestic W^P fragmenting	Block-diagonal domestic W^P
Transnational integration	A2: W^P changes endogenously	Elite cross-border linkages
Theory is part of structure	A3: no information network	W^I shapes beliefs about theory

Sunkel and Paz's book is the structuralist critique organized as a textbook. This chapter is the same critique organized as a table of matrix properties. The mathematics adds precision; the diagnosis was always theirs.

Further Reading

The welfare theorems and their assumptions: Mas-Colell et al. (1995), Chapter 16; Debreu (1959); Arrow (1951). For the assumptions' failure in developing economies: Stiglitz (1989), "Markets, Market Failures, and Development."

Endogenous production networks: Acemoglu et al. (2012a); Baqaee and Farhi (2019); Oberfield (2018), "A Theory of Input-Output Architecture"; Acemoglu and Azar (2020), "Endogenous Production Networks"; Hidalgo et al. (2007), "The Product Space Conditions the Development of Nations."

Information networks and belief dynamics: DeGroot (1974); Golub and Jackson (2010); Chandrasekhar et al. (2020), "Testing Models of Social Learning on Networks"; Acemoglu et al. (2011), "Bayesian Learning in Social Networks."

Endogenous preferences: Bowles (1998), "Endogenous Preferences: The Cultural Consequences of Markets and Other Economic Institutions"; Bisin and Verdier (2001), "The Economics of Cultural Transmission and the Dynamics of Preferences."

The structuralist critique: Sunkel and Paz (1970), *El subdesarrollo latinoamericano y la teoría del desarrollo*, México: Siglo XXI / ILPES (the definitive systematic treatment; the full text in Spanish is available through CEPAL's digital repository). Sunkel (1971), "Capitalismo transnacional y desintegración nacional en América Latina."

PART II

The Three Networks

Chapter 5

The Three Networks

Where the Structure Enters the Model

> *"Por dependencia entendemos una situación en la cual la economía de ciertos países está condicionada por el desarrollo y la expansión de otra economía a la cual la primera está sometida."*
>
> ["By dependence we mean a situation in which the economy of certain countries is conditioned by the development and expansion of another economy to which the former is subjected."]
> — Theotônio dos Santos, *La estructura de la dependencia* (1970)

> *"Tell me who you walk with, and I'll tell you who you are."*
> — Proverb, every language that has one

> *"Dime con quién andas y te diré tu vector de Perron."*
> — This book

Part I showed that the benchmark competitive model can be written as anonymous interaction — every agent interacts only with the market aggregate, $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$, of which $W^{\text{CE}} = \frac{1}{N}\mathbf{1}\mathbf{1}'$ is the equal-endowment case — with no utility externalities (Chapter 1, Section 1.7). Part II replaces that matrix with non-anonymous interaction structures and asks what they do.

This chapter introduces the three weight matrices that organize economic life:

1. W^U: the **utility network**. Who affects your wellbeing directly?

2. W^P: the **production network**. Who depends on whom for inputs and outputs?

3. W^I: the **information network**. Whose signals shape your beliefs and desires?

Every economic interaction passes through at least one of these networks; many pass through all three simultaneously. The benchmark general-equilibrium model abstracts from the first ($W^U = 0$), takes the second as technology, and treats beliefs and preferences as given rather than formed through the third. This chapter makes the three networks visible, distinct, and formal; Chapter 6 then derives power from them and Chapter 7 the bargaining allocation. Throughout, each layer is a row-stochastic matrix on the same set of agents (Chapter 1, Section 1.7).

5.1 The Utility Network W^U

5.1.1 Utility with Externalities

In Chapter 1, agent i's utility depended only on her own consumption: $U_i(x_i)$. We now let it depend on everyone else's consumption too, weighted by the utility network.

Definition 5.1 (Utility with Social Externalities)**.** *Agent i's utility is:*

$$V_i(x_1, \ldots, x_N) = U_i(x_i) + \gamma_U \sum_{j \neq i} W^U_{ij}\, U_j(x_j), \tag{5.1}$$

where:

- $U_i(x_i)$ *is agent i's private utility from her own consumption.*
- $W^U_{ij} \geq 0$ *is the weight that agent i places on agent j's utility. This is the* ***social externality****: how much j's wellbeing directly affects i.*
- $\gamma_U \geq 0$ *is the overall strength of social externalities.*

In matrix form, stacking the N agents:

$$\mathbf{V} = \mathbf{U} + \gamma_U W^U \mathbf{U} = (I + \gamma_U W^U)\mathbf{U}, \tag{5.2}$$

where $\mathbf{U} = (U_1(x_1), \ldots, U_N(x_N))'$ is the vector of private utilities and $\mathbf{V}$ is the vector of social utilities.

Intuition: What W^U does to utility

Without W^U (the benchmark model), your happiness depends only on your own consumption. With W^U, other people's happiness enters directly. If W^U_{ij} is large and positive, you are *altruistic* toward j: their consumption makes you happier. If we allowed $W^U_{ij} < 0$ (which some models do), you would be *envious*: their consumption makes you worse off.

The parameter γ_U controls how much you care about others overall. When $\gamma_U = 0$, we are back to the benchmark model regardless of W^U. When $\gamma_U > 0$, the structure of W^U matters: *who* you care about — not just *how much* — shapes the equilibrium.

Example 5.2 (Three Vendors on Calle Comercio)**.** *María ($i = 1$), Rosa ($j = 2$), Carlos ($j = 3$). Quadratic private utility: $U_i = x_i - \frac{1}{2}x_i^2$.*

$$W^U = \begin{pmatrix} 0 & 0.30 & 0.02 \\ 0.25 & 0 & 0.05 \\ 0.02 & 0.05 & 0 \end{pmatrix}. \tag{5.3}$$

María places weight 0.30 on Rosa (comadre bond) and only 0.02 on Carlos. Rosa places 0.25 on María and 0.05 on Carlos. Carlos barely cares about either.

With $\gamma_U = 0.5$ and allocations $x_1 = 0.6$, $x_2 = 0.4$, $x_3 = 0.8$:

$$V_1 = U_1(0.6) + 0.5[0.30 \cdot U_2(0.4) + 0.02 \cdot U_3(0.8)] \tag{5.4}$$

$$= 0.42 + 0.5[0.30 \times 0.32 + 0.02 \times 0.48] \tag{5.5}$$

$$= 0.42 + 0.5[0.096 + 0.0096] = 0.42 + 0.053 = 0.473. \tag{5.6}$$

María's social utility (0.473) is higher than her private utility (0.42) because her comadre Rosa exists and has positive consumption. The externality adds 12.6% to her well-being. This is small in levels but enormous in marginal terms: María's willingness to help Rosa (e.g., with a loan during a bad week) is proportional to $\gamma_U W^U_{12} = 0.15$, which is large enough to generate substantial mutual aid.

5.1.2 What W^U Is Not

Three common confusions:

W^U is not the market. The market is the price mechanism, which aggregates demands anonymously. W^U is the direct externality channel: j's consumption enters i's utility *without* passing through the price. Both channels operate simultaneously. Chapter 7 embeds both in general equilibrium.

W^U is not altruism alone. Altruism is $W^U_{ij} > 0$: you benefit from j's consumption. But W^U also captures status competition (j's consumption raises the bar for i), reference-dependent preferences (i evaluates her consumption relative to j's), and network externalities (j's adoption of a technology raises i's benefit from the same technology). All of these are direct externalities mediated by a weight matrix.

W^U is not W^I. The utility network says whose consumption affects your happiness. The information network says whose signals affect your beliefs. María cares about Rosa's wellbeing (W^U) AND learns about prices from Rosa's behavior (W^I). These are different channels with different matrices. A dictator might have high W^I centrality (everyone watches what the dictator does) but low W^U weight (nobody cares about the dictator's personal happiness). Chapter 8 develops W^I in detail.

5.1.3 Properties of W^U

Remark 5.3 (Basic Properties of the Utility Network). For the utility externality model (5.1):

1. **Diagonal is zero**: $W^U_{ii} = 0$ by convention (own utility is captured by U_i, not by the externality term).

2. **Raw weights vs. the interaction matrix**: in (5.1) the raw intensities matter, so the total externality weight $\sum_j W^U_{ij}$ varies across agents: a "social" agent (María) has a high row sum; an "isolated" agent has a low row sum. When the utility network is used as an interaction matrix — for its Perron vector, its spectral gap, or a composite (Section 5.6) — the book uses the row-normalized $D^{-1}W^U$, with an all-zero row replaced by a self-loop.

3. **Not necessarily symmetric**: $W^U_{ij} \neq W^U_{ji}$ in general. María may care more about Rosa than Rosa cares about María.

4. **The benchmark model is** $W^U = 0$: no externalities. Equivalently, $\gamma_U = 0$.

5. **Nonnegativity**: $W^U \geq 0$ (altruism or care). Envy ($W^U_{ij} < 0$) is excluded from the interaction matrix, since Perron–Frobenius needs nonnegative weights; it can be studied in the utility model itself.

Key Structural Insight

The distinction between $W^U = 0$ (benchmark model) and $W^U = \frac{1}{N}\mathbf{1}\mathbf{1}' - I$ (anonymous equal externalities) matters. The first says nobody's consumption affects anyone else. The second says *everyone's* consumption affects everyone else *equally*. Both are extreme; neither is realistic. The real W^U is sparse (most pairs have zero or near-zero weight), asymmetric (compadrazgo bonds are not symmetric), and clustered (your utility network is local: family, friends, neighbors, compadres).

5.2 The Production Network W^P

5.2.1 From Chapter 1 to Chapter 5

We have already met the input-output matrix A from Chapters 1 and 3. Chapter 1 introduced it as a technological object: a_{ij} is how much of good j is needed to produce one unit of good i, so row i is sector i's recipe. Chapter 3 embedded it in a growth model and showed that the block structure of the production network determines whether growth propagates. The production network W^P is its row-normalized version.

Here we develop W^P as one of three networks, clarify what it shares with W^U and W^I and how it differs, and show how all three interact in the same economy.

Definition 5.4 (Production Network)**.** *The **production network** W^P is the $N \times N$ row-stochastic matrix with $W^P_{ij} \geq 0$ the weight of agent (or sector) j in agent i's production: how much i depends on j's output.*

In the inter-sectoral interpretation, $W^P = D^{-1}A$ with $D = \text{diag}(A\mathbf{1})$: $W^P_{ij} = a_{ij}/\sum_k a_{ik}$ is the share of input j in sector i's intermediate costs (the input-share matrix of Acemoglu et al. 2012a). The technical coefficients a_{ij} themselves (units of good j per unit of good i) stay in A.

In the inter-agent interpretation, W^P_{ij} captures production spillovers: how much j's activity increases i's productivity. This includes knowledge spillovers, shared infrastructure, supply chain dependencies, and agglomeration effects.

The two interpretations come with different output equations, and the indices must be kept straight. With technical coefficients, the sectors j use $a_{ji}y_j$ units of good i as intermediate input, so the quantity balance is (Chapter 1)

$$\mathbf{y} = A'\mathbf{y} + \mathbf{d} \qquad \Rightarrow \qquad \mathbf{y} = (I - A')^{-1}\mathbf{d}, \tag{5.7}$$

with the Leontief inverse $(I - A')^{-1}$, which requires $\rho(A) < 1$; multipliers and linkages are computed from A, never from the row-normalized W^P. With spillovers, agent i's output rises with the output of those she depends on, $\mathbf{Y} = \mathbf{f}(\mathbf{K}, \mathbf{L}) + \gamma_P W^P \mathbf{Y}$, so $\mathbf{Y} = (I - \gamma_P W^P)^{-1} \mathbf{f}(\mathbf{K}, \mathbf{L})$ with $0 < \gamma_P < 1$.

5.2.2 W^P vs. W^U: Different Channels, Different Matrices

Example 5.5 (Why They Differ: The Gas Engineer)**.** *Consider Patricia, a petroleum engineer working for YPFB in Tarija. She interacts with two groups: her coworkers at the gas plant (production network) and her family and friends in Cochabamba (utility network).*

*Patricia's **production network** W^P: high weights on the gas plant operators, the pipeline maintenance crew, the Brazilian equipment supplier, the Argentine gas buyer. Low or zero weights on the quinoa farmer in Oruro, the api vendor in La Paz, her mother in Cochabamba. The production network is organized by the technology of gas extraction, not by social bonds.*

*Patricia's **utility network** W^U: high weights on her mother, her partner, her children, her close friends. Moderate weights on her coworkers (she likes some of them). Zero weights on the Argentine gas buyer (she has never met him). The utility network is organized by affection and social proximity, not by technological linkages.*

The two matrices have almost disjoint support: the people who matter for Patricia's production barely overlap with the people who matter for Patricia's happiness. This is typical. W^P and W^U are not the same matrix, and confusing them leads to analytical errors.

The most common error: assuming that strengthening production linkages (W^P) automatically strengthens social bonds (W^U) or changes aspirations (W^I). It does not. Bolivia can build a gas-to-fertilizer plant that links the gas sector to agriculture (W^P changes) without affecting whether the quinoa farmer's daughter aspires to be an engineer (W^I unchanged).

Property	W^U (Utility)	W^P (Production)
What it captures	Who affects your happiness	Who affects your output
Channel	Direct externality in U	Intermediate inputs / spillovers
Typical structure	Sparse, clustered, local	Sparser, sectoral, sometimes global
Symmetry	Often asymmetric	Often asymmetric
Diagonal	Zero by convention	Positive (own-sector recycling)
Structural amplifier	$(I + \gamma_U W^U)$	$(I - A')^{-1}$ (Leontief, from A); $(I - \gamma_P W^P)^{-1}$ (spillovers)
Benchmark model	$W^U = 0$	Technology, taken as given (I-O tables exist)

5.3 The Information Network W^I

5.3.1 The Missing Network

W^U has been studied (under the name "social preferences" or "other-regarding preferences") since at least Becker (1974). W^P has been studied (under the name "input-output analysis") since Leontief (1941). W^I — the network through which information, signals, and norms flow, shaping beliefs and eventually preferences — is the one the benchmark general-equilibrium model leaves out. It is far from absent in economics: a large literature studies social learning on networks, from DeGroot (1974) through herding models (Banerjee, 1992) and learning from neighbors (Bala and Goyal, 1998) to Golub and Jackson (2010). What this book adds is to put it in the same framework as W^U and W^P.

It is not W^U and it is not W^P. And, for the questions of Part III, it is arguably the most important of the three.

Definition 5.6 (Information Network)**.** *The **information network** W^I is the $N \times N$ row-stochastic matrix where $W^I_{ij} \geq 0$ is the weight that agent i places on agent j's signals when forming beliefs about the world. Row-stochasticity ($\sum_j W^I_{ij} = 1$ for all i) means that*

each agent's attention is a probability distribution over other agents: you allocate 100% of your attention, and W^I describes how.

Intuition: What W^I does

Think of W^I as your *attention allocation*. Each day, you observe the world — but not all of it. You observe a weighted sample of other people's behaviors, outcomes, and signals. The weights are W^I_{ij}. Your neighbor gets high weight (you see her every day). The president gets moderate weight (you see him on TV). A farmer in Beni gets zero weight (you don't know he exists).

The information you receive determines your beliefs. Your beliefs determine your choices. So W^I — who you pay attention to — shapes your economic behavior even though it never enters your utility function directly.

In the DeGroot learning model (developed in Chapter 8), repeated application of W^I drives beliefs toward a consensus determined by the left Perron eigenvector of W^I. The agents with the highest Perron weight are the ones whose initial beliefs dominate the long-run consensus. This is **epistemic power**: the ability to shape what others believe, not through coercion but through structural position in the information network.

5.3.2 DeGroot Learning: A Preview

Suppose each agent starts with an initial belief $b_i^{(0)}$ about some quantity (the return on education, the value of saving, the probability of success in the modern sector). At each period, agents update their beliefs by taking a weighted average of their neighbors' beliefs:

$$\mathbf{b}^{(t+1)} = W^I \mathbf{b}^{(t)}. \tag{5.8}$$

This is the DeGroot learning model. It is extremely simple: just multiply the belief vector by W^I at each step. Yet its implications are profound.

Proposition 5.7 (DeGroot Convergence)**.** *If W^I is row-stochastic, aperiodic, and strongly connected (every agent can reach every other agent through a chain of positive weights), then:*

$$\lim_{t\to\infty} (W^I)^t = \mathbf{1}\boldsymbol{\pi}', \tag{5.9}$$

*where $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_N)'$ is the **left Perron eigenvector** of W^I: the unique vector satisfying $\boldsymbol{\pi}' W^I = \boldsymbol{\pi}'$ with $\sum_i \pi_i = 1$.*

All agents converge to the same belief:

$$b_i^{(\infty)} = \boldsymbol{\pi}'\mathbf{b}^{(0)} = \sum_{j=1}^{N} \pi_j b_j^{(0)} \quad \textit{for all } i. \tag{5.10}$$

The long-run belief is a weighted average of the initial beliefs, with weights given by $\boldsymbol{\pi}$.

Proof. A strongly connected (irreducible) and aperiodic nonnegative matrix is primitive. By the Perron–Frobenius theorem for primitive matrices (Theorem 6.6, item 6; Seneta, 2006, Thms. 1.1–1.2), $(W^I)^t/\rho^t \to \mathbf{v}\boldsymbol{\pi}'$ with $\boldsymbol{\pi}'\mathbf{v} = 1$, and $\boldsymbol{\pi}$ is the unique left eigenvector for ρ up to scale. Row-stochasticity gives $W^I\mathbf{1} = \mathbf{1}$, so $\rho = 1$ and $\mathbf{v} = \mathbf{1}$ once $\boldsymbol{\pi}$ is normalized to $\mathbf{1}'\boldsymbol{\pi} = 1$. Then $\mathbf{b}^{(t)} = (W^I)^t\mathbf{b}^{(0)} \to \mathbf{1}\boldsymbol{\pi}'\mathbf{b}^{(0)}$. Chapter 8 restates this as Theorem 8.1; the model is DeGroot (1974)'s. □

Intuition: What this says in words

Everyone eventually agrees — but they agree on a *weighted average* of initial beliefs, not the simple average. The weights π_j are the Perron eigenvector entries: agents with high π_j have more influence on the long-run consensus. This is not because they shout louder or coerce others. It is because the network structure channels information flow through them. They are listened to more, directly or indirectly, by more people.

The Perron eigenvector $\boldsymbol{\pi}$ is the **structural power** of each agent in the information network. It is not a choice or an attribute of the agent. It is a property of the network topology — a property of W^I.

Example 5.8 (Luisa's Information Network)**.** *Luisa in Potosí,* $N = 5$ *reference agents: Mother (*$j = 1$*), School friend (*$j = 2$*), Cousin in Buenos Aires (*$j = 3$*), TikTok feed (*$j = 4$*), Teacher (*$j = 5$*).*

$$W^I_{Luisa} = \begin{pmatrix} 0.35 & 0.25 & 0.15 & 0.15 & 0.10 \end{pmatrix}. \tag{5.11}$$

Luisa allocates 35% of her attention to her mother (she sees her every day, trusts her judgment), 25% to her school friend (constant contact), 15% to her cousin in Buenos Aires (WhatsApp, intermittent but aspirational), 15% to TikTok (algorithmically curated, diverse signals), and 10% to her teacher.

Before the phone — the network a girl in her position would have had a generation earlier, labelled "1990" for concreteness: $W^I_{Luisa,\,1990} = (0.50, 0.30, 0, 0, 0.20)$*. Mother and friend dominated. The cousin in Buenos Aires had zero weight (no WhatsApp). TikTok did not exist. The information network was local: Potosí signals only.*

The shift from W^I_{1990} to W^I_{2020} is a major structural change in Luisa's life. It exposes her to signals about returns (her cousin's engineering career) that were previously invisible. Whether those signals change her θ (patience, investment in education) depends on the formation model in Chapter 10. But the precondition — the change in W^I — is what makes formation change possible.

Prebisch Connection: Information as structural power

In the global economy, W^I has a clear center-periphery structure. The center produces the signals that the periphery consumes: financial news from New York, technology standards from Silicon Valley, fashion from Paris, economic policy from Washington. The periphery does not produce signals that the center consumes. Entry $W^I_{\text{Bolivia, USA}}$ is large (Bolivian policymakers watch the Fed). Entry $W^I_{\text{USA, Bolivia}}$ is approximately zero (the Fed does not watch the Banco Central de Bolivia).

The information asymmetry compounds the production asymmetry. Not only does the periphery export primary commodities and import manufactures (W^P); it also exports attention and imports beliefs (W^I). The Perron eigenvector of the global W^I assigns almost all weight to center countries. The long-run consensus about "how economies should be run" converges to the initial beliefs of the center — which is how the Washington Consensus becomes consensus even in countries where its prescriptions are structurally inappropriate.

This is one formal reading of what dependency theorists described as cultural or intellectual dependence. It can be measured: it is the Perron eigenvector of W^I_{global}, and its concentration is N_{eff} (Definition 6.15).

5.4 The Three Networks Together

5.4.1 The Full Model

We now write the economy with all three networks active simultaneously.

Definition 5.9 (The Three-Network Economy)**.** *An economy with N agents, L goods, and three networks (W^U, W^P, W^I) is defined by:*

Preferences *(shaped by W^U and, indirectly, by W^I):*

$$V_i(\mathbf{x}) = U_i(x_i; \theta_i) + \gamma_U \sum_{j \neq i} W^U_{ij} U_j(x_j; \theta_j), \tag{5.12}$$

where θ_i is agent i's preference parameter, which in the formation model (Chapter 10) evolves according to W^I.

Production *(shaped by W^P; spillover form, with $0 < \gamma_P < 1$):*

$$\mathbf{Y} = (I - \gamma_P W^P)^{-1} \mathbf{f}(\mathbf{K}, \mathbf{L}). \tag{5.13}$$

Beliefs *(shaped by W^I):*

$$\mathbf{b}^{(t+1)} = W^I \mathbf{b}^{(t)}. \tag{5.14}$$

Market clearing: *$\sum_i x_{i\ell}^* = \bar{\omega}_\ell$ for each good ℓ.*

Key Structural Insight

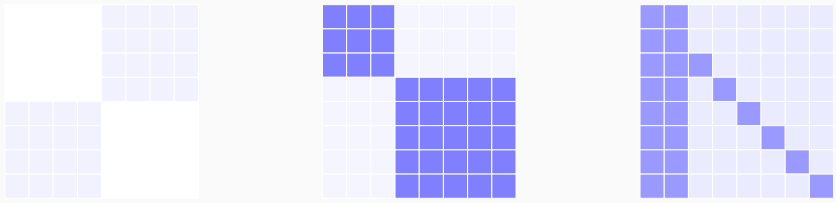
The three networks enter through three different channels:

Network	Channel	Object modified	Structural amplifier
W^U	Utility	What you want	$(I + \gamma_U W^U)$
W^P	Production	What you can make	$(I - \gamma_P W^P)^{-1}$
W^I	Information	What you believe	$(W^I)^t \to \mathbf{1}\boldsymbol{\pi}'$

The benchmark competitive model sets $W^U = 0$, takes technology from input-output data (the one network economists have always measured), and treats beliefs and preferences as given rather than formed through W^I. This book studies general (W^U, W^P, W^I).

Note the three amplifiers. The utility amplifier $(I + \gamma_U W^U)$ is a small perturbation of the identity: externalities add a correction to private utility. The production amplifier $(I - \gamma_P W^P)^{-1}$ is a Leontief-type inverse: potentially large, summing infinite rounds of indirect effects. The information amplifier $(W^I)^t$ converges to a rank-1 matrix (for primitive W^I): in the long run, everyone agrees, and the agreement reflects the Perron eigenvector. The three amplifiers plausibly operate on different timescales — utility effects at once, production effects over business cycles, information effects over generations — though that ordering is a modelling choice, not a result.

Visualization: Three networks, one economy, three heatmaps



W^U **(utility)**
Who you care about

W^P **(production)**
Who depends on your output

W^I **(information)**
Who you listen to

Schematic (shades show relative weights, not a calibrated matrix). Same eight agents, three different networks. W^U is clustered (two social groups, caring mostly within). W^P is block-diagonal (enclave: three agents in the modern sector, five in the traditional). W^I is star-shaped (two media sources dominate everyone's attention). The three heatmaps have different topologies — the agent who is central in one may be peripheral in another. The composite $W = \alpha_U W^U + \alpha_P W^P + \alpha_I W^I$ blends all three.

5.4.2 When the Networks Align

In the simplest case, all three networks are the same: $W^U = W^P = W^I = W$. Your friends are your business partners are your information sources. This is the world of a small village: the same 50 people appear in all three matrices.

When the networks align, the analysis simplifies dramatically: there is one weight matrix, one Perron eigenvector, one spectral gap. Power is power across all channels. The salteña vendor who is central in the utility network is also central in the production network and the information network. Life is simple and oppressive.

5.4.3 When the Networks Diverge

In a complex economy, the three networks diverge. Your production network (the global supply chain) has nothing to do with your utility network (your family in Cochabamba) or your information network (TikTok, the Banco Central, your teacher).

This divergence creates both opportunities and tensions:

Opportunity: You can be peripheral in W^P (a quinoa farmer in Oruro) but central in W^I (a TikTok influencer with 200,000 followers). Information power can compensate for production powerlessness — Luisa's phone gives her access to signals about the modern sector even though her production network is entirely traditional.

Tension: You can be central in W^P (the YPFB gas engineer) but peripheral in W^U (she moved to Tarija for work, leaving her family in Cochabamba). Economic success coexists with social isolation. The production network rewards one location; the utility network rewards another.

Structural trap: When W^P is block-diagonal (Pinto) AND W^I is block-diagonal (traditional-sector people attend only to traditional-sector signals), the two networks reinforce each other. Growth in the modern sector does not propagate through production (W^P disconnected) AND does not propagate through information (W^I disconnected). The trap is double-locked. Breaking it requires changing both matrices — which is why structural policy is harder than it looks.

5.5 Canonical Topologies

Any of the three networks can have any topology. But certain topologies appear again and again across economic settings. We catalog the most important ones and develop intuition for reading their structural properties.

(Readers should consult the interactive *Seeing the Structure* companion tool, which visualizes all topologies below for $N = 3$ to 100.)

5.5.1 Complete (Anonymous Symmetric)

$W = \frac{1}{N}\mathbf{1}\mathbf{1}'$. Every entry is $1/N$. Doubly stochastic, symmetric, rank 1, maximal spectral gap $\Gamma = 1$, equal Perron eigenvector $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$.

As W^U: universal, symmetric altruism. You care about everyone equally.

As W^P: every sector uses every other sector's output in equal proportion. No specialization, no comparative advantage, no structural heterogeneity. Unrealistic but useful as a benchmark.

As W^I: you listen to everyone equally. No opinion leaders, no media concentration, no filter bubbles. Information aggregates to the simple average of initial beliefs.

Economics: $W^{\text{CE}} = \frac{1}{N}\mathbf{1}\mathbf{1}'$ is the equal-endowment case of anonymous market interaction, $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$ with $\mathbf{s}$ the wealth shares. The First Welfare Theorem does not depend on it: it holds for every $\mathbf{s}$ and for any technology, given local nonsatiation, complete markets, price-taking and no externalities ($W^U = 0$) (Proposition 6.13). What the complete topology adds is *equal power*.

5.5.2 Star (Hub-Spoke)

One hub (h) connected to all spokes; spokes connected only to the hub.

$$W_{ih} \text{ large for all } i \neq h; \qquad W_{ij} \approx 0 \text{ for } i, j \neq h. \tag{5.15}$$

Perron eigenvector concentrates on the hub: if spokes place weight a on the hub and the hub keeps s, then $\pi_h = a/(a + 1 - s)$, independent of N (Proposition 6.11). The spectral gap depends on self-loops, not on concentration: a pure star (no self-loops) is bipartite and has $\Gamma = 0$; a lazy star has $\Gamma = 0.5$.

As W^U: everyone cares about the hub (celebrity, patron, chief), nobody cares about each other. Patron-client structures, populist politics.

As W^P: all production passes through a single firm or sector (the platform, the middleman, the colonial trading company). Amazon, the East India Company.

As W^I: one source broadcasts, all others listen. State media, monopoly news, papal authority. If the hub listens only to herself ($s = 1$), the long-run consensus equals the hub's initial belief: epistemic dictatorship — reached quickly, since such a network mixes fast.

5.5.3 Block-Diagonal (Dual Economy)

Two or more blocks with strong internal links and weak cross-block links.

$$W \approx \begin{pmatrix} W_{11} & \varepsilon \\ \varepsilon' & W_{22} \end{pmatrix}, \quad \|\varepsilon\| \ll \|W_{11}\|, \|W_{22}\|. \tag{5.16}$$

Weak cross-block coupling forces a small spectral gap: if every agent in block 1 places total weight ε_1 on block 2 and every agent in block 2 places ε_2 on block 1, then $\Gamma \leq \varepsilon_1 + \varepsilon_2$. The Perron eigenvector splits between the blocks in proportion to their *inflows*: block 1 holds exactly $\varepsilon_2/(\varepsilon_1 + \varepsilon_2)$ of total power (Proposition 6.12). Symmetric coupling gives equal power however small Γ is.

As W^U: two social groups that care about their own members but not about the other group. Ethnic enclaves, caste, class segregation.

As W^P: Pinto's structural heterogeneity. The modern sector and the traditional sector, barely connected. Growth does not propagate (Chapter 3). Bolivia's gas enclave is real, but measured at the level of 12 aggregate sectors it is not of this form: INE's 2019 supply-use table gives the domestic W^P a gap of $\Gamma = 0.467$, not a small one. The enclave shows up instead in the Perron vector and in where output goes: hydrocarbons have $\pi = 0.039$ and export 70% of their output (INE supply-use tables, base 2017, Instituto Nacional de Estadística (INE), Bolivia, 2026; author's

calculations, `data/compute/compute_bolivia_io.py`; Chapter 3, Section 3.4).

As W^I: two information ecosystems. Rural and urban. Spanish-language and Quechua-language. Fox News and MSNBC. Each group converges to its own consensus internally, but the two groups' consensuses may diverge. This is the formation-level mechanism of structural reproduction: if traditional-sector people attend only to traditional-sector signals, their beliefs about returns are calibrated to the traditional sector, which reproduces traditional-sector behavior, which keeps them in the traditional sector.

5.5.4 Core-Periphery (Prebisch)

A dense core and a sparse periphery, with asymmetric links: periphery depends on core, core barely depends on periphery.

$$W_{ij} \text{ large for } i \in \text{periphery}, j \in \text{core}; \qquad W_{ij} \approx 0 \text{ for } i \in \text{core}, j \in \text{periphery}. \tag{5.17}$$

Perron eigenvector assigns high weight to the core and low weight to the periphery: $\pi_{\text{core}} \gg \pi_{\text{periphery}}$.

As W^U: the periphery admires and envies the center; the center is indifferent to the periphery. Colonial relations, celebrity culture, academic prestige hierarchies.

As W^P: Prebisch's center-periphery. The periphery exports raw materials to the center and imports manufactures. Backward linkages of the center, computed from the technical coefficients A (column sums of the Leontief inverse), are high and those of the periphery low. Chapter 16 relates the terms of trade to this asymmetric structure.

As W^I: the periphery absorbs signals from the center (economic theory, cultural norms, consumption patterns, policy advice) but not vice versa. The long-run consensus converges to center beliefs. The Washington Consensus as a DeGroot equilibrium.

5.5.5 Other Topologies

Several additional topologies appear in applications:

Ring (supply chain): Each agent connects to neighbors. Sequential production. Shocks propagate slowly along the chain. Manufacturing assembly lines, agricultural value chains.

Hierarchy (tree): Top-down connections. Firms, bureaucracies, military, caste. Power typically falls with depth in the tree. Information flows down; surplus flows up.

Small-world: Local clusters plus random long-range shortcuts. Most social networks exhibit this structure (Watts and Strogatz, 1998). High clustering with short path lengths. Ideas diffuse faster than in a ring but less uniformly than in a complete graph.

Scale-free: Few super-connected hubs, many weakly connected nodes. Power-law degree distribution. Wealth, internet, citation networks. For a random walk on an undirected network π_i is proportional to degree, so the degree inequality becomes power inequality.

Random (Erdős-Rényi): Each link exists with probability p. No structure by design. Useful as a null model: any deviation from random IS structure.

5.6 The Composite Weight Matrix

In many applications, the three networks operate simultaneously and interact. We define the composite.

Definition 5.10 (Composite Weight Matrix)**.** *Let W^U, W^P, W^I be row-stochastic matrices on the* same *N agents (raw intensities K are first normalized, $W = D^{-1}K$). The* ***composite weight matrix*** *is:*

$$W = \alpha_U W^U + \alpha_P W^P + \alpha_I W^I, \quad \alpha_U + \alpha_P + \alpha_I = 1, \tag{5.18}$$

where $\alpha_U, \alpha_P, \alpha_I \geq 0$ are the relative importance of each channel. W is then row-stochastic. If the layers live on different node sets (sectors, media outlets, individuals), there is no composite.

The normalization is not cosmetic. Mixing raw layers with different scales makes $\boldsymbol{\pi}$ depend on units: in the La Paz example below, doubling the raw utility intensities moves the Perron vector of the unnormalized composite from

$$(0.311,\ 0.364,\ 0.325) \quad \text{to} \quad (0.327,\ 0.388,\ 0.286),$$

with no change in anyone's relationships.

The composite W is a single matrix that summarizes the overall structure of economic interaction. Its Perron eigenvector $\boldsymbol{\pi}(W)$ is the **power vector**: agent i's structural power is π_i, determined jointly by her position in all three networks.

Proposition 5.11 (Power from the Composite)**.** *If the composite W is irreducible, its*

left Perron eigenvector $\boldsymbol{\pi}$ satisfies:

$$\pi_i = \alpha_U \sum_j \pi_j W^U_{ji} + \alpha_P \sum_j \pi_j W^P_{ji} + \alpha_I \sum_j \pi_j W^I_{ji}. \tag{5.19}$$

An agent's power is the weighted sum of power directed toward her through all three channels: how much powerful agents care about her (utility channel), how much powerful agents depend on her output (production channel), and how much powerful agents listen to her signals (information channel).

Proof. W is row-stochastic (each layer is, and the α's sum to one), so $\rho(W) = 1$; since W is irreducible, the Perron–Frobenius theorem (Theorem 6.6, Part II) gives a unique positive $\boldsymbol{\pi}$ with $\boldsymbol{\pi}'W = \boldsymbol{\pi}'$ and $\mathbf{1}'\boldsymbol{\pi} = 1$. Write out the i-th entry of $\boldsymbol{\pi}'(\alpha_U W^U + \alpha_P W^P + \alpha_I W^I)$. □

Intuition: What determines power

Your structural power π_i is high if:

1. People who care about you (W^U_{ji} large) are themselves powerful.
2. People who depend on your output (W^P_{ji} large) are themselves powerful.
3. People who listen to your signals (W^I_{ji} large) are themselves powerful.

This is recursive: power begets power, through the Perron eigenvector's self-referential definition — one formal reading of "privilege reproduces itself." One more feature matters: weight that flows into an agent who sends little of it back accumulates there. In this reading, self-reliance counts as power, as the next example shows.

Example 5.12 (Power in La Paz)**.** *Consider three agents: a Cabinet Minister ($i = 1$), a Gas Executive ($i = 2$), and a Market Vendor ($i = 3$).* [ILLUSTRATIVE]

Utility network (raw intensities: who cares about whom):

$$K^U = \begin{pmatrix} 0 & 0.3 & 0.0 \\ 0.2 & 0 & 0.0 \\ 0.0 & 0.0 & 0 \end{pmatrix}. \tag{5.20}$$

The Minister and the Executive care about each other. Nobody cares about the Vendor, and she cares about nobody.

Production network (raw input coefficients: who depends on whose output):

$$K^P = \begin{pmatrix} 0.1 & 0.4 & 0.1 \\ 0.3 & 0.2 & 0.0 \\ 0.0 & 0.0 & 0.3 \end{pmatrix}. \tag{5.21}$$

The Minister depends on gas revenue (0.4 weight on Executive). The Executive depends on regulatory approval (0.3 weight on Minister). The Vendor is self-contained.

Information network (already row-stochastic: who listens to whom):

$$W^I = \begin{pmatrix} 0.3 & 0.5 & 0.2 \\ 0.4 & 0.3 & 0.3 \\ 0.2 & 0.1 & 0.7 \end{pmatrix}. \tag{5.22}$$

The Minister listens mostly to the Executive. The Executive listens to the Minister and somewhat to the Vendor (market prices as signal). The Vendor listens mostly to herself (local knowledge).

Row-normalizing the first two layers (the Vendor's all-zero utility row becomes a self-loop) gives

$$W^U = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad W^P = \begin{pmatrix} 1/6 & 2/3 & 1/6 \\ 0.6 & 0.4 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{5.23}$$

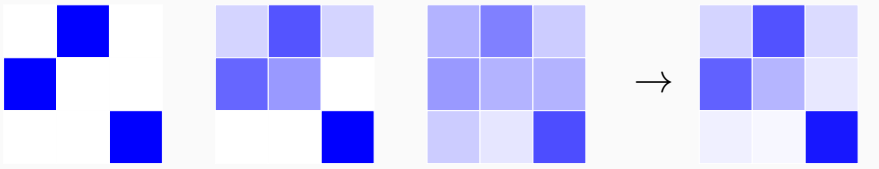
Composite ($\alpha_U = 0.2, \alpha_P = 0.5, \alpha_I = 0.3$):

$$W = 0.2W^U + 0.5W^P + 0.3W^I = \begin{pmatrix} 0.173 & 0.683 & 0.143 \\ 0.620 & 0.290 & 0.090 \\ 0.060 & 0.030 & 0.910 \end{pmatrix}. \tag{5.24}$$

The left Perron eigenvector (computed numerically): $\boldsymbol{\pi} \approx (0.211, 0.227, 0.562)$, with $N_{\text{eff}} = 2.43$. The Vendor has the most structural power in this reading. Nobody depends heavily on her, but everyone depends on her a little (the Minister's $W_{13} = 0.143$, the Executive's $W_{23} = 0.090$) and she sends almost nothing back ($W_{33} = 0.91$). Weight flows into her node and stays there. Even in the information layer alone she leads, $\boldsymbol{\pi}(W^I) = (0.281, 0.266, 0.453)$, because her column of W^I has the largest sum (1.2).

The lesson is about the measure as much as about La Paz. The Perron vector measures long-run influence — whose position the system converges toward — and that rewards autonomy as well as being depended upon. Whether autonomy of this kind is "power" in the political sense is an interpretive question the matrix cannot settle; Exercise 5.7 shows how strongly the answer depends on the weights α.

Visualization: La Paz: three networks and their composite



The La Paz economy: Minister, Gas Executive, Market Vendor (shade $= 100\,W_{ij}$, white $= 0$). Three row-stochastic layers blend into the composite. Column 2 of W^P is bright in the first two rows: the Minister and the Executive depend on the Executive's output; the Vendor does not. Column 3 of W^I has the largest sum: the Vendor is the most listened-to agent once her own weight is counted. The bright $(3,3)$ cell of the composite — the Vendor's self-reliance — is what gives her the largest Perron weight, $\boldsymbol{\pi} = (0.21, 0.23, 0.56)$.

5.7 What the Benchmark Model Assumes

We can now state precisely what the benchmark competitive model abstracts from, in the language of the three networks.

Remark 5.13 (The Benchmark as a Network Assumption). The benchmark Arrow–Debreu model:

1. sets $W^U = 0$ (no utility externalities);
2. takes production technology as given (the input-output matrix A, taken as data);
3. treats preferences and beliefs as exogenous, rather than formed through an information network W^I;
4. makes interaction anonymous: each agent interacts with the market aggregate, so the implied interaction matrix is $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$ with $\mathbf{s}$ the wealth shares (W^{CE} in the equal-endowment case).

Under local nonsatiation, complete markets and price-taking, (1) is what delivers the First Welfare Theorem; (4) holds for every wealth distribution, so the theorem says nothing about how unequal $\mathbf{s}$ — and hence $\boldsymbol{\pi}(W^{\text{mkt}}) = \mathbf{s}$ — may be.

Structural economics relaxes each element:

1. General W^U (utility externalities matter for welfare, efficiency, and policy).

2. General W^P (the topology of production determines growth propagation and structural change).

3. General W^I (information flows shape beliefs and preferences, reproducing or disrupting the structure).

4. Non-anonymous interaction, mediated by all three networks, whose Perron vector need not equal wealth shares.

This is the structural program. The rest of the book develops each relaxation: Chapter 6 derives power from W, Chapter 7 derives the bargaining equilibrium, Chapter 8 develops belief manipulation through W^I, and Chapters 9–12 develop the formation model where W^I shapes preferences.

5.8 Summary

1. The economy is organized by three networks: W^U (utility externalities), W^P (production dependence), and W^I (information flows). The benchmark Arrow–Debreu model abstracts from the first ($W^U = 0$), takes the second as technology, and treats beliefs as exogenous rather than formed through the third.

2. W^U captures who affects your happiness. It enters utility directly: $V_i = U_i + \gamma_U \sum_j W^U_{ij} U_j$. Utility interdependence, not unequal power, is what breaks the First Welfare Theorem.

3. W^P captures who depends on whose output: it is the row-normalized input-coefficient matrix, $W^P = D^{-1}A$. Multipliers and linkages come from A and the Leontief inverse $(I - A')^{-1}$, not from W^P.

4. W^I captures who listens to whom. It drives belief dynamics through DeGroot learning: $\mathbf{b}^{(t+1)} = W^I \mathbf{b}^{(t)}$, converging (for primitive W^I) to a consensus weighted by the Perron eigenvector $\boldsymbol{\pi}(W^I)$. This is epistemic power.

5. The composite $W = \alpha_U W^U + \alpha_P W^P + \alpha_I W^I$ of row-stochastic layers on the same agents summarizes the full structure; its Perron eigenvector is the power vector, recursive and sensitive to self-reliance. Canonical topologies (complete, star, block-diagonal, core-periphery, and others) differ on two separate coordinates: how integrated they are (Γ) and how concentrated power is (N_{eff}).

5.9 Exercises

5.1 Utility externalities: keeping up and falling behind. María sells api on Calle Comercio, between Doña Rosa's stall (empanadas) and Don Carlos's stall (newspapers). María and Rosa share a comadrazgo bond: Rosa is godmother to María's youngest daughter. When Rosa's sales are good, María is genuinely happy for her; when they collapse, María worries, lends her money, and feels the loss almost as her own. Don Carlos is just a neighbor; his fortunes do not move her. A stranger in Santa Cruz does not enter her utility at all. Example 5.2 formalizes this with $\gamma_U = 0.5$.

(a) Compute V_2 and V_3 (Rosa's and Carlos's social utilities).

(b) With the total $x_1 + x_2 + x_3 = 1.8$ fixed, compare the allocation that maximizes $\sum_i V_i$ with the one that maximizes $\sum_i U_i$. Who gets more when social utilities are counted? (Use the column sums $c_j = \sum_i W^U_{ij}$.)

(c) What happens as $\gamma_U \to 1$? As $\gamma_U \to 0$?

5.2 Three networks for a factory town. A town has $N = 4$ agents: Factory Owner (F), Worker (W), Shopkeeper (S), Mayor (M).

(a) Write plausible 4×4 matrices W^U, W^P and W^I, and row-normalize each one.

(b) Compute the composite W with equal weights $\alpha = 1/3$.

(c) Find the Perron eigenvector. Who has the most power? Explain why.

(d) Now suppose the factory closes. Which entries of W^P go to zero? How does the Perron eigenvector change?

5.3 DeGroot learning. For $N = 3$ with:

$$W^I = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.1 & 0.6 & 0.3 \\ 0.2 & 0.2 & 0.6 \end{pmatrix}. \tag{5.25}$$

(a) Verify W^I is row-stochastic.

(b) Compute $(W^I)^2$ and $(W^I)^5$ (use Python). Are the rows becoming more similar?

(c) Find the left Perron eigenvector $\boldsymbol{\pi}$. Which agent has the most epistemic power?

(d) Starting from $\mathbf{b}^{(0)} = (0, 0.5, 1)'$, compute $\mathbf{b}^{(t)}$ for $t = 1, 2, 5, 10, 50$. Verify convergence to $\boldsymbol{\pi}'\mathbf{b}^{(0)}$.

5.4 The phone as structural change. María's daughter Luisa is fifteen and lives in Potosí. On her phone she sees videos of university students in La Paz, posts from her cousin who studies engineering in Buenos Aires, and messages from school friends wondering whether to work in the market, attend the normal school in Sucre, or try for the public university in Oruro. Twenty years earlier she would have seen only what Potosí showed her: market vendors, miners, teachers. Luisa's pre-phone W^I and post-phone W^I are in Example 5.8. Extend to $N = 5$ agents (Mother, Friend, Cousin, TikTok, Teacher) where each agent also has a W^I row (Mother listens to: herself 0.6, friend 0.2, teacher 0.2; etc.).

(a) Construct the full 5×5 pre-phone and post-phone W^I.

(b) Compute the Perron eigenvector for both. How does epistemic power shift?

(c) Let initial beliefs about "returns to education" be

$$\mathbf{b}^{(0)} = (0.3,\ 0.3,\ 0.8,\ 0.6,\ 0.7)$$

(Mother and Friend think returns are low; Cousin, TikTok, Teacher think returns are high). What is the long-run consensus pre-phone? Post-phone?

(d) Interpret: how much does the phone change Luisa's long-run belief about education?

5.5 When networks align. Construct a 4×4 economy where $W^U = W^P = W^I$ (a village where everyone is simultaneously your friend, your supplier, and your information source).

(a) Compute the composite W. Note: it equals each component matrix.

(b) Now perturb: add one external production link (agent 1 begins importing from outside). How does W^P change? How does the composite change? What happens to the Perron eigenvector?

(c) Interpret: what does it mean for a village to enter the global supply chain?

5.6 Prebisch in W^I. Construct a 6×6 global W^I: agents 1–2 are "center" countries, agents 3–6 are "periphery" countries. Center listens mostly to itself; periphery listens mostly to center.

(a) Compute the Perron eigenvector. How much epistemic power do center countries have?

(b) Initial beliefs about "the optimal tariff policy": center believes $b = 0$ (free trade), periphery believes $b = 1$ (protection). What is the long-run consensus?

(c) What fraction of the consensus is determined by the center? Is this consistent with the Washington Consensus?

(d) What would it take (in terms of W^I entries) for the periphery's belief to survive?

5.7 Composite power in the La Paz model. Using the stylized three-agent model of Example 5.12:

(a) Vary α_P from 0 to 1 (holding α_U / α_I constant). Plot π_3 (Vendor's power) as a function of α_P. When does the Vendor have the most power? The least?

(b) Interpret: in what kind of economy does the market vendor matter most? When does she become invisible?

(c) Add a fourth agent: "Luisa (Student)." Construct plausible W^U, W^P, W^I rows for Luisa. How does her inclusion change the power structure?

(d) Replace the Vendor's self-loop in the normalized W^U by a uniform row $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. How much does π_3 change? What does this say about how "power" is attributed to self-reliant agents?

5.8 From story to matrix: the chain from gas well to school (data). A story often told about Bolivia: gas from a well in Tarija goes to a YPFB processing plant; one component feeds a thermoelectric plant in Santa Cruz; the electricity powers a flour mill in Cochabamba; the flour goes to a bakery, which sells bread to a miner in Potosí; the tin is exported, and the export revenue funds the school where María's daughter studies. The story adds that at each step the domestic link breaks and points abroad: the processing plant's equipment is imported, the power plant was built by a foreign firm, and the mill buys imported wheat.

(a) Write each link of the chain as an entry of A (technical coefficients) and of $W^P = D^{-1}A$. Which entries are domestic, which are imports?

(b) Using INE Bolivia's input-output tables (*Matriz Insumo-Producto*) or supply-use tables, check the story's claims: what shares of the relevant sectors' inputs are imported? (Treat every claim in the story as a hypothesis to test, not a fact.)

(c) Aggregate to a gas block and a traditional block and compute the cross-block weights $\varepsilon_1, \varepsilon_2$ of Proposition 6.12. Is the chain "mostly imaginary", as Pinto's structural heterogeneity would suggest?

Mathematical Appendices

Appendix 5A: Graph Theory for Economists

A **directed graph** (digraph) $G = (V, E)$ consists of a set of *vertices* (nodes, agents) $V = \{1, \ldots, N\}$ and a set of *edges* (links, connections) $E \subseteq V \times V$. An edge (i, j) means agent i is connected to agent j.

Every $N \times N$ weight matrix W defines a weighted directed graph: the vertices are agents $1, \ldots, N$, and the edge from i to j has weight W_{ij}. (Convention, used throughout the book: $W_{ij} > 0$ means i places weight on j — i depends on, cares about, or listens to j — drawn as an edge $i \to j$. Influence then flows against the arrows, from j to i.)

Key definitions. *Path*: a sequence of edges connecting i to j. The *length* of a path is the number of edges. A path of length k from i to j exists if and only if $(W^k)_{ij} > 0$: the (i, j) entry of the k-th power of W.

Connected: i can reach j through a path of some finite length. *Strongly connected*: every agent can reach every other agent. Equivalently, W is *irreducible*: $(I + W)^{N-1}$ has all positive entries.

Aperiodic: the greatest common divisor of the lengths of all cycles through any agent is 1. Sufficient condition, for irreducible W: some diagonal entry $W_{ii} > 0$ (some agent has a self-loop, i.e., some of their influence stays with themselves).

Why these matter: The Perron-Frobenius theorem (Chapter 6) requires W to be nonnegative. Strong connectivity ensures a unique Perron eigenvector. Aperiodicity ensures convergence of (W^t) to a limit (the consensus matrix $\mathbf{1}\boldsymbol{\pi}'$). Without strong connectivity, the network fragments into components, each with its own Perron eigenvector. Without aperiodicity, the system cycles rather than converging.

Degree and centrality. *In-degree*: $d_i^{\text{in}} = \sum_j W_{ji}$ (column sum): the total weight others place on i, i.e. how much influence i exerts. Not necessarily 1 even for row-stochastic matrices. (Same definition as Chapter 6.)

Out-degree: $d_i^{\text{out}} = \sum_j W_{ij}$ (row sum): the total weight i places on others. For row-stochastic W: always 1.

Eigenvector centrality: the Perron eigenvector $\boldsymbol{\pi}$ assigns each agent a centrality proportional to the centrality of agents who point toward her. This is the recursive "power" we have been developing. It is not the same as degree (a node can have high degree but low eigenvector centrality if its neighbors are peripheral).

Chapter 6 develops this fully.

Adjacency matrix vs. weight matrix. The *adjacency matrix* G is binary: $G_{ij} = 1$ if there is an edge from i to j, and 0 otherwise. (We write G, not A, to avoid a clash with the technical coefficients.) The *weight matrix* W generalizes this: $W_{ij} \geq 0$ is the strength of the edge. In this book, we always use weight matrices; a row-normalized adjacency matrix $D^{-1}G$ is the random walk on the graph.

Appendix 5B: Stochastic Matrices and the Perron Eigenvector

Row-stochastic matrices. A nonnegative matrix W is **row-stochastic** if $W\mathbf{1} = \mathbf{1}$: every row sums to 1. This means each row is a probability distribution. In the context of W^I, row-stochasticity means each agent allocates 100% of her attention across all agents (including possibly herself).

Key property. If W is row-stochastic, then $\rho(W) = 1$ and $\mathbf{1}$ is a right eigenvector: $W\mathbf{1} = 1 \cdot \mathbf{1}$.

The left Perron eigenvector. The left eigenvector corresponding to eigenvalue 1 is $\boldsymbol{\pi}$: the vector satisfying $\boldsymbol{\pi}'W = \boldsymbol{\pi}'$ with $\boldsymbol{\pi} \geq 0$ and $\mathbf{1}'\boldsymbol{\pi} = 1$. For a row-stochastic matrix, $\boldsymbol{\pi}$ is also the *stationary distribution* of the Markov chain defined by W: if you start at a random node drawn from $\boldsymbol{\pi}$ and take a random step according to W, you remain distributed according to $\boldsymbol{\pi}$.

Why left, not right? The right eigenvector of a row-stochastic matrix is always $\mathbf{1}$ — uninformative. The left eigenvector is the interesting object: it tells you which agents have the most influence in the long run. "Left Perron" is the power vector throughout this book.

Computing $\boldsymbol{\pi}$ by iteration. Start with any positive vector $\mathbf{v}^{(0)}$. Iterate: $\mathbf{v}^{(t+1)\prime} = \mathbf{v}^{(t)\prime}W$. Normalize: $\boldsymbol{\pi}^{(t)} = \mathbf{v}^{(t)}/\|\mathbf{v}^{(t)}\|_1$. If W is irreducible and aperiodic, $\boldsymbol{\pi}^{(t)} \to \boldsymbol{\pi}$ as $t \to \infty$. Convergence is geometric at rate $|\lambda_2(W)|$, the second-largest eigenvalue in magnitude. Faster convergence means larger spectral gap $\Gamma = 1 - |\lambda_2|$.

Example. For the 3×3 matrix from Exercise 5.3:

$$W^I = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.1 & 0.6 & 0.3 \\ 0.2 & 0.2 & 0.6 \end{pmatrix}. \tag{5.26}$$

Starting from $\mathbf{v}^{(0)} = (1, 1, 1)$:

t	π_1	π_2	π_3	Max change
0	0.333	0.333	0.333	—
1	0.267	0.367	0.367	0.067
2	0.243	0.373	0.383	0.023
5	0.233	0.373	0.395	0.001
∞	0.233	0.372	0.395	0

Agent 3 has the most epistemic power (0.395), although agents 2 and 3 have the same self-weight ($W^I_{22} = W^I_{33} = 0.6$) and the same column sum (1.1). The tie is broken by *who* listens: agent 2 listens to agent 3 ($W^I_{23} = 0.3$) more than agent 3 listens to agent 2 ($W^I_{32} = 0.2$), while agents 1 and 3 listen to each other equally ($W^I_{13} = W^I_{31} = 0.2$). Structural position, not self-confidence, determines power. The second eigenvalues are a complex pair of modulus 0.361, so $\Gamma = 0.639$.

Doubly stochastic matrices. If W is both row-stochastic AND column-stochastic ($\mathbf{1}'W = \mathbf{1}'$), then $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$: all agents have equal power. This is the competitive benchmark from Chapter 1 ($W^{CE} = \frac{1}{N}\mathbf{1}\mathbf{1}'$ is doubly stochastic). Any deviation from doubly stochastic generates power asymmetry.

Structuralist Case Study: Dos Santos's Three Regimes of the Weight Matrix

In 1970, Theotônio dos Santos published "The Structure of Dependence" in the *American Economic Review* — a six-page paper in the *Papers and Proceedings* session on the economics of imperialism. The paper's core contribution was a **taxonomy of three historical forms of dependence**, each corresponding to a specific phase of world capitalist development. Read through this chapter's framework, the three forms are three historical regimes of the weight matrix W.

Regime 1: Colonial dependence (trade-export). The metropole extracts raw materials from the colony through monopoly control of land, mines, and labor. The weight matrix is a **star graph**: all edges point to the colonial power. W^P has one bright column (the metropole) and $N - 1$ dim columns (the colonies). The colonies do not trade with each other — the off-diagonal entries among periphery nodes are zero. W^I is similarly star-shaped: information (prices, technology, legal frameworks) flows from the metropole to the colonies, not between colonies. Power is maximally concentrated: $\boldsymbol{\pi} \approx (1, 0, \ldots, 0)$ and $N_{\text{eff}} \approx 1$. Note what the spectral gap does here: a matrix in which every row attends to the metropole is close to rank one, $W \approx \mathbf{1}\mathbf{e}_1'$, and mixes almost instantly — $W = 0.95\,\mathbf{1}\mathbf{e}_1' + 0.05\,I$

has $\Gamma = 0.95$. The colonial economy is highly *integrated* around its center and maximally *concentrated* in power: the two coordinates of Chapter 6 point in opposite directions.

Regime 2: Financial-industrial dependence (late 19th century). Big capital invests in raw material and agricultural production for export to hegemonic centers. The star becomes a **hub-spoke with multiple hubs**: several industrial powers (Britain, France, the US) each have their own peripheral suppliers. W^P now has a few bright columns instead of one. The periphery begins to develop limited internal linkages (railways for commodity transport, ports, basic processing), adding small off-diagonal entries among peripheral nodes. But the dominant flows still point toward the center. W^I remains center-dominated: economic policy knowledge flows from London and New York.

Regime 3: Technological-industrial dependence (post-WWII). Multinational corporations invest in industries oriented toward the internal market of the underdeveloped countries. The topology becomes more complex: MNCs create **bridging nodes** that connect the domestic W^P to the global W^P through intra-firm trade. The domestic matrix acquires new blocks (the MNC-owned modern sector), but these blocks connect outward to headquarters rather than inward to the traditional domestic sector. This anticipates what Sunkel (1971) would call "transnational integration and national disintegration" — in matrix terms, a rewiring: cross-country edges among elite nodes strengthen while within-country edges between modern and traditional blocks weaken. The information network W^I becomes layered: the MNC subsidiary receives technical knowledge from headquarters ($W^I_{\text{subsidiary, HQ}}$ large) but does not transmit local knowledge back ($W^I_{\text{HQ, subsidiary}} \approx 0$).

Regime	**Topology of W**	**Power (N_{eff})**	**Integration (Γ)**	**This chapter**
Colonial	Star, close to rank one ($\mathbf{1}\mathbf{e}_1'$)	$\pi_{\text{metropole}} \approx 1$, $N_{\text{eff}} \approx 1$	High: Γ near 1	Star topology; Prop. 5.11
Financial-industrial	Multi-hub	$\pi_{\text{centers}} \gg \pi_{\text{periphery}}$, $N_{\text{eff}} \approx$ number of hubs	Low if the hubs' spheres barely interact	Block-diagonal topology; Prop. 6.12
Technological-industrial	Blocks with MNC bridges	$\pi_{\text{MNC}} > \pi_{\text{domestic elite}} \gg \pi_{\text{traditional}}$	Low domestically (modern vs. traditional block), high among elite nodes	Composite W, Def. 5.10; aligned and divergent networks

The taxonomy's power is its specificity: each regime generates a *different* heatmap and a different combination of power concentration (N_{eff}) and integration (Γ), even though all three are "dependent." The two coordinates must be read separately: the most concentrated regime (colonial) is also the most integrated. The colonial heatmap has one blazing column. The financial-industrial heatmap has several bright columns and dim periphery. The technological-industrial heatmap has a complex block structure with MNC bridging nodes creating cross-border links while domestic cross-block links remain dark.

Dos Santos's definition — "a situation in which the economy of certain countries is conditioned by the development and expansion of another economy" — is the statement that the Perron eigenvector of the global W assigns dominant weight to the conditioning economy. The taxonomy refines this: *how* the conditioning occurs (through which channel of W and with which topology) determines the specific developmental consequences. The colonial star produces extraction. The financial hub-spoke produces commodity dependence. The technological-industrial complex produces an exclusionary modernization — a modern sector that connects to the world but not to its own country.

The three networks of this chapter — W^U, W^P, W^I — offer one formalization of Dos Santos's taxonomy. His taxonomy describes what the matrices look like. The Perron eigenvector, and its concentration N_{eff}, tell you who benefits.

Further Reading

Social preferences and externalities: Fehr and Schmidt (1999), "A Theory of Fairness, Competition, and Cooperation." Sobel (2005), "Interdependent Preferences and Reciprocity." Bowles and Gintis (2011), *A Cooperative Species*.

Production networks: See readings from Chapters 1 and 3. The comprehensive reference is now Carvalho and Tahbaz-Salehi (2019), "Production Networks: A Primer."

DeGroot learning: DeGroot (1974), "Reaching a Consensus." Banerjee (1992), "A Simple Model of Herd Behavior." Bala and Goyal (1998), "Learning from Neighbours." Golub and Jackson (2010), "Naïve Learning in Social Networks and the Wisdom of Crowds." This is the key paper on convergence, influence, and the Perron eigenvector in learning models.

Network economics textbooks: Jackson (2008), *Social and Economic Networks*. Goyal (2023), *Networks: An Economics Approach*. Sargent and Stachurski (2024), *Economic Networks*.

The interaction of multiple networks: Kivelä et al. (2014), "Multilayer Networks." For the economic application: the composite W is related to the "multiplex" framework in network science.

Dependency and information: Cardoso and Faletto (1979), *Dependency and Development in Latin America*. For the modern formalization: Acemoglu et al. (2010), "Spread of (Mis)information in Social Networks."

Dependency theory: Dos Santos (1970), "The Structure of Dependence" (the six pages that define the taxonomy formalized in the case study above). Dos Santos (1978), *Imperialismo y Dependencia*, México: Era. Sunkel (1971), "Capitalismo transnacional y desintegración nacional."

Chapter 6

Centrality, Power, and the Perron Vector

Where Power Comes From

> *"En el capitalismo periférico, la apropiación del excedente económico está íntimamente ligada a las relaciones de poder entre los distintos estratos sociales."*
>
> ["In peripheral capitalism, the appropriation of the economic surplus is intimately linked to the power relations between different social strata."]
>
> — Raúl Prebisch, *Capitalismo Periférico: Crisis y Transformación* (1981)

> *"Power is not an institution, and not a structure; neither is it a certain strength we are endowed with; it is the name that one attributes to a complex strategical situation in a particular society."*
>
> — Michel Foucault, *The History of Sexuality* (1976)

> *"Power is the left Perron eigenvector of the weight matrix."*
>
> — This book

Foucault insisted that power is relational, not an attribute of individuals. This chapter takes that relational view and asks how much of it can be measured. Our answer is partial but precise: one dimension of power is a property of the network — specifically, of the row-stochastic weight matrix W that encodes who affects whom (Chapter 1, Section 1.7). It is computed from the Perron-Frobenius theorem.

This chapter does three things. First, it develops the concept of centrality — the idea that some positions in a network are more "important" than others —

through examples, building from simple degree counting to the recursive logic of eigenvector centrality. Second, it states and proves the Perron-Frobenius theorem, which guarantees that the centrality measure we want actually exists and is unique. Third, it derives the properties of the Perron eigenvector — the power vector $\boldsymbol{\pi}$ — and separates two things the spectral gap Γ is often taken to measure: the *integration* of the network (which Γ does measure) and the *concentration* of power (which it does not; that is measured by N_{eff} and $\text{Var}(\pi)$).

By the end, the student will know where power comes from (the network), how to compute it (the Perron eigenvector), how to measure its concentration (the effective number of agents N_{eff}), why the spectral gap is a separate coordinate (Proposition 6.16), and when power is equal (W doubly stochastic) — a distributional condition, not an efficiency condition.

6.1 Who Matters? Four Attempts at an Answer

6.1.1 A Small Economy

Consider five agents in a production network. Following the book's convention, W is row-stochastic: row i is the distribution of agent i's dependence (input-cost shares, $W^P = D^{-1}A$ in the notation of Chapter 1), so each row sums to one. [ILLUSTRATIVE]

$$W = \begin{pmatrix} 0.4 & 0.4 & 0.1 & 0.1 & 0.0 \\ 0.3 & 0.3 & 0.2 & 0.2 & 0.0 \\ 0.1 & 0.2 & 0.3 & 0.2 & 0.2 \\ 0.0 & 0.1 & 0.4 & 0.3 & 0.2 \\ 0.0 & 0.1 & 0.1 & 0.3 & 0.5 \end{pmatrix}. \tag{6.1}$$

Read row i: these are the weights that agent i places on others. Agent 1 depends heavily on agent 2 ($W_{12} = 0.4$). Agent 5 depends heavily on herself ($W_{55} = 0.5$) and on agent 4 ($W_{54} = 0.3$). Agent 3 is in the middle — connected to everyone.

Question: which agent is the most powerful?

Four different answers emerge from four different ways of measuring importance. All four are "correct" in their own domain. Only one is the right measure of structural power.

6.1.2 Attempt 1: Count the Connections (Degree Centrality)

The simplest measure: how many connections does each agent have?

Definition 6.1 (Degree Centrality)**.** *For a weight matrix W, agent i's **in-degree** is the column sum $d_i^{in} = \sum_j W_{ji}$ (total weight that others place on i). The **out-degree** is the row sum $d_i^{out} = \sum_j W_{ij}$ (total weight that i places on others).*

For our W: the column sums (in-degree) are:

Agent	1	2	3	4	5
In-degree d_i^{in}	0.80	1.10	1.10	1.10	0.90

Agents 2, 3 and 4 are tied at the top: each receives the most total weight from others. Agent 1 is least important by this measure.

Intuition: What degree gets right and wrong

Degree counts *direct* connections: how many people depend on you (in-degree) or how much you depend on others (out-degree). It is easy to compute and often correlates with importance. But it misses something critical: *who* depends on you matters. If you are depended upon by powerful agents, that should count more than being depended upon by peripheral agents. Degree treats all connections equally. A farmer who sells to the president and a farmer who sells to another farmer have the same in-degree, but they do not have the same power.

6.1.3 Attempt 2: Count the Paths (Katz Centrality)

Degree counts direct connections. A more sophisticated measure counts *all* paths — direct and indirect.

Definition 6.2 (Katz Centrality)**.** *Agent i's **Katz centrality** is:*

$$c_i^{Katz} = \sum_{k=1}^{\infty} \delta^k \sum_{j=1}^{N} (W^k)_{ji} = \Big(\big[(I - \delta W')^{-1} - I \big] \mathbf{1} \Big)_i, \tag{6.2}$$

where $\delta \in (0, 1/\rho(W))$ is a decay parameter that downweights longer paths ($\rho(W) = 1$ for row-stochastic W), and $(W^k)_{ji}$ is the total weight of all paths of length k from j to i (with the convention of Appendix 5A: an edge $j \to i$ of weight W_{ji} means j places weight on i).

Katz centrality says: agent i is important if many paths lead to her, directly or indirectly, weighted by length. The parameter δ ensures convergence of the Neumann series.

Intuition: The Leontief connection

In matrix form, $\mathbf{c}^{\text{Katz}}$ is the vector of column sums of $(I - \delta W)^{-1}$, minus one. This is the same Neumann series as the Leontief inverse of Chapter 1 — but there it is applied to the technical-coefficient matrix A, which is what output multipliers require. Applied to a row-stochastic W, it is a centrality measure, not a multiplier (all row sums of $(I - \delta W)^{-1}$ equal $1/(1-\delta)$).
Katz improves on degree by counting indirect connections. Its limitation is the analyst-chosen δ: short paths dominate when δ is small, long paths when δ is close to $1/\rho$. As $\delta \to 1/\rho$, the normalized Katz vector converges to the eigenvector centrality of Attempt 4 — the measure that counts paths of every length without discounting.

6.1.4 Attempt 3: Ask the Neighbors (Bonacich Centrality)

Definition 6.3 (Bonacich Centrality)**.** *Agent i's* ***Bonacich centrality*** *is the i-th component of:*

$$\mathbf{c}^{Bon} = (I - \beta W')^{-1} W' \mathbf{1} = (I - \beta W')^{-1} \mathbf{d}^{in}, \tag{6.3}$$

where $\beta \in (-1/\rho(W), 1/\rho(W))$ *is a parameter and* $\mathbf{d}^{in} = W'\mathbf{1}$ *is the in-degree vector.*

Bonacich centrality starts from degree (the $\mathbf{d}^{\text{in}}$ term) and then amplifies it through the Leontief-type multiplier $(I - \beta W')^{-1}$. As defined here it is Katz centrality rescaled: since $(I - \beta W')^{-1}\beta W' = (I - \beta W')^{-1} - I$,

$$\mathbf{c}^{\text{Bon}}(\beta) = \tfrac{1}{\beta}\, \mathbf{c}^{\text{Katz}}(\beta) \qquad (\beta \neq 0), \tag{6.4}$$

so for $\beta > 0$ the two rank agents identically. Bonacich's (1987) contribution is the interpretation of the parameter, including negative values.

The parameter β has a remarkable interpretation. When $\beta > 0$, an agent is more central if her neighbors are also central — this captures *cooperation* (your friends' power helps you). When $\beta < 0$, an agent is more central if her neighbors are *less* central — this captures *bargaining* (your power comes from your partners' weakness, because they have no outside options).

Ballester et al. (2006) proved the foundational result: in linear-quadratic games on networks, the Nash equilibrium action of each agent is proportional to her Bonacich centrality. This is the "Bonacich-Nash linkage" — the first clean mapping from network position to economic outcome.

Intuition: Getting closer

Katz and Bonacich already make the centrality of your neighbors matter, through the indirect paths. But both depend on a parameter that must be chosen by the analyst, and different values give different rankings. There is no "canonical" Bonacich centrality.

The eigenvector centrality of the next subsection is the limit of the normalized Katz/Bonacich vector as the parameter approaches $1/\rho(W)$. It is therefore not a different idea but the parameter-free end of the same family — determined entirely by the network. Choosing it is a modelling decision: it is the right object for long-run (DeGroot) influence, while Katz/Bonacich are the right objects for linear-quadratic games with a finite discount.

6.1.5 Attempt 4: The Recursive Definition (Eigenvector Centrality)

Here is the idea that resolves everything.

An agent is powerful if powerful agents depend on her.

This is circular. The power of agent i depends on the power of agents j who point to i, which depends on the power of agents k who point to j, and so on. We seem to need to know the answer before we can compute it.

But this circularity is not a bug. It is a feature. The circularity defines a *fixed-point equation*: we seek a vector $\boldsymbol{\pi}$ such that the power of each agent equals the total power directed toward her.

Definition 6.4 (Eigenvector Centrality — Left Perron Eigenvector)**.** *Agent i's **eigenvector centrality** (or **structural power**) is π_i, the i-th component of the vector $\boldsymbol{\pi}$ satisfying:*

$$\boldsymbol{\pi}'W = \rho(W) \cdot \boldsymbol{\pi}', \tag{6.5}$$

with $\boldsymbol{\pi} \geq 0$ and $\|\boldsymbol{\pi}\|_1 = \sum_i \pi_i = 1$. For the row-stochastic matrices used throughout this book $\rho(W) = 1$, so (6.5) reads $\boldsymbol{\pi}'W = \boldsymbol{\pi}'$.

Equivalently, for each agent:

$$\pi_i = \frac{1}{\rho(W)} \sum_{j=1}^{N} \pi_j W_{ji}. \tag{6.6}$$

Intuition: Reading the equation

Equation (6.6) says: agent i's power is proportional to the sum, over all agents j, of (a) j's power π_j times (b) how much j depends on i (that is, W_{ji}). You are powerful if powerful agents point toward you.

The factor $1/\rho(W)$ is the reciprocal of the Perron eigenvalue, not a normalization (it equals one when W is row-stochastic); the separate normalization $\sum_i \pi_i = 1$ fixes the scale of $\boldsymbol{\pi}$.

This is the recursive definition we wanted: power depends on the power of those who depend on you. The Perron-Frobenius theorem (next section) guarantees that this circular definition has a unique solution.

Example 6.5 (Computing Power for the Five-Agent Economy)**.** *For the matrix W in (6.1), the left Perron eigenvector (computed iteratively — see Appendix 6A) is:*

$$\boldsymbol{\pi} \approx (0.144,\ 0.208,\ 0.236,\ 0.227,\ 0.185), \tag{6.7}$$

with $N_{\text{eff}} = 1/\sum_i \pi_i^2 = 4.87$ (Definition 6.15).

Agent 3 is the most powerful *($\pi_3 = 0.236$), even though she tied with Agents 2 and 4 on in-degree. Why? Because Agent 3 is depended upon by Agent 4 ($W_{43} = 0.4$), who is himself powerful ($\pi_4 = 0.227$). Agent 2's largest dependent is Agent 1 ($W_{12} = 0.4$), who is the least powerful ($\pi_1 = 0.144$). Being depended upon by a weak agent counts for less: Agent 2 falls from a tie for first on in-degree to third on eigenvector centrality.*

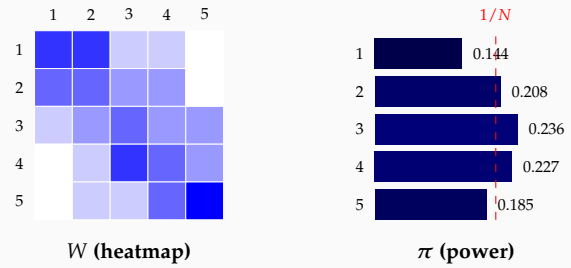
Agent 1 is the least powerful *($\pi_1 = 0.144$). She is peripheral: few agents depend on her, and those who do are not particularly powerful themselves.*

Compare the four centrality measures:

	Agent 1	*Agent 2*	*Agent 3*	*Agent 4*	*Agent 5*
In-degree	*0.80*	***1.10***	***1.10***	***1.10***	*0.90*
Katz ($\delta = 0.5$)	*0.77*	*1.08*	***1.13***	*1.11*	*0.90*
Bonacich ($\beta = 0.3$)	*1.12*	*1.56*	***1.60***	*1.58*	*1.28*
Eigenvector π_i	*0.144*	*0.208*	***0.236***	*0.227*	*0.185*

Katz, Bonacich and the eigenvector agree on the ranking $3 \succ 4 \succ 2 \succ 5 \succ 1$; in-degree cannot separate Agents 2, 3 and 4. At a common parameter value Katz and Bonacich must rank identically, because of (6.4); here the parameters differ ($\delta = 0.5$, $\beta = 0.3$) and the rankings happen to coincide. The eigenvector is the parameter-free limit of the family: the normalized Katz vector is $(0.152, 0.210, 0.231, 0.223, 0.185)$ at $\delta = 0.9$ and $(0.144, 0.208, 0.236, 0.227, 0.185)$ at $\delta = 0.999$.

Visualization: The five-agent economy: heatmap and power distribution



Left: the 5×5 weight matrix (6.1) as a heatmap (shade $= 200\, W_{ij}$; white $= 0$). Agent 3's column (column 3) receives weight from every row — she is depended upon broadly. Right: the Perron eigenvector. Agent 3 has the most power (0.236). The dashed red line marks $1/N = 0.20$ (equal power). Agents above the line are structurally advantaged; agents below it are disadvantaged. The distance from the red line, $\pi_i - 1/N$, is the power correction that enters the bargaining allocation (Proposition 7.3).

6.2 The Perron-Frobenius Theorem

The recursive definition of eigenvector centrality (6.5) raises an obvious question: does it have a solution? Is the solution unique? Is it positive (do all agents have nonnegative power)?

For a strongly connected network the answer to all three is yes, and the proof is the Perron-Frobenius theorem.

6.2.1 Statement

Theorem 6.6 (Perron-Frobenius)**.** *Let W be an $N \times N$ nonnegative matrix ($W_{ij} \geq 0$ for all i, j). Then:*

Part I (Perron, 1907): *If $W > 0$ (all entries strictly positive):*

1. *W has a real eigenvalue $\rho = \rho(W) > 0$ that is strictly larger in magnitude than all other eigenvalues: $|\lambda_k| < \rho$ for all $k \neq 1$.*

2. *The corresponding right eigenvector $\mathbf{v}$ can be chosen strictly positive: $v_i > 0$ for all i.*

3. *The corresponding left eigenvector $\boldsymbol{\pi}$ can be chosen strictly positive: $\pi_i > 0$ for all i.*

4. *Both $\mathbf{v}$ and $\boldsymbol{\pi}$ are unique up to normalization.*

Part II (Frobenius, 1912): *If $W \geq 0$ (nonnegative, some entries may be zero) and W is **irreducible** (the network is strongly connected — every agent can reach every other agent through some chain of positive weights):*

5. *Items (2)–(4) still hold, and $\rho > 0$ is a simple eigenvalue with $|\lambda_k| \leq \rho$ for all k. The strict dominance in (1) holds if and only if W is also **aperiodic** (the greatest common divisor of cycle lengths through any node is 1); an irreducible, aperiodic nonnegative matrix is called **primitive**. A periodic irreducible W — a bipartite star, an even ring without self-loops — has other eigenvalues of modulus ρ (e.g. $-\rho$).*

6. *If W is primitive, then $\frac{1}{\rho^t} W^t \to \mathbf{v}\boldsymbol{\pi}'$ as $t \to \infty$, with the normalization $\boldsymbol{\pi}'\mathbf{v} = 1$.*

Appendix 6.7 proves Part I and items (5)–(6) for primitive W, and the periodic case of item (5). For the irreducible case of items (2)–(4) and the simplicity of ρ, which are standard, see Seneta (2006, Thm. 1.5) or Horn and Johnson (2012, Thm. 8.4.4); for primitive matrices, Seneta (2006, Thms. 1.1–1.2).

Intuition: What each part says in words

(1) Dominant eigenvalue: Among all the eigenvalues of W, there is one — the spectral radius ρ — that is real, positive, and (for positive or primitive W) strictly larger in modulus than all others. This means the long-run behavior of the system is dominated by a single "mode."
(2)–(3) Positive eigenvectors: The right eigenvector $\mathbf{v}$ and left eigenvector $\boldsymbol{\pi}$ have all positive entries. No agent has zero power. No agent has negative power. Everyone matters, at least a little.
(4) Uniqueness: There is only one such $\boldsymbol{\pi}$ (up to scaling). Power is determined uniquely by the network. There is no ambiguity, no analyst choice, no free parameter.
(5) Extension to sparse matrices: Even if some entries of W are zero (not everyone is directly connected to everyone), existence, positivity and uniqueness of $\boldsymbol{\pi}$ survive as long as the network is *strongly connected*: there is a directed path between every ordered pair of agents. What can fail is strict dominance: a periodic network cycles.
(6) Convergence: If W is primitive and you repeatedly apply it to any initial vector — iterate beliefs through DeGroot learning, propagate influence through the network — the system converges to a state described by the Perron eigenvectors. In the long run, only $\boldsymbol{\pi}$ matters.

Key Structural Insight

The Perron-Frobenius theorem is why structural power is well-defined. Without it, the recursive definition $\pi_i = \frac{1}{\rho}\sum_j \pi_j W_{ji}$ might have no solution, multiple solutions, or solutions with negative entries (meaningless as a power measure). With it, the solution exists, is unique, and is positive. The existence of $\boldsymbol{\pi}$ is a theorem; reading it as "power" is an interpretation, which this book states separately in each context (DeGroot influence in Chapter 8, Nash weights by assumption in Chapter 7).

The theorem requires two conditions: nonnegativity ($W \geq 0$, which is natural — dependencies are nonnegative) and irreducibility (the network is connected — which is an empirical question). When the network is reducible (disconnected), it fragments into irreducible components, each with its own Perron eigenvector. This is the mathematical content of structural heterogeneity: a block-diagonal W has multiple Perron vectors, one per block.

6.2.2 Proof Sketch

The full proof is in Appendix 6.7. Here we give the key ideas for the positive case ($W > 0$).

Step 1: The map T. Define $T(\mathbf{x}) = W\mathbf{x}/\|W\mathbf{x}\|_1$ for $\mathbf{x} \geq 0$, $\mathbf{x} \neq 0$. This maps the open simplex $\Delta = \{\mathbf{x} > 0 : \|\mathbf{x}\|_1 = 1\}$ to itself (because $W > 0$ ensures $W\mathbf{x} > 0$).

Step 2: T is a contraction. On Δ with the Hilbert projective metric $d_H(\mathbf{x}, \mathbf{y}) = \ln \frac{\max_i x_i/y_i}{\min_i x_i/y_i}$, the map T satisfies $d_H(T(\mathbf{x}), T(\mathbf{y})) \leq \tau \cdot d_H(\mathbf{x}, \mathbf{y})$ where $\tau < 1$ depends on the ratio of the smallest to largest entry of W. This is Birkhoff's contraction theorem (1957).

Step 3: Fixed point. (Δ, d_H) is a complete metric space, so by the Banach fixed-point theorem T has a unique fixed point $\mathbf{v}^*$. This satisfies $W\mathbf{v}^* = \rho\mathbf{v}^*$ where $\rho = \|W\mathbf{v}^*\|_1$. So $\mathbf{v}^*$ is the right Perron eigenvector.

Step 4: Left eigenvector. Apply the same argument to W' (the transpose). The unique fixed point of $T'(\boldsymbol{\pi}) = W'\boldsymbol{\pi}/\|W'\boldsymbol{\pi}\|_1$ is $\boldsymbol{\pi}$, the left Perron eigenvector.

Step 5: Dominance and simplicity. Contraction alone only rules out competing *positive* eigenvectors; complex eigenvalues of modulus ρ need a separate argument, using $\boldsymbol{\pi} > 0$. It is given in Appendix 6B (Steps 6–7).

6.3 Properties of the Power Vector

6.3.1 Six Facts About Power

Throughout this section W is nonnegative, row-stochastic and irreducible, so that $\boldsymbol{\pi}$ exists and is unique (Theorem 6.6).

Remark 6.7 (Power Sums to One). $\sum_{i=1}^{N} \pi_i = 1$ by normalization. Power is a distribution: an agent's power π_i can be interpreted as her "share of total structural influence."

Proposition 6.8 (Equal Power iff Doubly Stochastic)**.** *Let W be row-stochastic and irreducible. Then $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$ (equal power for all agents) if and only if W is **doubly stochastic** (every column sum also equals 1).*

Proof. ($\Leftarrow$): If W is doubly stochastic, $\mathbf{1}'W = \mathbf{1}'$, so $\boldsymbol{\pi}' = \frac{1}{N}\mathbf{1}'$ satisfies $\boldsymbol{\pi}'W = \frac{1}{N}\mathbf{1}'W = \frac{1}{N}\mathbf{1}'$. By uniqueness of the Perron eigenvector (irreducibility), $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$.

($\Rightarrow$): If $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$, then $\frac{1}{N}\mathbf{1}'W = \frac{1}{N}\mathbf{1}'$ (because $\rho = 1$), so $\mathbf{1}'W = \mathbf{1}'$: column sums equal 1. Together with row-stochasticity, W is doubly stochastic. □

Remark 6.9 (Row-stochasticity is needed). For a general nonnegative irreducible matrix the correct condition is "all column sums are equal" (they then equal ρ), not "doubly stochastic". Example: $\begin{pmatrix} 0.2 & 0.6 \\ 0.4 & 0 \end{pmatrix}$ has row sums $(0.8, 0.4)$, column sums $(0.6, 0.6)$, $\rho = 0.6$ and $\boldsymbol{\pi} = (\frac{1}{2}, \frac{1}{2})$. This is one reason the book works with row-stochastic W whenever $\boldsymbol{\pi}$ is read as power.

Intuition: What equal power does and does not mean

Proposition 6.8 says: *power is equal if and only if the interaction matrix is doubly stochastic*. The benchmark $W^{\mathrm{CE}} = \frac{1}{N}\mathbf{1}\mathbf{1}'$ is doubly stochastic, so it has equal power — but so does every doubly stochastic W, including very fragmented ones (Example 6.17). Equal power is a statement about *column sums*: whether some agents receive more total weight than others.

It is *not* the condition for the welfare theorems. Those concern externalities and complete markets (Proposition 6.13). Unequal power is a distributional fact: it decides who gets what (Chapter 7), not whether the allocation is Pareto efficient.

Proposition 6.10 (Power is Monotone in Connections)**.** *Fix agents $i \neq k$ and an agent j (possibly $j = i$). Move weight $t \in (0, W_{jk}]$ in row j from k to i: $W_{ji} \mapsto W_{ji} + t$, $W_{jk} \mapsto W_{jk} - t$, all other entries unchanged. If the new matrix is still irreducible, π_i weakly increases. Being depended upon more heavily raises your power.*

Proof. Let Q be W with row and column i deleted. Because W is irreducible, i can be reached from every $l \neq i$, so $Q^n \to 0$ and $\rho(Q) < 1$. Let $\mathbf{h} = (I - Q)^{-1}\mathbf{1} = \sum_{n \geq 0} Q^n \mathbf{1}$ (the expected hitting times of i from $l \neq i$) and $m_i = 1 + \sum_{l \neq i} W_{il} h_l$ (the mean return time to i). Then $\pi_i = 1/m_i$ (Kac's formula; Levin et al., 2009, Prop. 1.14). Directly: let $\mathbf{g}$ equal h_l at $l \neq i$ and 0 at i. Then $(\mathbf{g} - W\mathbf{g})_l = 1$ for $l \neq i$ and $(\mathbf{g} - W\mathbf{g})_i = 1 - m_i$, i.e. $(I - W)\mathbf{g} = \mathbf{1} - m_i \mathbf{e}_i$; premultiplying by $\boldsymbol{\pi}'$ gives $0 = 1 - m_i \pi_i$. If $j \neq i$, the move lowers the entry Q_{jk} (as $k \neq i$) and raises only W_{ji}, which lies in the omitted column i; hence Q weakly decreases entrywise, $\mathbf{h} = \sum_n Q^n \mathbf{1}$ weakly decreases, and so does m_i. If $j = i$, $\mathbf{h}$ is unchanged and m_i changes by $-t\, h_k \leq 0$. In both cases m_i weakly falls, so π_i weakly rises. □

Proposition 6.11 (Power in Stars)**.** *Let agent h be the hub of a star with $N \geq 3$ agents. Each spoke places weight $a \in (0, 1]$ on the hub and $1 - a$ on herself; the hub keeps $s \in [0, 1)$ and spreads $1 - s$ equally over the spokes. Then*

$$\pi_h = \frac{a}{a + 1 - s}, \qquad \textit{spectrum}(W) = \{1,\ s - a,\ 1 - a\ (\textit{multiplicity } N - 2)\}, \quad (6.8)$$

so $\Gamma = 1 - \max(|s - a|, 1 - a)$. Both are independent of N. The hub's power approaches 1 only when $s \to 1$ (the hub ignores the spokes).

Proof. Stationarity across the cut between $\{h\}$ and the spokes gives $\pi_h(1 - s) = (1 - \pi_h)a$, hence $\pi_h = a/(a + 1 - s)$. Any $\mathbf{x}$ with $x_h = 0$ and $\sum_{\text{spokes}} x_i = 0$ satisfies $W\mathbf{x} = (1 - a)\mathbf{x}$, which gives $N - 2$ eigenvalues $1 - a$ (that subspace has dimension $N - 2$). The partition $\{h\}$, spokes is exactly lumpable: with M the $N \times 2$ membership matrix, $WM = M\bar{W}$ for the two-state chain $\bar{W} = \begin{pmatrix} s & 1 - s \\ a & 1 - a \end{pmatrix}$, so $\text{span}(M)$ is W-invariant and complementary to that subspace, and W acts on it as $\bar{W}$, with eigenvalues 1 and $\operatorname{tr}\bar{W} - 1 = s - a$. Since $a > 0$ and $s < 1$, W is irreducible and $\boldsymbol{\pi}$ is unique. □

The pure star ($a = 1$, $s = 0$) gives $\pi_h = \frac{1}{2}$ and $\Gamma = 0$, because it is bipartite (periodic), not because power is concentrated. Its lazy version ($a = s = \frac{1}{2}$) gives $\pi_h = \frac{1}{2}$ and $\Gamma = 0.5$.

Proposition 6.12 (Weakly Coupled Blocks)**.** *Partition the agents into blocks B_1 and B_2. Suppose every agent in B_1 places total weight $\varepsilon_1 \in (0, 1)$ on B_2 and every agent in B_2 places total weight $\varepsilon_2 \in (0, 1)$ on B_1. Then, exactly,*

$$\sum_{i \in B_1} \pi_i = \frac{\varepsilon_2}{\varepsilon_1 + \varepsilon_2}, \qquad 1 - \varepsilon_1 - \varepsilon_2 \in \textit{spectrum}(W), \quad \textit{so } \Gamma \leq \varepsilon_1 + \varepsilon_2. \quad (6.9)$$

Proof. Let M be the $N \times 2$ block-membership matrix. The hypothesis says $WM = M\bar{W}$ with $\bar{W} = \begin{pmatrix} 1-\varepsilon_1 & \varepsilon_1 \\ \varepsilon_2 & 1-\varepsilon_2 \end{pmatrix}$. Then $(\boldsymbol{\pi}'M)\bar{W} = \boldsymbol{\pi}'WM = \boldsymbol{\pi}'M$, so the block masses $\boldsymbol{\pi}'M$ are the stationary distribution of $\bar{W}$, namely $(\varepsilon_2, \varepsilon_1)/(\varepsilon_1 + \varepsilon_2)$. If $\bar{\mathbf{x}}$ is the eigenvector of $\bar{W}$ for $1 - \varepsilon_1 - \varepsilon_2$, then $W(M\bar{\mathbf{x}}) = M\bar{W}\bar{\mathbf{x}} = (1 - \varepsilon_1 - \varepsilon_2)M\bar{\mathbf{x}}$, and $M\bar{\mathbf{x}} \neq \mathbf{0}$. Since $\varepsilon_1 + \varepsilon_2 > 0$, this eigenvalue is not the simple Perron root 1, so $|1 - \varepsilon_1 - \varepsilon_2| \leq |\lambda_2|$ and $\Gamma \leq 1 - |1 - \varepsilon_1 - \varepsilon_2| \leq \varepsilon_1 + \varepsilon_2$. (This is exact lumpability in the sense of Kemeny and Snell, 1960, Thm. 6.3.2.) □

So weak coupling does two separate things. It makes Γ small (the blocks mix slowly), and it splits power between the blocks in proportion to the *inflows* $\varepsilon_2 : \varepsilon_1$ — not in proportion to block size or to any spectral radius. Symmetric coupling ($\varepsilon_1 = \varepsilon_2$, equal block sizes, doubly stochastic blocks) gives perfectly equal power with Γ as small as we like. As $\varepsilon \to 0$ the distribution *within* each block approaches that block's own Perron vector (Simon–Ando aggregation; Meyer, 2000). Exactly block-diagonal W is reducible: then each block has its own Perron vector and $\boldsymbol{\pi}$ is not unique.

Proposition 6.13 (The Welfare Theorem Revisited)**.** *In the Walrasian benchmark each agent interacts only with the market aggregate, and aggregate demand weights agents by their wealth shares* $\mathbf{s}$*. The implied interaction matrix is* $W^{mkt} = \mathbf{1}\mathbf{s}'$*: rank one, with* $\boldsymbol{\pi}(W^{mkt}) = \mathbf{s}$ *and* $\Gamma(W^{mkt}) = 1$*.* $W^{CE} = \frac{1}{N}\mathbf{1}\mathbf{1}'$ *is the equal-wealth special case. The First Welfare Theorem holds for* every $\mathbf{s}$*, under local nonsatiation, complete markets, price-taking and no externalities (*$W^U = 0$*).*

Proof. $\mathbf{s}'(\mathbf{1}\mathbf{s}') = (\mathbf{s}'\mathbf{1})\mathbf{s}' = \mathbf{s}'$; and any probability vector $\mathbf{q}$ with $\mathbf{q}'\mathbf{1}\mathbf{s}' = \mathbf{q}'$ satisfies $\mathbf{q}' = (\mathbf{q}'\mathbf{1})\mathbf{s}' = \mathbf{s}'$, so $\mathbf{s}$ is the unique stationary vector even if some $s_i = 0$ (when W^{mkt} is reducible). $\mathbf{1}\mathbf{s}'$ has rank one, so its eigenvalues are 1 and 0 ($N - 1$ times): $\Gamma = 1$. The First Welfare Theorem is the standard one (e.g. Mas-Colell et al., 1995, Prop. 16.C.1); its hypotheses do not involve $\mathbf{s}$. □

Hence unequal power, by itself, does not make an equilibrium inefficient: the Walrasian equilibrium with unequal wealth is Pareto efficient and has unequal $\boldsymbol{\pi}$. What breaks efficiency is utility interdependence ($W^U \neq 0$) and missing markets. What unequal power does is *distributional*: Chapter 7 shows how $\boldsymbol{\pi}$ shifts the division of a surplus, at a utilitarian (not Pareto) cost.

6.4 The Spectral Gap

6.4.1 Definition and Meaning

Definition 6.14 (Spectral Gap)**.** *For a row-stochastic W with eigenvalues ordered by modulus, $1 = \lambda_1, |\lambda_2| \geq |\lambda_3| \geq \cdots$, the **spectral gap** is*

$$\Gamma(W) = 1 - |\lambda_2(W)| \in [0, 1]. \tag{6.10}$$

This is the book's single definition of Γ.

What Γ measures: integration (mixing speed). For primitive W, deviations from the long-run state decay like $|\lambda_2|^t$: in DeGroot learning $\|\mathbf{b}^{(t)} - b^*\mathbf{1}\| \leq C|\lambda_2|^t$, with half-life $\ln 2/(-\ln|\lambda_2|)$. Large Γ means fast consensus and fast mixing; small Γ means slow convergence and persistent disagreement.

Periodicity caveat. $\Gamma = 0$ whenever W is reducible (several unit eigenvalues) *or* periodic (an eigenvalue of modulus one other than 1). A ring or a star *without* self-loops has $\Gamma = 0$ for the second reason: the dynamics oscillate rather than fail to integrate. When periodicity is an artefact of the drawing, use the **lazy** walk $\frac{1}{2}(I + W)$, which has the same $\boldsymbol{\pi}$.

The spectral gap is therefore informative about one dimension of structure. Read correctly, it says:

1. **Speed of convergence.** In DeGroot learning, beliefs converge at rate $|\lambda_2|^t$. Large Γ means fast convergence to consensus. Small Γ means slow convergence and persistent disagreement.

2. **Fragmentation.** Weak coupling between blocks forces a small gap: Proposition 6.12 gives $\Gamma \leq \varepsilon_1 + \varepsilon_2$, and for reversible W Cheeger's inequality $\Phi^2/2 \leq 1 - \lambda_2 \leq 2\Phi$ ties $1 - \lambda_2$ to the conductance Φ of the worst bottleneck. The converse fails: small Γ can come from periodicity rather than fragmentation.

3. **Not the concentration of power.** Γ says nothing about how unequal $\boldsymbol{\pi}$ is (Proposition 6.16 below). Concentration is measured separately (Definition 6.15).

4. **Not the strength of propagation.** How strongly shocks are amplified through production depends on the technical coefficients A and the Leontief inverse $(I - A')^{-1}$ (Chapter 1), not on Γ of the share matrix.

Definition 6.15 (Concentration of Power)**.** *For a power vector $\boldsymbol{\pi}$ in the simplex, the* ***effective number of agents*** *is*

$$N_{\mathit{eff}} = \frac{1}{\sum_{i=1}^{N} \pi_i^2} \in [1, N], \tag{6.11}$$

the inverse Herfindahl index of power. $N_{\mathit{eff}} = N$ iff $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$, and $N_{\mathit{eff}} = 1$ iff one agent holds all power. Two equivalent or complementary measures are the ***variance of power***

$$\mathrm{Var}(\boldsymbol{\pi}) = \frac{1}{N}\sum_{i=1}^{N}\Big(\pi_i - \frac{1}{N}\Big)^2 = \frac{1}{N}\Big(\frac{1}{N_{\mathit{eff}}} - \frac{1}{N}\Big) \in \Big[0, \tfrac{N-1}{N^2}\Big], \tag{6.12}$$

whose maximum is attained at $\boldsymbol{\pi} = \mathbf{e}_1$, and the ***power Gini*** *$G(\boldsymbol{\pi})$, the Gini coefficient of the entries of $\boldsymbol{\pi}$. These — not Γ — are what "unequal power" means in this book.*

Proposition 6.16 (Spectral Gap and Power Are Independent)**.** *For every $\boldsymbol{\pi}$ in the open simplex and every $\gamma \in (0, 1]$, the matrix*

$$W = \gamma\, \mathbf{1}\boldsymbol{\pi}' + (1-\gamma) I \tag{6.13}$$

is row-stochastic, irreducible and aperiodic, with $\boldsymbol{\pi}(W) = \boldsymbol{\pi}$ and $\Gamma(W) = \gamma$.

Proof. $W\mathbf{1} = \gamma\mathbf{1} + (1-\gamma)\mathbf{1} = \mathbf{1}$. Because $\boldsymbol{\pi} > 0$, every entry of W is positive ($\gamma\pi_j$ off the diagonal, $\gamma\pi_i + 1 - \gamma$ on it), so W is irreducible and aperiodic. Further $\boldsymbol{\pi}'W = \gamma\boldsymbol{\pi}' + (1-\gamma)\boldsymbol{\pi}' = \boldsymbol{\pi}'$, and since $\mathbf{1}\boldsymbol{\pi}'$ has eigenvalues $\{1, 0, \ldots, 0\}$, the spectrum of W is $\{1, 1-\gamma, \ldots, 1-\gamma\}$, so $\Gamma = \gamma$. □

Any degree of power concentration is compatible with any spectral gap. Three examples ($N = 10$): the near-dictatorship $\mathbf{1}\boldsymbol{\pi}'$ with $\pi_1 = 0.91$ has $\Gamma = 1$ and $N_{\mathrm{eff}} = 1.21$; the same $\boldsymbol{\pi}$ with $\gamma = 0.05$ has $\Gamma = 0.05$; and the doubly stochastic, weakly coupled two-block matrix of Example 6.17 has uniform $\boldsymbol{\pi}$ ($N_{\mathrm{eff}} = 10$) and $\Gamma = 0.02$. Along the path $W_t = (1-t)W + t\,\mathbf{1}\boldsymbol{\pi}'$, $\boldsymbol{\pi}$ stays fixed while Γ rises to 1: raising integration never, by itself, costs anyone power. The distance of W from W^{CE} therefore has (at least) two independent coordinates: the *power distance* $d_\pi = \frac{1}{2}\|\boldsymbol{\pi} - \frac{1}{N}\mathbf{1}\|_1$ (or N_{eff}/N) and the *integration deficit* $1 - \Gamma$.

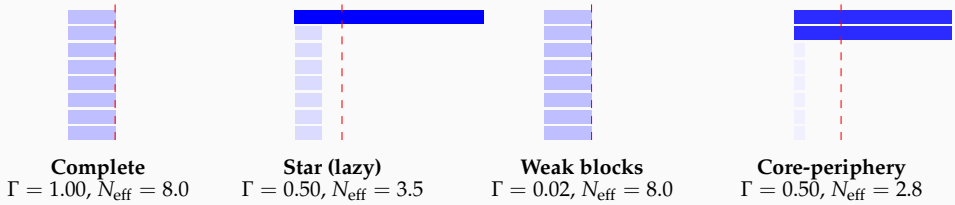
Example 6.17 (Spectral Gap and Concentration Across Topologies)**.** *For $N = 10$, with the following constructions (all row-stochastic). Complete: $\frac{1}{N}\mathbf{1}\mathbf{1}'$. Ring: lazy walk $\frac{1}{2}I + \frac{1}{4}(S + S')$ on the cycle (S the cyclic shift). Small-world: the cycle plus chords $\{1, 6\}$ and $\{3, 8\}$, row-normalized, lazy. Star: lazy pure star ($a = s = \frac{1}{2}$ in Proposition 6.11). Weak blocks: two blocks of five, weight $(1-\varepsilon)/5$ within and $\varepsilon/5$ across, $\varepsilon = 0.01$. Core-periphery: two core agents put 0.9 on the core and 0.1 spread over the periphery; each*

peripheral agent keeps 0.5 *and puts* 0.25 *on each core agent. Broadcast hub:* $\mathbf{1}\boldsymbol{\pi}'$ *with* $\pi_1 = 0.91$*, the rest equal.*

Topology	Γ	*Max* π_i	N_{eff}	$G(\boldsymbol{\pi})$	*Reading*
Complete	*1.000*	*0.100*	*10.00*	*0.00*	*integrated, equal*
Ring (lazy)	*0.095*	*0.100*	*10.00*	*0.00*	*slow mixing, equal*
Small-world (lazy)	*0.167*	*0.125*	*9.60*	*0.10*	*shortcuts speed mixing*
Star (lazy)	*0.500*	*0.500*	*3.60*	*0.40*	*integrated, concentrated*
Weak blocks	*0.020*	*0.100*	*10.00*	*0.00*	*fragmented, equal*
Core-periphery	*0.500*	*0.417*	*2.85*	*0.63*	*integrated, concentrated*
Broadcast hub	*1.000*	*0.910*	*1.21*	*0.81*	*fully integrated, near-dictator*

Without self-loops the ring and the star are periodic and have $\Gamma = 0$ *(with the same* $\boldsymbol{\pi}$*). The two columns move independently: the most fragmented network (weak blocks) has perfectly equal power, and the most integrated non-trivial one (broadcast hub) has the most concentrated power.*

Visualization: Four topologies, four power distributions



Power distributions ($\boldsymbol{\pi}$) for four of the constructions of Example 6.17, at $N = 8$ (bar length $\propto \pi_i$, shade $= 200\pi_i$). Dashed red line: $1/N = 0.125$ (equal power). Complete and weak blocks are both perfectly flat, although their spectral gaps are 1.00 and 0.02. Star and core-periphery have the same gap (0.50) but different concentration. The shape of $\boldsymbol{\pi}$ is read from N_{eff}; Γ reports how fast the network mixes, not how power is distributed.

6.4.2 The Spectral Gap as Development Diagnostic

Prebisch Connection: Two coordinates of the global economy

Combining the insights from Chapters 3 and 5, the global production share matrix W^P_{global} has to be read on two coordinates. Its spectral gap measures how well integrated the global economy is: whether shocks and demand mix quickly across all regions, or stay trapped in weakly coupled blocks. Its Perron vector, summarized by N_{eff}, measures how concentrated structural power is.

Prebisch's center–periphery argument is about the second coordinate. The periphery depends on the center ($W^P_{\text{periphery, center}}$ large) while the center barely depends on the periphery ($W^P_{\text{center, periphery}} \approx 0$). That asymmetry concentrates $\boldsymbol{\pi}$ on the center (the core-periphery row of Example 6.17: $N_{\text{eff}} = 2.85$ out of 10). It need not lower Γ at all — in that example $\Gamma = 0.5$, and if every country imported from the same core the matrix would be close to rank one, with Γ close to 1. A world can be tightly integrated and highly unequal in power at the same time.

Domestic structural heterogeneity (Chapter 3) is, in the theory, about the first coordinate: weak links between a modern and a traditional block would make Γ of the domestic W^P small (Proposition 6.12). The data do not show this for Bolivia. In INE's 2019 supply-use table, at 12 sectors on the ISIC-based aggregation, Bolivia's domestic W^P has $\Gamma = 0.475$. That is inside the range of the comparators in the OECD ICIO tables for 2019, from the USA (0.463) to Mexico (0.649), with Korea at 0.511. Its power Gini, 0.435, is *lower* than Korea's (0.658) or Germany's (0.611) (INE supply-use tables, base 2017, Instituto Nacional de Estadística (INE), Bolivia, 2026; OECD ICIO 2023, OECD, 2023; author's calculations, `data/compute/compute_oecd_io.py`). Across normalizations and aggregations Bolivia's 2019 Γ ranges from 0.33 to 0.69, so single values can be compared only on a common basis. Bolivia's enclave sits on the second coordinate instead: the hydrocarbons sector has a small Perron weight ($\pi = 0.039$ of 11 sectors) and exports 70% of its output (Chapter 3, Section 3.2). Nor is concentration of $\boldsymbol{\pi}$ in a domestic production network, by itself, a sign of oppression. In industrial economies $\boldsymbol{\pi}$ concentrates on manufacturing and business-service hubs, that is, on industrial depth. The distributional reading of $\boldsymbol{\pi}$ belongs to bargaining (Chapter 7, Assumption 7.A), and to the utility and information layers. Epistemic dependence (Chapter 5) is about the second: the Perron vector of the global W^I. The periphery's double trap is a statement on both coordinates, and the two have to be measured separately.

6.5 Power in Practice

6.5.1 Where Does Power Come From?

The Perron eigenvector gives us a precise answer. From equation (6.6):

$$\pi_i = \frac{1}{\rho} \sum_{j=1}^{N} \pi_j W_{ji}. \tag{6.14}$$

Your power is the weighted sum of the power of agents who depend on you, where the weights are how much they depend on you. This generates three *sources* of structural power:

1. Being needed (production power). If many agents use your output as an input (W^P_{ji} large for many j), you have production power. Saudi Arabia has production power in the global oil network. TSMC has production power in the semiconductor network.

2. Being cared about (social power). If powerful agents care about your well-being (W^U_{ji} large for powerful j), you have social power. The president's spouse has social power. A patron's favored client has social power.

3. Being listened to (epistemic power). If powerful agents pay attention to your signals (W^I_{ji} large for powerful j), you have epistemic power. The Federal Reserve chair has epistemic power. A TikTok influencer has epistemic power.

6.5.2 Power vs. Wealth

Power ($\boldsymbol{\pi}$) and wealth ($\mathbf{w}$) are not the same thing.

Wealth is what you have — your endowment ω_i, your capital K_i, your income Y_i. Wealth can be redistributed by transfers: tax the rich, give to the poor, and $\mathbf{w}$ changes.

Power is where you sit — your position in W. Holding W fixed, a transfer changes $\boldsymbol{\omega}$ but not $\boldsymbol{\pi}$: the Perron eigenvector is a property of the matrix, not of the endowment vector. (If W itself responds to the allocation — the feedback studied from Chapter 7 on — transfers can move $\boldsymbol{\pi}$ indirectly.)

Example 6.18 (A Rich Farmer and a Poor Minister)**.** *Suppose agent 1 is a wealthy farmer ($w_1 = 100$) and agent 2 is a poor government minister ($w_2 = 10$). In a 2×2 economy with:*

$$W = \begin{pmatrix} 0.8 & 0.2 \\ 0.9 & 0.1 \end{pmatrix}, \tag{6.15}$$

the farmer depends on himself ($W_{11} = 0.8$) and barely on the minister ($W_{12} = 0.2$). But the minister depends heavily on the farmer ($W_{21} = 0.9$) — because agricultural output feeds the government's tax base.

The rows sum to one ($0.8 + 0.2 = 0.9 + 0.1 = 1$), so W is row-stochastic and $\rho = 1$. The left Perron eigenvector solves $\boldsymbol{\pi}'W = \boldsymbol{\pi}'$. From the first column: $0.8\pi_1 + 0.9\pi_2 = \pi_1$, so $\pi_2/\pi_1 = 0.2/0.9 = 2/9$. With $\pi_1 + \pi_2 = 1$: $\pi_1 = 9/11 \approx 0.818$, $\pi_2 = 2/11 \approx 0.182$.

The farmer has 82% of the structural power despite the minister having the political title. Why? Because the minister depends on the farmer (for tax revenue) more than the farmer depends on the minister (for governance). The dependency is asymmetric, and asymmetry is power.

The farmer is rich AND powerful. The minister is poor AND less powerful. This is a structurally stable configuration: power and wealth reinforce each other. Chapter 13 will show that such configurations are fixed points of the structural reproduction process.

6.6 Summary

1. **Four centrality measures** capture different aspects of network importance: degree counts direct connections; Katz counts all paths, discounted; Bonacich is Katz rescaled, with an interpretable sign; and the eigenvector is recursive power. Eigenvector centrality is the parameter-free limit of the Katz family.

2. **The Perron-Frobenius theorem** guarantees that $\boldsymbol{\pi}$ exists, is unique, and is positive for any irreducible (strongly connected) nonnegative matrix; strict dominance of the Perron root, and convergence of W^t, need primitivity (aperiodicity as well).

3. **Power is equal if and only if a row-stochastic W is doubly stochastic.** This is a distributional statement. The First Welfare Theorem holds for the Walrasian $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$ with any wealth shares $\mathbf{s}$; efficiency fails through externalities ($W^U \neq 0$), not through unequal $\boldsymbol{\pi}$.

4. **Two coordinates, not one.** The spectral gap $\Gamma = 1 - |\lambda_2|$ measures integration (mixing speed, fragmentation); the concentration of power is measured by $N_{\text{eff}} = 1/\sum \pi_i^2$, $\text{Var}(\boldsymbol{\pi})$ or the power Gini. The two are independent (Proposition 6.16).

5. **Power comes from position**, not endowment. Holding W fixed, transfers change $\boldsymbol{\omega}$ but not $\boldsymbol{\pi}$; structural policy changes W and therefore $\boldsymbol{\pi}$, with sensitivities given by Lemma 6.19.

6.7 Exercises

6.1 Computing centralities. For the five-agent economy (6.1):

(a) Compute all four centrality measures: degree, Katz with $\delta = 0.5$, Bonacich with $\beta = 0.3$, and eigenvector. Use Python.

(b) Do they agree on the ranking? Where do they disagree?

(c) Now double W_{31} (Agent 3 depends more on Agent 1), taking the extra 0.1 from W_{33} so the row still sums to one. Recompute the eigenvector centrality. Does Agent 1's power rise? By how much? (Check your answer against Proposition 6.10.)

6.2 Doubly stochastic or not. Classify each matrix as doubly stochastic (equal power) or not (asymmetric power), and for those that are not, find the Perron eigenvector:

(a) $W = \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

(b) $W = \begin{pmatrix} 0.5 & 0.5 \\ 0.8 & 0.2 \end{pmatrix}$

(c) $W = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \end{pmatrix}$ (compare with (a): same $\boldsymbol{\pi}$? same Γ?)

(d) $W = \begin{pmatrix} 0.4 & 0.3 & 0.3 \\ 0.3 & 0.4 & 0.3 \\ 0.3 & 0.3 & 0.4 \end{pmatrix}$

6.3 The spectral gap and convergence. For each topology (complete, lazy star, lazy ring, weak blocks with $\varepsilon = 0.01$), built as in Example 6.17 with $N = 8$:

(a) Compute Γ and N_{eff}. Then compute Γ for the ring and the star *without* self-loops, and explain why it is zero.

(b) Starting from $\mathbf{b}^{(0)} = (1, 0, 0, 0, 0, 0, 0, 0)'$, iterate $\mathbf{b}^{(t+1)} = W\mathbf{b}^{(t)}$ for $t = 1, \ldots, 50$. Plot $\|\mathbf{b}^{(t)} - \boldsymbol{\pi}'\mathbf{b}^{(0)}\mathbf{1}\|$ (distance from consensus) against t.

(c) Verify that the convergence rate matches $|\lambda_2|^t$.

6.4 Bonacich with negative β. For a buyer-seller network:

$$W = \begin{pmatrix} 0 & 0 & 0.5 & 0.5 \\ 0 & 0 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0 & 0 \\ 0.5 & 0.5 & 0 & 0 \end{pmatrix}. \tag{6.16}$$

(a) Compute Bonacich centrality with $\beta = 0.3$ (cooperation) and $\beta = -0.3$ (bargaining).

(b) Under bargaining ($\beta < 0$), which agents have higher centrality? Why?

(c) Relate to Manea's (2011) bargaining result: in symmetric bipartite networks, all agents have equal bargaining power. Does the eigenvector centrality confirm this? Compute Γ for this W and explain the value (the network is bipartite).

6.5 Power iteration by hand. For $W = \begin{pmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{pmatrix}$:

(a) Start with $\mathbf{v}^{(0)} = (1, 1)$. Compute $\mathbf{v}^{(1)} = W'\mathbf{v}^{(0)}$, normalize. Repeat for $t = 2, 3, 4, 5$.

(b) At what iteration does $\mathbf{v}^{(t)}$ converge to two decimal places?

(c) Compute the exact Perron eigenvector algebraically (solve $\boldsymbol{\pi}'W = \boldsymbol{\pi}'$).

(d) Compare with the iterative result. Compute Γ and verify the convergence rate.

6.6 Structural power vs. formal power. An economy has 4 agents: President, General, Businesswoman, Farmer. Construct a plausible 4×4 weight matrix where the Farmer has the highest eigenvector centrality despite having the lowest formal rank. (Hint: make everyone's food supply depend on the Farmer, directly or indirectly.) Compute $\boldsymbol{\pi}$ and verify.

6.7 Power in a stylized dual economy. Using the two-sector stylized economy of Example 3.6 (Modern/gas and Traditional, $W^P = D^{-1}A_{\text{dual}}$):

(a) Compute the left Perron eigenvector. Which sector has the most structural power?

(b) Now suppose gas revenues are redistributed to the traditional sector (a transfer). Does $\boldsymbol{\pi}$ change? Why or why not?

(c) Now suppose a gas-to-fertilizer plant is built, adding a link from Gas to Traditional (the technical coefficient a_{TM}, Traditional's purchases from the modern sector, rises from 0.02 to 0.15). Recompute $W^P = D^{-1}A$ and $\boldsymbol{\pi}$. How does the traditional sector's power change?

(d) Which policy — the transfer or the plant — changes the power structure?

(e) *Data.* Download a published input-output table (INE Bolivia's *Matriz Insumo-Producto*, or a Latin American country in the OECD ICIO tables), aggregate it to two sectors (modern and traditional), form the share matrix $W^P = D^{-1}A$, and compute $\boldsymbol{\pi}$, N_{eff} and Γ. Compare with the stylized model, and with the 12-sector values for Bolivia in 2019 ($\Gamma = 0.467$, $N_{\text{eff}} = 6.48$ of 11; Chapter 3, Section 3.4). Does your two-sector aggregation make Bolivia look more or less integrated, and why?

Mathematical Appendices

Appendix 6A: Computing the Perron Eigenvector

Power iteration. The simplest algorithm:

1. Start with $\mathbf{v}^{(0)} = \frac{1}{N}\mathbf{1}$ (or any positive vector).
2. Iterate: $\tilde{\mathbf{v}}^{(t+1)} = W'\mathbf{v}^{(t)}$ (multiply by the transpose).
3. Normalize: $\mathbf{v}^{(t+1)} = \tilde{\mathbf{v}}^{(t+1)} / \|\tilde{\mathbf{v}}^{(t+1)}\|_1$.
4. Repeat until $\|\mathbf{v}^{(t+1)} - \mathbf{v}^{(t)}\|_\infty < \varepsilon$ (convergence tolerance).

The spectral radius is estimated as $\rho \approx \|\tilde{\mathbf{v}}^{(t+1)}\|_1 / \|\mathbf{v}^{(t)}\|_1$ (the growth factor before normalization).

Convergence rate. For primitive W the error after t iterations is $\|\mathbf{v}^{(t)} - \boldsymbol{\pi}\| = O(|\lambda_2/\rho|^t)$ (times a polynomial in t if λ_2 is defective). For row-stochastic W the rate is $|\lambda_2| = 1 - \Gamma$. For the complete graph ($\Gamma = 1$): convergence in one step. For a weakly coupled block graph ($\Gamma \approx 0.01$): each digit of accuracy requires $\ln 10/(-\ln 0.99) \approx 230$ iterations. For periodic W ($\Gamma = 0$) plain power iteration does not converge; iterate the lazy matrix $\frac{1}{2}(I + W)$, which has the same $\boldsymbol{\pi}$.

Python code.

```
import numpy as np

def perron_eigenvector(W, tol=1e-10, maxiter=1000):
    """Left Perron eigenvector of W."""
    n = W.shape[0]
    v = np.ones(n) / n
    for _ in range(maxiter):
        v_new = W.T @ v
        v_new = v_new / v_new.sum()
        if np.max(np.abs(v_new - v)) < tol:
            return v_new
        v = v_new
    return v

def spectral_gap(W):
```

```
    """Gamma = 1 - |lambda_2| for row-stochastic W."""
    eigenvalues = np.sort(np.abs(np.linalg.eigvals(W)))[::-1]
    return 1 - eigenvalues[1]


def n_eff(pi):
    """Effective number of agents, 1 / sum pi_i^2."""
    return 1 / np.sum(pi ** 2)
```

All numbers in this chapter are reproduced by `code/ch06.py`.

Computational complexity. Each iteration of power iteration requires one matrix-vector multiplication: $O(N^2)$ for a dense matrix, $O(\text{nnz})$ for a sparse matrix (where nnz is the number of nonzero entries). For most economic networks (sparse, $\text{nnz} \ll N^2$), this is efficient even for N in the millions.

Appendix 6B: Proof of Perron-Frobenius (Positive Case)

We prove Theorem 6.6 for the case $W > 0$ (all entries strictly positive) in Steps 1–7, extend it to primitive W in Step 8, and show in Step 9 why strict dominance fails for periodic W. The general irreducible case of items (2)–(4) needs additional machinery (Seneta, 2006, Thm. 1.5; Horn and Johnson, 2012, Thm. 8.4.4).

Setup. Let $W > 0$ be $N \times N$ with all entries strictly positive. Define the positive cone $\mathcal{K} = \{\mathbf{x} \in \mathbb{R}^N : x_i > 0 \text{ for all } i\}$ and the simplex $\Delta = \{\mathbf{x} \in \mathcal{K} : \|\mathbf{x}\|_1 = 1\}$.

Step 1: W maps the positive cone to itself. Since $W > 0$ and $\mathbf{x} > 0$, the product $W\mathbf{x}$ has all positive entries: $(W\mathbf{x})_i = \sum_j W_{ij} x_j > 0$.

Step 2: Hilbert projective metric. For $\mathbf{x}, \mathbf{y} \in \mathcal{K}$, define:

$$M(\mathbf{x}/\mathbf{y}) = \max_i \frac{x_i}{y_i}, \qquad m(\mathbf{x}/\mathbf{y}) = \min_i \frac{x_i}{y_i}, \qquad d_H(\mathbf{x}, \mathbf{y}) = \ln \frac{M(\mathbf{x}/\mathbf{y})}{m(\mathbf{x}/\mathbf{y})}. \tag{6.17}$$

This is a pseudo-metric on the positive cone ($d_H(\mathbf{x}, \mathbf{y}) = 0$ iff $\mathbf{x} = c\mathbf{y}$ for some $c > 0$). Restricted to the open simplex Δ (one representative per ray) it is a metric, and (Δ, d_H) is complete.

Step 3: Birkhoff's contraction. For $W > 0$, Birkhoff's theorem gives $d_H(W\mathbf{x}, W\mathbf{y}) \leq \tau(W)\, d_H(\mathbf{x}, \mathbf{y})$ with $\tau(W) = \frac{1-\sqrt{\phi(W)}}{1+\sqrt{\phi(W)}}$, where $\phi(W) = \min_{i,j,k,l} \frac{W_{ik} W_{jl}}{W_{jk} W_{il}} > 0$. (The cruder ratio $(\min_{ij} W_{ij} / \max_{ij} W_{ij})^2$ is a lower bound for $\phi(W)$, so it also yields a

valid $\tau < 1$.) Since $\phi(W) > 0$, $\tau(W) < 1$: the map $\mathbf{x} \mapsto W\mathbf{x}$ is a strict contraction in the Hilbert metric.

Step 4: Existence and uniqueness. The normalized map $T(\mathbf{x}) = W\mathbf{x}/\|W\mathbf{x}\|_1$ sends Δ into itself and satisfies $d_H(T\mathbf{x}, T\mathbf{y}) = d_H(W\mathbf{x}, W\mathbf{y})$, so it is a contraction on the complete metric space (Δ, d_H). By the Banach fixed-point theorem, T has a unique fixed point $\mathbf{v}^* \in \Delta$. This satisfies $W\mathbf{v}^* = \rho\mathbf{v}^*$ where $\rho = \|W\mathbf{v}^*\|_1 > 0$. Uniqueness in Δ means there is no other positive eigenvector, up to scale.

Step 5: Positivity and the left eigenvector. $\mathbf{v}^* \in \Delta$, so $\mathbf{v}^* > 0$. Applying Steps 1–4 to $W' > 0$ gives a unique $\boldsymbol{\pi} > 0$ in Δ with $W'\boldsymbol{\pi} = \rho'\boldsymbol{\pi}$. Since $\rho'\boldsymbol{\pi}'\mathbf{v}^* = \boldsymbol{\pi}'W\mathbf{v}^* = \rho\boldsymbol{\pi}'\mathbf{v}^*$ and $\boldsymbol{\pi}'\mathbf{v}^* > 0$, $\rho' = \rho$: $\boldsymbol{\pi}'W = \rho\boldsymbol{\pi}'$.

Step 6: Dominance. Let $W\mathbf{u} = \lambda\mathbf{u}$ with $\mathbf{u} \in \mathbb{C}^N \setminus \{0\}$ and $|\lambda| = \rho$. Entrywise, $\rho|\mathbf{u}| = |W\mathbf{u}| \leq W|\mathbf{u}|$. Multiply the nonnegative vector $W|\mathbf{u}| - \rho|\mathbf{u}|$ by $\boldsymbol{\pi}' > 0$: $\boldsymbol{\pi}'W|\mathbf{u}| - \rho\boldsymbol{\pi}'|\mathbf{u}| = 0$, so $W|\mathbf{u}| = \rho|\mathbf{u}|$. By Step 4, $|\mathbf{u}| = c\,\mathbf{v}^*$ with $c > 0$. Equality in $|\sum_j W_{ij}u_j| = \sum_j W_{ij}|u_j|$ with all $W_{ij} > 0$ forces all u_j to share one argument θ: $\mathbf{u} = e^{i\theta}c\,\mathbf{v}^*$. Then $\lambda\mathbf{u} = W\mathbf{u} = \rho\mathbf{u}$, so $\lambda = \rho$, and its eigenspace is spanned by $\mathbf{v}^*$. Hence every other eigenvalue has $|\lambda_k| < \rho$.

Step 7: Algebraic simplicity. If ρ were not algebraically simple, there would be $\mathbf{w}$ with $(W - \rho I)\mathbf{w} = \mathbf{v}^*$. Premultiplying by $\boldsymbol{\pi}'$ gives $0 = \boldsymbol{\pi}'\mathbf{v}^* > 0$, a contradiction.

This completes the proof for the positive case.

Step 8: Primitive matrices. Let W be irreducible and aperiodic. Then some power W^m is positive. (For each i the set of n with $(W^n)_{ii} > 0$ is closed under addition and has greatest common divisor 1, so it contains every $n \geq n_i$ for some n_i; if i reaches j in ℓ_{ij} steps, $(W^n)_{ij} > 0$ for all $n \geq n_i + \ell_{ij}$; take $m = \max_{i,j}(n_i + \ell_{ij})$. See Seneta, 2006, Ch. 1 and Appendix A.) Apply Steps 1–7 to W^m, whose Perron root is $\rho(W)^m$. If $W\mathbf{u} = \lambda\mathbf{u}$ with $|\lambda| = \rho$, then $W^m\mathbf{u} = \lambda^m\mathbf{u}$ with $|\lambda^m| = \rho^m$, so $\lambda^m = \rho^m$ and $\mathbf{u}$ is a multiple of the Perron vector $\mathbf{v}^*$ of W^m. Since $W\mathbf{v}^*$ is also an eigenvector of W^m for ρ^m, $W\mathbf{v}^* = c\,\mathbf{v}^*$ with $c > 0$, and $c = \rho$; hence $\lambda = \rho$. So ρ strictly dominates, and the algebraic simplicity of ρ^m for W^m gives the algebraic simplicity of ρ for W (each eigenvalue λ of W contributes λ^m to the spectrum of W^m, with the same algebraic multiplicity). This gives items (1)–(5) for primitive W. For item (6), write $P = \mathbf{v}\boldsymbol{\pi}'$ with $\boldsymbol{\pi}'\mathbf{v} = 1$. Then $WP = PW = \rho P$ and $P^2 = P$, so

$(W/\rho)^t - P = (W/\rho - P)^t$; the matrix $W/\rho - P$ has the eigenvalues λ_k/ρ ($k \geq 2$) and 0, all of modulus less than one, so its powers tend to zero.

Step 9: Periodic matrices. Let W be irreducible with period $d > 1$. Its nodes split into cyclic classes $C_0, \dots, C_{d-1}$ such that $W_{ij} > 0$ only if $i \in C_r$ and $j \in C_{r+1 \bmod d}$ (Seneta, 2006, Ch. 1). With $\omega = e^{2\pi \mathrm{i}/d}$ and $r(i)$ the class of i, set $x_i = \omega^{r(i)} v_i$, where $W\mathbf{v} = \rho\mathbf{v}$, $\mathbf{v} > 0$. Then $(W\mathbf{x})_i = \sum_j W_{ij}\omega^{r(i)+1} v_j = \omega\rho\, x_i$, so $\omega\rho \neq \rho$ is an eigenvalue of modulus ρ: strict dominance fails. Items (2)–(4) and the simplicity of ρ still hold (Seneta, 2006, Thm. 1.5). □

Appendix 6C: Perturbation of the Perron Vector and of the Spectral Gap

This lemma is the single sensitivity result used throughout the book (Chapters 7, 8, 13, 14, 15, 23 and 24).

Lemma 6.19 (Perturbation Lemma)**.** *Let W be irreducible and row-stochastic with Perron vector $\boldsymbol{\pi}$. Let E satisfy $E\mathbf{1} = \mathbf{0}$ and $E_{ij} \geq 0$ wherever $W_{ij} = 0$, so that $W(t) = W + tE$ is row-stochastic, nonnegative and irreducible for small $t \geq 0$.*

1. ***Perron vector.*** *With the fundamental matrix $Z := (I - W + \mathbf{1}\boldsymbol{\pi}')^{-1}$,*

$$\left.\frac{d\boldsymbol{\pi}'}{dt}\right|_{t=0} = \boldsymbol{\pi}' E Z. \tag{6.18}$$

   If W is diagonalizable with right eigenvectors $\mathbf{v}_k$ and left eigenvectors $\mathbf{u}_k'$ normalized biorthogonally ($\mathbf{u}_k'\mathbf{v}_l = \delta_{kl}$, $\mathbf{v}_1 = \mathbf{1}$, $\mathbf{u}_1 = \boldsymbol{\pi}$), equivalently

$$\frac{d\boldsymbol{\pi}'}{dt} = \sum_{k \geq 2} \frac{\boldsymbol{\pi}' E \mathbf{v}_k}{1 - \lambda_k}\, \mathbf{u}_k'. \tag{6.19}$$

2. ***Eigenvalues.*** *If λ is a simple eigenvalue of W with right vector $\mathbf{v}$, left vector $\mathbf{u}$ and $\mathbf{u}'\mathbf{v} = 1$, then $d\lambda/dt = \mathbf{u}' E \mathbf{v}$ and, for $\lambda \neq 0$,*

$$\frac{d|\lambda|}{dt} = \frac{\mathrm{Re}\big(\bar{\lambda}\, \mathbf{u}' E \mathbf{v}\big)}{|\lambda|}. \tag{6.20}$$

   Hence $\Gamma = 1 - |\lambda_2|$ is differentiable along E whenever λ_2 is simple and no eigenvalue other than λ_2 and its conjugate has modulus $|\lambda_2|$, with $d\Gamma/dt = -d|\lambda_2|/dt$.

Proof. (1) $\boldsymbol{\pi}(t)$ is the normalized left eigenvector of the simple eigenvalue 1 of $W(t)$, and it is differentiable in t. Indeed, let $B(t)$ be $I - W(t)$ with its first column replaced by $\mathbf{1}$. If $\mathbf{x}'B(t) = \mathbf{0}'$, then $\mathbf{x}'\mathbf{1} = 0$ and $\mathbf{x}'(I - W(t))$ vanishes in columns

$2, \dots, N$; since $(I - W(t))\mathbf{1} = \mathbf{0}$, its first entry vanishes too, so $\mathbf{x} \propto \boldsymbol{\pi}(t)$, and $\mathbf{x}'\mathbf{1} = 0$ forces $\mathbf{x} = \mathbf{0}$. Hence $B(t)$ is invertible and $\boldsymbol{\pi}(t)' = \mathbf{e}_1' B(t)^{-1}$, a rational function of t. Differentiating $\boldsymbol{\pi}(t)'W(t) = \boldsymbol{\pi}(t)'$ and $\boldsymbol{\pi}(t)'\mathbf{1} = 1$ at $t = 0$ gives $\dot{\boldsymbol{\pi}}'(I - W) = \boldsymbol{\pi}'E$ and $\dot{\boldsymbol{\pi}}'\mathbf{1} = 0$. The second identity gives $\dot{\boldsymbol{\pi}}'\mathbf{1}\boldsymbol{\pi}' = 0$, so $\dot{\boldsymbol{\pi}}'(I - W + \mathbf{1}\boldsymbol{\pi}') = \boldsymbol{\pi}'E$. The matrix $I - W + \mathbf{1}\boldsymbol{\pi}'$ is invertible: it maps $\mathbf{1}$ to $\mathbf{1}$, and on the W-invariant complement $\{\mathbf{x} : \boldsymbol{\pi}'\mathbf{x} = 0\}$ of $\operatorname{span}(\mathbf{1})$ it acts as $I - W$, whose eigenvalues there, $1 - \lambda_k$ ($k \geq 2$), are nonzero because the eigenvalue 1 is simple. This gives (6.18), with no diagonalizability needed. In the diagonalizable case $Z = \mathbf{1}\boldsymbol{\pi}' + \sum_{k\geq 2}(1 - \lambda_k)^{-1}\mathbf{v}_k\mathbf{u}_k'$, and $\boldsymbol{\pi}'E\mathbf{1} = 0$ gives (6.19).

(2) A simple root λ of $p(t, z) = \det(zI - W(t))$ has $\partial_z p \neq 0$, so by the implicit function theorem $\lambda(t)$ is differentiable near $t = 0$. Since $\lambda(t)I - W(t)$ has rank $N - 1$, a nonzero column of its adjugate is a right eigenvector $\mathbf{v}(t)$ that is differentiable in t (cf. Horn and Johnson, 2012, Sec. 6.3). Differentiating $W(t)\mathbf{v}(t) = \lambda(t)\mathbf{v}(t)$ gives $E\mathbf{v} + W\dot{\mathbf{v}} = \dot{\lambda}\mathbf{v} + \lambda\dot{\mathbf{v}}$; premultiplying by $\mathbf{u}'$ and using $\mathbf{u}'W = \lambda\mathbf{u}'$ and $\mathbf{u}'\mathbf{v} = 1$ gives $\dot{\lambda} = \mathbf{u}'E\mathbf{v}$. Finally $|\lambda|^2 = \lambda\bar{\lambda}$, so $2|\lambda|\,\frac{d|\lambda|}{dt} = 2\,\mathrm{Re}(\bar{\lambda}\dot{\lambda})$. □

Three consequences.

1. **A common error.** A formula of the form $\sum_{k\geq 2}(1 - \lambda_k)^{-1}\mathbf{u}_k(\mathbf{u}_k'E\mathbf{v}_1)$ is identically zero, because $E\mathbf{v}_1 = E\mathbf{1} = \mathbf{0}$ for every row-sum-preserving perturbation. The correct projection is onto $\boldsymbol{\pi}'E$, as in (6.19). Similarly, $\operatorname{sign}(\lambda)\,d\lambda$ is not the derivative of $|\lambda|$ when λ is complex; use (6.20).

2. **When is power cheap to move?** By (6.19), $\boldsymbol{\pi}$ is most sensitive when some λ_k is close to $+1$, i.e. when $1 - \lambda_2$ is small (weakly coupled blocks). Eigenvalues near -1 (periodicity) give $1 - \lambda_k \approx 2$ and create no sensitivity. So the cost of moving power by a given amount — through manipulation (Chapter 8) or escape (Chapter 7) — scales *with* $1 - \lambda_2$: it is smallest in fragmented networks, not largest.

3. **Where to add links to raise Γ.** Moving weight t in row j from k to i ($E = \mathbf{e}_j(\mathbf{e}_i - \mathbf{e}_k)'$) changes λ_2 by $t\,u_{2j}(v_{2i} - v_{2k})$, to first order. The largest entries of $|\mathbf{u}_2|$ and $|\mathbf{v}_2|$ lie deep *inside* the blocks, not on the bridge nodes. Example (`code/ch06.py`): two five-agent path-like blocks joined by a weak bridge, $\Gamma = 0.025$; moving self-weight into a link between the two block cores raises Γ at rate 0.182, while the same move between the two bridge nodes raises it at rate 0.109. Raising integration means connecting the cores of the blocks, not reinforcing the existing bridge.

Structuralist Case Study: Prebisch's *Capitalismo Periférico* and the Perron Eigenvector

In 1981, at the age of eighty, Raúl Prebisch published *Capitalismo Periférico: Crisis y Transformación* (México: Fondo de Cultura Económica, 344 pp.) — his final major work and, in many ways, his most theoretically ambitious. Where the 1949 manifesto diagnosed the center-periphery structure of international trade, *Capitalismo Periférico* turned inward, analyzing the **power structure within peripheral capitalism** and the mechanisms through which the economic surplus is appropriated by those at the top of the social hierarchy.

The book's argument maps directly onto this chapter's formal framework.

Part 1: The surplus and its appropriation. Prebisch's *excedente* is, roughly, the part of the fruits of rising productivity that is not transferred to the labor force but retained by the owners of the means of production. In a capitalist economy, the surplus is not distributed by a planner; it is *appropriated* through a power structure. The agents who appropriate the most surplus are not necessarily the most productive — they are the ones with the strongest structural position. In this book's formalization (an interpretation, not Prebisch's own notation) the division of a surplus Φ follows the allocation rule of Chapter 7, $x_i = \Omega/N + (\pi_i - 1/N)\Phi$ (Proposition 7.3): agents with $\pi_i > 1/N$ capture more than an equal share, regardless of their marginal product. Proposition 6.8 states the condition under which appropriation is equal: W must be doubly stochastic. Prebisch's book can be read as an argument that peripheral W is far from doubly stochastic.

Part 2: Power is relational, not individual. Prebisch explicitly rejected the neoclassical view that factor payments reflect marginal products. In peripheral capitalism, wages are held below marginal product not because of market imperfections (in the neoclassical sense) but because *the power structure of the labor market favors employers*. The employer's structural position — monopsony in thin labor markets, control over hiring and firing in the absence of unions, political connections that shape regulation — gives her a Perron weight $\pi_{\text{employer}} \gg 1/N$. The worker's structural position — limited outside options, geographic immobility, dependence on a single employer — gives $\pi_{\text{worker}} \ll 1/N$. The wage is determined not by the intersection of supply and demand curves but by the bargaining equilibrium of Chapter 7, with weights from the Perron eigenvector of this chapter.

This is exactly equation (6.6): $\pi_i = \frac{1}{\rho}\sum_j \pi_j W_{ji}$. The worker's power is low because the powerful agents who "point toward" her (her employers) do not depend heavily on her individually (W_{ji} small for any specific worker j). The employer's

power is high because workers depend heavily on her (W_{ij} large for many workers i). The recursion — power comes from being depended upon by the powerful — is the mathematical content of Prebisch's argument that "power relations between social strata" determine surplus appropriation.

Part 3: The political economy of structural reproduction. Prebisch argued that the power structure reproduces itself politically: those who appropriate the surplus use part of it to maintain the institutions (legal frameworks, education systems, media) that sustain their structural position. In the notation of Chapters 5 and 8: the powerful invest in W^I (shaping information flows to justify the existing distribution) and in W^P (maintaining production structures that keep their Perron weight high). How concentrated the resulting power is, is measured by N_{eff} (Definition 6.15); the spectral gap does not measure it (Proposition 6.16). How hard the structure is to dislodge is a question about the sensitivity of $\boldsymbol{\pi}$ to changes in W (Lemma 6.19), taken up in Chapters 7 and 13.

Prebisch's argument (1981)	**Formal object**	**This chapter**
Surplus appropriated by power	$\boldsymbol{\pi}(W)$ enters the allocation rule	Def. 6.4; Prop. 7.3
Power is relational, not individual	$\pi_i = \frac{1}{\rho}\sum_j \pi_j W_{ji}$ (recursive)	Eq. (6.6)
Equal appropriation iff balanced dependence	$\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$ iff W doubly stochastic	Prop. 6.8
Power concentrates in hubs	Star: $\pi_h = a/(a+1-s)$	Prop. 6.11
Concentration of power	N_{eff}, $\text{Var}(\boldsymbol{\pi})$; independent of Γ	Def. 6.15; Prop. 6.16
Power $\neq$ wealth	$\boldsymbol{\pi}$ is property of W, not $\boldsymbol{\omega}$	Ex. 6.18

Prebisch was eighty when he wrote this. He had spent three decades building institutions (CEPAL, UNCTAD, ILPES) and two decades watching those institutions' prescriptions — import substitution, regional integration, international commodity agreements — partially succeed and partially fail. *Capitalismo Periférico* was his attempt to explain why: the structural policies he had championed changed W^P (the production network) but did not change the *power structure* that determined who appropriated the gains. The Perron eigenvector shifted, but not enough, because the other networks (W^U, W^I) reproduced the old $\boldsymbol{\pi}$.

The Perron-Frobenius theorem guarantees that π exists and is unique for any strongly connected W. What Prebisch understood at eighty, without the theorem, is that π is the object that determines who gets what — and that changing it requires changing all three networks, not just one.

Further Reading

The Perron-Frobenius theorem: Horn and Johnson (2012), *Matrix Analysis*, Chapter 8 — the definitive treatment. Seneta (2006), *Non-Negative Matrices and Markov Chains* — for the probabilistic perspective. Meyer (2000), *Matrix Analysis and Applied Linear Algebra*, Chapter 8 — accessible.

Centrality measures: Bonacich (1987), "Power and Centrality: A Family of Measures." Ballester et al. (2006), "Who's Who in Networks." Jackson (2008), *Social and Economic Networks*, Chapter 2.

Spectral methods in economics: Golub (2025), "Eigenvalues in Microeconomics." Sargent and Stachurski (2024), *Economic Networks*, Chapters 2–3. Acemoglu et al. (2012a), "Network Origins of Aggregate Fluctuations," for the "influence vector" $(\alpha/N)\mathbf{1}'(I-(1-\alpha)W^P)^{-1}$ — a Katz/Leontief object whose $\alpha \to 0$ limit is the Perron vector of the input-share matrix. Kemeny and Snell (1960), *Finite Markov Chains*, and Meyer (1980), on the fundamental matrix used in Appendix 6C.

Birkhoff's contraction: Birkhoff (1957), "Extensions of Jentzsch's Theorem." Lemmens and Nussbaum (2012), *Nonlinear Perron-Frobenius Theory* — the modern reference.

The structuralist tradition on power: Prebisch (1981), *Capitalismo Periférico: Crisis y Transformación*, México: FCE (the primary source for the case study above). Marini (1973), *Dialéctica de la dependencia* (superexploitation as $\pi_{\text{worker}} \ll 1/N$). Quijano (2000a), "Colonialidad del poder, eurocentrismo y América Latina" (power as a multi-layer matrix).

Chapter 7

The Bargaining Equilibrium

Where Power Meets Prices

> *"El fundamento de la dependencia es la superexplotación del trabajo, es decir, el hecho de que el trabajo se remunera por debajo de su valor."*
>
> ["The foundation of dependence is the superexploitation of labor — that is, the fact that labor is remunerated below its value."]
> — Ruy Mauro Marini, *Dialéctica de la dependencia* (1973)

> *"$x_i^* = f(\boldsymbol{\pi}(W), \boldsymbol{\omega}, p)$. That is the whole chapter."*
> — This book

Chapter 1 derived the competitive equilibrium: every agent maximizes utility at market prices, markets clear, and the allocation is Pareto efficient. Chapter 6 showed that anonymous market interaction corresponds to $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$, with power equal to wealth shares — the First Welfare Theorem holds for any $\mathbf{s}$, so equal power is a distributional property, not a condition for efficiency — and that equal power means a doubly stochastic W (Proposition 6.8). This chapter asks: how is a surplus divided when interaction is not anonymous and power, read from W, is unequal?

The answer is the *bargaining equilibrium*: a generalized Nash bargaining solution whose weights are taken from the weight matrix W through the Perron eigenvector (Assumption 7.A). Power enters the allocation problem. The salteña vendor's compadrazgo bond, the gas executive's regulatory leverage, the girl in Potosí's information network — all of these shape W, which shapes $\boldsymbol{\pi}$, which shapes the allocation.

The main results: (1) with a transferable surplus, the bargaining allocation is equal division plus a power correction, $x_i = \Omega/N + (\pi_i - 1/N)\Phi$ (Proposition 7.3); (2) the bargaining equilibrium exists and is unique when there are gains from cooperation; (3) it is Pareto efficient for every $\boldsymbol{\pi}$, so the cost of unequal power is distributional — exactly $\frac{\kappa N}{2}\Phi^2 \operatorname{Var}(\boldsymbol{\pi})$ in utilitarian terms — and it decomposes exactly into power, externality, and interaction components; and (4) equal power is not the same as Walras.

7.1 From Competitive to Bargaining

7.1.1 Why Not Walras?

The Walrasian equilibrium assumes price-taking: every agent is so small relative to the market that she cannot affect prices, and interacts only with the market aggregate. Interaction is anonymous ($W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$). But in many real economies, interaction is not anonymous:

- A monopsony employer in a company town sets wages, not the market.
- A landlord with the only irrigated field in a valley sets the sharecropping rate.
- A country that supplies a large share of the world's lithium (Chile) affects the world price.
- A platform (Amazon, Uber) sets the terms of trade for millions of small sellers.

In each case, some agent has structural power — a position in W that gives her Perron weight $\pi_i > 1/N$ — and she uses that power to pull the allocation toward herself. The Walrasian model abstracts from this by making interaction anonymous.

The bargaining equilibrium keeps everything the Walrasian model has (preferences, endowments, production) and adds one object: the power vector $\boldsymbol{\pi}(W)$.

7.1.2 A Two-Person Example First

Before the general model, let us see the mechanism in the simplest possible case.

Example 7.1 (The Landlord and the Farmer)**.** *Two agents: a Landlord (L) and a Farmer (F). One good (output). The Landlord owns the land ($\omega_L = 10$); the Farmer owns labor*

($\omega_F = 2$). Together they can produce $Y = 20$ (more than the sum of their outside options, so there are gains from cooperation). Apart, the Landlord's best alternative use of the land yields 10 and the Farmer's best alternative (casual day labor) yields 2. These are outside options, not the production function evaluated with one factor at zero.

*The **surplus** from cooperation is $S = 20 - 10 - 2 = 8$. How is it divided?*

***Competitive answer:** Each factor is paid its marginal product. If the joint production function is $Y = AK^{0.5}L^{0.5}$ with $K = 10$, $L = 2$ and $A = \sqrt{20} \approx 4.47$ (so that $Y = 20$), then $MPK = 0.5 \times 20/10 = 1$ and $MPL = 0.5 \times 20/2 = 5$. The Farmer gets $5 \times 2 = 10$ and the Landlord gets $1 \times 10 = 10$. Equal split. The Farmer's income quintuples; the Landlord's stays the same. This assumes perfect competition in both factor markets.*

***Bargaining answer:** The surplus is divided according to power weights $\boldsymbol{\pi} = (\pi_L, \pi_F)$. The Nash bargaining solution gives each agent their disagreement payoff plus a share of the surplus:*

$$x_L = 10 + \pi_L \times 8 = 10 + 8\pi_L, \qquad x_F = 2 + \pi_F \times 8 = 2 + 8\pi_F. \tag{7.1}$$

If the Landlord has more power ($\pi_L = 0.7$, $\pi_F = 0.3$): $x_L = 15.6$, $x_F = 4.4$. The Farmer's income doubles (not quintuples). The Landlord captures most of the surplus.

If the Farmer has more power ($\pi_L = 0.3$, $\pi_F = 0.7$): $x_L = 12.4$, $x_F = 7.6$. The Farmer does much better.

If power is equal ($\pi_L = \pi_F = 0.5$): $x_L = 14$, $x_F = 6$. This is the equal split of the surplus — each agent gets her disagreement payoff plus half of S — and it is not the competitive answer (10, 10).

Scenario	*Landlord*	*Farmer*	*Surplus share to F*
Autarky	*10.0*	*2.0*	*—*
Competitive (MPL pricing)	*10.0*	*10.0*	*100%*
Bargaining ($\pi_L = 0.7$)	*15.6*	*4.4*	*30%*
Bargaining ($\pi_L = 0.5$)	*14.0*	*6.0*	*50%*
Bargaining ($\pi_L = 0.3$)	*12.4*	*7.6*	*70%*

Intuition: Why the competitive answer is a special case

The competitive equilibrium gives the Farmer 100% of the surplus — because competitive factor markets price labor at its marginal product, which is very high when labor is scarce (only 2 units of labor with 10 units of land). But this requires the Farmer to have an outside option that credibly threatens the Landlord: if the Farmer can walk away and sell her labor elsewhere at the competitive wage, the Landlord must match that wage.

In a competitive market, the "outside option" is the market itself: any buyer or seller can transact with any other buyer or seller. Interaction is anonymous ($W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$), so nobody has hold-up power.

In a real village, the Farmer's outside option may be very limited: the Landlord is the only employer within walking distance, the Farmer has no savings to relocate, and social norms (the compadrazgo bond, the patron-client relationship) create obligations that limit exit. The Farmer's position in W is peripheral: W_{FL} is large (the Farmer depends on the Landlord) but W_{LF} is small (the Landlord can replace the Farmer). This asymmetry gives the Landlord high π_L and allows her to capture most of the surplus.

The competitive benchmark abstracts from this. The bargaining model puts it back.

7.2 The General Bargaining Problem

7.2.1 Setup

N agents, L goods, preferences $U_i(x_i)$ (with externalities from W^U as in Chapter 5), endowments $\boldsymbol{\omega} = (\omega_1, \ldots, \omega_N)$, and a row-stochastic composite weight matrix W on the same N agents, with Perron eigenvector $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_N)'$.

Each agent's **disagreement utility** — what she gets if bargaining fails — is d_i. When the outside option is to consume her endowment,

$$d_i = V_i(\boldsymbol{\omega}) = U_i(\omega_i) + \gamma_U \sum_{j \neq i} W^U_{ij} U_j(\omega_j). \tag{7.2}$$

The **surplus** for agent i from any allocation $\mathbf{x} = (x_1, \ldots, x_N)$ is:

$$S_i(\mathbf{x}) = V_i(\mathbf{x}) - d_i. \tag{7.3}$$

Assumption 7.A (Power as Nash weights). *The bargaining weights are the Perron vector of the composite weight matrix: agent i's weight is $\pi_i = \pi_i(W)$.*

This is a modelling postulate, not a result. It says that the long-run influence structure of Chapter 6 also governs how a surplus is split. Strategic models of bargaining in networks do not, in general, deliver it: in Manea (2011) and Abreu and Manea (2012a,b) equilibrium payoffs depend on the network in ways that are not eigenvector centrality. Appendix 7B discusses what a microfoundation would have to deliver.

Definition 7.2 (The Bargaining Equilibrium)**.** *The **bargaining equilibrium** is the allocation* $\mathbf{x}^*$ *that solves:*

$$\max_{\mathbf{x}\geq 0} \sum_{i=1}^{N} \pi_i \ln S_i(\mathbf{x}) \qquad \textit{subject to} \qquad \sum_{i=1}^{N} x_{i\ell} = \bar{\omega}_\ell, \quad \ell = 1, \ldots, L, \tag{7.4}$$

over the feasible allocations with $S_i(\mathbf{x}) > 0$ *for all* i*, where* $\boldsymbol{\pi} = \boldsymbol{\pi}(W)$ *(Assumption 7.A).*

Intuition: What the bargaining problem maximizes

The objective (7.4) is the **weighted Nash product**: the weighted sum of log-surpluses. Each agent's log-surplus $\ln S_i$ is weighted by her power π_i. An agent with higher power has more "voting weight" in the allocation decision — her surplus matters more.

When $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$ (equal power), this is the standard symmetric Nash bargaining solution. When π_i varies, it is the asymmetric Nash bargaining solution with weights given by the network.

Why logarithms? The Nash axioms (invariance to affine utility transformations, Pareto efficiency, symmetry, independence of irrelevant alternatives) characterize the product $\prod_i S_i^{\pi_i}$ as the unique solution satisfying these properties. Taking logs converts the product to a sum, making the optimization problem concave.

The problem needs *gains from cooperation*: some feasible allocation must give every agent $S_i > 0$. In a one-good exchange economy with endowments as the disagreement point there is none — any reallocation hurts someone — so the only individually rational allocation is $\mathbf{x} = \boldsymbol{\omega}$ and there is nothing to bargain over. The economies below therefore have a joint surplus, as in the landlord–farmer example.

7.2.2 First-Order Conditions

For the case without utility externalities ($\gamma_U = 0$, so $V_i = U_i$ and $S_i = U_i(x_i) - d_i$), the Lagrangian is:

$$\mathcal{L} = \sum_{i=1}^{N} \pi_i \ln\big(U_i(x_i) - d_i\big) - \sum_{\ell=1}^{L} \mu_\ell \left(\sum_i x_{i\ell} - \bar{\omega}_\ell \right). \tag{7.5}$$

The first-order condition for $x_{i\ell}$ is:

$$\frac{\pi_i}{S_i(\mathbf{x}^*)} \cdot \frac{\partial U_i}{\partial x_{i\ell}} = \mu_\ell, \qquad \forall\, i, \ell. \tag{7.6}$$

Intuition: Reading the FOC

The left side is the "power-adjusted marginal utility": agent i's marginal utility of good ℓ, scaled by her power-to-surplus ratio π_i / S_i. At the optimum, this must be equal across all agents for each good.

Compare with the competitive FOC: $\frac{\partial U_i}{\partial x_{i\ell}} / \lambda_i = p_\ell$, where λ_i is agent i's marginal utility of wealth. In both cases marginal rates of substitution are equalized across agents — both allocations are Pareto efficient. What differs is *where on the contract curve* the allocation lands: in the bargaining case the factor π_i / S_i moves it toward the powerful.

An agent with high π_i relative to S_i (powerful but not yet receiving much surplus) gets a higher marginal weight — the solution gives her more. An agent with low π_i gets less, even if her marginal utility is high. This is the mechanism of structural inequality.

7.2.3 The Allocation Rule with Transferable Surplus

To build full intuition, we solve the model explicitly in the transferable-utility (TU) case of the landlord–farmer example: cooperation produces a joint output Ω that can be divided freely, and agent i's outside option is d_i.

Proposition 7.3 (The Allocation Rule)**.** *Let the bargaining be over a transferable output Ω, with surpluses $S_i = x_i - d_i$, disagreement payoffs satisfying $\sum_j d_j < \Omega$, and weights $\boldsymbol{\pi} > 0$ (Assumption 7.A). The unique bargaining equilibrium is*

$$x_i^* = d_i + \pi_i \Big(\Omega - \sum_j d_j \Big). \tag{7.7}$$

*If all disagreement payoffs are equal, $d_i = \bar{d}$, then with the **bargaining surplus** $\Phi :=$*

$\Omega - N\bar{d} > 0$,

$$x_i^* = \frac{\Omega}{N} + \left(\pi_i - \frac{1}{N}\right)\Phi. \tag{7.8}$$

Proof. This is the transferable-utility closed form of the asymmetric Nash solution (Kalai, 1977). The objective $\sum_i \pi_i \ln(x_i - d_i)$ is strictly concave on $\{x_i > d_i\}$ (as $\boldsymbol{\pi} > 0$) and tends to $-\infty$ at its boundary. The feasible set $\{\sum_i x_i = \Omega,\ x_i > d_i\}$ is nonempty because $\sum_j d_j < \Omega$, and bounded, since $d_i < x_i < \Omega - \sum_{j\neq i} d_j$. A maximizer therefore exists, lies in the interior, and is unique. With a linear constraint and a concave objective, the Lagrange conditions are necessary and sufficient: $\pi_i/(x_i - d_i) = \mu$, i.e. $x_i - d_i = \pi_i/\mu$. Summing and using $\sum_i \pi_i = 1$ gives $1/\mu = \Omega - \sum_j d_j$, which is (7.7). With $d_i = \bar{d}$, $x_i^* = \bar{d} + \pi_i\Phi = (\Omega - \Phi)/N + \pi_i\Phi$, which is (7.8). □

Example 7.4 (The Allocation Rule at Work)**.** *N agents, transferable joint output Ω, equal outside options $\bar{d}$, power $\boldsymbol{\pi}$. Each agent receives the equal-division amount Ω/N plus a correction $(\pi_i - 1/N)\Phi$: the deviation of her power from equal, times the surplus at stake. Powerful agents ($\pi_i > 1/N$) get more than equal division; powerless agents ($\pi_i < 1/N$) get less; when $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$ the correction vanishes. The landlord–farmer example is the case $N = 2$, $\Omega = 20$, $\boldsymbol{d} = (10, 2)$, where (7.7) gives $x_L = 10 + 8\pi_L$.*

To evaluate allocations we use a concave utility of consumption, $U(x) = x - \frac{\kappa}{2}x^2$, calibrated so that $\kappa x_i < 1$ for every allocation considered (agents are below satiation). Because $\sum_i (x_i^* - \Omega/N) = 0$, the linear terms cancel and the utilitarian loss relative to equal division is exactly

$$L_{power} := \sum_i \left[U\left(\tfrac{\Omega}{N}\right) - U(x_i^*)\right] = \frac{\kappa N}{2}\,\Phi^2\,\mathrm{Var}(\boldsymbol{\pi}). \tag{7.9}$$

Key Structural Insight

Equation (7.8) is the allocation rule of this book. It says: *the allocation is equal division plus a power correction.* The power correction is proportional to $\pi_i - 1/N$: how much agent i's structural power deviates from the egalitarian benchmark.

Its size is governed by two things: the surplus at stake, Φ, and the concentration of power, measured by $\mathrm{Var}(\boldsymbol{\pi})$ or N_{eff} (Definition 6.15). It is *not* governed by the spectral gap: by Proposition 6.16, any concentration of $\boldsymbol{\pi}$ is compatible with any Γ.

The entire distributional impact of social structure, in this model, is captured by the vector $(\pi_i - 1/N)_{i=1}^N$.

7.3 Properties of the Bargaining Equilibrium

Theorem 7.5 (Existence and Uniqueness)**.** *Suppose: (i) each U_i is continuous, increasing and strictly concave on a compact convex consumption set; (ii) $W^U \geq 0$ and $\gamma_U \geq 0$, so each V_i is concave in $\mathbf{x}$; (iii) $\boldsymbol{\pi} > 0$; and (iv)* ***gains from cooperation****: some feasible allocation gives $S_i(\mathbf{x}) > 0$ for every i. Then the bargaining equilibrium (7.4) exists and is unique.*

Proof. Let X^+ be the set of feasible allocations with $S_i > 0$ for all i; it is nonempty by (iv) and convex because each S_i is concave. On X^+ the objective $f(\mathbf{x}) = \sum_i \pi_i \ln S_i(\mathbf{x})$ is concave, and strictly concave: along any direction $\mathbf{h} \neq 0$ some $h_i \neq 0$, and $\ln S_i$ is strictly concave along $\mathbf{h}$ because U_i is strictly concave in x_i and ln is increasing and concave. Fix $\mathbf{x}^0 \in X^+$ and let F be the (compact) set of feasible allocations. Each S_j is continuous, hence bounded above on F. Let $\mathbf{x}^n$ be a sequence in the upper level set $\{\mathbf{x} \in X^+ : f(\mathbf{x}) \geq f(\mathbf{x}^0)\}$ converging to $\bar{\mathbf{x}} \in F$. If $S_i(\bar{\mathbf{x}}) = 0$ for some i, then $\pi_i \ln S_i(\mathbf{x}^n) \to -\infty$ while the other terms stay bounded above, so $f(\mathbf{x}^n) \to -\infty$, a contradiction. Hence $\bar{\mathbf{x}} \in X^+$ and, by continuity, $f(\bar{\mathbf{x}}) \geq f(\mathbf{x}^0)$. The level set is therefore closed in the compact F, hence compact, so f attains its maximum on it (and so on X^+); strict concavity makes the maximizer unique. Without (ii) — for instance with envy, $W^U_{ij} < 0$ — V_i need not be concave and uniqueness can fail. □

Proposition 7.6 (Equal Power Is Not Walras)**.** *If $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$ and the disagreement payoffs are equal, the TU bargaining equilibrium is equal division of the joint output, $x_i^* = \Omega/N$. The symmetric Nash bargaining solution does* not*, in general, coincide with the Walrasian equilibrium.*

Proof. The first statement is (7.8) with $\pi_i = 1/N$. For the second, take two agents with $U = \frac{1}{2}\ln x + \frac{1}{2}\ln y$ and endowments $(0.9, 0.3)$ and $(0.1, 0.7)$. The Walrasian equilibrium has price ratio one (each agent spends half her wealth on each good, and with $p_x = p_y$ demands sum to the endowment $(1, 1)$) and gives agent A the bundle $(0.600, 0.600)$. Equal Cobb–Douglas preferences and aggregate endowment $(1, 1)$ make the contract curve the diagonal $x_A = y_A = a$, along which $U_A = \ln a$ and $U_B = \ln(1 - a)$. The symmetric Nash solution is Pareto efficient, so it lies on this curve and maximizes $\ln(\ln a - d_A) + \ln(\ln(1 - a) - d_B)$ with $d_A = \frac{1}{2}\ln 0.27$ and $d_B = \frac{1}{2}\ln 0.07$; the maximizer is $a = 0.631 \neq 0.600$ (`code/ch07.py`). □

Intuition: What the competitive benchmark is

The Walrasian equilibrium is not "bargaining with equal power". It is anonymous interaction, $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$ with $\boldsymbol{\pi} = \mathbf{s}$ the wealth shares (Proposition 6.13), and it is Pareto efficient for every $\mathbf{s}$. The bargaining equilibrium is also Pareto efficient. The two are different ways of choosing a point on the contract curve; the structural model asks which point the network selects.

Proposition 7.7 (Direction of the Distributional Loss)**.** *Under the allocation rule* (7.8)*:*

1. *The bargaining allocation is Pareto efficient for every $\boldsymbol{\pi}$ (it maximizes an increasing function of the surpluses over the feasible set).*

2. *Agents with $\pi_i > 1/N$ receive more than equal division, and agents with $\pi_i < 1/N$ receive less.*

3. *The utilitarian loss relative to equal division is $L_{power} = \frac{\kappa N}{2}\Phi^2 \operatorname{Var}(\boldsymbol{\pi})$, which is increasing in the concentration of power:* $\operatorname{Var}(\boldsymbol{\pi}) = \frac{1}{N}\left(\frac{1}{N_{eff}} - \frac{1}{N}\right)$.

Proof. (1) Any allocation that raised some S_i without lowering another would raise the objective. (2) is immediate from (7.8) since $\Phi > 0$. (3) is (7.9), together with the identity (6.12). □

Intuition: A distributional cost, not an inefficiency

The bargaining equilibrium gives the powerful agent more than an equal share. Her marginal utility is then lower than the powerless agent's, so a utilitarian planner with concave U would transfer a unit from the powerful to the powerless. That transfer is not a Pareto improvement — the powerful agent loses — so the loss is *distributional*: it measures inequality through the lens of a utilitarian social welfare function.

This is the structural mechanism of inequality: not that the powerful agent is greedier or less deserving, but that the network gives her a position from which she can claim a larger share of the surplus. The source of the inequality is W, not preferences.

7.4 The Welfare Loss Decomposition

7.4.1 Total Welfare Loss

Let social welfare count interdependent utilities,

$$\mathcal{S}(\mathbf{x}) = \sum_i V_i(\mathbf{x}) = \sum_j (1 + \gamma_U c_j)\, U(x_j),$$

where $c_j = \sum_{i \neq j} W^U_{ij}$ is how much others care about agent j. Let $\mathbf{x}^B$ be the TU bargaining allocation (7.8) (bargaining is over transferable output; W^U enters only the evaluation), $\mathbf{x}^E = \frac{\Omega}{N}\mathbf{1}$ equal division, and $\mathbf{x}^{FB}$ the maximizer of $\mathcal{S}$ subject to $\sum_j x_j = \Omega$. The total welfare loss is $L = \mathcal{S}(\mathbf{x}^{FB}) - \mathcal{S}(\mathbf{x}^B) \geq 0$.

Theorem 7.8 (Welfare Loss Decomposition)**.** *With $U(x) = x - \frac{\kappa}{2}x^2$ and $\kappa x_j < 1$ throughout,*

$$L = L_{power} + L_{externality} + L_{interaction}, \tag{7.10}$$

exactly, where

1. $L_{power} = \frac{\kappa N}{2}\Phi^2 \operatorname{Var}(\boldsymbol{\pi})$*: the loss from unequal power, as in* (7.9)*;*
2. $L_{externality} = \mathcal{S}(\mathbf{x}^{FB}) - \mathcal{S}(\mathbf{x}^E) \geq 0$*: the gain the planner forgoes by not tilting consumption toward the agents others care about; for $\gamma_U > 0$ it is zero if and only if all c_j are equal (for $\gamma_U = 0$ it is always zero);*
3. *the interaction term*

$$\begin{aligned} L_{interaction} &= \gamma_U \sum_j c_j \left[U(\tfrac{\Omega}{N}) - U(x^B_j)\right] \\ &= -\gamma_U N\, U'(\tfrac{\Omega}{N})\, \Phi\, \operatorname{Cov}(c_j, \pi_j) + \gamma_U \tfrac{\kappa}{2}\Phi^2 \sum_j c_j \left(\pi_j - \tfrac{1}{N}\right)^2, \end{aligned}$$

with $\operatorname{Cov}(c_j, \pi_j) = \frac{1}{N}\sum_j (c_j - \bar{c})(\pi_j - \frac{1}{N})$ the covariance across agents (the same $\frac{1}{N}$ convention as $\operatorname{Var}(\boldsymbol{\pi})$).

Proof. Write $L = [\mathcal{S}(\mathbf{x}^{FB}) - \mathcal{S}(\mathbf{x}^E)] + [\mathcal{S}(\mathbf{x}^E) - \mathcal{S}(\mathbf{x}^B)]$. The first bracket is $L_{\text{externality}}$; it is nonnegative by optimality of $\mathbf{x}^{FB}$. Because $\mathcal{S}$ is strictly concave ($\kappa > 0$, $1 + \gamma_U c_j > 0$), $\mathbf{x}^{FB}$ is its unique maximizer on $\{\sum_j x_j = \Omega\}$, characterized by $(1 + \gamma_U c_j)(1 - \kappa x_j) = \mu$ for all j. So $L_{\text{externality}} = 0$ iff $\mathbf{x}^E = \mathbf{x}^{FB}$, iff these conditions hold at equal x_j with $1 - \kappa x_j > 0$, i.e. iff all $\gamma_U c_j$ are equal. The second bracket is $\sum_j [U(\Omega/N) - U(x^B_j)] + \gamma_U \sum_j c_j [U(\Omega/N) - U(x^B_j)] = L_{\text{power}} + L_{\text{interaction}}$. Finally, with $\delta_j = x^B_j - \Omega/N = (\pi_j - 1/N)\Phi$, $U(\Omega/N) - U(x^B_j) = -U'(\Omega/N)\delta_j +$

$\frac{\kappa}{2}\delta_j^2$ exactly. Summing with weights 1 gives L_{power} (the linear terms cancel because $\sum_j \delta_j = 0$), and summing with weights $\gamma_U c_j$ gives the second line of $L_{\text{interaction}}$, using $\sum_j c_j(\pi_j - 1/N) = \sum_j (c_j - \bar{c})(\pi_j - 1/N) = N\,\text{Cov}(c_j, \pi_j)$. □

Intuition: Three sources of structural welfare loss

The decomposition says: measured against a social welfare function that counts interdependent utilities, the cost of structure comes from three channels.

Power (L_{power}): unequal bargaining tilts the allocation. The cost is proportional to $\text{Var}(\boldsymbol{\pi})$. When $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$, this term vanishes.

Externalities ($L_{\text{externality}}$): when some agents' wellbeing matters more to others (unequal c_j), equal division is no longer the welfare optimum. This term exists even with equal power.

Interaction ($L_{\text{interaction}}$): its leading term has the sign of $-\text{Cov}(c_j, \pi_j)$ (since $U'(\Omega/N) > 0$). When the powerful are also the agents others care about — a tight elite whose members care about each other — the interaction term is *negative*: a welfare function that counts the elite's mutual affection makes their capture look *less* costly. That is a reason to evaluate structural outcomes with private utilities U, not with V (see Chapter 2), rather than a sign that elite cohesion is benign.

Prebisch Connection: The welfare cost of center-periphery structure

Applied to the global economy, the three components are:

L_{power}: the distributional cost of unequal division of the gains from trade. Center countries have higher π in the global W^P, so they capture more of the gains. The cost is proportional to $\text{Var}(\pi_i^{\text{global}})$.

$L_{\text{externality}}$: the cost of unpriced cross-country spillovers. Technology spillovers from center to periphery are not priced in global markets; environmental externalities from periphery resource extraction are not priced either.

$L_{\text{interaction}}$: whether the center's cohesion ($W^U_{\text{center, center}} \gg W^U_{\text{center, periphery}}$) raises or lowers the measured loss depends on the sign of $\text{Cov}(c_j, \pi_j)$ — and, as above, on whether the evaluation counts that cohesion as welfare.

The phrase "unequal exchange" is Arghiri Emmanuel's (1969). Prebisch's own mechanism was the secular deterioration of the periphery's terms of trade. In this framework both become statements about $\boldsymbol{\pi}$ and Φ that can be computed.

7.5 Computing the Bargaining Equilibrium

7.5.1 The Algorithm

For general preferences, the bargaining equilibrium is computed as follows:

Step 1: Compute the power vector $\boldsymbol{\pi}$ from the row-stochastic W (Chapter 6, power iteration).

Step 2: Compute disagreement utilities d_i for each agent, and check that some feasible allocation gives every agent $S_i > 0$.

Step 3: Solve (7.4) by interior-point methods or sequential quadratic programming. The objective is concave, so standard convex optimization tools apply.

In the TU case, Step 3 is the closed form (7.7).

7.5.2 Comparative Statics: What Changes the Allocation?

Proposition 7.9 (Comparative Statics)**.** *Under the allocation rule* (7.7)*:*

1. ***Changing outside options*** *(for instance through endowments):* $\partial x_i^* / \partial d_i = 1 - \pi_i > 0$ *and* $\partial x_i^* / \partial d_j = -\pi_i$ *for* $j \neq i$*. Raising agent* i*'s outside option strengthens her bargaining position even if* $\boldsymbol{\pi}$ *is unchanged.*

2. ***Changing the network*** *(*$W \to W + tE$*, with* W *irreducible and* E *as in Lemma 6.19):* $dx_i^*/dt = (\Omega - \sum_j d_j)\, d\pi_i/dt$ *(which is* $\Phi\, d\pi_i/dt$ *with equal outside options), with* $d\boldsymbol{\pi}'/dt = \boldsymbol{\pi}' E Z$ *at* $t = 0$*. Building a new production link can change who gets what without any transfer.*

3. ***Changing preferences or productivity*** *(e.g. through the formation mechanism of Chapter 10) acts through* Ω *and* $\boldsymbol{d}$*:* $\partial x_i^* / \partial \Omega = \pi_i$*.*

Proof. Differentiate (7.7), which is linear in $(\boldsymbol{d}, \Omega)$ for fixed $\boldsymbol{\pi}$ and linear in $\boldsymbol{\pi}$ for fixed $(\boldsymbol{d}, \Omega)$; $1 - \pi_i > 0$ because $\boldsymbol{\pi} > 0$ and $N \geq 2$. Item 2 then uses (6.18). □

Key Structural Insight

There are three levers for changing the allocation: endowments and outside options (transfers), networks (structural policy), and preferences (formation). The benchmark model emphasizes the first lever. Structural economics adds the other two.

This is why structural policy is different from redistribution. A transfer changes $\boldsymbol{d}$: it moves resources from one person to another. A structural policy changes W: it changes the network of dependencies, which changes power, which changes the allocation. A formation intervention changes W^I: it changes the information flows that shape preferences — and, because W^I is a layer of the composite W, it changes $\boldsymbol{\pi}$ as well.

A gas-to-fertilizer plant (Chapter 3) is a structural policy: it raises the technical coefficient a_{TM} (Traditional's purchases from the modern sector) from 0.02 to 0.15 in the A of Chapter 3's stylized dual economy (Example 3.6), which changes $W^P = D^{-1}A$, which changes $\boldsymbol{\pi}$, which changes the bargaining allocation. The Bono Juancito Pinto (conditional cash transfer) is a transfer: holding W fixed, it changes $\boldsymbol{d}$ but not $\boldsymbol{\pi}$. The phone in Luisa's hand (Chapter 5) is a formation intervention: it changes W^I, which changes beliefs, which eventually changes θ and therefore preferences.

All three change the allocation. The second and third change the power structure directly; a transfer changes it only if W responds to the allocation.

7.6 Two Worked Economies

Both economies have $N = 4$, a transferable joint output $\Omega = 20$, equal outside options $\bar{d} = 3$ (so $\Phi = 20 - 12 = 8$), and utility $U(x) = x - 0.05x^2$ ($\kappa = 0.1$, so $\kappa x_i < 1$ for every allocation below). Both production share matrices are row-stochastic. The numbers are [ILLUSTRATIVE] and reproduced by `code/ch07.py`.

7.6.1 Economy A: The Connected Village

Farmer (1), Miller (2), Baker (3), Consumer (4). Production network (strong internal linkages):

$$W^P = \begin{pmatrix} 0.6 & 0.2 & 0.1 & 0.1 \\ 0.2 & 0.5 & 0.2 & 0.1 \\ 0.1 & 0.3 & 0.4 & 0.2 \\ 0.1 & 0.1 & 0.2 & 0.6 \end{pmatrix}. \tag{7.11}$$

Left Perron eigenvector: $\boldsymbol{\pi} \approx (0.256, 0.282, 0.218, 0.244)$, $N_{\text{eff}} = 3.97$. Power is *nearly equal*: the column sums $(1.0, 1.1, 0.9, 1.0)$ are close to one, so W^P is close to doubly stochastic. $\Gamma = 0.47$.

Bargaining allocation (from (7.8)):

	Farmer	Miller	Baker	Consumer
Outside option d_i	3.00	3.00	3.00	3.00
Equal division (utilitarian optimum)	5.00	5.00	5.00	5.00
Bargaining	5.05	5.26	4.74	4.95

The Miller gets slightly more (most depended upon: column sum 1.1), the Baker slightly less (column sum 0.9). But the deviations are small. Distributional loss: $L_{\text{power}} = 0.0068$.

7.6.2 Economy B: The Gas Enclave

Gas Executive (1), Government Minister (2), Market Vendor (3), Quinoa Farmer (4). Production network: two blocks, with *asymmetric* dependence between them. Each Gas/Government row places total weight $\varepsilon_1 = 0.02$ on the traditional block; each Vendor/Farmer row places $\varepsilon_2 = 0.06$ on the enclave (fuel, transport, public services):

$$W^P = \begin{pmatrix} 0.48 & 0.50 & 0.01 & 0.01 \\ 0.40 & 0.58 & 0.01 & 0.01 \\ 0.03 & 0.03 & 0.64 & 0.30 \\ 0.03 & 0.03 & 0.30 & 0.64 \end{pmatrix}. \tag{7.12}$$

Left Perron eigenvector: $\boldsymbol{\pi} \approx (0.334, 0.416, 0.125, 0.125)$, $N_{\text{eff}} = 3.17$. Power is *unequal*: the Gas/Government block holds $0.75 = \varepsilon_2/(\varepsilon_1 + \varepsilon_2)$ of structural power (Proposition 6.12) despite being half the population. $\Gamma = 0.08$: the blocks mix slowly.

Bargaining allocation:

	Gas Exec	Minister	Vendor	Farmer
Outside option d_i	3.00	3.00	3.00	3.00
Equal division (utilitarian optimum)	5.00	5.00	5.00	5.00
Bargaining	5.67	6.33	4.00	4.00

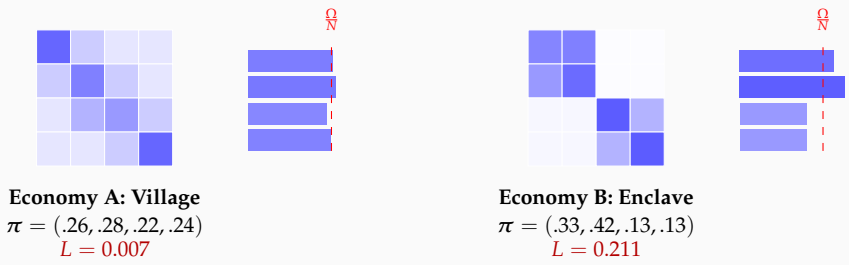
The Gas/Government block receives 12.0 units, 20% more than equal division; the Vendor/Farmer block receives 8.0, 20% less. From identical outside options, the structure produces inequality. Distributional loss: $L_{\text{power}} = 0.211$ — about 31 times the connected village.

Key Structural Insight

The two economies have *identical outside options, identical preferences, identical number of agents*. They differ only in W^P — the structure of production linkages. The connected village (Economy A) is nearly equal; the enclave economy (Economy B) has a large distributional loss and significant inequality. What produces the inequality is the *asymmetry* of the dependence between the blocks, not the block structure as such. If the Vendor and the Farmer also placed only 0.02 on the enclave (same within-block pattern otherwise), the two blocks would hold equal power, $\boldsymbol{\pi} \approx (0.223, 0.277, 0.250, 0.250)$, even though Γ would be smaller still (0.04). The matrix is the structure; the direction of dependence is the inequality.

Economy B is a stylized enclave with Bolivian names, not Bolivia's measured structure, and on one point the data run the other way. In INE's 2019 supply-use table the hydrocarbons sector is *peripheral* in the domestic W^P: its Perron weight is $\pi = 0.039$, below the uniform share $1/11 \approx 0.091$, because other domestic sectors spend only small shares of their input costs on it; 70% of its output is exported. Bolivia's production power Gini, 0.435, is lower than Korea's (0.658) or Germany's (0.611) (INE supply-use tables, base 2017, Instituto Nacional de Estadística (INE), Bolivia, 2026; OECD ICIO 2023, OECD, 2023; author's calculations, `data/compute/`). In a production network a large Perron weight marks a hub that many sectors buy from — in industrial economies, manufacturing and business services — so a concentrated $\boldsymbol{\pi}(W^P)$ is not by itself a sign of oppression. Reading $\boldsymbol{\pi}$ as bargaining power is Assumption 7.A, and it is most natural for the utility and information layers of the composite. The real enclave's rents reach the rest of the economy through the fiscal system (the direct hydrocarbons tax, IDH) rather than through production linkages.

Visualization: Two economies, two heatmaps, two allocations



Economy A: Village
$\pi = (.26, .28, .22, .24)$
$L = 0.007$

Economy B: Enclave
$\pi = (.33, .42, .13, .13)$
$L = 0.211$

Left pair: connected village — substantial off-diagonal weights, nearly flat allocation bars, distributional loss 0.007. Right pair: gas enclave — near-white cross-block cells (weights 0.01–0.03; the lower-left cells, the traditional block's dependence on the enclave, are three times the upper-right ones), allocation bars split into two tiers, loss 0.211 (about 31 times larger). Same outside options, same preferences. Only W differs. The dashed red line marks equal division Ω/N.

7.7 The Escape Problem

Can the powerless agents change the structure? The escape problem asks: what does it cost to move from the current W to a W' with more equal power?

Definition 7.10 (Escape Threshold)**.** *The **escape threshold** ε^* is the minimum row-stochastic perturbation to W needed to make every powerless agent strictly better off in the bargaining equilibrium:*

$$\varepsilon^* = \min_{\Delta W} \|\Delta W\|_F \quad s.t. \quad x_i^B(W + \Delta W) > x_i^B(W) \quad \forall\, i \in \mathcal{P}, \tag{7.13}$$

where $\mathcal{P}$ is the set of agents with below-average power ($\pi_i < 1/N$).

Proposition 7.11 (Escape Cost and Integration)**.** *Let W be irreducible and row-stochastic, with fundamental matrix $Z = (I - W + \mathbf{1}\boldsymbol{\pi}')^{-1}$. To first order in t, a perturbation tE (with E as in Lemma 6.19, so $E\mathbf{1} = 0$) that raises the total power of the group $\mathcal{P}$ by $\Delta > 0$ must satisfy*

$$t\,\|E\|_\infty \;\geq\; \frac{\Delta}{\|Z\|_\infty}, \qquad \|Z\|_\infty \geq \max_{k \geq 2} \frac{1}{|1 - \lambda_k|}. \tag{7.14}$$

The lower bound on the cost of escape is therefore smallest *when some λ_k is close to $+1$ — weakly coupled, fragmented networks — and of order Δ in well-mixed ones. In the two-block case of Proposition 6.12 the block masses are exactly $\varepsilon_2/(\varepsilon_1+\varepsilon_2)$ and $\varepsilon_1/(\varepsilon_1+\varepsilon_2)$, whose sensitivity to ε_1 is $\varepsilon_2/(\varepsilon_1+\varepsilon_2)^2$: it grows without bound as the coupling weakens.*

Proof. By Lemma 6.19, $d\boldsymbol{\pi}' = t\,\boldsymbol{\pi}' E Z$, so $\Delta \leq \|d\boldsymbol{\pi}\|_1 \leq t\,\|\boldsymbol{\pi}' E\|_1 \|Z\|_\infty \leq t\,\|E\|_\infty \|Z\|_\infty$, using $\|\mathbf{x}' A\|_1 \leq \|\mathbf{x}\|_1 \|A\|_\infty$ and $\|\boldsymbol{\pi}\|_1 = 1$. Any induced matrix norm bounds the

spectral radius, and Z has eigenvalues 1 and $1/(1-\lambda_k)$. The two-block formula is (6.9). □

In the weak-block example of `code/ch07.py` ($N = 4$, symmetric coupling ε), $\|Z\|_\infty = 2.5$, 10 and 50 for $\varepsilon = 0.2$, 0.05 and 0.01 ($\Gamma = 0.4$, 0.1, 0.02). In a star (Proposition 6.11) the hub's power $a/(a+1-s)$ responds smoothly to the spokes' dependence a: for $N = 5$, $a = 0.9$, $s = 0.5$, raising every spoke's power by 50% (from 0.089 to 0.134) requires lowering a to 0.43.

Intuition: Where the structural trap really is

To change the allocation, you must change $\boldsymbol{\pi}$. To change $\boldsymbol{\pi}$, you must change W. Mathematically, the Perron vector is *most* movable precisely in fragmented, weakly coupled networks: a few new cross-block links shift block power a great deal. So the difficulty of escape is not that $\boldsymbol{\pi}$ is spectrally rigid. The **structural trap** is political-economic: changing W requires resources (building links, forming coalitions, creating new production relationships), and those resources are allocated by the current W, which gives the most to the agents who benefit from the current structure — and who control the cheap levers. The trap is self-reinforcing, which is why structural change often requires an external shock (a new technology, a revolution, a commodity price collapse) rather than endogenous reform.

The escape problem connects to Chapter 14 (coalition formation) and Chapter 15 (operator design), where the cost of changing W is modelled explicitly.

7.8 Summary

1. The **bargaining equilibrium** solves $\max \sum_i \pi_i \ln S_i$ subject to feasibility, with weights $\boldsymbol{\pi}(W)$ by assumption (Assumption 7.A). It exists and is unique when there are gains from cooperation, and it is Pareto efficient for every $\boldsymbol{\pi}$; with equal power it is *not* in general the Walrasian equilibrium.

2. With a transferable surplus the **allocation rule** is $x_i = \Omega/N + (\pi_i - 1/N)\Phi$, with Φ the surplus over outside options (Proposition 7.3). Powerful agents consume more; powerless agents consume less.

3. The cost of unequal power is **distributional**: $L_{\text{power}} = \frac{\kappa N}{2}\Phi^2 \operatorname{Var}(\boldsymbol{\pi})$ exactly, and the total loss decomposes exactly into power, externality and interaction terms (Theorem 7.8).

4. Two economies with identical outside options and preferences but different W produce different allocations and different inequality; what matters is the direction of dependence between blocks, not block structure or Γ as such.

5. Three policy levers: transfers (change $\boldsymbol{d}$), structural policy (change W), formation intervention (change W^I, which also moves $\boldsymbol{\pi}$). The Perron vector is *most* sensitive to changes in W in weakly coupled networks (Proposition 7.11); the structural trap is political-economic, not spectral.

7.9 Exercises

7.1 Landlord-Farmer revisited. In Example 7.1, suppose the production function is $Y = K^{0.3}L^{0.7}$ (more labor-intensive).

(a) Recompute the competitive allocation (adjust A so that $Y = 20$). Who benefits from the higher labor share?

(b) Compute the bargaining allocation for $\boldsymbol{\pi} = (0.7, 0.3)$. Compare.

(c) Show that no $\boldsymbol{\pi}$ with $\pi_F \in [0, 1]$ reproduces the competitive allocation, and explain why: compare the Landlord's competitive payment with his outside option $d_L = 10$. (In the original example, which $\boldsymbol{\pi}$ reproduces the competitive allocation?)

7.2 The FOC in practice. Three agents jointly produce a transferable output $\Omega = 3$; their outside options are $\boldsymbol{d} = (0.6, 0.4, 0.2)$ and $\boldsymbol{\pi} = (0.5, 0.3, 0.2)$. Consumption is evaluated with $U(x) = x - \frac{\kappa}{2}x^2$, $\kappa = 0.25$.

(a) Explain why, if instead the agents simply held one good with endowments $(1.5, 1.0, 0.5)$ and those endowments were the disagreement point, there would be nothing to bargain over.

(b) Write the FOC system (7.6) explicitly for $S_i = x_i - d_i$.

(c) Solve numerically with `scipy.optimize`; check the result against (7.7) and verify $\kappa x_i < 1$.

(d) Compare with the equal-power case $\boldsymbol{\pi} = (1/3, 1/3, 1/3)$. Who gains, who loses?

7.3 Welfare decomposition. Using the two worked economies from Section 7.6:

(a) Compute L_{power} for both economies. Verify that Economy B is much larger.

(b) Now add utility externalities: $\gamma_U = 0.3$ with W^U equal to W^P with its diagonal set to zero (your production partners are also your friends). Compute the three terms of Theorem 7.8. Which component is largest?

(c) Is the interaction term positive or negative? Relate its sign to $\mathrm{Cov}(c_j, \pi_j)$ and interpret.

7.4 Structural policy vs. transfer. Economy B from Section 7.6 (the enclave). Two policies:

(a) **Transfer**: Take 1 unit of outside option from the Gas Executive and give it to the Farmer ($\boldsymbol{d}' = (2, 3, 3, 4)$). Compute the new bargaining allocation (7.7) and welfare.

(b) **Structural**: Set every cross-block entry of W^P to 0.05, taking the difference from the diagonal so rows still sum to one. Compute new $\boldsymbol{\pi}$, Γ and the new bargaining allocation. Compute welfare.

(c) Which policy produces more welfare gain? Which reduces inequality more? Which changes the power structure?

7.5 The escape threshold. For a star network with $N = 5$, hub weight $W_{ih} = a = 0.9$ for all spokes (each spoke keeps 0.1), and a hub that keeps $s = 0.5$ and spreads the rest equally (Proposition 6.11):

(a) Compute $\boldsymbol{\pi}$ and Γ.

(b) How much must W_{ih} decrease (i.e., how much must the spokes' dependence on the hub fall) for the most peripheral agent's power to rise by 50%?

(c) Relate to: a worker leaving a monopsony employer, a country diversifying exports away from a single commodity, a media consumer switching from state TV to independent sources.

7.6 Bolivia: the full model. Construct a 4×4 economy with Gas, Government, Urban Informal, Rural Traditional sectors. Assign plausible row-stochastic W^P, W^U, W^I on these four nodes, outside options, a joint surplus, and preferences.

(a) Compute the composite W and the Perron eigenvector.

(b) Compute the bargaining equilibrium.

(c) Simulate three policies: (i) gas revenue transfer to Rural Traditional, (ii) building a gas-to-fertilizer plant (structural), (iii) expanding rural

internet (formation via W^I). Which has the largest effect on the Rural Traditional sector's consumption?

7.7 Data: labor shares as bargaining outcomes. Download the labor share of income (compensation of employees over GDP) for Bolivia, Chile and Mexico from CEPALSTAT (ECLAC).

(a) Treat a two-agent economy (labor, capital) with the TU rule (7.7). State the outside options you assume and back out the implied π_{labor} for each country.

(b) How sensitive is the implied π_{labor} to the assumed outside options? What does this say about identifying $\boldsymbol{\pi}$ from distributional data alone?

Mathematical Appendices

Appendix 7A: The Allocation Rule and the Exact Power Loss

Why a one-good exchange economy cannot be used. Take one good, $U_i(x_i) = x_i - \frac{\kappa}{2}x_i^2$ below satiation, endowments $\boldsymbol{\omega}$ with $\sum_i \omega_i = \Omega$, and disagreement point $d_i = U_i(\omega_i)$. Since U_i is increasing, $S_i > 0$ requires $x_i > \omega_i$ for every i, which contradicts $\sum_i x_i = \Omega = \sum_i \omega_i$. The set of individually rational allocations is $\{\boldsymbol{\omega}\}$, every $S_i = 0$, and the Nash product is identically $-\infty$ off the endowment. Any "solution" obtained by expanding S_i around $\boldsymbol{\omega}$ (for instance with a negative multiplier) puts some $S_i < 0$ inside a logarithm. Bargaining needs a surplus to divide.

The TU problem. Cooperation produces a transferable output Ω; agent i's outside option is d_i, with $\sum_j d_j < \Omega$:

$$\max_{\mathbf{x}} \sum_{i=1}^{N} \pi_i \ln(x_i - d_i) \quad \text{s.t.} \quad \sum_i x_i = \Omega,\ x_i > d_i. \tag{7.15}$$

The first-order conditions $\pi_i/(x_i - d_i) = \mu$ and feasibility give $x_i = d_i + \pi_i(\Omega - \sum_j d_j)$; see the proof of Proposition 7.3. With $d_i = \bar{d}$ and $\Phi = \Omega - N\bar{d}$, $x_i = \Omega/N + (\pi_i - 1/N)\Phi$.

The exact power loss. Let $\delta_i = x_i - \Omega/N = (\pi_i - 1/N)\Phi$, so $\sum_i \delta_i = 0$. For $U(x) = x - \frac{\kappa}{2}x^2$, $U(\Omega/N) - U(\Omega/N + \delta_i) = -U'(\Omega/N)\delta_i + \frac{\kappa}{2}\delta_i^2$ exactly. Summing, the linear terms cancel:

$$L_{\text{power}} = \frac{\kappa}{2}\sum_i \delta_i^2 = \frac{\kappa}{2}\Phi^2 \sum_i \Big(\pi_i - \frac{1}{N}\Big)^2 = \frac{\kappa N}{2}\Phi^2 \operatorname{Var}(\boldsymbol{\pi}). \tag{7.16}$$

The calibration $\kappa x_i < 1$ keeps every agent below satiation, so that U is increasing on all allocations considered; Φ is the surplus over outside options, not $1/\kappa$.

Appendix 7B: The Nash Axioms and Network Bargaining

The Nash bargaining solution. Nash (1950) showed that four axioms uniquely characterize a bargaining solution:

1. **Pareto efficiency**: no Pareto improvement is possible.

2. **Symmetry**: if agents are identical, they get the same payoff.

3. **Invariance to affine transformations**: the solution is independent of the units in which utility is measured.

4. **Independence of irrelevant alternatives**: expanding the feasible set without changing the optimal point does not change the solution.

The unique solution satisfying (1)–(4) is $\mathbf{x}^* = \arg\max \prod_i (U_i(x_i) - d_i)$: the product of surpluses.

Asymmetric Nash bargaining. Kalai (1977) showed that replacing (2) with asymmetric weights $\boldsymbol{\pi}$ gives $\mathbf{x}^* = \arg\max \prod_i (U_i(x_i) - d_i)^{\pi_i}$: the weighted product. Taking logs: $\arg\max \sum_i \pi_i \ln(U_i(x_i) - d_i)$. This is equation (7.4).

Where do the weights come from? In Nash's original formulation, the weights are arbitrary. In Rubinstein's (1982) alternating-offers model, they come from relative patience (discount rates). In this book, they are set equal to the Perron eigenvector, $\pi_i = \pi_i(W)$ — Assumption 7.A. That is a modelling postulate: it ties the weights to the structure, but it is not derived from a bargaining game. A microfoundation would need a protocol in which an agent's bargaining strength is proportional to her long-run influence in W — for instance, recognition probabilities equal to the stationary frequency with which a random walk on W visits her. The strategic network-bargaining literature (Manea, 2011; Abreu and Manea, 2012a,b) shows that, with the usual protocols, limit payoffs depend on the network in other ways (through "oligopoly" subnetworks and outside options), not through eigenvector centrality. Readers should treat the results of this chapter as conditional on Assumption 7.A.

Structuralist Case Study: Marini's Superexploitation as a Bargaining Outcome

Ruy Mauro Marini (1932–1997) was a Brazilian sociologist and economist who spent most of his career in exile — in Mexico, Chile, and then Mexico again after the Pinochet coup — developing the most radical version of dependency theory. His 1973 book *Dialéctica de la dependencia* advanced a thesis that drew criticism from several directions and that can be given a formal reading through the bargaining equilibrium of this chapter: **the defining feature of peripheral capitalism is superexploitation** — the systematic payment of labor-power below its value. What follows is this book's formal reinterpretation, not Marini's own apparatus; where the two diverge, we say so.

Marini's argument has three parts.

Part 1: The terms-of-trade mechanism transfers surplus from periphery to center. Following Prebisch, Marini argued that unequal exchange in international trade transfers part of the surplus produced in the periphery to the center. But Marini went further: he asked what the peripheral capitalist does to *compensate* for this leakage. The answer: she increases the rate of exploitation of domestic labor, paying workers less than the value their labor produces. This is not a market failure in the neoclassical sense — it is the market working exactly as the power structure dictates.

In this chapter's framework: the global bargaining equilibrium gives the center a surplus share $\pi_{\text{center}}(W_{\text{global}})$, which is large because the global W^P is center-periphery asymmetric. The peripheral capitalist's share of the gains from trade is correspondingly small (though, by individual rationality, never below her outside option). To maintain her accumulation, she squeezes the domestic surplus by exploiting the one factor she controls: labor. The domestic bargaining equilibrium gives the worker the share $\pi_{\text{worker}}(W_{\text{domestic}})$ of the domestic surplus, which is small because the domestic W gives workers almost no structural power.

Part 2: Superexploitation means $\pi_{\text{worker}} \ll 1/N$. Marini defined superexploitation through three mechanisms: (i) increasing the intensity of work (more output per hour), (ii) extending the working day (more hours), and (iii) reducing the worker's consumption below the level needed for "normal" reproduction of labor power. In our reading, all three lower the worker's effective bargaining weight. In the notation of this chapter, the allocation rule gives the worker

$$x^*_{\text{worker}} = \frac{\Omega}{N} + \Big(\pi_{\text{worker}} - \frac{1}{N}\Big)\Phi.$$

When $\pi_{\text{worker}} \ll 1/N$, the power correction is large and negative, and the worker receives substantially less than equal division. The formal signature is $x^*_{\text{worker}} < \Omega/N$ driven by $\pi_{\text{worker}} \ll 1/N$. Note where the reinterpretation departs from Marini: his benchmark is the *value of labor-power* (the cost of its normal reproduction), not equal division; the model captures the mechanism (a weak bargaining position), not the Marxian value accounting.

Part 3: Superexploitation is structurally stable. Marini argued that superexploitation is not a transient aberration that development will cure. It is a *structural feature* of peripheral capitalism: the power structure that produces low wages also prevents the formation of a domestic mass market, which prevents ISI from generating internal demand, which prevents the structural transformation that would increase workers' Perron weight. The allocation reinforces the structure. This is the fixed-point argument from Chapter 2 (Furtado's thesis) applied to the labor

market: the equilibrium $(\boldsymbol{\pi}^*, \mathbf{x}^*)$ satisfies $\boldsymbol{\pi}^* = \boldsymbol{\pi}(W(\mathbf{x}^*))$ — the power structure produces the allocation that reproduces the power structure.

Marini's argument	**Formal object**	**This chapter**
Surplus appropriation	$x_i^* = \Omega/N + (\pi_i - 1/N)\Phi$	Prop. 7.3
Superexploitation	$\pi_{\text{worker}} \ll 1/N$	Eq. (7.8)
Transfer from periphery to center	Global $\boldsymbol{\pi}$: $\pi_{\text{center}} \gg 1/N$	Sec. 7.6
Domestic compensation via labor squeeze	Domestic $\boldsymbol{\pi}$: $\pi_{\text{worker}} \ll 1/N$	Ex. B
Structural stability	Fixed point: $W(\mathbf{x}^*) \to \boldsymbol{\pi}^* \to \mathbf{x}^*$	Sec. 7.7
Escape is blocked politically, not spectrally	$t\|E\|_\infty \geq \Delta/\|Z\|_\infty$	Prop. 7.11

The *Dialéctica* was controversial in its time; its best-known critique is Cardoso and Serra (1978), to which Marini replied in "Las razones del neodesarrollismo" (1978). The bargaining equilibrium gives one part of Marini's thesis a testable formulation, conditional on Assumption 7.A: the left Perron eigenvector of the domestic weight matrix assigns workers a share substantially below $1/N$. Compute W from input-output data, labor market surveys, and social network data. Compute $\boldsymbol{\pi}$ and N_{eff}. Compare π_{worker} with $1/N$. The distance measures the weakness of labor's bargaining position — the mechanism Marini emphasized, not his value-theoretic definition.

Marini wrote in exile. The military governments that exiled him did not change the bargaining problem. They changed who got to sit at the table — they pushed π_{worker} toward zero by force. The Perron eigenvector remained well-defined. It just became more concentrated.

Further Reading

Nash bargaining: Nash (1950), "The Bargaining Problem." Kalai (1977), "Nonsymmetric Nash Solutions and Replications of 2-Person Bargaining." Rubinstein (1982), "Perfect Equilibrium in a Bargaining Model."

Bargaining in networks: Manea (2011), "Bargaining in Stationary Networks." Abreu and Manea (2012b), "Markov Equilibria in a Model of Bargaining in Networks"; and Abreu and Manea (2012a), "Bargaining and Efficiency in Networks." Corominas-Bosch (2004), "Bargaining in a Network of Buyers and Sellers."

Power and allocation: on relationship-specific investments and bargaining in production networks, see Acemoglu and Tahbaz-Salehi (2025). This is the closest existing work to the model in this chapter.

Welfare decomposition: The decomposition (7.10) is an exact second-order expansion of quadratic utility around equal division, weighted by $1 + \gamma_U c_j$; it is bookkeeping, not a new result, and its interest rests on Assumption 7.A. Emmanuel (1969), *L'échange inégal*, for the term "unequal exchange." For related welfare loss analysis in network settings: Elliott et al. (2014), "Financial Networks and Contagion."

Dependency theory and labor: Marini (1973), *Dialéctica de la dependencia*, México: Era (the six-chapter essay formalized in the case study above; available in Spanish at `marini-escritos.unam.mx`). Cardoso and Serra (1978), "Las desventuras de la dialéctica de la dependencia," and Marini (1978), "Las razones del neodesarrollismo" — the critique and his response. Prebisch (1981), *Capitalismo Periférico*, Chapter 4, for the structuralist version of the same argument without the Marxist vocabulary.

Chapter 8

Belief Manipulation and Epistemic Power

Who Controls What You Believe

> *"[La colonialidad] se funda en la imposición de una clasificación racial/étnica de la población del mundo como piedra angular de dicho patrón de poder y opera en cada uno de los planos, ámbitos y dimensiones, materiales y subjetivas, de la existencia social cotidiana y a escala societal."*
>
> ["[Coloniality] is founded on the imposition of a racial/ethnic classification of the world's population as the cornerstone of that pattern of power, and it operates on every plane, domain, and dimension — material and subjective — of everyday social existence and at the societal scale."]
>
> — Aníbal Quijano, "Colonialidad del poder y clasificación social" (2000)

Chapters 5–7 developed the three-network model, computed power from the Perron eigenvector, and derived the bargaining equilibrium. But in all of these, beliefs were fixed: agents knew their preferences, their endowments, and their network position. This chapter removes that assumption.

In the real world, agents do not observe the state of the economy directly. They observe signals filtered through an information network W^I — conversations with friends, broadcasts from media, teachings from schools, patterns on social media. Their beliefs about the world are W^I-weighted averages of the signals they receive. When W^I is asymmetric, some agents' signals carry more weight than others in shaping the collective understanding of reality. This is *epistemic power*: the power to shape beliefs.

This chapter develops three results. First, the DeGroot model of belief dynamics: repeated averaging on W^I converges to a consensus determined by the left Perron eigenvector $\boldsymbol{\pi}^I$. Second, the cost of manipulation: an agent who can rewire W^I can steer the consensus toward any belief within the range of initial beliefs, at a cost that rises with the target's distance from the network's natural consensus and *falls* as the network becomes more weakly coupled. Third, the connection to preferences: when beliefs about returns shape investment decisions, epistemic power becomes economic power — the bridge to the formation model of Part III.

8.1 DeGroot Learning

8.1.1 The Model

There are N agents and a binary state of the world $\omega \in \{0, 1\}$ (e.g., "the return to education is high" vs. "the return to education is low"). Each agent i starts with an initial belief $b_i^{(0)} \in [0, 1]$: her prior probability that $\omega = 1$.

At each period $t = 1, 2, \ldots$, agent i updates her belief by taking a weighted average of her neighbors' beliefs:

$$b_i^{(t)} = \sum_{j=1}^{N} W_{ij}^I b_j^{(t-1)}. \tag{8.1}$$

In vector notation:

$$\mathbf{b}^{(t)} = W^I \mathbf{b}^{(t-1)} = (W^I)^t \mathbf{b}^{(0)}. \tag{8.2}$$

This is not Bayesian updating. It is naïve averaging: agent i treats her neighbors' beliefs as data, regardless of whether those beliefs are based on independent information. The neighbors do the same. Information gets double-counted, triple-counted, endlessly recirculated. The model is wrong as a description of optimal inference. As a description of how people actually form beliefs it does well: in network experiments it fits choices better than Bayesian benchmarks (Chandrasekhar et al., 2020).

Intuition: Why DeGroot, not Bayes?

Bayesian updating requires agents to know: (i) the data-generating process, (ii) the signal structure of every other agent, (iii) the network topology, and (iv) the prior distribution. In practice, the girl in Potosí does not know the distribution of returns to education. She does not know what signals her mother received. She does not know the network. She observes her mother's behavior, her teacher's advice, and her TikTok feed, and she averages. DeGroot captures this.

DeMarzo et al. (2003) motivate repeated averaging as boundedly rational "persuasion bias": agents fail to adjust for the repetition of information they have already heard. Molavi et al. (2018) give an axiomatic foundation for (log-)linear non-Bayesian updating. Golub and Jackson (2010) then ask when such naive learning is nonetheless *wise* (Section 8.2).

8.1.2 Convergence

Theorem 8.1 (DeGroot Convergence)**.** *If W^I is row-stochastic, irreducible, and aperiodic, then:*

$$\lim_{t\to\infty}(W^I)^t = \mathbf{1}(\boldsymbol{\pi}^I)', \tag{8.3}$$

where $\boldsymbol{\pi}^I$ is the left Perron eigenvector of W^I (with $\sum_i \pi_i^I = 1$). Consequently:

$$\lim_{t\to\infty}\mathbf{b}^{(t)} = \left(\sum_{j=1}^{N}\pi_j^I b_j^{(0)}\right)\cdot\mathbf{1} = b^*\cdot\mathbf{1}. \tag{8.4}$$

All agents converge to the same belief b^*: the $\boldsymbol{\pi}^I$-weighted average of initial beliefs.

Proof. An irreducible, aperiodic nonnegative matrix is primitive, so by Theorem 6.6 (item 6; Seneta, 2006, Thm. 1.2) $(W^I)^t/\rho^t \to \mathbf{v}(\boldsymbol{\pi}^I)'$ with $(\boldsymbol{\pi}^I)'\mathbf{v} = 1$. For row-stochastic W^I, $\rho = 1$ and $\mathbf{v} = \mathbf{1}$ (because $W^I\mathbf{1} = \mathbf{1}$), and $(\boldsymbol{\pi}^I)'\mathbf{1} = 1$. Hence $(W^I)^t \to \mathbf{1}(\boldsymbol{\pi}^I)'$ and $\mathbf{b}^{(t)} = (W^I)^t\mathbf{b}^{(0)} \to \mathbf{1}(\boldsymbol{\pi}^I)'\mathbf{b}^{(0)}$. No diagonalizability is needed. □

Irreducibility can be relaxed: the same conclusion holds whenever W^I has a single closed communicating class and that class is aperiodic; $\boldsymbol{\pi}^I$ is then zero outside the closed class (Golub and Jackson, 2010). If W^I has several closed classes, as in a block-diagonal W^I, there is no global consensus and $\boldsymbol{\pi}^I$ is not unique.

Definition 8.2 (Epistemic Power)**.** *Agent j's **epistemic power** is π_j^I: her weight in the long-run consensus. An agent with $\pi_j^I \gg 1/N$ has disproportionate influence on what*

the society ends up believing. An agent with $\pi_j^I \approx 0$ is epistemically irrelevant — her initial belief has negligible effect on the consensus, no matter how correct it is.

8.1.3 Speed of Convergence

Proposition 8.3 (Convergence Rate)**.** *Under the hypotheses of Theorem 8.1, the rate of convergence to consensus is:*

$$\|\mathbf{b}^{(t)} - b^*\mathbf{1}\| \leq C \cdot |\lambda_2|^t, \tag{8.5}$$

where $|\lambda_2|$ is the second-largest eigenvalue of W^I in modulus and C depends on $\mathbf{b}^{(0)}$ (if λ_2 is defective, multiply the bound by a polynomial in t). The half-life of disagreement is $t_{1/2} = \ln 2 / \ln(1/|\lambda_2|)$.

Proof. $\mathbf{b}^{(t)} - b^*\mathbf{1} = ((W^I)^t - \mathbf{1}(\boldsymbol{\pi}^I)')\mathbf{b}^{(0)} = (W^I - \mathbf{1}(\boldsymbol{\pi}^I)')^t\mathbf{b}^{(0)}$, since $W^I\mathbf{1}(\boldsymbol{\pi}^I)' = \mathbf{1}(\boldsymbol{\pi}^I)'W^I = (\mathbf{1}(\boldsymbol{\pi}^I)')^2 = \mathbf{1}(\boldsymbol{\pi}^I)'$. The matrix $W^I - \mathbf{1}(\boldsymbol{\pi}^I)'$ has the eigenvalues of W^I with 1 replaced by 0: it vanishes on $\operatorname{span}(\mathbf{1})$ and equals W^I on the invariant complement $\{\mathbf{x} : (\boldsymbol{\pi}^I)'\mathbf{x} = 0\}$, where the eigenvalues are $\lambda_2, \ldots, \lambda_N$ because 1 is simple. Its spectral radius is therefore $|\lambda_2| < 1$ (primitivity), and the Jordan form gives $\|(W^I - \mathbf{1}(\boldsymbol{\pi}^I)')^t\| \leq C' t^{m-1}|\lambda_2|^t$, with m the size of the largest Jordan block for an eigenvalue of modulus $|\lambda_2|$ ($m = 1$ if those eigenvalues are semisimple). □

The spectral gap $\Gamma^I = 1 - |\lambda_2|$ (Definition 6.14) governs the speed. Large Γ^I: consensus forms quickly (well-connected W^I). Small Γ^I: disagreement persists for many rounds (clustered or block-diagonal W^I).

Example 8.4 (Three Information Networks)**.** $N = 4$, *with initial beliefs* $\mathbf{b}^{(0)} = (0.9, 0.8, 0.2, 0.1)$*: agents 1–2 believe "high" and agents 3–4 believe "low".*

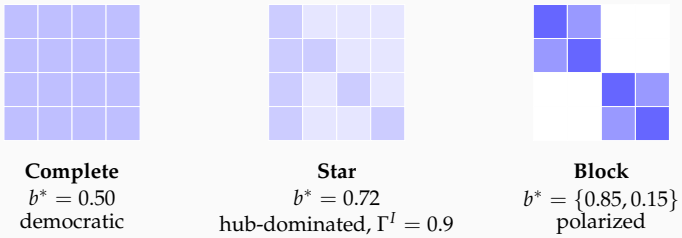
Case A: Complete. $W^I = \frac{1}{4}\mathbf{1}\mathbf{1}'$. $\boldsymbol{\pi}^I = (0.25, 0.25, 0.25, 0.25)$. *Consensus:* $b^* = 0.25(0.9 + 0.8 + 0.2 + 0.1) = 0.50$. *Convergence: instant (one round). The truth is averaged democratically.*

Case B: Star *(agent 1 is the hub).* $W^I = \begin{pmatrix} 0.7 & 0.1 & 0.1 & 0.1 \\ 0.6 & 0.2 & 0.1 & 0.1 \\ 0.6 & 0.1 & 0.2 & 0.1 \\ 0.6 & 0.1 & 0.1 & 0.2 \end{pmatrix}$. *Perron eigenvector:* $\boldsymbol{\pi}^I = (2/3, 1/9, 1/9, 1/9) \approx (0.667, 0.111, 0.111, 0.111)$, *from the balance condition* $0.3\,\pi_1 = 0.6\,(1 - \pi_1)$. *Consensus:* $b^* = \frac{2}{3}(0.9) + \frac{1}{9}(0.8 + 0.2 + 0.1) = 0.722$. *The hub's belief (0.9) dominates:* $N_{\text{eff}} = 2.08$. *All non-unit eigenvalues equal* 0.1, *so* $\Gamma^I = 0.9$ *and the half-life is* 0.30 *rounds. A hub that everyone listens to makes consensus fast: a large spectral gap and concentrated power go together here (Proposition 6.16).*

Case C: Block-diagonal. $W^I = \begin{pmatrix} 0.6 & 0.4 & 0.0 & 0.0 \\ 0.4 & 0.6 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.6 & 0.4 \\ 0.0 & 0.0 & 0.4 & 0.6 \end{pmatrix}$. *Two disconnected blocks:* W^I *is reducible,* $\Gamma^I = 0$*, and* $\boldsymbol{\pi}^I$ *is not unique. Each block converges internally: block 1 to* $0.5(0.9) + 0.5(0.8) = 0.85$*, block 2 to* $0.5(0.2) + 0.5(0.1) = 0.15$*. No global consensus. The society remains polarized.*

	b_1^*	b_2^*	b_3^*	b_4^*	Γ^I
Complete	*0.50*	*0.50*	*0.50*	*0.50*	*1.00*
Star	*0.72*	*0.72*	*0.72*	*0.72*	*0.90*
Block	*0.85*	*0.85*	*0.15*	*0.15*	*0.00*

Visualization: Three information networks, three belief trajectories



Complete
$b^* = 0.50$
democratic

Star
$b^* = 0.72$
hub-dominated, $\Gamma^I = 0.9$

Block
$b^* = \{0.85, 0.15\}$
polarized

Same initial beliefs (0.9, 0.8, 0.2, 0.1); shade $= 100\, W_{ij}^I$ (white $= 0$). Three different W^I's produce three different outcomes: democratic consensus, hub-dominated consensus, and permanent polarization. The heatmap determines the epistemic outcome. Note that the hub-dominated network converges fastest after the complete one: concentration of $\boldsymbol{\pi}^I$ and speed of consensus are different properties.

8.2 The Wisdom and Foolishness of Crowds

8.2.1 Wisdom: When Does the Consensus Get It Right?

Proposition 8.5 (Accuracy of the Consensus)**.** *Let* W^I *satisfy the hypotheses of Theorem 8.1 and be fixed (not dependent on the noise). Suppose the true state is* ω*, and agent* i*'s initial belief is* $b_i^{(0)} = \omega + \varepsilon_i$*, where the* ε_i *are independent, zero-mean noises with variances* σ_i^2*. The DeGroot consensus is:*

$$b^* = \omega + \sum_j \pi_j^I \varepsilon_j. \tag{8.6}$$

The consensus error is $e = b^* - \omega = \sum_j \pi_j^I \varepsilon_j$*, with:*

$$\mathrm{Var}(e) = \sum_{j=1}^{N} (\pi_j^I)^2 \sigma_j^2 \geq \frac{\sigma_{\min}^2}{N_{\mathit{eff}}}, \qquad \sigma_{\min}^2 = \min_j \sigma_j^2, \tag{8.7}$$

where $N_{\mathit{eff}} = 1/\sum_j (\pi_j^I)^2$ *is the effective number of agents of Definition 6.15. With equal variances* $\sigma_j^2 = \sigma^2$*,* $\mathrm{Var}(e) = \sigma^2/N_{\mathit{eff}}$ *exactly, and* $\mathrm{Var}(e) \geq \sigma^2/N$ *with equality iff* $\boldsymbol{\pi}^I = \frac{1}{N}\mathbf{1}$*.*

Proof. By Theorem 8.1, $b^* = \sum_j \pi_j^I(\omega + \varepsilon_j) = \omega + \sum_j \pi_j^I \varepsilon_j$, since $\sum_j \pi_j^I = 1$; so $E[e] = 0$. By independence, $\mathrm{Var}(e) = \mathrm{Var}(\sum_j \pi_j^I \varepsilon_j) = \sum_j (\pi_j^I)^2 \sigma_j^2 \geq \sigma_{\min}^2 \sum_j (\pi_j^I)^2 = \sigma_{\min}^2 / N_{\mathrm{eff}}$. With equal variances the inequality is an equality, and $\sum_j (\pi_j^I)^2 \geq 1/N$ (Cauchy–Schwarz, $\sum_j \pi_j^I = 1$) with equality iff $\pi_j^I = 1/N$. □

Golub and Jackson's (2010) theorem is the asymptotic counterpart: along a sequence of growing networks with independent, bounded-variance signals, the DeGroot consensus converges to the truth (the society is *wise*) if and only if the influence of the most influential agent vanishes, $\max_j \pi_j^I \to 0$.

Remark 8.6 (Effective sample size). Read epistemically, N_{eff} is the number of "independent voices" in the consensus. When $\boldsymbol{\pi}^I = \frac{1}{N}\mathbf{1}$: $N_{\mathrm{eff}} = N$ (every agent counts equally). When one agent holds almost all influence ($\boldsymbol{\pi}^I \approx \mathbf{e}_1$, as with the broadcast hub $\mathbf{1}\boldsymbol{\pi}'$ of Chapter 6): $N_{\mathrm{eff}} \approx 1$. A star need not be that extreme: the star of Example 8.4 has $N_{\mathrm{eff}} = 2.08$. The Herfindahl index of epistemic power is $H = 1/N_{\mathrm{eff}}$.

Intuition: The crowd is wise when power is equal

The wisdom-of-crowds effect requires many independent, equally precise signals to be aggregated with equal weight. (With unequal precisions the variance-minimizing weights are proportional to $1/\sigma_j^2$, not equal.) When W^I concentrates epistemic power, the consensus overweights a few agents' signals and underweights the rest. The effective sample size shrinks. The consensus becomes less accurate, not because individuals are dumber but because the network throws away most of the available information.

Suppose, for illustration, that 5% of the population holds 90% of epistemic weight, spread equally among them, and the rest share 10% equally. Then $N_{\mathrm{eff}} \approx 0.062N$: the consensus aggregates about sixteen times fewer independent voices than the population contains.

8.2.2 Foolishness: Misinformation in Asymmetric Networks

Proposition 8.7 (Misinformation Amplification)**.** *Let W^I satisfy the hypotheses of Theorem 8.1 — for instance a star with hub h — and let initial beliefs be as in Proposition 8.5, except that agent h has a biased initial belief, $b_h^{(0)} = \omega + \delta + \varepsilon_h$ for some bias $\delta \neq 0$. Then the consensus inherits the bias:*

$$b^* = \omega + \pi_h^I \delta + \sum_j \pi_j^I \varepsilon_j, \qquad E[b^*] - \omega = \pi_h^I \delta. \tag{8.8}$$

The bias in the consensus is proportional to π_h^I: the hub's epistemic power. When $\pi_h^I \to 1$, the consensus reproduces the hub's bias exactly.

Proof. By Theorem 8.1, $b^* = \sum_j \pi_j^I b_j^{(0)} = \omega \sum_j \pi_j^I + \pi_h^I \delta + \sum_j \pi_j^I \varepsilon_j$, and $\sum_j \pi_j^I = 1$. Taking expectations, with $E[\varepsilon_j] = 0$, gives the bias. □

This is the misinformation mechanism: a biased agent with high epistemic power contaminates the entire network. The contamination is permanent (it enters the long-run consensus) and is not self-correcting (DeGroot agents do not learn from prediction errors). The only correction is structural: change W^I to reduce the hub's epistemic power.

Prebisch Connection: The Washington Consensus as a DeGroot equilibrium

The Washington Consensus of the 1990s — fiscal discipline, trade liberalization, privatization, deregulation — spread through two channels. One was conditionality: IMF and World Bank lending was tied to policy reform. The DeGroot lens isolates a second channel: the information network W^I_{global} had a hub-spoke topology, with the IMF, World Bank, and US Treasury as dominant nodes, and peripheral policymakers updated their beliefs by listening to these institutions.

[ILLUSTRATIVE] Suppose $\pi^I_{\text{IMF}} = \pi^I_{\text{WB}} = \pi^I_{\text{UST}} = 0.2$, and the remaining 0.4 is spread over 200 other agents. The consensus b^* then sits close to the three hubs' shared priors: markets work, governments fail, structural adjustment is necessary. The other agents' priors (including the structuralists at CEPAL, who had different priors about the role of the state) carry little weight because their epistemic power π^I_j is small.

The point is not that markets never work. It is that such a consensus aggregates about $N_{\text{eff}} \approx 8.3$ independent voices in a world with hundreds of relevant economies. Much of the wisdom of the crowd is thrown away by the network topology.

8.3 Belief Manipulation

8.3.1 The Manipulation Problem

Suppose a *manipulator* — a government, a media company, a foreign power — wants to steer the consensus to a target belief b_T. She cannot change agents' initial beliefs directly, but she can change the information network W^I: by controlling media outlets, shaping algorithms, subsidizing certain voices, or silencing others.

Definition 8.8 (Manipulation Cost)**.** *Given the current network W^I with Perron eigenvector $\boldsymbol{\pi}^I$ and initial beliefs $\mathbf{b}^{(0)}$, the current consensus is $b^* = (\boldsymbol{\pi}^I)'\mathbf{b}^{(0)}$. The manipulator wants to achieve a target b_T. The **manipulation cost** is:*

$$C(b_T) = \min_{\Delta W^I} \left\{ \|\Delta W^I\|_F : (\boldsymbol{\pi}^I(W^I + \Delta W^I))'\mathbf{b}^{(0)} = b_T \right\}, \tag{8.9}$$

subject to $W^I + \Delta W^I$ remaining row-stochastic and nonnegative. Since any consensus is a convex combination of initial beliefs, only targets $b_T \in [\min_j b^{(0)}_j, \max_j b^{(0)}_j]$ are reachable; for other targets $C(b_T) = \infty$.

Proposition 8.9 (Manipulation Cost, First Order)**.** *Let W^I be irreducible and row-stochastic with fundamental matrix $Z = (I - W^I + \mathbf{1}(\boldsymbol{\pi}^I)')^{-1}$. A perturbation tE (with E as in Lemma 6.19, so $E\mathbf{1} = 0$) moves the consensus $b^* = (\boldsymbol{\pi}^I)'\mathbf{b}^{(0)}$, to first order in t, by*

$$db^* = t\,(\boldsymbol{\pi}^I)'EZ\big(\mathbf{b}^{(0)} - b^*\mathbf{1}\big), \qquad \text{so} \qquad t\,\|E\|_\infty \;\geq\; \frac{|b_T - b^*|}{\max_j |b_j^{(0)} - b^*|} \cdot \frac{1}{\|Z\|_\infty} \tag{8.10}$$

for any first-order move from b^* to b_T. Moreover $\|Z\|_\infty \geq \max_{k\geq 2} 1/|1-\lambda_k|$.

Proof. $b^* = (\boldsymbol{\pi}^I)'\mathbf{b}^{(0)}$, so by Lemma 6.19 $db^* = t(\boldsymbol{\pi}^I)'EZ\mathbf{b}^{(0)}$. Since $Z\mathbf{1} = \mathbf{1}$ and $E\mathbf{1} = 0$, $EZ\mathbf{1} = 0$, so $\mathbf{b}^{(0)}$ can be replaced by $\mathbf{b}^{(0)} - b^*\mathbf{1}$. Then

$$|db^*| \leq t\|(\boldsymbol{\pi}^I)'E\|_1\,\|Z\|_\infty\,\|\mathbf{b}^{(0)} - b^*\mathbf{1}\|_\infty \leq t\|E\|_\infty\|Z\|_\infty \max_j |b_j^{(0)} - b^*|.$$

The last claim holds because an induced norm bounds the spectral radius and Z has eigenvalues 1 and $1/(1-\lambda_k)$. □

Intuition: What makes manipulation cheap?

The cost of manipulation is low when:

(i) The target belief is close to the current consensus ($|b_T - b^*|$ small). Nudging is cheap.

(ii) Initial beliefs are diverse ($\max_j |b_j - b^*|$ large). The manipulator can amplify an existing minority voice rather than creating a new one.

(iii) The network is weakly coupled (λ_2 close to $+1$, so $\|Z\|$ is large). Small changes in the network produce large changes in $\boldsymbol{\pi}^I$ when some eigenvalue is close to one. The consensus is structurally fragile. Eigenvalues close to -1 (periodic networks, which also have small Γ^I) do not create this fragility.

In the two-block example of `code/ch08.py` (two pairs, beliefs 0.9 and 0.1, symmetric coupling ε), moving 0.01 of each traditional agent's attention to the other block shifts the consensus from 0.50 to 0.51 when $\varepsilon = 0.2$, but to 0.63 when $\varepsilon = 0.01$.

The most manipulable society is one with a fragmented information network (weak coupling), diverse initial beliefs, and a target close to an already-existing minority view. Echo chambers plus polarization plus a skilled manipulator: the conditions for epistemic capture.

8.3.2 The Algorithmic Manipulator

In the pre-digital era, manipulation required owning a newspaper or a radio station: physically controlling a hub in W^I. In the algorithmic era, manipulation operates through the *edges* of W^I rather than the nodes: the algorithm determines W^I_{ij} (what content agent i sees from agent j) for every pair simultaneously.

Proposition 8.10 (Algorithmic Control of W^I)**.** *A platform that controls the recommendation algorithm chooses W^I_{ij} for all i, j. Absent other constraints, this is equivalent to choosing $\boldsymbol{\pi}^I$ (and therefore b^*) directly: every $\boldsymbol{\pi}$ in the open simplex is attained, with any spectral gap $\gamma \in (0, 1]$. In practice the choice is constrained by user engagement (agents must find the recommended content sufficiently interesting to consume).*

Proof. $W^I = \gamma \mathbf{1}\boldsymbol{\pi}' + (1-\gamma)I$ has Perron vector $\boldsymbol{\pi}$ and $\Gamma^I = \gamma$ (Proposition 6.16). □

Example 8.11 (The Platform Economy)**.** *$N = 6$: three content creators (C1, C2, C3) and three consumers (U1, U2, U3). C1 produces pro-market content ($b_{C1} = 0.9$), C2 produces structuralist content ($b_{C2} = 0.3$), C3 produces entertainment ($b_{C3} = 0.5$).*

Consumers' priors are $(0.8, 0.5, 0.2)$ for (U1, U2, U3). [ILLUSTRATIVE]

***Organic** W^I: Each creator keeps weight 0.8 on her own line and gives 0.1 to each other creator; each consumer follows all three creators equally, $W^I_{U_k,C_j} = 1/3$, and places no weight on consumers. The creators form the only closed class, with equal weights, so $\boldsymbol{\pi}^I = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0, 0, 0)$: consumers' priors do not affect the long run. Consensus: $b^* = (0.9 + 0.3 + 0.5)/3 = 0.567$ for everyone. $\Gamma^I = 0.3$.*

***Algorithmic** W^I: The platform's engagement algorithm shows consumers content that confirms their priors, and creators stop engaging with each other (each reinforces only her own line). U1 (prior 0.8) gives 0.875 to C1 and 0.125 to C3; U3 (prior 0.2) gives 0.875 to C2 and 0.125 to C3; U2 sees only C3. Now each creator is a closed class of her own, and no single consensus forms. Instead there are two echo chambers, $b_{U1} = 0.85$ and $b_{U3} = 0.325$, around an entertainment audience at 0.5.*

The algorithm did not inject misinformation. It cut the links among the sources and sorted the audience, replacing one consensus with polarized ones. The spectral gap fell from 0.30 to 0 (three unit eigenvalues). Note that sorting the consumers alone would not have been enough: as long as the creators listen to each other, everyone still converges to 0.567.

8.4 From Beliefs to Preferences: The Bridge to Part III

Everything so far treats beliefs as a one-shot process: initial beliefs are updated once and converge. But beliefs about returns shape investment decisions, which

produce outcomes, which become signals, which update beliefs. The loop is dynamic.

Definition 8.12 (The Belief-Action-Signal Loop)**.** *Consider a sequence of generations $t = 0, 1, 2, \ldots$. In generation t:*

1. *Agents form beliefs about returns: $\hat{r}_{i,t} = \sum_j W^I_{ij} r_{j,t-1}$ (DeGroot on observed returns).*

2. *Agents choose actions based on beliefs: $a_{i,t} = f(\hat{r}_{i,t}, \theta_{i,t})$ (investment, education, labor supply).*

3. *Actions produce outcomes: $r_{i,t} = g(a_{i,t}, \mathbf{x}_t)$ (realized returns, depending on the allocation).*

4. *Outcomes become the next generation's signals.*

A steady state is a fixed point of the belief-action-signal map: $\hat{r}_i = \sum_j W^I_{ij} g(f(\hat{r}_j, \theta_j), \mathbf{x}^*)$. (Existence and stability depend on f and g; Part III studies them.)

Key Structural Insight

When W^I is block-diagonal, the belief-action-signal loop produces **two different steady states in the same economy**. Block 1 (modern sector) observes high returns, forms optimistic beliefs, invests heavily, achieves high returns — confirming the beliefs. Block 2 (traditional sector) observes low returns, forms pessimistic beliefs, underinvests, achieves low returns — confirming those beliefs too.

Both blocks are in equilibrium. Both have "rational" beliefs (their beliefs match the returns they observe). But the traditional sector's beliefs are only rational *conditional on the traditional sector's W^I*. If the girl in Potosí could observe the modern sector's returns (if W^I had cross-block entries), she would form different beliefs, make different investments, and achieve different outcomes.

The structure produces the beliefs that justify the structure.

This is the formation mechanism: W^I shapes beliefs, beliefs shape preferences (through θ), preferences shape the allocation, and the allocation confirms the beliefs. Part III (Chapters 9–12) develops this fully.

8.5 Summary

1. **DeGroot learning** on W^I converges to a consensus $b^* = (\boldsymbol{\pi}^I)'\mathbf{b}^{(0)}$: the Perron-weighted average of initial beliefs. Epistemic power π_j^I determines each agent's influence on the long-run consensus.

2. The **effective sample size** $N_{\text{eff}} = 1/\sum_j (\pi_j^I)^2$ measures the epistemic richness of the consensus: with equally precise independent signals the consensus error has variance σ^2/N_{eff}. Concentrated $\boldsymbol{\pi}^I$ wastes information: $N_{\text{eff}} \ll N$. Concentration and speed are distinct — a hub everyone listens to gives fast consensus and low N_{eff}.

3. **Misinformation amplification:** a biased hub contaminates the consensus in proportion to π_h^I. The contamination is permanent under DeGroot dynamics.

4. **Manipulation cost** is bounded below by $|b_T - b^*|/(\|Z\|_\infty \max_j |b_j^{(0)} - b^*|)$ and is infinite outside the range of initial beliefs. It is lowest in weakly coupled networks (λ_2 near $+1$, large $\|Z\|$), where small rewiring moves $\boldsymbol{\pi}^I$ a lot. Algorithmic platforms choose W^I directly; cutting links among sources creates echo chambers and drives Γ^I to zero.

5. The **belief-action-signal loop** connects epistemic power to economic outcomes: W^I shapes beliefs, beliefs shape actions, actions produce returns, returns become signals. When W^I is block-diagonal, the loop produces self-confirming pessimism in the powerless block.

8.6 Exercises

8.1 DeGroot by hand. For $W^I = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.1 & 0.6 & 0.3 \\ 0.3 & 0.2 & 0.5 \end{pmatrix}$ and $\mathbf{b}^{(0)} = (1.0, 0.0, 0.5)'$:

(a) Compute $\mathbf{b}^{(1)}$, $\mathbf{b}^{(2)}$, $\mathbf{b}^{(3)}$ by hand.

(b) Compute $\boldsymbol{\pi}^I$ (use the method from Chapter 6 or software).

(c) Verify that $\mathbf{b}^{(t)} \to b^*\mathbf{1}$ and check $b^* = (\boldsymbol{\pi}^I)'\mathbf{b}^{(0)}$.

(d) What is the half-life of disagreement? (Compute $|\lambda_2|$.)

8.2 Effective sample size. For $N = 100$, compute N_{eff} for:

(a) $\boldsymbol{\pi}^I = \frac{1}{100}\mathbf{1}$ (complete graph).

(b) $\pi_1^I = 0.5$, $\pi_j^I = 0.5/99$ for $j \geq 2$ (one dominant hub).

(c) $\pi_1^I = 0.9$, $\pi_j^I = 0.1/99$ for $j \geq 2$ (state media monopoly).

(d) Plot N_{eff} as a function of π_1^I (with the rest distributed uniformly). At what π_1^I does N_{eff} drop below 10?

8.3 Manipulation cost. Using the three-agent network from Exercise 8.1:

(a) The current consensus is b^*. A manipulator wants $b_T = 0.8$. Compute the required shift in $\boldsymbol{\pi}^I$.

(b) What is the cheapest edge to change? (Hint: which edge has the largest effect on $\boldsymbol{\pi}^I$ per unit of perturbation?)

(c) Compute Z and the lower bound of Proposition 8.9. Then mix W^I with the identity, $W_\alpha = \alpha W^I + (1-\alpha)I$ (same $\boldsymbol{\pi}^I$, eigenvalues $1 - \alpha(1-\lambda_k)$). How do Z and the bound change as α falls?

8.4 The algorithm as W^I designer. Consider $N = 8$: 4 liberals ($b_i^{(0)} = 0.8$) and 4 conservatives ($b_i^{(0)} = 0.2$).

(a) Under $W^I = \frac{1}{8}\mathbf{1}\mathbf{1}'$: what is the consensus? What is Γ^I?

(b) Under W^I = block-diagonal (liberals talk only to liberals, conservatives only to conservatives): what are the two "consensuses"? What is Γ^I?

(c) An engagement-maximizing algorithm assigns $W_{ij}^I \propto \exp(-\beta|b_i^{(0)} - b_j^{(0)}|)$ (agents see content similar to their prior). Compute W^I for $\beta = 0, 1, 5, 20$. Plot $\Gamma^I(\beta)$.

(d) Show that for every finite β the matrix is symmetric with equal row sums, hence doubly stochastic, so the consensus is 0.5 for every β: the society never splits into two consensuses. What diverges as β grows? Compute the half-life of disagreement as a function of β (show $\Gamma^I = 1 - \tanh(0.3\beta)$).

8.5 Bolivia's epistemic structure. As a stylized description, treat Bolivia as three media markets: La Paz (state media dominant), Santa Cruz (private media dominant), and Cochabamba (mixed). Model this as $N = 3$ blocks.

(a) Construct plausible W^I matrices for each block (who listens to whom within each city's media ecosystem).

(b) Compute $\boldsymbol{\pi}^I$ for the national composite.

(c) Under what initial beliefs $\mathbf{b}^{(0)}$ about economic policy (e.g., nationalization vs. liberalization) does the consensus favor nationalization? Liberalization?

(d) How much would it cost (in $\|\Delta W^I\|_F$) to shift the consensus from 0.6 (mild pro-nationalization) to 0.4 (mild pro-liberalization)?

(e) *Data.* Calibrate the within-city rows of W^I with survey shares of main news sources (for instance from Latinobarómetro's questions on how respondents get political information) and recompute $\boldsymbol{\pi}^I$ and N_{eff}.

8.6 The belief-action-signal loop. In a two-sector economy (Modern, Traditional):

(a) Returns are $r_M = 0.15$ and $r_T = 0.05$. Under $W^I_{\text{block}} = \text{diag}(W_{11}, W_{22})$ (each sector observes only its own returns): what beliefs form in each sector?

(b) An agent in the Traditional sector receives a smartphone (a new edge in W^I connecting her to the Modern sector). Her belief about the return to education changes from $\hat{r} = 0.05$ to $\hat{r}' = 0.5 \times 0.05 + 0.5 \times 0.15 = 0.10$. If investment is $a = \theta \cdot \hat{r}$, by how much does her investment increase?

(c) If 10% of Traditional-sector agents receive smartphones, what is the effect on the sector's average belief? On Γ^I?

(d) This is the formation mechanism of Part III. Write down the full loop: $W^I \to \hat{r} \to a \to r \to$ (next period's signal). Under what condition is the Traditional sector's low-return equilibrium stable?

8.7 Quijano in matrix form. Quijano (2000a,b) argued that the "coloniality of power" operates through racial classification that structures all domains of social existence. Formalize:

(a) Define W^I where the epistemic weights correlate with racial classification: European-descended agents have $\pi^I_j \gg 1/N$; Indigenous agents have $\pi^I_j \ll 1/N$.

(b) What consensus forms about the "value" of different forms of knowledge (Western science vs. traditional medicine, formal education vs. artisanal skill)?

(c) How does this consensus shape preferences (θ) regarding education, language, and economic aspiration?

(d) What structural change in W^I would Quijano's "decoloniality" require? Write it as a matrix perturbation.

Mathematical Appendices

Appendix 8A: Power Iteration and Convergence of $(W^I)^t$

Power iteration. To compute the left Perron eigenvector $\boldsymbol{\pi}^I$:

1. Start with any initial vector $\boldsymbol{\pi}^{(0)} > 0$ with $\sum_i \pi_i^{(0)} = 1$.
2. Iterate: $\boldsymbol{\pi}^{(t+1)\prime} = \boldsymbol{\pi}^{(t)\prime} W^I$, then normalize $\boldsymbol{\pi}^{(t+1)} \leftarrow \boldsymbol{\pi}^{(t+1)} / \|\boldsymbol{\pi}^{(t+1)}\|_1$.
3. Convergence: $\|\boldsymbol{\pi}^{(t)} - \boldsymbol{\pi}^I\| \leq C|\lambda_2/\lambda_1|^t$.

The convergence rate is $|\lambda_2/\lambda_1| = |\lambda_2|$ (since $\lambda_1 = 1$ for row-stochastic matrices). Large spectral gap = fast convergence.

Python implementation.

```
import numpy as np

def perron_left(W, tol=1e-12, max_iter=10000):
    """Left Perron eigenvector of row-stochastic W."""
    n = W.shape[0]
    pi = np.ones(n) / n
    for _ in range(max_iter):
        pi_new = pi @ W
        pi_new /= pi_new.sum()
        if np.max(np.abs(pi_new - pi)) < tol:
            return pi_new
        pi = pi_new
    return pi
```

Convergence of $(W^I)^t$. Theorem 8.1 states $(W^I)^t \to \mathbf{1}(\boldsymbol{\pi}^I)'$. The rate: $\|(W^I)^t - \mathbf{1}(\boldsymbol{\pi}^I)'\| \leq C|\lambda_2|^t$, where C depends on the condition number of the eigenvector matrix.

For practical purposes: if $|\lambda_2| = 0.7$, then after $t = 10$ rounds, the error is $0.7^{10} \approx 0.028$ (2.8%). After $t = 20$: $0.7^{20} \approx 0.0008$ (0.08%). Consensus to within 5% is reached in about $\ln 20/(-\ln|\lambda_2|) \approx 3/\Gamma^I$ rounds when Γ^I is small.

Appendix 8B: Sensitivity of $\boldsymbol{\pi}^I$ to Perturbations of W^I

The question. If we perturb W^I by a small ΔW^I, how much does $\boldsymbol{\pi}^I$ change?

First-order perturbation. The answer is Lemma 6.19 (Chapter 6, Appendix 6C). For a perturbation tE that keeps the rows summing to one ($E\mathbf{1} = 0$),

$$d(\boldsymbol{\pi}^I)' = t\,(\boldsymbol{\pi}^I)'EZ, \qquad Z = \left(I - W^I + \mathbf{1}(\boldsymbol{\pi}^I)'\right)^{-1}, \tag{8.11}$$

or, when W^I is diagonalizable with biorthogonal eigenvectors ($\mathbf{u}_k'\mathbf{v}_l = \delta_{kl}$),

$$d(\boldsymbol{\pi}^I)' = t\sum_{k=2}^{N} \frac{(\boldsymbol{\pi}^I)'E\mathbf{v}_k}{1-\lambda_k}\,\mathbf{u}_k'. \tag{8.12}$$

The projection is onto $(\boldsymbol{\pi}^I)'E$, not onto $E\mathbf{v}_1$: since $\mathbf{v}_1 = \mathbf{1}$, $E\mathbf{v}_1 = 0$ for every admissible perturbation, so a formula built on $\mathbf{u}_k'E\mathbf{v}_1$ would be identically zero.

The sensitivity is inversely proportional to $1 - \lambda_k$. When some λ_k is close to $+1$ (weakly coupled blocks), small perturbations in W^I produce large changes in $\boldsymbol{\pi}^I$. This is why manipulation is cheap in fragmented networks. Eigenvalues near -1 (periodicity) give $1 - \lambda_k \approx 2$ and do not create sensitivity, even though they also make Γ^I small.

Implication for manipulation. The cheapest manipulation targets the direction associated with the eigenvalue closest to one. In a weakly coupled two-block W^I, this is the between-block balance: by Proposition 6.12, the blocks' shares of epistemic power are exactly $\varepsilon_2/(\varepsilon_1+\varepsilon_2)$ and $\varepsilon_1/(\varepsilon_1+\varepsilon_2)$, so a small rewiring that shifts attention from one block to the other produces a large shift in epistemic power when $\varepsilon_1, \varepsilon_2$ are small.

Structuralist Case Study: Quijano's Coloniality and the Epistemic Perron Vector

Aníbal Quijano (1930–2018) was a Peruvian sociologist who introduced the notion of the coloniality of power in the early 1990s. His 2000 essays "Colonialidad del poder, eurocentrismo y América Latina" (Quijano 2000a, in Lander, ed., *La colonialidad del saber*) and "Colonialidad del poder y clasificación social" (Quijano, 2000b) are central texts of the decolonial turn in Latin American social theory. His argument was not about colonialism as a historical event but about **coloniality** as a persistent structure of power — a structure that survived independence and continues to organize knowledge, labor, and social classification in the postcolonial world.

Quijano's framework maps onto the three-network model of this book in a specific and illuminating way.

Part 1: Racial classification as a node attribute that constrains W. Quijano

argued that the colonial invention of "race" was not a byproduct of conquest but its *organizing principle*: a classification system that assigned social position on the basis of phenotype. In network terms: race is a **node attribute** that determines which edges exist in W^U, W^P, and W^I simultaneously. People of European descent form dense connections among themselves across all three networks; people classified as "Indian" or "Black" are confined to sparse, peripheral positions. The weight matrix is not random or purely economic — it is structured by a categorical attribute that was imposed by conquest and maintained by institutional inertia.

Part 2: Epistemic coloniality is π^I dominance. The deepest dimension of coloniality, for Quijano, is epistemic: the colonial powers did not merely control trade routes and labor — they controlled the *definition of knowledge*. European science, philosophy, and social theory were installed as the universal standard; all other forms of knowledge were classified as "traditional," "local," "pre-modern." In this chapter's notation: the global W^I assigns European intellectual traditions a Perron weight $\pi^I_{\text{Euro}} \gg 1/N$, while Andean, African, and Asian intellectual traditions receive $\pi^I \approx 0$. The DeGroot consensus on "what counts as knowledge" converges to the European prior, not because that prior is more accurate but because W^I is hub-dominated. The effective sample size for epistemology is close to one: little more than one tradition's knowledge enters the consensus.

Part 3: The coloniality of power operates across all three networks simultaneously. Quijano's distinctive contribution was insisting that race structures W^U (who you care about), W^P (who you trade with and work for), and W^I (whose knowledge you accept) *simultaneously and correlatively*. The Perron eigenvector rankings are *aligned across networks*: the agents who are powerful in W^U are also powerful in W^P and in W^I, and this alignment is not a coincidence but a structural feature maintained by the racial classification system. In the notation of Chapter 5: the three networks satisfy $\boldsymbol{\pi}^U \approx \boldsymbol{\pi}^P \approx \boldsymbol{\pi}^I$ — the "aligned networks" case. Alignment does not amplify concentration: row-stochastic layers that share a stationary vector $\boldsymbol{\pi}$ have a composite with that same $\boldsymbol{\pi}$, since $\boldsymbol{\pi}' \sum_k \alpha_k W^k = \boldsymbol{\pi}'$. What alignment removes is compensation: no channel offsets the others, so a group that is peripheral in one network is peripheral in all.

Quijano's argument	Formal object	This chapter
Race as organizing principle	Node attribute constraining W	All three networks
Epistemic coloniality	$\pi^I_{\text{Euro}} \gg 1/N$	Def. 8.2
One tradition dominates	$N_{\text{eff}} \approx 1$	Rem. 8.6
Consensus reproduces colonial priors	$b^* \approx b_{\text{Euro}}$	Thm. 8.1
Power aligned across networks	$\boldsymbol{\pi}^U \approx \boldsymbol{\pi}^P \approx \boldsymbol{\pi}^I$	Ch. 5, aligned case
Decoloniality as structural change	Rewire W^I to equalize $\boldsymbol{\pi}^I$	Prop. 8.9

Quijano's decolonial project is, in matrix terms, a call to *rewire* W^I *so that* $\boldsymbol{\pi}^I$ *equalizes across racial categories*. This does not mean all knowledge traditions are equally accurate (that is an empirical question). It means all traditions should have equal *voice* in the DeGroot process — equal epistemic power — so that the consensus reflects $N_{\text{eff}} = N$ rather than $N_{\text{eff}} = 1$. Proposition 8.5 tells us that this equalization would make the consensus more accurate, not less, as long as different traditions carry independent and comparably precise information (with unequal precisions, the accuracy-maximizing weights are proportional to precision — which is itself an empirical question, not something W^I should settle by fiat).

Quijano wrote in Spanish, from Lima. He did not have the Perron eigenvector. He had the diagnosis: that the pattern of power is maintained not primarily by guns but by the control of what counts as knowledge. This chapter gives that diagnosis its eigenvalue.

Further Reading

DeGroot learning: DeGroot (1974), "Reaching a Consensus." Golub and Jackson (2010), "Naïve Learning in Social Networks and the Wisdom of Crowds" — the key paper: naive learning is wise if and only if the most influential agent's weight vanishes. DeMarzo et al. (2003), "Persuasion Bias, Social Influence, and Unidimensional Opinions." Molavi et al. (2018), "A Theory of Non-Bayesian Social Learning." Chandrasekhar et al. (2020), "Testing Models of Social Learning on Networks."

Bayesian vs. naïve learning: Acemoglu et al. (2011), "Bayesian Learning in Social Networks." Jadbabaie et al. (2012), "Non-Bayesian Social Learning."

Misinformation and manipulation: Acemoglu et al. (2010), "Spread of (Mis)information in Social Networks." Candogan and Drakopoulos (2020), "Optimal Signaling of Content Accuracy." Allcott and Gentzkow (2017), "Social Media and Fake News in the 2016 Election."

Decolonial theory: Quijano (2000a), "Colonialidad del poder, eurocentrismo y América Latina," in Lander (ed.), *La colonialidad del saber*, Buenos Aires: CLACSO (the primary source). Quijano (2000b), "Colonialidad del poder y clasificación social" (source of the epigraph). Mignolo (2000), *Local Histories/Global Designs*. For the connection to economics: Ndikumana and Boyce (2011), *Africa's Odious Debts*, Chapter 1.

Algorithmic information networks: Ribeiro et al. (2020), "Auditing Radicalization Pathways on YouTube." Huszár et al. (2022), "Algorithmic Amplification of Politics on Twitter."

PART III

The Formation Model

Chapter 9

Where Preferences Come From

The Assumption Economics Is Afraid to Touch

> *"O mito do desenvolvimento econômico tem exercido, no plano das relações internacionais, uma função similar àquela desempenhada, no plano das relações de classe, pelos mitos criados pela ideologia da classe dominante."*
>
> ["The myth of economic development has exercised, in the plane of international relations, a function similar to that performed, in the plane of class relations, by the myths created by the ideology of the dominant class."]
>
> — Celso Furtado, *O Mito do Desenvolvimento Econômico* (1974)

Part II derived the allocation from the weight matrix: given $(W, \boldsymbol{\omega}, \boldsymbol{\theta})$, the bargaining equilibrium produces $\mathbf{x}^*$. The weights $\boldsymbol{\pi}(W)$ and endowments $\boldsymbol{\omega}$ were computed; the preference vector $\boldsymbol{\theta}$ was assumed.

This chapter removes the last exogenous object. It asks: where does $\boldsymbol{\theta}$ come from?

The answer proposed here is W^I: preference parameters are modelled as the output of a formation process driven by the outcomes an agent observes, and which outcomes she observes is set by the information network. A discount factor, an aspiration level, or a labor-supply norm then becomes a function of the W^I-weighted returns of an agent's reference group rather than a draw from a fixed distribution.

This is Part III's working hypothesis: **preference parameters that the benchmark model treats as primitives can be modelled as the output of a formation process mediated by W^I.** Section 9.1.4 asks how much of the evidence is about preferences rather than beliefs. Chapter 10 will write the formal model. Chapter 11 will measure the formation effect empirically. Chapter 12 will ask what

happens to welfare analysis when the welfare function is endogenous. This chapter lays the conceptual and empirical foundation.

9.1 The *De Gustibus* Assumption

9.1.1 What Economists Assume

In 1977, George Stigler and Gary Becker published "De Gustibus Non Est Disputandum" in the *American Economic Review* (Stigler and Becker, 1977). Their argument: economists should not try to explain tastes. Tastes are stable, exogenous, and uniform across agents (or at least randomly distributed). All observable differences in behavior can be explained by differences in prices and constraints, not differences in preferences.

This is assumption A6 from Chapter 4: $\boldsymbol{\theta}$ is fixed. It is the deepest assumption in the standard model — the one that makes all other analysis possible. If preferences are exogenous, the economist's job is to study how constraints (prices, endowments, institutions) map into outcomes. If preferences are endogenous, everything changes: the constraints shape the preferences that evaluate the outcomes that the constraints produce.

Definition 9.1 (The Preference Vector)**.** *The* ***preference vector*** $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_N)' \in \mathbb{R}^N_+$ *collects each agent's preference parameter. In different applications,* θ_i *represents:*

Context	θ_i ***represents***
Savings/investment	*Time discount factor (θ_i high = patient)*
Education	*Aspiration level (θ_i high = ambitious)*
Labor supply	*Work ethic / market orientation (θ_i high = more market hours)*
Entrepreneurship	*Risk tolerance (θ_i high = risk-seeking)*
Consumption	*Taste for quality / variety (θ_i high = diversified demand)*

Under A6: θ_i is drawn once and never changes. Under the formation model: $\theta_i = f(W^I, \boldsymbol{\theta}_{-i}, history)$.

9.1.2 Why Becker and Stigler Were Right (Methodologically)

Becker and Stigler's position was not about the real world. It was about methodology: if you allow preferences to be endogenous, you can "explain" any observed behavior by invoking the right preference change. The theory becomes unfalsifiable. Explaining high savings by "patient preferences" and then explaining low savings by "impatient preferences" is circular.

This is a legitimate concern. Any model of endogenous preferences must be disciplined by three requirements:

1. **The formation process must be specified**: not "preferences change" but "preferences change according to this equation with these observable inputs."

2. **The formation process must be falsifiable**: the equation must generate testable predictions distinct from those of a model with fixed preferences.

3. **The formation effect must be measurable**: we must be able to separate the part of behavior explained by preference change from the part explained by price/constraint change.

The formation model of Chapter 10 is built to meet the first two requirements; Chapter 11 asks how far the data can meet the third. The model's key assumption is that preferences evolve on a *different timescale* than prices — between generations rather than within them. Chapter 11 shows that this assumption helps but does not by itself deliver identification.

Becker himself later relaxed the stability assumption: in *Accounting for Tastes* (1996) and Becker and Murphy, *Social Economics* (2000), current choices depend on stocks of personal and social capital accumulated from past consumption and from the choices of one's peers. The formation equation below is close in spirit: the stock is θ_i and the peer input arrives through W^I.

9.1.3 What the Evidence Says

Five preference parameters that the benchmark model treats as exogenous. For each, evidence that it varies with the social and informational environment. The evidence is uneven: strong for aspirations and labor norms, thinner for time preference, and in several cases (flagged below) it concerns *beliefs* rather than tastes.

Time Discount (δ_i)

Standard assumption: δ_i is a deep parameter of the utility function, stable across time, drawn from a population distribution. Empirically:

Stylized Fact 9.2 (Patience Varies Systematically)**.**

1. *The Global Preferences Survey (Falk et al., 2018, QJE) measures patience with experimentally validated survey items in 76 countries. Patience varies substantially both across and within countries, and country-average patience correlates with savings, educational attainment and income.*

2. *Variation across countries is consistent with, but does not by itself establish, formation through W^I: it is equally consistent with selection, institutions, or genes. Direct causal evidence on network formation of time preference is thinner than for the parameters below.*

In the formation model of Section 9.2,

$$\delta_{i,t+1} = \rho_\delta \delta_{i,t} + (1 - \rho_\delta)\bar{\delta} + \mu_\delta \Big(\sum_j W^I_{ij} r_{j,t} - r^* \Big),$$

where $r_{j,t}$ is the return observed from agent j. Agents who observe high returns (through connections to the modern sector) form higher patience; agents who observe low returns (through connections to the traditional sector) discount the future more steeply.

Aspiration Level (a_i)

Standard assumption: aspirations are not modeled. Empirically:

Stylized Fact 9.3 (Network-Mediated Aspirations)**.**

1. *Randomly showing Ethiopian farmers a documentary about successful farmers in their region raised the farmers' aspirations, savings, and children's school enrollment six months later (Bernard et al., 2014). In this chapter's notation, the treatment was a change in W^I: the film added a new edge from the successful farmer (signal) to the viewing farmer (receiver).*

2. *Beaman et al. (2012, Science) found that randomly assigned exposure to female village leaders in West Bengal raised parents' and adolescent girls' aspirations for girls' education and careers, and girls' educational attainment. The treatment was a change in the composition of signals reaching the girl through W^I.*

3. *Ray (2006) developed a theory of "aspiration windows": the set of outcomes a person considers possible, determined by the outcomes observed in her reference group. In this chapter's notation, the aspiration window is the range of W^I-weighted signals.*

Labor Supply Elasticity (η_i)

Standard assumption: the elasticity of labor supply is a preference parameter governing the tradeoff between leisure and consumption. Empirically:

Stylized Fact 9.4 (Network-Mediated Labor Norms)**.**

1. *Female labor force participation varies widely across regions at similar income levels. Among second-generation American women, who all face US prices and institutions, labor supply is predicted by female participation in their parents' country of origin, conditional on individual and parental characteristics (Fernández, 2007; Fernández and Fogli, 2009). This is the "epidemiological" evidence of Chapter 11. It identifies a cultural component broadly defined (beliefs, norms or tastes), not a preference parameter specifically.*

2. *Fogli and Veldkamp (2011) document that the rise in US female participation spread geographically like an information diffusion process, and model it as women learning from nearby working mothers about the cost of maternal employment. The mechanism is learning: a belief, transmitted locally.*

3. *Bursztyn et al. (2020, AER) find that most young married men in Saudi Arabia privately support women working outside the home but underestimate other men's support. Correcting that belief raised husbands' willingness to help their wives search for jobs. What changed was a belief about the norm, not the men's own preference.*

Risk Tolerance (σ_i)

Standard assumption: risk aversion is stable and innate. Empirically:

Stylized Fact 9.5 (Experience-Dependent Risk Attitudes)**.**

1. *Malmendier and Nagel (2011, QJE): households that have experienced low stock-market returns over their lifetimes (with more weight on recent years) report lower willingness to take financial risk and invest less in stocks. This is own experience — in network terms, the diagonal of W^I — rather than a signal relayed by others.*

2. *Cameron and Shah (2015, JHR): individuals in Indonesian villages recently hit by floods or earthquakes are more risk-averse in incentivized gambles.*

3. *Dohmen et al. (2012, REStud): children's risk and trust attitudes are positively related to their parents' attitudes and to the prevailing attitudes in their region — a parental and a local channel of transmission. A parent–child correlation is not an estimate of the persistence parameter ρ below without further assumptions (Chapter 11).*

Taste for Variety / Quality (ν_i)

Standard assumption: consumers have CES preferences with elasticity σ as a fixed parameter. Empirically:

Stylized Fact 9.6 (Formed and Persistent Consumption Tastes)**.**

1. *Atkin (2013, AER): interstate migrants within India retain food tastes formed in their region of origin, even when those foods are relatively expensive at the destination. Tastes are formed early and persist.*

2. *Bronnenberg et al. (2012, AER): brand preferences for consumer packaged goods depend on where consumers lived early in life. Migrants' brand shares move part of the way toward their new location's shares soon after moving, but a substantial gap persists for decades.*

3. *The "demonstration effect" (Duesenberry, 1949; applied internationally by Nurkse, 1953) is the idea that consumers emulate the consumption of the groups they observe. Furtado (1974) made the international version central to his account of dependency: peripheral elites adopt centre consumption patterns transmitted through trade, media and education.*

9.1.4 Beliefs or Preferences?

Much of the evidence above concerns *beliefs* rather than tastes. Fernández (2013, *AER*) and Fogli and Veldkamp (2011) model the rise of female participation as learning about the cost of working. Bursztyn et al. (2020) correct a belief about a norm. Malmendier and Nagel's households update beliefs about stock returns. A model in which W^I transmits beliefs is the DeGroot model of Chapter 8, not a model of preference formation.

The distinction matters for two reasons.

1. **Identification.** Social learning (I copy you because your choice tells me something) and social utility (I copy you because I care about what you do) predict the same correlation. Separating them requires designs such as Bursztyn et al. (2014, *Econometrica*), who randomize whether a peer's choice is merely observed or also implemented.

2. **Welfare.** Correcting a mistaken belief improves an agent's choices by her own lights. Changing her tastes changes the lights by which choices are judged. Chapter 12 is only needed for the second case.

For a formal theory of preference change driven by changes in which considerations an agent finds motivationally salient, see Dietrich and List (2013).

Convention for Part III. θ_i is a *reduced-form behavioral parameter*: the part of i's behavior that is not explained by current prices and constraints. It may be a taste, a

belief about returns, or a perceived norm. Where the distinction matters — above all for welfare in Chapter 12 — we say which reading is in force.

9.2 The Formation Equation

9.2.1 The General Form

All five channels share a common mathematical structure. Agent i's preference parameter evolves according to:

Definition 9.7 (The Formation Equation)**.** *Let i index a **dynasty** (a family line or a group of agents with a common position) and $t = 0, 1, 2, \ldots$ index **generations**. One period is one generation, about 25 years. Then*

$$\theta_{i,t+1} = \underbrace{\rho\theta_{i,t}}_{\text{persistence}} + \underbrace{(1-\rho)\bar{\theta}}_{\text{anchor}} + \underbrace{\mu\left(\sum_j W^I_{ij} r_{j,t} - r^*\right)}_{\text{formation signal}}, \tag{9.1}$$

where:

- $\rho \in (0, 1)$: ***intergenerational persistence***. *The share of the parent generation's preference that the child generation inherits within the dynasty. High ρ: preferences change slowly (culture is sticky). Low ρ: preferences change fast (culture is fluid). Throughout Part III, ρ is per generation. A per-decade coefficient ρ_{10} corresponds to $\rho = \rho_{10}^{2.5}$.*

- $\bar{\theta}$: ***long-run anchor***. *The preference level that would prevail in isolation, absent any network signal. A biological or deep-cultural baseline.*

- $\mu > 0$: ***formation sensitivity***. *How strongly the network signal shifts preferences. Large μ: signals reshape preferences powerfully. Small μ: preferences are robust to signals.*

- $r_{j,t}$: ***observed return*** *of agent j in generation t, as seen by the children of generation t during their formative years: income, education outcome, consumption, social standing.*

- r^*: ***reference return***. *The benchmark against which observed returns are evaluated.*

- W^I_{ij}: ***information weight***. *How much of agent j's return agent i observes. W^I is row-stochastic, so the signal is a weighted average.*

The equation is linear, so it can push θ_i outside the range in which the parameter makes sense (a discount factor outside $(0, 1)$, say). Read it as a local approximation around the relevant range.

The formation signal $\sum_j W^I_{ij} r_{j,t} - r^*$ is the gap between the W^I-weighted average of observed returns and the reference. When the people you observe are doing well ($\sum_j W^I_{ij} r_{j,t} > r^*$), your preference shifts upward (more patient, more ambitious, more market-oriented). When they are doing poorly, your preference shifts downward.

Intuition: What the equation says in words

Your preferences are a mixture of three things: (1) who you were yesterday ($\rho\theta_{i,t}$), (2) a deep anchor that pulls you toward a baseline ($\bar{\theta}$), and (3) what you see in your network, compared to your expectations. If your network shows you high returns and your expectations are low, your preferences shift upward. If your network shows you low returns and your expectations are high, your preferences shift downward.

The relative strength of (1), (2), and (3) determines the dynamics. When ρ is large, yesterday's preferences dominate — culture is sticky. When μ is large, the network signal dominates — culture is malleable. The ratio $\mu/(1-\rho)$ is the key parameter: it measures how much of the long-run preference level is determined by the network versus the deep anchor. Chapter 10 formalizes this.

9.2.2 Two Timescales

The formation equation operates on a different timescale than the market. This is a modelling assumption, not a result:

Definition 9.8 (Two-Timescale System)**.** *In the combined system:*

1. ***Fast dynamics*** *(within a generation): given $\boldsymbol{\theta}_t$, the market clears and the bargaining equilibrium $\mathbf{x}^*(\boldsymbol{\theta}_t, W)$ is reached within a period. Returns $\mathbf{r}_t = g(\mathbf{x}^*_t)$ are realized.*

2. ***Slow dynamics*** *(between generations): the returns $\mathbf{r}_t$ enter the formation equation (9.1), producing $\boldsymbol{\theta}_{t+1}$. The new preferences generate a new bargaining equilibrium $\mathbf{x}^*_{t+1}$.*

The fast dynamics are a fixed-point problem in $\mathbf{x}$ given $\boldsymbol{\theta}$. The slow dynamics are a fixed-point problem in $\boldsymbol{\theta}$ given the function $\mathbf{x}^*(\boldsymbol{\theta})$.

The separation suggests an identification strategy. Within a generation, behavior responds to prices at (approximately) fixed preferences; across generations, preferences move. Chapter 11 shows that this does *not* by itself identify the formation parameters: θ is unobserved, prices are not stable across generations, and the cohort–year decomposition that implements the idea is not identified without further restrictions.

9.2.3 The Steady State

Proposition 9.9 (Formation Steady State)**.** *Let returns be $r_{j,t} = g_j(\boldsymbol{\theta}_t)$, and define the map $T(\boldsymbol{\theta}) := \rho\boldsymbol{\theta} + (1-\rho)\bar{\theta}\mathbf{1} + \mu(W^I\mathbf{g}(\boldsymbol{\theta}) - r^*\mathbf{1})$. Suppose either*

(a) $\mathbf{g}$ *is continuous and T maps a nonempty compact convex set $\Theta \subset \mathbb{R}^N$ into itself (this holds, for example, if every g_j is continuous and bounded); or*

(b) $\mathbf{g}$ *is affine, $\mathbf{g}(\boldsymbol{\theta}) = \mathbf{a} + G\boldsymbol{\theta}$, and $I - \frac{\mu}{1-\rho}W^IG$ is nonsingular.*

Then a steady state $\boldsymbol{\theta}^*$ exists (in case (b) it is unique), and it satisfies

$$\theta_i^* = \bar{\theta} + \frac{\mu}{1-\rho}\left(\sum_j W_{ij}^I g_j(\boldsymbol{\theta}^*) - r^*\right). \tag{9.2}$$

The steady-state preference deviates from the anchor $\bar{\theta}$ by an amount proportional to $\mu/(1-\rho)$ times the network-weighted return gap.

Proof. Set $\theta_{i,t+1} = \theta_{i,t} = \theta_i^*$ in (9.1): $\theta_i^* = \rho\theta_i^* + (1-\rho)\bar{\theta} + \mu(\sum_j W_{ij}^I r_j^* - r^*)$. Solve: $(1-\rho)\theta_i^* = (1-\rho)\bar{\theta} + \mu(\sum_j W_{ij}^I r_j^* - r^*)$, giving (9.2). Under (a), existence follows from Brouwer's fixed-point theorem applied to the continuous map $T : \Theta \to \Theta$. If $\mathbf{g}$ is bounded, say $|g_j| \le B$, then T maps the cube $\{\|\boldsymbol{\theta} - \bar{\theta}\mathbf{1}\|_\infty \le \frac{\mu}{1-\rho}(B + |r^*|)\}$ into itself. Under (b), the steady-state condition is the linear system $(I - \frac{\mu}{1-\rho}W^IG)\boldsymbol{\theta}^* = \bar{\theta}\mathbf{1} + \frac{\mu}{1-\rho}(W^I\mathbf{a} - r^*\mathbf{1})$, which has a unique solution when the matrix is nonsingular. No irreducibility of W^I is needed in either case. Note that the affine examples in this chapter are covered by (b), not by Brouwer: an unbounded linear g does not map a compact set into itself in general. □

Example 9.10 (Heterogeneous Steady States (Not Multiple Equilibria))**.** *$N = 2$ agents: Modern (M) and Traditional (T). Block-diagonal $W^I = I$: each agent observes only her own sector. $\bar{\theta} = 1$, $\rho = 0.8$, $\mu = 0.3$, $r^* = 0.1$.*

Returns: $r_M(\theta_M) = 0.05 + 0.1\theta_M$ (high ambition yields high returns in the modern sector), $r_T(\theta_T) = 0.03 + 0.02\theta_T$ (ambition yields little in the traditional sector — low complementarity).

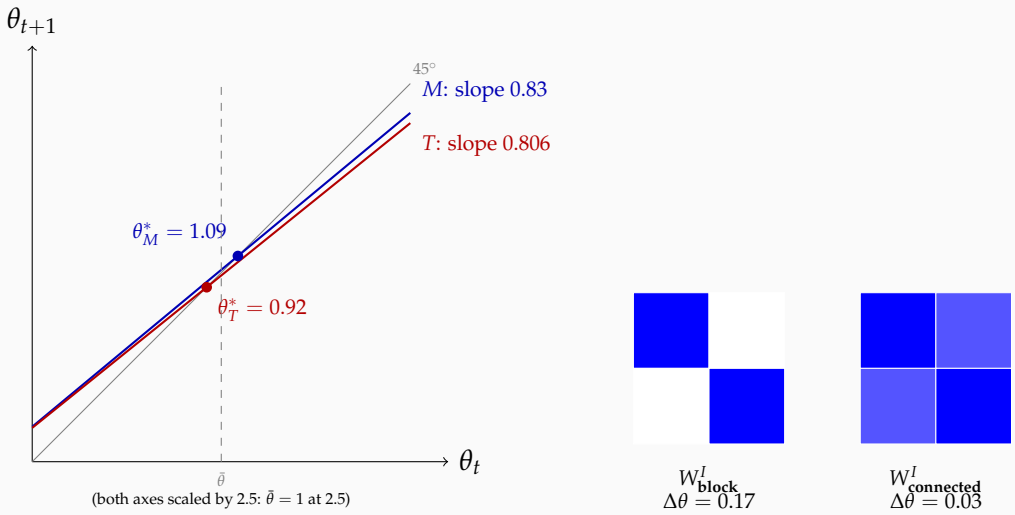
Steady state for M: $\theta_M^* = 1 + \frac{0.3}{0.2}(0.05 + 0.1\theta_M^* - 0.1) = 1 + 1.5(0.1\theta_M^* - 0.05)$. *Solving:* $\theta_M^* = 1 + 0.15\theta_M^* - 0.075$, *so* $0.85\theta_M^* = 0.925$, *giving* $\theta_M^* = 1.088$.

Steady state for T: $\theta_T^* = 1 + 1.5(0.02\theta_T^* + 0.03 - 0.1) = 1 + 0.03\theta_T^* - 0.105$. *Solving:* $0.97\theta_T^* = 0.895$, *giving* $\theta_T^* = 0.923$.

The Modern-sector agent ends up more ambitious ($\theta_M^* = 1.09$) *and the Traditional-sector agent less ambitious* ($\theta_T^* = 0.92$) *than the anchor* ($\bar{\theta} = 1$). *The difference* ($\Delta\theta = 0.166$) *comes from the heterogeneous return functions* $g_M \neq g_T$, *given that each agent observes only her own sector. With identical return functions the gap would be zero for any* W^I. *What* W^I *does is decide whose returns enter whose signal. Connecting the agents with* $W^I = \begin{pmatrix} 1-w & w \\ w & 1-w \end{pmatrix}$ *shrinks the gap:* 0.062 *at* $w = 0.3$, 0.031 *at* $w = 0.4$, *and* 0 *at* $w = 0.5$, *where both observe the same signal.*

This is heterogeneity, not multiplicity. Each agent's map is a contraction (slopes $\rho + \mu g_M' = 0.83$ *and* $\rho + \mu g_T' = 0.806$), *so each has a unique, globally stable steady state. Each steady state is consistent with the returns the agent observes: high ambition produces high returns, which sustain high ambition. Nothing here is self-reinforcing in the sense of a trap, however — a shock to* θ_T *dies out and* θ_T *returns to 0.923. Genuine multiplicity, where history selects among steady states, needs nonlinear returns (Chapter 10, Example 10.11).*

Visualization: Two agents, two steady states, one W^I



Left: phase diagram. Each map crosses the 45° line once with slope below 1, so each agent has a unique stable steady state; the Modern agent's lies above the anchor $\bar{\theta}$ and the Traditional agent's below it. Right: with $W^I = I$ the preference gap is 0.17; a connected W^I with weights 0.6/0.4 reduces it to 0.03 (0.031). The gap comes from different returns, and W^I decides how much of the other sector's returns each agent sees.

9.3 Persistence: Why the Past Lingers

The formation equation has an important dynamic property: after W^I changes, preferences adjust only gradually, and slowly if ρ is close to 1.

Proposition 9.11 (Persistence of Preferences)**.** *Suppose W^I changes from W^I_{old} to W^I_{new} at $t = 0$, starting from $\boldsymbol{\theta}_0 = \boldsymbol{\theta}^*_{old}$. Let $\boldsymbol{\theta}^*_{new}$ be a steady state of the new map, let $\mathbf{g}$ be differentiable there, and let*

$$J_{new} := \rho I + \mu W^I_{new} D\mathbf{g}(\boldsymbol{\theta}^*_{new}), \qquad \lambda := \rho(J_{new}) < 1. \tag{9.3}$$

Then for every $\varepsilon > 0$ there are constants $C, \delta > 0$ such that, whenever $\|\boldsymbol{\theta}^*_{old} - \boldsymbol{\theta}^*_{new}\| < \delta$,

$$\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*_{new}\| \le C \cdot (\lambda + \varepsilon)^t \cdot \|\boldsymbol{\theta}^*_{old} - \boldsymbol{\theta}^*_{new}\|. \tag{9.4}$$

If $\mathbf{g}$ is affine the bound holds globally. The asymptotic half-life of preference adjustment is

$$t_{1/2} = \frac{\ln 2}{\ln(1/\lambda)} \quad \text{generations.} \tag{9.5}$$

If returns are increasing in preferences ($D\mathbf{g} \geq 0$), then $\lambda \geq \rho$: complementarities slow adjustment down. Equality $\lambda = \rho$ holds when the formation signal does not respond to $\boldsymbol{\theta}$ ($D\mathbf{g} = 0$), and more generally exactly when $W^I_{new} D\mathbf{g}$ has spectral radius zero; it fails, for example, as soon as some agent observes herself ($W^I_{ii} > 0$) and her own return rises with her own preference ($\partial g_i / \partial \theta_i > 0$).

Proof. Write T for the new map and $\Delta_t = \boldsymbol{\theta}_t - \boldsymbol{\theta}^*_{\text{new}}$. Since $T(\boldsymbol{\theta}^*_{\text{new}}) = \boldsymbol{\theta}^*_{\text{new}}$ and $T(\boldsymbol{\theta}) - T(\boldsymbol{\theta}^*_{\text{new}}) = \rho\Delta + \mu W^I_{\text{new}}(\mathbf{g}(\boldsymbol{\theta}) - \mathbf{g}(\boldsymbol{\theta}^*_{\text{new}}))$, differentiability of $\mathbf{g}$ gives $\Delta_{t+1} = J_{\text{new}}\Delta_t + R(\Delta_t)$ with $\|R(\Delta)\| = o(\|\Delta\|)$. It suffices to prove the bound for $\varepsilon < 1 - \lambda$, since $(\lambda + \varepsilon)^t$ is increasing in ε. For such ε there is a vector norm $\|\cdot\|_\star$ whose induced matrix norm satisfies $\|J_{\text{new}}\|_\star \leq \lambda + \varepsilon/2$ (Horn and Johnson, 2012, Lemma 5.6.10). Let $c\|\cdot\| \leq \|\cdot\|_\star \leq c'\|\cdot\|$ (equivalence of norms on $\mathbb{R}^N$). Choose $\eta > 0$ such that $\|R(\Delta)\|_\star \leq (\varepsilon/2)\|\Delta\|_\star$ whenever $\|\Delta\|_\star < \eta$, and set $\delta := \eta / c'$. If $\|\Delta_0\| < \delta$ then $\|\Delta_0\|_\star < \eta$, and whenever $\|\Delta_t\|_\star < \eta$ we get $\|\Delta_{t+1}\|_\star \leq (\lambda + \varepsilon)\|\Delta_t\|_\star < \eta$ because $\lambda + \varepsilon < 1$. By induction $\|\Delta_t\|_\star \leq (\lambda + \varepsilon)^t \|\Delta_0\|_\star$ for all t, and (9.4) follows with $C := c'/c$. If $\mathbf{g}$ is affine, $R \equiv 0$, so no δ is needed and $\|\Delta_t\|_\star \leq \|J_{\text{new}}\|^t_\star \|\Delta_0\|_\star$ for every Δ_0. The half-life (9.5) solves $\lambda^t = 1/2$, the leading rate as $\varepsilon \to 0$.

Finally, if $D\mathbf{g} \geq 0$ then $A := \mu W^I_{\text{new}} D\mathbf{g} \geq 0$ (as $W^I \geq 0$ and $\mu > 0$), and the eigenvalues of $J_{\text{new}} = \rho I + A$ are $\rho + \alpha$ for α an eigenvalue of A. By Perron–Frobenius for nonnegative matrices (Horn and Johnson, 2012, Thm. 8.3.1), $\rho(A)$ is itself an eigenvalue of A, so $\lambda \geq \rho + \rho(A)$; conversely $|\rho + \alpha| \leq \rho + |\alpha| \leq \rho + \rho(A)$ for every eigenvalue α. Hence $\lambda = \rho + \rho(A) \geq \rho$, with equality iff $\rho(A) = 0$. Since $0 \leq \operatorname{diag}(A_{11}, \ldots, A_{NN}) \leq A$, monotonicity of the spectral radius (Horn and Johnson, 2012, Thm. 8.1.18) gives $\rho(A) \geq \max_i A_{ii}$, and $A_{ii} \geq \mu W^I_{ii} \partial g_i / \partial \theta_i$, which gives the final example. □

Example 9.12 (Persistence Rates)**.** ***Half-lives when the signal does not feed back ($\lambda = \rho$).*** *With $\rho = 0.9$, the half-life is $\ln 2 / \ln(10/9) \approx 6.6$ generations, or about 164 years at 25 years per generation.*

Suppose, hypothetically, that a country's W^I changed at $t = 0$ and $\lambda = 0.9$. One generation later 10% of the gap to the new steady state has closed; after two generations 19%; after four (a century) 34%. Preferences formed under the old structure would still account for most of the gap a century later.

$\rho = 0.7$: half-life $= \ln 2 / \ln(10/7) \approx 1.9$ generations (49 years), about 3.4 times faster than $\rho = 0.9$.

$\rho = 0.5$: half-life $= 1$ generation (25 years).

With feedback. *In Example 9.10, the Modern agent's map has slope $\lambda = \rho + \mu g'_M = 0.8 + 0.3 \times 0.1 = 0.83$. The half-life is therefore $\ln 2 / \ln(1/0.83) = 3.7$ generations, not*

the 3.1 *implied by* $\rho = 0.8$ *alone.*

Which value of ρ *applies is an empirical question (Chapter 11). The table is a menu, not an estimate.*

ρ	$t_{1/2}$ *(generations)*	$t_{1/2}$ *(years)*	*Speed*
0.95	*13.5*	*338*	*Glacial*
0.90	*6.6*	*164*	*Slow*
0.80	*3.1*	*78*	*Moderate*
0.70	*1.9*	*49*	*Visible*
0.50	*1.0*	*25*	*Fast*

Key Structural Insight

Persistence is one reason structural change can be slow. W^P can change within a decade (a road, a factory), and W^I within years (phones, a media campaign). But if λ is close to 1, preferences formed under the *old* W^I persist for generations.

This means: (1) structural policies that change W have a delayed effect on preferences — the payoff is intergenerational, (2) evaluating policies using *current* preferences underestimates their long-run impact, because current preferences were formed under the old structure, and (3) the welfare function itself changes over time — which is the problem of Chapter 12.

Persistence can explain why countries with similar current institutions and prices have different outcomes: their preference vectors $\boldsymbol{\theta}$ reflect different histories of W^I. With a unique stable steady state this is slow convergence, not hysteresis. True path dependence — where a temporary shock has permanent effects — requires multiple steady states (Chapter 10).

9.4 Enforcement vs. Formation

The standard model explains behavior through two channels: **prices** (you work more because wages rose) and **constraints** (you work more because the law changed). Both operate within a generation, holding $\boldsymbol{\theta}$ fixed. We call these **enforcement effects**: the structure enforces behavior through prices and constraints.

The formation model adds a third channel: **the structure changes $\boldsymbol{\theta}$**. You work more because your reference group works more, and observing them shifted your norm. This is the **formation effect**: the structure forms preferences.

Definition 9.13 (Enforcement, Formation, and the Ψ Ratio)**.** *Fix an outcome $y_i = y_i(\boldsymbol{\theta}, W, \boldsymbol{\omega})$ for agent i: a behavior, an allocation, or a return. The economy starts at a steady state $\boldsymbol{\theta}^0 = \boldsymbol{\theta}^*(W)$. At $t = 0$ a permanent perturbation ΔW is applied. The perturbation must be stated explicitly: which layer (W^P, W^I, ...) changes, how its rows remain stochastic, or which transfer $\Delta\boldsymbol{\omega}$ is made. Let $\boldsymbol{\theta}_T$ be the preference vector T generations later, generated by the formation equation under the new structure. Then:*

$$E_i(\Delta W) := y_i(\boldsymbol{\theta}^0, W + \Delta W) - y_i(\boldsymbol{\theta}^0, W), \tag{9.6}$$

$$F_i(\Delta W, T) := y_i(\boldsymbol{\theta}_T, W + \Delta W) - y_i(\boldsymbol{\theta}^0, W + \Delta W). \tag{9.7}$$

*E_i is the **enforcement effect** (the impact at fixed preferences); F_i is the **formation effect** by horizon T. The two effects add up to the total change, $E_i + F_i = y_i(\boldsymbol{\theta}_T, W + \Delta W) - y_i(\boldsymbol{\theta}^0, W)$. The Ψ **ratio** of agent i for perturbation ΔW at horizon T is*

$$\Psi_i(\Delta W, T) := \frac{F_i(\Delta W, T)}{E_i(\Delta W)}, \qquad \text{defined when } E_i(\Delta W) \neq 0. \tag{9.8}$$

The long-run ratio $\Psi_i(\Delta W, \infty)$ uses the new steady state, $\boldsymbol{\theta}_\infty = \boldsymbol{\theta}^*(W + \Delta W)$. The aggregate ratio is

$$\bar{\Psi}(\Delta W, T) := \frac{\sum_i |F_i(\Delta W, T)|}{\sum_i |E_i(\Delta W)|}.$$

Absolute values are needed when allocations sum to a fixed total, because then $\sum_i E_i = \sum_i F_i = 0$.

This is the only definition of Ψ used in the book. Chapter 10 computes it inside the formation model. Chapter 11 discusses statistics that are related to it but are not the same object.

Remark 9.14 (Reading Ψ).

1. Ψ is a property of a *perturbation, an outcome, an agent, and a horizon*, not of an economy. The same economy has different Ψ for a transfer and for a road.

2. $\Psi_i(\Delta W, 0) = 0$, and if $\mu = 0$ then $F_i \equiv 0$ for every perturbation: with exogenous preferences (A6) the benchmark model is complete.

3. $\Psi > 0$: formation reinforces enforcement. $\Psi > 1$: most of the change by horizon T comes from preference change. $\Psi < 0$: formation offsets enforcement. That is possible; nothing forces the two effects to share a sign.

4. An evaluation that ignores formation captures the share $1/(1 + \Psi)$ of the total effect at horizon T and misses the share $\Psi/(1 + \Psi)$ (for $\Psi > -1$).

5. When $E_i = 0$, Ψ_i is undefined. This happens, for example, for a pure information policy whose outcome does not depend on W^I at fixed preferences. Report F_i itself in that case.

Chapter 11 asks what data can reveal about Ψ. The short answer: less than one would like, and only under strong assumptions.

9.5 The Self-Confirming Trap

Proposition 9.15 (Block-wise Steady States)**.** *Let W^I be row-stochastic and block-diagonal with blocks $B_1, \dots, B_k$ (the weights within a block are arbitrary), and let the return functions be identical within each block and depend only on the block's own preferences: $g_j(\boldsymbol{\theta}) = h_b(\boldsymbol{\theta}_{B_b})$ for every $j \in B_b$. Suppose either (a) every h_b is continuous and bounded, or (b) $\mathbf{g}$ is affine and case (b) of Proposition 9.9 holds. Then a steady state exists in which every agent of block b has the same preference θ_b^*, satisfying*

$$\theta_b^* = \bar{\theta} + \frac{\mu}{1-\rho}\left(\bar{r}_b(\theta_b^*) - r^*\right), \tag{9.9}$$

where $\bar{r}_b(\theta_b^*)$ is the average return within block b at steady state. Each block's preference level is consistent with the returns it observes: agents in block b observe $\bar{r}_b$ and form the θ_b^* that those returns sustain.

Proof. With block-diagonal W^I, the formation equation decomposes: agents in block b observe only returns from block b. Let S be the set of block-constant vectors, $\boldsymbol{\theta} = \sum_b c_b \mathbf{1}_{B_b}$. At $\boldsymbol{\theta} \in S$ every $j \in B_b$ earns the same return $\bar{r}_b(c_b) := h_b(c_b \mathbf{1}_{B_b})$, so every $i \in B_b$ receives the signal $\sum_{j \in B_b} W^I_{ij} \bar{r}_b(c_b) - r^* = \bar{r}_b(c_b) - r^*$ (rows sum to one). Hence T maps S into S, and a block-constant steady state is a vector $(c_1, \dots, c_k)$ solving the k decoupled scalar equations $c_b = \bar{\theta} + \frac{\mu}{1-\rho}(\bar{r}_b(c_b) - r^*)$, which is (9.9).

(a) If $|h_b| \le B$, the continuous function $f_b(c) := c - \bar{\theta} - \frac{\mu}{1-\rho}(\bar{r}_b(c) - r^*)$ is negative at $c = \bar{\theta} - K$ and positive at $c = \bar{\theta} + K$ for $K > \frac{\mu}{1-\rho}(B + |r^*|)$, so it has a zero by the intermediate value theorem.

(b) If $\mathbf{g} = \mathbf{a} + G\boldsymbol{\theta}$, then $\bar{r}_b(c) = \alpha_b + \beta_b c$ is affine and $W^I G$ maps S into S, acting on block b's coordinate as multiplication by β_b. So each β_b is an eigenvalue of $W^I G$, and nonsingularity of $I - \frac{\mu}{1-\rho} W^I G$ implies $1 - \frac{\mu}{1-\rho}\beta_b \neq 0$. Each scalar equation then has a unique solution, and by uniqueness in Proposition 9.9(b) the resulting block-constant vector is *the* steady state. □

Intuition: The deepest trap

The block-wise steady state is one way to formalize the idea behind the phrase "culture of poverty" — without its pejorative connotation, and with the cause placed in the structure rather than in the people.

In the model, an agent in the traditional sector is not "poor because she has low aspirations." She has low aspirations because W^I shows her only traditional-sector returns. If she observed the modern sector's returns, she would form higher aspirations. Whether that helps her depends on whether those returns are *attainable* for her. In Example 9.10 her own return function is g_T, and an aspiration calibrated on g_M would overshoot what her sector rewards. The Prebisch box below is a case where the imported signal does harm. An edge in W^I is not good or bad in itself; what matters is whether the returns it carries are relevant to the observer.

So part of the problem lies in the network, not in the preferences. Change W^I *together with* access to the observed returns (the scholarship, the job network, the mentor in the modern sector), and the preferences change. Leave W^I and the return structure unchanged, and the preferences persist.

This is Furtado's "autonomous process of underdevelopment" (Chapter 2) applied to preferences: the structure produces the preferences that justify the structure.

Prebisch Connection: Demonstration effect and cultural dependency

Nurkse (1953) called it the international "demonstration effect", extending Duesenberry's (1949) idea: consumers in poorer countries emulate the consumption standards of richer ones, which depresses saving. Furtado (1974) made it a pillar of cultural dependency. Income concentration lets a peripheral minority reproduce centre consumption patterns, and that composition of demand then shapes what the periphery's industry produces and which techniques it imports. Prebisch's late work (Prebisch, 1981) attacked the same "privileged consumer society". In network terms, peripheral elites observe centre lifestyles through W^I (travel, media, education abroad) and form consumption preferences calibrated to centre income levels.

In the formation equation: the peripheral elite's W^I has cross-border edges to the center. They observe r_{center} (high consumption, high variety) and form θ^*_{elite} calibrated to those signals. But their income is peripheral, so the gap between aspirations and means is large. They consume rather than save. The modern-sector investment that would close the gap does not occur.

This is the mirror image of the traditional-sector case above. There, *too few* cross-block edges kept aspirations low; here, cross-border edges carry signals that are *not attainable* at peripheral incomes. The elite's preferences are consistent with what it observes, but the consequences are self-defeating: too little is invested. The formation equation by itself cannot say whether an edge helps or harms. That depends on whether the observed returns are attainable by the observer — a point that returns in Chapter 12.

9.6 Summary

1. **Five preference parameters** that the benchmark model treats as exogenous — time discount, aspiration level, labor supply, risk tolerance, taste for variety — vary systematically with the social and informational environment. Much of this evidence concerns beliefs and norms rather than tastes, so Part III treats θ as a reduced-form behavioral parameter.

2. The **formation equation** $\theta_{i,t+1} = \rho\theta_{i,t} + (1-\rho)\bar{\theta} + \mu(\sum_j W^I_{ij} r_{j,t} - r^*)$, with t indexing generations of a dynasty, has three components: intergenerational persistence (ρ), anchor ($\bar{\theta}$), and network signal. Steady states exist under stated conditions (bounded returns, or affine returns with a nonsingular system). Heterogeneous returns produce heterogeneous steady states; that is not multiplicity.

3. **Persistence**: after W^I changes, preferences converge at the local rate $\lambda = \rho(\rho I + \mu W^I D\mathbf{g})$, which is at least ρ when returns rise with preferences. The half-life is $\ln 2 / \ln(1/\lambda)$ generations: 6.6 generations (about 164 years) if $\lambda = 0.9$.

4. The Ψ **ratio** $\Psi_i(\Delta W, T) = F_i / E_i$ compares formation and enforcement effects for a stated perturbation, outcome and horizon. It is not a property of the economy, and it can be negative or undefined.

5. **Block-wise steady states**: with block-diagonal W^I, each block forms preferences consistent with the returns it observes. Whether adding an edge helps depends on whether the returns it carries are attainable by the observer — the traditional-sector case and Furtado's demonstration effect point in opposite directions.

9.7 Exercises

9.1 The formation equation by hand. For $\rho = 0.8$, $\bar{\theta} = 1$, $\mu = 0.2$, $r^* = 0.1$, and a constant return $r = 0.15$:

(a) Starting from $\theta_0 = 0.5$, compute $\theta_1, \theta_2, \ldots, \theta_{10}$ by hand.

(b) Compute the steady state θ^* analytically. Verify convergence.

(c) Plot the trajectory. How many periods to reach 90% of the steady state?

(d) Now set $\rho = 0.95$ and repeat. How does the convergence speed change?

9.2 Heterogeneous steady states. Reproduce Example 9.10:

(a) Verify the steady-state values $\theta^*_M = 1.088$ and $\theta^*_T = 0.923$.

(b) Now connect the blocks: $W^I = \begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{pmatrix}$. Compute the new steady states.

(c) How large must the cross-block weight be for the two steady-state preferences to coincide? Why does the gap vanish exactly there?

(d) Interpret: what real-world intervention corresponds to increasing W^I_{12}?

9.3 Persistence in female labor supply. Suppose the norm for female market labor is θ^F (the share of time allocated to market work). The formation equation: $\theta^F_{t+1} = 0.85\theta^F_t + 0.15 \times 0.5 + 0.1(\bar{r}_t - 0.1)$, where $\bar{r}_t = 0.3\theta^F_t$ (returns are proportional to participation rate — network effects in labor markets).

(a) Compute the steady state. Is it unique?

(b) Starting from $\theta_0^F = 0.2$ (low participation): simulate 50 generations. Plot the trajectory.

(c) At generation 10, WWII occurs: θ_{10}^F jumps to 0.6 (women enter factories). Simulate the post-war trajectory. Does θ^F return to the pre-war level?

(d) In this linear model the steady state is unique, so the war has *no* permanent effect. Show this. What feature would the model need for a temporary shock to have a permanent effect? (Hint: an S-shaped return function, Chapter 10.) Compare the empirical evidence on WWII mobilization and later female participation (e.g. Fernández et al., 2004, *QJE*, "Mothers and Sons").

9.4 Aspiration and the smartphone. Use the parameters and return functions of Example 9.10. An agent in the traditional sector has $\theta_0^T = 0.8$ (low aspiration). At $t = 5$, she gets a smartphone that adds a cross-block edge in W^I: she now observes modern-sector returns with weight 0.3.

(a) Compute the old steady state (block-diagonal W^I) and the new steady state (with the cross-block edge).

(b) Simulate the transition. How many generations to close half the aspiration gap?

(c) What if 100 agents get smartphones simultaneously? Does the aggregate effect equal 100 times the individual effect? (Hint: the returns function may change when many agents shift behavior.)

9.5 Bernard et al. (2014) as a W^I perturbation. In the Ethiopian aspiration experiment:

(a) Model the documentary as adding edges in W^I from "successful farmer" (the film subject) to the treated farmers. Write the perturbation ΔW^I.

(b) Compute the change in steady-state aspirations $\Delta\theta^*$ as a function of μ, ρ, and the film subject's return r_{film}.

(c) Suppose the documentary raised aspirations by 0.1 SD at six months (an illustrative figure). What would this imply about μ? What is the problem with mapping a six-month effect into a per-generation parameter?

(d) The effect on savings was larger than the effect on aspirations. Can the formation model explain this? (Hint: savings = $f(\theta)$ may be convex.)

9.6 The Ψ ratio for education. Use Definition 9.13 with outcome x and horizon $T = 1$. In a simple model where $x = \theta \cdot w/p$ (quantity demanded is proportional to θ times real income):

(a) A policy cuts the price of education by 10% (enforcement). Compute $\Delta x/x$.

(b) The same policy, by increasing returns to educated women in the girl's reference group, raises θ by 5% over one generation (formation). Compute the additional $\Delta x/x$.

(c) What is Ψ? Which effect is larger?

(d) Under what conditions does the formation effect exceed the enforcement effect ($\Psi > 1$)?

9.7 Furtado's demonstration effect. In a two-country model (Center, Periphery):

(a) The Peripheral elite observes the Center's consumption through W^I (travel, media). Model this as $W^I_{\text{elite, Center}} = 0.4$, $W^I_{\text{elite, Periphery}} = 0.6$.

(b) The Center's consumption-to-income ratio is 0.85 (high consumption). The Periphery's sustainable ratio is 0.70 (lower income, needs more saving). Let θ be the elite's consumption-to-income ratio, with formation equation (9.1), anchor $\bar{\theta} = 0.70$, reference $r^* = 0.70$, $\rho = 0.9$ and $\mu = 0.4$ (so $\mu/(1-\rho) = 4$). The observed "returns" are the two consumption ratios. Compute the elite's steady-state θ^*.

(c) At what $W^I_{\text{elite, Center}}$ does the elite's steady-state consumption-to-income ratio exceed 1 (chronic dissaving)?

(d) This is Furtado's "myth of development": the periphery consumes as if it were the center. What structural change would break the pattern?

Mathematical Appendices

Appendix 9A: Fixed Points of the Formation Mapping

The formation mapping. Define $\Phi : \mathbb{R}^N \to \mathbb{R}^N$ by:

$$\Phi_i(\boldsymbol{\theta}) = \bar{\theta} + \frac{\mu}{1-\rho}\left(\sum_j W^I_{ij} g_j(\boldsymbol{\theta}) - r^*\right). \tag{9.10}$$

The steady state $\boldsymbol{\theta}^*$ is a fixed point: $\Phi(\boldsymbol{\theta}^*) = \boldsymbol{\theta}^*$.

Existence. If each g_j is continuous and bounded by B, then Φ maps the compact convex cube $\Theta = \{\|\boldsymbol{\theta} - \bar{\theta}\mathbf{1}\|_\infty \le \frac{\mu}{1-\rho}(B + |r^*|)\}$ into itself, and Brouwer's theorem gives a fixed point. With unbounded (e.g. affine) g, Brouwer does not apply directly. In the affine case $\mathbf{g} = \mathbf{a} + G\boldsymbol{\theta}$, the fixed-point condition is linear, and a unique fixed point exists iff $I - \frac{\mu}{1-\rho}W^I G$ is nonsingular.

Uniqueness. Φ has a unique fixed point if it is a contraction: $\|\Phi(\boldsymbol{\theta}) - \Phi(\boldsymbol{\theta}')\|_\infty \le q\|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_\infty$ for some $q < 1$ on the whole domain (Banach). When g_j depends only on θ_j, a valid constant is $q = \frac{\mu}{1-\rho}\|W^I\|_\infty \cdot \max_j \sup_\theta |g_j'(\theta)|$, with $\|W^I\|_\infty = 1$ for row-stochastic W^I. Uniqueness is guaranteed when $\frac{\mu}{1-\rho}\max_j \sup|g_j'| < 1$. This condition is sufficient, not necessary.

Multiple equilibria. When $\frac{\mu}{1-\rho}$ is large and g_j has a steep enough segment (an S-shape), the contraction condition fails and multiple steady states *can* exist; Chapter 10 (Example 10.11) gives one with three. Example 9.10 is *not* such a case: there $q = 1.5 \times 0.1 = 0.15 < 1$, so each agent has a unique steady state, and the two values differ only because the return functions differ.

Appendix 9B: The Timescale Separation

The two-timescale system. The combined system is:

$$\text{Fast:}\quad \mathbf{x}_t = \mathbf{x}^*(\boldsymbol{\theta}_t, W) \quad \text{(within-period equilibrium)} \tag{9.11}$$

$$\text{Slow:}\quad \boldsymbol{\theta}_{t+1} = \rho\boldsymbol{\theta}_t + (1-\rho)\bar{\theta}\mathbf{1} + \mu(W^I\mathbf{g}(\mathbf{x}_t) - r^*\mathbf{1}) \quad \text{(formation)} \tag{9.12}$$

Slow–fast scaling. A clean slow–fast analysis separates the *speed* of preference adjustment from the *long-run sensitivity* to the signal. Write $\varepsilon := 1 - \rho$ and $\mu = \varepsilon\kappa$, so that

$$\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t = \varepsilon\left[(\bar{\theta}\mathbf{1} - \boldsymbol{\theta}_t) + \kappa\,(W^I\mathbf{g}(\mathbf{x}_t) - r^*\mathbf{1})\right].$$

The steady state depends on $\kappa = \mu/(1-\rho)$ but not on ε, and ε alone sets the speed. In the limit $\varepsilon \to 0$ with κ fixed, preferences are frozen on any finite horizon and only enforcement effects operate. Over $t \sim 1/\varepsilon$ generations they follow the ODE $\dot{\boldsymbol{\theta}} = \bar{\theta}\mathbf{1} - \boldsymbol{\theta} + \kappa(W^I\mathbf{g} - r^*\mathbf{1})$. Letting $\rho \to 1$ with μ fixed instead would send the steady-state deviation $\mu/(1-\rho)$ to infinity — one reason to report $\kappa = \mathcal{M}$ (Chapter 10) and ε separately.

Identification. The idea behind Chapter 11's two-step strategy: the fast dynamics would identify demand and supply parameters at fixed $\boldsymbol{\theta}$, and the slow dynamics would identify $(\rho, \mu, \bar{\theta})$ from across-cohort variation. Chapter 11 shows what this requires: a restriction on age, period and cohort effects; a way to deal with the reflection problem; and a measured W^I.

Structuralist Case Study: Furtado's *O Mito do Desenvolvimento* and Cultural Dependency

In 1974, while in exile and teaching in Paris, Celso Furtado published *O Mito do Desenvolvimento Econômico* (Furtado, 1974), a short book that pushed his analysis of underdevelopment into the domain of **culture, consumption patterns, and the social construction of desire**.

Furtado's earlier work (*Desenvolvimento e Subdesenvolvimento*, 1961) had analyzed the production structure — in this book's terms, the block-diagonal W^P. As the first head of SUDENE (from 1959) he had tried to change that structure through policy. *O Mito* asked a different question: *even if you change W^P, will it matter if the preferences of peripheral agents are calibrated to the center?*

The argument has three parts.

Part 1: The myth of development is the belief that peripheral countries can replicate center consumption patterns. Furtado argued that "development" as understood in the postwar period meant achieving center-level consumption: automobiles, appliances, industrialized food, suburban housing. This consumption pattern was not a natural aspiration but a *culturally constructed* one, transmitted to the periphery through W^I (advertising, education, media, elite travel). In the formation equation (9.1): the return $r_{j,t}$ observed by the peripheral elite is center-level consumption, filtered through W^I edges that connect peripheral elites to center lifestyles. The formation signal pushes θ_i (taste for consumption) upward, toward the center's steady state.

Part 2: Center consumption patterns at peripheral income levels produce structural distortion. If θ_i (consumption aspiration) is calibrated to center returns

but income is peripheral, the savings rate collapses: agents consume beyond their means, relying on debt, capital inflows, or natural resource rents. For Furtado the mechanism is distributive and structural. Income concentration allows a minority to reproduce centre consumption patterns. That composition of demand then directs industrialization toward goods and techniques designed for centre incomes, while the majority is excluded from the modern consumption pattern rather than merely "unambitious". In the model, block-wise steady states (Section 9.5) operate through the consumption channel: the elite's $\theta_{\text{consumption}}$ is centre-calibrated, saving and broad-based investment are low, and W^P stays block-diagonal. Furtado's emphasis on exclusion is a statement about *access* (who can attain which returns), which the formation equation alone does not represent.

Part 3: The ecological impossibility. Furtado's most prescient argument: even if every country could replicate center consumption, the planet cannot sustain it. The myth of development is a myth not only because the periphery cannot achieve it within the existing power structure, but because *nobody* can achieve it within the planet's ecological constraints. This was 1974, two years after *The Limits to Growth* (Meadows et al., 1972), which Furtado discusses explicitly. In the formation framework: the centre's $\boldsymbol{\theta}^*$ is not a feasible global steady state. It is a local configuration sustained by the centre's disproportionate claim on global resources (Chapter 16). This part of the argument is not formalized in this chapter.

Furtado's argument (1974)	**Formal object**	**This chapter**
Consumption aspirations are culturally formed	$\theta_{i,t+1} = f(W^I, r_{j,t})$	Def. 9.7
Peripheral elites observe center lifestyles	$W^I_{\text{elite, center}} > 0$	Sec. 9.2
High aspiration at low income = undersaving	θ_{cons} high, Y low $\to S$ low	Prebisch box
Underdevelopment reproduces itself culturally	Block-wise steady states	Prop. 9.15
Past structures persist in current preferences	Persistence at rate $\lambda \geq \rho$	Prop. 9.11
The myth is a myth for everyone	Center $\boldsymbol{\theta}^*$ not globally feasible	Not formalized here (Ch. 16, Ch. 26)

O Mito was Furtado at his most radical. He had started in 1961 by saying underdevelopment was a structural condition of W^P. By 1974 he was also saying that underdevelopment is reproduced through consumption patterns and culture — in this book's terms, through $\boldsymbol{\theta}$ as well as W^P. The production structure helps produce the preferences that sustain it. On this reading, changing W^P is necessary but not sufficient: W^I — the channel through which the structure becomes desire — and access to the returns it displays must change too.

He did not have the formation equation. He had the argument: that the most insidious form of dependence is not economic but *cultural* — the colonization of what you want. This chapter gives that argument its dynamics.

Further Reading

Endogenous preferences: Bowles (1998), "Endogenous Preferences: The Cultural Consequences of Markets and Other Economic Institutions." Bisin and Verdier (2001), "The Economics of Cultural Transmission and the Dynamics of Preferences." Bisin and Verdier (2011), "The Economics of Cultural Transmission and Socialization," in Benhabib, Bisin, and Jackson (eds.), *Handbook of Social Economics*.

Aspirations: Ray (2006), "Aspirations, Poverty, and Economic Change," in Banerjee, Bénabou, and Mookherjee (eds.), *Understanding Poverty*. Bernard et al. (2014), "The Future in Mind: Aspirations and Forward-Looking Behaviour in Rural Ethiopia," CSAE WP. Genicot and Ray (2017), "Aspirations and Inequality."

Cultural transmission and labor: Fernández (2013), "Cultural Change as Learning: The Evolution of Female Labor Force Participation over a Century." Fernández (2007), "Women, Work, and Culture." Fernández et al. (2004), "Mothers and Sons." Fogli and Veldkamp (2011), "Nature or Nurture? Learning and the Geography of Female Labor Force Participation." Fernández and Fogli (2009), "Culture: An Empirical Investigation."

Risk and experience: Malmendier and Nagel (2011), "Depression Babies: Do Macroeconomic Experiences Affect Risk Taking?"

The structuralist tradition on culture: Furtado (1974), *O Mito do Desenvolvimento Econômico*, Rio de Janeiro: Paz e Terra (the primary source). Furtado (1978), *Criatividade e Dependência na Civilização Industrial*, also from exile. Nurkse (1953), *Problems of Capital Formation in Underdeveloped Countries* (the international demonstration effect). Prebisch (1981), *Capitalismo Periférico*.

Beliefs versus tastes; social capital: Becker (1996), *Accounting for Tastes*. Becker and Murphy (2000), *Social Economics*. Bursztyn et al. (2014), "Understanding Mechanisms Underlying Peer Effects." Dietrich and List (2013), "Where Do Preferences

Come From?"

Chapter 10

The Formal Formation Model

The Complete Dynamic System

> *"As situações de dependência na América Latina contemporânea supõem um modo de articulação entre a dinâmica econômica e a ação social que é distinto tanto da expansão dos centros como daquela dos países coloniais."*
>
> ["The situations of dependence in contemporary Latin America presuppose a mode of articulation between economic dynamics and social action that is distinct both from the expansion of the centers and from that of the colonial countries."]
>
> — Fernando Henrique Cardoso and Enzo Faletto, *Dependência e Desenvolvimento na América Latina* (1969)

Chapter 9 introduced the formation equation: preferences evolve through W^I. Chapter 7 derived the bargaining equilibrium: allocations depend on power $\boldsymbol{\pi}(W)$ and preferences $\boldsymbol{\theta}$. This chapter closes the loop.

The formation model is a dynamical system with three equations:

1. The bargaining equilibrium determines allocations given power and preferences.

2. The return function maps allocations into observable outcomes.

3. The formation equation maps observed returns into next-period preferences.

The loop $\boldsymbol{\theta}_t \to \mathbf{x}_t^* \to \mathbf{r}_t \to \boldsymbol{\theta}_{t+1}$ is the complete model. The steady state $(\boldsymbol{\theta}^*, \mathbf{x}^*, \mathbf{r}^*)$ is a fixed point of this system. With nonlinear returns, multiple fixed points can coexist in the same economy with the same W (Section 10.3). Which

one obtains depends on history — on the initial $\boldsymbol{\theta}_0$, which was formed under a possibly different W^I.

This is the chapter that makes the book's title operational. An economy is not just a weight matrix W. It is a weight matrix, a preference vector formed by that matrix, and an allocation produced by the two together. Structure is not a constraint *on* agents with given preferences. Structure is the process that *produces* the preferences.

10.1 The Complete System

10.1.1 Three Equations

Definition 10.1 (The Formation Model)**.** *The **formation model** consists of three components operating in sequence within each generation t (one period is one generation, as in Chapter 9):*

Equation 1: The Bargaining Equilibrium. *Given power $\boldsymbol{\pi}$, preferences $\boldsymbol{\theta}_t$, and endowments $\boldsymbol{\omega}$:*

$$x^*_{i,t} = h_i(\boldsymbol{\pi}, \boldsymbol{\theta}_t, \boldsymbol{\omega}) = \omega_i + \pi_i \Phi + \phi\,(\theta_{i,t} - \bar{\theta}_t), \qquad \bar{\theta}_t = \frac{1}{N}\sum_j \theta_{j,t}. \tag{10.1}$$

*This is the transferable-utility allocation rule of Chapter 7 (Proposition 7.3), $x_i = d_i + \pi_i(S - \sum_j d_j)$. Here $\Phi = S - \sum_j d_j > 0$ is the bargaining surplus. The disagreement payoff is $d_i = \omega_i + \phi(\theta_{i,t} - \bar{\theta}_t)$: an agent who is more patient or ambitious than average ($\theta_{i,t} > \bar{\theta}_t$) has a better outside option, and $\phi \geq 0$ measures by how much. The preference term is a reduced-form **assumption**; it sums to zero, so $\sum_i x^*_{i,t} = \sum_i \omega_i + \Phi =: \Omega$. With equal endowments, the rule reduces to $x_i = \Omega/N + (\pi_i - 1/N)\Phi + \phi(\theta_i - \bar{\theta})$. Power $\boldsymbol{\pi}$ is the Perron vector of the bargaining layer (Chapter 7), which none of the policies in this chapter changes.*

Equation 2: The Return Function. *The allocation produces observable returns:*

$$r_{i,t} = g_i(\mathbf{x}^*_t) = \alpha_i + \beta_i x^*_{i,t} + \gamma_i \sum_j W^P_{ij} x^*_{j,t}, \tag{10.2}$$

where α_i is a sector-specific baseline, β_i is the private return to resources, and $\gamma_i \sum_j W^P_{ij} x^*_{j,t}$ is the network spillover: your return depends on the resources of your suppliers, weighted by the input shares W^P.

Equation 3: The Formation Equation *(Chapter 9). Observed returns update pref-*

erences for the next generation:

$$\theta_{i,t+1} = \rho\theta_{i,t} + (1-\rho)\theta^\circ + \mu\left(\sum_j W^I_{ij} r_{j,t} - r^*\right). \tag{10.3}$$

We write θ° for the anchor ($\bar{\theta}$ in Chapter 9), to keep it distinct from the cross-sectional mean $\bar{\theta}_t$.

The three equations form a map $T : \mathbb{R}^N \to \mathbb{R}^N$:

$$\boldsymbol{\theta}_{t+1} = T(\boldsymbol{\theta}_t) \equiv \rho\boldsymbol{\theta}_t + (1-\rho)\theta^\circ\mathbf{1} + \mu\left(W^I\mathbf{g}(\mathbf{h}(\boldsymbol{\pi}, \boldsymbol{\theta}_t, \boldsymbol{\omega})) - r^*\mathbf{1}\right). \tag{10.4}$$

The structural objects are the weight matrices W^P and W^I, the power vector $\boldsymbol{\pi}$, and the parameters

$$(\rho, \mu, \theta^\circ, r^*, \Phi, \phi, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}).$$

The dynamics are in $\boldsymbol{\theta}$. With the affine h and g above, T is affine. Nonlinear return functions (Section 10.3) make it nonlinear.

Intuition: The loop in words

Generation t: You arrive with preferences $\theta_{i,t}$ inherited from your parents' generation and your own early-life signals. Given those preferences and the power structure $\boldsymbol{\pi}$, the bargaining equilibrium determines your allocation $x^*_{i,t}$. Your allocation, combined with the production network, produces your return $r_{i,t}$. Your children observe your return (and your neighbors' returns, weighted by W^I). They form preferences $\theta_{i,t+1}$ based on what they saw. They enter the next generation's bargaining equilibrium. The loop continues.
The benchmark model runs this loop once (at $t = 0$, with $\boldsymbol{\theta}_0$ given) and stops. The formation model runs it forever.

10.1.2 The Linearized System

Let $C := \mathrm{diag}(\boldsymbol{\beta}) + \mathrm{diag}(\boldsymbol{\gamma})W^P$ be the Jacobian of the return function, and $P := I - \frac{1}{N}\mathbf{1}\mathbf{1}'$ the centering matrix. Because h depends on $\theta_i - \bar{\theta}$, its Jacobian is $D_{\boldsymbol{\theta}}\mathbf{h} = \phi P$, not a diagonal matrix. The Jacobian of T is therefore

$$J = DT = \rho I + \mu\phi\, W^I C P. \tag{10.5}$$

In the affine model J is constant. With nonlinear returns, C is replaced by $D\mathbf{g}$ evaluated at the steady state.

Proposition 10.2 (Structure of the Jacobian)**.** *1. The Jacobian decomposes as*

$$J = \underbrace{\rho I}_{\text{persistence}} + \underbrace{\mu\phi\, W^I \operatorname{diag}(\boldsymbol{\beta}) P}_{\text{direct formation}} + \underbrace{\mu\phi\, W^I \operatorname{diag}(\boldsymbol{\gamma}) W^P P}_{\text{network-amplified formation}}.$$

2. *$J\mathbf{1} = \rho\mathbf{1}$: ρ is always an eigenvalue of J, with eigenvector $\mathbf{1}$.*

3. *A uniform shift of all preferences, $\boldsymbol{\theta} \mapsto \boldsymbol{\theta} + c\mathbf{1}$, changes no allocation and no return. It decays at rate ρ and is never amplified by the loop. The loop can amplify only relative preferences.*

Proof. (1) Expand C. (2) $P\mathbf{1} = 0$. (3) $\mathbf{h}(\boldsymbol{\theta} + c\mathbf{1}) = \mathbf{h}(\boldsymbol{\theta})$ because h depends on $\theta_i - \bar{\theta}$. So the signal is unchanged, and $T(\boldsymbol{\theta} + c\mathbf{1}) = T(\boldsymbol{\theta}) + \rho c\mathbf{1}$. □

Part (3) is a substantive result, not a technicality. In this model there is no economy-wide "virtuous circle" of rising aspirations: preferences matter for allocations only relative to others', so a society-wide rise in ambition buys nothing in the bargaining equilibrium and is not self-reinforcing. Virtuous and vicious circles are *relative*: between blocks, classes, or regions.

10.2 Steady States

10.2.1 Existence and Characterization

A steady state is a fixed point $\boldsymbol{\theta}^* = T(\boldsymbol{\theta}^*)$. From (10.4):

$$(1-\rho)(\boldsymbol{\theta}^* - \theta^\circ\mathbf{1}) = \mu\left(W^I \mathbf{g}(\mathbf{h}(\boldsymbol{\pi}, \boldsymbol{\theta}^*, \boldsymbol{\omega})) - r^*\mathbf{1}\right). \tag{10.6}$$

Agent by agent:

$$\theta_i^* = \theta^\circ + \frac{\mu}{1-\rho}\left(\sum_j W_{ij}^I r_j^* - r^*\right). \tag{10.7}$$

The steady-state preference deviates from the anchor by $\frac{\mu}{1-\rho}$ times the W^I-weighted return gap. The ratio $\frac{\mu}{1-\rho}$ is the **formation multiplier**: how strongly the network signal moves long-run preferences, holding returns fixed.

Definition 10.3 (Formation Multiplier)**.** *The **formation multiplier** is:*

$$\mathcal{M} = \frac{\mu}{1-\rho}. \tag{10.8}$$

When $\mathcal{M}$ is large (strong signal sensitivity and/or persistent preferences), small differences in network-observed returns produce large differences in steady-state preferences.

When $\mathcal{M}$ is small, the anchor dominates and preferences are approximately exogenous. When returns themselves respond to preferences, the full long-run multiplier is the matrix $(I-J)^{-1}\mu$ of Theorem 10.12, not the scalar $\mathcal{M}$.

Theorem 10.4 (Existence, Uniqueness, Multiplicity)**.** 1. ***Affine case*** *(the model of Definition 10.1). $T(\boldsymbol{\theta}) = J\boldsymbol{\theta} + \mathbf{c}$. A steady state exists and is unique iff 1 is not an eigenvalue of J. If $\rho(J) < 1$, it is globally asymptotically stable.*

2. ***Nonlinear case.*** *If $\mathbf{g} \circ \mathbf{h}$ is continuous and T maps a nonempty compact convex set Θ into itself (for instance, if the returns are bounded), a steady state exists (Brouwer). If, in addition, T is continuously differentiable on an open set containing Θ and $\sup_{\boldsymbol{\theta}\in\Theta} \|DT(\boldsymbol{\theta})\| < 1$ for some induced matrix norm, the steady state in Θ is unique.*

3. ***Multiplicity requires nonlinearity.*** *An affine map has zero, one, or a continuum of fixed points, never exactly two or three. Several isolated steady states can arise only when returns are nonlinear in preferences and the uniqueness condition in (2) fails.*

Proof. (1) $\boldsymbol{\theta} = J\boldsymbol{\theta} + \mathbf{c}$ iff $(I-J)\boldsymbol{\theta} = \mathbf{c}$, which has a unique solution iff $I - J$ is nonsingular. If $\rho(J) < 1$, the deviations $\boldsymbol{\theta}_t - \boldsymbol{\theta}^* = J^t(\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*) \to 0$ from any start. (2) Brouwer gives existence. On a convex set, the mean-value inequality $T(\boldsymbol{\theta}) - T(\boldsymbol{\theta}') = \int_0^1 DT(\boldsymbol{\theta}' + s(\boldsymbol{\theta} - \boldsymbol{\theta}'))\, ds\, (\boldsymbol{\theta} - \boldsymbol{\theta}')$ gives $\|T(\boldsymbol{\theta}) - T(\boldsymbol{\theta}')\| \le \sup_\Theta \|DT\|\, \|\boldsymbol{\theta} - \boldsymbol{\theta}'\|$, so T is a contraction of the closed set Θ into itself and Banach's theorem gives uniqueness in Θ. (3) The solution set of $(I-J)\boldsymbol{\theta} = \mathbf{c}$ is empty, a point, or an affine subspace. □

A convenient sufficient condition for the uniqueness bound in the affine model (with $\boldsymbol{\beta}, \boldsymbol{\gamma} \ge 0$) uses $\|W^I\|_\infty = 1$, $\|C\|_\infty = \max_i(\beta_i + \gamma_i)$ and $\|P\|_\infty = 2(N-1)/N$:

$$\mu\phi \max_i(\beta_i + \gamma_i)\, \frac{2(N-1)}{N} < 1 - \rho \quad \Longrightarrow \quad \|J\|_\infty < 1. \tag{10.9}$$

It includes the network spillover γ, and it is sufficient, not necessary.

10.2.2 Integrated and Stratified Steady States

Proposition 10.5 (Integrated and Stratified Steady States)**.** *Let $\boldsymbol{\theta}^*$ be a steady state with returns $\mathbf{r}^*$, and let $\tau_1(W^I) := \frac{1}{2}\max_{i,j}\sum_k |W^I_{ik} - W^I_{jk}|$ be the Dobrushin coefficient of W^I.*

1. ***Integration compresses preferences.*** *For all i, j:*

$$|\theta_i^* - \theta_j^*| \;\le\; \mathcal{M}\,\tau_1(W^I)\,\big(\max_k r_k^* - \min_k r_k^*\big). \tag{10.10}$$

   If every agent's information row is similar ($\tau_1(W^I)$ small), steady-state preferences are nearly uniform, whatever the dispersion of returns.

2. ***Fragmentation stratifies preferences.*** *If W^I is block-diagonal, each agent's steady-state preference depends on the steady-state returns only through the returns of her own block (cf. Proposition 9.15); the returns themselves may still depend on other blocks through $\bar{\theta}$ and W^P. The economy fragments into k "cultures", each internally coherent.*

Proof. (1) By (10.7), $\theta_i^* - \theta_j^* = \mathcal{M}\sum_k (W_{ik}^I - W_{jk}^I) r_k^*$. The weights $W_{ik}^I - W_{jk}^I$ sum to zero, and their positive part sums to $\frac{1}{2}\sum_k |W_{ik}^I - W_{jk}^I| \le \tau_1(W^I)$. Hence the sum is at most $\tau_1(W^I)(\max_k r_k^* - \min_k r_k^*)$ in absolute value. (2) In (10.7), $W_{ij}^I = 0$ for j outside i's block. □

Note the measure: what compresses preferences is how *similar* the agents' information rows are (τ_1). Since $|\lambda_2(W^I)| \le \tau_1(W^I)$, a large spectral gap $\Gamma(W^I)$ is necessary for τ_1 to be small, but it is not sufficient.

Example 10.6 (A Stylized Bolivia: Three Blocks [ILLUSTRATIVE])**.** *$N = 9$: three blocks of 3 agents each. Block 1 (gas/government), block 2 (urban informal), block 3 (rural traditional). All parameters are illustrative: this is a stylized dual economy with Bolivian labels, not an estimate of Bolivia's structure.*

- *Returns: $\alpha = (0.06, 0.04, 0.03)$ and $\beta = (0.10, 0.07, 0.04)$ by block; $\gamma_i = 0.02$ for all i.*
- *W^P: within-block input shares $0.94/3$ each, cross-block 0.01 each.*
- *W^I: within-block weight $0.85/3$ each, cross-block 0.025 each. Both are row-stochastic.*
- *Power (per agent): $\pi = (0.15, 0.11, 0.0733)$ by block, summing to 1. $N_{\text{eff}}(\boldsymbol{\pi}) = 8.34$.*
- *Allocation: $\omega_i = 0.5$, $\Phi = 4.5$ (so $\Omega = 9$, $\Omega/N = 1$), $\phi = 1$.*
- *Formation: $\rho = 0.85$, $\mu = 0.2$ ($\mathcal{M} = 1.33$), $\theta^\circ = 1$, $r^* = 0.1$.*

The spectral radius is $\rho(J) = 0.866$, so the unique steady state is globally stable. Solving $(I - J)\boldsymbol{\theta}^* = \mathbf{c}$ (`code/ch10.py`):

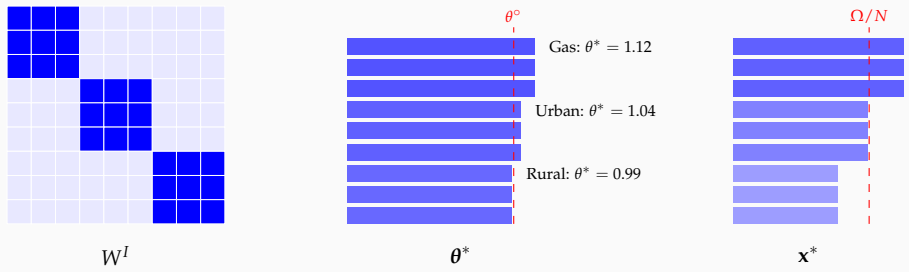
Steady-state preferences:

	Block 1 (Gas)	Block 2 (Urban)	Block 3 (Rural)
Return $\bar{r}_b^*$	0.209	0.129	0.076
Preference θ_b^*	1.125	1.041	0.987
Allocation $\bar{x}_b^*$	1.249	0.985	0.766
Power π_i (per agent)	0.150	0.110	0.073

Three internally coherent cultures in one economy. The gas block is more patient and ambitious ($\theta_1^* = 1.12$) because it observes high returns. The rural block is less so ($\theta_3^* = 0.99$) because it observes low returns. The urban block is in between. Each block's preferences are consistent with the returns it observes. But the rural block's low returns are partly caused by its low preferences (through ϕ), which are partly caused by its isolation in W^I. The preference gap is $\theta_1^* - \theta_3^* = 0.137$. It attains the bound (10.10) exactly ($\mathcal{M} = 1.33$, $\tau_1(W^I) = 0.775$, return range 0.133), because returns are constant within blocks.

The stylization departs from the data in two places. Its W^P is nearly block-diagonal ($\Gamma(W^P) = 0.09$), whereas Bolivia's measured domestic W^P is not: $\Gamma = 0.467$ on 12 sectors in 2019 (INE supply-use tables, base 2017, input shares; 0.33–0.69 across normalizations), in the range of Korea (0.511) and the US (0.463) on a common basis (Instituto Nacional de Estadística (INE), Bolivia, 2026; OECD, 2023). And the gas block's large π_i here is a bargaining weight (Assumption 7.A), not production centrality: in the measured W^P hydrocarbons are peripheral ($\pi = 0.039$), with 70% of their output exported. What the example illustrates — stratified preferences from a block-diagonal W^I — depends on the information network, not on $\Gamma(W^P)$.

Visualization: A stylized Bolivia: three blocks, three steady states



Left: W^I with three blocks (gas, urban, rural). Center: steady-state preferences — the gas block lies above the anchor (red), the rural block below. Right: steady-state allocations — the gas block gets 25% more than equal division, the rural block 23% less. Power and preferences push in the same direction: the structure produces preferences that widen the allocation gap produced by power.

10.3 Stability Analysis

10.3.1 Local Stability

Theorem 10.7 (Local Stability)**.** *Let T be continuously differentiable near a steady state $\boldsymbol{\theta}^*$, with Jacobian $J = DT(\boldsymbol{\theta}^*)$. If $\rho(J) < 1$, then $\boldsymbol{\theta}^*$ is locally asymptotically stable. If $\rho(J) > 1$, it is unstable. (If $\rho(J) = 1$, the linearization is inconclusive.)*

Proof. Write $T(\boldsymbol{\theta}^* + \Delta) - \boldsymbol{\theta}^* = J\Delta + R(\Delta)$ with $\|R(\Delta)\| = o(\|\Delta\|)$. For $\rho(J) < 1$, the argument in the proof of Proposition 9.11 applies verbatim and gives $\|\Delta_t\| \le C(\rho(J) + \varepsilon)^t \|\Delta_0\|$ near $\boldsymbol{\theta}^*$.

For $\nu := \rho(J) > 1$, let m be the largest modulus of an eigenvalue of J with modulus below ν ($m := 0$ if there is none), and pick $\max(m, 1) < a < b < \nu$. Split $\mathbb{R}^N = E_u \oplus E_s$ into the real J-invariant spectral subspaces for the eigenvalues of modulus ν and of modulus $\le m$, with restrictions J_u, J_s. Since $\rho(J_s) < a$ and $\rho(J_u^{-1}) = 1/\nu < 1/b$, there are norms on the two subspaces with $\|J_s x\| \le a\|x\|$ and $\|J_u x\| \ge b\|x\|$ (Horn and Johnson, 2012, Lemma 5.6.10); set $\|x\| := \max(\|x_u\|, \|x_s\|)$. Choose $\varepsilon > 0$ with $a + \varepsilon \le b - \varepsilon$ and $b - \varepsilon > 1$, and $\eta > 0$ with $\|R(\Delta)\| \le \varepsilon\|\Delta\|$ for $\|\Delta\| < \eta$. On the cone $K := \{\|\Delta_s\| \le \|\Delta_u\|\}$, where $\|\Delta\| = \|\Delta_u\|$, the next deviation Δ' satisfies $\|\Delta'_u\| \ge (b - \varepsilon)\|\Delta_u\|$ and $\|\Delta'_s\| \le a\|\Delta_s\| + \varepsilon\|\Delta_u\| \le (b - \varepsilon)\|\Delta_u\|$. So K is forward invariant within the η-ball and $\|\Delta_t\|$ grows at least like $(b - \varepsilon)^t$ there. Every nonzero $\Delta_0 \in K$, however small, therefore leaves the η-ball, and $\boldsymbol{\theta}^*$ is not stable. □

Proposition 10.8 (A Sufficient Condition for Stability)**.** *For any induced matrix norm, $\rho(J) \le \|J\|$. With the ∞-norm, in the affine model with $\boldsymbol{\beta}, \boldsymbol{\gamma} \ge 0$ and $\phi \ge 0$,*

$$\rho(J) \;\le\; \rho + \mu\phi\, \|W^I\|_\infty \, \|C\|_\infty \, \|P\|_\infty \;=\; \rho + \mu\phi\, \max_i(\beta_i + \gamma_i)\, \frac{2(N-1)}{N}. \tag{10.11}$$

Hence stability follows from

$$\mu\phi\, \max_i(\beta_i + \gamma_i)\, \frac{2(N-1)}{N} < 1 - \rho. \tag{10.12}$$

The condition is sufficient, not necessary.

Proof. Induced norms are submultiplicative and dominate the spectral radius, and the norm is subadditive, so $\|J\|_\infty \le \rho + \mu\phi\|W^I\|_\infty\|C\|_\infty\|P\|_\infty$. We have $\|W^I\|_\infty = 1$, because W^I is row-stochastic, and

$$\|C\|_\infty = \max_i \left(\beta_i + \gamma_i \sum_j W^P_{ij}\right) = \max_i(\beta_i + \gamma_i), \qquad \|P\|_\infty = \left(1 - \frac{1}{N}\right) + \frac{N-1}{N}.$$

□

Note that the spectral radius of a product cannot be bounded by the product of spectral radii, which is why the bound uses norms. In Example 10.6 the bound gives $\rho(J) \leq 0.893$; the actual value is 0.866.

Intuition: When is the steady state unstable?

The loop has four links:
μ: How strongly signals shift preferences.
W^I: How the signals are averaged (it never amplifies: $\|W^I\|_\infty = 1$).
β, γ: How strongly resources produce returns, directly and through suppliers.
ϕ: How strongly (relative) preferences translate into resources.
If the total feedback through the loop is weaker than the damping from persistence $(1 - \rho)$, the steady state is stable. If the loop gain at a steady state exceeds the damping in the sense $\rho(J) > 1$, a positive shock to a block's relative θ produces higher x, higher r, and an even larger increase in θ. The system then moves away from that steady state toward another one (or toward a corner). This is one formal reading of "virtuous circle" and "vicious circle" in development. By Proposition 10.2(3), the circles are relative, between groups.

10.3.2 Basin of Attraction

When multiple steady states exist, the **basin of attraction** of each determines which initial conditions lead to which long-run outcome.

Definition 10.9 (Basin of Attraction)**.** *The basin of attraction of steady state $\boldsymbol{\theta}^*$ is:*

$$\mathcal{B}(\boldsymbol{\theta}^*) = \left\{ \boldsymbol{\theta}_0 : \lim_{t \to \infty} T^t(\boldsymbol{\theta}_0) = \boldsymbol{\theta}^* \right\}. \tag{10.13}$$

Multiple steady states need nonlinear returns (Theorem 10.4(3)). The cleanest case is a single block, or one agent whose own preference drives her own returns, so that the dynamics reduce to a scalar map $\theta_{t+1} = T(\theta_t)$. In the full model a small block's own θ raises its own allocation by $\phi(1 - 1/N) \approx \phi$, so $r = g(\theta)$ is a reduced form of $g \circ h$.

Proposition 10.10 (Basin Structure for a Scalar Block)**.** *Let $T : \mathbb{R} \to \mathbb{R}$ be continuous and strictly increasing, with exactly three fixed points $\theta_L < \theta_U < \theta_H$, such that $T(\theta) > \theta$*

for $\theta < \theta_L$, $T(\theta) < \theta$ on (θ_L, θ_U), $T(\theta) > \theta$ on (θ_U, θ_H), and $T(\theta) < \theta$ for $\theta > \theta_H$. Then:

1. *θ_L and θ_H are stable, with basins $(-\infty, \theta_U)$ and (θ_U, ∞); θ_U is the basin boundary.*
2. *If T is continuously differentiable in (θ, a) and depends on a parameter a (e.g. the baseline return) with $\partial T/\partial a > 0$, and $T'(\theta_U) > 1$, then the threshold falls as a rises: $d\theta_U/da = -(\partial T/\partial a)/(T'(\theta_U) - 1) < 0$. Higher baseline returns make escape easier.*

Proof. (1) An increasing map on the line generates monotone trajectories. Starting in (θ_L, θ_U), $T(\theta) < \theta$, and since T is increasing, $T(\theta) > T(\theta_L) = \theta_L$; the trajectory decreases and is bounded below by θ_L, so it converges to a fixed point in $[\theta_L, \theta)$, which can only be θ_L (the limit is a fixed point by continuity of T). The other intervals are analogous; for instance, from $\theta < \theta_L$ the trajectory increases and is bounded above by $T(\theta_L) = \theta_L$. Stability in the sense of Lyapunov follows because each trajectory stays between its starting point and the fixed point it approaches. (2) Implicit function theorem applied to $T(\theta, a) - \theta = 0$ at θ_U. □

Example 10.11 (Poverty Trap as a Low-θ Basin [ILLUSTRATIVE])**.** *One block with S-shaped (logistic) returns in its own aspiration:*

$$r(\theta) = 0.04 + \frac{0.18}{1 + e^{-25(\theta - 1)}},$$

which rises from 0.04 to 0.22. Learning a little yields little; past a threshold, returns rise steeply; then they saturate. Formation: $\rho = 0.85$, $\mu = 0.25$, $\theta^\circ = 1$, $r^* = 0.12$.

The map $T(\theta) = 0.85\theta + 0.15 + 0.25\,(r(\theta) - 0.12)$ has three fixed points (computed numerically in `code/ch10.py`):

	θ^*_{low}	$\theta^*_{unstable}$	θ^*_{high}
Fixed point	*0.881*	*0.980*	*1.162*
Slope $T'(\theta^*)$	*0.902*	*1.115*	*0.869*
Return $r(\theta^*)$	*0.049*	*0.108*	*0.217*

The basin boundary is $\hat{\theta} = 0.980$ (Proposition 10.10). Agents who start below it converge to the poverty trap ($\theta^* = 0.881$: low aspiration, low returns). Agents who start above converge to the high steady state ($\theta^* = 1.162$).

The distance between the trap and the boundary, $\hat{\theta} - \theta^*_{low} = 0.099$, is the "aspiration gap" that must be bridged to escape. In the model, a one-time intervention that raises θ by more than 0.099 (an aspiration documentary, a scholarship, a mentor program) permanently shifts the trajectory to the high steady state; a smaller one decays back to the trap.

Here the contraction condition fails: $\mu \max r' = 0.28 > 1 - \rho = 0.15$. *Uniqueness is guaranteed only for* $\mu < 0.133$, *and three steady states appear for* μ *above about* 0.188 *(Appendix 10B).*

10.4 The Enforcement-Formation Decomposition

10.4.1 Formal Derivation

This section computes the objects of Definition 9.13 inside the formation model. A permanent perturbation (of W^P, W^I or $\boldsymbol{\omega}$) is applied at $t = 0$ to an economy at steady state $\boldsymbol{\theta}^0$. Let T_{new} be the new map and J_{new} its Jacobian.

Theorem 10.12 (The Formation Effect)**.** *Define the* ***signal shock*** $\mu\Delta\mathbf{s} := T_{new}(\boldsymbol{\theta}^0) - \boldsymbol{\theta}^0$*: the change in next generation's preferences produced by the new structure at the old preferences. In the affine model, if* $\rho(J_{new}) < 1$,

$$\boldsymbol{\theta}_T - \boldsymbol{\theta}^0 = \left(I - J_{new}^T\right)(I - J_{new})^{-1}\mu\Delta\mathbf{s}, \qquad \Delta\boldsymbol{\theta}^* := \boldsymbol{\theta}_\infty - \boldsymbol{\theta}^0 = (I - J_{new})^{-1}\mu\Delta\mathbf{s}. \tag{10.14}$$

With nonlinear returns the same formulas hold to first order in the size of the perturbation. The matrix $(I - J)^{-1}$ *is the* ***formation Leontief inverse****: it sums all rounds of the formation feedback loop, analogous to* $(I - A')^{-1}$ *in the production cascade.*

Proof. Let $\mathbf{d}_t = \boldsymbol{\theta}_t - \boldsymbol{\theta}^0$. Since T_{new} is affine, $\mathbf{d}_{t+1} = T_{\text{new}}(\boldsymbol{\theta}^0 + \mathbf{d}_t) - \boldsymbol{\theta}^0 = J_{\text{new}}\mathbf{d}_t + \mu\Delta\mathbf{s}$, with $\mathbf{d}_0 = 0$. Hence $\mathbf{d}_T = \sum_{k=0}^{T-1} J_{\text{new}}^k \mu\Delta\mathbf{s} = (I - J_{\text{new}}^T)(I - J_{\text{new}})^{-1}\mu\Delta\mathbf{s}$, which tends to $(I - J_{\text{new}})^{-1}\mu\Delta\mathbf{s}$ when $\rho(J_{\text{new}}) < 1$. In the nonlinear case, index the perturbation by its size s, T_s, with T_0 the old map, $(\boldsymbol{\theta}, s) \mapsto T_s(\boldsymbol{\theta})$ twice continuously differentiable and $\rho(DT_0(\boldsymbol{\theta}^0)) < 1$. Then $\mu\Delta\mathbf{s} = O(s)$ and $J_{\text{new}} = DT_0(\boldsymbol{\theta}^0) + O(s)$. For finite T, $\mathbf{d}_{t+1} = J_{\text{new}}\mathbf{d}_t + \mu\Delta\mathbf{s} + O(\|\mathbf{d}_t\|^2)$, and induction gives $\mathbf{d}_t = O(s)$ and hence $\mathbf{d}_T = \sum_{k<T} J_{\text{new}}^k \mu\Delta\mathbf{s} + O(s^2)$. For $T = \infty$, the implicit function theorem applied to $\boldsymbol{\theta} - T_s(\boldsymbol{\theta}) = 0$ at $(\boldsymbol{\theta}^0, 0)$ (where $I - DT_0$ is invertible) gives a steady state $\boldsymbol{\theta}^*(s)$ with $\boldsymbol{\theta}^*(s) - \boldsymbol{\theta}^0 = (I - J_{\text{new}})^{-1}\mu\Delta\mathbf{s} + O(s^2)$. □

Corollary 10.13 (Enforcement and Formation in the Model)**.** *For an outcome* $\mathbf{y}$ *that is affine in* $\boldsymbol{\theta}$*, with* $Y := D_{\boldsymbol{\theta}}\mathbf{y}$ *evaluated under the new structure* $W + \Delta W$ *(e.g.* $Y = \phi P$ *for allocations,* $Y = C_{new}\phi P$ *for returns), in the affine model with* $\rho(J_{new}) < 1$ *Definition 9.13 gives*

$$E_i = y_i(\boldsymbol{\theta}^0, W + \Delta W) - y_i(\boldsymbol{\theta}^0, W), \qquad F_i(T) = \left[Y\,(I - J_{new}^T)(I - J_{new})^{-1}\mu\Delta\mathbf{s}\right]_i. \tag{10.15}$$

Proof. Substitute Theorem 10.12 into Definition 9.13: $F_i = y_i(\boldsymbol{\theta}_T, \cdot) - y_i(\boldsymbol{\theta}^0, \cdot) = [Y(\boldsymbol{\theta}_T - \boldsymbol{\theta}^0)]_i$. □

Intuition: The formation Leontief inverse

The matrix $(I - J)^{-1} = I + J + J^2 + J^3 + \cdots$ sums the direct signal (I), the first-round feedback (J: higher relative θ produces higher r, which raises θ further), the second-round feedback (J^2), and so on. Just as the production Leontief inverse amplifies demand shocks through the production network, the formation Leontief inverse amplifies preference signals through the formation loop. The factor $(I - J^T)$ in (10.14) records that only T rounds have occurred by generation T.

When the spectral radius $\rho(J)$ is close to 1 (near the stability boundary), the formation Leontief inverse is large: small signals produce large long-run preference changes — but they take many generations to arrive.

10.4.2 The Ψ Ratio in a One-Loop Model

The general $\Psi_i(\Delta W, T)$ of Definition 9.13 is computed from Corollary 10.13. A closed form is available when the loop reduces to a scalar.

Proposition 10.14 (The Ψ Ratio in a One-Loop Model)**.** *Consider one agent (or a block treated as one agent) with outcome $x = e + \eta\theta$, return $r = \alpha + \beta x$, and the formation equation (10.3). A permanent policy shifts e by $\Delta e \neq 0$. Let $\kappa := \mu\beta\eta$ and $\lambda := \rho + \kappa$, with $|\lambda| < 1$ (only $\lambda \neq 1$ is needed for $\Psi(T)$). Then*

$$\Psi(T) = \kappa \frac{1 - \lambda^T}{1 - \lambda}, \qquad \Psi(\infty) = \frac{\mu\beta\eta}{1 - \rho - \mu\beta\eta}. \tag{10.16}$$

Limiting cases:

1. *$\mu = 0$ (no formation sensitivity): $\Psi \equiv 0$. The benchmark model is complete.*
2. *$\rho \to 0$ (no persistence): $\Psi(\infty) \to \frac{\mu\beta\eta}{1-\mu\beta\eta}$.*
3. *$\lambda \uparrow 1$, i.e. $\rho \uparrow 1 - \mu\beta\eta$ with $\kappa > 0$ (the stability boundary): $\Psi(\infty) \to \infty$. But at any finite horizon $\Psi(T) \leq \kappa T$: a near-unit root makes the long-run formation effect large and slow. For $\rho > 1 - \mu\beta\eta$ the steady state is unstable and $\Psi(\infty)$ is not defined.*

Proof. $E = \Delta e$. At θ^0 the signal shock is $\mu\beta\Delta e$ and the scalar Jacobian is λ. By Theorem 10.12, $\theta_T - \theta^0 = \mu\beta\Delta e\,(1 - \lambda^T)/(1 - \lambda)$, and $F = \eta(\theta_T - \theta^0)$. Dividing, $\Psi(T) = F/E = \mu\beta\eta(1 - \lambda^T)/(1 - \lambda)$; letting $T \to \infty$ (using $|\lambda| < 1$) gives $\Psi(\infty)$.

The bound uses $(1-\lambda^T)/(1-\lambda) = \sum_{k<T}\lambda^k \leq \sum_{k<T}|\lambda|^k \leq T$ for $|\lambda| \leq 1$, with $\kappa > 0$. □

Intuition: The magnitude of Ψ

By Remark 9.14, an evaluation that ignores formation captures the share $1/(1+\Psi)$ of the effect at horizon T. With $\Psi(\infty) = 2$, it captures one third of the long-run effect; with $\Psi = 0.1$, it captures 91%. Which case obtains is an empirical question about a *specific* perturbation and outcome. Chapter 11 shows that the statistics usually reported in the literature are related to, but not the same as, Ψ. This chapter therefore uses only model-computed values.

10.5 Comparative Dynamics

10.5.1 Three Policy Levers

The formation model distinguishes three types of structural intervention, each operating on a different component of the system. Throughout, $\boldsymbol{\pi}$ is held fixed (it belongs to the bargaining layer, which none of the three levers touches).

Proposition 10.15 (Three Policy Levers)**.** *Take the outcome to be returns, $y = \mathbf{r}$.*

1. ***Transfers** change $\boldsymbol{\omega}$. Enforcement: $E = C\Delta\boldsymbol{\omega}$, since allocations move one-for-one at fixed $\boldsymbol{\theta}$. Formation: the signal shock is $\mu\Delta\mathbf{s} = \mu W^I C\Delta\boldsymbol{\omega}$, which is first order. Transfers do have a formation effect, transmitted through the returns that others observe, and it is small when ϕ, β and γ are small.*

2. ***Structural policy** changes W^P (a road, a supplier program, industrial policy). Enforcement: $E = \operatorname{diag}(\boldsymbol{\gamma})\Delta W^P\mathbf{x}^0$. Formation: $\mu\Delta\mathbf{s} = \mu W^I \operatorname{diag}(\boldsymbol{\gamma})\Delta W^P\mathbf{x}^0$.*

3. ***Information policy** changes W^I (internet access, school exchange, mentor networks, media regulation). At fixed $\boldsymbol{\theta}$ neither allocations nor returns change, so $E = 0$ and Ψ is undefined. The whole effect is formation: $\mu\Delta\mathbf{s} = \mu\,\Delta W^I\,\mathbf{r}^0$.*

Proof. Differentiate (10.1)–(10.3) at fixed $\boldsymbol{\theta}^0$, and use Theorem 10.12. □

Evidence that transfers shift aspirations and investment beyond their direct effect — especially when the beneficiaries' reference groups are treated too — includes Macours and Vakis (2014, *EJ*) and Macours et al. (2012, *AEJ: Applied*) on a Nicaraguan cash-transfer program.

Example 10.16 (A Stylized Bolivia: Three Policies Compared [ILLUSTRATIVE])**.** *Use Example 10.6. Policy goal: reduce the preference gap between the gas block and the*

rural block (initially 0.137*). The outcome for* Ψ *is returns; the aggregate ratio* $\bar{\Psi}$ *is used throughout.*

Policy A — Transfer. *Move* 0.1 *of endowment from each gas agent to each rural agent.*

Policy B — Structural. *Raise cross-block production links between gas and rural agents from* 0.01 *to* 0.08 *each, lowering the within-block shares of those blocks to* $0.73/3$ *to keep* W^P *row-stochastic.*

Policy C — Information. *Rural agents' weight on each gas agent rises from* 0.025 *to* 0.075*, and their own-block weight falls from* 0.85 *to* 0.70 *(a targeted media program, school exchange, or mentor network).*

	$\Delta\theta^*_{gas}$	$\Delta\theta^*_{rural}$	$\Delta(\theta_1 - \theta_3)$	$\bar{\Psi}(\infty)$	$\bar{\Psi}(1)$
A: Transfer	-0.014	$+0.006$	-0.020 *(15%)*	*0.12*	*0.02*
B: Structural	-0.002	$+0.002$	-0.005 *(3%)*	*0.09*	*0.01*
C: Information	-0.001	$+0.027$	-0.028 *(21%)*	*undefined*	*undefined*

Transfer. The transfer moves returns immediately (enforcement) and, through what others observe, preferences over time. In this calibration the formation effect adds about 12% to the enforcement effect in the long run and 2% after one generation, and it narrows the preference gap by 15%.

Structural. With a small spillover coefficient ($\gamma = 0.02$*), the production link moves returns and preferences only slightly.*

Information. Rural agents now see more of the gas block's returns. Allocations and returns do not change at fixed preferences ($E = 0$*), so* Ψ *is undefined. The preference gap narrows by 21% — more than with either other lever — through formation alone. The only change is who sees what.*

Key Structural Insight

The three policy levers operate on three different objects: transfers change $\boldsymbol{\omega}$, structural policy changes W^P, and information policy changes W^I. They differ in *how* they work, not by a universal ranking:

	Object changed	Enforcement	Formation
Transfer	$\boldsymbol{\omega}$	Immediate	Through observed returns
Structural	$W^P \to \mathbf{r}$	Immediate	Through observed returns
Information	W^I	None	All of the effect; generations

Transfers and structural policy act at once and also, more slowly, on preferences. Information policy acts only through preferences, and so only over generations. In Example 10.16 it narrows the preference gap most, but which lever is most effective per unit of cost depends on parameters (ϕ, β, γ, μ, ρ) that must be estimated. An evaluation that holds preferences fixed (A6) assigns information policy no effect at all — which is why the channel is easy to miss.

10.6 Summary

1. The **formation model** is a three-equation dynamical system: bargaining equilibrium (the TU rule $x_i = \omega_i + \pi_i\Phi + \phi(\theta_i - \bar{\theta})$) $\to$ return function $\to$ formation equation, producing a map $\boldsymbol{\theta}_{t+1} = T(\boldsymbol{\theta}_t)$ with Jacobian $J = \rho I + \mu\phi W^I C P$. Since $J\mathbf{1} = \rho\mathbf{1}$, the loop amplifies only *relative* preferences.

2. Steady states satisfy $\theta_i^* = \theta^\circ + \mathcal{M}(\sum_j W_{ij}^I r_j^* - r^*)$ with $\mathcal{M} = \mu/(1-\rho)$. Preference dispersion is bounded by $\mathcal{M}\,\tau_1(W^I)$ times the return range: similar information rows compress preferences, and block-diagonal W^I stratifies them.

3. In the affine model the steady state is unique (when $I - J$ is nonsingular) and globally stable when $\rho(J) < 1$. A norm bound gives a sufficient condition. **Multiple steady states** require nonlinear (e.g. S-shaped) returns, and then a poverty trap is a low-θ basin, with an explicit escape threshold.

4. The **formation effect** of a perturbation is $Y(I - J^T)(I - J)^{-1}\mu\Delta\mathbf{s}$ at horizon T: the formation Leontief inverse applied to the signal shock. The Ψ ratio of Chapter 9 follows; in a one-loop model, $\Psi(\infty) = \mu\beta\eta/(1 - \rho - \mu\beta\eta)$.

5. **Three policy levers**: transfers and structural policy have enforcement effects and (first-order) formation effects through observed returns. Information policy has no enforcement effect (Ψ undefined) and works entirely through formation. In the stylized Bolivia calibration it narrows the preference gap most.

10.7 Exercises

10.1 The formation map by hand. For a single agent with reduced-form returns $r(\theta) = 0.05 + 0.1\theta$: $\rho = 0.8$, $\mu = 0.3$, $\theta^\circ = 1$, $r^* = 0.1$.

(a) Write the formation map $T(\theta)$ explicitly.

(b) Find all fixed points. (Solve a linear equation.)

(c) Compute $J = T'(\theta^*)$. Is the steady state stable?

(d) Starting from $\theta_0 = 0.5$, simulate 20 periods.

10.2 Why convex returns are not enough. For a single agent, $\rho = 0.85$, $\mu = 0.25$, $\theta^\circ = 1$, $r^* = 0.1$, $r(\theta) = 0.05 + 0.1\theta + 0.05\theta^2$:

(a) Write $T(\theta)$. Show that $T(\theta) - \theta$ is a quadratic in θ, so there are at most two fixed points.

(b) Find them, and compute $T'(\theta^*)$ at each. Which is stable?

(c) Describe the long-run outcome from $\theta_0 = 0.9$, $\theta_0 = 1.1$, $\theta_0 = 5$ and $\theta_0 = 9$. Is there a poverty trap?

(d) Explain why a poverty trap with two stable steady states needs returns whose slope first rises and then falls (an S-shape), as in Example 10.11.

10.3 The Jacobian. For $N = 3$, block-diagonal W^I (two blocks: agents 1–2 with weights $\frac{1}{2}, \frac{1}{2}$, and agent 3 alone), reduced-form returns $r_i = 0.1 + 0.1\theta_i$, $\rho = 0.8$, $\mu = 0.2$:

(a) Write the 3×3 Jacobian J explicitly.

(b) Compute its eigenvalues. Is the steady state stable?

(c) How do the eigenvalues change if μ doubles? At what μ does the steady state lose stability?

10.4 Three policies for the stylized Bolivia. Reproduce Example 10.16:

(a) Verify the three-block steady state.

(b) Compute the enforcement and formation effects for each policy.

(c) Which policy narrows the preference gap most? Which narrows the allocation gap most?

(d) Design a combined policy (all three levers). What is the total $\Delta(\theta_1 - \theta_3)$? Is it larger than the sum of the individual effects?

10.5 The formation Leontief inverse. For $N = 2$ with $J = \begin{pmatrix} 0.85 & 0.05 \\ 0.03 & 0.90 \end{pmatrix}$:

(a) Verify $\rho(J) < 1$.

(b) Compute $(I - J)^{-1}$ exactly.

(c) Compute the partial sum $I + J + J^2 + \cdots + J^{10}$. How close is it to the exact inverse?

(d) Interpret: a formation signal of $\Delta s = (0.01, 0)$ (only agent 1 sees a slightly higher return). What is $\Delta\boldsymbol{\theta}^* = (I - J)^{-1}\mu\Delta s$ with $\mu = 0.2$? Does agent 2's preference change even though her signal didn't?

10.6 Basin boundaries and policy design. In the poverty trap model of Example 10.11:

(a) A one-time aspiration intervention raises θ by δ. What is the minimum δ to escape the trap?

(b) A permanent subsidy raises α (baseline return) by $\Delta\alpha$. How does the basin boundary $\hat{\theta}$ shift?

(c) Which is cheaper: raising θ by δ once, or raising α permanently? Under what conditions does each dominate?

(d) Connect to Bernard et al. (2014): the aspiration documentary is a one-time $\Delta\theta$. When does it produce a permanent effect (escape) vs. a transient one (return to trap)?

10.7 Cardoso and Faletto's "situations." Cardoso and Faletto (1969) identified several "situations of dependency," each characterized by a specific alliance between domestic classes and foreign capital. Model three situations as different (W^P, W^I) configurations for a four-agent economy (foreign capital, domestic bourgeoisie, middle class, workers):

(a) **Enclave:** Foreign capital controls production; domestic classes are peripheral. Write W^P and W^I.

(b) **National populism:** Domestic bourgeoisie allied with workers against foreign capital. Write (W^P, W^I).

(c) **Associated dependent:** Domestic bourgeoisie allied with foreign capital against workers. Write (W^P, W^I).

(d) For each situation: compute $\boldsymbol{\pi}$, $\boldsymbol{\theta}^*$, and $\mathbf{x}^*$. Which "situation" produces the most equal allocation? The most unequal?

Mathematical Appendices

Appendix 10A: The Formation Leontief Inverse

Definition. The formation Leontief inverse is $(I - J)^{-1}$, where $J = DT(\boldsymbol{\theta}^*)$ is the Jacobian of the formation map at a stable steady state.

Neumann series. Since $\rho(J) < 1$:

$$(I - J)^{-1} = I + J + J^2 + J^3 + \cdots = \sum_{k=0}^{\infty} J^k. \tag{10.17}$$

Each term J^k has a formation interpretation: I is the direct signal (round 0); J is the first feedback (higher θ produces higher r, which raises θ); J^2 is the second feedback; and so on. The series converges because each round is damped by the factor $\rho(J)$.

Comparison with the production Leontief inverse. In Chapter 1, $(I - A')^{-1}$ sums the production cascade: direct demand + indirect demand + ... The formation Leontief inverse sums the preference cascade: direct signal + first-round feedback + ... The mathematical structure is identical. The timescale is different: production rounds happen within a period; formation rounds happen across generations.

Block structure. If J is approximately block-diagonal (because W^I and W^P are), then $(I - J)^{-1}$ is also approximately block-diagonal. The within-block entries of $(I - J)^{-1}$ can be large (strong within-block feedback), while the cross-block entries are small (weak cross-block feedback). This is the mathematical structure of block-wise steady states: within each block, signals feed back, but they barely cross block boundaries.

Appendix 10B: Bifurcation Analysis

The question. As the formation multiplier $\mathcal{M} = \mu/(1 - \rho)$ increases, the number and stability of steady states can change. A **bifurcation** is a qualitative change in the dynamical system as a parameter crosses a threshold.

Saddle-node bifurcation. The generic bifurcation in the scalar formation model. As $\mathcal{M}$ (or μ) increases past a critical value, two new steady states appear together, one stable and one unstable. Below it there is one stable steady state; above it there

are three (two stable, one unstable). This is the "poverty trap" bifurcation: the trap exists only when the formation multiplier is large enough.

The critical point is *not* given by a norm bound. It is the parameter value at which the map becomes tangent to the 45° line, i.e. at which some θ_c satisfies both

$$T(\theta_c) = \theta_c \qquad \text{and} \qquad T'(\theta_c) = 1. \tag{10.18}$$

The contraction condition $\mu \max_\theta |g'(\theta)| < 1 - \rho$ is only *sufficient* for uniqueness, so it gives a lower bound on the critical value. In Example 10.11 (holding $\rho = 0.85$), the contraction bound guarantees uniqueness for $\mu < 0.133$, while the saddle-node occurs at $\mu_c \approx 0.188$ ($\mathcal{M}_c \approx 1.25$). Between the two values the steady state is still unique, even though the map is not a contraction.

Below the critical value the steady state is unique, but it need not be close to the anchor: uniqueness is not exogeneity. Above it, the formation loop creates self-reinforcing dynamics and multiple steady states.

Pitchfork bifurcation. When the system has a symmetry (two identical blocks, say), the symmetric steady state ($\theta_1 = \theta_2$) can become unstable as $\mathcal{M}$ increases, splitting into two asymmetric steady states ($\theta_1 > \theta_2$ and $\theta_1 < \theta_2$). This is the "symmetry-breaking" bifurcation: identical populations can develop different cultures because the formation feedback amplifies small initial differences. In Example 10.11 with the reference return set to the logistic midpoint ($r^* = 0.13$), the map is symmetric about $\theta = 1$, and the symmetric point loses stability when $0.85 + \mu \cdot 1.125 = 1$, i.e. at $\mu = 0.133$: below it the symmetric point is the unique steady state, above it two new stable steady states appear on either side.

Structuralist Case Study: Cardoso and Faletto's "Situations of Dependency"

In 1969, Fernando Henrique Cardoso and Enzo Faletto published *Dependência e Desenvolvimento na América Latina* (Cardoso and Faletto, 1969) — a book that redefined dependency theory by insisting that dependence is not a single condition but a set of historically specific **situations**, each characterized by a particular articulation between domestic class structure and the international economy.

Their argument rejected both stagnationist readings of dependency (that the global structure dictates local outcomes) and the optimism of modernization theory. Instead, they proposed that outcomes depend on the *local configuration* of social forces — who is allied with whom, who controls which resources, and how domestic classes relate to foreign capital. In this book's language, that is a theory of **local network topology**.

A caveat comes first. Cardoso and Faletto were explicitly *anti-formalist*. Their "historical-structural" method treated dependency as a way of analyzing concrete situations, not as a formal theory to be tested. Palma (1978) put this distinction in his title, and Cardoso (1977, "The Consumption of Dependency Theory in the United States", *LARR*) criticized attempts to turn the approach into testable propositions. What follows is a *translation* into this chapter's model. It is useful for organizing comparisons, but its objects — a scalar θ, fixed matrices — are much thinner than the class projects and political conflicts they wrote about.

Part 1: Dependency is not a structure but a set of situations. Cardoso and Faletto distinguished several "situations of dependency" in Latin American history: (i) enclaves controlled by foreign capital (Bolivia's tin mines, Chilean copper, Venezuelan oil), (ii) nationally controlled export economies (Argentine beef, Brazilian coffee), (iii) import-substitution regimes with nationalist alliances (Perón's Argentina, Vargas's Brazil), (iv) the "new dependency" of the internationalized domestic market (post-1964 Brazil's alliance of the state, local capital and multinational capital), which Cardoso later named "associated-dependent development" (Cardoso, 1973). Each situation has a different W: different production linkages (W^P), different social bonds (W^U), and different information flows (W^I).

In the formation model: each "situation" is a different triple (W^P, W^U, W^I), producing a different composite W, a different Perron eigenvector $\boldsymbol{\pi}$, a different formation signal, and therefore a different steady-state $\boldsymbol{\theta}^*$. The same country (Brazil) passes through different "situations" over time — in the model, because the class alliances shift and rewire W.

Part 2: Class alliances determine the network. The distinctive analytical move: Cardoso and Faletto placed class alliances at the center. In the enclave situation, the alliance is between foreign capital and a thin domestic administrative class. In the national-populist situation, the alliance is between the domestic bourgeoisie and workers, against foreign capital. In the associated-dependent situation, the domestic bourgeoisie allies with foreign capital against workers.

Each alliance is a specific pattern of edges in W. The enclave has a star topology centered on foreign capital. The national-populist has a dense domestic cluster with weak foreign links. The associated-dependent has a cross-border elite cluster ($W^I_{\text{domestic bourgeoisie, MNC}} \gg 0$) and a disconnected worker block ($W^I_{\text{worker, elite}} \approx 0$).

Part 3: Situations shape political projects, not just allocations. For Cardoso and Faletto, each situation came with its own political projects and ideologies, carried by class alliances: oligarchic, populist, developmentalist. These are collective, contested projects, not individual preference parameters. The formation model captures only a thin shadow of them: each situation's W^I produces a different

steady-state $\boldsymbol{\theta}^*$. Read θ as, for example, the time horizon of investment, and the model says that an enclave W^I that isolates the domestic majority from modern-sector returns sustains low θ^* there, while a nationally integrated W^I supports higher θ^*. The claim that preferences are endogenous to the situation is the book's; the content of the ideologies is theirs.

Situation	**Alliance**	**Network topology**	**Formation channel (stylized)**
Enclave	Foreign capital + admin. class	Star, block-diagonal W^I	Majority isolated from modern returns
National-populist	Domestic burg. + workers	Dense domestic, weak foreign	Domestic returns widely observed
Associated-dependent	Domestic burg. + foreign capital	Elite cross-border, worker disconnect	Elite observes centre; workers do not

The formation channel differs across situations, because what each group can observe differs. In the enclave, the domestic majority does not see the modern sector's returns. In the associated-dependent situation, the elite's W^I connects to the centre, so the elite adopts centre consumption patterns and technocratic values (Furtado's mechanism, Chapter 9), while workers remain in a separate W^I block. Note that Ψ is a property of a *perturbation*, not of a situation (Definition 9.13). The comparable statement is that the same policy has different formation effects in different situations.

Cardoso and Faletto wrote historical sociology, deliberately. The formation model offers their "situations" a parameter space and their "alliances" a network topology — a vocabulary for comparison, not a replacement for the concrete historical analysis they insisted on.

Further Reading

The formation model and its relatives: The specific model of this chapter is the book's, but it has close antecedents. On cultural transmission and endogenous preferences: Bisin and Verdier (2001, 2011); Bowles (1998); Panebianco and Verdier (2017, *GEB*) on transmission in networks. On aspirations, segregation and multiple steady states: Mookherjee et al. (2010, *JEEA*); Genicot and Ray (2017); Dalton

et al. (2016, *EJ*). On reference consumption formed in a network and its interaction with market integration: Ghiglino and Goyal (2010, *JEEA*). On stratification through local interactions: Durlauf (1996); Bénabou (1996); Brock and Durlauf (2001). On learning as the mechanism: Fernández (2013, *AER*). The empirical counterpart of the enforcement–formation distinction is the epidemiological approach of Fernández and Fogli (2009); see Chapter 11. On transfers and aspirations: Macours and Vakis (2014, *EJ*).

Multiple equilibria in development: Azariadis and Stachurski (2005), "Poverty Traps," in *Handbook of Economic Growth*. Kraay and McKenzie (2014), "Do Poverty Traps Exist?" Genicot and Ray (2017), "Aspirations and Inequality."

Dynamical systems in economics: Acemoglu (2009), *Introduction to Modern Economic Growth* (the appendix on dynamical systems). Galor (2011), *Unified Growth Theory*.

Dependency theory: Cardoso and Faletto (1969), *Dependencia y desarrollo en América Latina*, México: Siglo XXI (the primary source; Portuguese edition *Dependência e Desenvolvimento na América Latina*, Rio de Janeiro: Zahar, 1970; English translation: University of California Press, 1979). Cardoso (1973), "Associated-Dependent Development: Theoretical and Practical Implications," in Stepan (ed.), *Authoritarian Brazil*. Cardoso (1977), "The Consumption of Dependency Theory in the United States." For the debate: Palma (1978), "Dependency: A Formal Theory of Underdevelopment or a Methodology for the Analysis of Concrete Situations of Underdevelopment?"

Chapter 11

The Ψ Ratio: Measuring Formation

Separating What the Structure Enforces from What It Forms

> *"The important thing in science is not so much to obtain new facts as to discover new ways of thinking about them."*
> — Attributed to William Lawrence Bragg

Chapters 9 and 10 built the formation model: preferences evolve through W^I, producing a dynamical system with enforcement and formation channels. The Ψ ratio (Definition 9.13) compares the formation effect of a stated perturbation with its enforcement effect. This chapter asks what data can reveal about it.

The leading strategy exploits the timescale separation of Chapter 9. Enforcement effects operate within a generation: a price change today shifts behavior today. Formation effects operate across generations: a structural change today shifts the signals young people observe, which shifts their preferences, which shifts their behavior *when they become adults*. The hope is that within-generation variation identifies enforcement and across-cohort variation identifies formation.

The chapter's conclusion is more cautious than that hope. The statistics used in practice — cohort–year variance ratios and epidemiological coefficients — are informative, but they are *not* Ψ. They map to it only under strong, stated restrictions. All numerical magnitudes in this chapter are [ILLUSTRATIVE] unless a source is given.

11.1 The Identification Problem

11.1.1 Why It Is Hard

The econometrician observes behavior $x_{i,t}$ — how much agent i works, saves, invests in education, or consumes. She wants to decompose $x_{i,t}$ into:

$$x_{i,t} = f(\underbrace{p_t, w_{i,t}, Z_{i,t}}_{\text{enforcement: prices, constraints}}, \underbrace{\theta_{i,t}}_{\text{preference}}). \tag{11.1}$$

The problem: $\theta_{i,t}$ is unobservable. It enters the behavior function alongside prices, wages, and constraints. If θ is correlated with prices (because W^I and W^P overlap, or because the same structural change affects both), the enforcement and formation effects are confounded.

Example 11.1 (The Confound)**.** *Bolivia opens a gas pipeline to a rural region. Two things happen simultaneously: (1) local wages rise (enforcement: the bargaining equilibrium shifts because W^P changed), and (2) young people observe higher returns to modern-sector work (formation: W^I signals shift, eventually changing θ). An econometrician who regresses labor supply on wages and omits θ attributes nothing to formation. Worse, she may attribute the formation effect to wages, biasing the wage coefficient upward.*

Proposition 11.2 (Omitted-Preference Bias)**.** *Suppose $x_{i,t} = \beta p_{i,t} + b\,\theta_{i,t} + u_{i,t}$ with $\mathbb{E}[u \mid p, \theta] = 0$, finite second moments and $\operatorname{Var}(p) > 0$, and the econometrician regresses x on p alone (with an intercept) in a random sample. Then*

$$\operatorname{plim} \hat{\beta} = \beta + b\,\frac{\operatorname{Cov}(p, \theta)}{\operatorname{Var}(p)}.$$

If the same structural change raises both prices and aspirations ($\operatorname{Cov}(p, \theta) > 0$) and aspirations raise the behavior ($b > 0$), then $\hat{\beta}$ is biased upward: the regression overstates enforcement and attributes formation to prices.

Proof. By the law of large numbers, $\operatorname{plim} \hat{\beta} = \operatorname{Cov}(x, p) / \operatorname{Var}(p)$. Substituting the model, $\operatorname{Cov}(x, p) = \beta \operatorname{Var}(p) + b \operatorname{Cov}(\theta, p) + \operatorname{Cov}(u, p)$, and $\operatorname{Cov}(u, p) = 0$ because $\mathbb{E}[u \mid p, \theta] = 0$. This is the omitted-variable-bias formula (Angrist and Pischke, 2009, §3.2.2). □

11.1.2 The Timescale Idea and Its Limits

The idea: enforcement and formation operate on different timescales.

Definition 11.3 (Timescale Separation in Data)**.**

1. ***Enforcement variation****: within a cohort, across time. Agents born in the same year face different prices at different points in their lives. This variation is meant to identify the behavioral response at fixed preferences, on the assumption that preferences were formed before the price change.*

2. ***Formation variation****: across cohorts, within a time period. Agents of different cohorts observed in the same year may have different preferences, because they were exposed to different W^I signals during their formative years. This variation is meant to identify the preference difference, holding current prices constant.*

The classification fails in both directions. Enforcement can be *cohort-specific*: the oral contraceptive pill changed women's career investments according to their age when legal access arrived (Goldin and Katz, 2002, *JPE*; Bailey, 2006, *QJE*), so a "price" shows up as a cohort effect. Formation can be *contemporaneous*: learning from neighbours (Fogli and Veldkamp, 2011) and corrections of perceived norms (Bursztyn et al., 2020) move behavior within a generation, so formation shows up as a year effect.

Definition 11.4 (Cohort–Year Specification and the Variance Ratio)**.** *Let $x_{i,c,t}$ be the behavior of agent i from cohort c at time t. The **cohort–year specification** is*

$$x_{i,c,t} = \underbrace{\alpha_t + \beta p_t + \gamma Z_{i,t}}_{\text{year effects}} + \underbrace{\delta_c}_{\text{cohort effect}} + \underbrace{\eta_{i,c,t}}_{\text{residual}}. \tag{11.2}$$

*The **cohort–year variance ratio** is the descriptive statistic*

$$\hat{\Psi}_{var} := \frac{\mathrm{Var}(\hat{\delta}_c)}{\mathrm{Var}(\hat{\alpha}_t + \hat{\beta} p_t)}. \tag{11.3}$$

*$\hat{\Psi}_{var}$ is **not** the Ψ ratio of Definition 9.13. It compares squared dispersions of two sets of fixed effects over two sample windows. Ψ compares the formation and enforcement effects of a single perturbation.*

Proposition 11.5 (Age–Period–Cohort Non-Identification)**.** *Let age be $a = t - c$. For any constant k, the decompositions*

$$(\alpha_t, \delta_c, A_a) \quad \text{and} \quad (\alpha_t - kt,\ \delta_c + kc,\ A_a + ka)$$

produce identical fitted values. So if age effects A_a are included (and behavior has a life-cycle profile, as labor supply does), linear trends in α_t and δ_c are not identified. $\hat{\Psi}_{var}$ then

depends on k: it can be made close to 0 *when the cohort effects are close to linear in c, and very large when the year effects are close to linear in t (in each case provided the other set of effects is not close to linear with the same slope k). If age effects are omitted, as in* (11.2)*, they are loaded onto the cohort and year effects.*

Proof. $(\alpha_t - kt) + (\delta_c + kc) + (A_a + k(t-c)) = \alpha_t + \delta_c + A_a$. $\operatorname{Var}(\delta_c + kc) / \operatorname{Var}(\alpha_t - kt)$ is a ratio of two nonnegative quadratics in k. The numerator $\operatorname{Var}(\delta_c) + 2k \operatorname{Cov}(\delta_c, c) + k^2 \operatorname{Var}(c)$ is minimized at $k_\delta := -\operatorname{Cov}(\delta_c, c) / \operatorname{Var}(c)$, minus the slope of the least-squares trend through δ_c; its minimum is the variance of the detrended δ_c, which is near 0 when δ_c is nearly linear. At k_δ the ratio is therefore near 0 unless the denominator is also near 0 there. Symmetrically, the denominator is minimized at $k_\alpha := \operatorname{Cov}(\alpha_t, t) / \operatorname{Var}(t)$ with minimum the variance of the detrended α_t, so the ratio is very large at k_α when α_t is nearly linear and the numerator is not near 0 there. (As $|k| \to \infty$ the ratio tends to $\operatorname{Var}(c) / \operatorname{Var}(t)$.) □

Any estimate of $\hat{\Psi}_{\text{var}}$ therefore requires an explicit identifying restriction. Common choices are Deaton's normalization (year effects orthogonal to a linear trend), equality of two adjacent age or cohort effects, or a proxy variable for one dimension. The restriction must be stated, and the sensitivity of $\hat{\Psi}_{\text{var}}$ to it reported (Hall, 1971; Heckman and Robb, 1985; Deaton and Paxson, 1994; McKenzie, 2006; Fosse and Winship, 2019). Deaton's normalization assigns all trends to age and cohort, so it mechanically tends to *raise* $\operatorname{Var}(\hat{\delta}_c)$ relative to $\operatorname{Var}(\hat{\alpha}_t)$. On the table of Exercise 11.1, the two-way fit gives $\hat{\Psi}_{\text{var}} = 8.0$. The observationally equivalent APC decompositions with $k = -0.10, -0.05, 0, +0.05$ per decade give $\hat{\Psi}_{\text{var}} = 0.03, 0.62, 8.0, 507$.

Proposition 11.6 (When the Variance Ratio Maps to Ψ)**.** *Suppose (i) the APC restriction used is correct; (ii) all year-effect variation is enforcement and all cohort-effect variation is formation; and (iii) a single structural index z drives both, with* $\alpha_t = e\, z_t$ *and* $\delta_c = f\, z_{\tau(c)}$*, where* $z_{\tau(c)}$ *is the index in cohort c's formative years, and the two sample windows have the same* $\operatorname{Var}(z) > 0$*, with* $e \neq 0$ *and the fixed effects consistently estimated. Then* $\hat{\Psi}_{var} \to (f/e)^2$*, and* $\sqrt{\hat{\Psi}_{var}}$ *estimates* $|\Psi(\Delta z, \infty)|$ *for a permanent change in z, in the population of interest.*

Proof. Under (i)–(iii), $\operatorname{Var}(\delta_c) / \operatorname{Var}(\alpha_t) = f^2 \operatorname{Var}(z) / (e^2 \operatorname{Var}(z))$. A permanent change Δz moves the outcome by $e\Delta z$ at fixed preferences and by $f\Delta z$ once a cohort has formed under it, so $\Psi(\Delta z, \infty) = f/e$. □

Each condition is strong, and none is testable from the cohort–year table alone. Outside these conditions, $\hat{\Psi}_{\text{var}}$ is a descriptive statistic about the relative dispersion of cohort and year effects. It is useful, but it is not Ψ. Note also that $\hat{\Psi}_{\text{var}} > 0$

in every finite sample: sampling noise makes $\mathrm{Var}(\hat{\delta}_c) > 0$ even when every $\delta_c = 0$. "$\hat{\Psi}_{\text{var}} > 0$" is therefore not evidence of formation. The relevant test is an F-test of $\delta_c = 0$ for all c, and the variance components should be bias-corrected (subtract the average sampling variance of the $\hat{\delta}_c$).

11.2 The Epidemiological Approach

11.2.1 The Idea

The cleanest evidence on cultural transmission comes from migrants (Fernández, 2011). A woman raised in country B by parents from country A faces country B's prices and constraints (enforcement) but may carry part of country A's culture (formation). Her behavior reveals the interaction of A's formation with B's enforcement.

Definition 11.7 (Epidemiological Regression)**.** *For female labor force participation of second-generation immigrant women (born and raised in country B, parents from country A):*

$$LFP_{i,A,B,t} = \alpha + \beta \cdot LFP_{A,t_0} + \gamma \cdot Z_{i,B,t} + \varepsilon_{i,t}, \tag{11.4}$$

where LFP_{A,t_0} is the female labor force participation rate in the parents' country of origin around the time of emigration, used as a proxy for the culture they brought.

The key coefficient is β: the sensitivity of daughters' LFP to origin-country LFP, controlling for current conditions $Z_{i,B,t}$. If $\beta = 0$, there is no evidence of a transmitted cultural component. If $\beta > 0$, the parents' cultural background affects the daughter's behavior even in a different country.

Proposition 11.8 (What the Epidemiological Coefficient Measures)**.** *Assume:*

1. *the origin rate reflects origin preferences and origin enforcement, $LFP_A = a + b\,\theta_A + e_A$, with a common constant a and e_A uncorrelated with θ_A;*

2. *the mother carries the origin preference, $\theta_m = \theta_A$ (no selection on θ into migration);*

3. *the daughter's preference follows the formation equation within the dynasty, $\theta_d = \rho\,\theta_m + (1-\rho)\theta^{\circ} + \mu s_B$, with a host-country signal s_B that is common to all daughters in B (and so absorbed by the constant);*

4. *the daughter's behavior is $LFP_d = a_B + b\,\theta_d + u_d$, with u_d uncorrelated with origin variables conditional on Z.*

All second moments below are conditional on Z (that is, after partialling Z out), and $\mathrm{Var}(LFP_A) > 0$. *Then the population coefficient in* (11.4) *is*

$$\beta = \rho\,\lambda, \qquad \lambda := \frac{b^2\,\mathrm{Var}(\theta_A)}{b^2\,\mathrm{Var}(\theta_A) + \mathrm{Var}(e_A)} \in [0,1]. \tag{11.5}$$

The coefficient understates ρ by the reliability λ of origin LFP as a proxy for origin culture. For a third generation (grandparents from A), $\beta_3 = \rho^2\lambda$, so $\beta_3/\beta_2 = \rho$ when λ is common to both generations.

Proof. We have

$$\mathrm{Cov}(\mathrm{LFP}_d, \mathrm{LFP}_A) = b\rho\ \mathrm{Cov}(\theta_A, a_A + b\theta_A + e_A) = \rho b^2\,\mathrm{Var}(\theta_A),$$

using assumptions 2–4 ($\theta_d = \rho\theta_A + \text{const}$ and $\mathrm{Cov}(u_d, \mathrm{LFP}_A) = 0$) and then assumption 1. Also $\mathrm{Var}(\mathrm{LFP}_A) = b^2\,\mathrm{Var}(\theta_A) + \mathrm{Var}(e_A)$. Divide. For the third generation, iterate assumption 3 once more (the host-born mother's signal is also common), so $\theta_d = \rho^2\theta_A + \text{const}$. □

Three caveats follow from the derivation.

- *Origin LFP includes origin enforcement* (e_A): wages, laws and childcare in country A. That is why $\beta < \rho$.
- *Bad controls.* Conditioning on the daughter's education, marital status or number of children removes channels through which culture operates. Those controls bias β toward zero as an estimate of the total cultural effect (Angrist and Pischke, 2009, §3.2.3).
- *What is transmitted.* β identifies a cultural component broadly defined — beliefs, norms, tastes, or ethnic-network access to jobs — not a preference parameter specifically (Chapter 9, Section 9.1.4).

Intuition: Why migrants help, and why they are not a clean experiment

Second-generation women in the same host labor market face the same prices and laws. If daughters of parents from high-participation countries work more, the difference is not explained by those prices. It may come from culture transmitted through the family W^I. But the women do *not* differ only in their parents' culture. They differ in parental human capital and wealth, in ethnic-community networks (a local W^P and W^I), and in the discrimination they face. The epidemiological design controls for what it can observe and interprets the rest with care.

11.2.2 What the Evidence Shows: Female Labor Force Participation

Stylized Fact 11.9 (Epidemiological Evidence on Female Work)**.** *Using US Census data on second-generation American women, Fernández and Fogli (2009, AEJ: Macro) and Fernández (2007, JEEA) find that female labor force participation in the parents' country of ancestry (measured around 1950) predicts the daughters' work behavior in the US, conditional on individual and parental characteristics. The results are robust to a range of controls for parental human capital. Fernández (2013, AER) models the twentieth-century rise in US female participation as learning about the costs of working.*

Example 11.10 (From Epidemiological Coefficients to ρ [ILLUSTRATIVE])**.** *Suppose the second-generation coefficient is $\hat{\beta}_2 = 0.40$ and the third-generation coefficient is $\hat{\beta}_3 = 0.25$ (illustrative values, not taken from the papers above). By Proposition 11.8, $\hat{\rho} = \hat{\beta}_3/\hat{\beta}_2 = 0.625$ per generation, and the reliability of origin LFP as a cultural proxy is $\hat{\lambda} = \hat{\beta}_2/\hat{\rho} = 0.64$. Reading $\hat{\beta}_2 = 0.40$ directly as ρ would understate persistence.*

Example 11.11 (A Cohort–Year Variance Ratio for Female LFP [ILLUSTRATIVE])**.** *Suppose that, under a stated APC restriction, the variance of female LFP across cohort–year cells in the US, 1940–2000, is 0.042; the year-effect component is $\mathrm{Var}(\hat{\alpha}_t) = 0.013$; and the cohort-effect component is $\mathrm{Var}(\hat{\delta}_c) = 0.027$. These are illustrative numbers. Then*

$$\hat{\Psi}_{var} = \frac{0.027}{0.013} = 2.08, \qquad \sqrt{\hat{\Psi}_{var}} = 1.44. \tag{11.6}$$

Cohort effects are about 1.44 times as dispersed as year effects (in standard deviations). This does not mean "for every 1 pp of LFP attributable to wages, another 2 pp are attributable to norms". That reading would need Proposition 11.6, and even then the effect ratio would be $\sqrt{2.08} = 1.44$, not 2.08. Under a different APC restriction, the same data could give a very different $\hat{\Psi}_{var}$ (Proposition 11.5).

11.3 Four Applications

Each application pairs what the literature documents (where a source exists) with an *illustrative* variance ratio. The variance ratios show how the statistic would be reported; they are not estimates.

11.3.1 Application 1: Savings Rates Across Countries

Cross-country differences in saving rates are not fully explained by interest rates, income and demographic structure; the residual is often attributed to "cultural differences" without being measured.

Stylized Fact 11.12 (Patience, Saving, and Culture)**.**

1. *In the Global Preferences Survey (Falk et al., 2018, QJE), country-average patience is positively correlated with saving and with educational attainment.*

2. *The epidemiological evidence on saving is mixed. Carroll et al. (1994, QJE) find little evidence that immigrants' saving behavior in Canada reflects saving rates in their countries of origin.*

3. [ILLUSTRATIVE] *A cohort–year decomposition with cohort share 57% and year share 43% of explained variance gives* $\hat{\Psi}_{var} = 1.3$.

11.3.2 Application 2: Educational Aspirations in Sub-Saharan Africa

Stylized Fact 11.13 (Formation in Education)**.**

1. *Parental, especially maternal, education is among the strongest correlates of children's schooling in household surveys such as the DHS. The correlation mixes genes, resources and formation.*

2. *The Bernard et al. (2014) documentary experiment changes* W^I *by randomized design. It is the type of evidence that identifies a formation effect directly, at a short horizon. Mapping a six-month effect into the per-generation* μ *of Chapter 9 requires assumptions about exposure and persistence that the experiment does not pin down.*

3. [ILLUSTRATIVE] *A variance ratio of* $\hat{\Psi}_{var} = 0.8$ *would be consistent with large enforcement changes in the period (school construction, fee abolition).*

11.3.3 Application 3: Risk-Taking After Financial Crises

Stylized Fact 11.14 (Formation in Risk Attitudes)**.**

1. *Malmendier and Nagel (2011, QJE) use the Survey of Consumer Finances. Households whose lifetime experienced stock returns are lower (weighting recent years more heavily) report lower willingness to take financial risks, are less likely to own stocks, and hold a smaller share of liquid assets in stocks.*

2. *This is formation from own experience, not from signals relayed by others (Chapter 9): the diagonal of* W^I.

3. [ILLUSTRATIVE] *With current wealth and returns accounting for most of the variation in portfolio shares, a variance ratio around* $\hat{\Psi}_{var} = 0.5$ *would be plausible.*

11.3.4 Application 4: Entrepreneurship Norms in Bolivia

Example 11.15 (Self-Employment in Bolivia [ILLUSTRATIVE]). *This application is a template for Exercise 11.4, which uses Bolivia's Encuesta de Hogares (INE). All numbers are hypothetical.*

1. *Suppose self-employment rates range from 30% to 65% across departments, after controlling for income, education, and sector.*

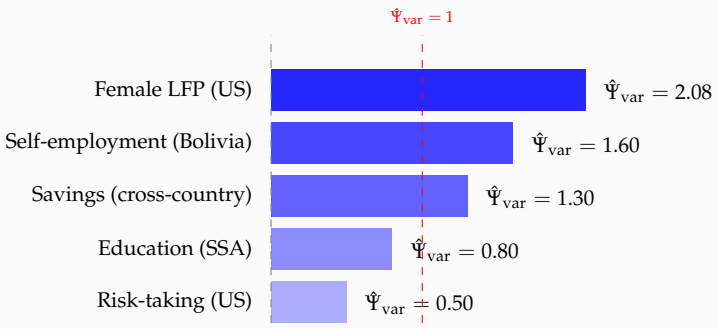
2. *Suppose departmental rates 20 years earlier predict current rates, consistent with transmission of self-employment norms — or with persistent local demand conditions, which the design must rule out.*

3. *Suppose internal migrants from high-self-employment departments keep higher self-employment rates in La Paz and Santa Cruz, controlling for current labor-market conditions (an internal epidemiological design).*

4. *Suppose a cohort–year decomposition, under a stated APC restriction, gives* $\hat{\Psi}_{var} = 1.6$.

For scale, the one national figure available here: 68.1% *of employed Bolivians were self-employed in 2019 (ILO modelled estimate, WDI* `SL.EMP.SELF.ZS`*; World Bank 2026b), above the top of the hypothetical departmental range. Departmental rates need the Encuesta de Hogares microdata, which are not used in this book.*

Visualization: $\hat{\Psi}_{\text{var}}$ across five illustrative applications



Five illustrative variance ratios. The dashed red line marks equal dispersion of cohort and year effects. The values show how the statistic is reported; they are not estimates. $\hat{\Psi}_{\text{var}} > 0$ holds by construction (it is a ratio of variances inflated by sampling noise), so a positive bar is not a rejection of A6. Each bar also depends on the APC restriction (Proposition 11.5).

11.4 The Network Disappears: Reflection and Measuring W^I

The designs above aggregate to cohorts or origin countries, so the information network W^I — the book's central object — drops out of the econometrics. Bringing it back raises the classic problems of social-interaction models.

At a steady state of the formation model with linearized returns, preferences satisfy

$$\boldsymbol{\theta} = a\mathbf{1} + \lambda\, W^I\boldsymbol{\theta} + W^I X\boldsymbol{\delta} + X\boldsymbol{\gamma} + \boldsymbol{\varepsilon}.$$

This is a linear-in-means (spatial-autoregressive) model. It has an *endogenous* effect (λ: my preference depends on my contacts' preferences, through their returns), *contextual* effects (δ: my contacts' characteristics) and possibly *correlated* effects (contacts share unobserved shocks, or chose each other because they are similar).

1. **Reflection** (Manski, 1993). When W^I averages over a whole group (cohort, region, origin country), group-mean behavior is regressed on group-mean behavior and endogenous and contextual effects cannot be separated. Regressions of cohort means on cohort means are exactly this case.

2. **Network identification** (Bramoullé et al., 2009). With an observed network and no correlated effects, the parameters are identified if I, W^I and $(W^I)^2$ are linearly independent (intransitive triads supply the instruments). A block-diagonal W^I with uniform within-block weights fails the condition, because $(W^I)^2 = W^I$. This is the pure block-diagonal case behind Chapter 10's original stratified economy (see `code/ch11.py`). Chapter 10's stated W^I, with cross-block weights 0.025, has three distinct eigenvalues $(1, 0.775, 0)$, so it satisfies the condition — but only weakly, since the cross-block entries are small. With group interactions and fixed effects, Lee (2007) shows that variation in group size is what identifies the model.

3. **Correlated effects and network formation.** If ties form on unobserved θ (homophily), W^I is endogenous. See Goldsmith-Pinkham and Imbens (2013, *JBES*) and the survey of Blume et al. (2011).

4. **Measuring W^I.** Full network surveys are expensive and usually sampled. Chandrasekhar and Lewis (2016) study estimation with sampled networks. Breza et al. (2020, *AER*) show how aggregated relational data can recover network statistics. de Paula et al. (2025) recover network ties from panel data on outcomes.

5. **Experiments on W^I.** Randomizing exposure — the Bernard et al. documentary (2014), Beaman et al.'s leader reservations (2012), partial-population designs (Moffitt, 2001), or separating learning from social utility (Bursztyn et al., 2014) — identifies formation effects of a specific ΔW^I, which is what Definition 9.13 asks for.

11.5 Estimating the Formation Parameters

11.5.1 From Moments to $(\hat{\rho}, \hat{\mu})$

The variance ratio is a reduced-form statistic. To recover the structural parameters of the formation equation (ρ, μ, θ°), we need more moments — and more assumptions.

Proposition 11.16 (Moments for the Formation Parameters)**.** *Let behavior be $x_g = a + b\,\theta_g + u_g$ in generation g of a dynasty, where u_g is i.i.d., independent of θ, and the year effects have been removed. Let θ follow the formation equation in a stationary environment, with the network signal uncorrelated across generations and with earlier preferences (no feedback from θ to the signal), so that $\theta_g = \rho\theta_{g-1} + const + \mu\,\varepsilon_g$ with ε_g white noise independent of θ_{g-1}. Write $R_\theta := b^2 \operatorname{Var}(\theta)/\operatorname{Var}(x) \in [0,1]$ for the share of behavioral variance due to preferences. Then:*

1. ***Parent–child correlation**: $\operatorname{Corr}(x_g, x_{g-1}) = \rho\, R_\theta$. On its own, this does not identify ρ: it confounds persistence with the preference share R_θ.*

2. ***Multigenerational correlations**: $\operatorname{Corr}(x_g, x_{g-2}) = \rho^2 R_\theta$, so the ratio of the grandparent–grandchild to the parent–child correlation identifies ρ.*

3. ***Response to a change in the environment**: after a permanent change in observed returns Δr, cohort effects move toward a new level at rate ρ, with long-run shift $b\,\mathcal{M}\,\Delta r$. Given ρ and b (e.g. under the normalization $b = 1$, which fixes the units of θ), this identifies μ if Δr is measured.*

4. ***Long-run level**: in a stable environment, the mean of θ identifies $\theta^\circ + \mathcal{M}(\bar{s} - r^*)$, which pins down the anchor θ° only once the signal level $\bar{s}$ and r^* are fixed.*

Proof. (1)–(2) With stationary AR(1) θ, iterating gives $\theta_g = \rho^k \theta_{g-k} + (\text{terms independent of } \theta_{g-k})$, so $\operatorname{Cov}(\theta_g, \theta_{g-k}) = \rho^k \operatorname{Var}(\theta)$. Since the u_g are i.i.d. and independent of θ,

$$\operatorname{Cov}(x_g, x_{g-k}) = b^2 \rho^k \operatorname{Var}(\theta);$$

divide by $\mathrm{Var}(x_g) = \mathrm{Var}(x_{g-k}) = \mathrm{Var}(x)$. The ratio in (2) is $\rho^2 R_\theta / (\rho R_\theta) = \rho$ when $\rho R_\theta \neq 0$. (3) With no feedback, the mean preference after the change obeys $\bar{\theta}_{g+1} - \theta^*_{\text{new}} = \rho(\bar{\theta}_g - \theta^*_{\text{new}})$, and by Proposition 9.9 $\theta^*_{\text{new}} - \theta^*_{\text{old}} = \mathcal{M}\Delta r$; multiply by b. (4) Take expectations in the steady-state condition (9.2). □

Example 11.17 (Estimating $(\hat{\rho}, \hat{\mu})$ for Female LFP [ILLUSTRATIVE])**.**

1. ***Correlations****: suppose the parent–child correlation of LFP (net of year effects) is* 0.35 *and the grandparent–grandchild correlation is* 0.217*. Then* $\hat{\rho} = 0.217/0.35 = 0.62$ *per generation, and* $\hat{R}_\theta = 0.35/0.62 = 0.565$*.*

2. ***Cross-cohort gradient****: suppose women born 1920–1930 had LFP* $\approx$ 0.30 *and women born 1950–1960 had LFP* $\approx$ 0.65*, under a stated APC restriction. The gradient is about* 0.012 *per birth year. Combined with a measured change in* W^I*-weighted returns, suppose this gives* $\hat{\mu} \approx 0.18$*.*

3. ***Long-run level****: suppose the 2000–2010 LFP level (*$\approx$ 0.72*) in a relatively stable environment gives* $\hat{\theta}^\circ \approx 0.68$ *after removing the formation-signal contribution.*

Parameter summary *(illustrative; no standard errors are reported because there is no estimation):*

Parameter	*Value*	*Interpretation*
$\hat{\rho}$	*0.62*	*Moderate persistence (half-life* $\approx$ 1.45 *generations)*
$\hat{\mu}$	*0.18*	*Moderate formation sensitivity*
$\hat{\theta}^\circ$	*0.68*	*Anchor: baseline LFP norm*
$\hat{\mathcal{M}}$	*0.47*	*Formation multiplier* $\mu/(1-\rho)$

Units matter. If the correlations were computed between birth decades rather than generations, 0.62 *would be a per-decade coefficient, corresponding to* $0.62^{2.5} = 0.30$ *per 25-year generation.*

11.5.2 The Structural Estimation

For the full formation model of Chapter 10, the structural parameters $(\rho, \mu, \theta^\circ, \boldsymbol{\beta}, \boldsymbol{\gamma})$ can in principle be estimated by GMM or maximum likelihood.

Definition 11.18 (Two-Step GMM for the Formation Model)**.** ***Step 1 (within generation).*** *Estimate the behavior function, with cohort effects* $\hat{\theta}_c$ *obtained under a stated APC restriction:*

$$\mathbb{E}\left[Z'_{i,t} \cdot (x_{i,c,t} - f(p_t, Z_{i,t}, \hat{\theta}_c))\right] = 0. \tag{11.7}$$

Step 2 (across generations). *Index cohorts so that c and $c+1$ are one generation apart (or rescale ρ). With instruments $\mathbf{z}_c$ dated $c-1$ or earlier, for example $\mathbf{z}_c = (1, \hat{\theta}_{c-1}, \hat{r}_{c-1})$,*

$$\mathbb{E}\left[\mathbf{z}_c\left(\hat{\theta}_{c+1} - \rho\hat{\theta}_c - (1-\rho)\theta^\circ - \mu(\hat{r}_c - r^*)\right)\right] = 0, \tag{11.8}$$

which is three moment conditions for three parameters. Here $\hat{r}_c$ is the average return observed by cohort c during its formative years.

Intuition: What the two steps require

Stage 1 uses within-period variation to estimate the response to prices and to extract cohort effects $\hat{\theta}_c$ as preference proxies. Stage 2 regresses $\hat{\theta}_{c+1}$ on $\hat{\theta}_c$ and the returns observed by cohort c. The two stages are *not* independent:

- $\hat{\theta}_c$ is a generated regressor, so second-stage standard errors need correction (Murphy–Topel, or a joint GMM).
- $\hat{\theta}_c$ is identified only up to the APC trend, and a linear trend in $\hat{\theta}_c$ changes the autoregressive coefficient. So $\hat{\rho}$ and $\hat{\mu}$ inherit the APC restriction.
- With few cohorts and a lagged dependent variable, cohort fixed effects in Stage 2 create Nickell bias, which is why Step 2 uses lagged instruments.
- Measurement error in $\hat{\theta}_c$ attenuates $\hat{\rho}$ unless instrumented.

11.6 Threats to Identification

11.6.1 Threat 1: Correlated Shocks

If a cohort-specific shock (e.g., a war that affects both preferences and prices for one generation) is not captured by the year effects, it contaminates $\hat{\delta}_c$.

Response: Control for cohort-level observables (cohort size, childhood GDP, childhood conflict exposure). Use the epidemiological design (migrants) as a robustness check: origin-country variation is uncorrelated with host-country shocks.

11.6.2 Threat 2: Selection into Migration

Migrants are not a random sample. If more entrepreneurial families emigrate, the epidemiological coefficient $\hat{\beta}$ reflects selection, not cultural transmission.

Response: Use second-generation immigrants (born in the host country — no migration decision of their own). Control for parental education, income, and occupation. Use variation in origin-country norms among immigrants from similar income levels.

11.6.3 Threat 3: Direct Transmission vs. Network Transmission

The formation model says preferences are formed through W^I (the network). The epidemiological design captures transmission through the family: the daughter's weight on her mother in W^I. This may understate the full formation effect if the weights on peers and media in W^I also contribute.

Response: Supplement with designs that capture non-family W^I: geographic peer effects (Fogli and Veldkamp, 2011), randomized exposure (Bernard et al., 2014), and media exposure variation (natural experiments from media deregulation).

11.6.4 Threat 4: Gene-Culture Confounding

Preferences may be partly genetically transmitted. If patience is heritable, the parent–child correlation reflects genes as well as W^I.

Response: Twin and adoption designs. Their message is sobering for family-transmission stories. Cesarini et al. (2009, *QJE*) estimate that about 20% of the variation in risk attitudes and giving in Swedish twins is genetic, with little role for the common family environment; most variation is individual-specific (including measurement error). Sacerdote (2007, *QJE*) studies Korean-American adoptees randomly assigned to families. He finds that adoptive parents' education has effects on adoptees' education and income that are much smaller than the corresponding parent–child gradients for biological children, though family environment matters more for outcomes such as drinking and smoking. Together these results suggest that *family* transmission is a modest part of formation; peers, schools, media and own experience — the rest of W^I — would have to carry most of it.

Proposition 11.19 (Sensitivity of the Variance Ratio [ILLUSTRATIVE])**.** *Suppose each threat removes a stated share of the cohort-effect variance, with the year-effect variance unchanged. Starting from the illustrative* $\hat{\Psi}_{var} = 2.08$ *of Example 11.11:*

	Lower	*Upper*
Point value	$\hat{\Psi}_{var} = 2.08$	
Correlated shocks (±10% of cohort variance)	*1.87*	*2.29*
Selection (−20%)	*1.66*	*2.08*
Gene–culture (−40%)	*1.25*	*2.08*
All three combined (multiplicative)	*0.90*	*2.29*

Under the combined worst case, $\hat{\Psi}_{var} < 1$. Whether cohort effects are more dispersed than year effects is not robust to plausible threats, even before the APC restriction is varied.

Proof. Each row multiplies the numerator by $(1 - s)$ for the stated share s; the combined row multiplies by $0.9 \times 0.8 \times 0.6$ (`code/ch11.py`). □

Selection and gene–culture confounding were defined for the epidemiological coefficient, not for the cohort–year decomposition. Applying them to $\hat{\Psi}_{\text{var}}$ assumes they contaminate the cohort effects in the same proportion. This is another reason to treat the table as a sensitivity exercise, not as bounds.

11.7 What Ψ Would Tell Us About Policy

The policy content belongs to Ψ itself, the causal ratio of Definition 9.13 for a specific perturbation and horizon — not to $\hat{\Psi}_{\text{var}}$.

Remark 11.20 (Reading Ψ for Policy).

1. $\Psi < 0.5$: enforcement dominates for that policy. Evaluations that hold preferences fixed miss less than a third of the effect (the share $\Psi/(1+\Psi)$).

2. $0.5 \leq \Psi \leq 1$: formation is substantial. Ignoring it misses 33–50% of the effect at the stated horizon.

3. $\Psi > 1$: formation dominates for that policy. Evaluations that hold preferences fixed miss most of the effect.

4. Claims that enforcement-only policies can *backfire* (reversal after enforcement is removed) need a mechanism, such as reactance or norm backlash, that the formation model of Chapter 10 does not contain.

Key Structural Insight

Ψ is a diagnostic for *one* intervention: it says how much of that intervention's effect comes through preferences, and how long it takes to arrive. It does *not* by itself say whether to invest in information or in prices. That comparison needs the total effect per unit of cost of each intervention, which can differ even when their Ψ ratios are equal. Measured honestly, the formation channel is often visible but rarely pinned down. The best evidence comes from designs that randomize W^I directly, and those identify Ψ for a specific change at a specific horizon.

Evaluations that hold preferences fixed cannot see formation at all. Estimating Ψ, however imprecisely, is what makes it visible.

Prebisch Connection: Structural adjustment and formation: a hypothesis

The Washington Consensus policies of the 1990s — trade liberalization, privatization, deregulation — worked mainly through enforcement levers: they changed prices and constraints, and their design largely took behavioral parameters as given.

A formation reading suggests a *hypothesis*, not a finding. Part of the gap between the expected and the realized supply response may reflect preferences and norms formed under the pre-reform W^I, which did not change when tariffs fell. Bolivia's 1985 programme, whose first task was to end hyperinflation, liberalized prices quickly. Whether behavior in the traditional sector responded slowly *because* of formation, rather than because of missing markets, credit, or infrastructure, is an open empirical question. The tools of this chapter could address it only with a stated APC restriction, a measured W^I, or a design that varies information exposure.

11.8 Summary

1. The **identification problem**: θ is unobserved and correlated with prices, so regressions on prices alone suffer omitted-preference bias. The timescale idea (enforcement in year effects, formation in cohort effects) fails in both directions and requires an explicit age–period–cohort restriction. On a simple table, observationally equivalent restrictions move the variance ratio from 0.03 to 507.

2. The **cohort–year variance ratio** $\hat{\Psi}_{\text{var}}$ is a descriptive statistic, not the Ψ of

Chapter 9. It maps to $|\Psi|^2$ only under strong conditions (correct APC restriction, clean assignment of effects, a single driving index). $\hat{\Psi}_{\text{var}} > 0$ is automatic, so a test of no cohort effects is needed.

3. The **epidemiological coefficient** equals $\rho\lambda$: persistence times the reliability of origin-country behavior as a proxy for origin culture. The ratio of third- to second-generation coefficients identifies ρ. Bad controls bias it toward zero, and it identifies culture broadly, not tastes.

4. **Bringing W^I back** raises the reflection problem (Manski), network identification conditions (Bramoullé–Djebbari–Fortin: uniform block-diagonal W^I fails them), homophily, and the problem of measuring W^I itself. Experiments that randomize W^I identify Ψ for a specific change.

5. **Structural parameters and robustness**: parent–child correlations confound ρ with the preference share of variance; multigenerational correlations separate them. Two-step GMM needs lagged instruments, generated-regressor corrections, and generation-spaced cohorts. Under plausible threats, the illustrative $\hat{\Psi}_{\text{var}} = 2.08$ can fall below 1, and policy conclusions should rest on Ψ for a specific intervention.

11.9 Exercises

11.1 Cohort-year decomposition by hand. Given the following female LFP rates (rows = cohorts born in decades, columns = observation years):

	1980	1990	2000
Born 1940s	0.38	0.42	0.44
Born 1950s	0.50	0.55	0.58
Born 1960s	0.60	0.66	0.72

(a) Estimate year effects $\hat{\alpha}_t$ (average across cohorts for each year).

(b) Estimate cohort effects $\hat{\delta}_c$ (average across years for each cohort, minus grand mean).

(c) Compute $\hat{\Psi}_{\text{var}} = \text{Var}(\hat{\delta}_c) / \text{Var}(\hat{\alpha}_t)$.

(d) Which is larger: the cohort effect or the year effect?

(e) Compute each cohort's age in each year. Add the linear trend k per decade to the cohort effects, subtract it from the year effects, and add

it to an age effect (Proposition 11.5). Verify that the fitted values do not change, and recompute $\hat{\Psi}_{\text{var}}$ for $k = -0.10, -0.05, 0.05$. What does your answer to (d) depend on?

11.2 Epidemiological design. A researcher has data on second-generation immigrant women in Argentina from 10 origin countries. Origin-country female LFP ranges from 0.20 (origin A) to 0.75 (origin J). All women face the same Argentine labor market.

(a) Write the epidemiological regression.

(b) The estimated $\hat{\beta} = 0.35$. Interpret.

(c) The residual variance (unexplained by origin) is $\sigma^2 = 0.04$. Assuming the ten origin rates are evenly spaced, compute the share of the variance of daughters' LFP that is explained by origin. Why is this not Ψ?

(d) Under what conditions does selection bias inflate $\hat{\beta}$?

11.3 What adoption designs can and cannot tell you. Sacerdote (2007) studies Korean-American adoptees who were assigned to families quasi-randomly, and compares them with the families' biological children. Suppose (illustratively) that the gradient of children's years of schooling in the mother's years of schooling is 0.10 for adoptees and 0.30 for biological children in the same families.

(a) Why can this design *not* estimate the effect of the adoptees' biological parents?

(b) Under an additive model (gradient = environment + genes), what share of the biological-child gradient is attributable to family environment?

(c) If θ (here: taste for schooling) were transmitted only through the family environment, what would the adoptee gradient say about ρ? What additional assumption about the mapping from θ to schooling do you need?

(d) Cesarini et al. (2009) find little role for the common family environment in risk attitudes. Is that consistent with a large formation effect through *non-family* W^I? What design would test it?

11.4 Bolivia's self-employment variance ratio. Using Bolivia's *Encuesta de Hogares* (INE, microdata 2000–2019):

(a) Construct cohort-year cells (5-year cohorts $\times$ survey years 2000–2019).

(b) Estimate year effects (labor market conditions) and cohort effects (norm persistence) for self-employment rates.

(c) State your APC restriction and compute $\hat{\Psi}_{\text{var}}$. Then show how it changes under an alternative restriction.

(d) Compare $\hat{\Psi}_{\text{var}}$ for La Paz vs. rural Altiplano. Can this comparison tell you whether formation is stronger where W^I is more block-diagonal? What else would you need?

11.5 The WWII natural experiment. Take as approximate, illustrative figures that US female LFP rose from $\sim$28% in 1940 to $\sim$36% in 1945 (mobilization), and that after the war it did not return to 28% but settled at $\sim$33% by 1950 and continued rising. (Check the figures against BLS and Census series before using them.)

(a) Decompose the 1940–1950 change into enforcement (wartime labor demand, then demobilization) and formation (daughters of wartime workers formed different norms).

(b) The enforcement effect should reverse after 1945. Does it? By how much?

(c) The formation effect should persist. Does it? Through which cohorts?

(d) Which perturbation, outcome and horizon would define a Ψ here (Definition 9.13)? Compare the war (temporary) with a permanent change in W^I.

11.6 Sensitivity analysis. Use the one-loop model of Chapter 10 (Proposition 10.14), $\Psi(\infty) = \mu\beta\eta/(1-\rho-\mu\beta\eta)$, with the illustrative values $\hat{\rho} = 0.62$, $\hat{\mu} = 0.18$ and $\beta\eta = 0.5$:

(a) Compute $\Psi(\infty)$ under $\rho = 0.50, 0.62, 0.75, 0.90$. Plot $\Psi(\rho)$.

(b) Compute $\Psi(\infty)$ under $\mu = 0.10, 0.18, 0.30, 0.50$ at $\rho = 0.62$. Is the steady state stable for all four values? At what μ does it become unstable?

(c) Which parameter does Ψ depend on more sensitively near the values given?

(d) At what ρ does $\Psi(\infty)$ diverge? (This is the stability boundary from Chapter 10.) Show that $\Psi(T) \leq \mu\beta\eta\, T$ at every finite horizon. What does this mean empirically?

11.7 Designing an aspiration intervention. You are designing a program to raise female labor force participation in a (hypothetical) Bolivian community where current LFP is 25% and the target is 50%. (Nationally, Bolivia's female LFP was 66.2% of women aged 15+ in 2019, ILO modelled estimate, WDI `SL.TLF.CACT.FE.ZS`; World Bank 2026b.)

(a) Suppose $\Psi(\infty) = 2.1$ for a childcare subsidy. If you invest entirely in enforcement (childcare subsidies, wage top-ups), how much LFP increase can you expect?

(b) If you invest entirely in formation (mentor programs, media campaigns, aspiration workshops): how much?

(c) What is the optimal split of a fixed budget between enforcement and formation? (Hint: Ψ alone does not answer this; you need the total effect per unit of cost of each instrument.)

(d) If the formation loop converges at rate $\lambda = 0.62$ per generation, how many generations until 90% of the long-run formation effect is realized?

Mathematical Appendices

Appendix 11A: The Cohort-Year Regression

The model. Consider a panel of N individuals observed across T years, belonging to C cohorts (defined by birth decade). The behavior of individual i in cohort c at time t:

$$x_{i,c,t} = \alpha + \sum_{t'} \alpha_{t'} \mathbf{1}[t = t'] + \sum_{c'} \delta_{c'} \mathbf{1}[c = c'] + \beta Z_{i,t} + \varepsilon_{i,c,t}, \tag{11.9}$$

where $\alpha_{t'}$ are year fixed effects and $\delta_{c'}$ are cohort fixed effects. Age effects can be added, but age $=$ year $-$ cohort is a linear identity. With all three sets of effects, one linear restriction is needed, not the deletion of one set. Deaton (1997, §2.7) uses year effects orthogonal to a trend and summing to zero (see Proposition 11.5).

The variance ratio. The year component is $E_{\text{var}} = \text{Var}(\hat{\alpha}_t + \hat{\beta} Z)$: the variance explained by time-varying factors. The cohort component is $F_{\text{var}} = \text{Var}(\hat{\delta}_c)$: the variance explained by cohort membership. $\hat{\Psi}_{\text{var}} = F_{\text{var}} / E_{\text{var}}$ is the statistic of Definition 11.4, not the Ψ of Chapter 9.

Bias and testing. $\mathbb{E}[\text{Var}(\hat{\delta}_c)] = \text{Var}(\delta_c) + \overline{\text{Var}}(\hat{\delta}_c - \delta_c)$, so the raw variance overstates dispersion. Subtract the average sampling variance of the estimated effects. Test $H_0 : \delta_c = 0 \; \forall c$ with an F-test rather than by checking $\hat{\Psi}_{\text{var}} > 0$.

Standard errors. The delta method gives

$$\text{SE}(\hat{\Psi}_{\text{var}}) \approx \hat{\Psi}_{\text{var}} \sqrt{\frac{\text{SE}^2(\hat{F}_{\text{var}})}{\hat{F}_{\text{var}}^2} + \frac{\text{SE}^2(\hat{E}_{\text{var}})}{\hat{E}_{\text{var}}^2} - 2\frac{\widehat{\text{Cov}}(\hat{F}_{\text{var}}, \hat{E}_{\text{var}})}{\hat{F}_{\text{var}} \hat{E}_{\text{var}}}}. \tag{11.10}$$

It is unreliable when the denominator is close to zero or the variance components are near their boundary. A cluster bootstrap over cohorts and years is preferable. In every case the standard error is conditional on the APC restriction.

Appendix 11B: The Epidemiological Estimator

Setup. Individual i from origin o in host country h at time t:

$$x_{i,o,h,t} = \alpha_h + \alpha_t + \beta \cdot \bar{x}_{o,t_0} + \gamma Z_{i,h,t} + \varepsilon_{i,o,h,t}, \tag{11.11}$$

where $\bar{x}_{o,t_0}$ is the average behavior in origin country o at the time of parental emigration.

Identification. $\hat{\beta}$ is identified under: (1) $\text{Cov}(\bar{x}_{o,t_0}, \varepsilon_{i,o,h,t} | Z, \alpha_h, \alpha_t) = 0$ (origin-country norms uncorrelated with host-country unobservables), (2) no direct effect of origin-country conditions on the child except through the transmitted cultural component, and (3) controls Z that are not themselves outcomes of culture (no bad controls).

Connection to ρ. By Proposition 11.8, for a person whose family emigrated g generations ago ($g = 1$ for the second generation), $\beta_g = \rho^g \lambda$, where $\lambda \leq 1$ is the reliability of $\bar{x}_o$ as a proxy for origin culture. The level of $\hat{\beta}$ confounds ρ and λ. The *ratio* of successive generations' coefficients identifies ρ when λ is common to both.

An R^2 ratio is not Ψ. One can compare the variance explained by origin, $\hat{\beta}^2 \,\text{Var}(\bar{x}_o)$, with the variance explained by host-country conditions, $\hat{\gamma}^2 \,\text{Var}(Z_h)$. This is another descriptive statistic, not Ψ. With a single host country, $\text{Var}(Z_h)$ reflects individual covariates rather than prices, and the ratio is not interpretable as formation relative to enforcement.

Further Reading

The epidemiological approach: Fernández (2011), "Does Culture Matter?" in Benhabib, Bisin, and Jackson (eds.), *Handbook of Social Economics*. Fernández (2013), "Cultural Change as Learning." Fernández and Fogli (2009), "Culture: An Empirical Investigation of Beliefs, Work, and Fertility." Fernández (2007), "Women, Work, and Culture." Carroll et al. (1994), "Are There Cultural Effects on Saving? Some Cross-Sectional Evidence."

Age–period–cohort identification: Deaton (1997), *The Analysis of Household Surveys*, Chapter 2. Hall (1971); Heckman and Robb (1985); Deaton and Paxson (1994); McKenzie (2006); Fosse and Winship (2019). Goldin and Katz (2002), "The Power of the Pill"; Bailey (2006), on cohort-specific enforcement.

Social interactions and networks: Manski (1993), "Identification of Endogenous Social Effects: The Reflection Problem." Bramoullé et al. (2009), "Identification of Peer Effects through Social Networks." Lee (2007). Blume et al. (2011), in *Handbook of Social Economics*. Goldsmith-Pinkham and Imbens (2013). Breza et al. (2020). Bursztyn et al. (2014). Fogli and Veldkamp (2011), "Nature or Nurture?"

Formation experiments: Bernard et al. (2014), "The Future in Mind," CSAE Working Paper. Beaman et al. (2012), "Female Leadership Raises Aspirations and Educational Attainment for Girls."

Global Preferences Survey: Falk et al. (2018), "Global Evidence on Economic Preferences." The comprehensive cross-country dataset on patience, risk tolerance, altruism, and trust.

Gene-culture interaction: Cesarini et al. (2009), "Genetic Variation in Preferences for Giving and Risk Taking." Sacerdote (2007), "How Large Are the Effects from Changes in Family Environment?"

Structural adjustment: Williamson (1990), "What Washington Means by Policy Reform." Rodrik (2006), "Goodbye Washington Consensus, Hello Washington Confusion." For Bolivia: Morales and Sachs (1990), "Bolivia's Economic Crisis," in Sachs (ed.), *Developing Country Debt and Economic Performance*, vol. 2 (NBER).

Chapter 12

Welfare Under Endogenous Preferences

When the Ruler Is Made of Rubber

> *"The deprived people tend to come to terms with their deprivation because of the sheer necessity of survival, and they may, as a result, lack the courage to demand any radical change, and may even adjust their desires and expectations to what they unambitiously see as feasible."*
> — Amartya Sen, *Development as Freedom* (1999)

Parts I and II evaluated welfare using fixed preferences: given $\boldsymbol{\theta}$, a welfare function ranks allocations, and the bargaining allocation $x_i = \Omega/N + (\pi_i - 1/N)\Phi$ falls short of the utilitarian benchmark by the distributional loss $L_{\text{power}} = (\kappa N/2)\Phi^2 \operatorname{Var}(\boldsymbol{\pi})$ (Chapter 7, Equation (7.9)). Part III made $\boldsymbol{\theta}$ endogenous: the structure produces the preferences.

This chapter confronts the consequence: *if the structure produces the preferences, and the preferences are the yardstick for evaluating the structure, the yardstick is made of rubber.*

Consider an agent in the traditional sector who reports moderate satisfaction with a low allocation. Her aspirations were formed under a block-diagonal W^I that showed her only traditional-sector outcomes, so a modest allocation meets them. Under a different W^I she would have formed higher aspirations and would judge the same allocation as falling short. Which aspirations are the "right" ones for welfare evaluation?

The problem has no easy solution. This chapter develops four approaches and states what each can and cannot do. It identifies what can be ranked robustly and what cannot. It then says plainly where a value judgment is unavoidable.

Which reading of θ is in force. Throughout this chapter θ_i is an *aspiration level* a_i, a reference point for one's own outcome (Chapter 9, Definition 9.1, "education" row). Utility is reference-dependent, in the spirit of Genicot and Ray (2017) and of the relative-income literature (Clark et al., 2008). Richer models of aspirations or reference consumption formed through a social network or reference group are Mookherjee et al. (2010) and Ghiglino and Goyal (2010); this chapter uses the simplest linear–quadratic version:

$$s_i = v(x_i - a_i), \qquad v(z) = z - \frac{\kappa}{2}z^2, \qquad \kappa z < 1 \text{ on the relevant range}, \tag{12.1}$$

so v is increasing and concave and every allocation in the examples lies below satiation. Aspirations are formed by the formation equation with the observed outcomes as signal. Its steady state, with anchor 0 and reference 0, is

$$\mathbf{a} = \mathcal{M}\, W^I \mathbf{x}, \qquad 0 \leq \mathcal{M} < 1. \tag{12.2}$$

An aspiration is a taste-like reference point, not a belief. If θ were a mistaken belief about returns, correcting it would be uncontroversially good (Chapter 9, Section 9.1.4), and most of this chapter would be unnecessary.

12.1 The Circularity Problem

12.1.1 Statement

Definition 12.1 (The Welfare Circularity)**.** *In the formation model:*

1. *The structure ($\boldsymbol{\pi}$, W^P) determines the allocation $\mathbf{x}^*$ (through the bargaining equilibrium).*
2. *The information network W^I determines the aspirations $\mathbf{a}^*$ (through the formation equation).*
3. *Standard welfare analysis evaluates $\mathbf{x}^*$ using $s_i = v(x_i^* - a_i^*)$: the allocation is evaluated by aspirations that the same structure produced.*

The circularity: the structure produces both the object being evaluated ($\mathbf{x}^*$) and the criterion for evaluation ($\mathbf{a}^*$). A structure that produces low allocations and low aspirations can look acceptable by its own standards.

Proposition 12.2 (The Satisfaction Paradox)**.** *Take two blocks of equal size with allocations $x_M > x_T$ (constant within blocks). Aspirations follow* (12.2)*, and v is as in* (12.1)*,*

with all arguments below satiation. Compare a pure block-diagonal W^I (each agent observes only her own block, uniformly) with the integrated $W^I = W^{CE}$. The cross-block satisfaction gap satisfies

$$\frac{s_M - s_T \big|_{\textit{block-diagonal}}}{s_M - s_T \big|_{\textit{integrated}}} = 1 - \mathcal{M}. \tag{12.3}$$

The allocation gap $x_M - x_T$ is the same in both cases. As $\mathcal{M} \to 1$, reported satisfaction in the low-allocation block approaches that in the high-allocation block, even though $x_T < x_M$.

Proof. For quadratic v, $v(z_1) - v(z_2) = (z_1 - z_2)\,[1 - \kappa(z_1 + z_2)/2]$. Block-diagonal: $z_b = (1 - \mathcal{M})x_b$. So the gap is $(1 - \mathcal{M})(x_M - x_T)\,[1 - \kappa(1 - \mathcal{M})(x_M + x_T)/2]$. Integrated: $a = \mathcal{M}\bar{x}$ with $\bar{x} = (x_M + x_T)/2$, so $z_b = x_b - \mathcal{M}\bar{x}$. The gap is $(x_M - x_T)\,[1 - \kappa(x_M + x_T - 2\mathcal{M}\bar{x})/2] = (x_M - x_T)\,[1 - \kappa(1 - \mathcal{M})(x_M + x_T)/2]$. The brackets coincide, and they are positive because each $\kappa z_b < 1$, so the denominator is nonzero ($x_M \neq x_T$) and the ratio is $1 - \mathcal{M}$. □

Intuition: The uncomfortable truth

The standard welfare criterion respects preferences. If the agent in the traditional sector reports satisfaction, the economist records satisfaction. This is preference sovereignty, a foundation of welfare economics since Pareto.

Preference sovereignty becomes problematic when preferences are endogenous to the structure. In the model, the agent is not dissatisfied *because* the structure limited what she observes, which limited her aspirations. Her satisfaction is then a signal both of well-being and of how her aspirations were formed. This is Sen's "adaptive preferences" argument, formalized in one specific way. It is *one* candidate explanation of why measured satisfaction differences across groups can be small relative to consumption differences. It is *one* reason why welfare analysis that takes **a** as given can be biased toward the status quo.

12.1.2 A Worked Example

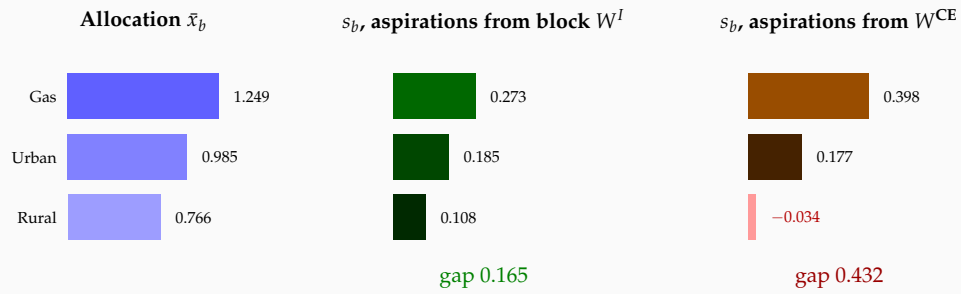
Example 12.3 (Satisfaction in Three Sectors [ILLUSTRATIVE])**.** *Take the steady-state allocations of Chapter 10's stylized Bolivia economy (Example 10.6): $\bar{x}_b = (1.249, 0.985, 0.766)$, with its W^I (within-block weight 0.85, cross-block 0.025 each). Aspirations $\mathbf{a} = 0.8\,W^I\mathbf{x}$ (e.g. $\rho = 0.75$, $\mu = 0.2$), and $v(z) = z - 0.25z^2$ ($\kappa = 0.5$; all $\kappa z < 1$).*

		Block W^I		*Integrated* (W^{CE})	
	$\bar{x}_b$	a_b	s_b	a_b	s_b
Gas (block 1)	*1.249*	*0.954*	*0.273*	*0.800*	*0.398*
Urban (block 2)	*0.985*	*0.791*	*0.185*	*0.800*	*0.177*
Rural (block 3)	*0.766*	*0.655*	*0.108*	*0.800*	−0.034
Gas − rural gap	*0.482*		*0.165*		*0.432*

With aspirations formed in the near-block-diagonal W^I, the gas–rural satisfaction gap is 0.165. With aspirations formed in an integrated network (everyone observes the economy-wide average), it is 0.432, and the rural block's satisfaction is negative: its allocation falls short of what it would aspire to if it saw the whole economy. The endogenous aspirations compress the measured gap to 38% of its integrated size. In the two-block, pure block-diagonal case of Proposition 12.2 the factor would be $1 - \mathcal{M} = 0.2$.

The structure produces contentment that masks deprivation, but only relative to an evaluative standard — here, aspirations formed in an integrated network. Choosing that standard is the subject of Section 12.2.

Visualization: The satisfaction paradox: endogenous vs. integrated aspirations



Left: allocations (identical in both evaluations). Center: satisfaction with aspirations formed in the near-block-diagonal W^I — the gap looks moderate because rural aspirations adjusted downward. Right: satisfaction with aspirations formed in an integrated network — the rural allocation falls short of its aspiration (negative s). Endogenous aspirations compress the measured gap (`code/ch12.py`).

12.2 Four Approaches to the Problem

There is no consensus on how to evaluate welfare when preferences are endogenous. The classic treatments (von Weizsäcker, 1971; Hammond, 1976; Pollak, 1978) already showed that rankings depend on which preference "vintage" is used. This

section presents four approaches, ordered from most conservative (closest to standard welfare economics) to most radical (furthest from preference sovereignty).

12.2.1 Approach 1: Laundered Preferences

Definition 12.4 (Laundered Aspirations)**.** *Fix a reference allocation $\mathbf{x}^{ref}$ and a "just" information network W^I_{just}. Agent i's **laundered aspiration** is the aspiration she would hold under W^I_{just}:*

$$\hat{a}_i = \mathcal{M} \sum_j W^I_{just,ij}\, x^{ref}_j. \tag{12.4}$$

It is held fixed while different allocations are compared. Welfare is $\hat{W}(\mathbf{x}) = \sum_i v(x_i - \hat{a}_i)$.

The term "laundering" is Goodin's (1986), who used it for filtering out preferences that should not count, such as malicious or antisocial ones. The counterfactual-network version used here is closer to *full-information* or informed-desire accounts of well-being (Brandt, 1979; Harsanyi, 1977), with the twist that the counterfactual is specified as a network.

Proposition 12.5 (Properties of Laundered Welfare)**.** *1. If $W^I_{just} = W^{CE}$, then $\hat{a}_i = \mathcal{M}\bar{x}^{ref}$ for all i: all agents share the same laundered aspiration.*

2. The laundered welfare function can rank allocations differently from the endogenous one when $W^I \neq W^I_{just}$, because the endogenous aspirations move with the allocation being evaluated while laundered ones do not (Section 12.3).

3. The laundered approach requires specifying W^I_{just} and $\mathbf{x}^{ref}$. Both are normative choices, not empirical ones.

Proof. (1) Rows of W^{CE} are identical. (2) An instance: $N = 2$, $\mathcal{M} = 0.8$, $\kappa = 0.5$, everyone watches agent 2 ($W^I = \mathbf{1}\mathbf{e}_2'$), $W^I_{\mathrm{just}} = W^{\mathrm{CE}}$ and $\mathbf{x}^{\mathrm{ref}} = \mathbf{x} = (1.5, 0.5)$, so $\hat{a}_i = 0.8$. Compare $\mathbf{x}$ with the equal split $\mathbf{x}' = (1, 1)$. Endogenously, $\mathbf{x} - \mathbf{a}^*(\mathbf{x}) = (1.1, 0.1)$ and $\mathbf{x}' - \mathbf{a}^*(\mathbf{x}') = (0.2, 0.2)$, so $W_1(\mathbf{x}) = 0.895 > W_1(\mathbf{x}') = 0.38$. With laundered aspirations, $\mathbf{x} - \hat{\mathbf{a}} = (0.7, -0.3)$ and $\mathbf{x}' - \hat{\mathbf{a}} = (0.2, 0.2)$, so $\hat{W}(\mathbf{x}) = 0.255 < \hat{W}(\mathbf{x}') = 0.38$. All arguments satisfy $\kappa z < 1$ (`code/ch12.py`). (3) This restates what the definition requires as inputs. □

Intuition: Strength and weakness

Strength: laundered aspirations keep the idea that people's own evaluations matter, while correcting for what the actual W^I hid from them. The correction is computable from the formation model.
Weakness: who decides what a "just" W^I looks like? Choosing W^I_{just} is itself a value judgment. If $W^I_{\text{just}} = W^{\text{CE}}$, we are saying that the right information structure is one where everyone observes everyone equally. That may be defensible, but it is not neutral. "Full information" is not the same as uniform weights either: a fully informed agent would weight the outcomes that are relevant and *attainable for her*, not the population average.

12.2.2 Approach 2: Opportunity-Based Welfare (Sen's Capabilities)

Definition 12.6 (Capability Set)**.** *Agent i commands the **entitlement** $e_i = \omega_i + \pi_i \Phi$ under the bargaining structure (Chapter 10). She converts resources into a functioning (being educated, being well nourished, working in a chosen occupation) at a **conversion factor** $c_i \in (0, 1]$, which depends on health, infrastructure, social norms, and what she knows about available options (her row of W^I). Her **capability set** is the set of achievable functionings:*

$$\mathcal{O}_i = \{f \in \mathbb{R}_+ : f \leq c_i\, e_i\} = [0,\, c_i e_i]. \tag{12.5}$$

The critical move: $\mathcal{O}_i$ is defined by entitlements and conversion factors, *not* by preferences. An agent who chooses a modest functioning from a large set has the capability; an agent whose set is small does not, whatever her aspirations. Sen's example: a person who fasts and a person who starves may have the same nutrition, but only the first has the capability to be well nourished.

Proposition 12.7 (Capability in the Formation Model)**.**

1. *$\mathcal{O}_i$ does not depend on a_i: aspirations do not enter the capability set.*

2. *$\mathcal{O}_i = \mathcal{O}_j$ iff $c_i e_i = c_j e_j$. With equal endowments and conversion factors and $\Phi > 0$, equal capability requires equal power $\pi_i = \pi_j$, but not equal aspirations.*

3. *With common c and ω, the sum $\sum_i c\, e_i = c\, \Omega$ is the same for every $\boldsymbol{\pi}$. So a utilitarian sum of capabilities cannot detect power concentration. The minimum capability $\min_i c\, e_i$ is maximized exactly at $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$.*

4. *Information policy can enlarge capability sets by raising conversion factors c_i (knowing which schools, jobs or programs exist), even when it changes neither power nor endowments.*

Proof. (1)–(2) Definition. (3) $\sum_i e_i = \sum_i \omega_i + \Phi \sum_i \pi_i = \Omega$. And $\min_i c\, e_i = c(\omega + \Phi \min_i \pi_i)$ with $c, \Phi > 0$, while $\min_i \pi_i \leq 1/N$, with equality iff $\boldsymbol{\pi}$ is uniform, because the π_i sum to 1. (4) Definition of c_i. □

Part (3) is why the capability approach resists summation. Sen's capability sets are vectors of freedoms, not quantities to be added. Measuring freedom by the size of a choice set is itself contested: Pattanaik and Xu (1990) axiomatize ranking opportunity sets by their cardinality, and Sen (1991) shows why that ignores the value of the options. Sugden (2004, *AER*) proposes an opportunity criterion that does not require coherent preferences.

Intuition: Sen in matrix notation

Sen (1999) argued that well-being should be measured not by what people have (allocations) or by how satisfied they are (utility), but by what they are *free to do and be* (capabilities). In the formation model, the capability set is constrained by two structural objects: her power π_i, which limits her entitlement, and her information network W^I, which limits her conversion factor — what she knows she can do with her resources.

Capability deprivation is therefore a *joint* constraint on power and information. You can be capability-deprived because you lack power (a structural constraint) or because you lack information about options (an epistemic constraint). Low aspirations are a separate matter. They do not shrink the capability set, but they may lead an agent to choose little from it, which is why adaptive preferences make satisfaction an unreliable guide.

12.2.3 Approach 3: Multi-Criteria Welfare

Definition 12.8 (Multi-Criteria Welfare)**.** *Evaluate welfare under several aspiration vectors simultaneously:*

$$\mathbf{W}(\mathbf{x}) = \big(W_1(\mathbf{x}, \mathbf{a}^*(\mathbf{x})),\ W_2(\mathbf{x}, \hat{\mathbf{a}}),\ W_3(\mathbf{x}, \mathbf{0}),\ W_4(\mathbf{x}, \mathbf{a}_{full})\big)\,, \tag{12.6}$$

where $W_k(\mathbf{x}, \mathbf{a}) = \sum_i v(x_i - a_i)$ *and:*

- W_1*: welfare under actual (endogenous) aspirations,* $\mathbf{a}^*(\mathbf{x}) = \mathcal{M} W^I \mathbf{x}$*, which move with the allocation evaluated.*
- W_2*: welfare under laundered aspirations (fixed).*
- W_3*: welfare under the anchor (no aspiration effect).*

- W_4: *welfare under "full-information" aspirations (fixed), formed under a network the evaluator regards as fully informative. If that network is taken to be W^{CE}, then $W_4 = W_2$ with $W^I_{just} = W^{CE}$; the two criteria differ only if they use different networks.*

An allocation $\mathbf{x}$ *is* ***robustly superior*** *to* $\mathbf{x}'$ *if* $W_k(\mathbf{x}) \geq W_k(\mathbf{x}')$ *for all* k*, with strict inequality for at least one.*

This is Sen's *intersection quasi-ordering* (Sen, 1970): rank only what every admissible criterion ranks the same way, and leave the rest unranked. Its choice-theoretic analogue is Bernheim and Rangel's (2009) unambiguous choice relation. There, W^I plays the role of an *ancillary condition* — a feature of the decision environment that changes choices without changing the feasible set — and $\mathbf{x}$ is ranked above $\mathbf{x}'$ only if it is chosen over $\mathbf{x}'$ under every welfare-relevant ancillary condition.

Proposition 12.9 (What Is Robustly Ranked)**.** *Let v be strictly increasing and $0 \leq \mathcal{M} < 1$.*

1. *If $\mathbf{x} \geq \mathbf{x}'$ componentwise with $\mathbf{x} \neq \mathbf{x}'$, then $\mathbf{x}$ is ranked above $\mathbf{x}'$ by every fixed-aspiration criterion (W_2, W_3, W_4).*

2. *$\mathbf{x}$ is ranked above $\mathbf{x}'$ by W_1 for every strictly increasing v if $(I - \mathcal{M}W^I)(\mathbf{x} - \mathbf{x}') \geq 0$, with $\neq$. This is a stronger requirement: raising others' outcomes raises my aspirations.*

3. *The multi-criteria order is therefore a strict but incomplete partial order. Transfers between agents — the core of distributional questions — are generally not robustly ranked.*

Proof. (1) Each $v(x_i - a_i)$ is strictly increasing in x_i for fixed a_i, so every term weakly rises and at least one strictly rises. (2) $x_i - a_i^*(\mathbf{x}) = [(I - \mathcal{M}W^I)\mathbf{x}]_i$, so the condition makes every argument weakly larger and at least one strictly larger. (3) Robust superiority is componentwise dominance of the vector $\mathbf{W}(\mathbf{x})$ with one strict inequality, which is irreflexive and transitive. A transfer lowers some x_i, so (1) does not apply to it, and the instance in the proof of Proposition 12.5(2) is a transfer ranked oppositely by W_1 and W_2; so the order is incomplete. □

12.2.4 Approach 4: Process-Based Welfare

Definition 12.10 (Process Criteria)**.** *Evaluate not the allocation but the* ***structure that produces allocations and aspirations****, using a vector of structural statistics:*

$$\big(N_{\mathit{eff}}(\boldsymbol{\pi}),\ N_{\mathit{eff}}(\boldsymbol{\pi}^I),\ \Gamma(W^I)\big), \tag{12.7}$$

where $N_{\mathit{eff}}(\boldsymbol{\pi}) = 1/\sum_i \pi_i^2$ measures how dispersed bargaining power is (Chapter 6, Definition 6.15); $N_{\mathit{eff}}(\boldsymbol{\pi}^I)$ measures how many voices enter the information consensus (Chapter 8); and $\Gamma(W^I)$ measures how integrated the information network is (Chapter 6, Definition 6.14). A structure is ***process-superior*** *if it is weakly better on every coordinate and strictly better on one.*

Intuition: Judging the structure, not the outcome

Process welfare sidesteps the circularity by evaluating the structure directly. A "good" structure has dispersed power (high $N_{\text{eff}}(\boldsymbol{\pi})$), plural information (high $N_{\text{eff}}(\boldsymbol{\pi}^I)$), and an integrated information network (high Γ). The three coordinates are logically independent. In particular, Γ says nothing about who has power (Proposition 6.16).

This does not dissolve the circularity; it moves it. Choosing these statistics — and declaring dispersion and integration good — *is* the value judgment. It is a defensible one, but it is not derived from the agents' preferences.

12.3 What Can and Cannot Be Ranked

Earlier drafts of this chapter stated a "Structural Welfare Theorem": that higher Γ, lower Δ and higher N_{eff} are welfare-improving under all four approaches. That statement is false. Its proof relied on "higher Γ implies more equal $\boldsymbol{\pi}$", which fails (Proposition 6.16), and on an algebra error (Appendix 12B). This section states what can be proved.

Proposition 12.11 (Fixed-Evaluation Ranking Is by Power Concentration)**.** *Let $x_i = \Omega/N + (\pi_i - 1/N)\Phi$ (Chapter 7, Proposition 7.3), and evaluate with $\sum_i v(x_i - \hat{a})$ for a common fixed aspiration $\hat{a}$ (W_2 with $W^I_{\mathit{just}} = W^{CE}$, W_3, or W_4). Then*

$$\sum_i v(x_i - \hat{a}) = N\, v\Big(\tfrac{\Omega}{N} - \hat{a}\Big) - \frac{\kappa N}{2}\Phi^2\ \mathrm{Var}(\boldsymbol{\pi}). \tag{12.8}$$

Hence, for equal $(\Omega, \Phi, \kappa, \hat{a})$ with $\kappa > 0$ and $\Phi \neq 0$, W_A is ranked above W_B iff $\mathrm{Var}(\boldsymbol{\pi}_A) < \mathrm{Var}(\boldsymbol{\pi}_B)$, i.e. iff $N_{\mathit{eff}}(\boldsymbol{\pi}_A) > N_{\mathit{eff}}(\boldsymbol{\pi}_B)$. The spectral gap plays no role.

Proof. Let $z_i = x_i - \hat{a}$. Then $\sum_i z_i = \Omega - N\hat{a}$ and $\sum_i z_i^2 = N(\Omega/N - \hat{a})^2 + N\,\mathrm{Var}(\mathbf{x})$, with $\mathrm{Var}(\mathbf{x}) = \Phi^2\,\mathrm{Var}(\boldsymbol{\pi})$. Substitute into $\sum_i (z_i - \frac{\kappa}{2} z_i^2)$: the result is $N(\Omega/N - \hat{a}) - \frac{\kappa N}{2}(\Omega/N - \hat{a})^2 - \frac{\kappa N}{2}\Phi^2\,\mathrm{Var}(\boldsymbol{\pi})$, which is (12.8). Finally, $\mathrm{Var}(\boldsymbol{\pi}) = \frac{1}{N}(1/N_{\mathrm{eff}} - 1/N)$ is decreasing in N_{eff}. □

Remark 12.12 (Γ Is Not a Distributional Criterion). Take $N = 3$, $W_A = \mathbf{1}\boldsymbol{\pi}_A'$ with $\boldsymbol{\pi}_A = (0.5, 0.3, 0.2)$, and $W_B = \frac{1}{2}W^{\mathrm{CE}} + \frac{1}{2}I$. Then $\Gamma(W_A) = 1 > \Gamma(W_B) = 0.5$, but $\boldsymbol{\pi}_B$ is uniform while $\mathrm{Var}(\boldsymbol{\pi}_A) = 0.0156$ ($N_{\mathrm{eff}} = 2.63$). With $\Omega = 3$, $\Phi = 1$ and $\kappa = 0.5$ (all $\kappa x_i < 1$), the fixed-evaluation loss is 0.0117 under W_A and 0 under W_B. The economy with the larger spectral gap is ranked lower. This is an instance of Proposition 6.16.

Proposition 12.13 (Endogenous Evaluation Depends on Who Is Watched)**.** *Let $v(z) = z$ (the linear case) and $\mathbf{a} = \mathcal{M}W^I\mathbf{x}$. Let $\mathbf{c} = (W^I)'\mathbf{1}$ be the column sums of W^I, i.e. how much each agent is watched. Then*

$$W_1(\mathbf{x}) = \sum_i (x_i - a_i) = \Omega - \mathcal{M}\sum_j c_j x_j. \tag{12.9}$$

At a given allocation, endogenous welfare is lower the more attention is concentrated on agents with high allocations. If W^I is doubly stochastic ($\mathbf{c} = \mathbf{1}$), $W_1 = (1 - \mathcal{M})\Omega$, whatever the distribution.

Proof. $\sum_i (W^I\mathbf{x})_i = \mathbf{1}'W^I\mathbf{x} = \mathbf{c}'\mathbf{x}$. □

In the stylized Bolivia allocation with $\mathcal{M} = 0.8$, $W_1 = 1.80$. If every agent moves 30% of her attention to the gas block (a "media hub" showing elite lifestyles), W_1 falls to 1.26 at the same allocation. By the endogenous criterion, the information change made everyone worse off. By the fixed criteria, nothing changed. By the process criterion, $N_{\mathrm{eff}}(\boldsymbol{\pi}^I)$ fell. This is Furtado's demonstration effect (Chapter 9) in welfare terms.

Example 12.14 (The Criteria Disagree [ILLUSTRATIVE])**.** *Two agents: a landlord with $\pi = 0.8$ and current aspiration $a_L = 4$, and a worker with $\pi = 0.2$ and $a_W = 1.5$ (taken as given; they are not of the form $\mathcal{M}W^I\mathbf{x}$). $\Omega = 10$, $\Phi = 4$, $v(z) = z - 0.05z^2$. The bargaining allocation is $\mathbf{x} = (6.2, 3.8)$. Equalizing power gives $\mathbf{x}' = (5, 5)$ (aspirations held at their current values).*

Criterion	$\pi = (0.8, 0.2)$	$\pi = (0.5, 0.5)$
Current aspirations, held fixed, sum	*3.99*	*3.84*
W_2: laundered $\hat{a} = 3$, sum	*3.46*	*3.60*
Rawlsian, current aspirations ($\min_i s_i$)	*1.96*	*0.95*
Capability, $\min_i e_i$	*3.80*	*5.00*

The criteria based on current aspirations rank the unequal structure higher; the laundered and capability criteria rank the equal one higher. Exercise 12.7 works through it.

Key Structural Insight

What the formation model can deliver is a *partial* ordering. Under any criterion that fixes aspirations, structures with less concentrated power ($N_{\text{eff}}(\pi)$ higher) rank higher, whatever their spectral gap. Under the endogenous criterion, the ranking also depends on whom the information network makes people watch. The criteria agree on componentwise improvements that do not raise aspirations faster than outcomes, and they generally disagree on redistribution.

The circularity is therefore not solved; it is made explicit. Choosing a criterion — endogenous aspirations, laundered ones, capabilities, or structural process statistics — *is* the value judgment. The model's contribution is to show exactly where each criterion leads, and which conclusions survive every choice.

12.4 Three Uncomfortable Implications

12.4.1 Implication 1: Satisfaction Surveys Can Be Structurally Biased

Corollary 12.15 (Satisfaction Bias)**.** *1. In the setting of Proposition 12.2, the slope of reported satisfaction on allocation across the two blocks, $(s_M - s_T)/(x_M - x_T)$, is $(1 - \mathcal{M})$ times the slope with integrated aspirations.*

2. Let allocations vary within blocks, with W^I block-diagonal and uniform within blocks, so every agent of block b has aspiration $a_b = \mathcal{M}\bar{x}_b$. Then within a block $s_i - s_j = v(x_i - a_b) - v(x_j - a_b)$: satisfaction differences reflect allocation differences only, with no compression factor. For linear v the within-block slope is 1, while the slope across block means is $1 - \mathcal{M}$ (and 1 with integrated aspirations).

Proof. (1) Proposition 12.2. (2) Within a block all agents share a_b. For linear v, mean satisfaction in block b is $\bar{x}_b - \mathcal{M}\bar{x}_b = (1 - \mathcal{M})\bar{x}_b$, and with integrated aspirations

it is $\bar{x}_b - \mathcal{M}\bar{x}$. □

If reference groups are national (blocks are countries), the model produces the pattern Easterlin (1974) described: a strong income–satisfaction gradient within countries and a compressed one across countries. The pattern itself is disputed. Stevenson and Wolfers (2008) find similar within- and cross-country gradients in log income. The formation model offers one mechanism, not a settled fact.

12.4.2 Implication 2: Evaluation Depends on the Vintage of Preferences

Proposition 12.16 (Evaluation and Preference Vintage)**.** *Let v be strictly increasing. A policy that changes W^I at $t = 0$ can be evaluated with the aspirations held at $t = 0$ (old vintage) or with those held at $t = T$ (after adjustment to the new W^I):*

$$W_0 = \sum_i v(x_{T,i} - a_{old,i}) \qquad \textit{versus} \qquad W_T = \sum_i v(x_{T,i} - a_{new,i}). \tag{12.10}$$

The two generally differ. In particular, if $\mathbf{a}_{new} \geq \mathbf{a}_{old}$ with $\mathbf{a}_{new} \neq \mathbf{a}_{old}$, then $W_T < W_0$. An information policy that shows people higher outcomes, $(W^I_{new}\mathbf{x})_i \geq (W^I_{old}\mathbf{x})_i$ for all i with strict inequality for some i, raises aspirations $\mathbf{a} = \mathcal{M}W^I\mathbf{x}$ (for $\mathcal{M} > 0$), so at a fixed allocation it lowers measured satisfaction under the new vintage.

Proof. v is strictly increasing, so the satisfaction of agent i weakly falls when a_i weakly rises at fixed x_i, and strictly falls for an agent whose a_i strictly rises. Summing gives $W_T < W_0$. The last claim follows because $\mathbf{a}_{\text{new}} - \mathbf{a}_{\text{old}} = \mathcal{M}(W^I_{\text{new}} - W^I_{\text{old}})\mathbf{x}$. □

This is not time inconsistency in the Strotz sense (a planner's own plan ceasing to be optimal later). It is the classic problem of changing tastes: which self, or which generation's tastes, should evaluate the policy (von Weizsäcker, 1971; Hammond, 1976).

Example 12.17 (Information Policy and Satisfaction [ILLUSTRATIVE])**.** *Use Chapter 10's Policy C (Example 10.16): rural agents' weight on each gas agent rises from* 0.025 *to* 0.075*. With $\mathcal{M} = 0.8$ and allocations held at their initial values, the rural aspiration rises from* 0.655 *to* 0.713*. Rural satisfaction then falls from* 0.108 *to* 0.053*.*

A cost–benefit analysis using new-vintage satisfaction records a loss. One using old-vintage aspirations records no change. A capability analysis records a gain, if the exposure raises the rural conversion factor c_i (knowledge of schools and jobs), and that gain eventually raises allocations as preferences and investment adjust. Standard evaluation by current satisfaction can therefore undervalue — or even penalize — formation interventions.

12.4.3 Implication 3: Consent Is Not Sufficient

Proposition 12.18 (Endogenous Aspirations Make Consent Harder)**.** *Let v be strictly increasing, $0 \leq \mathcal{M} < 1$, and let W^I be row-stochastic. Call a change $\mathbf{x} \to \mathbf{x}'$ **endogenously Pareto-improving** if $v(x'_i - a^*_i(\mathbf{x}')) \geq v(x_i - a^*_i(\mathbf{x}))$ for all i, with strict inequality for some i, where $\mathbf{a}^*(\mathbf{x}) = \mathcal{M}W^I\mathbf{x}$. Call it **laundered Pareto-improving** if the same holds with fixed aspirations $\hat{\mathbf{a}}$. Then every endogenously Pareto-improving change is laundered Pareto-improving. The converse fails whenever $\mathcal{M} > 0$ and some agent observes another ($W^I_{ij} > 0$ for some $i \neq j$).*

Proof. Because v is strictly increasing, endogenous improvement means $(I - \mathcal{M}W^I)(\mathbf{x}' - \mathbf{x}) \geq 0$, with $\neq$. Since $\mathcal{M} < 1$ and W^I is row-stochastic, $\rho(\mathcal{M}W^I) = \mathcal{M} < 1$ and $(I - \mathcal{M}W^I)^{-1} = \sum_k \mathcal{M}^k (W^I)^k \geq 0$, so $\mathbf{x}' - \mathbf{x} \geq 0$, and $\mathbf{x}' \neq \mathbf{x}$ because the inverse is nonsingular. For fixed $\hat{\mathbf{a}}$ this is a laundered improvement: every $v(x_i - \hat{a}_i)$ weakly rises and the one with $x'_i > x_i$ strictly rises. Conversely, raise only x_j by $\varepsilon > 0$. This is a laundered improvement, but agent i with $W^I_{ij} > 0$ sees her aspiration rise by $\mathcal{M}W^I_{ij}\varepsilon > 0$ and her satisfaction fall. For example, with two agents, $W^I = \frac{1}{2}\mathbf{1}\mathbf{1}'$ and $\mathcal{M} = 0.8$, raising x_2 by 0.1 changes the linear satisfaction terms by $(-0.04, +0.06)$. □

The nesting runs in a specific direction. Under endogenous aspirations, improvements for others are partly experienced as losses, so *fewer* changes command unanimous consent. The status quo is harder to dislodge by consent. And the absence of demand for change does not show that the current structure is welfare-maximizing: in the model it can reflect aspirations that the structure itself formed. This depends on the preference model. With other specifications, such as tastes that scale marginal utility, the two Pareto sets need not be nested at all.

Prebisch Connection: Peripheral capitalism and legitimacy

Prebisch (1981, *Capitalismo periférico*) argued that peripheral capitalism tends toward *crisis*. A privileged consumer society imitating the centre absorbs the surplus. Redistributive pressure from organized groups then produces inflationary struggle, and eventually the use of force to restore the system.

The formation model speaks to a complementary question: why redistributive pressure can stay latent for long periods. If aspirations in the traditional sector are formed under a W^I that shows only traditional-sector outcomes, the demand for change is muted. Proposition 12.18 shows that consent under endogenous aspirations is harder to obtain. When W^I opens up — urbanization, schooling, media — aspirations rise. In Prebisch's terms, the redistributive struggle intensifies.

This is not an argument for paternalism. It is an argument for *structural transparency*: before reading welfare off revealed preferences, ask how W^I formed them. Is information plural ($N_{\text{eff}}(\boldsymbol{\pi}^I)$ close to N)? Do people observe returns they can attain?

12.5 Summary

1. The **welfare circularity**: the structure produces both the allocation and the aspirations used to evaluate it. With reference-dependent utility $v(x-a)$ and aspirations $\mathbf{a} = \mathcal{M}W^I\mathbf{x}$, segregated information compresses the measured satisfaction gap between blocks by the factor $1-\mathcal{M}$ (the satisfaction paradox), while allocation gaps are unchanged.

2. **Four approaches**: laundered aspirations (evaluate under a "just" W^I), capabilities (feasible functionings given entitlements $e_i = \omega_i + \pi_i\Phi$ and conversion factors, independent of aspirations), multi-criteria welfare (Sen's intersection quasi-ordering; Bernheim–Rangel), and process criteria ($N_{\text{eff}}(\boldsymbol{\pi})$, $N_{\text{eff}}(\boldsymbol{\pi}^I)$, $\Gamma(W^I)$).

3. **What can be ranked**: under any common fixed aspiration, welfare equals the benchmark minus $(\kappa N/2)\Phi^2\,\text{Var}(\boldsymbol{\pi})$, so structures rank by power concentration and not by Γ. Under endogenous aspirations, welfare at a given allocation falls the more attention is concentrated on the rich. The earlier "Structural Welfare Theorem" is false.

4. **Three implications**: satisfaction surveys can be compressed across reference groups; evaluations depend on the vintage of aspirations (information policy

can lower measured satisfaction while expanding capabilities); and endogenous aspirations make unanimous consent *harder*, since every endogenous Pareto improvement is a laundered one but not conversely.

5. The circularity is made explicit, not solved: **choosing the criterion is the value judgment**. The practical recommendation is structural transparency — ask how W^I formed the preferences before reading welfare off them.

12.6 Exercises

12.1 The satisfaction paradox. Using Example 12.3, compute block satisfaction $s_b = v(\bar{x}_b - a_b)$ with $\mathbf{a} = \mathcal{M}W^I\mathbf{x}$ for $\mathcal{M} = 0, 0.4, 0.8$. Plot the gas-minus-rural satisfaction gap against $\mathcal{M}$. Then verify Proposition 12.2 for a two-block economy with pure block-diagonal W^I.

12.2 Laundered aspirations. For the stylized Bolivia three-block economy, compute laundered aspirations $\hat{a}_i$ under $W^I_{\text{just}} = \frac{1}{9}\mathbf{1}\mathbf{1}'$ at the current allocation. Which block has negative laundered satisfaction? Compare the ranking of the current allocation and of an equal allocation $\bar{x}_b = 1$ under endogenous and laundered aspirations. Does the ranking change?

12.3 Capability measurement. For $N = 2$, $\pi_1 = 0.7$, $\pi_2 = 0.3$, $\omega_1 = \omega_2 = 5$, $\Phi = 2$: compute the entitlements e_i and the capability sets $\mathcal{O}_i$ with $c_1 = c_2 = 1$. Then suppose agent 2's information network hides the options available to her, so that $c_2 = 0.8$. Which agent has the larger capability set, and by how much? Why does varying her aspiration a_2 not change your answer?

12.4 Γ is not a welfare criterion. Construct two 4×4 row-stochastic matrices with $\Gamma(W_A) > \Gamma(W_B)$ but $\text{Var}(\boldsymbol{\pi}_A) > \text{Var}(\boldsymbol{\pi}_B)$. (Hint: $W = \gamma\mathbf{1}\boldsymbol{\pi}' + (1-\gamma)I$ has Perron vector $\boldsymbol{\pi}$ and spectral gap γ.) Using Proposition 12.11, compute the fixed-evaluation loss for each and show that the ranking follows $N_{\text{eff}}(\boldsymbol{\pi})$, not Γ.

12.5 The Easterlin pattern. Simulate 10 countries, each a block with its own allocation distribution, under three information structures: national reference groups (block-diagonal W^I across countries), global reference groups (W^{CE}), and partial integration. With $\mathbf{a} = \mathcal{M}W^I\mathbf{x}$, compute (a) the cross-country slope of average satisfaction on average income and (b) the within-

country slope. Under which structure does the model reproduce Easterlin's pattern? What would Stevenson and Wolfers' evidence imply for W^I?

12.6 Evaluation and preference vintage. Reproduce Example 12.17. Then let aspirations adjust gradually, $\mathbf{a}_{t+1} = \rho\mathbf{a}_t + \mu W^I_{\text{new}}\mathbf{x}_t$ with $\mu/(1-\rho) = 0.8$, while rural allocations rise by 0.02 per generation. Compute rural satisfaction for $t = 0, \ldots, 20$ under (i) old-vintage, (ii) current-vintage, and (iii) new-steady-state aspirations. When does the policy first look welfare-improving under each? How does the answer depend on ρ?

12.7 Sen vs. Bentham. Work through Example 12.14: a landlord ($\pi = 0.8$, aspiration $a_L = 4$) and a worker ($\pi = 0.2$, $a_W = 1.5$), $\Omega = 10$, $\Phi = 4$, $v(z) = z - 0.05z^2$.

(a) Compute the bargaining allocation and check that every $\kappa(x_i - a_i) < 1$.

(b) Compute welfare under: (i) Bentham with current aspirations, (ii) laundered Bentham with $\hat{a} = 3$, (iii) Rawlsian with current aspirations ($\min_i s_i$), (iv) capability ($\min_i e_i$, with $c = 1$).

(c) A policy equalizes power: $\pi_1 = \pi_2 = 0.5$. Which criteria declare the policy an improvement? Which declare it a deterioration?

(d) The landlord opposes the policy. Under which criterion is his opposition "valid" (i.e., the policy makes him worse off)? What would change if aspirations adjusted to the new allocation?

Mathematical Appendices

Appendix 12A: Welfare Functions with Endogenous Preferences

The standard approach. Social welfare is $W(\mathbf{x}) = \sum_i \alpha_i U_i(x_i, \theta_i)$, where α_i are welfare weights and θ_i are preferences. Both α_i and θ_i are taken as given. The welfare function is well-defined and the ranking is complete (any two allocations can be compared).

The problem with endogeneity. When $\theta_i = \theta_i^*(W^I, \mathbf{x})$ depends on the information structure (and, through observed outcomes, on the allocation), the welfare function becomes $W(\mathbf{x}, W^I) = \sum_i \alpha_i U_i(x_i, \theta_i^*(W^I, \mathbf{x}))$. This is a function of *both* the allocation and the information structure. Two allocations cannot be compared without specifying W^I, and two allocations under different W^I's are evaluated by different welfare functions.

The four approaches formalized (with $s_i = v(x_i - a_i)$).

1. **Laundered**: $W^L(\mathbf{x}) = \sum_i v(x_i - \hat{a}_i)$, with $\hat{\mathbf{a}} = \mathcal{M} W^I_{\text{just}} \mathbf{x}^{\text{ref}}$ fixed.
2. **Capability**: $\mathcal{O}_i = [0, c_i e_i]$ with $e_i = \omega_i + \pi_i \Phi$. Aggregate with a criterion that is sensitive to distribution (e.g. $\min_i c_i e_i$), not with a sum.
3. **Multi-criteria**: $\mathbf{W}(\mathbf{x}) = (W^1(\mathbf{x}, \mathbf{a}^1), \ldots, W^K(\mathbf{x}, \mathbf{a}^K))$ — a vector of welfare evaluations under K aspiration specifications, ranked by intersection.
4. **Process**: $\big(N_{\text{eff}}(\boldsymbol{\pi}), N_{\text{eff}}(\boldsymbol{\pi}^I), \Gamma(W^I)\big)$ — evaluate the structure directly, by vector dominance.

Appendix 12B: Why the "Structural Welfare Theorem" Fails

An earlier draft claimed that $\Gamma_A > \Gamma_B$, smaller deviation from double stochasticity, and $N_{\text{eff}}(W^I_A) > N_{\text{eff}}(W^I_B)$ imply that W_A is better under all four approaches. Each step of its proof fails.

Capability and laundered welfare. Both relied on "higher Γ implies smaller $\text{Var}(\boldsymbol{\pi})$". That is false: Γ and $\boldsymbol{\pi}$ are independent (Proposition 6.16; Remark 12.12). The capability step also used "Jensen, since h is concave in π_i", but h is linear in π_i. With capability defined as maximal attainable allocation, the sum over agents is constant in $\boldsymbol{\pi}$ (Proposition 12.7(3)). The correct fixed-evaluation statement, with $N_{\text{eff}}(\boldsymbol{\pi})$ in place of Γ, is Proposition 12.11.

Endogenous welfare. The proof used $U_i = \theta_i x_i - \frac{\kappa}{2} x_i^2$ at $x_i = (\theta_i + d_i)/\kappa$, with $d_i = \pi_i - 1/N$. The correct identity is

$$U_i = \frac{\theta_i^2}{2\kappa} - \frac{d_i^2}{2\kappa},$$

because the cross-term cancels. The draft's $+\theta_i d_i/\kappa$ was wrong. Summing, $\sum_i U_i = \frac{N}{2\kappa}(\bar{\theta}^2 + \mathrm{Var}(\theta)) - \frac{N}{2\kappa}\mathrm{Var}(\pi)$. More homogeneous preferences (smaller $\mathrm{Var}(\theta)$) *lower* this sum, the opposite of what the proof needed. The claim that $\bar{\theta}$ rises with connectivity is also false in general. In Example 9.10, connecting the two agents *lowers* mean θ^* from 1.0055 to 1.0000 (`code/ch09.py`). Finally, with $U = \theta x - \frac{\kappa}{2} x^2$ a higher θ *raises* utility at a given x. This form therefore encodes the opposite of adaptive preferences, which is why this chapter uses reference-dependent utility instead.

Process welfare. Item (d) held "by assumption", which is a tautology: it restated the hypotheses as the conclusion.

What survives. Under fixed common aspirations, less concentrated power is better (Proposition 12.11). Under endogenous aspirations, the ranking depends on whom W^I makes people watch (Proposition 12.13). No single structural statistic ranks economies under all four approaches.

Structuralist Case Study: Sen's Capabilities and the Formation Model

Amartya Sen's *Development as Freedom* (1999) is the most influential critique of preference-based welfare economics in the development literature. Its central argument is that development should be measured not by income or utility but by **capabilities** — the substantive freedoms people have to lead the lives they have reason to value. The argument was aimed squarely at the problem this chapter addresses: the inadequacy of preference satisfaction as a welfare criterion when preferences are shaped by deprivation.

Sen's framework maps onto the formation model in three specific ways — each with a caveat about what the mapping leaves out.

Part 1: Adaptive preferences as a failure of welfare economics. Sen identified "adaptive preferences" — the phenomenon where the deprived adjust their expectations downward, reporting satisfaction with objectively inadequate conditions — as the fundamental challenge to utilitarian welfare economics. A woman who has never been to school does not desire schooling; a worker who has never

had a union does not miss collective bargaining; a society that has never experienced democracy does not demand it. Utility-based welfare analysis concludes: no deprivation, because no unsatisfied desire.

In the formation model, this is Proposition 12.2. The traditional sector's aspiration $a_T^* < a_M^*$ because W^I is block-diagonal, and the satisfaction gap is compressed relative to the integrated benchmark by the factor $1 - \mathcal{M}$, while the allocation gap is unchanged. In the model, that compression *is* the adaptive preference, and it is measurable once $\mathcal{M}$ is. Sen's concept is broader — it includes resignation, internalized norms about what one is entitled to, and lack of "courage to demand" — and the scalar aspiration captures only the first of these.

Part 2: Capabilities as an alternative evaluative space. Sen proposed measuring development by what people are *free to do and be* ("functionings" and "capabilities") rather than by what they have or how they feel. A person who chooses to fast has the capability to eat; a person who starves does not. The distinction matters because the first person's "preference" for fasting is exercised from a position of freedom; the second's "preference" for accepting hunger is formed by necessity.

In Definition 12.6, the capability set $\mathcal{O}_i = [0, c_i e_i]$ formalizes this. It is determined by entitlements ($e_i = \omega_i + \pi_i \Phi$) and conversion factors (c_i), not by preferences. The faster and the starving person may choose the same nutrition, but only the first has it in her set. The structural constraints on capability are twofold: power (π_i) limits what you can *get*, and information (W^I, through c_i) limits what you know you can *do* with it. Aspirations shape what is *chosen* from the set, not the set itself.

Part 3: Development as structural change. Sen argued that development should expand capabilities, not maximize utility. In the formation model, expanding the capabilities of the least advantaged means raising their entitlements (less concentrated power: higher $N_{\text{eff}}(\boldsymbol{\pi})$; Proposition 12.7(3)) and their conversion factors (more plural and relevant information: higher $N_{\text{eff}}(\boldsymbol{\pi}^I)$, and access to returns they can attain). These are structural properties of the bargaining layer and of W^I. The spectral gap Γ is not among them: it measures integration, not the distribution of power. Note, too, that summing capabilities across agents hides exactly the distributional question Sen cared about.

Sen's concept	Formal object	This chapter
Adaptive preferences	$a_T^* < a_M^*$ from block-diagonal W^I; gap compressed by $1 - \mathcal{M}$	Prop. 12.2
Satisfaction paradox	$s_T^* \approx s_M^*$ as $\mathcal{M} \to 1$ despite $x_T^* < x_M^*$	Ex. 12.3
Capability set	$\mathcal{O}_i = [0, c_i e_i]$, $e_i = \omega_i + \pi_i \Phi$	Def. 12.6
Freedom vs. choice	$\mathcal{O}_i$ independent of a_i; aspirations act on choice within the set	Prop. 12.7
Capability deprivation	Low π_i (entitlement) and poor information (conversion)	Joint structural constraint
Partial comparability	Intersection quasi-ordering over criteria	Prop. 12.9
Development as freedom	Raise $N_{\text{eff}}(\boldsymbol{\pi})$ and the conversion factors of the least advantaged	Props. 12.11, 12.7

Sen was not averse to formalization — much of his social-choice work is formal. What he resisted was reducing the capability approach to a fixed list of capabilities or a single metric. The formation model respects that: capability is one of four criteria, and the multi-criteria approach is Sen's own intersection quasi-ordering. What the model adds is a mechanism and some measurable quantities: the aspiration compression factor $1 - \mathcal{M}$, and the distinction between entitlements, conversion factors and aspirations. It also adds the structural statistics ($N_{\text{eff}}(\boldsymbol{\pi})$, $N_{\text{eff}}(\boldsymbol{\pi}^I)$) that bear on them. Critics of adaptive-preference arguments warn against using them to override people's own judgments (Nussbaum, 2001; Khader, 2011). That is why the chapter ends with structural transparency rather than with a planner's verdict.

Further Reading

Adaptive preferences: Sen (1999), *Development as Freedom*, Chapters 1–3. Nussbaum (2000), *Women and Human Development*, Chapter 1. Nussbaum (2001), "Adaptive Preferences and Women's Options." Khader (2011), *Adaptive Preferences and*

Women's Empowerment. Elster (1983), *Sour Grapes*, Chapter 3 (the original philosophical treatment). On aspirations: Genicot and Ray (2017); Dalton et al. (2016), "Poverty and Aspirations Failure."

Welfare with endogenous preferences: Stigler and Becker (1977), "De Gustibus Non Est Disputandum" (the opposing view). Goodin (1986), "Laundering Preferences," in Elster and Hylland (eds.), *Foundations of Social Choice Theory*. Bowles (1998), "Endogenous Preferences." von Weizsäcker (1971), "Notes on Endogenous Change of Tastes." Hammond (1976), "Endogenous Tastes and Stable Long-Run Choice." Pollak (1978), "Endogenous Tastes in Demand and Welfare Analysis." Brandt (1979), *A Theory of the Good and the Right*; Harsanyi (1977) on "true" (full-information) preferences.

The capability approach: Sen (1970), *Collective Choice and Social Welfare* (partial comparability and intersection quasi-orderings). Sen (1985), *Commodities and Capabilities*. Pattanaik and Xu (1990), "On Ranking Opportunity Sets in Terms of Freedom of Choice." Sen (1991), "Welfare, Preference and Freedom." Alkire (2002), *Valuing Freedoms*. Robeyns (2005), "The Capability Approach: A Theoretical Survey."

The Easterlin paradox: Easterlin (1974), "Does Economic Growth Improve the Human Lot?" Stevenson and Wolfers (2008), "Economic Growth and Subjective Well-Being." Clark et al. (2008), "Relative Income, Happiness, and Utility."

Policy evaluation under endogenous preferences: Bernheim and Rangel (2009), "Beyond Revealed Preference: Choice-Theoretic Foundations for Behavioral Welfare Economics." Sugden (2004), "The Opportunity Criterion: Consumer Sovereignty Without the Assumption of Coherent Preferences." Macours and Vakis (2014), on transfers, aspirations and social interactions.

PART IV

Dynamics of Structural Change

Chapter 13

How Networks Evolve

The Dynamics of Structural Change

> *"La internacionalización de la economía tiene como contrapartida la desintegración nacional: al mismo tiempo que se van creando vínculos intensos entre los núcleos dinámicos de los países subdesarrollados y el sistema económico internacional, se van desarticulando los vínculos entre esos núcleos y el resto de la economía nacional."*
>
> ["The internationalization of the economy has as its counterpart national disintegration: at the same time that intense links are created between the dynamic nuclei of underdeveloped countries and the international economic system, the links between those nuclei and the rest of the national economy are dismantled."]
> — Osvaldo Sunkel, "Capitalismo transnacional y desintegración nacional" (1971)

Parts I–III treated the weight matrices (W^U, W^P, W^I) as given. We asked: given the network, what is the equilibrium allocation, the power structure, the preference formation? But the networks themselves change. Trade routes open and close. Firms enter and exit. Social media platforms rewire who talks to whom. The question of Part IV is: *how does W evolve, and what determines whether it evolves toward integration or fragmentation?*

This chapter develops the general theory of network evolution. The central result: W changes through three mechanisms — edge creation (new links form), edge destruction (existing links break), and edge reweighting (the strength of existing links changes). Each mechanism is governed by the current state of the economy: the allocation, the returns, and the preferences. This creates a **co-evolution**:

W shapes $(\mathbf{x}^*, \boldsymbol{\theta}^*)$ (the static model of Parts I–III), and $(\mathbf{x}^*, \boldsymbol{\theta}^*)$ shapes the next period's W. The steady state of the co-evolutionary system is the **structural equilibrium**: a weight matrix that reproduces itself through the allocation and preferences it generates.

13.1 Three Mechanisms of Network Change

13.1.1 Edge Creation

New edges form when agents establish new connections. In W^P: a firm opens a supply contract with a new input provider. In W^U: a migrant forms a friendship in a new city. In W^I: a student enrolls in a university and begins receiving signals from a new reference group.

Definition 13.1 (Edge Creation Process)**.** *The probability that a new edge (i, j) forms in period $t+1$ depends on three factors through a logistic link, so that it always lies in $(0, 1)$:*

$$\Pr(\Delta W_{ij,t+1} > 0 \mid W_t) = \sigma\Big(p_0 + p_1 \underbrace{a(i, j; W_t)}_{\textit{proximity}} + p_2 \underbrace{s(i, j; \mathbf{x}_t^*)}_{\textit{surplus}} + p_3 \underbrace{q(i, j; W_t^I)}_{\textit{information}}\Big), \tag{13.1}$$

with $\sigma(z) = 1/(1+e^{-z})$ and $p_1, p_2, p_3 \geq 0$, where:

- $a(i, j; W_t)$: ***proximity in the existing network***, *a decreasing function of the network distance $d(i, j; W_t)$ (Appendix 13.8), e.g. $a = 1/d$. Agents form links more easily with friends of friends (triadic closure). The probability of connecting to j increases if i and j share a neighbor. This is the relatedness mechanism: new connections form near existing ones.*

- $s(i, j; \mathbf{x}_t^*)$: ***expected surplus from the connection.*** *i connects to j if the connection generates value — a profitable trade, a useful signal, a beneficial relationship. The surplus depends on the current allocation (which determines prices and returns).*

- $q(i, j; W_t^I)$: ***information.*** *i can only form a link with j if i knows about j. Knowledge of potential partners is filtered through W^I: you connect to people you have heard of, which are people in your information network's reach.*

Intuition: Why proximity dominates

The three factors create a strong path dependence. New edges form near existing edges (proximity), where the surplus is predictable (information), and where the connection can be sustained (surplus). This means W_{t+1} is similar to W_t: the network evolves gradually, not by random rewiring. The principle of relatedness (Hidalgo et al., 2007) — the empirical law that regions diversify into activities related to their existing activities — is a consequence of this proximity mechanism applied to W^P.

13.1.2 Edge Destruction

Existing edges break when the surplus from maintaining them turns negative, or when an exogenous shock disrupts the connection.

Definition 13.2 (Edge Destruction Process)**.** *An existing edge (i, j) is destroyed in period $t + 1$ with probability:*

$$\Pr(W_{ij,t+1} = 0 \mid W_{ij,t} > 0) = \delta_0 + \delta_1 \cdot \underbrace{\max(0, -s(i, j; \mathbf{x}_t^*))}_{\text{negative surplus}} + \delta_2 \cdot \underbrace{\sigma_{ij,t}}_{\text{exogenous shock}}, \quad (13.2)$$

where δ_0 is the baseline destruction rate (natural turnover), $s < 0$ means the link is unprofitable, and $\sigma_{ij,t}$ is an exogenous shock (a crisis, a policy change, a natural disaster that severs a trade route).

Example 13.3 (Bolivia's Gas Nationalization as Edge Destruction and Creation)**.** *Bolivia's nationalization of gas in 2006 was a simultaneous edge destruction and edge creation event in W^P:*

Destruction: *The edges from domestic gas production to foreign MNC headquarters ($W^P_{\text{Bolivia, MNC}}$) were weakened. Foreign companies lost operational control. The profits channel (surplus flowing from Bolivia to MNC shareholders) was partially severed.*

Creation: *New edges from gas production to the domestic treasury ($W^P_{\text{gas, treasury}}$) were strengthened. Gas revenues began flowing through domestic fiscal channels to domestic social programs.*

Net effect on Γ: *Small, and not upward. The nationalization increased some within-country flows (gas revenue to social spending) but did not create production linkages between the gas sector and the rest of the economy. The INE input-output tables allow a check (12 sectors, domestic flows, input shares; base 1990): $\Gamma(W^P)$ was 0.518 in 2006 and 0.484 in 2013. It drifted down slightly, and it was never low: Bolivia's domestic W^P is not block-diagonal, and its Γ is in the range of Korea's and the US's on a common*

basis (Chapter 16). The enclave shows up elsewhere. Hydrocarbons' power share in W^P rose with the boom, from 0.064 *in 2006 to* 0.085 *in 2013, but in 2019 it was only* 0.039 *(base 2017, not strictly comparable with base 1990); the sector exports* 70% *of its output, buys* 0.225 *of domestic intermediate inputs per unit of output, and pays* 6% *of its value added as compensation of employees (Instituto Nacional de Estadística (INE), Bolivia, 2026); author's calculations (`data/compute/compute_bolivia_io.py`). Its rents reach the economy through the treasury (the direct hydrocarbons tax, IDH), not through production edges. Strictly, the gas-to-treasury channel is a* fiscal *linkage in Hirschman's (1977) sense rather than a production edge; modelling it inside W^P requires a node set that includes the government as a sector.*

13.1.3 Edge Reweighting

The strength of existing edges changes continuously as the economic environment evolves.

Definition 13.4 (Edge Reweighting)**.** *The weight of an existing edge evolves according to:*

$$\tilde{W}_{ij,t+1} = W_{ij,t} + \eta \cdot \frac{\partial s(i,j)}{\partial W_{ij}} + \nu_{ij,t}, \tag{13.3}$$

where $\eta > 0$ governs the speed of adjustment and $\nu_{ij,t}$ is noise. Profitable connections strengthen ($\partial s/\partial W > 0$); unprofitable connections weaken. The updated row is then clipped at zero and renormalized to sum to one (Definition 13.5).

13.2 The General Network Evolution Equation

Combining creation, destruction, and reweighting, the network evolves according to:

Definition 13.5 (Network Evolution Equation)**.**

$$W_{t+1} = \mathcal{N}\Big(\underbrace{(\mathbf{1}\mathbf{1}' - D_t) \circ W_t}_{\text{surviving edges}} + \underbrace{C_t \circ \tilde{W}^{new}_{t+1}}_{\text{new edges}} + \underbrace{\eta \cdot S_t}_{\text{reweighting}} \Big), \tag{13.4}$$

*where $\circ$ is the Hadamard (element-wise) product, $\mathbf{1}\mathbf{1}'$ is the all-ones matrix, D_t is the $N \times N$ destruction probability matrix, C_t is the creation probability matrix, $\tilde{W}^{new}_{t+1}$ is the weight assigned to newly created edges, and $S_t = [\partial s(i,j)/\partial W_{ij}]$ is the surplus gradient matrix. The operator $\mathcal{N}$ clips negative entries to zero and divides each row by its sum, so W_{t+1} is again row-stochastic. Written with the probabilities D_t, C_t (rather than realized draws), (13.4) is the **mean-field** (expected) evolution map; the realized network fluctuates around it.*

Definition 13.6 (The Co-Evolutionary System)**.** *The complete structural economics model is a co-evolutionary system in* $(W_t, \boldsymbol{\theta}_t)$*:*

$$\boldsymbol{\theta}_{t+1} = T(\boldsymbol{\theta}_t; W_t) \quad \textit{(formation map (10.4))}, \tag{13.5}$$

$$W_{t+1} = \mathcal{E}(W_t; \mathbf{x}^*(\boldsymbol{\theta}_t, W_t)) \quad \textit{(network evolution map (13.4))}, \tag{13.6}$$

where $\mathbf{x}^*(\boldsymbol{\theta}_t, W_t)$ *is the bargaining equilibrium at period* t*. (The formation map is Chapter 10's* T*, with anchor* θ°*; its dependence on the network is written out as* $T(\cdot\,; W)$*. We write* $\mathcal{E}$ *for the network evolution map, rather than a second* T*, to keep the two maps apart.* Φ *is the bargaining surplus of Proposition 7.3, and* Ψ *is reserved for the formation-to-enforcement ratio of Definition 9.13.)*

The fast variable is $\mathbf{x}^*$ *(clears within a period). The medium variable is* $\boldsymbol{\theta}$ *(adjusts across generations). The slow variable, absent deliberate intervention, is* W *(evolves over years to decades). Three timescales, nested.*

Intuition: Three timescales

Markets clear in months. Preferences form in generations. Left to proximity-driven link formation, structures evolve over decades; coalitions and policy (Chapters 14–15) can move parts of W much faster. This ordering is what makes the analysis tractable: at any given timescale, the slower variables are approximately constant and the faster variables are in approximate equilibrium.

When we analyzed the bargaining equilibrium (Chapters 6–7), we held W and $\boldsymbol{\theta}$ fixed — both are slow relative to market clearing. When we analyzed the formation model (Chapters 9–10), we held W fixed — it is slow relative to preference formation. Now we let W move, completing the picture.

13.3 Structural Equilibrium

13.3.1 Definition

Definition 13.7 (Structural Equilibrium)**.** *A* ***structural equilibrium*** $(W^*, \boldsymbol{\theta}^*, \mathbf{x}^*)$ *is a network, a preference profile and an allocation that satisfy three conditions:*

1. $\mathbf{x}^* = h(\boldsymbol{\pi}(W^*), \boldsymbol{\theta}^*, \boldsymbol{\omega})$ *(bargaining equilibrium),*
2. $\boldsymbol{\theta}^* = T(\boldsymbol{\theta}^*; W^*)$ *(formation steady state),*
3. $W^* = \mathcal{E}(W^*; \mathbf{x}^*)$ *(network steady state).*

All three equations are simultaneously satisfied. The weight matrix produces the allocation and preferences that reproduce the weight matrix.

Theorem 13.8 (Existence of Structural Equilibrium)**.** *Fix $\varepsilon \in (0, 1/N)$ and let $\mathcal{W}_\varepsilon = \{W \in \mathbb{R}^{N\times N} : W\mathbf{1} = \mathbf{1},\ W_{ij} \geq \varepsilon\}$. Let $\Theta \subset \mathbb{R}^N$ be compact and convex. Assume (i) $T(\cdot\,; W)$ maps Θ into Θ; (ii) the mean-field evolution map $\mathcal{E}$ (including the renormalization $\mathcal{N}$) maps $\mathcal{W}_\varepsilon \times \mathbb{R}^N$ into $\mathcal{W}_\varepsilon$ (for instance, by mixing its output with weight $N\varepsilon$ on W^{CE}); (iii) T, $\mathcal{E}$ and h are continuous. Then at least one structural equilibrium with $W^* \in \mathcal{W}_\varepsilon$ exists.*

Proof. Every $W \in \mathcal{W}_\varepsilon$ is strictly positive, hence primitive, so its Perron root 1 is simple and $\boldsymbol{\pi}(W)$ is a continuous (indeed analytic) function on $\mathcal{W}_\varepsilon$. (On the full set of row-stochastic matrices this fails: at reducible W, such as an exactly block-diagonal matrix, $\boldsymbol{\pi}$ is not unique.) The composite map

$$\Xi : (W, \boldsymbol{\theta}) \mapsto \Big(\mathcal{E}\big(W; h(\boldsymbol{\pi}(W), T(\boldsymbol{\theta}; W), \boldsymbol{\omega})\big),\ T(\boldsymbol{\theta}; W)\Big)$$

is therefore continuous and, by (i)–(ii), maps the nonempty compact convex set $\mathcal{W}_\varepsilon \times \Theta$ into itself. Brouwer's fixed-point theorem gives a fixed point $(W^*, \boldsymbol{\theta}^*)$; set $\mathbf{x}^* = h(\boldsymbol{\pi}(W^*), \boldsymbol{\theta}^*, \boldsymbol{\omega})$. □

The theorem is about the mean-field (expected) map. A fixed point of the stochastic process itself is a stationary distribution over networks, not a single W^*.

13.3.2 Multiple Structural Equilibria

Proposition 13.9 (Multiplicity)**.** *The co-evolutionary system can admit multiple structural equilibria: there are parameter values for which two distinct structural equilibria coexist, each locally stable under the two-timescale reduced dynamics of Appendix 13.8 (with $\boldsymbol{\theta}$ held at its formation steady state $\boldsymbol{\theta}^*(W)$).*

Proof. Example 13.10 exhibits one such parameterization. There, for each c, $\boldsymbol{\theta}^*(c)$ is the formation steady state and $\mathbf{x}^* = h(\boldsymbol{\pi}(W(c)), \boldsymbol{\theta}^*(c), \boldsymbol{\omega})$, so conditions 1–2 of Definition 13.7 hold at every c, and condition 3 holds exactly at the fixed points of $c \mapsto (1-\eta)c + \eta F(c)$, i.e. where $F(c) = c$. The fixed points $c_A = 0.072$ and $c_B = 0.643$ are distinct and have $0 \leq F' < 1$, hence are locally stable by Proposition 13.14 (`code/ch13.py`). □

Multiplicity requires a complementarity (here: cross-block links raise the surplus from further cross-block links); with a contraction it is impossible. Two important types:

The integrated equilibrium. W^* has large Γ (well-connected), $\boldsymbol{\theta}^*$ is less dispersed across blocks, and allocation gaps are smaller. Cross-block edges are sustained by positive surplus. This equilibrium is self-reinforcing: the connected structure produces high returns for all agents, which sustains the preferences that support the connections, which maintains the structure.

The fragmented equilibrium. W^* is approximately block-diagonal (small Γ), $\boldsymbol{\theta}^*$ is stratified (different preferences per block), and allocation gaps are larger. Cross-block edges do not form because the surplus is too low (the traditional block's agents cannot afford the investment needed to link to the modern block). This equilibrium is also self-reinforcing: the disconnected structure produces low returns in the traditional block, which forms low aspirations, which prevents investment in cross-block links, which maintains disconnection.

Neither type need differ in the distribution of power: Γ and $\boldsymbol{\pi}$ are independent coordinates (Proposition 6.16), and in the example below $\boldsymbol{\pi}$ is identical in both equilibria.

Example 13.10 (Two Structural Equilibria in a Stylized Bolivia [ILLUSTRATIVE])**.** *$N = 6$: Gas block (agents 1–2), Urban block (agents 3–4), Rural block (agents 5–6). The labels are Bolivian; the parameters are illustrative. To make every step computable we restrict the evolution map to a one-parameter family. Let B be within-block averaging ($B_{ij} = 0.5$ if i, j are in the same block, 0 otherwise) and let $\mathbf{1}\mathbf{s}'$ be "open" links, in which every agent weights partners by their attractiveness $\mathbf{s} = (0.25, 0.25, 0.15, 0.15, 0.10, 0.10)'$. The network is*

$$W(c) = (1 - c)\, B + c\, \mathbf{1}\mathbf{s}', \qquad c \in [0.05, 0.65], \tag{13.7}$$

which is row-stochastic, asymmetric and strictly positive, so $W(c) \in \mathcal{W}_\varepsilon$. The three maps are:

- ***Bargaining*** *h: $x_i = \Omega/N + (\pi_i - 1/N)\Phi + (\theta_i - \bar{\theta})$ with $\Omega = 6$ and bargaining surplus $\Phi = 1$ (the allocation rule of Proposition 7.3 plus the preference term of Chapter 10).*
- ***Formation*** *T (Chapter 10) with $W^I = W(c)$, block returns $r = 0.20$ (gas), 0.12 (urban), 0.06 (rural), $\rho = 0.85$, $\mu = 0.2$, $\theta^\circ = 1$, $r^* = 0.10$. Its steady state is*

$$\boldsymbol{\theta}^*(c) = \theta^\circ \mathbf{1} + \tfrac{\mu}{1-\rho}\big(W(c)\mathbf{r} - r^*\mathbf{1}\big).$$

- ***Evolution*** *$\mathcal{E}$: cross-block intensity moves toward the logistic creation probability (13.1), $c_{t+1} = (1 - \eta)c_t + \eta F(c_t)$ with $\eta = 0.05$ and*

$$F(c) = 0.05 + 0.60\,\sigma\big(12\,(c - 0.30) + 12\,(\theta_5^*(c) - 1)\big).$$

The first term is proximity (more cross-block links make further ones "nearer"); the second is surplus, which rises with rural aspirations θ_5^*.

Structural equilibria are the fixed points $F(c) = c$. There are three (`code/ch13.py`): $c_A = 0.072$ (stable, $F'(c_A) = 0.28$), $\hat{c} = 0.282$ (unstable, $F'(\hat{c}) = 1.91$), and $c_B = 0.643$ (stable, $F'(c_B) = 0.10$).

Equilibrium A (Fragmented):

$$W_A^* = W(0.072) = \begin{pmatrix} 0.482 & 0.482 & 0.011 & 0.011 & 0.007 & 0.007 \\ 0.482 & 0.482 & 0.011 & 0.011 & 0.007 & 0.007 \\ 0.018 & 0.018 & 0.475 & 0.475 & 0.007 & 0.007 \\ 0.018 & 0.018 & 0.475 & 0.475 & 0.007 & 0.007 \\ 0.018 & 0.018 & 0.011 & 0.011 & 0.471 & 0.471 \\ 0.018 & 0.018 & 0.011 & 0.011 & 0.471 & 0.471 \end{pmatrix}. \tag{13.8}$$

$\Gamma_A = 0.072$, *max allocation gap* $(x_{\max} - x_{\min})/\bar{x} = 32\%$, *with*

$$\boldsymbol{\theta}^* = (1.128, 1.128, 1.029, 1.029, 0.955, 0.955),$$
$$\mathbf{x}^* = (1.174, 1.174, 0.975, 0.975, 0.851, 0.851).$$

Equilibrium B (Integrated):

$$W_B^* = W(0.643) = \begin{pmatrix} 0.339 & 0.339 & 0.096 & 0.096 & 0.064 & 0.064 \\ 0.339 & 0.339 & 0.096 & 0.096 & 0.064 & 0.064 \\ 0.161 & 0.161 & 0.275 & 0.275 & 0.064 & 0.064 \\ 0.161 & 0.161 & 0.275 & 0.275 & 0.064 & 0.064 \\ 0.161 & 0.161 & 0.096 & 0.096 & 0.243 & 0.243 \\ 0.161 & 0.161 & 0.096 & 0.096 & 0.243 & 0.243 \end{pmatrix}. \tag{13.9}$$

$\Gamma_B = 0.643$, *max allocation gap* 22%, *with*

$$\boldsymbol{\theta}^* = (1.089, 1.089, 1.051, 1.051, 1.022, 1.022),$$
$$\mathbf{x}^* = (1.118, 1.118, 0.980, 0.980, 0.902, 0.902).$$

In both equilibria $\boldsymbol{\pi} = \mathbf{s} = (0.25, 0.25, 0.15, 0.15, 0.10, 0.10)$ and $N_{eff} = 5.26$: the family (13.7) has $\Gamma(W(c)) = c$ exactly while $\boldsymbol{\pi}(W(c)) = \mathbf{s}$ for every c (Proposition 6.16). Integration changes preferences and the allocation gap through W^I; it does not redistribute power. Same economy, same parameters, two self-reinforcing structures. Which obtains depends on history: starting below $\hat{c} = 0.282$ the economy converges to A, above it to B.

Neither equilibrium is a description of Bolivia. Its measured domestic production net-

work has $\Gamma(W^P) = 0.567$ *(1990),* 0.494 *(2000),* 0.518 *(2006) and* 0.484 *(2013) on INE's base 1990, and* 0.467 *in 2019 on base 2017 (12 sectors, input shares), close to Korea's* 0.511 *on a common basis (Instituto Nacional de Estadística (INE), Bolivia, 2026; OECD, 2023). If Bolivia is fragmented in the sense of equilibrium A, the fragmentation does not show in aggregate* $\Gamma(W^P)$*; it would have to lie in* W^I *or* W^U*, or at finer sector detail.*

Visualization: Two structural equilibria: fragmented and integrated



Fragmented W_A^*
$\Gamma = 0.072$, gap $= 32\%$

Integrated W_B^*
$\Gamma = 0.643$, gap $= 22\%$

Same economy, same parameters. Brightness is W_{ij} relative to the largest entry of each matrix. Left: the fragmented equilibrium — three bright blocks, dark cross-block. Right: the integrated equilibrium — blocks still visible but cross-block entries are bright, and the gas columns (1–2) are brighter than the rural columns (5–6) in both, because $\boldsymbol{\pi} = \mathbf{s}$ in both. Both are self-reinforcing. Which one you get depends on where you start.

13.4 The Principle of Relatedness

The edge creation process (Definition 13.1) implies that new edges form near existing edges. This has a powerful consequence: the network's evolution is path-dependent.

Conjecture 13.11 (Path Dependence of Network Evolution)**.** *If edge creation is dominated by proximity* ($p_1 \gg p_2, p_3$ *in* (13.1))*, then:*

1. W_{t+1} *is structurally similar to* W_t*: new edges connect agents who are already close in the network.*

2. *The spectral gap* Γ_t *changes slowly. If* $\lambda_2(W_t)$ *is simple and no other eigenvalue (except its conjugate) shares its modulus, then locally* $|\Gamma_{t+1} - \Gamma_t| \leq C \cdot \|W_{t+1} - W_t\|$ *(Lemma 6.19). At a defective eigenvalue only a Hölder bound* $C\|W_{t+1} - W_t\|^{1/m}$ *holds (m the size of the Jordan block), so "slowly" needs this hypothesis.*

3. *Block-diagonal structures are absorbing: if* W_t *is approximately block-diagonal, new edges form within blocks (high proximity) rather than between blocks (low proximity), reinforcing the block-diagonal structure.*

4. *Escaping a block-diagonal structure requires either (a) a large exogenous shock that creates cross-block edges, or (b) deliberate policy that overrides the proximity mechanism.*

Why it is plausible. (1) follows from the proximity term dominating: the creation probability is increasing in $a(i, j; W_t)$, which is high for agents already close in W_t. (2) under the stated hypothesis is first-order eigenvalue perturbation. (3)–(4): in a block-diagonal W_t, within-block distances are small and cross-block distances are large, so edge creation favors within-block links. Example 13.10 is an instance: below $\hat{c}$ the dynamics return to the fragmented equilibrium. Turning (3)–(4) into a theorem requires a specific stochastic model of creation (e.g. a stochastic block model with a proximity kernel).

Prebisch Connection: The principle of relatedness as structural lock-in

The principle of relatedness (Hidalgo et al., 2007; Hidalgo, 2021) states that the probability a region enters a new economic activity increases with the number of related activities already present. In the product space, countries diversify into products near their current exports, not into distant products. This is Conjecture 13.11 applied to W^P. The production network evolves through proximity-dominated edge creation: you can produce what is "near" what you already produce. Bolivia produces gas (a product on the periphery of the product space, few connections to other products). The nearest products are other extractive commodities (mining, basic chemicals). The distant products are machinery, electronics, pharmaceuticals — the dense core of the product space.

The principle of relatedness is simultaneously the mechanism of industrial deepening (in the center: you diversify into related products, building an ever-denser W^P) and the mechanism of structural lock-in (in the periphery: you diversify into related commodities, staying on the sparse periphery of W^P). The same mechanism produces integration in the center and fragmentation in the periphery. This is not a paradox — it is a property of the proximity kernel: diversification follows edges, and the edge structure is dense in the core and sparse on the periphery.

13.5 Integration and Disintegration

13.5.1 When Does Γ Increase?

For a set of agents S with $\pi(S) = \sum_{i \in S} \pi_i \leq 1/2$, the **conductance** of the cut (S, S^c) is the probability flow across it, $\sum_{i \in S, j \notin S} \pi_i W_{ij} / \pi(S)$; the conductance $\phi(W)$ of W is the minimum over such S.

Proposition 13.12 (A Sufficient Condition for Structural Integration)**.** *Let W_t and W_{t+1} be irreducible, reversible with respect to their stationary distributions ($\pi_i W_{ij} = \pi_j W_{ji}$) and lazy ($W_{ii} \geq 1/2$). If*

$$\phi(W_{t+1}) > 2\sqrt{\phi(W_t)}, \tag{13.10}$$

then $\Gamma_{t+1} > \Gamma_t$: raising the weakest cross-block flow enough raises integration.

Proof. A lazy reversible W can be written $W = \frac{1}{2}(I + P)$ with $P = 2W - I$ non-negative, row-stochastic and reversible with respect to the same $\boldsymbol{\pi}$. The eigenvalues of P are real and lie in $[-1, 1]$, so those of W are real and lie in $[0, 1]$; hence $|\lambda_2(W)| = \lambda_2(W)$ and $\Gamma = 1 - \lambda_2$. Cheeger's inequality for reversible chains (Levin et al., 2009, Thm. 13.14), due to Lawler and Sokal (1988) and Sinclair and Jerrum (1989), gives $\phi^2/2 \leq 1 - \lambda_2 \leq 2\phi$, with ϕ the conductance defined above. Hence $\Gamma_{t+1} \geq \phi(W_{t+1})^2/2 > 2\phi(W_t) \geq \Gamma_t$. □

The condition is sufficient, not necessary. There is no "if and only if" in terms of cross-block weight alone: in a two-block matrix with fixed cross-block weight 0.04 per row, raising the within-block off-diagonal weight from 0.005 to 0.02 raises Γ from 0.050 to 0.080, because the slowest mode was *within* a block (`code/ch13.py`). Economic mechanisms that raise cross-block conductance:

1. **Cross-block surplus is positive**: $s(i, j) > 0$ for $i \in$ block b, $j \in$ block b' ($b \neq b'$). Trade between blocks is profitable.

2. **Information reaches across blocks**: $q(i, j; W^I) > 0$ for cross-block pairs. Agents in block b know about opportunities in block b'.

3. **Proximity barriers are overcome**: either by exogenous shocks that create cross-block edges (a road, a migration wave, a new trade route), or by policy that subsidizes cross-block links.

13.5.2 When Does Γ Decrease?

Remark 13.13 (Mechanisms of Structural Disintegration). By the same Cheeger bounds, Γ falls when the weakest cross-block flow collapses (for lazy reversible W, whenever $\phi(W_{t+1}) < \phi(W_t)^2/4$). Typical mechanisms:

1. **Cross-block edges are destroyed faster than created**: an external shock severs trade routes, a policy closes borders, a financial crisis disrupts supply chains.

2. **Within-block edges strengthen at the expense of cross-block edges**: agents retrench to familiar partners under uncertainty.

3. **Selective integration**: some agents in each block form cross-border links (the elite), while the remaining agents lose cross-block links. Define the *domestic* Γ as the spectral gap of the domestic principal submatrix of W with its rows renormalized. The domestic Γ falls; the global Γ may rise or fall, depending on where the rerouted weight comes from (Exercise **13.5**).

Key Structural Insight

Mechanism 3 is Sunkel's thesis: *transnational integration and national disintegration are the same process.* When the gas executive in La Paz connects to the global commodity market ($W^P_{\text{executive, global}}$ strengthens), his connection to the quinoa farmer weakens ($W^P_{\text{executive, farmer}}$ weakens — his supply chains are now global, not domestic). The domestic Γ decreases (more fragmented national economy). The global Γ can increase, but need not: in Exercise **13.5** it falls from 0.600 to 0.560, and over random diagonal-dominant starting matrices it rises in only 57% of cases. The same edge creation (gas executive $\to$ global market) is disintegration at the national scale; whether it is also integration at the world scale is an empirical question, not an identity.

13.5.3 A Phase Diagram

Proposition 13.14 (Phase Diagram for Γ in the Reduced Model)**.** *In Example 13.10, $\Gamma_t = c_t$ and*

$$\Gamma_{t+1} - \Gamma_t = \eta \left(F(\Gamma_t) - \Gamma_t\right), \tag{13.11}$$

where F combines the integration force (surplus and proximity of cross-block links, increasing in Γ) and the fragmentation force (the gap between F and the current level). Let $\eta \in (0,1)$ and $F' \geq 0$ (as in the example, where F is increasing in c). A fixed point Γ^* is locally asymptotically stable if $F'(\Gamma^*) < 1$ and unstable if $F'(\Gamma^*) > 1$.

Three regimes:

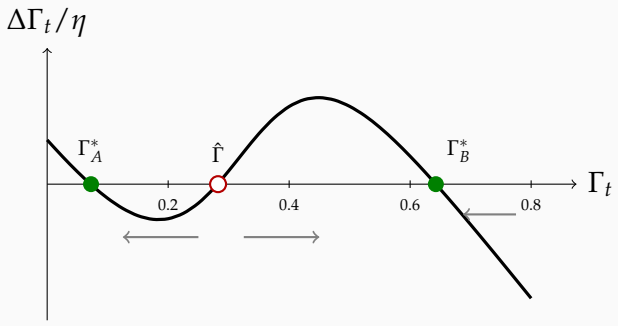
1. ***Integration:*** *$F(\Gamma) > \Gamma$. The economy moves toward the integrated equilibrium. $\Gamma \uparrow$.*

2. ***Fragmentation:*** *$F(\Gamma) < \Gamma$. The economy moves toward the fragmented equilibrium. $\Gamma \downarrow$.*

3. ***Lock-in:*** *near a stable fixed point, $F(\Gamma) \approx \Gamma$ and small shocks die out. The economy is trapped in its current structural configuration.*

Proof. $\Gamma(W(c)) = c$ because $W(c)$ has eigenvalues 1, $1-c$ (twice) and 0 (three times). To see this, let $\mathbf{e}_b$ be the indicator of block b. Then $B\mathbf{e}_b = \mathbf{e}_b$; write $s_b := \mathbf{s}'\mathbf{e}_b$ and $\mathbf{s}_{\text{blk}} = (s_b)_b$, which sums to one. So $W(c)\mathbf{e}_b = (1-c)\mathbf{e}_b + c\, s_b \mathbf{1}$: the span of the three block indicators is invariant, and on it $W(c)$ acts as $(1-c)I_3 + c\,\mathbf{1}\mathbf{s}'_{\text{blk}}$, whose eigenvalues are 1 and $1-c$ (twice). On the three-dimensional complement of vectors that sum to zero within each block, $B\mathbf{v} = 0$ and $\mathbf{s}'\mathbf{v} = 0$ (since $\mathbf{s}$ is constant within blocks), so $W(c)\mathbf{v} = 0$. Hence $|\lambda_2| = 1-c$ and, for $c \in (0,1]$, $\Gamma = c$; substituting $c_t = \Gamma_t$ in the evolution rule gives (13.11). The map $g(c) = (1-\eta)c + \eta F(c)$ has $g'(c) = 1 - \eta + \eta F'(c) \geq 1 - \eta > 0$. If $F'(\Gamma^*) < 1$ then $g'(\Gamma^*) \in (0,1)$, and the fixed point is locally asymptotically stable; if $F'(\Gamma^*) > 1$ then $g'(\Gamma^*) > 1$ and it is unstable. (At $F'(\Gamma^*) = 1$ the linearization is inconclusive.) The three regimes are the sign of $\Gamma_{t+1} - \Gamma_t = \eta(F(\Gamma_t) - \Gamma_t)$. □

Visualization: Phase diagram: integration vs. fragmentation forces



The net force $F(\Gamma) - \Gamma$ (thick curve) is S-shaped and crosses zero three times: at $\Gamma_A^* = 0.072$ (stable, fragmented), $\hat{\Gamma} = 0.282$ (unstable threshold) and $\Gamma_B^* = 0.643$ (stable, integrated). Below $\hat{\Gamma}$ the economy is drawn back to the trap; above it, toward integration. Moving the economy across $\hat{\Gamma}$ is structural change (Definition 13.15).

13.6 Structural Change as Transition Between Equilibria

Definition 13.15 (Structural Change)**.** ***Structural change*** *is a transition from one structural equilibrium* $(W_A^*, \boldsymbol{\theta}_A^*, \mathbf{x}_A^*)$ *to another* $(W_B^*, \boldsymbol{\theta}_B^*, \mathbf{x}_B^*)$*. The transition occurs when a shock or policy moves the system out of the basin of attraction of A and into the basin of attraction of B.*

Definition 13.16 (The Cost of Structural Change)**.** *The minimum "push" required to transition from the fragmented equilibrium A to the integrated equilibrium B is:*

$$\varepsilon^* = \inf\{\|W - W_A^*\|_F : W \in \mathcal{B}(W_B^*)\}, \tag{13.12}$$

the Frobenius distance from the current equilibrium to the basin $\mathcal{B}(W_B^*)$ *of the target equilibrium (holding* $\boldsymbol{\theta}$ *at its steady state).*

In Example 13.10 the basin of B is $\{W(c) : c > \hat{c}\}$, so $\varepsilon^* = (\hat{c} - c_A)\,\|\mathbf{1}\mathbf{s}' - B\|_F = 0.210 \times 1.463 = 0.307$. We conjecture that, more generally, this cost is:

1. **Large when ρ is large**: sticky preferences resist change, so the preference component of the trap is hard to escape.

2. **Large when proximity dominates edge creation**: the network's own dynamics pull it back toward the fragmented equilibrium.

3. **Small when s_{cross} is large**: cross-block trade is profitable, so once enough cross-block edges exist, they sustain themselves.

Example 13.17 (Four Paths to Structural Change)**.**

Mechanism	***Historical example***	***Network change***	ε^*
Exogenous shock	*WWII (US female LFP; much of the wartime entry reversed by 1950, Goldin 1991)*	*Mass entry of women into* W^P	*Large, one-time*
Infrastructure	*Trans-Siberian railway*	*Physical cross-block edge*	*Medium, permanent*
Information shock	*Mobile phones (Africa)*	W^I *gains cross-block edges*	*Small per edge, cumulative*
Policy	*South Korea's export-oriented industrialization and heavy-and-chemical-industry drive*	*Deliberate creation of* W^P *edges*	*Large, sustained*

13.7 Summary

1. Networks evolve through three mechanisms: **edge creation** (logistic in proximity, surplus and information; path-dependent), **edge destruction** (surplus-driven or shock-driven), and **edge reweighting** (continuous adjustment to surplus), combined in a renormalized mean-field evolution map $\mathcal{E}$.

2. The **co-evolutionary system** has three nested timescales: market clearing is fast, preference formation is slower, and network evolution is slowest. A structural equilibrium is a fixed point of all three. It exists by Brouwer once W is kept strictly positive (there, $\boldsymbol{\pi}(W)$ is continuous).

3. **Multiple structural equilibria** can coexist when cross-block links are complementary: in the worked example, a fragmented ($\Gamma = 0.072$) and an integrated ($\Gamma = 0.643$) equilibrium with the *same* power vector $\boldsymbol{\pi}$ but different preference dispersion and allocation gaps. Integration and power are separate coordinates.

4. The **principle of relatedness** — the empirical regularity that diversification follows existing links — is what proximity-dominated edge creation predicts

(Conjecture 13.11): structural deepening in the center, lock-in in the periphery. Sunkel's thesis adds that **integration and disintegration** can occur at different scales at once: selective elite integration lowers the domestic Γ, while the global Γ may move either way.

5. **Structural change** is a transition between equilibria, requiring a "push" across the basin boundary ($\varepsilon^* = 0.307$ in the example). The push can come from shocks, infrastructure, information, or policy.

13.8 Exercises

13.1 Edge creation by hand. For a 4×4 economy with

$$W_t = \begin{pmatrix} 0.6 & 0.38 & 0.01 & 0.01 \\ 0.38 & 0.6 & 0.01 & 0.01 \\ 0.01 & 0.01 & 0.6 & 0.38 \\ 0.01 & 0.01 & 0.38 & 0.6 \end{pmatrix} \quad (\Gamma = 0.040) :$$

(a) Compute the network distance between all pairs. Which pairs are "closest"?

(b) If a new edge forms with probability $\Pr \propto 1/d(i,j)$: what is the most likely new edge? Is it within-block or cross-block?

(c) If a policy subsidizes cross-block edges (doubles the creation probability for pairs in different blocks): what is now the most likely new edge?

(d) Simulate 20 periods of evolution with and without the subsidy. Plot Γ_t in both cases.

13.2 Structural equilibria. For the stylized Bolivia two-equilibrium example (Example 13.10):

(a) Verify that both W_A^* and W_B^* are structural equilibria (check all three conditions from Definition 13.7: $\boldsymbol{\theta}^* = T(\boldsymbol{\theta}^*; W)$, $\mathbf{x}^* = h(\cdot)$, and $F(c) = c$). Show that $\Gamma(W(c)) = c$ and $\boldsymbol{\pi}(W(c)) = \mathbf{s}$ for every c.

(b) Compute the basin boundary $\hat{c}$ by solving $F(c) = c$ on (c_A, c_B). Starting from W_A^*, what is the minimum Frobenius push ε^* that moves the system into the basin of W_B^*? (Answer: $\hat{c} = 0.282$, $\varepsilon^* = 0.307$.)

(c) Lower the proximity weight in F from 12 to 8 (keeping the surplus weight at 12). How do c_A, $\hat{c}$, c_B and ε^* change?

(d) What is the welfare gain from the transition? Use all four criteria from Chapter 12.

13.3 Bolivia's gas nationalization. Model the 2006 nationalization as simultaneous edge destruction ($W^P_{\text{Bolivia, MNC}} \to 0$) and edge creation ($W^P_{\text{gas, treasury}} \uparrow$):

(a) Using a 4×4 model (MNC, Gas Company, Government, Traditional), write W^P_{before} and W^P_{after}.

(b) Compute Γ before and after. Did it increase or decrease?

(c) Compute $\boldsymbol{\pi}$ before and after. Who gained power? Who lost?

(d) Did the nationalization change W^I? If so, in which direction?

13.4 The principle of relatedness. Consider a product space with 6 products arranged in a ring: $\phi_{p,p+1} = 0.3$ (adjacent products are related), $\phi_{p,p+2} = 0.05$ (distant products weakly related), $\phi_{p,p+3} = 0$ (opposite products unrelated).

(a) A country currently produces products 1 and 2. Compute relatedness density ω_{cp} for all 6 products.

(b) Under the principle of relatedness: which product is most likely to be entered next? (Products 3 and 6 tie, with $\omega = 0.50$.)

(c) If the policy goal is to reach product 5 (on the opposite side of the ring): how many "steps" are needed? How many generations (at one step per generation)?

(d) This is the product-space distance. Connect to the escape threshold ε^* from Definition 13.16.

(e) *Data.* Repeat (a)–(b) for Bolivia using the product-space proximity matrix and Bolivia's export basket from the Atlas of Economic Complexity (`atlas.cid.harvard.edu`) for a recent year. Which five products have the highest relatedness density?

13.5 Sunkel's integration/disintegration. Model a three-node world (Center, Elite of the periphery, Mass of the periphery), with

$$W^P_{\text{before}} = \begin{pmatrix} 0.6 & 0.2 & 0.2 \\ 0.2 & 0.6 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{pmatrix}.$$

A multinational corporation creates a link between Center and Periphery-Elite: W^P_{12} rises by 0.15, taken from W^P_{11}; W^P_{23} falls by 0.05, moved to W^P_{21} (Elite resources shift from domestic to global).

(a) Write W^P_{after} and check that it is row-stochastic.

(b) Compute the domestic Γ (the Elite–Mass submatrix with rows renormalized) before and after. (Answer: $0.500 \to 0.450$.)

(c) Compute the global Γ (all three) before and after. (Answer: $0.600 \to 0.560$.)

(d) Sunkel's prediction is that global Γ rises while domestic Γ falls. Does it hold here? Find a W^P_{before} (and a compensation rule) for which it does, and explain what the prediction requires of where the rerouted weight comes from.

13.6 WWII as structural change. Model US female LFP as a 2-block economy (Men, Women):

(a) Pre-war: W^P is block-diagonal (women in domestic production, men in market production). Women's $\theta^*_F = 0.3$ (low market orientation).

(b) WWII shock: women enter W^P_{market} (large cross-block edge creation). Compute immediate effect on $\boldsymbol{\theta}$.

(c) Post-war: the edge partially reverses (some women leave market). But children of wartime workers have W^I edges to the market. Simulate the formation dynamics for 3 generations.

(d) Does θ^*_F return to 0.3? Why or why not? Connect to Fernández (2013).

13.7 Network evolution simulation. Write a Python program that simulates the co-evolutionary system for $N = 8$, starting from a random initial W_0:

(a) Implement edge creation (proximity-dominated), destruction (surplus-driven), and reweighting.

(b) Run 100 simulations with different random W_0's. Classify the final W^* as "integrated" ($\Gamma > 0.3$) or "fragmented" ($\Gamma < 0.1$).

(c) What fraction converge to each type? How does the fraction depend on p_1 (proximity parameter)?

(d) At what p_1 does the fraction of fragmented equilibria exceed 50%?

Mathematical Appendices

Appendix 13A: Network Distance and Proximity

Geodesic distance. The shortest-path distance $d(i, j; W)$ between agents i and j in the weighted network W is the minimum number of edges in any path from j to i (unweighted) or the minimum sum of edge inverses $\sum_{e \in \text{path}} 1/W_e$ (weighted).

Effective resistance distance. For undirected networks, the effective resistance between i and j is $R_{ij} = (e_i - e_j)' L^+ (e_i - e_j)$, where L is the graph Laplacian $L = D - W$ (D is the degree matrix) and L^+ is its pseudoinverse. This distance accounts for multiple paths between i and j, not just the shortest.

Proximity as inverse distance. In the product space literature, proximity between products p and p' is measured by co-export frequency: $\phi_{pp'} = \min(\Pr(M_{cp} = 1 \mid M_{cp'} = 1), \Pr(M_{cp'} = 1 \mid M_{cp} = 1))$. This is the conditional probability of exporting p given that you export p' (or vice versa). High $\phi_{pp'}$ means the products require similar capabilities and tend to be co-produced.

Appendix 13B: Stability of Structural Equilibria

The Jacobian of the co-evolutionary system. At a structural equilibrium $(W^*, \boldsymbol{\theta}^*)$, the linearized system has a block Jacobian:

$$\mathcal{J} = \begin{pmatrix} \partial T/\partial \boldsymbol{\theta} & \partial T/\partial W \\ \partial \mathcal{E}/\partial \boldsymbol{\theta} & \partial \mathcal{E}/\partial W \end{pmatrix}_{(W^*, \boldsymbol{\theta}^*)}. \tag{13.13}$$

Stability requires all eigenvalues of $\mathcal{J}$ to have modulus less than 1. The upper-left block $\partial T/\partial \boldsymbol{\theta}$ is the formation Jacobian $J = \rho I + \mu \phi W^I CP$ of Proposition 10.2. The lower-right block $\partial \mathcal{E}/\partial W$ governs network self-reinforcement. The off-diagonal blocks capture the cross-effects: how $\boldsymbol{\theta}$ changes affect W (through the surplus, which depends on preferences), and how W changes affect $\boldsymbol{\theta}$ (through the formation signal, which depends on the network).

Two-timescale approximation. When $\eta \ll 1 - \rho$ (network evolution is much slower than preference formation), the system can be analyzed in two stages: (1) for each fixed W, find the formation steady state $\boldsymbol{\theta}^*(W)$, (2) study the dynamics of W under the assumption that $\boldsymbol{\theta}$ instantly adjusts to $\boldsymbol{\theta}^*(W)$. The reduced dynamics are $W_{t+1} = \mathcal{E}(W_t; \mathbf{x}^*(\boldsymbol{\theta}^*(W_t), W_t))$, a system in W alone. Example 13.10 is of this form, further restricted to the one-parameter family $W(c)$.

Transitions are products of matrices. Along a transition W_t changes every period, and the per-period gap $\Gamma(W_t)$ does not control the product $W_t \cdots W_1$ that governs how far shocks propagate: two matrices with $\Gamma = 0.850$ and 0.776 can have a product with $\Gamma = 0.042$. Use the Dobrushin coefficient τ_1, which is submultiplicative, for time-varying networks (Chapter 23).

Structuralist Case Study: Sunkel's Integration/Disintegration Thesis

Osvaldo Sunkel's 1971 essay "Capitalismo transnacional y desintegración nacional en América Latina" (Sunkel, 1971, *Estudios Internacionales*, 4(16), pp. 3–61) advanced a thesis that was ahead of its time by forty years and maps directly onto the network evolution theory of this chapter: **the same process that integrates the world economy disintegrates the national economies of the periphery**.

Sunkel's argument has three parts.

Part 1: Transnational integration creates cross-border elite linkages. Multinational corporations, international financial institutions, and globally connected professional classes form dense networks that cross national borders. In the notation of this chapter: the edge creation process (Definition 13.1) generates new edges between elite agents in different countries. The global W^P becomes denser among elite nodes. The proximity mechanism (Conjecture 13.11) reinforces these links: once an elite network exists, new members are drawn from agents who are already "close" to existing members — shared education (Harvard, LSE), shared language (English), shared cultural consumption (the same media, the same conferences). The global elite converges to a single information network $W^I_{\text{global elite}}$ with high N_{eff} and a shared consensus.

Part 2: National disintegration destroys domestic cross-block linkages. Simultaneously, the elite's domestic connections weaken. The gas executive in La Paz trades with Houston, not with the quinoa farmer in Oruro. His information network connects him to the *Wall Street Journal*, not to a community radio station. His children attend international schools, not public schools. Each of these shifts is an edge in W^P, W^U, or W^I that moves from domestic-cross-block to global-elite. The domestic matrix loses cross-block entries. The domestic Γ decreases.

This is Remark 13.13, mechanism 3: selective integration. The agents who form cross-border links are precisely the agents who previously served as bridges between domestic blocks. When they exit the domestic network (or reduce their domestic engagement), the domestic blocks lose their only connections. The blocks drift apart.

Part 3: Many processes, one process. Sunkel and Paz (1970) had argued that

development and underdevelopment are two faces of a single historical process. The 1971 essay identified, in our terms, what that process is: the rewiring of W that simultaneously integrates elites across borders and disintegrates national economies within borders. Every edge created in W_{global} between an elite node and the international system is, implicitly, an edge weakened in W_{domestic} between that elite node and the non-elite population. The zero-sum character is not about resources (growth can be positive-sum) but about *attention, links, and structural position*: an agent has a finite capacity for connections, and global connections displace domestic ones.

Sunkel's argument (1971)	**Formal object**	**This chapter**
Transnational integration	Cross-border edge creation	Def. 13.1
National disintegration	Domestic cross-block edge destruction	Def. 13.2
Simultaneous process	$\Gamma_{\text{domestic}} \downarrow$; Γ_{global} may rise	Rem. 13.13(3), Ex. **13.5**
Selective integration	Elite bridges exit domestic network	Sec. 13.5
Many processes, one process	One matrix rewiring, multiple observables	Key Insight
Path dependence	Proximity-dominated edge creation	Conj. 13.11

Sunkel wrote in 1971, before "globalization" entered common usage. The multinational corporations he described were oil companies and mineral extractors; the "transnational elite" was the alliance of foreign capital and domestic technocrats. Fifty years later, the MNCs are tech platforms, the transnational elite is the global professional class, and the disintegration is called "populist backlash" rather than "dependency." The matrix is different. The mechanism is the same.

Further Reading

Network evolution: Jackson and Rogers (2007), "Meeting Strangers and Friends of Friends." Barabási and Albert (1999), "Emergence of Scaling in Random Networks." For endogenous network formation with strategic agents: Bala and Goyal (2000), "A Noncooperative Model of Network Formation."

The principle of relatedness: Hidalgo et al. (2007), "The Product Space Conditions the Development of Nations." Hidalgo (2021), "Economic Complexity Theory and Applications." Hausmann and Klinger (2007), "The Structure of the Product Space and the Evolution of Comparative Advantage," CID Working Paper.

Path dependence in economic geography: Martin and Sunley (2006), "Path Dependence and Regional Economic Evolution." Boschma and Frenken (2011), "The Emerging Empirics of Evolutionary Economic Geography."

The structuralist tradition on structural change: Sunkel (1971), "Capitalismo transnacional y desintegración nacional en América Latina." Sunkel (1993), *Development from Within: Toward a Neostructuralist Approach for Latin America*, Lynne Rienner. Ocampo (2009), "Latin America and the Global Financial Crisis." On fiscal and other linkages from staples: Hirschman (1977), "A Generalized Linkage Approach to Development, with Special Reference to Staples." On the reversal of wartime female employment: Goldin (1991), "The Role of World War II in the Rise of Women's Employment."

Chapter 14

Coalitions and Class Alliances

Who Allies with Whom to Change the Matrix

> *"Não basta vincular de forma abstrata os modos de produção e as formas de exploração às fases do desenvolvimento capitalista. É necessário levar em conta as situações em que se dão as alianças entre grupos e classes que atuam no sentido de manter uma dada estrutura de domínio ou de promover a sua transformação."*
>
> ["It is not enough to link in abstract fashion the modes of production and forms of exploitation to the phases of capitalist development. It is necessary to take into account the situations in which the alliances between groups and classes act to maintain a given structure of domination or to promote its transformation."]
>
> — Fernando Henrique Cardoso and Enzo Faletto, *Dependência e Desenvolvimento na América Latina* (1969)

Chapter 13 showed how the weight matrix W evolves through edge creation, destruction, and reweighting. But it treated the evolution as a stochastic process — edges form or break according to surplus and proximity, as if the network rewires itself.

In practice, structural change is not automatic. It is *political*. Agents form coalitions to change W in their favor — to strengthen the edges that increase their power and weaken the edges that benefit their rivals. Trade policy, labor regulation, land reform, media ownership, education access, border control — all of these are instruments through which coalitions reshape the weight matrix.

This chapter introduces coalitions into the structural model. The key question: which coalitions form, and what do they do to W? The answer: a coalition $S \subseteq$

$\{1, \ldots, N\}$ forms if its members collectively gain from the structural change they can enact. The gain is measured by $\Delta\boldsymbol{\pi}$: the change in the Perron eigenvector. A coalition is viable if every member gains power ($\Delta\pi_i > 0$ for all $i \in S$) and the coalition can implement the required ΔW.

14.1 Coalitions as Network Designers

14.1.1 The Coalition Problem

Definition 14.1 (Coalition)**.** *A **coalition** is a subset $S \subseteq \{1, \ldots, N\}$ of agents who jointly choose a structural change ΔW to maximize their collective payoff (Definition 14.3):*

$$\max_{\Delta W \in \mathcal{F}(S)} \sum_{i \in S} v_i(S; \Delta W), \tag{14.1}$$

*where $\mathcal{F}(S)$ is the **feasibility set**: the set of structural changes the coalition can implement given its current power. Maximizing collective* power $\sum_{i \in S} \pi_i$ *is the special case in which ΔW does not change total surplus; for the grand coalition that objective is identically* 1 *and selects nothing.*

The feasibility set depends on what the coalition controls. An agent who controls a factory can rewire her production edges (ΔW^P). An agent who controls a media outlet can rewire information edges (ΔW^I). An agent who controls the state can use regulation, taxation, and subsidies to rewire multiple layers of W simultaneously. The more powerful the coalition, the larger $\mathcal{F}(S)$.

Definition 14.2 (Power-Bounded Feasibility)**.** *We assume that the feasibility set of coalition S is bounded by the coalition's current power:*

$$\mathcal{F}(S) = \left\{ \Delta W : \|\Delta W\|_F \leq \phi\left(\sum_{i \in S} \pi_i(W) \right) \right\}, \tag{14.2}$$

where $\phi(\cdot)$ is an increasing function: coalitions with more power can implement larger structural changes. The rich get richer: the powerful can change the structure more, which makes them more powerful, which expands their feasibility set.

Intuition: The political Coase theorem fails

Acemoglu (2003) asked: why don't societies always adopt efficient institutions? If the efficient structure benefits everyone (or at least generates a large enough surplus to compensate losers), why doesn't a grand coalition implement it?

The answer in this framework: the feasibility set depends on *current* power. The coalition that would benefit from the efficient W (the powerless majority) lacks the power to implement ΔW. The coalition that has the power (the powerful minority) has no incentive to implement it — the efficient W would reduce their power. The political Coase theorem fails because the enforcement mechanism is the current W, and W is not neutral.

14.1.2 The Payoff from a Coalition

Definition 14.3 (Coalition Payoff)**.** *The payoff to agent i from joining coalition S that implements ΔW_S is:*

$$v_i(S) = x_i^*(W + \Delta W_S, \boldsymbol{\theta}^*_{W+\Delta W_S}) - x_i^*(W, \boldsymbol{\theta}^*_W). \tag{14.3}$$

This includes both the enforcement effect (change in allocation at fixed preferences) and the formation effect (long-run change in allocation from preference adjustment):

$$v_i(S) = \underbrace{\Delta x_i^{enf}}_{enforcement} + \underbrace{\Delta x_i^{form}}_{formation} = E_i(\Delta W_S) + F_i(\Delta W_S). \tag{14.4}$$

Proposition 14.4 (Surplus and Power Components of the Payoff)**.** *Let W and $W + \Delta W_S$ be irreducible and row-stochastic, so that $\boldsymbol{\pi}$ is defined at both. Under the allocation rule $x_i = \Omega(W)/N + (\pi_i(W) - 1/N)\,\Phi$ (Proposition 7.3), with total output $\Omega(W)$ allowed to depend on W and the bargaining surplus Φ held fixed, the enforcement payoff (preferences fixed at $\boldsymbol{\theta}^*_W$) is exactly*

$$E_i(\Delta W_S) = \underbrace{\frac{\Delta\Omega}{N}}_{surplus} + \underbrace{\Phi\,\Delta\pi_i(W, \Delta W_S)}_{power}, \tag{14.5}$$

where $\Delta\Omega = \Omega(W + \Delta W_S) - \Omega(W)$ and $\Delta\pi_i = \pi_i(W + \Delta W_S) - \pi_i(W)$. If, in addition, we assume *that the formation effect is proportional to the enforcement effect, with a ratio Ψ common to all agents — the formation-to-enforcement ratio $\Psi_i(\Delta W_S, T)$ of the*

coalition's own perturbation ΔW_S (Definition 9.13) —

$$F_i(\Delta W_S) \approx \Psi\, E_i(\Delta W_S), \tag{14.6}$$

then the total payoff is

$$v_i(S) \approx (1+\Psi)\left(\frac{\Delta\Omega}{N} + \Phi\,\Delta\pi_i\right). \tag{14.7}$$

Proof. Subtract the allocation rule at W from the rule at $W + \Delta W_S$, at fixed preferences: $E_i = [\Omega(W+\Delta W_S) - \Omega(W)]/N + \Phi\,[\pi_i(W+\Delta W_S) - \pi_i(W)]$, since the $-\Phi/N$ terms cancel. Then (14.7) follows from $v_i = E_i + F_i$ in (14.4) and the assumed (14.6); it is exact if (14.6) holds with equality. □

Summing over all agents, $\sum_i v_i \approx (1+\Psi)\Delta\Omega$, because $\sum_i \Delta\pi_i = 0$. The power component is *exactly constant-sum*: one agent's gain in π is another's loss. Only the surplus component $\Delta\Omega$ can make everyone better off.

Key Structural Insight

The distributional part of the coalition payoff is determined by $\Delta\boldsymbol{\pi}$: the change in the power vector. Every political conflict — every trade war, labor dispute, media battle, land reform, nationalization — has a constant-sum fight over $\Delta\boldsymbol{\pi}$ inside it, alongside a possibly positive-sum change in total surplus $\Delta\Omega$. The coalition that wins is the one that can implement a ΔW that shifts $\boldsymbol{\pi}$ in its favor without destroying too much surplus.

The formation amplifier $(1+\Psi)$ is crucial: when $\Psi > 0$, a structural change has long-run effects beyond the immediate power shift, because the new W eventually changes preferences. A coalition that controls W^I (media, education) benefits from the formation amplifier: not only does it gain power immediately, but it shapes the preferences of the next generation to *accept* the new power structure. This is the deepest form of structural power: the power to make your power seem natural.

14.2 Which Coalitions Form?

14.2.1 Stability Concepts

Definition 14.5 (Stable Coalition Structure)**.** *A partition of agents into coalitions $\mathcal{S} = \{S_1, S_2, \ldots, S_k\}$ is **stable** if:*

1. ***Internal stability***: *No member of any coalition S_m would prefer to defect (to leave S_m and either join another coalition or stand alone). Writing $v_i(R)$ for i's payoff when coalition R implements ΔW_R (whether or not $i \in R$): for all $i \in S_m$, $v_i(S_m) \geq v_i(S_m \setminus \{i\})$, $v_i(S_m) \geq v_i(\{i\})$, and $v_i(S_m) \geq v_i(S' \cup \{i\})$ for every rival coalition S' that would admit i.*

2. ***External stability***: *No coalition S_m would accept a new member from outside. For all $j \notin S_m$: if j wants to join S_m, then at least one existing member of S_m would be made worse off.*

3. ***Coalition rationality***: *Each coalition implements the ΔW_S that maximizes its collective payoff given the other coalitions' actions.*

Remark 14.6 (Existence is not guaranteed). A stable coalition structure in the sense of Definition 14.5 need not exist. Compact feasibility sets and continuous payoffs do not suffice: coalition-proof Nash equilibria can fail to exist (Bernheim et al., 1987), and hedonic coalition-formation games can have no stable partition. Sufficient conditions exist for special classes, e.g. the "top-coalition" property of Banerjee et al. (2001) or the conditions in Bogomolnaia and Jackson (2002). Whether the structural power game satisfies any of them depends on $\mathcal{F}$ and $\Omega(\cdot)$, and is open.

Conjecture 14.7 (Coalition Formation)**.** *In the structural model with N agents and the payoffs of Proposition 14.4:*

1. *The grand coalition $S = \{1, \ldots, N\}$ chooses the surplus-maximizing structure $W^\sharp = \arg\max_W \Omega(W)$ within $\mathcal{F}(\{1, \ldots, N\})$ (it is indifferent to $\Delta\boldsymbol{\pi}$, which sums to zero). It is stable only if the surplus gain $(1 + \Psi)\Delta\Omega$ can be shared so as to compensate every agent whose power falls. This requires side payments — which may not be credible if the current W is the enforcement mechanism (Acemoglu's point).*

2. *Typically, stable partitions involve 2–3 coalitions that balance each other's power.*

Discussion. (1) The grand coalition's objective is $\sum_i v_i = (1 + \Psi)\Delta\Omega$, so its choice depends only on surplus. An agent with $\Phi\Delta\pi_i < -\Delta\Omega/N$ loses and must be compensated by a transfer. The transfer must be credible — but credible transfers require an enforcement mechanism, and the enforcement mechanism is the current W, which the grand coalition proposes to change. (2) With 2–3 coalitions, each has enough power to implement local ΔW but not enough to achieve the grand coalition's global restructuring. Neither claim is proved here.

14.2.2 The Class Alliance as a Coalition

Definition 14.8 (Class Alliance)**.** *A **class alliance** is a coalition formed between agents from different structural positions (blocks of W) who share a common interest in a specific ΔW. The alliance is stable if all members gain from the structural change, despite occupying different positions in the current structure.*

Example 14.9 (Four Class Alliances in Latin America [ILLUSTRATIVE])**.** *Four agents: Foreign Capital (F), Domestic Bourgeoisie (B), Middle Class (M), Workers (W). Current power: $\boldsymbol{\pi} = (0.35, 0.30, 0.20, 0.15)$. The $\Delta\boldsymbol{\pi}$ vectors below are illustrative; each sums to zero, as it must (`code/ch14.py`). The four-alliance typology is this book's stylization, not Cardoso and Faletto's own classification (see the case study).*

Alliance 1: Oligarchic-Export *($\{F, B\}$)**.** Foreign capital and the export fraction of the domestic bourgeoisie (landowners, mine owners) export raw materials. They strengthen W^P_{FB} (supply chain links) and W^I_{FB} (shared business culture). The structural change reinforces the export enclave:*

$$\Delta\boldsymbol{\pi} \approx (+0.05, +0.03, -0.04, -0.04). \tag{14.8}$$

F and B gain; M and W lose. The enclave deepens.

Alliance 2: National-Populist *($\{B, M, W\}$)**.** Domestic bourgeoisie, middle class, and workers unite against foreign capital. They implement ISI: tariffs weaken $W^P_{F,domestic}$ (fewer imported inputs) and strengthen $W^P_{domestic}$ (local supply chains). New domestic W^I channels (national media, public education):*

$$\Delta\boldsymbol{\pi} \approx (-0.10, +0.02, +0.05, +0.03). \tag{14.9}$$

F loses heavily; all domestic classes gain. Perón's Argentina, Vargas's Brazil, the MNR in Bolivia.

Alliance 3: Techno-Bureaucratic *($\{F, B, M\}$ **excluding** W)**.** Post-coup modernization. Foreign capital, domestic bourgeoisie, and the technical middle class ally to implement "efficient" structural adjustment. They strengthen W^P_{FB} and W^P_{BM} while weakening labor protections (W^U_{WM} and W^I_{WM} edges severed):*

$$\Delta\boldsymbol{\pi} \approx (+0.03, +0.04, +0.02, -0.09). \tag{14.10}$$

Workers bear all the cost. Brazil post-1964, Chile post-1973, Bolivia 1985.

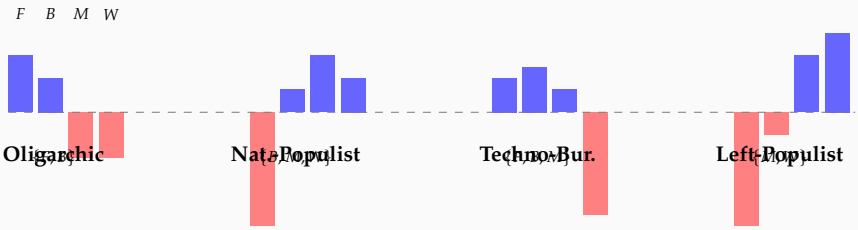
Alliance 4: Left-Populist *($\{M, W\}$)**.** Middle class and workers unite against foreign capital. The domestic bourgeoisie loses power under this program ($\Delta\pi_B = -0.02$), so by internal stability (Proposition 14.10) it joins only if compensated through the surplus term*

$\Delta\Omega/N$ *or a side payment. They nationalize key sectors* (ΔW^P*: gas/mining edges redirect from F to state), expand public* W^I *(state media, education reform), and strengthen labor linkages* (W^U_{MW}*):*

$$\Delta\boldsymbol{\pi} \approx (-0.10, -0.02, +0.05, +0.07). \tag{14.11}$$

F loses most; W gains most. Evo Morales's Bolivia, Chávez's Venezuela.

Visualization: Four alliances, four $\Delta\pi$ vectors



Four bars per alliance: $\Delta\pi_F$, $\Delta\pi_B$, $\Delta\pi_M$, $\Delta\pi_W$. Blue bars (above zero): the agent gains power. Red bars (below): she loses. Each alliance is defined by who gains and who pays. Within each group the bars sum to zero: power is constant-sum, so what one class gains another loses.

14.3 The Stability of Alliances

14.3.1 When Alliances Hold

Proposition 14.10 (Alliance Stability Conditions)**.** *Suppose* $\Delta\Omega = 0$ *for the policies considered (pure redistribution of power), that* $\Phi > 0$ *and* $\Psi > -1$ *is common to these policies, so that* $v_i = (1+\Psi)\Phi\,\Delta\pi_i$ *with a positive factor (Proposition 14.4), and normalize the stand-alone payoff to zero,* $v_i(\{i\}) = 0$*. Alliance S is internally stable (Definition 14.5) if and only if*

$$\Delta\pi_i(W, \Delta W_S) \geq \max\left\{0,\ \Delta\pi_i(W, \Delta W_{S\setminus\{i\}})\right\} \quad \textit{for all } i \in S, \tag{14.12}$$

AND no member can achieve a higher $\Delta\pi_i$ *by defecting to a rival alliance* S'*:*

$$\Delta\pi_i(W, \Delta W_S) \geq \Delta\pi_i(W, \Delta W_{S'\cup\{i\}}) \quad \textit{for all feasible } S'. \tag{14.13}$$

Proof. With $v_i(R) = (1+\Psi)\Phi\,\Delta\pi_i(W, \Delta W_R)$ for every coalition R, and $(1+\Psi)\Phi > 0$, each inequality $v_i(S) \geq v_i(R)$ in Definition 14.5 is equivalent to $\Delta\pi_i(W, \Delta W_S) \geq \Delta\pi_i(W, \Delta W_R)$. Taking $R = \{i\}$ (payoff 0) and $R = S \setminus \{i\}$ gives (14.12); taking $R = S' \cup \{i\}$ gives (14.13). □

Example 14.11 (Why the National-Populist Alliance Breaks)**.** *Consider Alliance 2 ($\{B, M, W\}$). Initially stable: all three domestic classes gain from ISI. But as ISI succeeds, the domestic bourgeoisie grows — her π_B increases. At some threshold, she can earn a higher $\Delta\pi_B$ by switching to Alliance 3 ($\{F, B, M\}$): partnering with foreign capital to "open" the economy.*

The switch occurs when:

$$\Delta\pi_B(\Delta W_{\{F,B,M\}}) > \Delta\pi_B(\Delta W_{\{B,M,W\}}). \tag{14.14}$$

In words: when the domestic bourgeoisie is strong enough that partnering with foreign capital (and cutting out workers) yields a higher power gain than continuing the national-populist alliance. This is the transition from ISI populism to techno-bureaucratic modernization — the story of Brazil 1964, Argentina 1976, Chile 1973.

Phase	*Alliance*	π_B	$\Delta\pi_B$ *from the alliance*
Early ISI	$\{B, M, W\}$	*0.20*	$+0.05$
Mature ISI	$\{B, M, W\}$	*0.30*	$+0.02$ *(diminishing)*
Post-ISI crisis	*Switch to* $\{F, B, M\}$	*0.30*	$+0.04$

The mature-ISI and post-ISI rows use the $\Delta\pi_B$ of Alliances 2 and 3 in Example 14.9, where $\pi_B = 0.30$; the early-ISI row is illustrative. The alliance breaks because its own success changes the power structure, which changes the payoffs, which changes the optimal alliance. The matrix W evolved (Chapter 13), and the evolution shifted the coalition incentives.

14.3.2 When Alliances Cycle

Conjecture 14.12 (Alliance Cycling)**.** *In the co-evolutionary system with coalition formation, the following cycle can occur:*

1. *Phase 1: National-populist alliance ($\{B, M, W\}$) implements ISI. π_B rises. Domestic Γ increases.*

2. *Phase 2: B defects to techno-bureaucratic alliance ($\{F, B, M\}$). Liberalization. π_W falls. W^I becomes more center-dominated.*

3. *Phase 3: Workers and middle class form left-populist alliance ($\{M, W\}$, sometimes $+B$). Nationalization. π_F falls. Domestic Γ increases again.*

4. *Phase 4: Resource rents create corruption/inefficiency. B defects again to $\{F, B\}$ or $\{F, B, M\}$. Return to Phase 2.*

*The cycle is the **Latin American pendulum**: ISI → liberalization → populism → liberalization → ⋯*

Why it is plausible. Each phase changes W, which changes $\boldsymbol{\pi}$, which changes the coalition payoffs $\Delta\pi_i(S)$, which destabilizes the current alliance and triggers a switch. The cycle is sustained because no single structural equilibrium satisfies all agents simultaneously: the efficient W requires the grand coalition, but the grand coalition requires credible commitment, which requires the enforcement mechanism that the coalition proposes to change. A proof would need an explicit $\mathcal{F}$, $\Omega(\cdot)$ and $\boldsymbol{\pi}(\cdot)$ for which the switching sequence can be computed (Exercise **14.3**). When the power component is constant-sum, an empty core (Proposition 14.14) makes some form of instability unavoidable, but not this particular sequence.

Prebisch Connection: The Latin American pendulum

The cycling conjecture formalizes what economic historians of Latin America have observed for a century: the oscillation between inward-oriented and outward-oriented development strategies, between populist inclusion and technocratic exclusion, between nationalization and liberalization.

The standard explanation is "policy failure": populism leads to fiscal crisis, which triggers liberalization, which leads to inequality, which triggers populism. This explanation blames the cycle on bad policy.

The structural explanation is different: the cycle is endogenous to the coevolutionary system. Each alliance changes W in a way that eventually destabilizes itself. ISI strengthens the bourgeoisie, who then defects. Liberalization weakens workers, who then rebel. Nationalization creates rents, which corrupt the coalition. The cycle is not a sequence of mistakes. It is the dynamics of a system that has no stable grand coalition because the enforcement mechanism is the object being contested.

Bolivia's cycle: MNR revolution (1952, Alliance 2) $\rightarrow$ military rule (from 1964; under Barrientos it rested on the Military–Peasant Pact, so it fits Alliance 3 only loosely) $\rightarrow$ democratic opening (1982) and structural adjustment (1985, Alliance 3 variant) $\rightarrow$ resource nationalism (2006, Alliance 4) $\rightarrow$ political crisis (2019, unstable). Each transition is a coalition switch driven by $\Delta\pi$. One part of Conjecture 14.12 fails in the Bolivian data: Phase 3 did not raise domestic Γ. On INE's input-output tables (12 sectors, domestic flows, input shares), $\Gamma(W^P)$ was 0.518 in 2006 and 0.484 in 2013 (Instituto Nacional de Estadística (INE), Bolivia, 2026); author's calculations (`data/compute/compute_bolivia_io.py`). What the nationalization changed was the distribution of rents, through fiscal channels, not the integration of the production network.

14.4 The Structural Power Game

14.4.1 Formalization

Definition 14.13 (The Structural Power Game)**.** *The **structural power game** $\mathcal{G} = (N, \{A_i\}, \{v_i\}, W)$ is defined by:*

- *N players (agents or classes).*
- *Action set A_i: the set of coalitions agent i can join.*

- *Payoff* $v_i(S) = (1+\Psi)\big(\Delta\Omega/N + \Phi\,\Delta\pi_i(W, \Delta W_S)\big)$*: the allocation change from the coalition's structural policy (Proposition 14.4).*

- *The current weight matrix* W*, which determines both the payoffs and the feasibility sets.*

Define the transferable-utility (TU) version of the game by

$$v(S) = \max_{\Delta W \in \mathcal{F}(S)} \sum_{i \in S} v_i(S; \Delta W),$$

the most that coalition S can secure for its members on its own (with $v(S) \geq 0$, since $\Delta W = 0$ is feasible).

Proposition 14.14 (Properties of the Structural Power Game)**.** *1.* ***Constant-sum power.*** $\sum_i v_i = (1+\Psi)\Delta\Omega$*. If* $\Psi > -1$*, the game is constant-sum if and only if no feasible* ΔW *changes total surplus; it is positive-sum only through* $\Delta\Omega$*. Any* ΔW *that increases* π_i *for some* i *decreases* π_j *for some* j*.*

2. ***Empty core without surplus.*** *If* $\Delta\Omega \equiv 0$*, the core is empty as soon as some coalition can gain, i.e.* $v(S) > 0$ *for some* S*.*

3. ***Path dependence.*** *The current* W *determines the feasibility sets, which determine which coalitions can form, which determines the next* W*.*

Proof. (1) Sum the payoff of Definition 14.13 over i and use $\sum_i \Delta\pi_i = 0$ (both $\boldsymbol{\pi}(W)$ and $\boldsymbol{\pi}(W + \Delta W)$ sum to one). With $1 + \Psi > 0$ the sum vanishes for every feasible ΔW exactly when $\Delta\Omega = 0$ for every feasible ΔW. If $\Delta\pi_i > 0$ then $\sum_{j \neq i} \Delta\pi_j < 0$, so some $\Delta\pi_j < 0$. (2) With $\Delta\Omega \equiv 0$, $v(N) = \max \sum_i v_i = 0$ (the argument is unchanged if the maximum is replaced by a supremum). A core allocation needs $x(N) = v(N) = 0$, $x(S) \geq v(S) > 0$ and $x(N \setminus S) \geq v(N \setminus S) \geq 0$, so $x(N) > 0$: a contradiction. (3) is immediate from Definition 14.2. □

Concentrated power is not what empties the core: any power-shifting opportunity does, once surplus is held fixed (in the toy version of `code/ch14.py`, six coalitions have $v(S) > 0$). With $\Delta\Omega > 0$ the core can be nonempty; by the Bondareva–Shapley theorem it is nonempty if and only if the TU game is balanced. Superadditivity, $v(S \cup T) \geq v(S) + v(T)$ for disjoint S, T, fails when $\mathcal{F}(S \cup T)$ is smaller than the combined reach of S and T acting separately, which Definition 14.2 does not rule out.

14.4.2 Equilibrium Alliance Structures

Conjecture 14.15 (Alliance Regions [ILLUSTRATIVE])**.** *In the structural power game with $N = 4$ (Foreign Capital, Domestic Bourgeoisie, Middle Class, Workers), the alliance that forms is determined by (π_F, π_W) through four exhaustive regions (thresholds illustrative):*

1. *$\pi_F \geq 0.40$ (foreign capital dominant): $\{F, B\}$ (oligarchic-export). Workers and middle class cannot form a winning counter-coalition because $\mathcal{F}(\{M, W\})$ is too small.*

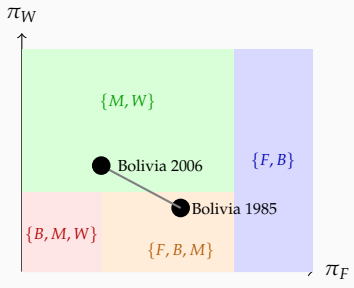
2. *$\pi_F < 0.40$ and $\pi_W \geq 0.15$ (organized labor): $\{M, W\}$ (left-populist), possibly joined by B when compensated.*

3. *$0.15 \leq \pi_F < 0.40$ and $\pi_W < 0.15$ (workers weak, foreign capital substantial): $\{F, B, M\}$ (techno-bureaucratic).*

4. *$\pi_F < 0.15$ and $\pi_W < 0.15$ (domestic bourgeoisie and middle class dominant): $\{B, M, W\}$ (national-populist).*

The chapter's baseline $\boldsymbol{\pi} = (0.35, 0.30, 0.20, 0.15)$ lies in region 2. The selection is not unique in general: near the boundaries more than one alliance can be stable, and which one forms is a matter of political contingency.

Visualization: Alliance stability regions in the (π_F, π_W) plane



Each region corresponds to the alliance predicted by Conjecture 14.15, as a function of (π_F, π_W) (with π_B and π_M filling the remainder). Thresholds and Bolivian coordinates are illustrative. Bolivia moved from the techno-bureaucratic region (1985: π_F high, π_W low) to the left-populist region (2006: π_F lower, π_W higher). The arrow is one step of the Latin American pendulum.

14.5 Structural Policy as Coalition Output

Every structural policy is a specific ΔW chosen by a specific coalition. The table of policies from Chapter 10 (transfers, structural policy, information policy) can now

be mapped to coalitions:

Definition 14.16 (Policy as Coalition Output: a Typology)**.**

Policy	ΔW	***Typical coalition***	$\Delta\pi$ ***pattern***
Trade liberalization	$W^P_{domestic} \downarrow$, $W^P_{global} \uparrow$	$\{F, B\}$ *or* $\{F, B, M\}$	$\pi_F \uparrow$, $\pi_W \downarrow$
ISI / industrial policy	$W^P_{domestic} \uparrow$, $W^P_{global} \downarrow$	$\{B, M, W\}$	$\pi_B \uparrow$, $\pi_F \downarrow$
Nationalization	$W^P_F \to W^P_{state}$	$\{M, W\}$ *or* $\{B, M, W\}$	$\pi_F \downarrow$, $\pi_M \uparrow$
Land reform	$W^P_{landlord} \to W^P_{farmer}$	$\{M, W\}$	$\pi_{landlord} \downarrow$
Media regulation	ΔW^I *(de-concentrate)*	$\{M, W\}$ *or* $\{B, M, W\}$	$N_{eff} \uparrow$
Education expansion	ΔW^I *(cross-block edges)*	*Variable*	Ψ*-amplified*
Cash transfers	$\Delta\boldsymbol{\omega}$ *(not* ΔW*)*	*Any*	$\Delta\boldsymbol{\pi} = 0$

Cash transfers (last row) do not change W within the period and therefore do not change $\boldsymbol{\pi}$ on impact. This is the Transfer Trap from Chapter 2, now with a political explanation: every coalition can agree on transfers (they don't threaten anyone's structural power), but no coalition will implement the ΔW that reduces its own $\boldsymbol{\pi}$. Over time the statement weakens: in the co-evolutionary system of Chapter 13, $W_{t+1} = \mathcal{E}(W_t; \mathbf{x}^*_t)$ depends on the allocation, so transfers that change $\mathbf{x}^*$ can change future links and hence future $\boldsymbol{\pi}$.

14.6 Summary

1. **Coalitions** are subsets of agents who jointly choose ΔW to maximize their collective payoff. The feasibility set $\mathcal{F}(S)$ depends on current power: the powerful can change more.

2. The **coalition payoff** is $v_i(S) \approx (1 + \Psi)(\Delta\Omega/N + \Phi\,\Delta\pi_i)$: a surplus component shared by all and a power component that is exactly constant-sum, both amplified by the formation amplifier $(1 + \Psi)$, where Ψ is the ratio of Definition 9.13 for the coalition's perturbation. Without the surplus term the game is constant-sum, the grand coalition is indifferent among all ΔW, and the core is empty whenever any coalition can gain power.

3. **Four Latin American alliances** (a stylization): $\{F, B\}$ (oligarchic export), $\{B, M, W\}$ (national populist), $\{F, B, M\}$ (technobureaucratic) and $\{M, W\}$ (left populist). Each implements a different ΔW with a different $\Delta\boldsymbol{\pi}$; which one forms is conjectured to depend on (π_F, π_W).

4. **Alliance cycling** (conjectured): ISI strengthens B, who defects $\to$ liberalization weakens W, who rebels $\to$ populism $\to$ crisis $\to$ repeat. Each alliance's success changes the power structure that destabilizes it.

5. The **political Coase theorem fails** because the enforcement mechanism is the current W. The grand coalition requires credible commitment, which requires an institution that the coalition proposes to change. Cash transfers are the policy every coalition can agree on, because on impact they change $\boldsymbol{\omega}$ but not $\boldsymbol{\pi}$.

14.7 Exercises

14.1 Coalition payoffs. For $N = 4$ with $\boldsymbol{\pi} = (0.35, 0.30, 0.20, 0.15)$, total output $\Omega = 4$, bargaining surplus $\Phi = 2$, and $\kappa = 0.5$ for welfare evaluation:

(a) Compute the current bargaining allocation $\mathbf{x}^*$ and check $\kappa x_i < 1$. (Answer: $\mathbf{x}^* = (1.2, 1.1, 0.9, 0.8)$.)

(b) Coalition $\{3, 4\}$ (workers and middle class) implements ΔW that shifts $\boldsymbol{\pi}$ to $(0.25, 0.25, 0.25, 0.25)$ with $\Delta\Omega = 0$ and $\Psi = 0$. Compute $v_3(\{3, 4\})$ and $v_4(\{3, 4\})$. Are both positive? (Answer: 0.10 and 0.20.)

(c) Can $\{3, 4\}$ implement this ΔW? Check feasibility: $\|\Delta W\|_F \leq \phi(\pi_3 + \pi_4) = \phi(0.35)$.

(d) Agent 2 considers defecting from $\{1, 2\}$ to $\{2, 3, 4\}$. Using the $\Delta\boldsymbol{\pi}$ of Alliances 1 and 2 in Example 14.9, compute $\Delta\pi_2$ under both coalitions. Which does she prefer? (Answer: $+0.03$ vs. $+0.02$; she stays.)

14.2 Alliance stability. For Example 14.9: verify that in each alliance all members gain, and use (14.13), with the other three alliances as the rivals, to check which alliances are internally stable; identify the conditions under which each breaks. Why would B not join Alliance 4 without compensation?

14.3 The Latin American pendulum. Specify $\mathcal{F}(S)$, $\Omega(W)$ and a 4×4 initial W, and simulate the cycling conjecture for 8 periods. Start with Alliance 2

($\{B, M, W\}$). At each period: (a) compute the coalition's ΔW and new $\boldsymbol{\pi}$, (b) check whether any member wants to defect, (c) if so, form the new alliance. Plot $\boldsymbol{\pi}_t$ over time. Does it cycle?

14.4 Bolivia 1952–2019. Map Bolivia's political history to the four-alliance framework:

(a) MNR Revolution (1952): which alliance? What ΔW?

(b) Military governments (1964–1982): alliance switch to what?

(c) Structural adjustment (1985): which alliance implements the NEP?

(d) Evo Morales (2006): alliance switch to what? What ΔW^P (nationalization)?

(e) Political crisis (2019): alliance breakdown. Which conditions from Proposition 14.10 failed?

14.5 Why cash transfers are universal. Show formally that, holding W fixed within the period, $\Delta\boldsymbol{\pi}(\Delta\boldsymbol{\omega}) = 0$: transfers do not change the Perron eigenvector. Then use the evolution map $W_{t+1} = \mathcal{E}(W_t; \mathbf{x}_t^*)$ of Chapter 13 (Definition 13.6) to show how transfers can change $\boldsymbol{\pi}_{t+1}$. Explain why this makes transfers the *least controversial* policy tool: no agent's structural power is threatened. Then explain why this makes transfers the *least effective* tool for structural change.

14.6 The formation amplifier in politics. Compare two alliances with the same $\Delta\boldsymbol{\pi}$ but different Ψ:

(a) Alliance A controls W^P (production policy); its perturbation ΔW_A has $\Psi_A := \Psi(\Delta W_A, T) \approx 1$.

(b) Alliance B controls W^I (media, education); its perturbation ΔW_B has $\Psi_B := \Psi(\Delta W_B, T) \approx 3$.

(c) With $\Delta\Omega = 0$, compute the total payoff $v_i = (1 + \Psi)\Phi\,\Delta\pi_i$ for each (Definition 9.13: Ψ belongs to the perturbation, not to the alliance or the layer as such). Which alliance is more valuable to its members?

(d) Which type of alliance is harder to form? (Hint: controlling W^I may require controlling the state first, which requires power, which requires an alliance…)

14.7 Empty core. Construct a 3-agent structural power game with $\Delta\Omega \equiv 0$ in which coalition $\{1, 2\}$ can raise $\pi_1 + \pi_2$. Show that the core is empty

(Proposition 14.14). Now add a surplus term: for which values of $\Delta\Omega$ does the core become nonempty? Use the Bondareva–Shapley theorem.

Mathematical Appendices

Appendix 14A: Cooperative Game Theory for Structural Change

The transferable-utility game. A TU game is (N, v) where $v : 2^N \to \mathbb{R}$ assigns a value to each coalition. The **core** is the set of payoff vectors $\mathbf{x}$ with $\sum_{i \in S} x_i \geq v(S)$ for all S and $\sum_i x_i = v(N)$. The core is non-empty if and only if the game is balanced (the Bondareva–Shapley theorem; Bondareva, 1963; Shapley, 1967).

Application to structural change. The TU version $v(S)$ defined before Proposition 14.14 is a TU game, but the structural power game is not a *standard* one because: (1) the value of a coalition depends on the current W (state-dependent), (2) the feasibility set depends on power (endogenous constraints), and (3) the formation amplifier $(1 + \Psi)$ makes payoffs time-dependent (they include long-run preference effects). Nevertheless, the core concept provides useful predictions about which alliance structures are stable. When the power component is the only one ($\Delta\Omega \equiv 0$), the game is constant-sum and its core is empty as soon as any $v(S) > 0$.

The Shapley value. The Shapley value $\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(N-|S|-1)!}{N!} [v(S \cup \{i\}) - v(S)]$ gives each agent her average marginal contribution across all possible coalitions. In the structural power game, ϕ_i measures agent i's "structural leverage" — how much she improves any coalition she joins. Agents with high Shapley values are the "kingmakers" whose alliance decisions determine the political equilibrium.

Appendix 14B: Sensitivity of π to ΔW

First-order perturbation. Let W be irreducible and row-stochastic with Perron vector $\boldsymbol{\pi}$, and let ΔW satisfy $\Delta W \mathbf{1} = 0$ (rows stay stochastic) and keep $W + \Delta W$ nonnegative. By Lemma 6.19,

$$\Delta \boldsymbol{\pi}' \approx \boldsymbol{\pi}' \, \Delta W \, Z, \qquad Z = (I - W + \mathbf{1}\boldsymbol{\pi}')^{-1}, \tag{14.15}$$

equivalently $\Delta \boldsymbol{\pi}' \approx \sum_{k \geq 2} \frac{\boldsymbol{\pi}' \Delta W \mathbf{v}_k}{1 - \lambda_k} \mathbf{u}_k'$ with biorthogonal left and right eigenvectors ($\mathbf{u}_k' \mathbf{v}_l = \delta_{kl}$). An expansion that projects ΔW onto $\mathbf{v}_1 = \mathbf{1}$ instead is identically zero, because $\Delta W \mathbf{1} = 0$; `code/ch14.py` checks both against a finite difference.

Implication for coalitions. The "cheapest" $\Delta \boldsymbol{\pi}$ is achieved by perturbing W along the mode whose eigenvalue λ_k is closest to $+1$ (weakly coupled blocks); eigenvalues near -1 (periodicity) do not create sensitivity. In weakly coupled economies,

small perturbations of the cross-block entries produce large changes in the block masses of $\boldsymbol{\pi}$ — coalitions need little power to shift *who has power*. This does not contradict the lock-in of Chapter 13: what is cheap to move is $\boldsymbol{\pi}$; what is hard to move, under proximity-driven link formation, is the integration Γ itself.

Structuralist Case Study: Cardoso and Faletto's Class Alliances

This is Cardoso and Faletto's second appearance in the book. In Chapter 10, their "situations of dependency" were mapped to network topologies — different configurations of (W^P, W^U, W^I) producing different structural equilibria. Here, their analysis of class alliances provides the political mechanism that determines *which* topology obtains.

The distinctive contribution of *Dependência e Desenvolvimento* (Cardoso and Faletto, 1969) was the insistence that economic outcomes cannot be read off from structural position alone. Two countries with identical W^P (same export structure, same degree of foreign penetration) can have radically different development trajectories because the *class alliances* differ. Brazil's post-1964 trajectory (techno-bureaucratic $\{F, B, M\}$) and Argentina's post-1946 trajectory (national-populist $\{B, M, W\}$) cannot be explained by their trade structures (both were semi-industrialized primary exporters). They can be explained by the alliances.

Part 1: The alliance determines ΔW, not vice versa. The standard structuralist analysis (Prebisch, Furtado, Pinto) starts from W and derives the allocation. Cardoso and Faletto reversed the causation for the political layer: the alliance determines which ΔW is implemented, and therefore which W_{t+1} obtains. Different alliances produce different structural changes even from the same starting W. In this chapter's notation: the coalition S is the independent variable; ΔW_S is the dependent variable. Political agency enters the model through coalition formation, not through the network evolution equation's stochastic mechanics.

Part 2: Alliances are constrained by but not determined by structure. Cardoso and Faletto rejected mechanical determinism: the structure constrains which alliances are feasible (through $\mathcal{F}(S)$) but does not uniquely determine which alliance forms. In the same structural situation, different political choices can lead to different alliances (Brazil chose techno-bureaucratic in 1964; Peru chose military-populist in 1968; Chile chose democratic-socialist in 1970 — all from similar structural positions). The model of this chapter captures this through the possible non-uniqueness of stable alliances near the boundaries of Conjecture 14.15, and through the absence of any general existence or uniqueness result (Remark 14.6): the selection among alliances depends on political contingency.

Cardoso and Faletto's own categories. Their typology is not the four alliances above (which are this book's stylization) but a classification of situations of dependency by who controls the export sector: *enclave economies*, in which foreign capital controls the export sector and its links to the rest of the economy are thin, versus economies in which the export sector is under *national control*; and, in the later phase, the *internationalization of the internal market*, in which foreign capital produces for the domestic market in association with domestic groups — what Cardoso later called associated-dependent development. In this chapter's notation, enclave vs. national control is a statement about which block of W^P the export node belongs to and who appropriates its π; internationalization of the internal market is Alliance 3's ΔW.

Part 3: The alliance is the development strategy. The deepest insight: Cardoso and Faletto argued that "the development strategy" is not chosen by a benevolent planner who evaluates alternatives on efficiency grounds. The strategy is the ΔW that the winning coalition implements to maximize its collective payoff, of which $\Delta\boldsymbol{\pi}$ is the distributional core. ISI was not chosen because it was "efficient" — it was chosen because it was the structural policy preferred by the $\{B, M, W\}$ coalition. Liberalization was not chosen because it was "efficient" — it was chosen because it was preferred by the $\{F, B, M\}$ coalition. The policy debate between ISI and liberalization is, in this framework, a debate between coalitions with different $\Delta\boldsymbol{\pi}$ interests disguised as a debate about economic theory.

Cardoso & Faletto's argument	Formal object	This chapter
Class alliances shape the structure	Coalition S chooses ΔW_S	Def. 14.1
Alliances are constrained by structure	$\mathcal{F}(S)$ depends on $\sum_{i \in S} \pi_i$	Def. 14.2
Same structure, different alliances	Non-unique stable alliances	Conj. 14.15, Rem. 14.6
Development strategy = coalition output	ΔW maximizes coalition $\sum_{i \in S} v_i$	Def. 14.1
Enclave vs. national control	Block membership of the export node	Case study
Pendulum: ISI $\to$ liberalization $\to$	Alliance cycling from co-evolutionary dynamics	Conj. 14.12
Political Coase theorem fails	Enforcement mechanism is the current W	Def. 14.2

Cardoso launched the Plano Real (monetary stabilization) in 1993–94 as Finance Minister under Itamar Franco, and as President (1995–2002) pursued a broadly liberal economic program — closer to Alliance 3 ($\{F, B, M\}$) than to the national-populist program his academic work had analyzed. Whether this proves or disproves the model is an exercise the reader should not skip (Exercise 14.4 is related but more general). What the model predicts is that the alliance a politician joins depends on the current π, not on the politician's academic publications. When π changes, the optimal alliance changes, and the development strategy changes with it.

Further Reading

Coalition formation: Bernheim et al. (1987), "Coalition-Proof Nash Equilibria." Ray and Vohra (1997), "Equilibrium Binding Agreements." Banerjee et al. (2001), "Core in a Simple Coalition Formation Game." Bogomolnaia and Jackson (2002), "The Stability of Hedonic Coalition Structures." Acemoglu et al. (2012b), "Dynamics and Stability of Constitutions, Coalitions, and Clubs."

The political economy of development: Acemoglu (2003), "Why Not a Political Coase Theorem?" Acemoglu and Robinson (2006), *Economic Origins of Dictatorship and Democracy*. Bardhan (2005), *Scarcity, Conflicts, and Cooperation*.

Class alliances in Latin America: Cardoso and Faletto (1969), *Dependência e Desenvolvimento*, especially Chapters 4–6 (the empirical alliance analysis for Argentina, Brazil, Chile, Mexico). O'Donnell (1973), *Modernization and Bureaucratic-Authoritarianism*, for the analysis of the transition from populist to bureaucratic-authoritarian regimes. Collier and Collier (1991), *Shaping the Political Arena*, for the long-run consequences of labor incorporation strategies.

Bolivia's political economy: Klein (2011), *A Concise History of Bolivia*, Chapters 8–12. Dunkerley (1984), *Rebellion in the Veins*. Webber (2011), *From Rebellion to Reform in Bolivia*.

Chapter 15

Operator Design

Engineering the Weight Matrix

> *"O processo de substituição de importações pode ser entendido como um processo de desenvolvimento 'parcial' e 'fechado' que, respondendo às restrições do comércio exterior, procurou repetir aceleradamente, em condições históricas distintas, a experiência de industrialização dos países desenvolvidos."*
>
> ["The process of import substitution can be understood as a 'partial' and 'closed' development process that, responding to foreign trade constraints, sought to repeat at an accelerated pace, under distinct historical conditions, the industrialization experience of the developed countries."]
>
> — Maria da Conceição Tavares, "Auge e declínio do processo de substituição de importações no Brasil" (1964; in Tavares 1972)

Chapter 13 showed how networks evolve stochastically. Chapter 14 showed who decides the evolution. This chapter asks: given a development objective, *how should the planner design* ΔW*?*

The answer is operator design: the systematic construction of structural changes that move W toward a target. The "operator" is the policy instrument — a tariff, an infrastructure investment, an education reform, a media regulation — and the "design" is the choice of which entries of W to modify, by how much, and in what sequence.

This chapter treats structural policy as an engineering problem. The decision variable is ΔW; the constraints are the feasibility set $\mathcal{F}(S)$ from Chapter 14 and row-stochasticity. The objective must say what the planner wants, and the conventions of Chapter 1 (Section 1.7) separate two different targets: *power*, measured

by the concentration of $\boldsymbol{\pi}$ ($\mathrm{Var}(\boldsymbol{\pi})$, the power Gini, N_{eff} or $d_\pi = \frac{1}{2}\|\boldsymbol{\pi} - \mathbf{1}/N\|_1$), and *integration*, measured by the spectral gap Γ (the speed at which shocks and signals mix). The two are independent (Proposition 6.16), and a third target, output, is measured from the technical-coefficient matrix A, not from W.

This is where the product space meets the weight matrix, and where ISI becomes matrix algebra. It is also where the book meets a rigorous literature it should build on: optimal industrial policy in production networks (Liu, 2019) and the optimization of the spectral gap (Boyd, Diaconis, and Xiao, 2004).

15.1 The Structural Policy Problem

15.1.1 Three Objectives

Definition 15.1 (The Operator Design Problem)**.** *Given the current weight matrix W, a budget $B > 0$ on the size of the intervention, measured in a norm $\|\cdot\|$ (Frobenius or ℓ_1, below), and a feasibility set $\mathcal{F}$, the **operator design problem** is:*

$$\max_{\Delta W \in \mathcal{F}} \Omega(\Delta W) \quad \textit{subject to} \quad \|\Delta W\| \le B, \quad W + \Delta W \ge 0, \quad (W + \Delta W)\mathbf{1} = \mathbf{1}, \tag{15.1}$$

where $\Omega(\Delta W)$ is the objective function and the constraints ensure that $W + \Delta W$ remains a valid row-stochastic nonnegative matrix.

Three natural objectives:

1. **Maximize** Γ: increase the spectral gap, making the economy more integrated — shocks, prices and signals propagate across blocks faster. This objective says nothing about who holds power.

$$\Omega_1(\Delta W) = \Gamma(W + \Delta W) - \Gamma(W). \tag{15.2}$$

2. **Minimize** $\mathrm{Var}(\boldsymbol{\pi})$ (or the power Gini, or d_π): reduce power inequality directly.

$$\Omega_2(\Delta W) = -\left[\mathrm{Var}(\boldsymbol{\pi}(W + \Delta W)) - \mathrm{Var}(\boldsymbol{\pi}(W))\right]. \tag{15.3}$$

3. **Maximize total output**: invest where the Leontief multiplier is largest. Output is computed from the technical coefficients A (row i is sector i's recipe; balance $\mathbf{y} = A'\mathbf{y} + \mathbf{d}$), with an intervention ΔA on input coefficients:

$$\Omega_3(\Delta A) = \mathbf{1}'(I - (A + \Delta A)')^{-1}\mathbf{d} - \mathbf{1}'(I - A')^{-1}\mathbf{d}. \tag{15.4}$$

It cannot be computed from the row-stochastic $W^P = D^{-1}A$: for every row-stochastic W, $(I - \gamma W)^{-1}\mathbf{1} = \mathbf{1}/(1-\gamma)$, so with $\mathbf{d} = \mathbf{1}$ the "multiplier" $\mathbf{1}'(I - \gamma W)^{-1}\mathbf{1} = N/(1-\gamma)$ is the same for every feasible ΔW.

Intuition: Three objectives, three answers

The three objectives do *not* point in the same direction. Adding cross-block edges raises Γ; whether it lowers $\mathrm{Var}(\boldsymbol{\pi})$ depends on whether the new edge sends weight toward the low-power block; and whether it raises output depends on the input intensity of the supplier. Example 15.6 shows a cross-block edge that wins on Γ and on power but loses on the multiplier. And Proposition 15.2 shows that Γ can be pushed all the way to 1 without moving $\boldsymbol{\pi}$ at all.

Proposition 15.2 (Integration Without Redistribution)**.** *Let W be irreducible and row-stochastic with Perron vector $\boldsymbol{\pi}$ and second-largest eigenvalue modulus $|\lambda_2|$. For $t \in [0,1]$, the matrix $W_t = (1-t)W + t\,\mathbf{1}\boldsymbol{\pi}'$ is row-stochastic, has $\boldsymbol{\pi}(W_t) = \boldsymbol{\pi}$, and has $\Gamma(W_t) = 1 - (1-t)|\lambda_2|$.*

Proof. W_t is a convex combination of row-stochastic matrices, hence row-stochastic, and $\boldsymbol{\pi}'W_t = (1-t)\boldsymbol{\pi}' + t\boldsymbol{\pi}' = \boldsymbol{\pi}'$. For $t < 1$, W_t has at least the support of W, so it is irreducible; for $t > 0$ it is strictly positive. Either way its stationary vector is unique, so $\boldsymbol{\pi}(W_t) = \boldsymbol{\pi}$. For the spectrum, let $H = \{\mathbf{v} : \boldsymbol{\pi}'\mathbf{v} = 0\}$. H is W-invariant, because $\boldsymbol{\pi}'W\mathbf{v} = \boldsymbol{\pi}'\mathbf{v}$, and $\mathbb{R}^N = \mathrm{span}(\mathbf{1}) \oplus H$ since $\boldsymbol{\pi}'\mathbf{1} = 1$. On H, $\mathbf{1}\boldsymbol{\pi}'\mathbf{v} = 0$, so $W_t = (1-t)W$ there, while $W_t\mathbf{1} = \mathbf{1}$. Because 1 is a simple eigenvalue of the irreducible W, the eigenvalues of $W|_H$ are $\lambda_2, \ldots, \lambda_N$, the non-Perron eigenvalues of W counted with multiplicity. So the spectrum of W_t is $\{1\} \cup \{(1-t)\lambda_k\}_{k \geq 2}$, and $\Gamma(W_t) = 1 - (1-t)|\lambda_2|$. This holds for periodic W too, where $|\lambda_2| = 1$ and $\Gamma(W_t) = t$. The construction is the same as Proposition 6.16, applied along a path. □

For a random 4×4 matrix in `code/ch15.py`, Γ rises from 0.541 to 0.656, 0.771 and 0.885 at $t = 0.25, 0.5, 0.75$ while $\mathrm{Var}(\boldsymbol{\pi})$ stays at 0.0066. Raising integration never *requires* anyone to give up power. Power redistribution is a separate objective, Ω_2, with its own political cost (Section 15.5).

15.1.2 The First-Order Solution

By Lemma 6.19, if λ_2 is simple and no other eigenvalue except its conjugate has modulus $|\lambda_2|$, then Γ is differentiable and, for a perturbation in direction E,

$$\frac{d\Gamma(W+tE)}{dt}\Big|_{t=0} = -\frac{\mathrm{Re}\big(\bar{\lambda}_2\, \mathbf{u}_2' E \mathbf{v}_2\big)}{|\lambda_2|} = \sum_{i,j} G_{ij} E_{ij}, \qquad G_{ij} := -\frac{\mathrm{Re}\big(\bar{\lambda}_2\, u_{2,i} v_{2,j}\big)}{|\lambda_2|}, \tag{15.5}$$

with left and right eigenvectors normalized biorthogonally ($\mathbf{u}_k'\mathbf{v}_l = \delta_{kl}$). Feasible directions satisfy $E\mathbf{1} = 0$, so the entries of G matter only through differences within a row: moving weight from (i,ℓ) to (i,j) changes Γ at rate $G_{ij} - G_{i\ell}$.

Proposition 15.3 (Optimal First-Order Intervention)**.** *Maximize the linearized objective $\sum_{ij} G_{ij}\Delta W_{ij}$ subject to $\Delta W\mathbf{1} = 0$ and the budget, ignoring the nonnegativity constraint (valid for small B when the relevant entries of W are positive).*

1. *ℓ_1 **budget** ($\sum_{ij}|\Delta W_{ij}| \le B$): the optimum moves $B/2$ of weight within a single row, from the entry with the smallest $G_{i\ell}$ to the entry with the largest G_{ij}, in the row i where $G_{ij} - G_{i\ell}$ is largest:*

$$(i^*, j^*, \ell^*) = \arg\max_{i,\,j,\,\ell}\ (G_{ij} - G_{i\ell}). \tag{15.6}$$

2. ***Frobenius budget*** *($\|\Delta W\|_F \le B$): the optimum spreads over all entries, $\Delta W^* = B\,\tilde{G}/\|\tilde{G}\|_F$, where $\tilde{G} = G - \frac{1}{N}G\mathbf{1}\mathbf{1}'$ is G with each row centered.*

Proof. The feasible set is a polytope (case 1) or a Euclidean ball (case 2) intersected with the subspace $\{E : E\mathbf{1} = 0\}$. (1) Take any feasible E and any row i. Its positive and negative parts have equal mass $m_i = \frac{1}{2}\sum_j |E_{ij}|$, because the row sums to zero. Pairing positive with negative mass, as in a transport plan, writes the row as $\sum_{j,\ell} c^i_{j\ell}(\mathbf{e}_j - \mathbf{e}_\ell)'$ with $c^i_{j\ell} \ge 0$ and $\sum_{j,\ell} c^i_{j\ell} = m_i$. Hence

$$\textstyle\sum_{ij} G_{ij}E_{ij} = \sum_i \sum_{j,\ell} c^i_{j\ell}(G_{ij} - G_{i\ell}) \le \Big(\sum_i m_i\Big)\max_{i,j,\ell}(G_{ij} - G_{i\ell}) \le \frac{B}{2}\,(G_{i^*j^*} - G_{i^*\ell^*}),$$

using $\sum_i m_i = \frac{1}{2}\|E\|_1 \le B/2$ and $\max_{i,j,\ell}(G_{ij} - G_{i\ell}) \ge 0$. The single move $E^* = \frac{B}{2}\mathbf{e}_{i^*}(\mathbf{e}_{j^*} - \mathbf{e}_{\ell^*})'$ attains the bound. (2) $\{E : E\mathbf{1} = 0\}$ is a linear subspace, and the orthogonal projection onto it in the Frobenius inner product is $E \mapsto E - \frac{1}{N}E\mathbf{1}\mathbf{1}'$. So for feasible E, $\langle G, E\rangle_F = \langle \tilde{G}, E\rangle_F \le \|\tilde{G}\|_F\, B$ by Cauchy–Schwarz, with equality at $E = B\tilde{G}/\|\tilde{G}\|_F$ (if $\tilde{G} = 0$, every feasible E is optimal). □

The single-edge answer is therefore an artefact of the ℓ_1 budget. With the same Frobenius norm $B = 0.05$, the projected-gradient step raises Γ by 0.053 in the

example of `code/ch15.py`, against 0.041 for the best single move. For symmetric (reversible) W on a fixed graph, maximizing Γ exactly — not just to first order — is a convex problem (the fastest-mixing Markov chain of Boyd, Diaconis, and Xiao, 2004, solvable as a semidefinite program). For non-symmetric W, $|\lambda_2|$ is neither convex nor Lipschitz, and only local methods are available.

Intuition: Which edge to add: connect the cores

Where are the large entries of G? The second eigenvectors $\mathbf{u}_2$ and $\mathbf{v}_2$ describe the slowest mode of the matrix: they are roughly constant, with opposite signs, on the two weakly connected blocks, and close to zero on agents who sit *between* the blocks. So $|u_{2,i}|$ and $|v_{2,j}|$ are largest *deep inside* the blocks, not on the boundary.

When $\lambda_2 > 0$, $G_{ij} > 0$ exactly when $u_{2,i}$ and $v_{2,j}$ have opposite signs: the best new edge runs from an agent deep in block A to an agent deep in block B. In the six-agent example of `code/ch15.py` (two blocks and one bridge agent), the scaled second left eigenvector is $(1, 1, -0.04, -0.65, -0.65, -0.65)$: the bridge agent has an entry near zero. The best edge runs from a core agent of block A to a core agent of block B ($d\Gamma/dt = 0.252$); the best edge into the bridge agent achieves only 0.032. The optimal structural intervention is not the "bridge builder" who links boundary agents, but a direct link between the cores of the two blocks. This is the Fiedler-vector heuristic of Ghosh and Boyd (2006) for growing well-connected graphs (add the edge that maximizes $(v_{2,i} - v_{2,j})^2$), transposed to non-symmetric W.

15.2 Five Operator Types

Every structural policy is an operator $\mathcal{L} : W \mapsto W + \Delta W$ that modifies the weight matrix. We classify five types by the layer of W they target and the type of edge they create.

15.2.1 Operator 1: Import Substitution (Replace Foreign Edges with Domestic Edges)

Definition 15.4 (ISI Operator). *The **import-substitution operator** $\mathcal{L}_{ISI}$ modifies W^P by:*

$$\mathcal{L}_{ISI} : W^P_{ij} \to W^P_{ij} + \delta^{dom}_{ij} - \delta^{for}_{ij}, \tag{15.7}$$

where $\delta^{dom}_{ij} > 0$ for domestic (i, j) pairs (new domestic supply chains) and $\delta^{for}_{ij} > 0$ for foreign (i, j) pairs (reduced import dependency).

Proposition 15.5 (Effect of ISI on Γ, to First Order). *1.* ***Global*** Γ**.** *Let W^P be row-stochastic on a node set that includes the foreign suppliers, and suppose its λ_2 satisfies the hypotheses of (15.5). Let the ISI operator be balanced row by row, $\sum_j \delta^{dom}_{ij} = \sum_j \delta^{for}_{ij}$ for every i, so that $E = \delta^{dom} - \delta^{for}$ has $E\mathbf{1} = 0$, and let $\delta^{for}_{ij} > 0$ only where $W^P_{ij} > 0$. Scale the operator by $s > 0$. If*

$$\sum_{(i,j)\in domestic} \delta^{dom}_{ij} G_{ij} > \sum_{(i,j)\in foreign} \delta^{for}_{ij} G_{ij}, \tag{15.8}$$

with G the gradient of $\Gamma(W^P)$ in (15.5), then $\Gamma(W^P + sE) > \Gamma(W^P)$ for all sufficiently small s. If the reverse strict inequality holds, Γ falls for small s.

2. ***Domestic*** Γ**.** *Let K be the domestic block of W^P (or of the domestic coefficients A), and let $\hat{K} = \operatorname{diag}(K\mathbf{1})^{-1}K$ be its row-renormalized version; suppose $\lambda_2(\hat{K})$ satisfies the same hypotheses, and let $\hat{G}$ be the gradient (15.5) at $\hat{K}$. Then the severed foreign edges drop out of $\hat{K}$ altogether. To first order the domestic Γ moves in the direction*

$$\hat{E} = \operatorname{diag}(K\mathbf{1})^{-1}\big(\delta^{dom} - \operatorname{diag}(\delta^{dom}\mathbf{1})\,\hat{K}\big), \qquad \hat{E}\mathbf{1} = 0, \tag{15.9}$$

and it rises for small s if $\sum_{ij} \hat{G}_{ij}\hat{E}_{ij} > 0$.

That is: the Γ-increasing effect of new domestic edges must exceed the Γ-decreasing effect of severed foreign edges, or, for the domestic Γ, the new domestic edges must point along the gradient once the row has been renormalized. This condition typically fails if the new domestic edges are within-block (connecting agents who were already connected) rather than cross-block (connecting the modern and traditional sectors).

Proof. (1) By Lemma 6.19 and (15.5), $\Gamma(W^P + sE) = \Gamma(W^P) + s\sum_{ij} G_{ij}E_{ij} + o(s)$. The linear coefficient is the left side of (15.8) minus the right side, and its sign fixes the sign of the change for small s. Nonnegativity holds for small s because δ^{for} is supported where $W^P > 0$. (2) The domestic block after the operator is $K + s\delta^{\text{dom}}$; the foreign entries are not in it. Differentiating $\hat{K}(s) = \operatorname{diag}((K + s\delta^{\text{dom}})\mathbf{1})^{-1}(K + s\delta^{\text{dom}})$ at $s = 0$ gives $\hat{E}$ in (15.9), and $\hat{E}\mathbf{1} = \operatorname{diag}(K\mathbf{1})^{-1}(\delta^{\text{dom}}\mathbf{1} - \delta^{\text{dom}}\mathbf{1}) = 0$. Then apply (15.5) at $\hat{K}$ in the direction $\hat{E}$. □

Example 15.6 (ISI That Works vs. ISI That Doesn't). *$N = 4$: Modern (M1, M2), Traditional (T1, T2). Domestic technical coefficients (row i is sector i's recipe; the rest of each unit of output is imports and value added) and the input-share matrix $W^P = D^{-1}A$*

are

$$A = \begin{pmatrix} 0.33 & 0.25 & 0.01 & 0.01 \\ 0.25 & 0.33 & 0.01 & 0.01 \\ 0.012 & 0.012 & 0.15 & 0.126 \\ 0.012 & 0.012 & 0.126 & 0.15 \end{pmatrix}, \qquad W^P = \begin{pmatrix} 0.550 & 0.417 & 0.017 & 0.017 \\ 0.417 & 0.550 & 0.017 & 0.017 \\ 0.040 & 0.040 & 0.500 & 0.420 \\ 0.040 & 0.040 & 0.420 & 0.500 \end{pmatrix}.$$

W^P is nearly block-diagonal, with $\Gamma = 0.113$ and $\boldsymbol{\pi} = (0.353, 0.353, 0.147, 0.147)$: the traditional sectors buy more from the modern block than they sell to it, so the modern sectors hold more power. Total output with $\mathbf{d} = \mathbf{1}$ is $\mathbf{1}'(I - A')^{-1}\mathbf{d} = 7.826$.

Both policies replace imported inputs of M1 with 0.05 of domestic input per unit of M1's output.

***ISI-A (within-block):** the substitute comes from the modern sector: $\Delta a_{M1,M2} = +0.05$. This strengthens the within-block link. Effect on Γ: -0.001. The modern sector densifies internally without connecting to the traditional sector.*

***ISI-B (cross-block):** the substitute comes from the traditional sector: $\Delta a_{M1,T1} = +0.05$. This creates a cross-block link. Effect on Γ: $+0.037$. The traditional sector now supplies the modern sector, creating the linkage that makes shocks and demand propagate across blocks.*

	Target	$\Delta\Gamma$	$\Delta \operatorname{Var}(\boldsymbol{\pi})$	Δ *total output*
ISI-A (within)	$a_{M1,M2}$	-0.001	$+0.0010$	$+0.312$
ISI-B (cross)	$a_{M1,T1}$	$+0.037$	-0.0103	$+0.181$

ISI-B dominates on integration and on power: it raises Γ and nearly eliminates the power gap ($\boldsymbol{\pi}$ becomes $(0.265, 0.268, 0.245, 0.223)$). ISI-A wins on the output multiplier, because the modern supplier is itself input-intensive. Same budget, same tariff, different structural effects — and no single objective ranks them (`code/ch15.py`).

15.2.2 Operator 2: Infrastructure (Create Physical Edges)

Definition 15.7 (Infrastructure Operator)**.** *The **infrastructure operator** $\mathcal{L}_{infra}$ creates new edges in W^P (and possibly W^I) by reducing the cost of interaction between previously disconnected agents:*

$$\mathcal{L}_{infra} : W^P_{ij} \to W^P_{ij} + \delta_{ij}(\textit{road, port, fiber optic, ...}). \tag{15.10}$$

Infrastructure is the literal "edge creation" operator: a road between two regions creates an edge in W^P. The operator design problem for infrastructure is: which road to build?

Corollary 15.8 (Optimal Road Placement)**.** *To first order and under an ℓ_1 budget, the road that maximizes Γ creates the edge (i^*, j^*) with the largest $G_{ij} - G_{i\ell}$ among pairs with $W^P_{ij} \approx 0$ and $W^P_{i\ell} > 0$ (Proposition 15.3): typically a link between the cores of the two most weakly connected blocks.*

Proof. Restrict the ℓ_1 problem of Proposition 15.3 to directions whose positive entries lie on absent edges ($W^P_{ij} \approx 0$) and whose negative entries lie where $W^P_{i\ell} > 0$, so that $W^P + E \geq 0$ for small B. The transport decomposition in that proof uses only pairs (j, ℓ) from these supports, so the bound and its attainment hold with the maximum taken over the admissible triples (i, j, ℓ). "Typically the cores" is not part of the claim; it is the heuristic explained after Proposition 15.3. □

Prebisch Connection: Hirschman's linkage argument as operator design

Hirschman (1958) was the leading critic of the balanced-growth "big push" (Rosenstein-Rodan, 1943). He argued instead for unbalanced growth: invest in sectors with strong backward and forward linkages, so that the pressures they create induce further investment. He measured linkages with direct input-output coefficients (Chenery and Watanabe, 1958); the Leontief-inverse indices — column sums of $(I - A')^{-1}$ for backward linkages — are Rasmussen's (1956). Both are computed from the technical coefficients A, never from the row-stochastic W^P, whose Leontief inverse has identical row sums for every sector.

Hirschman's answer and the spectral-gap answer (Proposition 15.3) do not always agree. Hirschman targets the sector with the strongest linkages; the spectral-gap criterion targets the edge that connects the cores of weakly coupled blocks. When the high-linkage sector is in the already-dense core, Hirschman's investment deepens the core without bridging to the periphery; the spectral-gap criterion bridges, but by itself ignores output (Example 15.6).

Hirschman himself singled out mining and other primary-export enclaves as *low*-linkage activities, and later (1977) added fiscal and consumption linkages to the list. For Bolivia his framework does not say "invest in gas"; it says that gas generates mainly a fiscal linkage, and asks what that revenue is spent on. The INE supply-use tables bear him out: in 2019 hydrocarbons bought 0.225 of domestic intermediate inputs per unit of output (manufacturing: 0.558), exported 70% of their output, paid 6% of their value added as compensation of employees, and had a power share of only 0.039 in the domestic W^P (Instituto Nacional de Estadística (INE), Bolivia, 2026); author's calculations (`data/compute/compute_bolivia_io.py`). The spectral-gap criterion adds: invest in the *edge between gas and the traditional sector* — the processing plant, the local supply contract, the vocational training program that connects quinoa farmers to the gas-sector labor market.

15.2.3 Operator 3: Education and Training (Create Capability Edges)

Definition 15.9 (Human Capital Operator)**.** *The **human capital operator** $\mathcal{L}_{HC}$ modifies W^P by expanding the set of production activities an agent can perform:*

$$\mathcal{L}_{HC} : W^P_{ij} \to W^P_{ij} + \delta_{ij}(\text{if agent } i \text{ acquires capability for activity } j). \tag{15.11}$$

Simultaneously, it modifies W^I by exposing the agent to new information:

$$\mathcal{L}_{HC} : W^I_{ij} \to W^I_{ij} + \delta'_{ij}(\textit{if agent } i \textit{ enters school/training with agent } j). \tag{15.12}$$

Education is a dual operator: it creates edges in BOTH W^P (new production capabilities) and W^I (new information sources). This is why education plausibly has a Ψ ratio (Definition 9.13) greater than 1: the production edge is the enforcement effect, the information edge is the formation effect. Education changes what you can do AND what you want.

15.2.4 Operator 4: Information Policy (Rewire W^I)

Definition 15.10 (Information Operator)**.** *The **information operator** $\mathcal{L}_{info}$ modifies W^I without changing W^P:*

$$\mathcal{L}_{info} : W^I_{ij} \to W^I_{ij} + \delta_{ij}(\textit{media, internet, mentor program, ...}). \tag{15.13}$$

Information policy does not change production or endowments. It changes who sees what. Its effect is entirely through the formation channel — but as Chapter 10 showed, the formation channel can be the dominant one ($\Psi > 1$).

Example 15.11 (The Smartphone as Information Operator [ILLUSTRATIVE])**.** *A program distributes smartphones to 1,000 rural households. Each phone adds edges from the rural block to the urban/modern block in W^I:*

$$\Delta W^I_{rural_i, urban_j} \approx 0.05 \quad \textit{for each connected pair.} \tag{15.14}$$

Immediate effect: *Zero. No change in W^P, no change in endowments, no change in prices. The standard CBA finds no benefit.*

Formation effect (one generation): *Rural agents now observe modern-sector returns. Their steady-state preference shifts from $\theta^*_T = 0.92$ toward $\theta^*_M = 1.09$. New steady state (with $\rho = 0.85$): $\theta^*_T = 0.96$ (partial convergence). Investment increases. Returns increase. The next generation's θ shifts further.*

Long-run effect (three generations): *$\theta^*_T \to 1.01$. The preference gap $1.09 - 0.92 = 0.17$ closes by 0.09, i.e. by 53% (24% after one generation). The rest of the allocation gap requires also changing W^P and $\boldsymbol{\pi}$.*

Ψ for this intervention: *undefined. The enforcement effect is zero, so the ratio Ψ of formation to enforcement has a zero denominator: this is a pure formation intervention. Standard evaluation gives benefit $= 0$. The formation model gives a long-run benefit of $0.09 \times N_{treated} \times (\textit{income gain per unit } \Delta\theta)$.*

15.2.5 Operator 5: Institutional Design (Change the Rules of Edge Formation)

Definition 15.12 (Institutional Operator)**.** *The **institutional operator** $\mathcal{L}_{inst}$ modifies the parameters of the edge creation/destruction process from Chapter 13:*

$$\mathcal{L}_{inst} : (p_0, p_1, p_2, p_3, \delta_0, \delta_1) \to (p_0', p_1', p_2', p_3', \delta_0', \delta_1'). \tag{15.15}$$

Unlike the other operators, which add or remove specific edges, the institutional operator changes the rules by which edges form and break.

Example 15.13 (Property Rights as Institutional Operator)**.** *Secure property rights reduce the destruction rate of production edges: $\delta_1 \downarrow$ (profitable links are less likely to be severed by expropriation). This makes cross-block investment less risky, which increases the creation rate of cross-block edges ($p_2 \uparrow$, because the surplus from connection is now more likely to be realized). The institutional operator does not directly add any edge. It changes the dynamics of edge formation, and the network evolves toward higher Γ over time.*

15.3 The Sequencing Problem

Conjecture 15.14 (Sequencing [ILLUSTRATIVE])**.** *When the budget allows multiple interventions over time, a natural heuristic is the greedy rule*

$$\mathcal{L}_t^* = \underset{\mathcal{L} \in \{ISI,\, infra,\, HC,\, info,\, inst\}}{\arg\max} \frac{\Delta\Omega(\mathcal{L} \mid W_t)}{cost(\mathcal{L})}, \tag{15.16}$$

where Ω is an objective defined on all *layers (e.g. a weighted combination of $\Gamma(W^P)$, $\mathrm{Var}(\boldsymbol{\pi})$ and the formation gain, since the information operator leaves W^P, and hence $\Gamma(W^P)$, unchanged), with the dynamic constraint that each intervention changes W_t and therefore the marginal effect of subsequent ones. A greedy rule is not optimal in general; it is optimal for special objectives (e.g. when marginal gains are diminishing in the submodular sense). We conjecture that, for plausible values of Ψ, the resulting sequence is:*

1. ***First, information policy*** *($\mathcal{L}_{info}$): cheap per unit of formation gain (it operates on W^I, whose perturbations have, illustratively, the highest Ψ), and it relaxes the political constraint by shifting preferences toward structural change.*
2. ***Second, institutional reform*** *($\mathcal{L}_{inst}$): reduces the destruction rate of future edges, making subsequent investments more durable.*
3. ***Third, education/training*** *($\mathcal{L}_{HC}$): dual operator, creates edges in both W^P and W^I.*

4. ***Fourth, infrastructure*** *($\mathcal{L}_{infra}$): creates physical W^P edges. Expensive, but by this point the institutional and informational preconditions are in place.*

5. ***Fifth, ISI/industrial policy*** *($\mathcal{L}_{ISI}$): the most expensive and politically fraught, but by this point the political coalition has shifted (because W^I changed preferences in step 1, which changed coalition incentives per Chapter 14).*

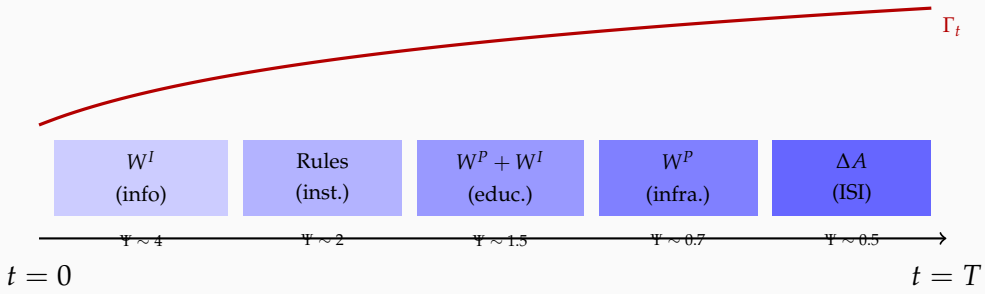
Key Structural Insight

The sequencing conjecture inverts the conventional wisdom. The standard approach starts with the "hard" infrastructure and industrial policy, then adds the "soft" information and institutional elements later. The formation model suggests the opposite: start with information (change what people see), then institutions (change the rules), then education (change what people can do), then infrastructure (change the physical connections), and finally industrial policy (change the production structure).

The reason is the formation amplifier. Each step's effectiveness depends on the Ψ ratio. With illustrative values — information $\Psi \sim$ 3–5, institutions $\Psi \sim$ 1–2, infrastructure $\Psi \sim$ 0.5–1 — starting with the high-Ψ interventions produces the preference changes that make the low-Ψ interventions politically feasible and economically effective. Estimating these Ψ values (Chapter 11) is what would turn the conjecture into a policy rule.

History is suggestive, not decisive. South Korea's sequence is usually summarized as land reform and mass education first, export promotion in the 1960s, and the heavy-and-chemical-industry drive in the 1970s — education and institutions before industrial policy, though not in exactly the order of the conjecture. Bolivia's gas nationalization was industrial policy without prior information or institutional change: it changed ownership and the fiscal destination of gas rents but not W^I, and the formation preconditions were not in place. By the INE input-output tables it did not change the domestic production network much either: $\Gamma(W^P)$ was 0.518 in 2006 and 0.484 in 2013 (Instituto Nacional de Estadística (INE), Bolivia, 2026).

Visualization: Conjectured sequencing: Ψ-ordered operators



Operators ordered by (illustrative) Ψ: highest-Ψ first. The red curve is schematic and shows Γ_t rising — steeply at first (cheap information interventions with large Ψ) and more gradually later (expensive industrial policy with smaller Ψ). Start soft, finish hard.

15.4 The Product Space as an Operator Map

The product space (Hidalgo et al., 2007) can now be interpreted as a map of feasible operators on W^P.

Stylized Fact 15.15 (Product Space as Feasibility Constraint)**.** *The probability that a country develops comparative advantage in product p' is increasing in its relatedness density $\sum_p M_{cp}\phi_{pp'} / \sum_p \phi_{pp'}$, where $\phi_{pp'}$ is the co-export proximity of products (Hidalgo et al., 2007). Products in the dense core of the product space (machinery, chemicals, electronics) are feasible targets for economies that already produce related products. Products on the sparse periphery (minerals, agricultural commodities) offer few diversification paths.*

The operator-design interpretation needs care. The product space is a network of *capability similarity* between products, built from co-export patterns; it is not the input-output network W^P, and its edges are not input flows. It enters operator design as a constraint on which ISI operators are *feasible* (likely to succeed), not as a map of $\partial\Gamma/\partial W^P$. Whether a new core product also raises $\Gamma(W^P)$ depends on its input-output links, which must be read from A.

Example 15.16 (Bolivia in the Product Space [ILLUSTRATIVE])**.** *Bolivia's current export basket: natural gas, zinc, gold, soy, quinoa — all on the sparse periphery of the product space. Relatedness density to the dense core (machinery, electronics, pharmaceuticals): low; we use $\omega_{c,core} \approx 0.08$ for illustration (compute the actual value in Exercise **15.5**). The sourced complexity figures agree. Bolivia's ECI was -1.30 in 2019 (rank 131), and the product complexity index of its main export lines is low: natural gas (HS 2711) -2.23,*

zinc ores (2608) −1.67, soya beans (1201) −1.81 (The Growth Lab at Harvard University, 2026). Fuels were 32.1% and ores and metals 28.7% of merchandise exports (World Bank, 2026b). The periphery here is a position in the product space and in export orientation. Bolivia's measured domestic W^P has no weakly coupled block in the spectral sense ($\Gamma = 0.467$; Chapter 17), so the $\Delta\Gamma$ figures below are for a stylized dual economy.

Conventional ISI: *Target products adjacent in the product space — gas processing, zinc smelting, soy flour. These increase domestic value added but do not connect to the dense core. They add edges within the peripheral block of W^P. Illustratively, $\Delta\Gamma \approx 0.02$. Modest.*

Leapfrog strategy: *Target products in the dense core — pharmaceuticals, automotive parts, food processing machinery. These are distant in the product space (low relatedness density). The probability of success is low unless accompanied by massive human capital investment ($\mathcal{L}_{HC}$) and information rewiring ($\mathcal{L}_{info}$). But if successful: $\Delta\Gamma \approx 0.15$. Transformative.*

The sequencing solution: *(1) Information policy first: expose Bolivian engineers and technicians to the dense core (scholarships, technical exchanges, FDI in assembly operations). (2) Then education: build the capabilities (vocational training in the target sectors). (3) Then infrastructure: connect the production zones. (4) Then ISI: tariff protection for infant industries with cross-block linkages. This is the order of Conjecture 15.14, adapted to Bolivian conditions.*

15.5 The Optimal ΔW Under Political Constraints

Definition 15.17 (Politically Constrained Operator Design)**.** *Let S^* be the winning coalition (Chapter 14). For an objective Ω_k ($k = 1, 2$), the politically constrained problem is*

$$\Delta W^*_{con} = \underset{\Delta W \in \mathcal{F}(S^*)}{\arg\max}\ \Omega_k(\Delta W) \quad \textit{subject to} \quad \Delta\pi_i(W, \Delta W) \geq 0 \ \textit{for all } i \in S^*, \tag{15.17}$$

*together with the budget and row-stochasticity constraints of (15.1). The **political cost of structural reform** is*

$$C_{political} = \Omega^*_{k,unconstrained} - \Omega^*_{k,constrained} \geq 0, \tag{15.18}$$

which is nonnegative because the constrained feasible set is a subset of the unconstrained one.

Proposition 15.18 (Integration Carries No Political Cost)**.** *Let W be irreducible and row-stochastic, and suppose $\mathcal{F}(S^*)$ contains every $t(\mathbf{1}\boldsymbol{\pi}' - W)$ with $\|t(\mathbf{1}\boldsymbol{\pi}' - W)\| \leq B$*

(for instance, $\mathcal{F}(S^*)$ is a norm ball of radius at least B). For the integration objective Ω_1 (maximize Γ), the direction $E = \mathbf{1}\boldsymbol{\pi}' - W$ satisfies the political constraint with equality for every coalition. Hence the constrained optimum of $\Gamma(W + \Delta W)$ is at least $1 - (1 - \bar{t})|\lambda_2|$, i.e. $\Omega^*_{1,constrained} \geq \bar{t}\,|\lambda_2|$, with $\bar{t} = \min\{1, B/\|\mathbf{1}\boldsymbol{\pi}' - W\|\}$. If the budget allows $\bar{t} = 1$, both the constrained and the unconstrained optimum equal $\Gamma = 1$, so $C_{political} = 0$.

Proof. Along $W + tE = (1 - t)W + t\,\mathbf{1}\boldsymbol{\pi}'$, Proposition 15.2 gives $\Delta\boldsymbol{\pi} = 0$ and $\Gamma = 1 - (1 - t)|\lambda_2|$. The path is row-stochastic and nonnegative, and $\|tE\| \leq B$ for $t \leq \bar{t}$, so $t = \bar{t}$ is feasible for the constrained problem. Since $|\lambda_2| \geq 0$, $\Gamma \leq 1$ always; at $\bar{t} = 1$ the constrained value 1 therefore equals the unconstrained optimum. □

So no coalition ever has to give up power to let the economy integrate: a "political cost" measured in Γ is an artefact of reading Γ as power. The political constraint bites on *redistribution*. (Moving a stationary distribution by choosing edges, the unconstrained version of Ω_2, is the PageRank-optimization problem studied by de Kerchove et al., 2008 and Csáji et al., 2014.) For the power objective Ω_2 we conjecture that the cost is:

1. **Large when power is concentrated**: the winning coalition is small and powerful, and protecting each member's π_i rules out most variance-reducing ΔW.

2. **Small when power is equal**: the winning coalition is broad (near the grand coalition), and the political constraint is weak.

3. **Zero** if W is already doubly stochastic ($\boldsymbol{\pi}$ uniform, so there is nothing to redistribute), or if the reform raises every member's π_i (rare, since $\sum_i \Delta\pi_i = 0$).

Intuition: Why redistributive reform is slow [ILLUSTRATIVE]

The political cost explains why reform that redistributes power is slow even when the social returns are large. The unconstrained optimum for Ω_2 might be a road connecting the gas enclave to the traditional economy that lowers the power Gini by 0.20. The gas/government coalition blocks it because it would reduce their power: $\Delta\pi_{\text{gas}} < 0$. The constrained optimum is a road that connects two urban areas, acceptable to the coalition, lowering the power Gini by 0.05. The gap of 0.15 is the political cost. The same coalition has no reason to block an intervention that raises integration while leaving $\boldsymbol{\pi}$ unchanged (Proposition 15.18).

The only way to close the gap is to change the coalition. Chapter 14 showed that coalitions change when $\boldsymbol{\pi}$ changes. And $\boldsymbol{\pi}$ changes slowly because W changes slowly (Chapter 13). The circle: W is hard to change because the powerful block change $\rightarrow$ the powerful are powerful because of $W \rightarrow W$ is hard to change. Breaking the circle requires an exogenous shock, an information intervention that shifts preferences and coalition incentives, or the slow accumulation of cross-block edges that eventually tips the power balance.

15.6 Summary

1. The **operator design problem** chooses ΔW subject to a budget, feasibility and row-stochasticity. Its objective must say what is targeted: *integration* (Γ), *power* ($\text{Var}(\boldsymbol{\pi})$, power Gini, d_π), or *output* (computed from A, not from W). These targets are independent: Γ can be raised to 1 along a path that leaves $\boldsymbol{\pi}$ unchanged.

2. The **first-order solution** uses the complex-safe gradient of $|\lambda_2|$. Under an ℓ_1 budget it moves weight within one row (a single new edge); under a Frobenius budget it spreads over all entries. The best new edges connect the *cores* of weakly coupled blocks, not their boundary agents.

3. **Five operator types**: ISI (replace foreign with domestic edges), infrastructure (physical edges), education (dual $W^P + W^I$ edges), information policy (W^I only), and institutional reform (change the rules of edge dynamics). **ISI that works** for integration and power creates cross-block edges (modern $\leftrightarrow$ traditional); ISI that deepens only the modern block can still win on the output multiplier.

4. **Sequencing** is conjectured to be Ψ-ordered: start with high-Ψ operators (information, institutions, education) that create the preconditions for low-Ψ operators (infrastructure, industrial policy). The product space constrains which ISI operators are feasible.

5. The **political constraint** never prevents integration (no coalition need lose power for Γ to rise); it bites on redistribution. The political cost of reducing $\mathrm{Var}(\boldsymbol{\pi})$ is $\Omega^*_{2,\mathrm{unc}} - \Omega^*_{2,\mathrm{con}}$.

15.7 Exercises

15.1 Optimal edge selection. For a 4×4 nearly block-diagonal W^P (two blocks of 2, cross-block weight 0.02 in each row; an exactly block-diagonal matrix has a repeated unit eigenvalue, so Γ is not differentiable there): compute the gradient G_{ij} of (15.5) for all (i, j) pairs. Which move of weight within a row maximizes $\Delta\Gamma$ per unit of ℓ_1 budget? Is it within-block or cross-block?

15.2 ISI-A vs. ISI-B. Reproduce Example 15.6: starting from the printed A, compute the effect on Γ, $\mathrm{Var}(\boldsymbol{\pi})$, and total output $\mathbf{1}'(I - A')^{-1}\mathbf{1}$ of (a) $\Delta a_{M1,M2} = 0.05$, (b) $\Delta a_{M1,T1} = 0.05$. Which objective does each win? Find an input intensity for the traditional sectors at which ISI-B also wins on output.

15.3 The smartphone operator. Reproduce Example 15.11: for a two-block economy, add $\Delta W^I = 0.05$ cross-block edges. Simulate the formation dynamics for 5 generations. Compute the steady-state $\Delta\theta$ and Δx. Why is Ψ not defined for this intervention?

15.4 Sequencing. For a stylized 6×6 Bolivia model (gas, urban, rural blocks), compare:

(a) Sequence A: ISI first, then education, then information.

(b) Sequence B: Information first, then education, then ISI.

(c) Simulate both sequences for 10 periods. Plot Γ_t for each. Which sequence reaches $\Gamma = 0.30$ first? (The target refers to the stylized model; Bolivia's measured domestic $\Gamma(W^P)$ is already 0.467.)

15.5 Bolivia in the product space. Using OEC data (`oec.world`) for Bolivia:

(a) List Bolivia's top 10 exports and their position in the product space (core vs. periphery).

(b) Compute relatedness density to three target products: lithium batteries, food processing machinery, pharmaceutical intermediates.

(c) For each target: what capability edges (ΔW^P) are needed? What information edges (ΔW^I)?

(d) Estimate the sequencing cost for reaching each target (order of magnitude).

15.6 Hirschman vs. spectral gap. For the Bolivia I-O table (from Exercise 1.7; the 12-sector domestic table for 2019 built from INE's supply-use tables is in `data/processed/bolivia_Z/`):

(a) Compute backward linkages (column sums of $(I - A')^{-1}$) from the technical coefficients A, and check that the same computation on $W^P = D^{-1}A$ is uninformative.

(b) Compute the gradient G_{ij} of $\Gamma(W^P)$ for all edges.

(c) Do the two criteria agree on where to invest? Where do they disagree?

(d) When would Hirschman's criterion lead to the wrong investment?

15.7 Political cost. For a stylized $N = 4$ Bolivia model: the winning coalition is {Gas, Government} ($\pi_1 + \pi_2 = 0.65$). (a) Show that for the objective Γ the political cost is zero once the budget reaches $\|\mathbf{1}\boldsymbol{\pi}' - W\|$ (Proposition 15.18). (b) For the objective $\mathrm{Var}(\boldsymbol{\pi})$, compute the unconstrained optimum and the constrained optimum (subject to $\Delta\pi_1 \geq 0, \Delta\pi_2 \geq 0$) under a budget $B = 0.1$, numerically. What is the political cost? What coalition change would reduce it?

Mathematical Appendices

Appendix 15A: Sensitivity of Γ to Edge Perturbations

Hypotheses. W is irreducible and row-stochastic, with eigenvalues ordered $1 = \lambda_1, |\lambda_2| \geq \cdots \geq |\lambda_N|$ and $\Gamma = 1 - |\lambda_2|$. λ_2 is simple, and no eigenvalue other than λ_2 and its conjugate has modulus $|\lambda_2|$. (Otherwise Γ is the maximum of several smooth branches and has only one-sided directional derivatives; this happens, e.g., at an exactly block-diagonal W or when $\lambda_2 = -\lambda_3$.)

Eigenvalue perturbation. By Lemma 6.19, for a perturbation $W + tE$,

$$\frac{d\lambda_k}{dt} = \mathbf{u}_k' E \mathbf{v}_k, \qquad \frac{d|\lambda_k|}{dt} = \frac{\mathrm{Re}\left(\bar{\lambda}_k \, \mathbf{u}_k' E \mathbf{v}_k\right)}{|\lambda_k|}, \tag{15.19}$$

where $\mathbf{u}_k, \mathbf{v}_k$ are left and right eigenvectors, normalized biorthogonally ($\mathbf{u}_k' \mathbf{v}_l = \delta_{kl}$, e.g. the rows of V^{-1}). The expression $-\mathrm{sign}(\lambda_2)\, d\lambda_2$ is correct only for real λ_2; for complex λ_2 it is wrong. In `code/ch15.py`, $\lambda_2 = -0.004 + 0.216i$: the finite difference and the formula above both give $d\Gamma/dt = -0.2213$, while the sign formula gives 0.3492.

The compensation direction. A "partial derivative" $\partial\Gamma/\partial W_{ij}$ is not defined on the set of row-stochastic matrices; only directional derivatives along E with $E\mathbf{1} = 0$ are. The direction must be stated. Two common choices:

- *Pairwise:* $E = \mathbf{e}_i(\mathbf{e}_j - \mathbf{e}_\ell)'$ moves weight from (i, ℓ) to (i, j); the rate is $G_{ij} - G_{i\ell}$, with G as in (15.5). Feasible if $W_{i\ell} > 0$.

- *Uniform:* $E_{ij} = 1$, $E_{i\ell} = -1/(N-1)$ for $\ell \neq j$; the rate is $G_{ij} - \frac{1}{N-1}\sum_{\ell \neq j} G_{i\ell}$. Feasible only if every $W_{i\ell} > 0$ — often violated in sparse input-output matrices, where only one-sided derivatives exist.

Interpretation. With $\lambda_2 > 0$, $G_{ij} = -\lambda_2 u_{2,i} v_{2,j}/|\lambda_2| = -u_{2,i} v_{2,j}$: the rate is large and positive when $u_{2,i}$ and $v_{2,j}$ are large in magnitude and of opposite sign. Both vectors are nearly constant on each weakly coupled block and close to zero on agents between blocks, so the optimal edge connects an agent deep in one block to an agent deep in the other — the cores, not the boundary.

Appendix 15B: Constrained Optimization on the Stochastic Matrix Space

The constraint set. The space of $N \times N$ row-stochastic nonnegative matrices is a convex polytope: $\mathcal{W} = \{W \geq 0 : W\mathbf{1} = \mathbf{1}\}$. The operator design problem is a constrained optimization on $\mathcal{W}$.

Projected gradient. The gradient of $\Gamma(W)$ may point outside $\mathcal{W}$ (violating nonnegativity or row-stochasticity). The projected gradient method iterates:

$$W_{t+1} = \text{Proj}_{\mathcal{W}} \left(W_t + \eta \nabla_W \Gamma(W_t) \right), \tag{15.20}$$

where $\text{Proj}_{\mathcal{W}}$ is the Euclidean projection onto $\mathcal{W}$. It acts row by row: each row is projected onto the probability simplex by sorting its entries and subtracting a common threshold, then clipping at zero (Duchi et al., 2008). "Set negatives to zero and renormalize" is a different, non-Euclidean map, and does not give a projected-gradient method. The gradient itself exists only under the hypotheses of Appendix 15A.

Sinkhorn projection. When the target is a doubly stochastic matrix — equivalently, uniform power $\boldsymbol{\pi} = \mathbf{1}/N$ (Chapter 6); W^{CE} is one such matrix, but not the only one — Sinkhorn's algorithm (alternating row and column normalization) converges, if W has total support, to the doubly stochastic matrix $D_1 W D_2$ that is closest to W in KL-divergence among those with the same zero pattern. It is a power-equalizing operator; it need not raise Γ.

Exact optimization of the gap. For symmetric W supported on a fixed graph, the second-largest eigenvalue modulus is a convex function of W, and minimizing it (maximizing Γ) is the fastest-mixing Markov chain problem, solvable as a semidefinite program (Boyd, Diaconis, and Xiao, 2004). For non-symmetric W, $|\lambda_2|$ is neither convex nor Lipschitz, and the first-order methods above are local.

Structuralist Case Study: Tavares's ISI as Structural Engineering

Maria da Conceição Tavares (1930–2024) was a Portuguese-born Brazilian economist who became one of the most influential structuralist thinkers of the twentieth century. Her essay on the rise and decline of import substitution in Brazil, first published in Spanish in CEPAL's *Boletín Económico de América Latina* (1964) and reprinted in Portuguese as "Auge e declínio do processo de substituição de importações no Brasil" in her collection *Da substituição de importações ao capitalismo*

financeiro (1972), provided the definitive analytical autopsy of Brazilian ISI — not as a policy failure but as a **specific operator applied to W^P that succeeded at its immediate objective and failed at its structural one**.

Tavares's argument has three parts. Parts 1 and 2 are hers; Part 3 is this book's reinterpretation of her diagnosis in the chapter's framework.

Part 1: ISI was a response to a balance-of-payments constraint, not a development strategy. Tavares argued that Brazil's ISI was not a deliberate plan to industrialize but a response to *estrangulamento externo* — the recurrent external strangulation of the capacity to import: the Great Depression collapsed export revenues, WWII interrupted imports, and the Korean War destabilized commodity prices. Each crisis forced the domestic production of goods previously imported. In operator-design notation: each crisis was an exogenous edge destruction in W^P_{global} (severed import edges), which triggered compensating edge creation in W^P_{domestic} (new domestic supply chains). The ISI operator $\mathcal{L}_{\text{ISI}}$ was applied reactively, not optimally.

Part 2: Each round of substitution recreated the constraint. The import-substitution process successfully built a modern industrial sector (automobiles, steel, chemicals, machinery). But each round of substitution shifted the composition of imports toward intermediate and capital goods that the new industries needed, so that imports became less compressible and the external constraint returned in a new form. Tavares also stressed the limits set by the size and distribution of the internal market and by financing. The new industrial sector depended on *itself* and on imported capital goods, and connected weakly to the pre-existing traditional economy. In operator-design terms: ISI applied $\mathcal{L}_{\text{ISI}}$ of type A (within-block), not type B (cross-block). The modern block densified internally ($W^P_{M,M}$ increased), but the edges between the modern and traditional blocks ($W^P_{M,T}$) remained dark. Pinto's structural heterogeneity was not resolved by ISI — it was *replicated at a higher level of complexity*. The tripartite economy (modern/intermediate/traditional) became a quadripartite economy (ISI-modern/export-modern/intermediate/traditional).

Part 3 (our reinterpretation): The "decline" of ISI as a structural property. Tavares did not use input-output multipliers or spectral language. In this chapter's terms, her diagnosis says that ISI's growth slowed not because of "inefficiency" (the later neoliberal critique) but because the domestic multiplier ran into a limit: once the easy substitutions were made (consumer goods, light industry), the remaining imports (capital goods, intermediate inputs) required capabilities that could only be acquired through cross-block linkages that ISI had not created. The Leontief inverse $(I - A')^{-1}$ had bright entries *within* the modern

block but dark entries *between* the modern and traditional blocks, so the multiplier plateaued, and the integration of the domestic economy (Γ of W^P) barely rose. Tavares herself did not conclude that stagnation was permanent: with Serra ("Más allá del estancamiento," 1971) she argued against the stagnationist thesis, showing how a concentrated income distribution could sustain a new growth pattern — a reminder that integration and distribution are separate coordinates.

Tavares's argument (1964/1972)	**Formal object**	**This chapter**
ISI as response to external strangulation	Reactive $\mathcal{L}_{\text{ISI}}$, not optimal	Def. 15.4
Constraint recreated by import composition	Foreign edges shift to capital goods	Def. 15.4
New sector without cross-links	ISI-A (within-block), not ISI-B (cross)	Ex. 15.6
Structural heterogeneity replicated	Γ_{domestic} barely rises	Prop. 15.5
Multiplier plateau (our reading)	$(I - A')^{-1}$ with dark cross-entries	Ex. 15.6
Cross-block ISI needed	ISI-B: modern inputs from traditional sector	Ex. 15.6

Tavares's essay appeared in 1964 and was reprinted in the 1972 collection. She later served as a federal deputy in Brazil and continued to influence economic policy until her death in 2024. Read through this chapter's lens, her diagnosis of ISI is that it ran into limits not because it was wrong in principle but because the operator was applied to the wrong entries of the matrix, under a binding external constraint. The operator-design framework of this chapter is, in a sense, the engineering manual that Tavares's diagnosis called for: which edges should the operator target, for which objective, in what sequence, and under what political constraints?

Further Reading

Industrial policy: Liu (2019), "Industrial Policies in Production Networks" — the rigorous version of this chapter's question: which sectors to subsidize, derived from the Leontief inverse. Juhász, Lane, and Rodrik (2024), "The New Economics of Industrial Policy." Lane (2025), "Manufacturing Revolutions: Industrial Policy and Industrialization in South Korea." Rodrik (2004), "Industrial Policy for

the Twenty-First Century." Rodrik (2008), "Normalizing Industrial Policy." Hausmann and Rodrik (2003), "Economic Development as Self-Discovery." Hausmann, Hwang, and Rodrik (2007), "What You Export Matters." Chang (2002), *Kicking Away the Ladder*.

The product space and structural transformation: Hidalgo et al. (2007), "The Product Space Conditions the Development of Nations." Hidalgo and Hausmann (2009), "The Building Blocks of Economic Complexity." Hausmann et al. (2014), *The Atlas of Economic Complexity*.

Optimal network design: Jackson (2008), *Social and Economic Networks*, Chapter 11. For spectral methods in network design: Boyd, Diaconis, and Xiao (2004), "Fastest Mixing Markov Chain on a Graph." For projection onto the simplex: Duchi, Shalev-Shwartz, Singer, and Chandra (2008), "Efficient Projections onto the ℓ_1-Ball for Learning in High Dimensions."

Linkages: Hirschman (1958), *The Strategy of Economic Development*. Rasmussen (1956), *Studies in Inter-Sectoral Relations*. Chenery and Watanabe (1958), "International Comparisons of the Structure of Production." Rosenstein-Rodan (1943), "Problems of Industrialisation of Eastern and South-Eastern Europe."

The structuralist tradition on ISI: Tavares (1964), "Auge y declinación del proceso de sustitución de importaciones en el Brasil." Tavares and Serra (1971), "Más allá del estancamiento." Tavares (1972), *Da substituição de importações ao capitalismo financeiro* (the primary source). Furtado (1959), *Formação Econômica do Brasil*, Chapters 30–33. Hirschman (1968), "The Political Economy of Import-Substituting Industrialization in Latin America."

South Korea's industrial policy: Amsden (1989), *Asia's Next Giant: South Korea and Late Industrialization*. Wade (1990), *Governing the Market*. For the reciprocal control mechanisms (the state disciplines firms, not just subsidizes them): Amsden's concept maps directly to $\mathcal{L}_{\text{inst}}$ with δ_1 (destruction of inefficient edges) as the enforcement mechanism.

PART V

Three Scales

Chapter 16

The Global Weight Matrix

Center, Periphery, and the Spectral Structure of World Trade

> *"Las ventajas del progreso técnico se han concentrado principalmente en los centros industriales, sin propagarse a la periferia del sistema económico mundial en medida suficiente."*
>
> ["The advantages of technical progress have concentrated principally in the industrial centers, without propagating to the periphery of the world economic system in sufficient measure."]
> — Raúl Prebisch, *El desarrollo económico de la América Latina y algunos de sus principales problemas* (1949)

> *"Development depends not so much on finding optimal combinations for given resources and factors of production as on calling forth and enlisting for development purposes resources and abilities that are hidden, scattered, or badly utilized."*
> — Albert O. Hirschman, *The Strategy of Economic Development* (1958)

Part V applies the framework to three scales: global (this chapter), national (Chapter 17), and individual (Chapter 18). At each scale, the same mathematics operates: the weight matrix W, the Perron eigenvector $\boldsymbol{\pi}$, the spectral gap Γ, and the formation model. What changes is the data.

This chapter takes W^P_{global}: the weight matrix of international trade, where countries are nodes and trade flows are edges. Its Perron vector measures how concentrated structural power is among countries — Prebisch's center-periphery hypothesis, in one of its dimensions. Its spectral gap measures something else: how fast shocks mix across the world economy. The Economic Complexity Index

(Hidalgo and Hausmann, 2009) is a spectral object too, but of a different matrix, built from countries' export baskets rather than from bilateral trade. Hirschman's linkages are column and row sums of the Leontief inverse of the inter-country *coefficient* matrix A, not of W.

Everything in Chapters 1–15 applies. The only difference is that N is now 200 countries rather than 4 agents, and W is estimated from trade data rather than assumed.

16.1 The Global Production Network

16.1.1 From Bilateral Trade to the Weight Matrix

Definition 16.1 (Global Production Weight Matrix)**.** *Let $K_{cc'}$ be the value (in USD) of country c's imports from country c', for $c \neq c'$. The **global production weight matrix** is $W^P = D^{-1}K$ with $D = \mathrm{diag}(K\mathbf{1})$:*

$$W^P_{cc'} = \frac{K_{cc'}}{\sum_{c''} K_{cc''}}. \tag{16.1}$$

Entry $W^P_{cc'}$ is the share of country c's imports that comes from country c', so W^P is row-stochastic. Two treatments of the diagonal must be distinguished:

- Imports only *(BACI, COMTRADE): $K_{cc} = 0$, so $W^P_{cc} = 0$.*
- Absorption shares *(inter-country input-output tables such as OECD ICIO): K_{cc} is domestic purchases, so W^P_{cc} is the domestic share of absorption, typically far larger than any import share. This W^P is strongly diagonal, and its Γ is much smaller than that of the imports-only matrix.*

Every reported Γ or $\boldsymbol{\pi}$ must state the source, year, country aggregation and diagonal treatment.

For $N \approx 200$ countries and data from the BACI trade database (imports only) or an inter-country input-output table (absorption shares), W^P_{global} is a row-stochastic matrix of that size. Aggregating countries into regions changes both Γ and $\boldsymbol{\pi}$ (Exercise **16.1**).

Remark 16.2 (Properties of W^P_{global}). The numbers in this remark are for 2019, from the Growth Lab bilateral goods-trade data (HS92), imports only (zero diagonal), 230 economies (The Growth Lab at Harvard University, 2026); author's calculations (`data/compute/compute_global_wp.py`). Exercise **16.1** asks you to recompute them from BACI.

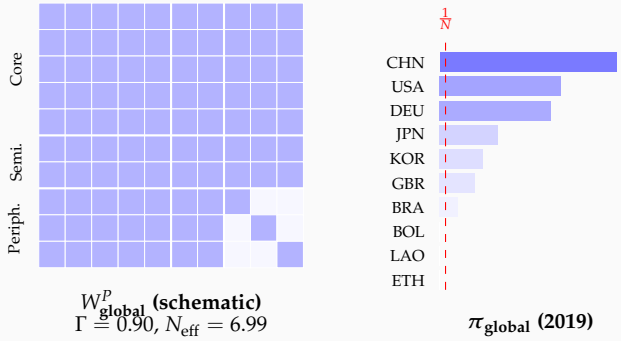
1. **Not doubly stochastic.** The column sums of any row-stochastic $N \times N$ matrix average exactly 1; here they vary by orders of magnitude, because large exporters appear in almost every country's import basket and small exporters in few. The column sum of the US is the sum, over importers, of the US share of their imports: 24.2 in 2019 (China 30.7, Germany 11.1). Bolivia's is 0.10. The world does not depend on Bolivia the way Bolivia depends on the world.

2. **Core-periphery topology.** A dense core of industrialized countries (high mutual trade) and a sparse periphery of most of the rest (trade mostly with the core, rarely with each other). The size of the core depends on the threshold used to define it.

3. **Concentrated Perron vector.** China has $\pi = 0.130$, the US 0.089 and Germany 0.082. The top 10 countries (China, the US, Germany, Japan, France, Korea, the Netherlands, Italy, the UK and Russia) hold 51.9% of structural power, and $\pi_{\text{Bolivia}} = 0.00053$. With $N = 230$, $N_{\text{eff}} = 22.6$ and the power Gini is 0.85. Concentration is what N_{eff} and $\text{Var}(\boldsymbol{\pi})$ measure.

4. **The spectral gap measures frictions, not power.** A core-periphery structure in which everyone buys mainly from the core is close to rank one, and rank-one matrices have $\Gamma = 1$ for any $\boldsymbol{\pi}$. In frictionless gravity, every importer's shares equal the exporters' world shares $\mathbf{s}$: $W^P = \mathbf{1}\mathbf{s}'$, with $\Gamma = 1$ and $\boldsymbol{\pi} = \mathbf{s}$, as concentrated as world exports. The measured $\boldsymbol{\pi}$ is indeed close to $\mathbf{s}$ (China 0.130 against an export share of 0.142; the US 0.089 against 0.091; Bolivia 0.00053 against 0.00046), but the measured gap is $\Gamma_{\text{global}} = 0.452$, far below 1. That gap reflects distance, trade costs and regional blocs — how far the world is from anonymous market interaction (Chapter 1, Section 1.7) — not how unequal power is (Proposition 6.16).

Visualization: The global W^P: a 10×10 schematic, and the measured Perron vector



Left: schematic global W^P (row-normalized cell values in `code/ch16.py`). The upper-left block (core) is bright: dense mutual trade. The lower-right (periphery) is dark except for column entries toward the core: the periphery depends on the core, not on each other. The lower-left rectangle is brighter than the upper-right: asymmetric dependence. This hub structure is close to rank one, so its spectral gap is *large* ($\Gamma = 0.90$): everyone mixes quickly through the core. Its power is concentrated: the five core countries hold 0.163 each, the periphery 0.016 each. The measured 2019 matrix has $\Gamma = 0.452$: distance and regional blocs keep it further from rank one than the schematic. Right: the measured Perron vector for ten of the 230 economies. The dashed red line marks $1/N = 0.0043$. China has 30 times the "fair share" of structural power (0.130×230), the US 20 times. Bolivia (0.00053) is barely visible.

Source (right panel): Growth Lab bilateral goods trade, HS92, 2019 (The Growth Lab at Harvard University, 2026); imports-only W^P; author's calculations (`data/compute/compute_global_wp.py`). Left panel: schematic.

16.1.2 The Global Leontief Inverse

Definition 16.3 (Global Leontief Inverse and Linkages)**.** *Let A be the inter-country input-output coefficient matrix, with the Chapter 1 convention: $a_{cc'}$ is the input from country c' needed per unit of country c's output (row c is c's recipe), and $\mathbf{y} = A'\mathbf{y} + \mathbf{d}$. The **global Leontief inverse** is $L = (I - A')^{-1}$, which exists with $L \geq I$ when $\rho(A) < 1$. Entry $\ell_{cc'}$ is the gross output generated in country c by a unit of final demand for country c''s output, including all rounds of indirect effects through the global supply chain.*

1. *Country c''s **backward linkage** is $BL_{c'} = \sum_c \ell_{cc'}$ (column sum): the total gross output generated by one unit of final demand for c'. High BL: c' uses many intermediate inputs, domestic or foreign.*

2. *Country c's* ***forward linkage*** *is* $FL_c = \sum_{c'} \ell_{cc'}$ *(row sum): how much of c's output is required, directly and indirectly, when every country's final demand rises by one unit. (The Ghosh supply-side inverse is the usual alternative for forward linkages.)*

3. *The* ***key sector*** *(Rasmussen, 1956) has both* $BL > \bar{BL}$ *and* $FL > \bar{FL}$*: it generates demand upstream AND supplies inputs downstream.*

Since $L \geq I$*, every BL and FL is at least* 1*, and* $\bar{BL} = \bar{FL}$ *(both equal the sum of all entries of L divided by N). These indices must be computed from A: for the row-stochastic* W^P*,* $(I - \gamma W^P)^{-1}\mathbf{1} = \mathbf{1}/(1-\gamma)$*, so every row sum would be the same.*

Example 16.4 (Global Linkages [ILLUSTRATIVE])**.** *Illustrative country-level linkages, of the kind one obtains by aggregating an inter-country input-output table to countries (Bolivia and Ethiopia are not separate economies in OECD ICIO, which places them in the Rest of the World; for them a table such as Eora would be needed):*

Country	*BL*	*FL*	*Type*
United States	*2.45*	*1.98*	*High/High: key sector (global hub)*
China	*2.68*	*2.31*	*High/High: manufacturing hub*
Germany	*2.51*	*2.12*	*High/High: machinery hub*
Japan	*2.33*	*1.85*	*High/High*
Bolivia	*1.28*	*1.12*	*Low/Low: peripheral enclave*
Ethiopia	*1.15*	*1.08*	*Low/Low*
Average (all economies)	*1.85*	*1.85*	*equal by construction*

The center has BL and FL both above average: it generates demand AND supplies inputs. The periphery has both below average: its demand doesn't propagate, and its output isn't widely used. In this illustration, a BL of 1.28 *would mean that one dollar of final demand for the country's output generates only $1.28 of gross output worldwide, including the dollar itself (vs. $2.68 for China).*

The Bolivia and Ethiopia rows are placeholders, not estimates. What the INE supply-use tables do give is Bolivia's domestic *linkages (2019, base 2017, 12 sectors, domestic flows at basic prices; not comparable with the global figures above): the sector mean is* $\bar{BL} = 1.445$*, and the low-linkage activity is the export enclave, not the economy as a whole. Hydrocarbons have* $BL = 1.34$ *and* $FL = 1.20$*, both below the mean; they buy* 0.225 *of domestic intermediate inputs per unit of output (manufacturing:* 0.558*) and export* 70% *of their output (Instituto Nacional de Estadística (INE), Bolivia, 2026); author's calculations (*`data/compute/compute_bolivia_io.py`*).*

16.2 Economic Complexity as Spectral Measurement

16.2.1 From Trade Data to the Specialization Matrix

The Economic Complexity Index (ECI) is a dimensionality-reduction method that extracts the leading non-trivial spectral structure from the country–product export matrix. It uses no bilateral information — who sells to whom — so it is not a function of W^P_{global}; it is a spectral summary of a different matrix, built from what each country exports.

Definition 16.5 (Specialization Matrix and RCA)**.** *Let X_{cp} be the export value of product p from country c. The **Revealed Comparative Advantage** is:*

$$R_{cp} = \frac{X_{cp}/X_c}{X_p/X}, \tag{16.2}$$

where $X_c = \sum_p X_{cp}$, $X_p = \sum_c X_{cp}$, and $X = \sum_{c,p} X_{cp}$. A country has comparative advantage in product p if $R_{cp} > 1$. The binary specialization matrix: $M_{cp} = \mathbf{1}[R_{cp} \geq 1]$.

Definition 16.6 (The Proximity Matrix and Product Space)**.** *The **proximity** between products p and p' is:*

$$\phi_{pp'} = \frac{\sum_c M_{cp} M_{cp'}}{\max(M_p, M_{p'})}, \tag{16.3}$$

*where $M_p = \sum_c M_{cp}$ is the ubiquity of product p. The **product space** is the network with products as nodes and $\phi_{pp'}$ as edge weights.*

The product space is a network of *capability similarity* between products: two products are close when the same countries tend to export both. It is not an input-output network, and it is not W^P at the product level. Its topology — dense core of machinery/chemicals/electronics, sparse periphery of raw materials — was first mapped for about 775 products (Hidalgo et al., 2007) and is now drawn over 5,000+.

16.2.2 The ECI as the Second Eigenvector

Definition 16.7 (Economic Complexity Index)**.** *Define the country-country matrix:*

$$\tilde{M}_{cc'} = \sum_p \frac{M_{cp} M_{c'p}}{M_c \cdot M_p}, \tag{16.4}$$

where $M_c = \sum_p M_{cp}$ is diversity. $\tilde{M}_{cc'}$ measures the similarity between countries c and c' based on their shared products, weighted by product ubiquity. This matrix is row-stochastic.

*The **ECI** is the second eigenvector of $\tilde{M}$:*

$$\tilde{M}\mathbf{v}_2 = \lambda_2 \mathbf{v}_2, \qquad ECI_c = \frac{v_{2,c} - mean(v_2)}{stdev(v_2)}. \tag{16.5}$$

*The first eigenvector is trivial (**1**, since $\tilde{M}$ is row-stochastic). The second eigenvector is the leading non-trivial spectral structure of the export matrix; its sign is chosen so that ECI correlates positively with diversity.*

Proposition 16.8 (Spectral Facts About $\tilde{M}$)**.** *Let $D_c = \mathrm{diag}(M_c)$ and $D_p = \mathrm{diag}(M_p)$, with all diversities and ubiquities positive. Then:*

1. *$\tilde{M} = D_c^{-1} M D_p^{-1} M'$ is row-stochastic, and its eigenvalues are real and lie in $[0, 1]$: they are the squared singular values of $S = D_c^{-1/2} M D_p^{-1/2}$.*
2. *The diversity shares $M_c / \sum_{c'} M_{c'}$ form a stationary (left Perron) vector of $\tilde{M}$; if $\tilde{M}$ is irreducible (the bipartite country–product graph of M is connected), it is the unique one, $\boldsymbol{\pi}(\tilde{M})_c = M_c / \sum_{c'} M_{c'}$. In the "power" reading of this book, the structural power of a country in $\tilde{M}$ is its diversity share.*
3. *ECI is a z-score, so its cross-country variance is 1 by construction and carries no information about inequality. (ECI is well defined, up to the sign convention, when $\lambda_2(\tilde{M})$ is simple.)*

These facts are known: Mealy et al. (2019) show that ECI is the second eigenvector of the same $\tilde{M}$ and interpret it as a spectral-clustering coordinate.

Proof. (1) Row sums: $\sum_{c'} \tilde{M}_{cc'} = \sum_p \frac{M_{cp}}{M_c} \sum_{c'} \frac{M_{c'p}}{M_p} = 1$. Also

$$SS' = D_c^{-1/2} M D_p^{-1} M' D_c^{-1/2}, \qquad \tilde{M} = D_c^{-1/2} (SS') D_c^{1/2},$$

so $\tilde{M}$ is similar to the positive semidefinite matrix SS', whose eigenvalues are the squared singular values of S, real and nonnegative. Since $\tilde{M}$ is nonnegative and row-stochastic, its spectral radius is 1, so they are at most 1. (2) For each c',

$$\sum_c M_c \tilde{M}_{cc'} = \sum_p \frac{M_{c'p}}{M_p} \sum_c M_{cp} = \sum_p M_{c'p} = M_{c'}.$$

Uniqueness under irreducibility is Perron–Frobenius (Seneta, 2006, Thm. 1.5). (3) By definition (16.5). `code/ch16.py` checks (1)–(2) numerically. □

Interpretation: ECI orders countries along the principal axis of differentiation in export baskets. High ECI = diverse, complex exports (machinery, chemicals, pharmaceuticals). Low ECI = concentrated, simple exports (oil, minerals, raw

agriculture). $\lambda_2(\tilde{M})$ close to 1 means that axis is strong (countries split sharply into two groups of baskets); close to 0 means baskets are similar. The gap $1 - \lambda_2(\tilde{M})$ is the spectral gap of $\tilde{M}$, a different matrix from W^P_{global}; it says nothing about the integration of world trade.

Intuition: What the ECI measures, in this book's language

The ECI measures where a country sits on the second eigenvector of the export-similarity matrix $\tilde{M}$ — the axis that captures the most information about who exports what. A country with ECI > 0 exports what the diversified core exports; a country with ECI < 0 exports what the periphery exports. Its link to power is indirect: in $\tilde{M}$ itself, power is diversity.
Japan is at the top of the ranking: diversified and complex (ECI 2.05, rank 1 in 2019; Korea 1.81, rank 2). Bolivia is near the bottom: concentrated and simple (ECI -1.30, rank 131), and it has fallen since 2006 (-1.21) (The Growth Lab at Harvard University, 2026). The gap between them is a property of export baskets, which shapes — but is not the same thing as — the country's position in W^P_{global}.

16.2.3 ECI Predicts Growth: A Structural Explanation

Empirical Result 16.9 (ECI Predicts Growth)**.** *ECI predicts future growth of GDP per capita, controlling for current income (Hidalgo and Hausmann, 2009; Hausmann et al., 2014).*

Conjecture 16.10 (A Structural Explanation)**.** *The predictive power of ECI works through three channels:*

1. *High ECI goes with stronger domestic inter-sectoral linkages, so shocks and demand propagate across sectors (Chapter 3); this should show up as a higher domestic $\Gamma(W^P)$ and higher multipliers computed from A.*

2. *High ECI goes with a diverse product portfolio, hence more independent signals in W^I and a higher N_{eff} (Chapter 8).*

3. *High ECI goes with a dense position in the product space, hence many feasible diversification paths (Chapter 13).*

The standard explanation is that ECI proxies for human capital and institutions. The conjecture is that it also proxies for the integration of the domestic

production network. Unlike the standard explanation, this one is testable with national input-output tables: compute domestic Γ and multipliers and check whether they mediate the ECI–growth relation (Exercise **16.6**).

A first look does not support channel 1 at the 12-sector level. On a common basis (12 ISIC-based sectors, domestic flows, input shares, 2019), Bolivia's domestic $\Gamma(W^P)$ is 0.475 with an ECI of -1.30, against Korea's 0.511 with an ECI of 1.81; the US has 0.463, Germany 0.544, Chile 0.557, China 0.582 and Mexico 0.649 (Instituto Nacional de Estadística (INE), Bolivia, 2026; OECD, 2023); author's calculations (`data/compute/compute_oecd_io.py`). Bolivia's own Γ ranges from 0.33 to 0.69 across normalizations, a wider spread than the cross-country one. Aggregate Γ does not separate these development paths. Whether multipliers, or Γ at finer sector detail, do so is the part of channel 1 that remains open.

16.3 ECI and Inequality

Empirical Result 16.11 (ECI and Inequality)**.** *Countries with high ECI have lower income inequality (Gini), controlling for GDP per capita (Hartmann et al., 2017, who report robustness across specifications and data sources).*

Remark 16.12 (What the framework does and does not explain). It is tempting to explain this association with a "spectral bound": high ECI $\Rightarrow$ high $\Gamma \Rightarrow$ low $\mathrm{Var}(\boldsymbol{\pi})$ $\Rightarrow$ low inequality. That chain fails at its second link. There is no bound of $\mathrm{Var}(\boldsymbol{\pi})$ by Γ: the matrix $W = \gamma \mathbf{1}\boldsymbol{\pi}' + (1-\gamma)I$ has $\Gamma = \gamma$ and Perron vector $\boldsymbol{\pi}$ for *any* $\boldsymbol{\pi}$ (Proposition 6.16); at $\gamma = 1$ a near-dictatorship has $\Gamma = 1$. The chain also mixes levels: $\boldsymbol{\pi}$ of W^P_{global} describes inequality *between* countries, while the Gini coefficient describes inequality *within* them.

What the framework does supply is the exact link between *domestic* power concentration and distributional loss under the allocation rule of Chapter 7: $L_{\text{power}} = \frac{\kappa N}{2}\Phi^2 \mathrm{Var}(\boldsymbol{\pi})$, with $\mathrm{Var}(\boldsymbol{\pi}) = \frac{1}{N}\left(\frac{1}{N_{\text{eff}}} - \frac{1}{N}\right)$. A structural explanation of the ECI–inequality association therefore has to run through the domestic power vector — for example, a diversified production structure spreading $\boldsymbol{\pi}$ across more sectors and groups (higher N_{eff}) — and that is an empirical hypothesis, testable with the diagnostics of Chapter 17.

For the production network the hypothesis meets an obstacle in the data. On the common 12-sector basis of 2019, sectoral power in W^P is *less* concentrated in Bolivia (power Gini 0.435, $N_{\text{eff}} = 6.67$ of 11) than in Korea (0.658, 3.62) or Germany (0.611, 3.89) (Instituto Nacional de Estadística (INE), Bolivia, 2026; OECD, 2023). In industrial economies $\boldsymbol{\pi}(W^P)$ concentrates on manufacturing and business-service

hubs, which is industrial depth, not oppression. A distributional channel, if there is one, runs through bargaining weights over groups (W^U, W^I, under Assumption 7.A), not through the sectoral $\boldsymbol{\pi}$ of W^P.

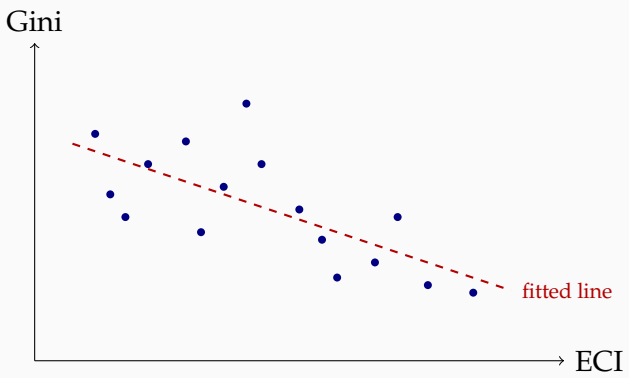
Stylized Fact 16.13 (Bolivia, 2005–2019: equity gains without complexity gains)**.** *Bolivia's Gini of per-capita household income fell from* 0.600 *in 2005 to* 0.426 *in 2019, and extreme poverty from* 38.2% *(2005) to* 15.2% *(2018, on the old poverty lines; INE, Encuesta de Hogares, cuadros 3.06.01.02–03). Over the same years its ECI fell from* -1.21 *(2006) to* -1.30 *(2019) (The Growth Lab at Harvard University, 2026), and its domestic* $\Gamma(W^P)$ *stayed roughly flat (*0.518 *in 2006 and* 0.484 *in 2013, 12 sectors, INE base 1990; Instituto Nacional de Estadística (INE), Bolivia 2026; author's calculations,* `data/compute/compute_bolivia_io.py`*). A cross-country association between complexity and inequality is not a law for change within a country: in Bolivia, large equity gains did not require rising ECI or rising* Γ*.*

Prebisch Connection: Fajnzylber's casillero vacío

In 1990, Fernando Fajnzylber published "Industrialización en América Latina: de la 'caja negra' al 'casillero vacío' " documenting that no Latin American country occupied the cell of a 2×2 table that combines high growth with equity — the "casillero vacío" (empty box). Several countries outside the region, such as South Korea, did occupy it. The empty box is defined in the space of *growth* $\times$ *equity*, not of complexity $\times$ inequality.

Chapter 17 develops Fajnzylber's argument as its case study. Its link to this chapter is the empirical association above: both high growth (Empirical Result 16.9) and lower inequality (Empirical Result 16.11) go with high ECI, which is why raising complexity — changing the production structure through the operators of Chapter 15 — is the structuralist route into the box. It is not the only route on record. Bolivia grew at 4.52% a year over 2006–2019 (World Bank, 2026b) while its Gini fell from 0.592 to 0.426 and its ECI declined (Stylized Fact 16.13). Whether that record is durable without structural change is the question Chapter 17 takes up.

Visualization: Complexity and inequality: a schematic



Schematic only (illustrative, anonymous points): the negative association between complexity and inequality documented by Hartmann et al. (2017). There is no spectral "fence": the dispersion around the line is large, and nothing in the framework rules out low-complexity, low-inequality outcomes. Bolivia moved toward that quadrant over 2005–2019 (Stylized Fact 16.13). Fajnzylber's casillero vacío lives in a different plane, growth × equity (Chapter 17).

16.4 Relatedness and the Global W^P Dynamics

16.4.1 The Principle of Relatedness as Global Path Dependence

Definition 16.14 (Relatedness Density)**.** *Country c's* ***relatedness density*** *to product p is:*

$$\omega_{cp} = \frac{\sum_{p'} M_{cp'} \phi_{pp'}}{\sum_{p'} \phi_{pp'}}. \tag{16.6}$$

This is the fraction of related products that c already produces. High ω_{cp}: country c has the capabilities to produce p. Low ω_{cp}: the capabilities are missing.

Stylized Fact 16.15 (The Principle of Relatedness)**.** *The probability that country c develops RCA in product p (enters the product) is increasing in ω_{cp} (Hidalgo et al., 2007). Countries diversify into products related to their current exports, not into random products.*

This is what the path-dependence conjecture of Chapter 13 predicts when edge creation is proximity-dominated ($p_1 \gg p_2, p_3$): new edges form near existing edges. The global consequence: countries on the dense core of the product space (high ω_{cp} for many products) diversify rapidly and raise their ECI; countries on the sparse periphery (low ω_{cp} for core products) diversify slowly or not at all, and remain at low ECI.

Example 16.16 (Relatedness Density: Bolivia vs. South Korea [ILLUSTRATIVE]).

Target product	$\omega_{Bolivia}$	ω_{Korea}
Lithium batteries	*0.04*	*0.52*
Pharmaceutical intermediates	*0.03*	*0.41*
Automotive parts	*0.02*	*0.68*
Processed food machinery	*0.08*	*0.45*
Average (all core products)	*0.06*	*0.55*

The ω values are placeholders, not estimates: computing them needs the country $\times$ product RCA data and the product-space proximities, which are not used in this book. In these illustrative numbers Korea's average relatedness to core products is 9 times Bolivia's. (Korea already has comparative advantage in several of these products; its column illustrates the density of a country inside the core.) Korea can diversify into products like these with high probability; Bolivia cannot, without the sequenced operators from Chapter 15 (information $\to$ institutions $\to$ education $\to$ infrastructure $\to$ industrial policy).

The complexity data that are available point the same way. In 2019 the ECI was -1.30 for Bolivia and 1.81 for Korea. The product complexity index (PCI) of electric accumulators (HS 8507, which includes lithium-ion batteries) was 0.55, against -0.57 for lithium carbonate (HS 2836) and -2.23 for natural gas (HS 2711) (The Growth Lab at Harvard University, 2026).

16.5 The Terms of Trade: A Spectral Conjecture

Conjecture 16.17 (Prebisch–Singer as a Spectral Hypothesis). *Apply the allocation rule of Chapter 7 (Proposition 7.3) to countries, reading $\boldsymbol{\pi}(W^P_{global})$ as bargaining weights (an assumption, not a result):*

$$x_c^* = \frac{\Omega_{global}}{N} + \Big(\pi_c(W^P_{global}) - \frac{1}{N}\Big)\Phi. \tag{16.7}$$

Countries with high structural power ($\pi_c > 1/N$) receive more than an equal share of global output; countries with low power ($\pi_c < 1/N$) receive less. The conjecture is that the terms-of-trade deterioration of primary exporters is the dynamic counterpart: as the global W^P evolves (Chapter 13), the core's π rises (the core adds dense linkages) and the periphery's π falls.

Two caveats keep this a conjecture. First, x_c is a share of output, not a relative price: a theory of the terms of trade needs prices, which (16.7) does not contain. Second, Prebisch's own mechanisms are different and more specific: low income elasticities of demand for primary goods, and asymmetric behaviour of wages and

prices over the cycle, so that technical progress in the center is absorbed as higher incomes and in the periphery as lower prices. The empirical record on the long-run trend is itself debated (Grilli and Yang, 1988; Harvey et al., 2010; Arezki et al., 2014). A spectral version would have to derive relative prices from W and show that they move with $\boldsymbol{\pi}$.

Example 16.18 (Bolivia's Terms of Trade)**.** *Bolivia's commodity export prices (gas, tin, zinc, soy) have experienced long-run terms-of-trade deterioration interspersed with commodity booms (1970s, 2003–2014). Because W^P_{global} is built from trade* values*, a price boom does change it: Bolivia's shares in its partners' imports rise, and so does $\pi_{Bolivia}$. The trade data show exactly this: $\pi_{Bolivia}$ was* 0.00017 *in 2000,* 0.00047 *in 2006 and* 0.00071 *in 2014, then fell back to* 0.00051 *in 2018 and* 0.00053 *in 2019. Fuels were* 48.8% *of merchandise exports in 2006,* 54.5% *in 2013 and* 32.1% *in 2019. The boom changed only the weights on existing edges — no new products, no new partners — and the gain reversed with prices. After booms, $x_{Bolivia}$ returns toward the pre-boom trajectory.*

The gas nationalization of 2006 rerouted gas revenue from MNCs to the treasury — a fiscal linkage — but it did not add cross-sector domestic production linkages. Bolivia's ECI was −1.21 *in 2006 and* −1.46 *in 2018 (*−1.30 *in 2019): over twelve years of gas revenue redistributed through fiscal channels, the export structure did not diversify; by this measure it became less complex.*

Sources: $\pi_{Bolivia}$, imports-only trade W^P from Growth Lab bilateral trade (HS92); ECI, Growth Lab HS92 (The Growth Lab at Harvard University, 2026); author's calculations (`data/compute/compute_global_wp.py`, `compute_eci_facts.py`). Fuel export shares: WDI `TX.VAL.FUEL.ZS.UN` (World Bank, 2026b).

16.6 The Three Networks at Global Scale

The global economy has three networks, just as the domestic economy does.

Remark 16.19 (Global Three-Network Structure).

1. W^P_{global}: the trade/production network. Measured by ICIO, BACI, or COMTRADE data. Core-periphery topology. (ECI is not computed from it, but from export baskets.)

2. W^I_{global}: the global information network. Measured (imperfectly) by media flows, academic citation patterns, policy diffusion. Hub-spoke topology centered on the US and Europe. The Washington Consensus as DeGroot equilibrium (Chapter 8). N_{eff} of global economic policy, counting institutions as nodes: illustratively 3–5 (IMF, World Bank, US Treasury, ECB, a few think tanks).

3. W^U_{global}: the global utility network. Harder to measure. Includes foreign aid (revealed altruism), migration patterns (revealed social bonds), and cultural

consumption (whose stories do you watch?). Also hub-spoke, correlated with W^I.

The three networks are **aligned at the global scale**: the countries that are central in W^P (trade hubs) are also central in W^I (information hubs) and W^U (cultural hubs). If the three layers are defined on the same node set (countries), this alignment reinforces the concentration of the composite Perron eigenvector (Chapter 5): the center is powerful in all three dimensions simultaneously.

Key Structural Insight

The global spectral structure tells the same story at all three layers. W^P: the center produces the goods. W^I: the center produces the ideas. W^U: the center produces the culture. The periphery consumes all three and produces the raw materials. The Perron eigenvector is aligned across layers. Reducing that concentration — raising N_{eff} of the global composite — is a distributional objective, separate from integration: the world is already integrated through its core (Γ is high in a hub structure; the 2019 trade matrix has $\Gamma_{\text{global}} = 0.452$), and more integration on the same hub structure leaves power where it is. The current structure reproduces itself through the formation model (Chapter 10): peripheral preferences are formed under a W^I that transmits center consumption patterns and economic doctrines. The formation loop operates at the global scale exactly as it does at the domestic scale.

16.7 Summary

1. The **global production weight matrix** $W^P_{\text{global}} = D^{-1}K$ is estimated from trade data; its values depend on the source, year, aggregation and diagonal treatment. Its column sums average 1 and vary by orders of magnitude (US 24.2, Bolivia 0.10 in 2019); its Perron vector is concentrated ($\pi_{\text{China}} = 0.130$, $\pi_{\text{US}} = 0.089$, $\pi_{\text{Bolivia}} = 0.00053$; the top 10 hold 51.9%) and tracks world export shares, as gravity predicts.

2. A core-periphery hub structure has a *large* spectral gap (the schematic has $\Gamma = 0.90$; frictionless gravity has $\Gamma = 1$; the measured 2019 matrix has $\Gamma = 0.452$, the distance reflecting trade frictions). Γ measures trade frictions and mixing; power concentration is measured by $\boldsymbol{\pi}$, N_{eff} and $\text{Var}(\boldsymbol{\pi})$.

3. The **ECI** is the second eigenvector of the export-similarity matrix $\tilde{M}$, not a function of W^P_{global}. $\tilde{M}$ has real eigenvalues in $[0, 1]$ and a Perron vector proportional to diversity. ECI predicts growth and lower inequality empirically;

the channels through domestic integration and domestic power are conjectures, and no spectral bound links Γ to inequality. Bolivia (ECI -1.30 in 2019, falling since 2006) has a domestic Γ in the range of Korea's and the US's, and cut its Gini from 0.600 to 0.426 over 2005–2019.

4. **Relatedness density** is the empirical counterpart of proximity-dominated edge creation (Chapter 13). Countries on the dense core diversify rapidly; countries on the sparse periphery are locked in. Linkages (BL, $FL \geq 1$) come from the coefficient matrix A, not from W.

5. The **Prebisch–Singer hypothesis** has a spectral version — the core's π rises as the global W^P evolves — that remains a conjecture until relative prices are derived from W. The three global networks are **aligned**: the center dominates production, information, and culture simultaneously.

16.8 Exercises

16.1 Build the global W^P. Using BACI trade data (available at `cepii.fr`). (The 2019 values in Remark 16.2 were computed in the same way from the Growth Lab's HS92 bilateral data; use them as a check.)

(a) Download bilateral trade data for 2018. Keep all countries, and also aggregate to 50 countries (rest of world as one node).

(b) Construct $W^P_{cc'}$ (Definition 16.1). Verify row-stochasticity.

(c) Compute column sums. Which country has the largest? Smallest?

(d) Compute $\boldsymbol{\pi}$ (power iteration, Appendix 6A). Plot π_c vs. GDP per capita. What is the correlation?

(e) Compute Γ and N_{eff} for the full and the 50-country matrix. How much does aggregation change each? Compare Γ with the frictionless-gravity benchmark $\mathbf{1}\mathbf{s}'$ ($\Gamma = 1$), where $\mathbf{s}$ is world export shares, and $\boldsymbol{\pi}$ with $\mathbf{s}$.

16.2 Compute the ECI. Using the Atlas of Economic Complexity (`atlas.cid.harvard.edu`) or OEC (`oec.world`):

(a) Download the ECI ranking for 2018. Identify the top 10 and bottom 10.

(b) Plot ECI vs. GDP per capita. What is the correlation? Is it stronger or weaker than π_c vs. GDP per capita from Exercise 16.1?

(c) Plot ECI vs. Gini coefficient. Is the low-ECI, low-Gini quadrant sparsely populated? (This is not Fajnzylber's casillero vacío, which is defined in growth $\times$ equity space; see Chapter 17.)

(d) Where is Bolivia? Where is South Korea? What is the ECI gap?

16.3 Relatedness density. Using OEC product-space data:

(a) Compute Bolivia's relatedness density to: lithium batteries, processed food, pharmaceutical intermediates.

(b) Compare with Chile, Peru, and Colombia.

(c) Which country has the highest average relatedness to core products? Why?

(d) Using Chapter 15's sequencing framework: what operator sequence would Bolivia need to reach $\omega > 0.3$ for lithium batteries?

16.4 Prebisch-Singer. Download Bolivia's terms-of-trade index (World Bank WDI or CEPAL):

(a) Plot the terms-of-trade index, 1960–2023.

(b) Identify the commodity booms (1970s, 2003–2014). During booms: did Bolivia's ECI change?

(c) Using BACI, recompute π_{Bolivia} for a boom year and a bust year, first with current values and then with trade valued at constant (base-year) prices. How much of the change in π_{Bolivia} is a price effect and how much a change in quantities and partners?

16.5 Global W^I. Use data on academic citations (Scopus) or media reach (Reuters, AP, AFP) to estimate a proxy for W^I_{global}:

(a) Construct a 20×20 matrix where $W^I_{cc'}$ is the share of academic citations in country c that cite papers from country c'.

(b) Compute $\boldsymbol{\pi}^I$. Which countries have the highest epistemic power?

(c) Compute N_{eff}. How many "independent voices" shape the global academic consensus?

(d) Compare $\boldsymbol{\pi}^I$ with $\boldsymbol{\pi}^P$ (from Exercise 16.1). Are they correlated? What does alignment mean for the composite power vector?

16.6 Integration, power and inequality. For 30–50 countries with national input-output tables (OECD ICIO):

(a) Compute the domestic Γ_c and N_{eff} of the domestic $W^P = D^{-1}A$ for each country.

(b) Show, by constructing $W = \gamma \mathbf{1}\boldsymbol{\pi}' + (1-\gamma)I$, that no function of Γ alone can bound $\text{Var}(\boldsymbol{\pi})$.

(c) Regress the Gini coefficient on ECI, then add Γ_c and N_{eff}. Which, if any, mediates the ECI–inequality association of Hartmann et al. (2017)?

(d) Which countries have much lower inequality than their complexity would predict? What do they have in common?

16.7 Bolivia's structural position. Synthesize all the data from this chapter's exercises:

(a) Bolivia's π_{global}, ECI, relatedness density to core products, terms-of-trade trajectory, and W^I position.

(b) Using the operator-design framework of Chapter 15: design a 20-year structural transformation plan for Bolivia. Which operators, in what sequence, targeting which products?

(c) Estimate the cost (order of magnitude) and the expected ΔECI, $\Delta\Gamma$, ΔGini.

(d) Is the plan politically feasible given Bolivia's current coalition structure (Chapter 14)?

Mathematical Appendices

Appendix 16A: The ECI as a Spectral Method

Derivation. Start with the binary specialization matrix M ($C \times P$). Define diversity $M_c = \sum_p M_{cp}$ and ubiquity $M_p = \sum_c M_{cp}$. The coupled equations:

$$K_c = \frac{1}{M_c} \sum_p M_{cp} K_p, \qquad K_p = \frac{1}{M_p} \sum_c M_{cp} K_c. \tag{16.8}$$

Substituting: $K_c = \sum_{c'} \tilde{M}_{cc'} K_{c'}$ where $\tilde{M}_{cc'} = \sum_p M_{cp} M_{c'p} / (M_c M_p)$. The solutions are the eigenvectors of $\tilde{M}$. Since $\tilde{M}$ is row-stochastic, the first eigenvector is $\mathbf{1}$ (trivial). The ECI is the second eigenvector, Z-score normalized.

Connection to SVD. Let $S = D_c^{-1/2} M D_p^{-1/2}$ with SVD $S = U \Sigma V'$. Then $\tilde{M} = D_c^{-1} M D_p^{-1} M' = D_c^{-1/2} (SS') D_c^{1/2} = D_c^{-1/2} U \Sigma^2 U' D_c^{1/2}$, so the eigenvalues of $\tilde{M}$ are the squared singular values of S and its right eigenvectors are $D_c^{-1/2} U$. The ECI is therefore, up to z-scoring, $D_c^{-1/2}$ times the second left singular vector of S. This is the connection to dimensionality reduction: the ECI summarizes the leading factor that explains the geography of economic activities.

Connection to this book's Γ. The spectral gap of $\tilde{M}$, $1 - \lambda_2(\tilde{M})$, is a property of export baskets. It is not a proxy for the spectral gap of W^P_{global}, which depends on bilateral flows that M does not contain. A small gap of $\tilde{M}$ means that one axis of differentiation (the ECI axis) dominates the similarity structure; a large one, that baskets are similar or that several weak axes compete.

Appendix 16B: The Global Leontief Inverse from ICIO Data

The ICIO table. The OECD Inter-Country Input-Output table reports intermediate flows $Z_{(c's'),(cs)}$ from sector s' of country c' to sector s of country c (rows are suppliers, the OECD convention), together with gross output X_{cs}. In this book's Chapter 1 convention the coefficient matrix has rows as recipes: $a_{(cs),(c's')} = Z_{(c's'),(cs)} / X_{cs}$, the input from $(c's')$ per unit of output of (cs). Its dimension is $(N_c N_s)^2$; the number of economies and industries depends on the release, so state it.

Two different country-level objects. (i) For *power and integration*, aggregate flows, not coefficients: country c's purchases from c' are $K_{cc'} = \sum_{s,s'} Z_{(c's'),(cs)}$, and $W^P = D^{-1} K$ as in Definition 16.1 (with the domestic block on the diagonal, so this is the

absorption-share treatment). (ii) For *multipliers and linkages*, compute the Leontief inverse $L = (I - A')^{-1}$ at full sector detail and aggregate its entries with output weights; never take the Leontief inverse of the row-stochastic W^P.

Computation. The global Leontief inverse for the full ICIO table is a matrix with a few thousand rows; modern computers handle this in seconds. Column sums give backward linkages and row sums forward linkages (Definition 16.3); all are at least 1.

Structuralist Case Study: Prebisch and Hirschman on the Global W^P

This chapter's case study pairs the two most influential structuralist contributions to the analysis of the global economic structure: Prebisch's center-periphery hypothesis and Hirschman's linkage theory. Together they provide the theoretical foundations for the empirical program of economic complexity.

Prebisch (1949): The global W^P is asymmetric. Prebisch's manifesto asserted that the global economy has a structure — that the pattern of trade is not the symmetric exchange of classical trade theory but an asymmetric system where the center exports manufactures (high linkages, complex, diversified) and the periphery exports primary commodities (low linkages, simple, concentrated). In this chapter's notation: W^P_{global} has core-periphery topology and $\boldsymbol{\pi}_{\text{center}} \gg \boldsymbol{\pi}_{\text{periphery}}$ (low N_{eff}). Prebisch's asymmetry is a statement about the *distribution* of $\boldsymbol{\pi}$, not about a small Γ: a hub structure integrates the world quickly while concentrating power. The ECI is an empirical measure of where each country sits on the related axis of export structure.

Prebisch also predicted that the terms of trade of primary exporters would deteriorate over time (the Prebisch–Singer hypothesis), through low income elasticities of demand for primary goods and the asymmetric behaviour of wages and prices over the cycle. Conjecture 16.17 proposes a spectral counterpart: as the global W^P evolves, the core adds dense linkages (increasing π_{center}) while the periphery adds only sparse commodity edges. Whether the terms-of-trade trend is the shadow of the Perron eigenvector is an open question, not a result.

Hirschman (1958): Linkages are the mechanism. Where Prebisch described the asymmetry, Hirschman identified the mechanism: backward and forward linkages. Industries with strong linkages generate demand for their suppliers (backward) and provide inputs for their customers (forward). The development strategy should target high-linkage industries, whose investment creates pressure for further investment upstream and downstream. Hirschman measured linkages

with the direct coefficients of Chenery and Watanabe (1958); the Leontief-inverse "key sector" indices are Rasmussen's (1956).

In this chapter's notation, backward linkages are the column sums of $L = (I - A')^{-1}$ (Definition 16.3), and the key-sector criterion is $BL > \bar{BL}$ AND $FL > \bar{FL}$. This is where the BOOK.md mapping "linkages as Katz centrality" holds, with two qualifications: $\mathbf{1}'L = \sum_{k\geq 0} \mathbf{1}'(A')^k$ is a Katz–Bonacich centrality with attenuation 1 that *includes* the $k = 0$ term (so every linkage is at least 1), and it converges only when $\rho(A) < 1$, the Hawkins–Simon condition. Rasmussen's indices further divide by the average. The product space (Hidalgo et al., 2007) maps a different kind of relatedness — shared capabilities revealed by co-exports, not input-output flows — and is best read as a capability-based complement to Hirschman's input-output linkages.

The synthesis: ECI = Prebisch's position + Hirschman's mechanism. The ECI measures a center-periphery position in export structure (where a country sits on the second eigenvector of $\tilde{M}$). The product space measures a capability analogue of Hirschman's linkage structure (which products are connected to which, and how densely). Relatedness density measures the feasibility of Hirschman's strategy: can this country reach the high-linkage products from where it currently stands?

Concept	**Prebisch (1949)**	**Hirschman (1958)**
Core claim	Global economy is asymmetric	Linkages drive development
Formal object	$\boldsymbol{\pi}_{\text{center}} \gg \boldsymbol{\pi}_{\text{periphery}}$	$BL_p = \sum_c \ell_{cp}$
Empirical measure	N_{eff} of W^P_{global}; ECI	Column sums of $(I - A')^{-1}$
Policy implication	Structural change, not just trade	Target high-linkage sectors
This chapter	Rem. 16.2, Conj. 16.17	Def. 16.3, Ex. 16.4
Product space role	Country position on the map	The map itself

Prebisch wrote the 1949 study at CEPAL in Santiago. Hirschman wrote *Strategy* at Yale, after four years as an economic adviser in Colombia (1952–56). Neither had trade data for 5,000 products, dimensionality-reduction algorithms, or the spectral theory needed to formalize their intuitions. What they had was the intuition that the global economy has a structure, that the structure determines who benefits from trade, and that the structure can be changed by policy. The ECI and the

product space are the empirical realization of their program. This chapter gives both programs their eigenvalues — and shows which eigenvalue answers which question.

Further Reading

Economic complexity: Hidalgo and Hausmann (2009), "The Building Blocks of Economic Complexity." Hausmann et al. (2014), *The Atlas of Economic Complexity*. Hidalgo (2021), "Economic Complexity Theory and Applications." Data: `oec.world` and `atlas.cid.harvard.edu`; the ECI and bilateral-trade figures in this chapter are from the Growth Lab's Dataverse releases (The Growth Lab at Harvard University, 2026).

Relatedness and the product space: Hidalgo et al. (2007), "The Product Space Conditions the Development of Nations." Hausmann and Klinger (2007), "The Structure of the Product Space."

ECI and inequality: Hartmann et al. (2017), "Linking Economic Complexity, Institutions, and Income Inequality." Fajnzylber is discussed in Chapter 17.

Terms of trade: Grilli and Yang (1988), "Primary Commodity Prices, Manufactured Goods Prices, and the Terms of Trade of Developing Countries." Harvey, Kellard, Madsen, and Wohar (2010), "The Prebisch-Singer Hypothesis: Four Centuries of Evidence." Arezki, Hadri, Loungani, and Rao (2014), "Testing the Prebisch-Singer Hypothesis since 1650."

Linkages and production networks: Rasmussen (1956), *Studies in Inter-Sectoral Relations*. Chenery and Watanabe (1958), "International Comparisons of the Structure of Production." Acemoglu, Carvalho, Ozdaglar, and Tahbaz-Salehi (2012a), "The Network Origins of Aggregate Fluctuations" — the influence vector, a Katz-type centrality of the input-share matrix.

Global input-output: OECD ICIO tables (`https://www.oecd.org/sti/ind/inter-country-input-output-tables.htm`); the comparator Γ values in this chapter use the 2023 edition (OECD, 2023). Timmer et al. (2015), "An Illustrated User Guide to the World Input-Output Database."

The structuralist tradition: Prebisch (1949), the manifesto. Hirschman (1958), *The Strategy of Economic Development*, Chapters 6–7. Singer (1950), "The Distribution of Gains between Investing and Borrowing Countries."

Chapter 17

Domestic Structural Diagnostics

Reading the National Heatmap

> *"La heterogeneidad estructural se manifiesta en la coexistencia de actividades económicas con niveles de productividad correspondientes a distintas épocas históricas."*
>
> ["Structural heterogeneity manifests itself in the coexistence of economic activities with productivity levels corresponding to distinct historical epochs."]
> — Aníbal Pinto, "Naturaleza e implicaciones de la 'heterogeneidad estructural' de la América Latina" (1970)

> *"El casillero vacío no es producto del azar ni de la fatalidad: es resultado de la naturaleza del patrón de industrialización vigente."*
>
> ["The empty box is not the product of chance or fate: it is the result of the nature of the prevailing industrialization pattern."]
> — Fernando Fajnzylber, "Industrialización en América Latina" (1990)

Chapter 16 applied the framework at the global scale: the Perron vector of W^P_{global} measures how concentrated power is among countries, the ECI measures where a country sits in the space of export baskets, and the product space maps the feasible diversification paths. This chapter turns inward: how do you read the structural health of a *national* economy?

The diagnostic toolkit has five instruments, each derived from the weight matrix:

1. The domestic spectral gap $\Gamma(W^P_{\text{domestic}})$: how integrated are the sectors (how fast do shocks mix)?

2. The between-block variance: how unequal are the blocks?
3. The domestic Perron eigenvector $\boldsymbol{\pi}_{\text{domestic}}$: who has structural power within the country?
4. The domestic $N_{\text{eff}}(W^I)$: how democratic is the information structure?
5. The domestic Ψ ratio: how much does the formation channel matter?

Each instrument is computable from available data: input-output tables, household surveys, media data, and cohort-year panels. Together they provide a structural health check — a way to diagnose where the economy departs from the benchmark $W^{\text{CE}} = \frac{1}{N}\mathbf{1}\mathbf{1}'$ along its two independent coordinates (Chapter 1, Section 1.7): the concentration of power ($G(\boldsymbol{\pi})$, N_{eff}) and the integration deficit ($1 - \Gamma$).

Bolivia is the case study throughout. Two of the five instruments — Γ and the Perron vector of W^P — are computed here from INE's supply–use tables (Instituto Nacional de Estadística (INE), Bolivia, 2026), with comparators from the OECD's Inter-Country Input-Output tables (OECD, 2023, 2021); employment comes from ILOSTAT (International Labour Organization, 2026), complexity from the Growth Lab (The Growth Lab at Harvard University, 2026), and distribution from INE and the World Bank (World Bank, 2026b). The other three instruments (σ_B, $N_{\text{eff}}(W^I)$, Ψ) need household-survey microdata, audience data and cohort panels that are not used here, and their Bolivian values remain [ILLUSTRATIVE]. Every number is reproduced by `code/ch17.py`, which reads the sourced values from `data/processed`.

The data do not confirm the picture that the structuralist tradition leads one to expect. Bolivia's production network is not unusually fragmented: its Γ lies inside the range of Korea, Germany, the United States, Chile, China and Mexico, and its sectoral power is *less* concentrated than theirs. The enclave is real, but it shows up in the Perron weight and the export orientation of hydrocarbons, not in Γ. Where Bolivia is distinctive is in complexity (a low and falling ECI) and in distribution (a large fall in inequality between 2005 and 2019 while Γ stayed roughly flat). The chapter reads the instruments in that light.

17.1 Instrument 1: The Domestic Spectral Gap

17.1.1 Estimation from I-O Tables

Definition 17.1 (Domestic Spectral Gap)**.** *Let A be the domestic technical-coefficient matrix with the Chapter 1 convention ($a_{ss'}$ = input from sector s' per unit of output of*

sector s; Appendix 17.9). The domestic production weight matrix is the matrix of input-cost shares, $W^P = D^{-1}A$ with $D = \mathrm{diag}(A\mathbf{1})$:

$$W^P_{ss'} = \frac{a_{ss'}}{\sum_{s''} a_{ss''}}, \tag{17.1}$$

*where s, s' index domestic sectors; every term in row s shares the denominator X_s, so the row is sector s's input mix. The **domestic spectral gap** is $\Gamma(W^P_{domestic}) = 1 - |\lambda_2(W^P)|$. Report also the normalization used and the level of aggregation: both change Γ.*

Example 17.2 (Bolivia's Domestic Γ, 2019)**.** *INE's supply–use tables for 2019 (base 2017), converted to a domestic industry-by-industry table at basic prices and aggregated to the book's 12 sectors. Linkages are column and row sums of $L = (I - A')^{-1}$ computed from the coefficients A, so all are at least 1 and their means coincide. "Key" sectors have both normalized linkages ($BL/\overline{BL}$, $FL/\overline{FL}$) at least 1; "backward" or "forward" sectors have only one; "weak" sectors have neither:*

Sector	*Output share*	*BL*	*FL*	*Type*
Hydrocarbons	*0.047*	*1.34*	*1.20*	*Weak*
Mining	*0.045*	*1.36*	*1.30*	*Weak*
Public administration	*0.107*	*1.24*	*1.02*	*Weak*
Manufacturing	*0.208*	*1.86*	*2.21*	*Key sector*
Construction	*0.093*	*1.57*	*1.13*	*Backward*
Commerce	*0.078*	*1.42*	*1.58*	*Forward*
Transport	*0.063*	*1.54*	*1.52*	*Key sector*
Financial services	*0.041*	*1.43*	*1.57*	*Forward*
Agriculture	*0.084*	*1.55*	*1.70*	*Key sector*
Other services	*0.144*	*1.36*	*2.02*	*Forward*
Personal services	*0.087*	*1.66*	*1.10*	*Backward*
Domestic services	*0.004*	*1.00*	*1.00*	*Weak (no intermediate flows)*
Average		*1.44*	*1.44*	

Source: INE, Cuadros de Oferta y Utilización 2019, base 2017 (Instituto Nacional de Estadística (INE), Bolivia, 2026); domestic flows by import proportionality, basic prices, fixed product-sales structure (Eurostat model D); author's calculations (`data/compute/compute_bolivia_io.py`, `code/ch17.py`).

Domestic services (households as employers) neither buy nor sell intermediate inputs, so their row of W^P is undefined; W^P is 11×11. Its spectrum:

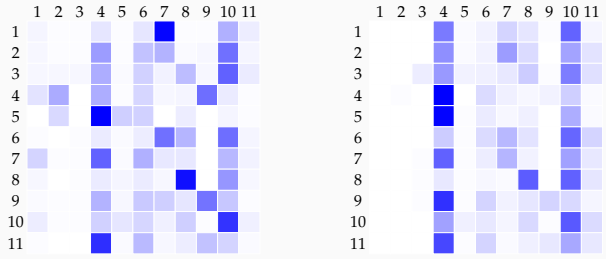
- $\lambda_1 = 1$ *(row-stochastic).*
- $\lambda_2 = 0.533$ *(real) and* $|\lambda_3| = 0.313$*: no eigenvalue close to 1, so there is no sign of weakly coupled blocks at this level of aggregation.*

- $\Gamma = 1 - 0.533 = 0.467$.

Bolivia's domestic $\Gamma = 0.467$ *is not small. The number depends on the normalization and the aggregation: normalized as sales shares (rows = sellers) the same flows give* $\Gamma = 0.694$*; at INE's 39-industry detail, input shares give* 0.410*. Across the two aggregations, input or sales shares, and with or without the diagonal, Bolivia's 2019* Γ *ranges over* 0.33–0.69*. Any single* Γ *used to compare economies should be read against that range.*

For comparison on a like-for-like basis (the same 12 groups built from ISIC Rev. 4, domestic flows, input shares, 2019), Bolivia's $\Gamma = 0.475$*; South Korea's is* 0.511*, Germany's* 0.544*, the United States'* 0.463*, Chile's* 0.557*, China's* 0.582 *and Mexico's* 0.649 *(OECD, 2023). Bolivia sits inside the comparator range, above the United States. Aggregate* Γ *does not separate Bolivia from the industrial economies.*

Visualization: Domestic W^P, 2019: Bolivia and South Korea



Bolivia W^P (INE) $\Gamma = 0.475$ | **South Korea W^P (ICIO)** $\Gamma = 0.511$

Row s is sector s's input mix ($W^P = D^{-1}A$, input shares); colour intensity is $W_{ss'}$ relative to each matrix's largest entry (Bolivia 0.60, construction's purchases from manufacturing; Korea 0.66, manufacturing's purchases from itself). Sectors: 1 hydrocarbons, 2 mining, 3 public administration, 4 manufacturing, 5 construction, 6 commerce, 7 transport, 8 financial services, 9 agriculture, 10 other services, 11 personal services; domestic services has no intermediate flows and is omitted. Both use the same ISIC-based 12-sector grouping, domestic flows at basic prices. Neither matrix is block-diagonal. In both, the strongest columns are manufacturing (4) and other services (10), the two hubs most sectors buy from; Korea's manufacturing column is stronger still, which is why its Perron vector is more concentrated. Bolivia's hydrocarbons column (1) is nearly white: few domestic sectors buy from it, because 70% of its output is exported. Its own row is dominated by one supplier, transport. Bolivia's Γ is 7% below Korea's, not 3.5 times below.

Sources: INE supply–use tables 2019 (Instituto Nacional de Estadística (INE), Bolivia, 2026); OECD ICIO 2023, Korea domestic block (OECD, 2023); author's calculations (`code/ch17.py`).

17.2 Instrument 2: Between-Block Variance

Definition 17.3 (Between-Block Variance)**.** *Partition the economy into k blocks (identified from the spectral decomposition of W^P or from sectoral groupings). Let $\bar{y}_b$ be the mean income (or output) per worker in block b, N_b the employment in block b, and $\bar{y}$ the national average. The **between-block variance** is:*

$$V_B = \sum_{b=1}^{k} \frac{N_b}{N} (\bar{y}_b - \bar{y})^2. \tag{17.2}$$

*The **between-block share** of total inequality is $\sigma_B = V_B / V_{total}$. Measured with income it describes income inequality; measured with output per worker it is closer to Pinto's productivity strata. The two need not coincide.*

Proposition 17.4 (Between-Block Share as Structural Inequality)**.** *Let N individuals with incomes y_i be partitioned into blocks of sizes N_b, with block means $\bar{y}_b$, overall mean $\bar{y} = \frac{1}{N}\sum_i y_i$, V_B as in (17.2), and within-block variance $V_W = \sum_b \frac{N_b}{N} \cdot \frac{1}{N_b} \sum_{i \in b} (y_i - \bar{y}_b)^2$. With quadratic utility $U(y) = y - \frac{\kappa}{2} y^2$ (and $\kappa y < 1$ for all y, so that U is increasing on the relevant range), the utilitarian loss from dispersion decomposes exactly as*

$$\sum_i \left[U(\bar{y}) - U(y_i) \right] = \tfrac{\kappa}{2} N (V_B + V_W),$$

*so the between-block part is $L_{block} = \frac{\kappa}{2} N V_B$ (Chapter 2's L_{block}). With $V_{total} = V_B + V_W$, the between-block share σ_B is the fraction of this loss that is **between blocks** — associated with the disconnection of the blocks — as opposed to **within blocks**.*

Proof. For each i, $U(\bar{y}) - U(y_i) = (1 - \kappa\bar{y})(\bar{y} - y_i) + \frac{\kappa}{2}(y_i - \bar{y})^2$, which is exact because U is quadratic. Summing over i, the linear term vanishes because $\bar{y}$ is the mean of the y_i. For the quadratic term, write $y_i - \bar{y} = (y_i - \bar{y}_b) + (\bar{y}_b - \bar{y})$ for i in block b; the cross term sums to zero within each block, so $\sum_i (y_i - \bar{y})^2 = \sum_b \sum_{i \in b} (y_i - \bar{y}_b)^2 + \sum_b N_b (\bar{y}_b - \bar{y})^2 = N(V_W + V_B)$ (the law of total variance). □

When σ_B is large: most inequality is between blocks. Transfers *across* blocks (pensions, conditional transfers to rural households) reduce V_B directly; transfers within blocks reduce only V_W. What transfers do not change is the productivity gap that regenerates V_B each period; cross-block linkages act on that.

When σ_B is small: most inequality is within blocks. The economy's problem is not structural heterogeneity but within-sector inequality (which may still have structural causes at a finer scale).

Example 17.5 (Bolivia's Between-Block Inequality [ILLUSTRATIVE])**.** *Illustrative inputs, of the kind one would compute from Bolivia's Encuesta de Hogares (EH) 2019 (Exercise 17.2). Three blocks: (1) Hydrocarbons/Mining/Government ($N_1 = 0.08N$, $\bar{y}_1 = 18{,}500$ Bs/month), (2) Formal urban ($N_2 = 0.35N$, $\bar{y}_2 = 5{,}200$ Bs/month), (3) Informal/Rural ($N_3 = 0.57N$, $\bar{y}_3 = 1{,}800$ Bs/month).*

National mean: $\bar{y} = 0.08 \times 18{,}500 + 0.35 \times 5{,}200 + 0.57 \times 1{,}800 = 4{,}326$ *Bs.*

Between-block variance:

$$V_B = 0.08(18{,}500 - 4{,}326)^2 + 0.35(5{,}200 - 4{,}326)^2 + 0.57(1{,}800 - 4{,}326)^2 \quad (17.3)$$

$$= 0.08 \times 200.8M + 0.35 \times 0.76M + 0.57 \times 6.38M \quad (17.4)$$

$$= 16.06M + 0.27M + 3.64M = 19.97M. \quad (17.5)$$

Assume a total variance $V_{total} = 32M$. (The three blocks are a substantive partition, by sector and formality, not one read off W^P: at the 12-sector level INE's W^P shows no weakly coupled blocks; Example 17.2. If Pinto's strata are present, they cut across sectors.)

Between-block share: $\sigma_B = 19.97/32 = 0.62$. Sixty-two percent of income inequality in this illustration is between blocks. Transfers within blocks address only the remaining 38%; transfers across blocks act on the 62% directly, but only while they last.

For comparison (illustrative): South Korea's $\sigma_B \approx 0.20$ (most inequality is within sectors); Brazil's $\sigma_B \approx 0.45$ (intermediate); the US's $\sigma_B \approx 0.25$. σ_B rises mechanically as the partition is refined, so comparisons require the same number and type of blocks.

Prebisch Connection: Pinto's heterogeneity measured

Pinto (1970) described structural heterogeneity qualitatively: the coexistence of a "modern pole" with productivity levels comparable to the industrial countries and a "primitive stratum" with colonial-era productivity. The between-block variance puts a number on it.

In the illustration, Bolivia's modern pole (gas/mining/government) has average income 10 times the primitive stratum (informal/rural): $\bar{y}_1/\bar{y}_3 = 18{,}500/1{,}800 = 10.3$. Pinto reported much larger *productivity* gaps between the modern and primitive strata for Latin America as a whole. Income gaps are smaller than productivity gaps because of transfers, remittances and self-employment income. The between-block share $\sigma_B = 0.62$ says that most of this inequality is between strata. Cross-block transfers can reduce it while they last; durable reduction requires structural change: cross-block linkages that equalize productivity across blocks.

17.3 Instrument 3: The Domestic Perron Eigenvector

Definition 17.6 (Domestic Power Distribution)**.** *The **domestic power distribution** is the left Perron eigenvector $\boldsymbol{\pi}$ of the domestic composite weight matrix $W = \alpha_U W^U + \alpha_P W^P + \alpha_I W^I$. The composite exists only if all three layers are row-stochastic on the* same *node set; in practice W^P is measured on sectors, W^I on media outlets or groups, and W^U on households, so each layer's $\boldsymbol{\pi}$ is reported separately. The **power Gini** is the Gini coefficient of $\pi_1, \ldots, \pi_N$.*

Example 17.7 (Bolivia's Domestic $\boldsymbol{\pi}$, 2019)**.** *The Perron vector of the 11×11 W^P of Example 17.2, with employment shares from ILOSTAT's tabulation of the 2019 household survey. ILOSTAT reports ISIC Rev. 4 sections, so the employment rows are mapped to the book's sectors approximately: it does not separate hydrocarbons from other mining, nor public from private education and health (counted here in other services).*

Sector	*ISIC*	*Emp. share*	π_s *(power)*	$\pi_s / share_s$
Hydrocarbons	*B*	*(in B)*	*0.039*	—
Mining	*B*	*(in B)*	*0.037*	—
Hydrocarbons and mining	*B*	*0.013*	*0.077*	*5.7*
Public administration	*O*	*0.026*	*0.005*	*0.19*
Manufacturing	*C*	*0.112*	*0.159*	*1.42*
Construction	*F*	*0.079*	*0.029*	*0.37*
Commerce	*G*	*0.171*	*0.095*	*0.55*
Transport	*H*	*0.071*	*0.092*	*1.30*
Financial services	*K*	*0.007*	*0.162*	*23.4*
Agriculture	*A*	*0.274*	*0.091*	*0.33*
Other services	*D,E,J,L–N,P,Q*	*0.121*	*0.270*	*2.23*
Personal services	*I, R, S*	*0.107*	*0.021*	*0.20*
Domestic services	*T*	*0.019*	*— (no W^P row)*	—
Total		*1.000*	*1.000*	

Sources: π: INE supply–use tables 2019, base 2017, domestic, input shares (Instituto Nacional de Estadística (INE), Bolivia, 2026); employment: ILOSTAT, DF_EMP_TEMP_SEX_ECO_NB, 2019 (International Labour Organization, 2026); author's calculations (`data/compute/compute_bolivia_io.py`, `code/ch17.py`).

Power in Bolivia's production network is spread fairly evenly: $G(\boldsymbol{\pi}) = 0.445$ and $N_{\mathit{eff}} = 6.48$ of a possible 11. On the like-for-like ISIC basis of Example 17.2 Bolivia's power Gini is 0.435, lower *than South Korea's* (0.658)*, Germany's* (0.611)*, and those of the United States* (0.614)*, Chile* (0.568)*, China* (0.622) *and Mexico* (0.608)*. The largest weights belong to other services* (0.270*: utilities, communications, real estate, business*

*services), financial services (*0.162*) and manufacturing (*0.159*): the hubs from which most sectors buy their inputs. In Korea manufacturing alone holds* $\pi = 0.434$*; in Germany other services hold* 0.440*. In industrial economies* $\boldsymbol{\pi}(W^P)$ *concentrates on manufacturing and business-service hubs, so a high power Gini in the production network is a mark of industrial depth, not by itself a sign of oppression.*

Hydrocarbons holds $\pi = 0.039$*, well below the uniform share* $1/11 \approx 0.091$*: the export sector sits at the periphery of the domestic network. Agriculture, with* $\pi = 0.091$*, carries more than twice as much. The power-employment ratio is highest in financial services (*23.4*, with 0.7% of employment) and in hydrocarbons and mining together (*5.7*, with 1.3%), and lowest in public administration, personal services and agriculture.*

Intuition: What the power-employment ratio means — and does not

Because $\boldsymbol{\pi}$ is computed from domestic input-cost shares, a sector's π_s reflects how much of the rest of the economy's intermediate purchases come from it, directly and recursively. Exports, taxes and royalties do not enter W^P. That is why hydrocarbons, which matter so much for foreign exchange and public revenue, carry little weight here. In 2019, 70% of the sector's output was exported; it bought only 0.225 of domestic intermediate inputs per unit of output (manufacturing: 0.558); and compensation of employees was 6% of its value added (INE supply–use tables; author's calculations). This is the enclave of Pinto (1970) and Prebisch, measured correctly: its rents reach the rest of the economy through the fiscal system (the direct hydrocarbons tax, IDH) and the exchange rate, not through production linkages. The enclave is visible in $\boldsymbol{\pi}$ and in export orientation, not in Γ.

Dividing by the employment share then mostly measures *capital intensity and labor productivity*: finance, and hydrocarbons and mining, employ few workers per unit of value. The ratio is therefore not a measure of the "structural privilege" of an individual worker in those sectors.

The farmer in the altiplano is the mirror image. Her sector's weight is not small — agriculture supplies a third of manufacturing's domestic intermediate inputs (food processing) and a seventh of personal services' (restaurants) — but it is shared among 27% of the workforce, so its power-employment ratio is 0.33.

If sectoral $\boldsymbol{\pi}$ is read as bargaining weight (Assumption 7.A), it enters the allocation rule $x_s^* = \Omega/N + (\pi_s - 1/N)\Phi$ (Proposition 7.3), with $N = 11$ sectors, and the gap $\pi_{\text{gas}} - \pi_{\text{agr}} = -0.052$: agriculture, not gas, would receive more than an equal sector share. Per worker the order reverses, because agriculture divides its share among about 21 times as many workers as hydrocarbons and mining combined. Neither reading is a measure of who captures the gas rents. $\boldsymbol{\pi}(W^P)$ records who supplies whom; the distributional reading of $\boldsymbol{\pi}$ belongs to the bargaining layers (W^U, W^I) of Chapter 7, where Assumption 7.A is a natural postulate.

17.4 Instrument 4: The Domestic $N_{\text{eff}}(W^I)$

Definition 17.8 (Domestic Information Effective Sample Size)**.** *Given a proxy for the domestic information network W^I (from media data, social network data, or education*

enrollment data), the ***domestic*** N_{eff} *is:*

$$N_{eff}(W^I) = \frac{1}{\sum_j (\pi_j^I)^2}, \tag{17.6}$$

where $\boldsymbol{\pi}^I$ *is the Perron eigenvector of* W^I *(Definition 6.15).* N_{eff} *lies between* 1 *and the number of* nodes *of* W^I*. It is therefore only comparable across countries when* W^I *is built on the same kind and number of nodes (outlets, outlet groups, or population groups), or after dividing by that number.*

Example 17.9 (Bolivia's N_{eff} [ILLUSTRATIVE])**.** *Approximating* W^I *on four nodes — Bolivia's main media ecosystems — with illustrative weights:*

1. *State media (Radio Patria Nueva, Bolivia TV, Cambio): dominant in the altiplano and valleys.* $\pi^I \approx 0.34$*.*
2. *Private commercial media (Unitel, Red Uno, Página Siete, El Deber): dominant in Santa Cruz and urban centers.* $\pi^I \approx 0.37$*.*
3. *Community/indigenous media (community radios, e.g. the ERBOL network): rural areas.* $\pi^I \approx 0.12$*.*
4. *Social media (Facebook, TikTok, WhatsApp): growing, cross-cutting.* $\pi^I \approx 0.17$*.*

$$N_{eff} = \frac{1}{0.34^2 + 0.37^2 + 0.12^2 + 0.17^2} = \frac{1}{0.2958} = 3.38 \approx 3.4.$$

Bolivia's information environment has approximately 3 independent voices out of a maximum of 4 on this node set. Dividing by the population (11.67 million in 2019; World Bank, 2026b) would mix node sets: the population is not the set of nodes of this W^I*. What the number says is that, at the level of ecosystems, the consensus is dominated by two sources.*

Two caveats. First, using audience shares $\mathbf{s}$ *as* $\boldsymbol{\pi}^I$ *is exact only if every node attends to the same mix,* $W^I = \mathbf{1}\mathbf{s}'$*; that matrix has* $\Gamma(W^I) = 1$ *(`code/ch17.py`). Concentration and integration are different questions here too. Second, with more nodes (individual outlets, or regions* $\times$ *outlets)* N_{eff} *would be larger. Illustrative comparisons on outlet-level node sets: the US* $N_{eff} \approx$ 15–20*, Scandinavian countries* $N_{eff} \approx$ 8–12*; these are not comparable with Bolivia's* 3.4 *on four ecosystems.*

17.5 Instrument 5: The Domestic Ψ Ratio

Definition 17.10 (Domestic Ψ)**.** *The domestic variance ratio for a target behavior (female LFP, savings, education enrollment) is estimated from the cohort-year decomposition of*

Chapter 11:

$$\hat{\Psi}_{domestic} = \frac{\text{Var}(\hat{\delta}_c)}{\text{Var}(\hat{\alpha}_t)}, \tag{17.7}$$

where $\hat{\delta}_c$ is the cohort effect and $\hat{\alpha}_t$ is the year effect. Two cautions from Chapter 11 apply. (i) With age effects in the model, linear trends in cohort and year effects are not identified (the age–period–cohort problem), so the ratio depends on the identifying restriction, which must be stated and varied. (ii) The variance ratio equals the formation-to-enforcement ratio Ψ only under additional assumptions (all cohort variation is formation, all year variation is enforcement, a common driving index).

Example 17.11 (Bolivia's Ψ for Self-Employment [ILLUSTRATIVE])**.** *Following the hypothetical template of Chapter 11: suppose Bolivia's self-employment rate varies across departments — nationally it was 68.1% of employment in 2019 (ILO modelled estimate, World Bank, 2026b); departmental rates require EH microdata — and that a cohort-year decomposition of EH data (2000–2019), under a stated APC restriction, gives:*

$\text{Var}(\hat{\delta}_c) = 0.028$ *(cohort component: norms about self-employment differ across generations).* $\text{Var}(\hat{\alpha}_t) = 0.018$ *(year component: labor market conditions change across time).*

$\hat{\Psi} = 0.028/0.018 = 1.56$.

Under the mapping assumptions of Chapter 11, the formation channel (norms transmitted through W^I) is 56% larger than the enforcement channel (prices and labor market conditions). Standard labor market policy (wage subsidies, credit access, training) captures at most $1/(1+1.56) = 39\%$ of the total effect. The majority operates through the formation channel.

Behavior	$\hat{\Psi}_{Bolivia}$	*Dominant channel*
Self-employment	*1.56*	*Formation*
Female LFP	≈ 1.8	*Formation*
Savings rate	≈ 1.1	*Formation (slight)*
Education enrollment	≈ 0.7	*Enforcement (slight)*

17.6 The Structural Health Dashboard

Definition 17.12 (Structural Health Dashboard)**.** *The **structural health dashboard** for a national economy reports five numbers:*

Instrument	*Symbol*	*Benchmark*	*Interpretation*
Spectral gap	Γ	$\to 1$	*Sectoral integration*
Between-block share	σ_B	$\to 0$	*Between-block inequality*
Power Gini	$G(\boldsymbol{\pi})$	$\to 0$	*Power concentration*
Information N_{eff}	$N_{eff}(W^I)$	$\to N$	*Information democracy*
Formation ratio	Ψ	—	*Formation channel strength*

Here N in the N_{eff} row is the number of nodes of W^I, and Γ measures mixing speed. The first four instruments have clear benchmarks (W^{CE} has $\Gamma = 1$, uniform $\boldsymbol{\pi}$ and $N_{eff} = N$; $\sigma_B \to 0$ also requires equal endowments). They measure two independent coordinates — integration (Γ) and concentration ($G(\boldsymbol{\pi})$, N_{eff}) — plus between-block inequality, which depends on endowments as well as on W. The fifth (Ψ) is descriptive: it tells you how much the formation channel matters, not whether it is "too large" or "too small."

Example 17.13 (Bolivia's Dashboard, 2019 (partly [ILLUSTRATIVE]))**.**

Instrument	*Bolivia*	*South Korea*	*Benchmark*	*Status*
Γ	*0.475*	*0.511*	*1.00*	*data*
σ_B	*0.62*	*0.20*	*0.00*	*illustrative*
$G(\boldsymbol{\pi})$	*0.435*	*0.658*	*0.00*	*data*
$N_{eff}(W^I)$	*3.4*	≈ 12	N	*illustrative*
Ψ *(self-employment)*	*1.56*	≈ 0.6	—	*illustrative*
Memo: ECI (Chapter 16)	-1.30	1.81	—	*data*

Sources: Γ and $G(\boldsymbol{\pi})$: domestic W^P, input shares, ISIC-based 12 sectors, 2019; INE supply–use tables (Instituto Nacional de Estadística (INE), Bolivia, 2026) and OECD ICIO 2023 (OECD, 2023); author's calculations (`data/compute/`, `code/ch17.py`). Bolivia's Γ ranges over 0.33–0.69 across normalizations (Example 17.2). ECI: Growth Lab, HS92 (The Growth Lab at Harvard University, 2026). The rows marked illustrative need household-survey microdata, audience data and cohort panels.

The two production-network instruments, now measured, do not place Bolivia far from Korea. Its spectral gap is 7% below Korea's, and its power Gini is lower *— Korea's is 1.5 times Bolivia's, because Korea's input purchases concentrate on its manufacturing hub. On these two instruments Bolivia is not an outlier. Where Bolivia does stand apart is in complexity: its ECI of -1.30 against Korea's 1.81 is a gap of more than three standard deviations of the cross-country distribution. The three remaining rows are illustrative: in them Bolivia's between-block share is 3 times Korea's, its information N_{eff} about 3.5 times smaller (meaningful only on comparable node sets; Section 17.4), and its Ψ ratio 2.6 times larger. Whether those gaps are real is an empirical question that the EH microdata and media data would settle.*

The dashboard is the structural physician's chart. It tells you what is wrong, how

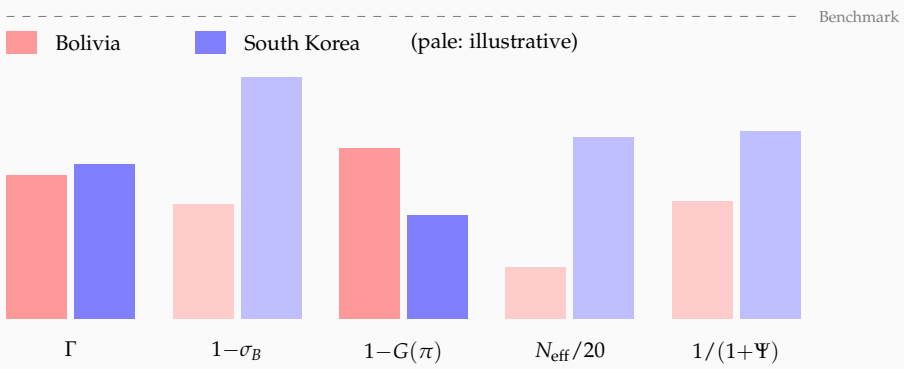
severe, and where to intervene (Chapter 15's operators target integration, power or output separately) — and, as here, what is not *wrong.*

Key Structural Insight

The instruments are not redundant, and no single number summarizes them. Γ measures integration: how fast shocks and signals mix across sectors. $G(\boldsymbol{\pi})$ and N_{eff} measure concentration of power, which is a different coordinate: any power distribution is compatible with any spectral gap (Proposition 6.16). A doubly stochastic, weakly coupled economy has $G(\boldsymbol{\pi}) = 0$ and $\Gamma \approx 0$; an economy in which every sector buys from one dominant supplier has $\Gamma \approx 1$ and high $G(\boldsymbol{\pi})$. σ_B adds endowments and productivity. Ψ measures how much any departure feeds back into preference formation.

Bolivia's measured numbers make the point concrete: an economy with a low and falling ECI and a textbook export enclave has a production-network Γ and $G(\boldsymbol{\pi})$ within the range of industrial economies. Aggregate Γ does not separate development paths, and a concentrated $\boldsymbol{\pi}(W^P)$ is as much a sign of industrial hubs as of exclusion. The dashboard is useful precisely because the instruments respond to different operators: Γ responds most to cross-block ΔW^P (structural policy); $G(\boldsymbol{\pi})$ to who supplies whom; N_{eff} to ΔW^I (information policy); Ψ tells you which lever matters most.

Visualization: Structural health: Bolivia vs. South Korea (bar chart)



Values from Example 17.13 (`code/ch17.py`); dark bars are data (2019, INE and OECD ICIO, like-for-like), pale bars are illustrative. The first four instruments are rescaled so that the benchmark is 1.0 (top); N_{eff} is divided by 20 as a common scale, which presumes comparable node sets. The fifth bar, $1/(1+\Psi)$, is the enforcement share and has no benchmark: it is shown for scale only. On the measured instruments Bolivia and Korea are close on integration, and Bolivia is *closer* to the benchmark on power concentration. The gaps measure different things — integration, between-block inequality, power concentration, information concentration — and should not be added up.

17.7 Structural Change Over Time

Definition 17.14 (Tracking Structural Health)**.** *The dashboard can be computed for multiple years to track structural change. The trajectory of* $(\Gamma_t, \sigma_{B,t}, G(\boldsymbol{\pi}_t), N_{eff,t})$ *over time reveals whether the economy is integrating or fragmenting* (Γ_t)*, and whether power is concentrating or dispersing* ($G(\boldsymbol{\pi}_t)$, $N_{eff,t}$)*.*

Example 17.15 (Bolivia 1990–2019: Structural Trajectory)**.** *INE publishes input-output tables for 1988–2009 and 2011–2014 (base 1990) and supply–use tables for 2017–2024 (base 2017). Each year is processed as in Example 17.2 (book's 12 sectors, domestic flows, input shares):*

Year	*Base*	*Sectors in W^P*	Γ	$G(\boldsymbol{\pi})$	$\pi_{hydrocarbons}$
1990	*1990*	*10*	*0.567*	*0.420*	*0.037*
2000	*1990*	*10*	*0.494*	*0.420*	*0.036*
2006 (nationalization)	*1990*	*11*	*0.518*	*0.424*	*0.064*
2011	*1990*	*11*	*0.478*	*0.369*	*0.088*
2013	*1990*	*11*	*0.484*	*0.371*	*0.085*
2017	*2017*	*11*	*0.506*	*0.446*	*0.045*
2019	*2017*	*11*	*0.467*	*0.445*	*0.039*

Source: INE, Matrices de Insumo Producto (base 1990) and Cuadros de Oferta y Utilización (base 2017) (Instituto Nacional de Estadística (INE), Bolivia, 2026); author's calculations (`data/compute/compute_bolivia_io.py`, `code/ch17.py`). The change of base (rule) brings a new classification and SNA vintage: do not compute changes across it. In 1990 and 2000 the table has no intermediate sales of public administration, so W^P has 10 sectors. There is no table for 2010.

The data tell a story of slow disintegration, not of an integrating economy. On base 1990, Γ fell from 0.567 (1990) to 0.484 (2013), a structural velocity of -0.036 per decade. On base 2017 it fell again, from 0.506 (2017) to 0.467 (2019). Korea's Γ also fell, from 0.626 (1995, OECD ICIO 2021 edition) to 0.511 (2019, 2023 edition). Over the period in which one economy became an industrial power and the other remained a commodity exporter, their aggregate Γ moved in the same direction. Power in W^P did not concentrate either: $G(\boldsymbol{\pi})$ was 0.42 in 1990 and 0.37 in 2013. What moved was the weight of hydrocarbons, which more than doubled with the gas boom, from 0.037 (1990) to 0.088 (2011), close to the uniform share $1/11 \approx 0.091$, and stood at 0.039 in 2019 on the new base, when 70% of the sector's output was exported.

The other two coordinates cannot be tracked with the data used here: σ_B needs EH microdata and $N_{eff}(W^I)$ audience data. Illustratively, σ_B might have fallen from about 0.68 to 0.62 and N_{eff} risen from about 2.1 to 3.4 (one more independent voice, part of it from social media after 2010) over 1990–2019 [ILLUSTRATIVE].

The gas nationalization of 2006 is barely visible in the structural indicators. It changed $\boldsymbol{\omega}$ (revenue distribution) dramatically: the state's take rose sharply — the hydrocarbons law of 2005 had already raised it to 50%, and the 2006 decree set 82% for the largest fields during a transition period. But Γ did not jump at 2006: it was 0.502 in 2005, 0.518 in 2006 and 0.504 in 2009. The gas sector remained an enclave in the precise sense of Example 17.7. The structural diagnosis: Bolivia redistributed within the existing structure without changing the structure.

Prebisch Connection: The paradox of redistribution without structural change

Bolivia under Morales (2006–2019) saw one of the largest falls in inequality in the region. Extreme poverty fell from 38.2% of the population in 2005 to 15.2% in 2018 (INE's earlier poverty lines; 13.2% in 2019 on the updated lines). The Gini of per-capita household income fell from 0.600 (2005) to 0.426 (2019). GDP growth averaged 4.52% a year over 2006–2019 (INE, Encuesta de Hogares; World Bank, 2026b). Over the same years Γ was roughly flat, drifting down, and the ECI fell from -1.21 (2006) to -1.46 (2018) and -1.30 (2019) (The Growth Lab at Harvard University, 2026).

How did inequality fall so much? Decompositions attribute most of the fall to labor income at the bottom of the distribution — faster earnings growth in low-skill, informal and rural activities during the commodity boom, together with minimum-wage increases — and a smaller part to public transfers (Vargas and Garriga, 2015). Distribution changed a great deal while W^P changed little. This is what the framework predicts once its two coordinates are kept apart: the allocation depends on endowments and on surplus as well as on $\boldsymbol{\pi}$, and $\boldsymbol{\pi}$ is a separate coordinate from Γ. It is also evidence against any claim that equity gains *require* a high Γ or a positive ECI: Bolivia's came with neither rising.

The Fajnzylber diagnosis: by his criterion — growth *and* equity — Bolivia came close to the casillero vacío during the boom. The structural question is whether it can stay there when commodity prices fall. Fajnzylber's answer, developed in the case study below, is that only a change in the pattern of industrialization (the operators of Chapter 15) makes growth with equity durable; redistribution on an unchanged production structure depends on the price cycle. What the nationalization did not create was not aggregate integration, which Bolivia already had, but linkages from the export enclave into the domestic network and a more complex export basket.

17.8 Summary

1. Five **diagnostic instruments**: Γ (integration), σ_B (between-block share), $G(\boldsymbol{\pi})$ (power Gini), N_{eff} (information concentration), and Ψ (formation ratio). All are computable from available data, once the normalization of W^P, the aggregation level, the block partition and the node set of W^I are stated.

2. Bolivia's measured **production network** is not fragmented: $\Gamma = 0.467$ in

2019 (INE, 12 sectors; 0.33–0.69 across normalizations), inside the range of Korea (0.511), Germany (0.544) and the United States (0.463) on a like-for-like basis, and its power Gini (0.435) is lower than theirs. The enclave is measured by $\boldsymbol{\pi}$ and export orientation: hydrocarbons hold $\pi = 0.039$ while exporting 70% of output. Bolivia is distinctive in complexity (ECI -1.30). The other instruments remain illustrative: $\sigma_B = 0.62$, $N_{\text{eff}} = 3.4$ on four ecosystems, $\Psi = 1.56$.

3. **Structural change is slow, and Γ drifted down**: from 0.567 (1990) to 0.484 (2013) in Bolivia, as Korea's fell from 0.626 (1995) to 0.511 (2019). Aggregate Γ does not separate development paths. Meanwhile inequality fell sharply (Gini 0.600 $\rightarrow$ 0.426, 2005–2019), mainly through labor income at the bottom: equity gains did not require a rising Γ or ECI. The gas nationalization changed revenues but not structure.

4. The instruments are **not redundant**: integration (Γ) and power concentration ($G(\boldsymbol{\pi})$, N_{eff}) are independent coordinates, and σ_B also depends on endowments. No single number — in particular not Γ — summarizes the distance from W^{CE}.

5. **Policy diagnostic**: compare each instrument to its benchmark, and choose the operator that acts on that instrument (Chapter 15), in the conjectured Ψ-ordered sequence.

17.9 Exercises

17.1 Compute Bolivia's Γ. Download Bolivia's supply–use or input–output tables from INE (Bolivia is not a separate economy in OECD ICIO). Build A with the Chapter 1 convention (Appendix 17.9) and aggregate to 8–12 sectors. Construct $W^P = D^{-1}A$. Compute the eigenvalues, Γ and $\boldsymbol{\pi}$. Repeat with the sales-share normalization and with a different aggregation. How much do Γ and $G(\boldsymbol{\pi})$ change? Compare with Example 17.2 and with the range 0.33–0.69.

17.2 Between-block variance. Using Bolivia's EH 2019:

(a) Define three blocks (gas/mining/government, formal urban, informal/rural).

(b) Compute $\bar{y}_b$ for each. Compute V_B and σ_B.

(c) Now try two blocks (formal/informal) and four blocks (gas, formal manufacturing, urban services, rural). How does σ_B change with the number of blocks?

(d) Which partition gives the highest σ_B? Show that refining a partition can never lower σ_B. What does this tell you about comparing σ_B across countries, and about the "natural" block structure?

17.3 Power-employment ratio. Reproduce Example 17.7:

(a) Compute $\boldsymbol{\pi}$ from the domestic W^P.

(b) Compute the power-employment ratio π_s/share_s for each sector.

(c) Which sector has the highest ratio? Least? How much of the ranking is explained by capital intensity (value added per worker)?

(d) A worker moves from agriculture ($\pi/\text{share} = 0.33$) to manufacturing ($\pi/\text{share} = 1.42$). If each sector's share $x_s^* = \Omega/N + (\pi_s - 1/N)\Phi$ (Chapter 7) is divided equally among its workers, by how much does her per-worker allocation change? What does the calculation assume?

17.4 Information democracy. Estimate N_{eff} for your country:

(a) Identify the major media outlets / information sources and estimate their audience shares.

(b) Approximate $\boldsymbol{\pi}^I$ from audience shares (as a proxy).

(c) Compute N_{eff}. How many "independent voices" does your country have?

(d) Compare with Bolivia's $N_{\text{eff}} = 3.4$, first on the same four-ecosystem node set, then on an outlet-level node set. How does the comparison change?

17.5 Bolivia's structural trajectory. Reproduce Example 17.15 from INE's tables (base 1990: 1988–2009 and 2011–2014; base 2017: 2017–2024), then extend it with EH data:

(a) Compute Γ, $G(\boldsymbol{\pi})$ and $\pi_{\text{hydrocarbons}}$ for every available year. Which years look like outliers (try 2014), and why must changes not be computed across the change of base?

(b) Compute σ_B for 2000, 2005, 2010, 2015, 2019. Plot the trajectory.

(c) Is there a structural break at 2006 (nationalization)? At 2009 (new constitution)?

(d) Project the trend forward: at the current rate, when will σ_B fall below 0.40?

17.6 Dashboard comparison. Choose three Latin American countries (e.g., Bolivia, Chile, Costa Rica). For each:

(a) Estimate Γ, σ_B, $G(\boldsymbol{\pi})$ from I-O and household survey data.

(b) Estimate N_{eff} from media data, on a common node set.

(c) Plot the three-country dashboard. Which country is structurally healthiest?

(d) Using Chapter 15's sequencing framework: what operator would you recommend for the country with the largest structural deficit?

17.7 Bolivia's escape plan. Synthesize the entire book for Bolivia:

(a) Current position: dashboard from Example 17.13. ECI $= -1.30$ in 2019, down from -1.21 in 2006 (Chapter 16). Product-space position: sparse periphery.

(b) Binding constraints: which instrument is furthest from benchmark? Remember that Γ (integration) and $G(\boldsymbol{\pi})$ (power) are independent coordinates: an operator that raises Γ need not lower $G(\boldsymbol{\pi})$ (Chapter 15).

(c) Operator sequence (Chapter 15): information $\to$ institutions $\to$ education $\to$ infrastructure $\to$ ISI-B.

(d) Political feasibility (Chapter 14): which coalition would support this sequence? Is the current coalition ($\{M, W\}$ or $\{B, M, W\}$) compatible?

(e) Expected trajectory: using the formation model (Chapter 10) with $\rho = 0.85$, $\mu = 0.2$: simulate 30 years. Plot the dashboard trajectory. Bolivia's Γ is already in the comparator range (Example 17.2): which instrument would you target instead, and how long does it take to move?

Mathematical Appendices

Appendix 17A: Estimating W^P from I-O Tables

The data. A national I-O table reports $Z_{ss'}$: the intermediate consumption of sector s's output used by sector s' as input (rows are sellers, the usual statistical convention). Total output of sector s is $X_s = \sum_{s'} Z_{ss'} + Y_s$ (intermediate sales + final demand). State whether the table is a supply–use table or a symmetric table, product-by-product or industry-by-industry, domestic or total flows (including imports), and at basic or purchasers' prices: each choice changes A.

Technical coefficients. With the book's Chapter 1 convention (row s is sector s's recipe), the technical coefficient is $a_{ss'} = Z_{s's}/X_s$: the input from s' per unit of output of s. The matrix $A = [a_{ss'}]$ is the transpose of the column-normalized table familiar from the statistical convention. Multipliers and linkages are computed from $L = (I - A')^{-1}$.

From A to W^P: two normalizations. (i) *Input shares* (used in this chapter): $W^P_{ss'} = a_{ss'}/\sum_{s''} a_{ss''} = Z_{s's}/\sum_{s''} Z_{s''s}$, the share of sector s's intermediate purchases that comes from s'. Every entry of row s shares the denominator, so the row is a probability vector. (ii) *Sales shares*: $B_{ss'} = Z_{ss'}/\sum_{s''} Z_{ss''}$, the share of sector s's intermediate *sales* that go to s'. The allocation coefficients $Z_{ss'}/X_s$ themselves are not row-stochastic (rows sum to the intermediate share of output); they must be normalized over intermediate sales. The two normalizations answer different questions — who depends on whom as a supplier, vs. who depends on whom as a customer — and give different Γ and $\boldsymbol{\pi}$: in Example 17.2, $\Gamma = 0.467$ vs. 0.694. Report both.

Spectral gap. Compute the eigenvalues of W^P using standard numerical linear algebra (e.g., `numpy.linalg.eig` in Python or `eigen` in R). The spectral gap is $\Gamma = 1 - |\lambda_2|$.

Practical considerations. (1) I-O tables are typically available at 30–100 sector detail. Aggregating to 8–15 sectors preserves the block structure while reducing noise. (2) The spectral gap is sensitive to small entries: sectors with negligible intermediate flows should be aggregated or excluded, and a sector with none (Bolivia's domestic services) must be dropped, since its row of W^P is undefined. (3) Comparing Γ across countries requires comparable sector classifications (ISIC

Rev. 4 is standard) and the same level of aggregation, since Γ changes with aggregation.

Appendix 17B: Estimating σ_B from Household Surveys

The data. Household surveys (Bolivia's EH, Brazil's PNAD, Mexico's ENIGH) report individual income and sector of employment. Block assignment: classify each individual into a block based on sector, formality status, or geographic location.

Computation. The between-block variance: $V_B = \sum_b (N_b/N)(\bar{y}_b - \bar{y})^2$. The within-block variance: $V_W = \sum_b (N_b/N) \operatorname{Var}_b(y)$. Total: $V_T = V_B + V_W$. The between-block share: $\sigma_B = V_B / V_T$.

Choice of blocks. The "natural" partition can be identified from the spectral decomposition of W^P: when λ_2 is real, the sign of the second eigenvector v_2 partitions sectors into two groups (for non-symmetric W^P, use the real part, or the eigenvectors of a symmetrized matrix) (the two sides of the principal axis of differentiation). For $k > 2$ blocks, use k-means clustering on the first $k-1$ eigenvectors (spectral clustering). Alternatively, use substantive knowledge (formal/informal, modern/traditional, urban/rural).

Relationship to Γ. The between-block share σ_B is related to Γ through the eigenvalue gap: when $\lambda_2 \approx 1$ (small Γ, for a reason other than periodicity), the two blocks are nearly independent, and σ_B can be large. When $\lambda_2 \ll 1$ (large Γ), the blocks are integrated, and σ_B tends to be small. The relationship is not deterministic: σ_B also depends on the endowment distribution, not just the network structure.

Structuralist Case Study: Pinto and Fajnzylber — Diagnosis and Prognosis

This chapter's case study pairs the two structuralist thinkers whose work most directly anticipates the diagnostic toolkit developed here: Pinto's structural heterogeneity (the diagnosis) and Fajnzylber's casillero vacío (the prognosis).

Pinto (1970): The diagnosis. Pinto's essay "Naturaleza e implicaciones de la 'heterogeneidad estructural' de la América Latina" provided the first systematic description of what this chapter calls the between-block variance. He identified three productivity strata — primitive, intermediate, and modern — coexisting within a single economy, with very large productivity gaps between the modern

and primitive strata. Pinto described the strata, estimated their relative sizes and shares of product, and argued that ISI had failed to integrate them.

In this chapter's notation, Pinto's three strata would correspond to three blocks of W^P with weak cross-block entries (Γ small). For Bolivia, INE's tables do not show such blocks at the 12-sector level ($\Gamma = 0.467$, no eigenvalue near 1; Example 17.2). If the strata exist, they lie within sectors — formal and informal commerce, commercial and peasant agriculture — and an aggregated W^P cannot see them; σ_B from household surveys, or a much finer table, can. His productivity gap corresponds to $\bar{y}_1/\bar{y}_3$ measured with output per worker; measured with income it is smaller. The failure of ISI to integrate the strata corresponds to Γ not increasing under ISI (Chapter 15: ISI applied the operator to within-block entries). σ_B is an income-based analogue of Pinto's heterogeneity, computable from any household survey; a productivity-based version needs output per worker by stratum.

Fajnzylber (1990): The prognosis. Twenty years after Pinto's diagnosis, Fajnzylber asked: what does structural heterogeneity imply for the joint trajectory of growth and inequality? His answer was the casillero vacío. In a 2×2 table of growth (dynamism) against equity, no Latin American country occupied the cell of growth *with* equity: some grew without equity, others had equity without growth. The "empty box" was occupied by countries outside the region (South Korea among them) but by no Latin American country.

Fajnzylber's explanation was structural: the prevailing industrialization pattern — a truncated industrial core without an endogenous capacity for technical progress, led by transnational firms and oriented to imitative consumption — could not deliver both. His prescription was "competitividad auténtica" (genuine competitiveness based on knowledge, technology and institutional quality) as opposed to "spurious" competitiveness based on low wages, devaluation or resource extraction — a distinction he drew in his 1988 article on international competitiveness.

In this chapter's notation, the box requires two things at once, measured by independent instruments: *dynamism*, whose structural basis is an integrated, complex production structure (high Γ, high ECI, high multipliers), and *equity*, whose structural basis is dispersed power and small between-block gaps (low $G(\boldsymbol{\pi})$, low σ_B). Because Γ and $\boldsymbol{\pi}$ are independent coordinates, neither follows from the other, which is one way to read why the box is hard to enter: a country can move along one coordinate without moving along the other. Bolivia during the gas boom moved toward the box on equity and growth without changing its production structure, and with a falling ECI; Fajnzylber's warning is that such a position is not durable.

The dashboard of this chapter operationalizes Pinto's diagnosis (σ_B measures heterogeneity) and gives Fajnzylber's prognosis two measurable coordinates (Γ for integration, $G(\boldsymbol{\pi})$ and σ_B for equity).

Concept	**Pinto (1970)**	**Fajnzylber (1990)**
Core claim	Productivity strata coexist	No LA country has growth + equity
Formal object	σ_B (between-block variance)	Growth $\times$ equity cell
Diagnosis	Three strata, σ_B large	Casillero vacío
Mechanism	Weak cross-block linkages in W^P	Truncated industrial core; integration and equity are separate coordinates
Dashboard instrument	σ_B, $G(\boldsymbol{\pi})$	Γ, ECI (dynamism); $G(\boldsymbol{\pi})$, σ_B (equity)
Policy implication	Build cross-block linkages	Raise ECI ("competitividad auténtica")
Bolivia 2019	$\sigma_B = 0.62$ (illustrative); no blocks visible in W^P	$\Gamma = 0.467$, ECI $= -1.30$

Pinto diagnosed the disease. Fajnzylber predicted its consequence. This chapter provides the laboratory: five instruments that measure the disease, track its evolution, and identify the structural interventions that could cure it. The instruments are computable from publicly available data (I-O tables, household surveys, trade data, media concentration indices). Any country can run the diagnostic. Run on INE's tables, it does not confirm Pinto's and Fajnzylber's prose in the form one might expect: Bolivia's $\Gamma = 0.467$ and $G(\boldsymbol{\pi}) = 0.445$ are those of an integrated production network with dispersed sectoral power. The heterogeneity they described has to be sought where the data point: in the export enclave (hydrocarbons' $\pi = 0.039$ with 70% of output exported), in complexity (ECI -1.30), and within sectors, which is what σ_B from the EH microdata (Exercise 17.2) would measure.

Further Reading

Structural heterogeneity: Pinto (1970), "Naturaleza e implicaciones de la 'heterogeneidad estructural'." CEPAL (2010), *Time for Equality*, Chapter 2. Infante and

Sunkel (2009), "Chile: hacia un desarrollo inclusivo."

The casillero vacío: Fajnzylber (1990), "Industrialización en América Latina: de la 'caja negra' al 'casillero vacío' ." Fajnzylber (1983), *La industrialización trunca de América Latina*. Fajnzylber (1988), "Competitividad internacional: evolución y lecciones." CEPAL (2012b), *Structural Change for Equality*.

Structural diagnostics methods: CEPAL's Structural Change Index (various publications). UNIDO's Competitive Industrial Performance Index. The Economic Complexity Observatory (`oec.world`).

Bolivia's inequality decline: Vargas and Garriga (2015), "Explaining Inequality and Poverty Reduction in Bolivia."

Bolivia data: INE (Instituto Nacional de Estadística): I-O and supply–use tables (Instituto Nacional de Estadística (INE), Bolivia, 2026), Encuesta de Hogares. ILOSTAT for employment by sector and sex (International Labour Organization, 2026); OECD ICIO for comparators (OECD, 2023, 2021); the Growth Lab's Atlas data for complexity (The Growth Lab at Harvard University, 2026). UDAPE (Unidad de Análisis de Políticas Sociales y Económicas): social indicators. The World Bank's Development Indicators for cross-country comparison.

Chapter 18

The Individual Scale

Gender, Race, and the Edges You Are Not Allowed to Form

> *"La clasificación social de la población mundial según la idea de raza es, sin duda, el más eficaz instrumento de dominación social inventado en los últimos 500 años."*
>
> ["The social classification of the world population according to the idea of race is, without a doubt, the most effective instrument of social domination invented in the last 500 years."]
> — Aníbal Quijano, "Colonialidad del poder" (2000)

> *"La superexplotación del trabajo constituye el principio fundamental de la economía subdesarrollada."*
>
> ["The superexploitation of labor constitutes the fundamental principle of the underdeveloped economy."]
> — Ruy Mauro Marini, *Dialéctica de la dependencia* (1973)

Chapter 16 analyzed the global weight matrix (countries as nodes). Chapter 17 analyzed the domestic weight matrix (sectors as nodes). This chapter completes Part V by zooming to the finest scale: individuals as nodes.

At the individual scale, a new element enters the model: **node attributes**. Each agent i carries observable characteristics — gender, race, ethnicity, caste, language, nationality — that constrain which edges she can form in W^U, W^P, and W^I. These constraints are not preferences. They are not technology. They are structural features of the weight matrix itself: certain edges are suppressed or amplified by the social classification system.

The central result: node attributes create **assortative mixing** — the tendency for agents with the same attribute to connect preferentially. When assortative mixing is strong, W becomes block-diagonal with blocks defined by attributes (men/women, white/indigenous, formal/informal). The entire machinery of this book — the spectral gap, the Perron eigenvector, the formation model, the welfare loss — applies, but now the blocks are defined by identity rather than sector. Two results organise the chapter. Assortative mixing lowers Γ (slower integration across groups), but by itself it does *not* concentrate power: with symmetric homophily the Perron vector is proportional to degree, so power gaps come from group size and from asymmetric opportunity structures, not from λ alone.

18.1 Node Attributes and the Constrained Weight Matrix

18.1.1 Formalization

Definition 18.1 (Node Attributes)**.** *Each agent i has a vector of **node attributes** $\mathbf{a}_i = (a_{i1}, a_{i2}, \ldots, a_{iK})$, where $a_{ik} \in \{0, 1, \ldots, C_k\}$ for categorical attributes (gender, race, language) or $a_{ik} \in \mathbb{R}$ for continuous attributes (age, education level). The attribute vector is exogenous to the model (for now — Chapter 27 considers algorithmic classification).*

Definition 18.2 (Attribute-Constrained Weight Matrix)**.** *Node attributes constrain W through an **assortative mixing function** $\phi : \mathbf{a}_i \times \mathbf{a}_j \to [0, 1]$ that scales the maximum feasible edge weight:*

$$W_{ij} \leq \phi(\mathbf{a}_i, \mathbf{a}_j) \cdot W_{ij}^{\max}, \tag{18.1}$$

where $W_{ij}^{\max}$ is the edge weight that would prevail absent the attribute constraint (from pure economic surplus, proximity, and information as in Chapter 13). When $\phi(\mathbf{a}_i, \mathbf{a}_j) = 1$ for all pairs: attributes do not constrain the network. When $\phi = 0$ for some pairs: those edges are forbidden.

Example 18.3 (Gender as an Attribute Constraint)**.** *In a labor market with strong gender norms: $\phi_{gender}(M, M) = 1$, $\phi_{gender}(F, F) = 1$, $\phi_{gender}(M, F) = \phi_{gender}(F, M) = \lambda_g$, where $\lambda_g \in [0, 1]$ measures gender openness. When $\lambda_g = 1$: no gender barrier. When $\lambda_g = 0$: complete gender segregation. When $\lambda_g = 0.3$ (an illustrative value used throughout this chapter): cross-gender professional linkages are 70% weaker than same-gender ones.*

The effect on W^P: women's production edges are confined to female-dominated sectors. W^P becomes approximately block-diagonal by gender, with within-gender blocks brighter than cross-gender entries. The spectral gap Γ drops: shocks and information cross the gender boundary slowly (Proposition 18.4). Whether women's Perron weight falls below

N_F/N is a separate question. It does not follow from $\lambda_g < 1$; it requires an asymmetry between the groups (Proposition 18.6).

The effect on W^I: women's information edges are confined to female-dominated social circles. W^I becomes gender-assortative. A girl whose information network is gender-assortative observes mostly female outcomes: low returns to education (because women in her network face constrained W^P, reducing their realized returns). Her formation signal is W^I-weighted, and W^I shows her a biased sample. She forms low aspirations — not because she is less capable but because her information network IS the constraint.

Proposition 18.4 (Assortative Mixing Reduces Γ)**.** *Let the raw interaction intensities be $K_{ij} = \phi(\mathbf{a}_i, \mathbf{a}_j)\, K_{ij}^{\max} \geq 0$, with every row of K nonzero, and $W = D^{-1}K$ with $D = \operatorname{diag}(K\mathbf{1})$.*

(a) ***Two equal groups (equitable partition).*** *Suppose there are two groups of n agents, $K_{ij}^{\max} = a > 0$ within groups (including $i = j$) and $K_{ij}^{\max} = c > 0$ across groups, and $\phi = 1$ within, $\lambda_k \in [0,1]$ across. Then*

$$\lambda_2(W) = \frac{a - \lambda_k c}{a + \lambda_k c}, \qquad \Gamma(W) = 1 - \frac{|a - \lambda_k c|}{a + \lambda_k c}, \tag{18.2}$$

and all other eigenvalues are 0. If $c \leq a$ (unconstrained cross-group links are no stronger than within-group links), then $\Gamma = 2\lambda_k c/(a + \lambda_k c)$ is strictly increasing in λ_k on $[0,1]$.

(b) ***General symmetric case.*** *Let $K^{\max}$ and ϕ be symmetric, and let S be a set of agents such that $\phi = \lambda_k$ on every pair with one member in S and the other in S^c (for instance, S is one class of attribute k, with $\phi_k = 1$ within classes and λ_k across them). Then*

$$\Gamma(W) \;\leq\; 2\,\frac{\lambda_k C_S}{\min(A_S, A_{S^c}) + \lambda_k C_S},$$

where $C_S = \sum_{i \in S,\, j \notin S} K_{ij}^{\max}$ is the unconstrained cross intensity and $A_S = \sum_{i,j \in S} K_{ij}$ the (constrained) intensity inside S; when $\phi = 1$ inside S, $A_S = \sum_{i,j \in S} K_{ij}^{\max}$. If A_S and A_{S^c} stay bounded away from zero, strong assortative mixing ($\lambda_k \to 0$) therefore forces $\Gamma \to 0$.

Proof. (a) Every row of K sums to $n(a + \lambda_k c)$, so $W_{ij} = \alpha/n$ within groups and β/n across, with $\alpha = a/(a + \lambda_k c)$ and $\beta = \lambda_k c/(a + \lambda_k c) = 1 - \alpha$. Rows within a group are identical, so the partition into groups is equitable: with V the $2n \times 2$ group-indicator matrix, $WV = VQ$ for the quotient $Q = \begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix}$. Hence $\mathbf{1}$ and the vector equal to $+1$ on one group and -1 on the other are eigenvectors of W

with eigenvalues 1 and $\alpha - \beta$. Every vector that sums to zero within each group is mapped to 0, and these vectors span the remaining $2n - 2$ dimensions. So the spectrum is $\{1, \alpha - \beta, 0, \ldots, 0\}$, which gives (18.2). If $c \leq a$ then $\lambda_k c \leq a$, so $|\alpha - \beta| = (a - \lambda_k c)/(a + \lambda_k c)$ and $\Gamma = 2\lambda_k c/(a + \lambda_k c)$, whose derivative in λ_k is $2ac/(a + \lambda_k c)^2 > 0$.

(b) Let $d_i = \sum_j K_{ij}$, $\mathrm{vol}(X) = \sum_{i \in X} d_i$ and $\pi_i = d_i/\mathrm{vol}(\mathrm{all})$. Symmetry of K gives $\pi_i W_{ij} = K_{ij}/\mathrm{vol}(\mathrm{all}) = \pi_j W_{ji}$, so W is reversible with respect to π, and its eigenvalues are real ($D^{1/2} W D^{-1/2} = D^{-1/2} K D^{-1/2}$ is symmetric). Order them $1 = \lambda_1 \geq \lambda_2^{\uparrow} \geq \cdots$. The variational characterization of the second eigenvalue of a reversible chain (Levin et al., 2009, ch. 13) is

$$1 - \lambda_2^{\uparrow} = \min_{f \text{ non-constant}} \frac{\frac{1}{2} \sum_{i,j} \pi_i W_{ij} (f_i - f_j)^2}{\mathrm{Var}_\pi(f)}.$$

Take $f = \mathbf{1}_S$. The numerator is $Q(S, S^c) := \sum_{i \in S, j \notin S} \pi_i W_{ij}$ and the denominator is $\pi(S)\pi(S^c) \geq \frac{1}{2} \min(\pi(S), \pi(S^c))$. Hence $1 - \lambda_2^{\uparrow} \leq 2\, Q(S, S^c)/\min(\pi(S), \pi(S^c))$, which is the upper half of Cheeger's inequality (Levin et al., 2009, Thm. 13.14). By the hypothesis on ϕ, $\mathrm{vol}(\mathrm{all})\, Q(S, S^c) = \sum_{i \in S, j \notin S} K_{ij} = \lambda_k C_S$ and $\mathrm{vol}(S) = A_S + \lambda_k C_S$, $\mathrm{vol}(S^c) = A_{S^c} + \lambda_k C_S$. Finally $\Gamma = 1 - \max_{l \geq 2} |\lambda_l| \leq 1 - |\lambda_2^{\uparrow}| \leq 1 - \lambda_2^{\uparrow}$. (If $\lambda_k = 0$, W is reducible and $\Gamma = 0$; the bound holds trivially.) □

Remark 18.5 (What the proposition does not say).

1. The hypothesis $c \leq a$ in (a) matters. With $a = 0.05$ and $c = 1$ (agents interact mainly *across* groups, so W is close to bipartite), $\Gamma = 0.095$ at $\lambda_k = 1$, 0.40 at $\lambda_k = 0.2$ and 1 at $\lambda_k = 0.05$: suppressing cross-group edges *raises* Γ.

2. In case (a), W is doubly stochastic, so $\boldsymbol{\pi} = \mathbf{1}/N$ for every $\lambda_k > 0$. Γ falls from 1 to 0.46 as λ_k falls from 1 to 0.3, while power stays exactly equal. Γ measures integration, not the distribution of power (Proposition 6.16).

(Numbers: `code/ch18.py`.)

18.2 Gender

18.2.1 Where a Gender Power Gap Can Come From

Proposition 18.6 (Homophily and Group Power). *Let $K^{\max}$ and ϕ be symmetric, $K = \phi \circ K^{\max}$, and $W = D^{-1}K$ irreducible. Define the per-capita power ratio of groups F and M as*

$$\frac{\pi_{avg}^F}{\pi_{avg}^M} = \frac{\sum_{j \in F} \pi_j / N_F}{\sum_{j \in M} \pi_j / N_M}. \tag{18.3}$$

(i) $\pi_i = d_i / \sum_j d_j$, *where* $d_i = \sum_j K_{ij}$ *is* i*'s total (constrained) interaction intensity. Power is proportional to degree.*

(ii) ***Group size.*** *Suppose* $K_{ij}^{\max} = 1$ *for all pairs (including* $i = j$*), two groups have population shares* p *and* $1 - p$*, and* $\phi = 1$ *within and* $\lambda \in (0, 1]$ *across groups. Then*

$$\frac{\pi_{avg}^{min}}{\pi_{avg}^{maj}} = \frac{p + \lambda(1-p)}{(1-p) + \lambda p}, \tag{18.4}$$

which is < 1 *if and only if* $\lambda < 1$ *and* $p < \frac{1}{2}$*. With equal group sizes (*$p = \frac{1}{2}$*) the ratio is* 1 *for every* λ*.*

(iii) ***Asymmetric opportunity structure.*** *With two equal groups, within-group intensities* $a_F, a_M > 0$ *(including* $i = j$*), unconstrained cross intensity* $c > 0$ *and* $\lambda_g \in (0, 1]$*,*

$$\frac{\pi_{avg}^{F}}{\pi_{avg}^{M}} = \frac{a_F + \lambda_g c}{a_M + \lambda_g c},$$

which is < 1 *iff* $a_F < a_M$*. Lowering* λ_g *widens the gap, towards* a_F / a_M*.*

Proof. (i) Symmetry of K gives $d'D^{-1}K = \mathbf{1}'K = (K'\mathbf{1})' = (K\mathbf{1})' = d'$, so $d / \sum_j d_j$ is a stationary vector, and it is the only one because W is irreducible. This is the textbook stationary distribution of a random walk on a weighted undirected graph (Levin et al., 2009, §9.1). (ii) With N agents, a minority member has $d = N[p + \lambda(1-p)]$ and a majority member $d = N[(1-p) + \lambda p]$; apply (i). The ratio is below one iff $p + \lambda(1-p) < (1-p) + \lambda p$, i.e. iff $(1-2p)(1-\lambda) > 0$. (iii) A member of group g has $d = n(a_g + \lambda_g c)$; apply (i). The derivative of the ratio in λ_g is $c(a_M - a_F)/(a_M + \lambda_g c)^2$, positive when $a_F < a_M$. □

Part (ii) is a degree ratio. That homophily combined with minority size puts a minority at a ranking disadvantage is known from network science, for growing networks with preferential attachment (Avin et al., 2015; Karimi et al., 2018).

A gender gap at roughly equal population shares therefore cannot come from homophily alone. It requires the asymmetry in (iii): women channelled into sectors with fewer inter-sectoral linkages (domestic work, informal services), or weaker connections to the high-linkage core. For an ethnic minority, group size (ii) already produces a gap. With $p = 0.41$ and $\lambda_r = 0.35$, formula (18.4) gives 0.84.

Example 18.7 (Bolivia's Gender Structure (partly [ILLUSTRATIVE]))**.** *Employment by sex in 2019, from ILOSTAT's tabulation of Bolivia's household survey (ISIC Rev. 4 sections):*

	ISIC	Male employment	Female employment
Gas/Mining	*B*	*93.5%*	*6.5%*
Manufacturing	*C*	*55.9%*	*44.1%*
Construction	*F*	*96.9%*	*3.1%*
Commerce	*G*	*38.6%*	*61.4%*
Agriculture	*A*	*54.0%*	*46.0%*
Domestic service	*T*	*6.1%*	*93.9%*
Government (public administration)	*O*	*65.7%*	*34.3%*

Source: ILOSTAT, DF_EMP_TEMP_SEX_ECO_NB, Bolivia 2019 (International Labour Organization, 2026); author's calculations (`code/ch18.py`). "Government" is ISIC section O only; public education and health are in sections P and Q.

Chapter 17's sectoral power table (Example 17.7) gives power per worker *in Bolivia's 2019 production network. The pattern by gender is mixed. The two most female sectors are at the bottom: personal services (69% female) has $\pi/\text{share} = 0.20$, and domestic service (94% female, household production) has no intermediate flows at all, so it is outside W^P — marginal in the strongest sense. Hydrocarbons and mining (6.5% female) has a high ratio (5.7). This is the asymmetry of Proposition 18.6(iii). But the highest ratio is in financial services (23.4), where 47% of workers are women, and male-dominated construction (0.37) and public administration (0.19) are near the bottom. Assigning each sector's π equally to its workers (and none to domestic service), women's per-capita power in W^P is 0.86 of men's (author's calculations, `code/ch18.py`): the sectoral composition of employment produces a gap, but a modest one. Gender norms enter twice: through ϕ_{gender}, and through the occupational channelling that shapes the opportunity structure $K^{\max}$; the sectoral data capture only the second, at the coarse level of 11 sectors.*

For illustration, take $\pi^F_{avg}/\pi^M_{avg} = 0.58$: women hold 58% of men's structural power per capita. This is a much larger gap than the sectoral composition alone produces (0.86); the difference would have to come from women's positions within sectors (occupation, firm size, formality) or from the information and utility layers, which the data used here do not measure. Part (iii) of the proposition reproduces 0.58 with equal group sizes, $\lambda_g = 0.3$, $c = a_M$ and a female within-block intensity of $a_F = 0.454\, a_M$. At the same densities with $\lambda_g = 1$ the ratio would be 0.73, so in this calibration the attribute constraint accounts for only part of the gap. Whether the structural-position gap explains the gender earnings gap is a hypothesis to test, not a result of the model.

18.2.2 The Formation Effect of Gender Norms

Proposition 18.8 (Gender Norms as a Formation Constraint)**.** *Gender norms constrain W^I by gender: women observe disproportionately female outcomes. The formation*

equation for female agents:

$$\theta^F_{i,t+1} = \rho\theta^F_{i,t} + (1-\rho)\bar{\theta} + \mu\left(\sum_j W^I_{ij} r_{j,t} - r^*\right), \tag{18.5}$$

where W^I_{ij} is gender-assortative ($W^I_{F,F} \gg W^I_{F,M}$). The signal $\sum_j W^I_{ij} r_{j,t}$ is a biased sample: it overweights female returns (low, if women's opportunity structure is constrained) and underweights male returns (high).

Let $0 \leq \rho < 1$, let male agents follow the same equation, and hold returns fixed at group values $\bar{r}^F, \bar{r}^M$. Then $\theta^F_{i,t}$ converges to its steady state from any initial condition, and the steady-state gender preference gap is:

$$\theta^{*F} - \theta^{*M} = \frac{\mu}{1-\rho}\left(\bar{s}^F - \bar{s}^M\right) = \frac{\mu}{1-\rho} \cdot \frac{1-\lambda_g}{1+\lambda_g}\left(\bar{r}^F - \bar{r}^M\right), \tag{18.6}$$

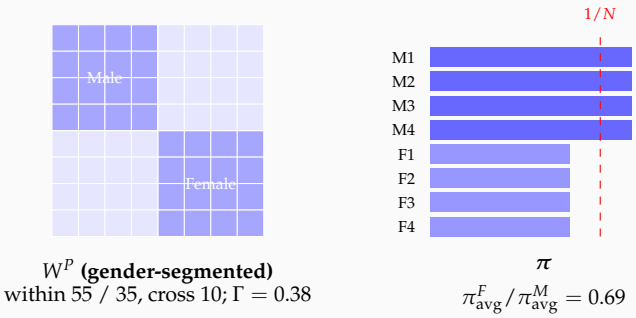
where $\bar{s}^g = \sum_j W^I_{ij} \bar{r}_j$ is the signal observed by a member i of group g (assumed the same for all members of g). The second equality holds for two equal groups with W^I built as in Proposition 18.4(a) with $a = c$: the own-group weight is $1/(1+\lambda_g)$. The gap vanishes if returns are equal, or *if $\lambda_g = 1$ (everyone sees the same signal). The preference gap IS the aspiration gap: women form lower aspirations because they observe lower returns, which are lower because their structural position is constrained by gender norms.*

Proof. With returns fixed, the signal of agent i in group g is the constant $\bar{s}^g$, so (18.5) is the scalar affine recursion $\theta_{t+1} = \rho\theta_t + (1-\rho)\bar{\theta} + \mu(\bar{s}^g - r^*)$. Its unique fixed point is $\theta^{*g} = \bar{\theta} + \frac{\mu}{1-\rho}(\bar{s}^g - r^*)$, and $\theta_{t+1} - \theta^{*g} = \rho(\theta_t - \theta^{*g})$, so $\theta_t \to \theta^{*g}$ because $\rho < 1$. Subtracting the two groups' fixed points gives the first equality in (18.6). With two equal groups and W^I as in Proposition 18.4(a) with $a = c$, every row puts total weight $1/(1+\lambda_g)$ on its own group and $\lambda_g/(1+\lambda_g)$ on the other. Hence $\bar{s}^F = \frac{\bar{r}^F + \lambda_g \bar{r}^M}{1+\lambda_g}$, $\bar{s}^M = \frac{\bar{r}^M + \lambda_g \bar{r}^F}{1+\lambda_g}$ and $\bar{s}^F - \bar{s}^M = \frac{1-\lambda_g}{1+\lambda_g}(\bar{r}^F - \bar{r}^M)$. □

Key Structural Insight

The gender gap operates through all three channels simultaneously:
Power channel (W^P): Gender norms channel women into low-π sectors. $\pi^F_{\text{avg}} < \pi^M_{\text{avg}}$. The bargaining allocation gives women less.
Information channel (W^I): Gender-assortative W^I shows women biased signals (low returns). Formation produces lower aspirations. The self-confirming trap: low aspirations → low investment → low returns → confirms low aspirations.
Utility channel (W^U): Gender norms may constrain who cares about whom. In patriarchal structures: men's W^U weights on women are low ("women's welfare doesn't enter the social welfare function"). This reduces women's externality centrality c^F_j, exacerbating the welfare loss.
Addressing the gender gap through any single channel is insufficient. Raising wages (enforcement on W^P) without changing norms (W^I) produces a partial effect with Ψ-attenuated formation. Changing norms (W^I) without opening sectors (W^P) produces aspiration frustration. The optimal intervention changes all three simultaneously — which is why comprehensive gender programs (education + labor market access + media representation + legal reform) outperform single-instrument policies.

Visualization: Gender as block-diagonal structure



W^P **(gender-segmented)**
within 55 / 35, cross 10; $\Gamma = 0.38$

π
$\pi^F_{\text{avg}} / \pi^M_{\text{avg}} = 0.69$

A stylized example, not Bolivia's measured structure (Example 18.7). Left: raw intensities K, brighter in the male block (55: denser linkages in male-dominated sectors) and dimmer in the female block (35: sparser linkages in female-dominated sectors). Cross-gender entries are dark (10). Right: the Perron vector of $W = D^{-1}K$. Since K is symmetric, $\pi_i \propto d_i$ (Proposition 18.6): $\pi = 0.148$ for each man and 0.102 for each woman, against $1/N = 0.125$. The gap comes from the unequal block densities. With equal densities the same cross weight would give uniform π.

18.3 Race and Ethnicity

18.3.1 Race as a Multi-Layer Constraint

Definition 18.9 (Racial Assortative Mixing)**.** *Let agents be classified into R racial/ethnic groups. The racial mixing function is:*

$$\phi_{race}(r_i, r_j) = \begin{cases} 1 & \text{if } r_i = r_j, \\ \lambda_r & \text{if } r_i \neq r_j, \end{cases} \tag{18.7}$$

where $\lambda_r \in [0, 1]$ is the cross-racial openness parameter. Racial classification, like gender norms, is socially constructed (Quijano, 2000a). The parameter λ_r is therefore a property of the social system, not of biology.

Conjecture 18.10 (Race Constrains All Three Networks Simultaneously)**.** *In many societies, racial assortative mixing operates on W^U, W^P, and W^I simultaneously, with positively correlated λ_r values (illustrative ranges, not estimates):*

1. *W^U: interracial social bonds are weaker ($\lambda_r^U \approx$ 0.2–0.4). Who you care about is structured by race.*
2. *W^P: labor markets are racially segmented ($\lambda_r^P \approx$ 0.3–0.6). Racial minorities are channeled into low-π sectors.*
3. *W^I: media, education, and social networks are racially homophilous ($\lambda_r^I \approx$ 0.2–0.5). The signals you receive come mostly from your racial group.*

Equal values of λ_r across layers do not by themselves align the Perron vectors. With symmetric ϕ, π reflects degree (Proposition 18.6). What aligns them is that the same *groups are smaller, or have the sparser opportunity structure, in every layer. When that holds, racial minorities have low power in W^U, in W^P and in W^I at once. This is our formal reading of Quijano's thesis that coloniality structures every sphere of social existence: assortative mixing combined with the same asymmetry in every layer.*

Example 18.11 (Bolivia's Ethnic Structure [ILLUSTRATIVE])**.** *In Bolivia's 2012 Census, about 41% of the population aged 15 and over declared belonging (pertenencia) to an indigenous nation or people, Quechua and Aymara being the largest. The remaining 59% declared no such belonging. The census offered no "mestizo" category, an omission that was publicly contested.*

Illustrative λ_r values for Bolivia (the evidence column lists the kind of data that could identify each value; the numbers are not estimated from it):

Network	λ_r *(indig.–non-indig.)*	*Possible evidence*
W^U	≈ 0.25	*Intermarriage rates (census)*
W^P	≈ 0.35	*Occupational segregation*
W^I	≈ 0.30	*Language segregation in media*

Illustrative per-capita power index by ethnicity (relative units):

	Population share	π_{avg} *(per-capita index)*
Indigenous	*0.41*	*0.008*
Non-indigenous	*0.59*	*0.014*
Ratio	*0.69*	*0.57*

Indigenous Bolivians are 41% of the population but would hold about 28% of structural power ($0.41 \times 0.008 / (0.41 \times 0.008 + 0.59 \times 0.014) = 0.284$*). The power–population ratio is* $0.284/0.41 = 0.69$*, and each indigenous person has* $0.008/0.014 = 0.57$ *of the structural power of a non-indigenous person. Homophily alone, at* $\lambda_r^P = 0.35$*, gives only* 0.84 *(formula* (18.4)*). The rest of the gap would have to come from the sectoral opportunity structure. Combined with the gender ratio of Example 18.7, an indigenous woman would hold about* $0.57 \times 0.58 = 0.33$ *of a non-indigenous man's per-capita power (if the two factors combine multiplicatively; see Section 18.3.2).*

18.3.2 Intersectionality: the Separable Baseline and the Interaction Term

Definition 18.12 (Separable Mixing and the Intersectional Interaction)**.** *When several attributes* $(a_1, \ldots, a_K)$ *constrain* W *through mixing functions* $\phi_1, \ldots, \phi_K$*, the* ***separable baseline*** *is*

$$\phi^{sep}(\mathbf{a}_i, \mathbf{a}_j) = \prod_{k=1}^{K} \phi_k(a_{ik}, a_{jk}). \tag{18.8}$$

The ***intersectional interaction*** *is the departure from it:*

$$\iota(\mathbf{a}_i, \mathbf{a}_j) = \phi(\mathbf{a}_i, \mathbf{a}_j) - \prod_{k=1}^{K} \phi_k(a_{ik}, a_{jk}). \tag{18.9}$$

Under the separable baseline, agents who differ on two attributes face $\lambda_1\lambda_2$, and on three $\lambda_1\lambda_2\lambda_3$. With the illustrative values of this chapter:

$$\phi^{\text{sep}}_{\text{indigenous woman, non-indigenous man}} = \lambda_r \cdot \lambda_g \approx 0.35 \times 0.30 = 0.105. \tag{18.10}$$

Her cross-group edge weight to a non-indigenous man is 10.5% of the unconstrained maximum, in W^P, W^I and W^U alike: the blocks are nested.

Two cautions.

1. Edge suppression is not a power gap. With symmetric ϕ and population-weighted opportunities, Exercise 18.3 shows a per-capita power ratio of 0.84 for the indigenous woman relative to the non-indigenous man, not 0.105. That 0.84 is entirely a group-size effect.

2. The product form is the *no-interaction* case. In logs it is additive: $\log \phi^{\text{sep}} = \sum_k \log \phi_k$. Crenshaw's point is precisely that the experience at an intersection is *not* decomposable into single-axis effects. In this notation, intersectionality lives in $\iota \neq 0$.

When separability fails. Separability is plausible when gender and racial constraints operate through unrelated mechanisms. It fails when they interact, for example when racialised stereotypes about gender roles differ across groups. The interaction can be:

- **reinforcing** ($\iota < 0$): the constraint on indigenous women is worse than the product predicts;

- **compensating** ($\iota > 0$): for example, women's market and solidarity networks that partially offset the double constraint.

The reinforcing case produces even lower Perron weights for agents disadvantaged on several axes; the compensating case produces less extreme outcomes. Which case holds is an empirical question: estimate the full ϕ on the cross-classified groups and test $\iota = 0$.

Intuition: Products, logs and interactions

Take three constraints: gender (λ_g), race (λ_r) and class (λ_c). The separable baseline gives $\lambda_g \lambda_r \lambda_c = 0.3 \times 0.35 \times 0.4 = 0.042$: an indigenous working-class woman has about 4% of the unconstrained edge weight to a non-indigenous middle-class man. This is equivalent to an additive model in log edge weights, so "multiplicative" and "additive" are not rival theories here. Both describe a world without intersectional interaction. What the formal model adds is a way to state the intersectional claim precisely ($\iota \neq 0$) and to test it.

18.4 The Formation Trap at the Individual Scale

Proposition 18.13 (Attribute-Mediated Self-Confirming Preferences)**.** *Consider two equal groups with exogenous, constant mean returns $\bar{r}_{adv} > \bar{r}_{disadv}$. The information matrix W^I gives every member of a group total own-group weight $1/(1+\lambda_k)$ and cross-group weight $\lambda_k/(1+\lambda_k)$. Let $0 \le \rho < 1$. Under the formation equation (18.5), applied to both groups, steady-state preferences are*

$$\theta_g^* = \bar{\theta} + \frac{\mu}{1-\rho}\left(\bar{s}_g - r^*\right), \qquad \bar{s}_g = \tfrac{1}{1+\lambda_k}\bar{r}_g + \tfrac{\lambda_k}{1+\lambda_k}\bar{r}_{-g}, \tag{18.11}$$

so that

$$\Delta\theta^* = \theta^*_{adv} - \theta^*_{disadv} = \mathcal{M}\,\frac{1-\lambda_k}{1+\lambda_k}\,(\bar{r}_{adv} - \bar{r}_{disadv}), \qquad \mathcal{M} = \frac{\mu}{1-\rho}.$$

The preference gap is:

1. *Proportional to the formation multiplier $\mathcal{M}$: sticky preferences amplify the signal gap into a larger preference gap.*

2. *Proportional to the return gap, scaled by $(1-\lambda_k)/(1+\lambda_k)$. This factor equals the second eigenvalue of the group-level W^I (Proposition 18.4(a) with $a = c$) and vanishes at $\lambda_k = 1$.*

3. *Self-confirming: each group's preferences are consistent with its W^I signal. If returns depend on θ (Exercise 18.4), the gap feeds back on itself. (This item interprets the steady state; it is not an additional claim.)*

Proof. This is Proposition 18.8 with the groups relabelled. Set $\theta_{t+1} = \theta_t$ in (18.5). Then $(1-\rho)\theta_g^* = (1-\rho)\bar{\theta} + \mu(\bar{s}_g - r^*)$, a unique fixed point since $\rho < 1$, and it is globally stable because deviations shrink by the factor ρ each period. Subtract the two groups' equations and use $\bar{s}_{\text{adv}} - \bar{s}_{\text{disadv}} = \frac{1}{1+\lambda_k}(\bar{r}_{\text{adv}} - \bar{r}_{\text{disadv}}) - \frac{\lambda_k}{1+\lambda_k}(\bar{r}_{\text{adv}} - \bar{r}_{\text{disadv}}) = \frac{1-\lambda_k}{1+\lambda_k}(\bar{r}_{\text{adv}} - \bar{r}_{\text{disadv}})$. For item 2, the group-level W^I is $\frac{1}{1+\lambda_k}\begin{pmatrix} 1 & \lambda_k \\ \lambda_k & 1 \end{pmatrix}$, whose eigenvalues are 1 and $\frac{1-\lambda_k}{1+\lambda_k}$. □

Example 18.14 (Gender Aspiration Gap in Bolivia (partly [ILLUSTRATIVE]))**.** *Account ownership is from the World Bank's Global Findex database, as published in the World Development Indicators. The business saving and borrowing rows are illustrative figures of the kind Findex reports; replace them with the published Bolivia tabulations, using the database's own indicator definitions, before use.*

	Women	*Men*
"Saved for business" (past year; illustrative)	*12%*	*22%*
"Borrowed for business" (past year; illustrative)	*8%*	*18%*
Has an account, age 15+ (2021)	*63.3%*	*74.0%*
Has an account, age 15+ (2024)	*53.5%*	*60.3%*

Source for account ownership: Global Findex via WDI, FX.OWN.TOTL.FE.ZS and FX.OWN.TOTL.MA.ZS (World Bank, 2026b).

The standard explanation: women have lower returns to business (enforcement: discrimination, credit constraints). The formation explanation: women observe lower returns in their network (W^I is gender-assortative), form lower business aspirations ($\theta^{*F}_{business} < \theta^{*M}_{business}$), invest less, earn lower returns, confirming the aspirations.

Suppose (illustratively) that the Chapter 11 cohort-year method, applied to female business ownership, gave

$$\Psi_{gender,\ business} = 2.3. \tag{18.12}$$

The formation channel would then be 2.3 times the enforcement channel. Removing credit constraints (enforcement) addresses 30% of the gap. Changing what women see — role models, media representation, cross-gender mentoring (information policy on W^I) — addresses 70%.

18.5 The Welfare Cost of Attribute Constraints

Definition 18.15 (Welfare Cost of an Attribute Constraint)**.** *The welfare loss associated with attribute constraint k (mixing parameter λ_k) is*

$$L(\lambda_k) = \underbrace{\frac{\kappa N}{2}\,\Phi^2\ \mathrm{Var}(\pi(\lambda_k))}_{L_{power}(\lambda_k)} + L_{formation}(\lambda_k). \tag{18.13}$$

*L_{power} is the exact distributional (utilitarian) loss under the allocation rule $x_i = \Omega/N + (\pi_i - 1/N)\Phi$ with quadratic utility (Proposition 7.3). $L_{formation}$ is the additional loss from the preference gap produced by the formation model (Chapter 12). It is not one of the three terms of Theorem 7.8: the externality and interaction terms, which depend on W^U through the c_j, are set aside here. The **cost of discrimination** is*

$$\Delta L = L(\lambda_k) - L(1), \tag{18.14}$$

the part of the loss attributable to the attribute constraint itself, holding the opportunity structure $K^{\max}$ fixed.

Example 18.16 (The Cost of Gender Discrimination [ILLUSTRATIVE])**.** *Take two equal groups and the calibration of Example 18.7: $\pi^F_{avg}/\pi^M_{avg} = 0.58$ at $\lambda_g = 0.3$, and 0.73 at $\lambda_g = 1$ with the same densities. Let $\kappa = 0.5$, per-capita endowment $\Omega/N = 1$ and per-capita bargaining surplus $\Phi/N = 1$.*

With per-capita ratio r, men have $N\pi_i = 2/(1+r)$ and women $2r/(1+r)$, so $N^2\,\mathrm{Var}(\pi) = \big((1-r)/(1+r)\big)^2$. For $r = 0.58$ this is 0.0707. Note that $\mathrm{Var}(\pi)$ itself is of order $1/N^2$: at $N = 11$ million its maximum possible value is $(N-1)/N^2 \approx 9 \times 10^{-8}$.

Allocations are $x_M = 1.266$ and $x_F = 0.734$, both with $\kappa x < 1$. The per-capita power loss is

$$\frac{L_{power}}{N} = \frac{\kappa}{2}\Big(\frac{\Phi}{N}\Big)^2 N^2\,\mathrm{Var}(\pi) = 0.25 \times 0.0707 = 0.0177,$$

which is 2.4% of $U(\Omega/N) = 0.75$. At $\lambda_g = 1$ ($r = 0.73$) the loss would be 0.0062. The cost of discrimination in the power channel is therefore 0.0114 per capita, about 65% of the power loss. The rest reflects the occupational opportunity structure.

If, as an illustrative assumption, the formation loss equals Ψ times the power loss with $\Psi = 2.3$, then $L_{total} = (1+\Psi)L_{power} = 3.3\,L_{power}$. An estimate that ignored formation would understate the cost by that factor. (Numbers: `code/ch18.py`.)

18.6 Policy: Changing λ

Remark 18.17 (Policy Instruments for Attribute Constraints).

Policy	**Target**	**Effect on λ**	Ψ
Affirmative action (quotas)	W^P	$\lambda^P_g \uparrow$ directly	Low
Equal-pay legislation	W^P (prices)	$\lambda^P_g \uparrow$ indirectly	Low
Mentoring programs	W^I	$\lambda^I_g \uparrow$	High
Media representation	W^I	$\lambda^I_g \uparrow$	High
Education coeducation	$W^I + W^U$	$\lambda^I_g, \lambda^U_g \uparrow$	High
Anti-discrimination law	Institutional	All $\lambda \uparrow$ via norms	Medium

The sequencing logic of Section 15.3 applies: high-Ψ interventions first (W^I policies: mentoring, media, education) then low-Ψ (W^P policies: quotas, equal pay). This is because changing W^I shifts the aspirations of the next generation, making the enforcement interventions more effective when they arrive. The Ψ column is a qualitative ranking, not an estimate.

18.7 Summary

1. **Node attributes** (gender, race, ethnicity, caste) constrain W through assortative mixing functions $\phi(\mathbf{a}_i, \mathbf{a}_j) \leq 1$. Lower ϕ means weaker cross-group edges and a more block-diagonal W.

2. Assortative mixing **reduces** Γ, i.e. integration across groups. This is exact for an equitable two-block partition when within-group links are at least as strong as cross-group links, and holds in general through the Cheeger bound. It does not by itself concentrate power: in the symmetric case π is uniform for every λ.

3. With symmetric homophily, **power is proportional to degree**. A group power gap comes from group size (minority ratio 0.84 at $p = 0.41$, $\lambda = 0.35$) or from an asymmetric opportunity structure. It is the latter that can produce a gender gap at equal population shares (illustrative ratio 0.58).

4. **Intersectionality** is the interaction term $\iota = \phi - \prod_k \phi_k$. The product of single-attribute constraints (e.g. $0.35 \times 0.30 = 0.105$ for the edge weight) is the separable, no-interaction baseline, and edge suppression is not the same as a power gap.

5. The **formation trap** operates through assortative W^I: the preference gap is $\mathcal{M}\,\frac{1-\lambda}{1+\lambda}$ times the return gap. Disadvantaged groups observe biased signals, form lower aspirations, invest less and earn less, which confirms the signal. Under the illustrative assumption $L_{\text{formation}} = \Psi L_{\text{power}}$, the welfare cost of discrimination is $(1 + \Psi)$ times its power component.

18.8 Exercises

18.1 Assortative mixing and Γ. For a 6×6 economy (3 men, 3 women): construct W^P with $\lambda_g = 1, 0.7, 0.4, 0.1, 0$. Compute $\Gamma(\lambda_g)$ for each. Plot Γ vs. λ_g. Is the relationship linear?

18.2 Bolivia's gender power gap. Using Bolivia's EH 2019:

(a) Classify sectors as male-dominated (>60% male), female-dominated (>60% female), and mixed.

(b) Compute the average Leontief multiplier for each group. Which has the highest?

(c) Approximate $\pi^F_{\text{avg}}/\pi^M_{\text{avg}}$ from the sectoral Perron weights and gender composition.

(d) How much would the gender power gap close if women entered male-dominated sectors at 40% (up from about 10% in gas/mining, construction and public administration in 2019; Example 18.7)?

18.3 Intersectionality. For $N = 4$ group-agents: Agent 1 (non-indigenous man), Agent 2 (non-indigenous woman), Agent 3 (indigenous man), Agent 4 (indigenous woman), with population shares $n = (0.295, 0.295, 0.205, 0.205)$. Let $\lambda_g = 0.3$ and $\lambda_r = 0.35$.

(a) Write the separable constraint matrix $\phi(\mathbf{a}_i, \mathbf{a}_j)$ for all 16 pairs.

(b) First take $W^{\max} = \frac{1}{4}\mathbf{1}\mathbf{1}'$ and $W = D^{-1}(\phi \circ W^{\max})$. Compute $\boldsymbol{\pi}$ and show that it is uniform. Why? (Hint: row sums of ϕ; Proposition 18.6.)

(c) Now weight opportunities by population, $K_{ij} = \phi_{ij}\, n_j$ (each group-agent stands for a group of size n_j), and $W = D^{-1}K$. Compute $\boldsymbol{\pi}$ and the per-capita power π_i/n_i. Verify that the two indigenous groups have the lowest per-capita power, and that gender makes no difference.

(d) Compute the per-capita ratio $(\pi_4/n_4)/(\pi_1/n_1)$. Compare it with $\lambda_g\lambda_r$ and with formula (18.4). What does this say about edge suppression versus power?

18.4 The formation trap. Consider two equal groups (advantaged A, disadvantaged D) with $\lambda = 0.25$, so that after row normalisation the own-group weight is $1/(1+\lambda) = 0.8$. Let $\rho = 0.85$, $\mu = 0.3$, $\bar{\theta} = 1$ and $r^* = 0.1$.

(a) Set up the formation model: $\theta^A_{t+1} = \rho\theta^A + (1-\rho)\bar{\theta} + \mu(0.8r^A + 0.2r^D - r^*)$, and similarly for D. Returns are $r^A = 0.05 + 0.1\theta^A$ and $r^D = 0.03 + 0.1\theta^D$: D's constrained opportunity structure costs it 0.02.

(b) Compute both steady states and $\Delta\theta^* = \theta^{*A} - \theta^{*D}$.

(c) Now set $\lambda = 0.5$ (own weight $2/3$). Compute the new steady states. By how much does $\Delta\theta^*$ shrink?

(d) At what λ does $\Delta\theta^*$ fall below 0.01?

(e) Show that if both groups had $r^g = 0.05 + 0.1\theta^g$, then $\Delta\theta^* = 0$ for every λ. Assortative information alone does not create a gap; it propagates one.

18.5 The cost of discrimination. Using the illustrative values of Example 18.16 ($\lambda_g = 0.3$, $\Psi = 2.3$): compute the welfare loss from gender discrimination as a fraction of GDP (you will need to assume a relationship between the welfare loss L and GDP). Compare your answer with the order of magnitude of income losses from gender gaps in labor markets estimated in the literature (e.g., Cuberes and Teignier, 2016).

18.6 Optimal intervention design. For the gender gap in Bolivia:

(a) Design three interventions: (a) quota (ΔW^P), (b) mentoring program (ΔW^I), (c) media campaign (ΔW^I).

(b) For each: estimate $\Delta\lambda_g$ and $\Delta\Gamma$.

(c) Using $\Psi = 2.3$: compute the total effect (enforcement + formation) for each.

(d) Sequence them optimally (Chapter 15). Which goes first?

18.7 Aspirations data exercise. Using a household survey with a module on educational aspirations (for example a UNICEF MICS round for a country where one is available, or a national survey):

(a) Compute the aspiration gap: the difference in educational aspirations ("how far in school do you want your child to go?") between indigenous and non-indigenous mothers, controlling for income and urban/rural.

(b) Is the gap consistent with the formation model's prediction? (Larger gap where W^I is more assortative.)

(c) Compare the aspiration gap in urban La Paz (where λ_r may be higher due to mixing) vs. rural Altiplano (where λ_r is lower). Does the gap vary as predicted?

Mathematical Appendices

Appendix 18A: Assortative Mixing and Spectral Gap

Setup. Partition N agents into G groups based on attribute k. Raw intensities are $K_{ij} = a$ within groups and $K_{ij} = \lambda_k c$ across groups, and $W = D^{-1}K$.

Eigenvalues. For $G = 2$ groups of equal size n, the group-level quotient is $\begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix}$ with $\alpha = a/(a + \lambda_k c)$ and $\beta = 1 - \alpha$. Hence $\lambda_1 = 1$ and $\lambda_2 = (a - \lambda_k c)/(a + \lambda_k c)$; the remaining $N - 2$ eigenvalues are 0. With $a = c$ this gives $\lambda_2 = (1 - \lambda_k)/(1 + \lambda_k)$ and $\Gamma = 2\lambda_k/(1 + \lambda_k)$: 0.18, 0.46, 0.67 and 1 at $\lambda_k = 0.1$, 0.3, 0.5 and 1. This is monotonically increasing in λ_k, as in Proposition 18.4(a). For $c > a$ the absolute value $|\lambda_2|$ is not monotone and the conclusion can reverse (Remark 18.5). The Cheeger bound of part (b) is tight in this example: $2\lambda_k c/(a + \lambda_k c)$ equals Γ.

Perron eigenvector. With symmetric K, $\pi \propto K\mathbf{1}$ (Proposition 18.6). With unequal group sizes or unequal internal densities, π assigns more per-capita weight to the larger group or the denser one, which formalizes the "structural advantage" of the dominant group. Without such an asymmetry, homophily leaves π uniform.

Appendix 18B: Estimating Mixing Parameters

The separable baseline and the intersectional interaction term, formerly in this appendix, are now in Section 18.3.2.

Empirical estimation. The mixing parameters λ_k can be estimated from social network data (intermarriage rates, occupational segregation indices, media consumption patterns) or from labor market outcomes using the structure: wage gap $= f(\lambda_g, \lambda_r, \ldots)$ with the Perron eigenvector providing the functional form. To test separability, estimate ϕ on the cross-classified groups (gender $\times$ ethnicity) and test $\iota = 0$ in (18.9); a log-linear model of edge weights with an interaction term is the natural specification.

Structuralist Case Study: Quijano and Marini — Race, Gender, and the Weight Matrix

This chapter's case study pairs the two structuralist thinkers who brought race and labor exploitation to the center of structural analysis: Quijano's coloniality of

power and Marini's superexploitation. Together they provide the theory for what this chapter formalizes as attribute-constrained weight matrices.

Quijano (2000a): Race as the master attribute. Quijano's essay "Colonialidad del poder" argued that the colonial invention of race was the foundational structuring principle of the modern world-system — more fundamental than class, nationality, or gender. The racial classification imposed during the conquest (European/Indian/Black/Mestizo) did not merely describe physical differences; it *organized* the entire pattern of social interaction. In this chapter's notation: race is the node attribute with the broadest scope — it constrains W^U, W^P, and W^I simultaneously, with low λ_r values across all three layers (Conjecture 18.10). What makes the classification effective as an "instrument of domination" is not homophily alone, which by itself leaves power equal between equal-sized groups (Proposition 18.6). It is homophily combined with an opportunity structure that is unequal in the same direction in every layer.

Quijano's second claim was that coloniality *survived* independence. The formal colonial apparatus was dismantled, but the racial classification continued to structure social interaction. In the formation model, racial assortative mixing ($\lambda_r < 1$) propagates return gaps into self-confirming preference gaps (Proposition 18.13). Indigenous communities that observe mainly the returns available in their own constrained opportunity structure form aspirations calibrated to those returns.

How long would such gaps take to fade? With half-life $\ln 2/\ln(1/\rho)$ generations (Chapter 9), $\rho = 0.85$–0.90 gives 4.3–6.6 generations, or roughly 110–200 years. Five centuries are therefore 2.5–4.7 half-lives. Initial conditions alone would have decayed to between 4% and 18% of their original size. The persistence Quijano describes therefore cannot rest on slow decay. In the model it requires that λ_r and the unequal opportunity structure be *reproduced* each generation, which is exactly his thesis that coloniality is an ongoing pattern of power, not a residue.

Marini (1973): superexploitation, amplified by race. For Marini, superexploitation means that labor-power is paid below its value, through intensified work, a longer working day and wages below what workers need to reproduce themselves. It is a Marxian value concept, not a statement about marginal products. In this book's bargaining reading (an interpretation, not Marini's own apparatus; Chapter 7), it corresponds to a weak structural position of workers, $\pi_{\text{worker}} \ll 1/N$. This chapter adds the racial dimension: within the worker class, racial minorities face an additional constraint ($\lambda_r < 1$) and a more restricted opportunity structure, which further reduce their Perron weight. Indigenous workers are *superexploited among the superexploited*: $\pi_{\text{indigenous worker}} < \pi_{\text{non-indigenous worker}} \ll 1/N$.

Combining Marini's class analysis with Quijano's racial analysis gives the intersectional structure of Section 18.3.2. Under the separable baseline, an indigenous working-class woman faces the product $\lambda_r \lambda_g \lambda_c$ of three edge constraints. Whether her position is worse still — whether the axes interact ($\iota < 0$) — is the empirical content of the intersectional claim.

Concept	**Quijano (2000)**	**Marini (1973)**
Core claim	Race structures all interaction	Labor-power paid below its value
Formal object	$\lambda_r < 1$ plus asymmetric $K^{\max}$ across W^U, W^P, W^I	$\pi_{\text{worker}} \ll 1/N$ (bargaining reading)
Mechanism	Assortative mixing, aligned	Asymmetric W^P, weak labor position
Persistence	Coloniality reproduced each generation	Superexploitation is structural
Formation link	Racial W^I produces self-confirming	Low $\pi \to$ low returns $\to$ low θ
Intersectionality	Separable baseline $\prod_k \phi_k$; interaction ι	Superexploitation within superexploitation
Bolivia (illustrative)	$\lambda_r \approx 0.35$, $\pi^{\text{indig}} / \pi^{\text{non-indig}} = 0.57$	—

Quijano wrote from Lima. Marini wrote from exile. Neither had the Perron eigenvector. But both understood that the social classification system — race for Quijano, class for Marini — is not a label attached to individuals. It is a structural property of the weight matrix: a constraint on which edges can exist, which determines who has power, which determines who gets what, which reproduces the classification. The circularity is the structure. This chapter gives it its eigenvalues.

Further Reading

Gender and networks: Beaman et al. (2012), "Female Leadership Raises Aspirations." Bursztyn et al. (2020), "Misperceived Social Norms: Women Working Outside the Home in Saudi Arabia." Cuberes and Teignier (2016), "Aggregate Effects of Gender Gaps in the Labor Market."

Race and economic structure: Darity and Mason (1998), "Evidence on Discrimination in Employment." Lang and Lehmann (2012), "Racial Discrimination in the Labor Market: Theory and Empirics." For Bolivia: Jiménez Pozo,

Landa Casazola, and Yañez Aguilar (2006), "Bolivia: Indicadores de equidad étnica."

Intersectionality: Crenshaw (1989), "Demarginalizing the Intersection of Race and Sex."

Decolonial theory: Quijano (2000a), "Colonialidad del poder, eurocentrismo y América Latina." Mignolo (2000), *Local Histories/Global Designs*. Walsh (2007), "Interculturalidad y colonialidad del poder."

Assortative mixing in networks: Newman (2003), "Mixing Patterns in Networks." Currarini, Jackson, and Pin (2009), "An Economic Model of Friendship." Golub and Jackson (2012), "How Homophily Affects the Speed of Learning and Best-Response Dynamics" — the spectral link between homophily and integration used in Proposition 18.4.

Chapter 19

Scale Invariance

Five Processes, One Process

> *"El desarrollo, el subdesarrollo, la dependencia, la marginalidad y la desigualdad espacial son manifestaciones diversas de un proceso único."*
>
> ["Development, underdevelopment, dependence, marginality, and spatial inequality are diverse manifestations of a single process."]
> — Osvaldo Sunkel and Pedro Paz, *El subdesarrollo latinoamericano y la teoría del desarrollo* (1970)

Chapter 16 analyzed the global W. Chapter 17 analyzed the domestic W. Chapter 18 analyzed the individual-level W. This chapter asks in what precise sense they are the same structure at different magnifications.

The working hypothesis: at every scale, the weight matrix has the same qualitative form — a dense core, a sparse periphery, weak cross-block linkages, a concentrated Perron eigenvector, a small spectral gap. We call this **structural isomorphism** across scales (Definition 19.3). It is a hypothesis about the recursive structure of social organization: the mechanisms that produce center-periphery at the global scale (proximity-dominated edge creation, assortative mixing, self-confirming preferences) may also operate at the national and individual scales. For Bolivia the data contradict part of it: the global and national matrices that can be measured are not weakly coupled ($\Gamma \approx 0.45$ at both; Section 19.1). Two exact results discipline the hypothesis. Aggregation mechanically raises the spectral gap (Proposition 19.5), so Γ values at different scales are not directly comparable. And individual power factors exactly into country, sector and within-sector shares (Proposition 19.7).

Sunkel and Paz (1970) stated the underlying idea in prose: development, underdevelopment, dependence, marginality, and spatial inequality are manifestations of a single process. This chapter gives that claim a precise matrix vocabulary.

19.1 The Three Scales

19.1.1 Notation

Let the subscript G, N, I denote the global, national, and individual scales:

	Global (G)	**National** (N)	**Individual** (I)
Nodes	Countries	Sectors	Persons
W^P	Bilateral trade	I-O table	Employment network
W^I	Media/academic flows	Domestic media	Social network
W^U	Aid/migration	Community ties	Friendships
Nodes n_s	≈ 200	≈ 12–100	$\approx 10^6$–10^8
Data source	BACI, ICIO	INE I-O table	Household surveys

Each scale has its own weight matrices (W_s^P, W_s^I, W_s^U) for $s \in \{G, N, I\}$, its own composite W_s, its own Perron eigenvector $\boldsymbol{\pi}_s$, its own spectral gap Γ_s, and its own formation dynamics $(\rho_s, \mu_s, \bar{\theta}_{0,s})$. (The subscript N means "national"; the number of nodes at scale s is written n_s.)

19.1.2 The Same Structure at Every Scale

Conjecture 19.1 (Structural Isomorphism Across Scales)**.** *At all three scales, the weight matrix satisfies:*

1. ***Not doubly stochastic**: column sums vary widely.* $\max_j \sum_i W_{ij} / \min_j \sum_i W_{ij} \gg 1$.

2. ***Core-periphery topology**: a dense core of well-connected nodes and a sparse periphery.*

3. ***Small spectral gap**:* $\Gamma_s \ll 1$ *for* $s \in \{G, N_{Bolivia}, I_{Bolivia}\}$.

4. ***Concentrated Perron eigenvector**:* $\max_i \pi_{s,i} / \min_i \pi_{s,i} \gg 1$.

5. ***Block-diagonal approximation**: W_s is approximately block-diagonal, with k_s blocks.*

Items 3 and 4 are logically independent: Γ *measures integration and* N_{eff} *or* $\mathrm{Var}(\pi)$ *measure concentration (Proposition 6.16).*

At the two scales where Bolivia's matrices can be measured, item 3 fails: the global trade matrix has $\Gamma_G = 0.452$ and Bolivia's domestic production matrix $\Gamma_N = 0.467$ (Example 19.2). Neither is small, and at the national scale no block structure is visible (item 5). Items 1 and 4 hold at both (column sums and Perron weights vary widely). The conjecture survives, if at all, in a weaker form: unequal column sums and concentrated $\boldsymbol{\pi}$ without fragmentation.

Example 19.2 (Bolivia at Three Scales (partly [ILLUSTRATIVE])). *The* Γ *and* $\max \pi / \min \pi$ *rows of the global and national columns are measured: the global column from bilateral goods trade among 230 economies in 2019 (Chapter 16; The Growth Lab at Harvard University, 2026), the national column from INE's 2019 supply–use tables (Chapter 17; Instituto Nacional de Estadística (INE), Bolivia, 2026). The other rows are illustrative, and the individual column is* hypothetical*: no data set records the person-level W for* 10^6–10^8 *people.*

Property	*Global*	*National*	*Individual*
Γ	*0.452*	*0.467*	*(*≈ 0.10*)*
$\max \pi / \min \pi$	*193 (US/Laos)*	*53 (other serv./pub. adm.)*	*(*≈ 50*)*
Number of blocks	*3 (core/semi/per.)*	*none visible (*$\lambda_2 = 0.53$*)*	*3+ (gender*$\times$*ethn.)*
σ_B	≈ 0.55	*0.62*	*(*≈ 0.50*)*
$N_{eff}(\pi^I)$	*3–5*	*3.4*	—

Sources: global: Growth Lab bilateral trade, HS92, 2019, imports-only W (The Growth Lab at Harvard University, 2026); national: INE supply–use tables 2019, domestic, input shares, 11 sectors (Instituto Nacional de Estadística (INE), Bolivia, 2026); author's calculations (`data/compute/`, `code/ch19.py`*). The US/Laos pair is the book's; the global minimum is far smaller. Domestic services has no* W^P *row, so the national minimum is public administration (*$\pi = 0.0051$*).* σ_B*,* N_{eff}*, the global block count and the individual column are illustrative.*

*The values should not be over-read. The domestic and global spectral gaps (*0.467 *and* 0.452*) are almost equal, and neither is small; at the national scale Bolivia's matrix mixes about as fast as those of the comparator economies (Chapter 17). Spectral gaps of matrices of very different size and aggregation are not directly comparable in any case: aggregation alone raises* Γ *(Proposition 19.5), so a lower* Γ *at a finer scale is partly mechanical, and the national value ranges over* 0.33–0.69 *across normalizations (Chapter 17). The illustrative between-block share of inequality is 55–62% at the two scales.* N_{eff} *is defined on the Perron vector of* W^I *at each scale, so it is bounded by the number of nodes; it is reported only where that vector exists. (The number of reference persons per individual is a different object, the row-level* $1 / \sum_j W_{ij}^2$*, and is not comparable.)*

19.2 Structural Isomorphism and Aggregation

Definition 19.3 (Structural Isomorphism Across Scales)**.** *The matrices (W_G, W_N, W_I) are called **structurally isomorphic** if each is approximately block-diagonal with a dense core and a sparse periphery, and if at each scale:*

(a) cross-block coupling is weak (small conductance), and hence Γ_s is small;

(b) power is concentrated: $N_{eff}(\pi_s)/n_s$ is well below 1;

(c) preferences are stratified: $\theta^*_{s,core} > \theta^*_{s,periphery}$;

(d) allocations are unequal: $x^*_{s,core} > x^*_{s,periphery}$.

Conjecture 19.4 (A Common Mechanism)**.** *Suppose that at every scale:*

1. *edge formation is proximity-dominated ($p_1 \gg p_2, p_3$ in the creation process of Definition 13.1);*
2. *returns are complementary ($\partial^2 g/\partial x_i \partial x_j > 0$ for neighbors in W^P);*
3. *preferences are formed through assortative W^I ($\lambda_k < 1$ for some attribute k; Proposition 18.13).*

Then the co-evolutionary dynamics of Chapter 13 (Section 13.2) select, for an open set of parameters at every scale, a structurally isomorphic equilibrium in the sense of Definition 19.3.

The conjecture is not a theorem. The co-evolutionary map has multiple equilibria (Proposition 13.9), and which one is selected depends on parameters that differ across scales. "Same functional form" does not imply "same qualitative fixed point".

What *can* be proved is how the spectral gap and the Perron vector behave under aggregation. This is what any comparison across scales has to control for.

Proposition 19.5 (Aggregation Raises the Spectral Gap)**.** *Let W be irreducible and row-stochastic on n nodes, and let $\mathcal{P} = \{B_1, \ldots, B_k\}$, $k \geq 2$, be a partition into nonempty blocks with indicator matrix V ($n \times k$). Suppose W is **exactly lumpable** with respect to $\mathcal{P}$ (Kemeny and Snell, 1960): $\sum_{j \in B'} W_{ij}$ is the same for all $i \in B$, for every pair of blocks B, B'. Call this common value $\bar{W}_{BB'}$. Then:*

1. *$\bar{W}$ is row-stochastic and irreducible, and $WV = V\bar{W}$;*
2. *every eigenvalue of $\bar{W}$ is an eigenvalue of W, so $\Gamma(\bar{W}) \geq \Gamma(W)$;*

3. *the Perron vector aggregates by summation:* $\bar{\pi}_B = \sum_{i \in B} \pi_i$.

Proof. (1) For $i \in B$, $(WV)_{iB'} = \sum_{j \in B'} W_{ij} = \bar{W}_{BB'} = (V\bar{W})_{iB'}$, and $\sum_{B'} \bar{W}_{BB'} = \sum_j W_{ij} = 1$. If $\bar{W}_{BB'} = 0$, no node of B has an edge into B'. A path $i \to \cdots \to j$ in W from B to B' therefore visits a sequence of blocks with positive entries of $\bar{W}$ between consecutive ones, so $\bar{W}$ is irreducible. (2) If $\bar{W}x = \mu x$ with $x \neq 0$, then $W(Vx) = V\bar{W}x = \mu\, Vx$, and $Vx \neq 0$ because the blocks are nonempty. So every eigenvalue of $\bar{W}$ is an eigenvalue of W. Both matrices are irreducible and stochastic, so 1 is a simple eigenvalue of each (Seneta, 2006, Thm. 1.5), and the second-largest modulus of each is the largest modulus among its eigenvalues other than 1. Since $\sigma(\bar{W}) \setminus \{1\} \subseteq \sigma(W) \setminus \{1\}$, $|\lambda_2(\bar{W})| \leq |\lambda_2(W)|$. (3) $\pi' V\bar{W} = \pi' WV = \pi' V$, and $\pi' V\mathbf{1} = \pi'\mathbf{1} = 1$. By irreducibility of $\bar{W}$ its stationary vector is unique, so $\bar{\pi} = \pi' V$. □

Two implications follow.

- Under exact lumpability $\Gamma_I \leq \Gamma_N$. A finer scale *always* looks at least as fragmented as the coarse one, whatever the economics, so an ordering such as $\Gamma_I < \Gamma_N$ in Example 19.2 is no evidence of a "more fragmented" individual scale.

- Aggregation (Definition 19.6) must sum, not average, Perron weights.

Scale comparisons of Γ are meaningful only within a fixed node set, or after an explicit aggregation.

Intuition: Why the same structure everywhere

The same three forces operate at every scale:

Proximity: Countries trade with trading partners of trading partners (global). Sectors buy from suppliers of suppliers (national). People connect with friends of friends (individual). At each scale, proximity makes new edges form near existing ones, producing path dependence and block-diagonal structures.

Complementarity: Rich countries trade more with rich countries, amplifying the core's density (global). Productive sectors buy more from productive sectors (national). High-skill workers hire high-skill workers (individual). At each scale, complementarity reinforces the dense core and starves the sparse periphery.

Assortative formation: Countries observe countries like themselves and converge on similar development doctrines (global W^I). Sectors influence sectors in their block (national W^I). People form aspirations from their reference group (individual W^I). At each scale, assortative information flow produces self-confirming preference stratification.

If the conjecture is right, removing any one of these forces breaks the isomorphism. All three are plausibly present at every scale of social organization, from the UN General Assembly to the village market, which is why the same mathematics is worth applying everywhere. Whether it produces the same structure is an empirical question.

19.3 The Nesting Structure

The three scales are not independent. They are nested: the individual-level W_I aggregates to produce the national-level W_N, which aggregates to produce the global-level W_G.

Definition 19.6 (Nested Weight Matrices)**.** *Let $\mathcal{P}_s$ be the partition of individuals into sectors, and $\mathcal{P}_c$ be the partition of individuals into countries. The national weight matrix W_N is the block-aggregation of W_I:*

$$W^N_{ss'} = \frac{\sum_{i \in s, j \in s'} W^I_{ij}}{\sum_{i \in s} \sum_j W^I_{ij}}, \tag{19.1}$$

and the global weight matrix W_G is the block-aggregation of W_I across countries:

$$W^G_{cc'} = \frac{\sum_{i\in c, j\in c'} W^I_{ij}}{\sum_{i\in c}\sum_j W^I_{ij}}. \tag{19.2}$$

These unweighted aggregates equal the lumped matrix of Proposition 19.5 when W_I is exactly lumpable. In general, the aggregate that preserves power is π-weighted, $\tilde{W}_{BB'} = \sum_{i\in B}\pi_i \sum_{j\in B'} W_{ij} / \sum_{i\in B}\pi_i$, whose Perron vector is $\bar{\pi}_B = \sum_{i\in B}\pi_i$. (Here W^I denotes the individual-scale matrix, not the information layer.)

Proposition 19.7 (Power Nesting). *Let W_I be irreducible and row-stochastic, and partition individuals into $k \geq 2$ blocks B (countries, or sectors within a country). For each block let $S_B = W_{BB} + W_{B*}(I - W_{**})^{-1}W_{*B}$ be its* stochastic complement, *where $*$ denotes all nodes outside B. Then $I - W_{**}$ is invertible, S_B is irreducible and row-stochastic, and with $s^{(B)}$ the Perron vector of S_B and ξ the Perron vector of the $k \times k$ coupling matrix $C_{BB'} = s^{(B)\prime} W_{BB'}\mathbf{1}$ (itself irreducible and row-stochastic), exactly*

$$\pi_{I,i} = \xi_{B(i)}\, s_i^{(B(i))}, \qquad \xi_B = \sum_{i\in B}\pi_{I,i}.$$

Applying the same identity inside each country c, to S_c and its partition into sectors, gives the nested factorization

$$\pi_{I,i} = \underbrace{\pi_{G,c(i)}}_{\text{country share}} \cdot \underbrace{\pi_{N,s(i)|c(i)}}_{\text{sector share in } c} \cdot \underbrace{\pi_{within,i|s(i),c(i)}}_{\text{individual share in } s}, \tag{19.3}$$

where each factor is a share that sums to one over its level: $\pi_{G,c} = \sum_{i\in c}\pi_{I,i}$ is the country's total power (the Perron vector of the π-weighted aggregate of Definition 19.6), $\pi_{N,\cdot|c}$ is the Perron vector of the sector coupling matrix built from S_c, and $\pi_{within,\cdot|s,c}$ is the Perron vector of the stochastic complement of sector s in S_c. These factors coincide with the Perron vectors of the unweighted aggregates (19.1)–(19.2) only in special cases, such as exact lumpability.

Proof. This is Meyer's (1989) uncoupling theorem for stochastic complementation, applied once per level. We give the argument. Write W for W_I and π for π_I, and fix a block B.

$I - W_{**}$ is invertible. $(W^n_{**}\mathbf{1})_i$ is the probability that the chain started at $i \in *$ stays outside B for n steps. By irreducibility there is a path of positive probability from i into B of length at most n, so this probability is below one for every i. Hence $\|W^n_{**}\|_\infty < 1$, $\rho(W_{**}) < 1$, and $(I - W_{**})^{-1} = \sum_{t\geq 0} W^t_{**} \geq 0$.

S_B is stochastic and irreducible. $S_B \geq 0$, and $W_{*B}\mathbf{1} = (I - W_{**})\mathbf{1}$ gives $S_B\mathbf{1} =$

$W_{BB}\mathbf{1} + W_{B*}\mathbf{1} = \mathbf{1}$. Entry $(S_B)_{ij}$ is the probability that the chain leaving $i \in B$ next enters B at j. Cutting a path of W from i to $j \in B$ at its visits to B shows that S_B has a positive path from i to j.

Uncoupling. Partition $\pi = (\pi_B, \pi_*)$. The equations $\pi' W = \pi'$ read $\pi'_B = \pi'_B W_{BB} + \pi'_* W_{*B}$ and $\pi'_* = \pi'_B W_{B*} + \pi'_* W_{**}$. The second gives $\pi'_* = \pi'_B W_{B*}(I - W_{**})^{-1}$, and substituting into the first gives $\pi'_B = \pi'_B S_B$. Since S_B is irreducible its stationary vector is unique, so $\pi_B = \xi_B\, s^{(B)}$ with $\xi_B = \pi'_B \mathbf{1} > 0$.

Coupling. $C \geq 0$ and $\sum_{B'} C_{BB'} = s^{(B)\prime} W_{B\cdot}\mathbf{1} = s^{(B)\prime}\mathbf{1} = 1$. Because $s^{(B)} = \pi_B/\xi_B > 0$, $C_{BB'} > 0$ iff $W_{BB'} \neq 0$, so C inherits irreducibility from W. Finally $\sum_B \xi_B C_{BB'} = \sum_B \pi'_B W_{BB'}\mathbf{1} = \pi'_{B'}\mathbf{1} = \xi_{B'}$, so ξ is the (unique) Perron vector of C.

Nesting. Apply the result to W with the partition into countries: $\pi_i = \pi_{G,c(i)}\, s_i^{(c(i))}$. Since S_c is irreducible and stochastic, apply it again to S_c with the partition of c into sectors: $s_i^{(c)} = \pi_{N,s(i)|c}\, \pi_{\text{within},i|s(i),c}$. (If c has a single sector, the middle factor is 1.) `code/ch19.py` verifies both levels numerically. □

Remark 19.8 (Simon–Ando: when the factors are local). The identity is exact but the stochastic complements depend on the whole matrix. When cross-block coupling is weak, $W_I = W_0 + \varepsilon C$ with W_0 block-diagonal and each diagonal block of W_0 irreducible and stochastic (individuals interact mostly within their sector, and sectors mostly within their country), Simon and Ando (1961) show that $s^{(B)}$ is within $O(\varepsilon)$ of the Perron vector of the diagonal block of W_0 alone. Each factor can then be measured at its own scale. In the code's example, the maximum error is 2×10^{-2} at $\varepsilon = 0.1$ and 3×10^{-4} at $\varepsilon = 0.001$. The exact Kronecker case of Appendix 19A is a special case that additionally requires the *same* sector matrix in every country.

Per-capita comparisons. Because the factors are shares, per-person comparisons must divide by group size. If $e_{s|c}$ is the sector's employment share and L_c the country's workforce, then for an individual with within-sector per-capita weight w_i (mean one in the sector),

$$\pi_{I,i} = \frac{\pi_{G,c}}{L_c} \cdot \frac{\pi_{N,s|c}}{e_{s|c}} \cdot w_i.$$

The three per-capita factors are country power per worker, sector power per worker (relative to the national average) and the individual's within-sector position.

Example 19.9 (An Indigenous Woman in Potosí (partly [ILLUSTRATIVE])). *Agent i: an indigenous woman working in informal personal services in Potosí, Bolivia. Agent j: a non-indigenous male gas executive in Santa Cruz. The country and sector factors use*

the 2019 data of Chapters 16 and 17; the within-sector factor is Chapter 18's illustrative calibration.

Country factor. *Both are Bolivian, so it cancels.*

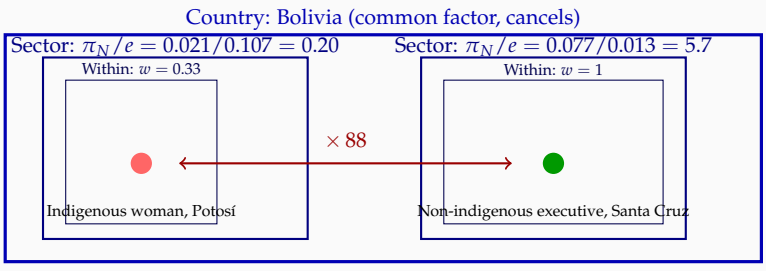
Sector factor. *In Chapter 17's table (Example 17.7), personal services hold* $\pi_N = 0.021$ *of national production-network power with 10.7% of employment (*0.20 *per worker). Employment in hydrocarbons is reported only together with mining: the two hold* 0.077 *with 1.3% of employment (*5.7 *per worker). Ratio:* $5.74/0.198 \approx 29$.

Within-sector factor. *By Chapter 18's illustrative calibration, an indigenous woman holds* $0.57 \times 0.58 = 0.33$ *of a non-indigenous man's per-capita power. Ratio:* $1/0.33 = 3.03$.

Total. $\pi_{I,j}/\pi_{I,i} \approx 29 \times 3.03 \approx 88$. *The sector factor still dominates, but not because the gas sector is central in* W^P*: its own* π *is only* 0.039. *Power per worker is high in hydrocarbons and mining because they employ very few people per unit of value — capital intensity, as Chapter 17 stresses — and low in personal services because the sector is peripheral and labour-intensive.*

A German engineer. *Germany holds* $\pi_G = 0.082$ *and Bolivia* 0.00053 *(Chapter 16, 2019), a ratio of 157.* Per worker, *with labour forces of 43.98 and 5.95 million (2019; World Bank, 2026b), the country factor is* $157 \times 5.95/43.98 = 21.2$. *Put the engineer in a sector whose power share equals its employment share (factor* 1, *against* 0.20 *for personal services) and give her within-sector factor* 1. *The compound ratio to the Potosí vendor is then* $21.2 \times 5.05 \times 3.03 \approx 325$. *Forgetting to divide country power by workforce size would inflate this to about* 2,400. *(Numbers:* `code/ch19.py`*.)*

Visualization: Power nesting: three scales, one agent



Two Bolivians in the same country. The outer box (country) is common to both and cancels. The middle boxes give sector power per worker in 2019 (INE and ILOSTAT; Chapter 17): 5.7 for hydrocarbons and mining against 0.20 for personal services, a factor of about 29. The inner boxes give the within-sector factor (illustrative, Chapter 18): 1 against 0.33, a factor of 3. The compound ratio is about 88×. Against a German engineer in an average sector it is about 325×, once country power is expressed per worker.

19.4 Five Processes, One Matrix Property

Sunkel argued that five phenomena are manifestations of a single process. The following definition gives each a counterpart as a property of the multi-scale weight matrix. It is a dictionary, not a theorem: it fixes what we will *mean* by each phenomenon in matrix terms. Integration (Γ) and the distribution of power (N_{eff}, $\text{Var}(\pi)$) are kept as separate coordinates, because they are independent (Proposition 6.16).

Definition 19.10 (Five Processes as Matrix Properties)**.** 1. ***Development*** *(at any scale):* Γ_s *is large and increasing. The weight matrix is becoming more connected, and growth propagates across nodes. If, in addition,* $N_{eff}(\pi_s)$ *is rising, power is also deconcentrating.*

2. ***Underdevelopment****:* Γ_s *is small and stagnant because cross-block coupling is weak. The weight matrix is nearly block-diagonal, and growth does not propagate. This is not the absence of development — it is a specific structural equilibrium (the fragmented fixed point of Chapter 13). It typically coexists with concentrated* π_s*, but that is a second, independent property.*

3. ***Dependence****: The off-diagonal block of* W^P *is asymmetric: the periphery's row entries (what it imports from the core) are large, but the core's row entries (what it imports from the periphery) are small. In Perron terms:* $\pi_{periphery} \ll 1/N$*. The periphery needs the core more than the core needs the periphery.*

4. ***Marginality****: Some nodes have* $\pi_i \approx 0$*: almost no one places weight on them. Their* columns *in W are nearly zero (their rows still sum to one: they depend on others). They exist in the economy but carry almost no weight in the network. Examples: Bolivia's domestic-services (household-production) sector, which has no intermediate flows at all and so drops out of* W^P *(Chapter 17), or the Potosí vendor of Example 19.9, whose per-capita power is about 1% of a gas executive's.*

5. ***Spatial inequality (in integration)****: The spectral gap varies across regional submatrices, each renormalized to be row-stochastic:* $\Gamma_{Santa\ Cruz} > \Gamma_{Potosí}$*. Some regions have integrated* W^P *(Santa Cruz: agro-industry, services, some manufacturing). Others have fragmented* W^P *(Potosí: mining enclave plus subsistence agriculture).* $\text{Var}_r(\Gamma_r)$ *measures regional disparity in* integration*; regional income inequality is a different quantity that it may help explain.*

All five are properties of the ***same*** *multi-scale weight matrix* (W_G, W_N, W_I)*:*

$$\textit{Development} = \textit{high}\ \Gamma, \quad \textit{Under-development} = \textit{low}\ \Gamma, \quad \textit{Dependence} = \textit{asymmetric}\ W, \tag{19.4}$$

$$\textit{Marginality} = \pi_i \approx 0, \quad \textit{Spatial inequality} = \mathrm{Var}_r(\Gamma_r). \tag{19.5}$$

Key Structural Insight

Sunkel's claim — five processes, one process — becomes, in this vocabulary, Conjecture 19.4. The "single process" would be the co-evolutionary dynamics of Chapter 13: proximity-dominated edge formation under complementary returns and assortative information. If the conjecture holds, this process simultaneously produces:

- Dense cores at every scale (development of the center).
- Sparse peripheries at every scale (underdevelopment of the periphery).
- Asymmetric cross-block edges (dependence).
- Zero-weight nodes (marginality).
- Geographic variation in Γ (spatial inequality).

On this reading the five phenomena have one explanation: the block structure of W at multiple scales, produced by a single dynamic mechanism. Interventions aimed at one symptom (transfers for marginality, trade policy for dependence, regional spending for spatial inequality) leave the mechanism in place. A structural intervention targets the mechanism: it raises cross-block coupling (Γ) at the binding scale, *and* it addresses the concentration of π directly. The two are separate targets.

19.5 Cross-Scale Feedback

Conjecture 19.11 (Cross-Scale Reinforcement)**.** *The three scales reinforce each other:*

1. ***Global*** $\rightarrow$ ***national****: A country's position in* W_G *(low* π_G*) constrains its domestic* Γ_N*. Countries with low global power face balance-of-payments constraints, commodity-dependent export structures, and external conditionality that can crowd out the domestic structural investments needed to raise* Γ_N*.*
2. ***National*** $\rightarrow$ ***individual****: Low domestic* Γ_N *means block-diagonal* W_N*, which channels individuals from disadvantaged groups into low-*π *sectors. The individual-level*

attribute constraints (Chapter 18) are enforced by the national structure: a woman can only work in sectors that the national W^P makes available to her.

3. ***Individual → national**: When large groups (women, indigenous people) have low individual π_I, the aggregate national Γ_N is depressed: the talent pool is under-utilized, cross-group linkages are absent, and the national product space is sparse. Individual discrimination reduces national structural health.*

4. ***National → global**: Countries with low Γ_N produce simple, concentrated exports (low ECI), which keeps them on the periphery of the global W_G (low π_G). Domestic fragmentation reproduces global peripherality.*

The four feedback loops form a vicious circle: low global position → low domestic Γ → low individual π → low national Γ → low global position. Breaking the circle at any one scale has limited effect unless the other scales respond. If the loops are strong, comprehensive structural reform (all scales, all networks) is needed, and piecemeal interventions produce piecemeal results. Each link is a testable hypothesis; none is established here.

Example 19.12 (Bolivia's Vicious Circle (partly [ILLUSTRATIVE])). *The five links, on the 2019 data of Chapters 16–17 and the illustrative values of Chapter 18:*

1. ***Global**: Bolivia holds $\pi_G = 0.00053$ in the 2019 trade matrix. Its exports sit on the periphery of the product space, with an ECI of -1.30, down from -1.21 in 2006 (The Growth Lab at Harvard University, 2026).*

2. ***National**: here the data break the circle as stated. Bolivia's $\Gamma_N = 0.467$ (2019) is within the range of the industrial economies, and W^P shows no weakly coupled blocks. What is national and peripheral is the export sector: hydrocarbons hold $\pi = 0.039$ in W^P while exporting 70% of output. ($\sigma_B = 0.62$ remains illustrative.)*

3. ***Individual**: Indigenous women hold about one-third of a non-indigenous man's per-capita power (Chapter 18, illustrative). Separable gender × ethnicity edge constraint: $\phi \approx 0.105$.*

4. ***Back to national**: With 41% of the population (indigenous) and 50% (women) facing attribute constraints, the national talent pool is fragmented and cross-group edges are suppressed. If this link operates, it does so on what the economy produces and exports, not on aggregate Γ_N, which is not low.*

5. ***Back to global**: With a simple export basket, Bolivia struggles to diversify (the principle of relatedness keeps it on the commodity periphery). π_G stays low.*

If the circle is self-reinforcing, it runs through the export enclave and complexity, not through domestic fragmentation. The gas nationalization of 2006 attempted to break it at the global-to-national link (capturing more resource rents). But without linking the enclave to the domestic network, addressing the national-to-individual link (attribute constraints) or the individual-to-national link (formation channel), the rents were redistributed within the existing structure. On Chapter 17's INE trajectory, Γ_N *was* 0.518 *in 2006 and* 0.467 *in 2019 (on a new base, not strictly comparable), while the ECI fell.*

19.6 The Scale at Which to Intervene

Definition 19.13 (The Binding Scale)**.** *Let* L_{total} *be the total welfare loss across all scales, decreasing in each* Γ_s*, and let* c_s *be the marginal cost of raising* Γ_s*. The* ***binding scale*** *is the one where a unit of resources reduces the loss most:*

$$s^* = \arg\max_{s \in \{G,N,I\}} \frac{1}{c_s} \left(-\frac{\partial L_{total}}{\partial \Gamma_s} \right). \tag{19.6}$$

The binding scale is *not* simply the scale with the smallest Γ. Across scales the Γ values are not directly comparable (Proposition 19.5), and the marginal loss reduction depends on the cross-scale feedbacks.

Remark 19.14 (Three illustrative cases).

- **Bolivia.** The chapter's judgement is that the **national** scale binds, but the data narrow what that can mean. Bolivia's measured $\Gamma_N = 0.467$ is in the comparator range (Korea 0.511, the United States 0.463), so a low aggregate Γ_N is not what binds. If the national scale binds, it is through the export enclave: a sector that buys 0.225 of domestic inputs per unit of output, pays 6% of its value added as wages, exports 70% of its output and holds $\pi = 0.039$ in W^P (Chapter 17). The operator sequence of Chapter 15 should then target the enclave's domestic linkages and the complexity of the export basket (ECI -1.30), not aggregate Γ_N. Whether the global scale binds instead is an open question that these data cannot settle.

- **South Korea** ($\Gamma_N = 0.511$ in 2019, OECD ICIO; illustrative $\Gamma_I \approx 0.35$ because of remaining gender gaps). The binding scale may be **individual**: attribute constraints are the largest remaining source of weak integration.

- **Uruguay**, a small open economy (illustrative $\Gamma_N \approx 0.30$, since Uruguay is not in the OECD ICIO tables; $\pi_G = 0.00055$ in the 2019 trade matrix). The binding scale may be **global**: the global position constrains diversification.

Prebisch Connection: Why Prebisch was right about industrialization but incomplete

Prebisch (1949) argued that the periphery should industrialize to escape the deteriorating terms of trade. His prescription operated at the global scale: change the periphery's position in W_G by adding manufacturing exports. He was right that the global scale matters. But he did not address the national and individual scales: even with manufacturing, a block-diagonal domestic W_N and attribute-constrained W_I prevent the structural integration needed for industrialization to propagate growth broadly.

South Korea industrialized — through export discipline rather than the inward-looking ISI usually associated with ECLA — and also addressed the other scales: land reform (national: broke the landlord block), education (individual: reduced human-capital attribute constraints), export discipline (global: moved up the product space). The three scales were addressed simultaneously. Bolivia followed only the global prescription (ISI, then gas nationalization) without the national and individual reforms. The result: Bolivia's global position did not improve — its ECI fell from -1.21 in 2006 to -1.46 in 2018 and stood at -1.30 in 2019 (The Growth Lab at Harvard University, 2026) — while its domestic Γ drifted down (Chapter 17) and the individual structures, on the illustrative numbers, barely moved.

The lesson: structural change must be multi-scale. Addressing one scale without the others is like brightening one block of the heatmap while leaving the rest dark.

19.7 Summary

1. **Structural isomorphism** across scales (core-periphery topology, weak cross-block coupling, concentrated π) is a definition plus a conjecture about a common mechanism (proximity, complementarity, assortative formation), not a theorem. For Bolivia the data contradict the small-Γ part: $\Gamma = 0.452$ for the 2019 global trade matrix and 0.467 for the national production matrix; the individual-scale value is hypothetical.

2. **Aggregation raises** Γ: under exact lumpability, the coarse matrix's eigenvalues are a subset of the fine matrix's, so $\Gamma_{\text{coarse}} \geq \Gamma_{\text{fine}}$ and Perron weights aggregate by summation. Cross-scale comparisons of Γ are partly mechanical.

3. Individual power is **nested** exactly, through stochastic complementation: $\pi_{I,i} = \pi_G \cdot \pi_{N|c} \cdot \pi_{\text{within}|s,c}$, with factors that are local under weak coupling (Simon–Ando). Per-capita comparisons must divide by group size. The Potosí vendor has about 1/88 of a gas executive's per-capita power (sector factor from 2019 data, within-sector factor illustrative), and the sector factor dominates.

4. **Five processes, one process** (Sunkel) becomes a dictionary: development = high and rising Γ; underdevelopment = low Γ from weak coupling; dependence = asymmetric cross-block W; marginality = $\pi_i \approx 0$; spatial inequality in integration = $\text{Var}_r(\Gamma_r)$. "One process" is the conjectured common mechanism.

5. **Cross-scale feedback** may lock the scales into a vicious circle (conjectured). The **binding scale** maximizes the loss reduction per unit cost; it is not simply the scale with the smallest Γ. For Bolivia the chapter argues it is the national scale.

19.8 Exercises

19.1 Three-scale comparison. Using Bolivia data, recompute the global and national columns of Example 19.2: Γ, $\max \pi / \min \pi$ and σ_B. For the individual scale, describe what network data would be needed and how you would sample it. Using Proposition 19.5, explain why a lower Γ at the finer scale is not, by itself, evidence of more fragmentation.

19.2 Power nesting. Choose three Bolivian agents: (a) a non-indigenous male in gas, (b) an indigenous male in agriculture, (c) an indigenous female in informal personal services. Compute each agent's per-capita power with the per-capita nesting formula, dividing sector power by employment share. What are the pairwise ratios? Which factor dominates?

19.3 The global comparison. Using the nesting formula: compute per-capita π_I for a German female engineer, a Brazilian factory worker, and the indigenous woman in Potosí, expressing country power per worker. What are the ratios? Which scale contributes most to the gap? How much does the answer change if you forget to divide by workforce size?

19.4 Five processes as matrix properties. For a 6×6 economy with three blocks: construct W and compute each of Sunkel's five phenomena. Verify that all

five are visible in the same matrix.

19.5 Cross-scale feedback simulation. Construct a three-scale model (G: 3 countries; N: 3 sectors per country; I: 2 agents per sector). Implement the nesting structure. Simulate the co-evolutionary dynamics with cross-scale feedback. Does the vicious circle emerge? At what parameter values does it break?

19.6 The binding scale. For three countries (Bolivia, South Korea, Uruguay): estimate Γ_G (from bilateral trade data), Γ_N (from I-O table), and approximate Γ_I (from household survey data on occupational segregation). Which scale is binding for each? Does the answer match the policy history?

19.7 Multi-scale structural plan for Bolivia.

(a) Global: target 3 products on the product space (Chapter 15).

(b) National: design the operator sequence for Γ_N (Chapter 15, information $\to$ education $\to$ infrastructure $\to$ ISI).

(c) Individual: design the $\Delta\lambda_g$, $\Delta\lambda_r$ interventions (Chapter 18).

(d) Estimate the combined $\Delta\Gamma$ at each scale and the timeline.

(e) Check political feasibility (Chapter 14): which coalition supports this plan?

Mathematical Appendices

Appendix 19A: The Kronecker Factorization

Setup. Consider an N-agent economy where agents are grouped into C countries, each with S sectors and K individuals per sector ($N = CSK$). Suppose the individual-level matrix factors *exactly* as $W_I = W_G \otimes W_N \otimes W_{\text{within}}$, with each factor row-stochastic and primitive (irreducible and aperiodic), so that W_I is primitive too. This is a strong assumption: it requires the same sectoral matrix W_N in every country and the same within-sector matrix in every sector. Then

$$\boldsymbol{\pi}_I = \boldsymbol{\pi}_G \otimes \boldsymbol{\pi}_N \otimes \boldsymbol{\pi}_{\text{within}}, \tag{19.7}$$

because $(\pi_G \otimes \pi_N \otimes \pi_w)'(W_G \otimes W_N \otimes W_w) = (\pi_G' W_G) \otimes (\pi_N' W_N) \otimes (\pi_w' W_w)$. The eigenvalues of a Kronecker product are the products of the factors' eigenvalues, and all have modulus at most one. The second-largest modulus is therefore $\max(|\lambda_2(W_G)|, |\lambda_2(W_N)|, |\lambda_2(W_{\text{within}})|)$, so

$$\Gamma_I = \min(\Gamma_G, \Gamma_N, \Gamma_{\text{within}}). \tag{19.8}$$

In this special case the scale with the smallest spectral gap is the "bottleneck" in integration.

Beyond the Kronecker case. Real matrices have scale-specific structure (different sector matrices in different countries) and cross-scale interactions (a gas executive in Bolivia trades with both Bolivian farmers and foreign companies). The general, exact statement is the stochastic-complementation identity of Proposition 19.7. Its factors are approximately local when coupling across blocks is weak (Simon–Ando). The sensitivity of π to the coupling is governed by the perturbation lemma (Lemma 6.19): it is large when an eigenvalue is close to $+1$, i.e. precisely when the blocks are weakly coupled.

Appendix 19B: Spectral Gap of Nested Block Matrices

Two-level nesting. Consider a matrix with k blocks at level 1, each containing m sub-blocks at level 2. Within level-2 sub-blocks: weight w_2. Between level-2 sub-blocks within the same level-1 block: weight w_1. Between level-1 blocks: weight w_0.

Let each sub-block have n nodes, $N = kmn$, and let $d = nw_2 + (m-1)nw_1 + (k-1)mnw_0$ be the common row sum; $W = K/d$. Because the partitions are

equitable, the spectrum is exact:

1. Between level-1 blocks: $\lambda = 1 - w_0/\bar{w}$, with $\bar{w} = d/N$ the mean entry of K (multiplicity $k - 1$).

2. Between sub-blocks within a block: $\lambda = n(w_2 - w_1)/d$ (multiplicity $k(m - 1)$).

3. Within sub-blocks: $\lambda = 0$ (multiplicity $N - km$).

Example ($k = m = 3$, $n = 4$, $w_2 = 1$, $w_1 = 0.3$, $w_0 = 0.05$): the eigenvalues are 1, 0.763 ($\times 2$), 0.368 ($\times 6$) and 0 ($\times 27$), so $\Gamma = 0.237$. The spectral gap is set by the weakest coupling — here the coupling between level-1 blocks. (Numbers: `code/ch19.py`.)

Structuralist Case Study: Sunkel's "Five Processes, One Process"

Osvaldo Sunkel's third appearance in this book (after Chapter 4 with Paz, and Chapter 13 on integration/disintegration). Here we address his deepest claim: that development, underdevelopment, dependence, marginality, and spatial inequality are not five problems but five symptoms of one structural condition.

Sunkel developed this argument in *El subdesarrollo latinoamericano y la teoría del desarrollo* (1970, with Pedro Paz) and in "Capitalismo transnacional y desintegración nacional en América Latina" (1971).

The formal translation. Definition 19.10 gives each of the five phenomena a counterpart as a property of the same multi-scale weight matrix. Development is high Γ. Underdevelopment is low Γ from weak coupling. Dependence is asymmetric W. Marginality is $\pi_i \approx 0$. Spatial inequality (in integration) is $\mathrm{Var}_r(\Gamma_r)$. That they share a single *cause* is the content of Conjecture 19.4, a claim the translation makes testable but does not prove.

The policy implication. If the five phenomena have one cause (the block structure of W), the structural response targets that cause: raise cross-block coupling at the binding scale, and address the concentration of π, which is a separate target. The sub-field fragmentation of development economics generates sub-field-specific policy recommendations: trade policy for dependence, growth policy for underdevelopment, social policy for marginality, fiscal transfers for spatial inequality. Each addresses one symptom without addressing the cause. The structural approach generates one recommendation: brighten the dark off-diagonal entries of W at the binding scale (Definition 19.13).

Why this matters for Bolivia. Bolivia has active policies addressing all five phenomena separately: gas revenue redistribution (underdevelopment), trade agreements (dependence), social transfers (marginality), regional investment (spatial inequality), and recent growth (development). On INE's tables none has raised Γ: it drifted down from 0.567 (1990) to 0.484 (2013) (Chapter 17) — but it was never low. What none of them changed is the export enclave's isolation within W^P (hydrocarbons' $\pi = 0.039$) and the complexity of exports (ECI -1.30 in 2019). The structural reading is that none targeted the weight matrix directly.

Sunkel's claim (1970)	**Formal object**	**This chapter**
Five processes, one process	Five properties of W	Def. 19.10
Same structure at every scale	Structural isomorphism; Γ comparable only after aggregation	Def. 19.3, Prop. 19.5
The single process	Co-evolutionary dynamics	Conj. 19.4
Nesting	$\pi_I = \pi_G \cdot \pi_N \cdot \pi_{\text{within}}$	Prop. 19.7
Cross-scale feedback	Vicious circle	Conj. 19.11
Policy: unified, not fragmented	Raise coupling at the binding scale	Def. 19.13

Sunkel wrote in Santiago, at ILPES, before the 1973 coup. The five processes he described — development for Santiago's elite, underdevelopment for the rural south, dependence on US copper companies, marginality of the Mapuche, spatial inequality between the capital and the periphery — were all visible from his window. Fifty years later, similar processes operate at the same scales. The matrix has different entries. Whether the structure is the same, and whether the same mechanism produces it at every scale, is what Conjecture 19.4 puts to the data.

Further Reading

Scale invariance in social systems: West (2017), *Scale: The Universal Laws of Growth, Innovation, Sustainability, and the Pace of Life in Organisms, Cities, Economies, and Companies*. Bettencourt et al. (2007), "Growth, Innovation, Scaling, and the Pace of Life in Cities."

Multi-scale economic structure: Hausmann and Hidalgo (2011), "The Network Structure of Economic Output." For the nesting of local, national, and global production networks: Baldwin and Lopez-Gonzalez (2015), "Supply-Chain Trade."

The structuralist synthesis: Sunkel and Paz (1970), *El subdesarrollo latinoamericano y la teoría del desarrollo*. Sunkel (1993), *Development from Within*. CEPAL (2012b), *Structural Change for Equality*, the institutional codification of the multi-scale structural approach.

Cross-scale interactions: Acemoglu and Robinson (2012), *Why Nations Fail*, Chapters 3–5 (how colonial institutions at the global scale shape domestic institutions). For the aggregation theory used in this chapter: Simon and Ando (1961), "Aggregation of Variables in Dynamic Systems"; Kemeny and Snell (1960), *Finite Markov Chains* (lumpability); Meyer (1989), "Stochastic Complementation, Uncoupling Markov Chains, and the Theory of Nearly Reducible Systems."

PART VI

Measurement

Chapter 20

Social Accounting Matrices as Weight Matrices

The Data Infrastructure of Structural Economics

> *"A structuralist model of the economy rests on three pillars: a social accounting matrix to describe the flows, a set of closure rules to determine which variables adjust, and a theory of distribution to explain who gets what."*
> — Lance Taylor, *Structuralist Macroeconomics* (1983)

Parts I–V developed the theory. Part VI builds the measurement infrastructure. The question is no longer "what does W imply?" but "where does W come from?"

The answer, for the production network W^P, is the Social Accounting Matrix (SAM): a square matrix that records every flow between every pair of accounts in the economy. The SAM is the natural data object for W^P, but two different matrices can be built from it, and they should not be confused. Normalizing *columns* gives expenditure shares: the coefficient matrix behind multipliers and, for the production block, the input-cost shares $W^P = D^{-1}A$ of Chapter 1. Normalizing *rows* gives receipt shares, a row-stochastic matrix over all accounts, whose Perron vector turns out to be nothing more than account size (Proposition 20.4).

This chapter translates the framework into SAM language, builds an illustrative SAM for a stylized Bolivia-like economy, checks its gas sector against INE's supply-use tables, and shows which of Chapter 17's diagnostics a SAM can deliver and which it cannot. The remaining chapters in Part VI extend the measurement to W^I (Chapter 21), W^U (Chapter 22), and time series (Chapter 23).

20.1 The SAM as the Complete Flow Matrix

20.1.1 Definition

Definition 20.1 (Social Accounting Matrix)**.** *A **Social Accounting Matrix** is an $M \times M$ matrix T where:*

1. *Each row and column represents an **account**: a production sector, a factor of production (labor, capital), a household group, a government account, a capital account, or the rest of the world.*
2. *Entry T_{ij} records the total payment from account j (column) to account i (row) during the accounting period.*
3. *Row sums equal column sums for every account: $\sum_j T_{ij} = \sum_j T_{ji}$ for all i. The SAM is balanced by construction (every dollar paid by one account is received by another).*

The SAM generalizes the I-O table in two directions: it includes factors and institutions (not just production sectors), and it records the *complete* circular flow of income — from production to factors to households to consumption to production.

Definition 20.2 (SAM Accounts — Standard Classification)**.**

Account type	***Examples***	***Role in W***
Production activities	*Agriculture, Manufacturing, Gas*	*W^P entries*
Commodities	*Goods and services (supply/demand)*	*Market clearing*
Factors	*Labor (by type), Capital*	*Factor payments*
Households	*By income quintile, urban/rural, ethnicity*	*W^U entries*
Government	*Tax, transfers, public spending*	*Policy instruments*
Capital account	*Savings, investment*	*Accumulation*
Rest of world	*Exports, imports, remittances*	*Global linkage*

20.1.2 From the SAM to W^P

Definition 20.3 (The SAM-Derived Weight Matrix)**.** *Row-normalize the SAM to obtain the weight matrix:*

$$W_{ij} = \frac{T_{ij}}{\sum_k T_{ik}}. \tag{20.1}$$

Entry W_{ij} is the share of account i's total receipts that comes from account j. This is a row-stochastic matrix (a receipt-share, or supply-side, matrix in the sense of Ghosh). The expenditure-share matrix used for multipliers normalizes columns instead (Definition 20.10).

Proposition 20.4 (The Perron Vector of a Balanced SAM Is Account Size)**.** *Let the SAM $T \geq 0$ be balanced, with account totals $y = T\mathbf{1} = T'\mathbf{1} > 0$, and irreducible (every account is linked to every other by some chain of payments). Then the Perron vector of $W = \operatorname{diag}(y)^{-1}T$ is*

$$\pi = \frac{y}{\mathbf{1}'y}.$$

Proof. $y' \operatorname{diag}(y)^{-1}T = \mathbf{1}'T = (T'\mathbf{1})' = y'$, so $y/\mathbf{1}'y$ is a stationary vector of W. W has the same zero pattern as T, so it is irreducible and its stationary vector is unique (Seneta, 2006, Thm. 1.5). □

The result is not new. It is the closed-system observation of Bródy (1970), and Wirkierman, Bianchi, and Torriero (2021) state it explicitly for a closed input–output system mapped into a regular Markov chain: the stationary probabilities coincide with normalized activity levels. So for a balanced SAM, "structural power" read off the row-normalized SAM is just the size of each account's turnover. Its Gini is the Gini of account totals. This is a useful check, not a finding: any claim about power that rests on this π is a claim about size. Distributional power must come from a model of bargaining or price formation (Chapter 7), not from the accounting identity.

Proposition 20.5 (The SAM Contains W^P as a Submatrix)**.** *The production-activities block T_{PP} of the SAM yields the I-O matrix. In Chapter 1's orientation (row i is i's recipe), $a_{ij} = T_{ji}/y_i$, i.e. $A = (T_{PP} \operatorname{diag}(y_P)^{-1})'$, and $W^P = D^{-1}A$ is the matrix of input-cost shares. Row-normalizing T_{PP} itself gives* receipt *shares, a different matrix. The full SAM extends W^P by including factor payments, household income, and government transfers — the channels through which income circulates.*

The full SAM-derived W is larger and richer than W^P alone:

1. *W^P captures inter-sectoral production linkages only.*
2. *The full W also captures factor-payment linkages (which sectors pay which factors), distributional linkages (which factors go to which households), and fiscal linkages (government taxation and transfers).*

The additional accounts can raise or lower the spectral gap; there is no general ordering (Example 20.8).

Proof. By Definition 20.1, T_{ji} for production accounts i, j is the payment of activity i to activity j, i.e. i's purchases of input j, and y_i is i's gross output. In value units, T_{ji}/y_i is the input of j per unit of i's output, which is a_{ij} in Chapter 1's orientation. Hence $A = (T_{PP} \operatorname{diag}(y_P)^{-1})'$, and when every activity buys some intermediate

input ($A\mathbf{1} > 0$), $W^P = D^{-1}A$ with $D = \text{diag}(A\mathbf{1})$ has entries $T_{ji} / \sum_l T_{li}$, the input-cost shares. Row-normalizing T_{PP} instead divides by $\sum_l T_{il}$, receipts from other activities, which differs unless T_{PP} is symmetric. The last sentence is Proposition 20.9(3). □

20.2 Constructing Bolivia's SAM

20.2.1 Data Sources

Example 20.6 (Bolivia SAM — Data Assembly)**.** *The sources one would use to build a Bolivian SAM:*

SAM block	*Data source*	*Detail*
Production × Production	*INE supply-use tables 2019*	*42 products × 39 ind.*
Production × Factors	*Same, value-added block*	*Value added by sector*
Factors × Households	*EH 2019*	*Wages, profits by HH type*
Households × Commodities	*EH 2019 (consumption)*	*Expenditure by HH type*
Government × all	*Fiscal data (MEFP)*	*Tax, transfers, spending*
Rest of world	*Balance of payments*	*Exports, imports, FDI*

INE publishes supply-use tables annually for 2017–2024 on the base-2017 national accounts, and symmetric 35 × 35 input–output matrices for 1988–2009 and 2011–2014 on base 1990 (Instituto Nacional de Estadística (INE), Bolivia, 2026). There is no 2016 table. The supply-use tables do not separate domestic from imported use, so an industry-by-industry block of domestic flows has to be derived from them (net taxes and margins removed, imports allocated proportionally, fixed product-sales structure); `data/compute/compute_bolivia_io.py` *does this and is the source of every Bolivian I-O number quoted below. INE does not publish a SAM.*

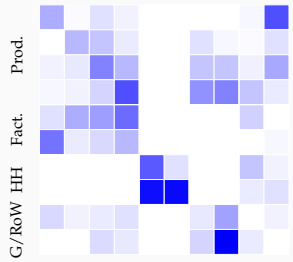
20.2.2 The SAM Structure

Example 20.7 (Bolivia SAM, Aggregated to 10 × 10 [ILLUSTRATIVE])**.** *Aggregate to 10 accounts: (1) Gas/Mining, (2) Agriculture, (3) Manufacturing, (4) Services, (5) Labor, (6) Capital, (7) Poor Households, (8) Non-Poor Households, (9) Government, (10) Rest of World.*

	Gas	*Agr*	*Mfg*	*Svc*	*Lab*	*Cap*	*HH-P*	*HH-NP*	*Gov*	*RoW*
Gas	2.1	0.1	0.8	0.3					0.2	4.5
Agr	0.0	1.8	1.5	0.5			0.8	0.2	0.1	0.8
Mfg	0.3	0.6	3.2	1.8			1.5	1.5	0.4	2.2
Svc	0.2	0.3	1.1	4.5			2.8	3.2	1.5	0.5
Lab	0.8	2.1	2.5	3.8					1.2	
Cap	3.6	0.5	1.0	1.8						0.2
HH-P					4.2	0.8			1.5	0.3
HH-NP					6.2	6.3			0.5	0.8
Gov	1.0	0.3	0.5	0.8			0.6	2.4		0.3
RoW	0.0	0.0	0.9	0.6			1.1	6.5	0.5	
Total	8.0	5.7	11.5	14.1	10.4	7.1	6.8	13.8	5.9	9.6

Illustrative values in billions of Bs, not estimates. Empty cells are zero. Each column records an account's payments, each row its receipts, and every row total equals the corresponding column total (last row). Household columns record consumption of domestic goods, direct taxes, and payments to RoW. RoW payments are imports plus, since this simplified SAM has no separate savings–investment account, net saving placed abroad. Without the household-consumption cells, seven accounts would not balance; `code/ch20.py` *checks the balance. The account sizes are not scaled to Bolivia's national accounts; the real hydrocarbons figures are in the Prebisch box after Example 20.13.*

Visualization: Bolivia's SAM as a heatmap



Bolivia SAM (illustrative)

The SAM of Example 20.7 as a heatmap: stronger blue means a larger flow; white means zero. The strongest cells are factor income received by households (Lab and Cap → HH-NP), non-poor households' payments to RoW, and gas exports (Gas row, RoW column). The production-to-production block (upper left) holds the I-O linkages. The household-consumption cells (production rows, household columns) close the circular flow. The faint cells are the weak links: capital income reaching poor households (0.8 of 7.1) and the gas sector's domestic intermediate purchases.

20.3 Running the Diagnostics on the SAM

20.3.1 Spectral Analysis of the Full SAM

Example 20.8 (Spectral Decomposition of Bolivia's SAM-Derived W [ILLUSTRATIVE])**.** *Row-normalize the* 10×10 *SAM of Example 20.7 to get* W*. Its eigenvalues, ordered by modulus:*

Eigenvalue	λ_k	$\lvert\lambda_k\rvert$
λ_1	1	*1.000*
$\lambda_{2,3}$	$0.013 \pm 0.598i$	*0.598*
λ_4	-0.486	*0.486*
λ_5	0.400	*0.400*
$\lambda_{6,7}$	$-0.064 \pm 0.232i$	*0.241*
$\lambda_{8,9}$	$0.202 \pm 0.017i$	*0.202*
λ_{10}	-0.041	*0.041*
$\Gamma = 1 - \lvert\lambda_2\rvert$		*0.402*

Two features stand out. First, the leading non-unit eigenvalues are a complex *pair. A pure three-step cycle (production → factors → households → production) has eigenvalues equal to the cube roots of unity, all of modulus 1, and hence* $\Gamma = 0$*, because it is periodic. Intra-block flows (I-O purchases) and leakages to government and RoW damp the rotation, leaving* $\lvert\lambda_2\rvert = 0.60$*. In a SAM, a small* Γ *can therefore reflect the circularity of the income flow rather than fragmentation into blocks. Second, the full-SAM gap (*0.402*) is* larger *than that of its own production block:* 0.391 *for the row-normalized* T_{PP}*, and* 0.248 *for the input-cost shares* $W^P = D^{-1}A$*. Adding accounts does not, in general, lower* Γ*. (Numbers:* `code/ch20.py`*.)*

Proposition 20.9 (What the SAM Adds to the I-O Table)**.**

1. *The I-O table captures only* W^P *(production linkages). The SAM captures the entire circular flow, including distributional channels: the separation between labor and capital income, between poor and non-poor households, and between fiscal and non-fiscal flows.*

2. *For a balanced SAM, the Perron vector of the row-normalized SAM is account size (Proposition 20.4). Distributional information lies in the* coefficients *(who pays whom), which the multiplier decomposition below exploits.*

3. *The spectral gap of the full SAM bears no general ordering to that of the I-O block. Complex eigenvalues signal circular flow, not blocks: if the row-normalized balanced SAM* W *has a non-real eigenvalue, then* $T \neq T'$*, i.e. some pair of accounts has*

unequal flows in the two directions. Weakly coupled blocks with symmetric flows produce only real eigenvalues.

Proof. (2) is Proposition 20.4. (3) Example 20.8 has $\Gamma_{\text{SAM}} > \Gamma_{\text{I-O}}$. A SAM whose production block is fully mixing ($\Gamma_{\text{I-O}} = 1$) but whose circular flow is nearly periodic gives the opposite ordering (`code/ch20.py` constructs one with $\Gamma_{\text{SAM}} = 0.05$). For the eigenvalue claim, suppose $T = T'$. With $Y = \operatorname{diag}(y)$, $Y^{1/2} W Y^{-1/2} = Y^{-1/2} T Y^{-1/2}$ is symmetric, so W is similar to a symmetric matrix and all its eigenvalues are real. Contrapositively, a non-real eigenvalue implies $T \neq T'$. The converse does not hold: an asymmetric T can have a real spectrum. (1) is a statement about which flows are recorded. □

20.3.2 The Structural Multiplier Matrix

Definition 20.10 (SAM Multiplier Matrix)**.** *Partition the SAM accounts into endogenous (e: production, factors, households) and exogenous (x: government, capital account, rest of world). The SAM coefficient matrix for endogenous accounts:*

$$A_n = T_{ee} \cdot diag(\mathbf{y}_e)^{-1}, \tag{20.2}$$

*where T_{ee} is the endogenous sub-matrix and $\mathbf{y}_e$ is the vector of total receipts for endogenous accounts. The **SAM multiplier matrix**:*

$$M_{SAM} = (I - A_n)^{-1}. \tag{20.3}$$

This is the generalized Leontief inverse. Entry m_{ij} measures the total increase in account i's income from a one-unit injection into account j, tracing the entire circular flow: production → factors → households → consumption → production → ...

Proposition 20.11 (Pyatt–Round Multiplicative Decomposition)**.** *Order the endogenous accounts as production (P), factors (F) and households (H), and assume $\rho(A_n) < 1$ (every endogenous account leaks, directly or indirectly, to the exogenous accounts). Their coefficients have the circular-flow pattern*

$$A_n = \begin{pmatrix} A_{PP} & 0 & A_{PH} \\ A_{FP} & 0 & 0 \\ 0 & A_{HF} & A_{HH} \end{pmatrix} : \tag{20.4}$$

production receives from production (intermediates) and from households (consumption), factors receive from production (value added), and households receive from factors (factor

income) and from each other (transfers). Let $\tilde{A} = \operatorname{diag}(A_{PP}, 0, A_{HH})$ *be the within-block part and define*

$$M_1 = (I - \tilde{A})^{-1}, \qquad A^* = M_1(A_n - \tilde{A}),$$
$$M_2 = I + A^* + A^{*2}, \qquad M_3 = (I - A^{*3})^{-1}.$$

Then all three matrices exist and are nonnegative, M_3 *is block-diagonal, and (Pyatt and Round, 1979)*

$$M_{SAM} = M_3\, M_2\, M_1. \tag{20.5}$$

- M_1 *is the* ***transfer*** *(own-block) multiplier: intermediate purchases within production and transfers among households.*
- M_2 *is the* ***open-loop*** *(cross-block) multiplier: how an injection in one block reaches the other two.*
- M_3 *is the* ***closed-loop*** *(circular) multiplier: the injection returns to its block of origin after a full circuit.*

Proof. Existence. $A_n \geq 0$ and $\rho(A_n) < 1$, so $M_{\text{SAM}} = \sum_t A_n^t \geq 0$. Since $0 \leq \tilde{A} \leq A_n$, $0 \leq \tilde{A}^t \leq A_n^t$ for every t, so $\sum_t \tilde{A}^t$ converges, $\rho(\tilde{A}) < 1$, and $M_1 = \sum_t \tilde{A}^t \geq 0$. Hence $A^* = M_1(A_n - \tilde{A}) \geq 0$. To see that $\rho(A^*) < 1$, use the identity $I - A_n = (I - \tilde{A})(I - A^*)$ (shown below). It gives $I - A^* = M_1(I - A_n)$, which is invertible, and $(I - A^*)M_{\text{SAM}} = M_1$. So for every m

$$\Big(\sum_{t=0}^{m} A^{*t}\Big) M_1 = \Big(\sum_{t=0}^{m} A^{*t}\Big)(I - A^*) M_{\text{SAM}} = (I - A^{*\,m+1}) M_{\text{SAM}} \leq M_{\text{SAM}},$$

because $A^{*\,m+1} M_{\text{SAM}} \geq 0$. Multiplying on the right by $A_n - \tilde{A} \geq 0$ gives $\sum_{t=1}^{m+1} A^{*t} \leq M_{\text{SAM}}(A_n - \tilde{A})$ for all m. The partial sums of a nonnegative series are bounded, so $A^{*t} \to 0$ and $\rho(A^*) < 1$ (this is Varga's regular-splitting argument). Hence $\rho(A^{*3}) < 1$ and $M_3 = \sum_t A^{*3t} \geq 0$.

Factorization. $I - A_n = (I - \tilde{A}) - (A_n - \tilde{A}) = (I - \tilde{A})\big(I - (I - \tilde{A})^{-1}(A_n - \tilde{A})\big) = (I - \tilde{A})(I - A^*)$, so $M_{\text{SAM}} = (I - A^*)^{-1} M_1$. For any square matrix, $(I - A^*)(I + A^* + A^{*2}) = I - A^{*3}$. Since A^* commutes with A^{*3} and both $I - A^*$ and $I - A^{*3}$ are invertible, $(I - A^*)^{-1} = (I - A^{*3})^{-1}(I + A^* + A^{*2}) = M_3 M_2$.

Structure. $A_n - \tilde{A}$ has nonzero blocks only in positions (P, H), (F, P) and (H, F) (pattern (20.4)), and M_1 is block-diagonal, so A^* has the same block-cyclic pattern of period three. Then A^{*3}, and hence M_3, is block-diagonal: each block's own circuit through the other two. □

Remark 20.12 (Additive form for reporting). Column sums of a product of matrices are not products of column sums, so the multiplicative factors cannot be reported as scalars that multiply. Stone's additive form can:

$$M_{\text{SAM}} = I + \underbrace{(M_1 - I)}_{\text{own-block}} + \underbrace{(M_2 - I)M_1}_{\text{open-loop}} + \underbrace{(M_3 - I)M_2M_1}_{\text{closed-loop}},$$

and the column sums of the four terms add up to the total multiplier.

Intuition: The three effects in structural language

- The own-block effect lives in the diagonal blocks of the heatmap: inter-industry purchases.
- The open-loop effect is the off-diagonal path: how a stimulus in production reaches households through factor payments.
- The closed-loop effect is the stimulus returning to production through consumption demand.

When cross-block coefficients are small (weak linkages from production to factors to households — the Pinto diagnostic), the open- and closed-loop effects are small and a stimulus stays where it starts. These are statements about the multiplier coefficients, not about Γ of the row-normalized SAM. That gap can be small precisely *because* the circular flow is strong (Example 20.8).

Example 20.13 (Bolivia's SAM Multipliers [ILLUSTRATIVE])**.** *Column sums of the additive decomposition for the SAM of Example 20.7. Government and RoW are exogenous.*

Injection into:	*Gas*	*Agr*	*Mfg*	*Svc*	*Poor HH*	*Non-Poor HH*
Initial injection	*1.00*	*1.00*	*1.00*	*1.00*	*1.00*	*1.00*
Own-block	*0.53*	*0.99*	*1.16*	*1.03*	*0.00*	*0.00*
Open-loop	*1.62*	*1.73*	*1.52*	*1.59*	*2.14*	*1.01*
Closed-loop	*2.02*	*2.48*	*2.13*	*2.23*	*2.28*	*1.06*
Total	*5.18*	*6.20*	*5.82*	*5.85*	*5.42*	*3.08*

The levels are high because this simplified SAM has no savings account: the only leakages are taxes and payments abroad. Key findings:

1. *Gas has the lowest total multiplier among the production sectors (5.18): 11% below manufacturing and 16% below agriculture. Its own-block effect is the smallest (0.53), because gas buys few domestic intermediates. Its circular-flow effects are also weaker: 45% of its output pays capital, and only 11% of capital income reaches poor households. The real tables make the same point more sharply (see the box below).*

2. *Agriculture, not manufacturing, has the highest multiplier (6.20). It pays a large share of its output to labor (37%), and labor income reaches poor households at a rate of 40%.*

3. *Money given to poor households has a much higher multiplier (5.42) than money given to non-poor households (3.08). In this SAM the non-poor send 47% of their spending abroad, against 16% for the poor.*

4. *Multiplying the column sums of M_1, M_2, M_3 is not a valid decomposition: for gas it would give 5.69 instead of the true 5.18.*

(Numbers: `code/ch20.py`*.)*

Prebisch Connection: Why gas revenue doesn't develop Bolivia

The SAM multiplier makes visible what Chapter 17's dashboard suggested: in this illustrative SAM, Bolivia's gas sector behaves as an enclave. Its total multiplier is the lowest of the four production sectors. Gas revenue flows mainly to capital owners, and capital income reaches poor households only weakly (0.8 of 7.1). The circular flow is thinner where it passes through the distribution of income.

The SAM is a toy, but its gas column can be checked against INE's 2019 supply-use tables (base 2017), and the real enclave is starker than the toy's:

- *Export orientation.* Hydrocarbons export 70.1% of their output (toy SAM: 56%).
- *Domestic linkages.* They buy 0.225 of domestic intermediate inputs per unit of output, against 0.558 for manufacturing (toy SAM: 0.325 against 0.574).
- *Who is paid.* Compensation of employees is 6.0% of the sector's value added (toy SAM: labor receives 18% of the gas sector's factor payments).
- *Size.* The sector is 4.7% of gross output and 6.1% of gross value added. The toy's gas account, a fifth of production, overstates it about fourfold.
- *Position in W^P.* In the 12-sector domestic input-share matrix, the hydrocarbons sector's Perron weight is $\pi = 0.039$. The sector sits at the periphery of the domestic network, not at its center (Chapter 17).

This is the enclave of Pinto and Prebisch, measured by export orientation, domestic input intensity and π, not by Γ. The rents are captured fiscally (IDH and royalties, Chapter 22) rather than through linkages or wages.

Source: INE supply-use tables, base 2017, 2019 (Instituto Nacional de Estadística (INE), Bolivia, 2026); author's calculations (`data/compute/compute_bolivia_io.py`, `data/compute/compute_bolivia_facts.py`). Toy-SAM ratios: `code/ch20.py`.

This is a central theme of Taylor (1983): in demand-driven structuralist models, the functional and size distribution of income shapes the multiplier, because different recipients have different spending and import propensities. In the SAM, it shows up as a strong Gas-to-Capital cell and a faint Capital-to-Poor-HH cell.

The policy reading needs care. An injection multiplier is not the return on investing in a sector. Investment demand falls mostly on capital goods, many of them imported, and the comparison of 5.18 with 5.82 does not by itself rank investment projects. What the decomposition does show is *where* the circular flow is thin, which is where Chapter 15's operators should act.

20.4 The SAM as a Diagnostic Platform

Remark 20.14 (Which Diagnostics a SAM Delivers). For the illustrative SAM of Example 20.7:

Instrument	SAM computation	Value
Γ	Spectral gap of row-normalized SAM	0.402
$G(\boldsymbol{\pi})$	Gini of Perron vector = Gini of account totals	0.18
Open-loop ratio	Open-loop multiplier, gas / mfg	1.06
Direct leakage	Gas payments to Gov + RoW, share of gas output	0.125
σ_B	Needs household microdata, not the SAM	(Ch. 17: 0.62)

A SAM supports the integration and multiplier diagnostics. It does not by itself deliver between-group inequality, which needs within-group dispersion from household surveys. Nor does it deliver information concentration (N_{eff}) or the Ψ ratio. The Gini of the SAM's Perron vector measures dispersion of account size (Proposition 20.4), not of bargaining power.

20.4.1 Counterfactual SAMs

Definition 20.15 (Counterfactual SAM Analysis)**.** *A **counterfactual SAM** is a modified SAM $T' = T + \Delta T$ that represents a hypothetical structural change. For the endogenous accounts it must preserve balance, $\Delta T\mathbf{1} = \Delta T'\mathbf{1}$. The simplest consistent way to build one is to change the coefficient matrix $A_n \to A_n'$, hold the exogenous injections x fixed, and recompute activity $y_e' = (I - A_n')^{-1}x$. Government and RoW then absorb the change in their balances.*

The structural effect of the intervention is:

$$\Delta\Gamma = \Gamma(W(T')) - \Gamma(W(T)), \qquad \Delta M = M(T') - M(T). \tag{20.6}$$

Example 20.16 (Counterfactual: Redirecting Gas Value Added [ILLUSTRATIVE])**.** *Current SAM: Gas → Capital 3.6B Bs, Gas → Labor 0.8B Bs. Capital pays 0.8B Bs to poor and 6.3B Bs to non-poor households. Labor pays 4.2B Bs and 6.2B Bs respectively, so 40% of labor income reaches poor households.*

Counterfactual: shift 1B Bs of gas value added from capital to labor, i.e. raise the labor coefficient of the gas column by $1/8$ and lower its capital coefficient by the same amount. Then solve the multiplier model for the new activity levels.

	Current SAM	*Counterfactual*
Γ_{SAM}	*0.4018*	*0.4026*
Total multiplier of gas	*5.18*	*5.30*
Poor households' share of household income	*0.330*	*0.344*

The reallocation is 23% of gas value added (19% if production taxes are included). It raises the gas multiplier by 2.3% and the poor's share of household income by 1.4 points. Γ *barely moves (*$+0.0008$*). That is what one should expect:* Γ *of the row-normalized SAM measures how fast the circular flow mixes across accounts, not how income is distributed. Distributional counterfactuals should be read off the multipliers and the household accounts, not off* Γ*. (Numbers:* `code/ch20.py`*.)*

20.5 Summary

1. The **SAM** is a balanced square matrix recording every flow between every account in the economy: columns are payments, rows are receipts. Column normalization gives the coefficient matrix for multipliers (and, for the production block, $W^P = D^{-1}A$). Row normalization gives receipt shares.

2. For a balanced SAM the Perron vector of the row-normalized matrix is **account size**, $\pi = y/\mathbf{1}'y$. The full-SAM spectral gap has no general ordering relative to the I-O block's (illustrative SAM: 0.40 against 0.39). Its leading eigenvalues are complex, reflecting the circular flow rather than fragmentation.

3. The **SAM multiplier matrix** $(I - A_n)^{-1}$ factors as $M_3M_2M_1$: closed-loop, open-loop and transfer multipliers (Pyatt–Round). Stone's additive form reports their contributions as column sums that add up.

4. **Bolivia's gas enclave** (illustrative): gas has the lowest multiplier of the production sectors (5.18, against 5.82 for manufacturing and 6.20 for agriculture). It buys few domestic inputs, and its income goes mainly to capital, which reaches poor households weakly. INE's 2019 tables confirm the pattern: hydrocarbons export 70.1% of output, buy 0.225 of domestic inputs per unit of output (manufacturing 0.558), pay 6.0% of value added as compensation, and carry a Perron weight of only 0.039 in the domestic W^P. The enclave is measured by these numbers, not by Γ.

5. **Counterfactual SAMs** evaluate structural interventions through the multiplier model. Shifting 1B Bs of gas value added from capital to labor raises

the gas multiplier by 2.3% and the poor's income share by 1.4 points, while leaving Γ essentially unchanged. A SAM supports integration and multiplier diagnostics, but not between-group inequality or information concentration.

20.6 Exercises

20.1 Build a SAM. Using INE's supply-use tables (base 2017), the EH, and fiscal data: construct a 10×10 SAM following the template in Example 20.7. Verify that it balances.

20.2 Row-normalize and compute Γ. From your SAM: compute W, its eigenvalues, and Γ_{SAM}. Compare with Γ of the production block alone. Which is smaller in your SAM? Are the leading eigenvalues complex, and what does that say about the circular flow? Verify that the Perron vector equals the vector of account totals.

20.3 SAM multipliers. Compute $(I - A_n)^{-1}$ for your SAM. Decompose it with Pyatt–Round ($M = M_3 M_2 M_1$) and report Stone's additive components. Which sector has the highest total multiplier? Which has the largest open-loop component?

20.4 The gas enclave in the SAM. Trace the circular flow from a 1B Bs injection into gas: Gas $\to$ Capital $\to$? $\to$? $\to$? At each stage, compute the leakage rate (the fraction that exits the circular flow through imports, savings, or foreign profit repatriation). At what stage does the flow die?

20.5 Counterfactual: land reform. Modify the SAM to represent a hypothetical land reform: shift 10% of capital income (Cap $\to$ HH-NP) to poor households (Cap $\to$ HH-P). Recompute Γ, the multipliers, and the poor's share of household income. Which of these moves most, and why?

20.6 Counterfactual: education spending. Increase government spending on education (account 9 $\to$ account 5, labor) by 0.5B Bs, financed by a tax on gas (account 1 $\to$ account 9). Recompute the multipliers. Is this reallocation multiplier-improving?

20.7 Cross-country SAM comparison. Using published SAMs for Bolivia, Chile, and South Korea:

(a) Compute Γ_{SAM} for each.

(b) Compute the open-loop and closed-loop multipliers of the manufacturing sector in each.

(c) Which country has the strongest circular flow? Why?

(d) Relate the SAM diagnostics to the dashboard from Chapter 17.

Mathematical Appendices

Appendix 20A: SAM Balancing

The problem. Raw data from I-O tables, household surveys, and fiscal accounts rarely balance exactly: row sums and column sums disagree because the data sources are independent with different coverage, timing, and definitions.

Cross-entropy balancing (Robinson, Cattaneo, and El-Said, 2001). Given a prior (unbalanced) SAM T^0 with column coefficients $\bar{a}_{ij} = T^0_{ij} / \sum_k T^0_{kj}$, find coefficients a_{ij} and totals y that minimize the cross-entropy distance to the prior coefficients, subject to balance:

$$\min_{a \geq 0,\, y} \sum_{ij} a_{ij} \ln \frac{a_{ij}}{\bar{a}_{ij}}, \qquad \text{s.t. } \sum_i a_{ij} = 1, \quad \sum_j a_{ij} y_j = y_i. \tag{20.7}$$

The solution stays as close as possible to the prior's structure while achieving exact balance; known totals or other data can be added as further constraints.

RAS method. The simpler bi-proportional adjustment: alternately scale rows and columns until they match *given* target totals. This is the same alternating-normalization idea as Sinkhorn's algorithm (Appendix 15B), which targets unit row and column sums. Unlike that problem, SAM balancing usually does not know the common totals in advance (it only requires row total = column total), so RAS needs target totals from outside data.

Appendix 20B: The Pyatt-Round Decomposition

Setup. Order the endogenous accounts as production (1), factors (2) and households (3). Rows receive from columns, so the coefficient matrix has the circular-flow pattern

$$A_n = \begin{pmatrix} A_{11} & 0 & A_{13} \\ A_{21} & 0 & 0 \\ 0 & A_{32} & A_{33} \end{pmatrix}. \tag{20.8}$$

Production receives from production (intermediates, A_{11}) and households (consumption, A_{13}). Factors receive from production (value added, A_{21}). Households receive from factors (factor income, A_{32}) and from other households (transfers, A_{33}). The zero blocks reflect the accounting structure: factors do not buy goods, production does not pay households directly, and households do not pay factors directly.

Decomposition. Let $\tilde{A} = \text{diag}(A_{11}, 0, A_{33})$ and $A^* = (I - \tilde{A})^{-1}(A_n - \tilde{A})$. Then A^* is block-cyclic of period three, and

$$M = (I - A_n)^{-1} = \underbrace{(I - A^{*3})^{-1}}_{M_3}\ \underbrace{(I + A^* + A^{*2})}_{M_2}\ \underbrace{(I - \tilde{A})^{-1}}_{M_1}.$$

The proof is in Proposition 20.11. M_3 is block-diagonal: each block's own circuit through the other two. Stone's additive form $M = I + (M_1 - I) + (M_2 - I)M_1 + (M_3 - I)M_2M_1$ is the one to report as column sums. `code/ch20.py` verifies both identities on the chapter's SAM.

Structuralist Case Study: Taylor's Structuralist Macroeconomics

Lance Taylor's *Structuralist Macroeconomics: Applicable Models for the Third World* (1983) is a foundational text of applied structuralist macroeconomics. It built macro models on SAMs — the accounting framework developed by Stone, by Pyatt and Thorbecke (1976) and by Pyatt and Round (1979) — and used them to capture the structural heterogeneity, distributional flows and institutional constraints that benchmark macro models abstract from.

Taylor's contribution to this book's framework is methodological. The SAM is not just a data table but a *modeling platform*: it encodes the economy's complete flow structure, and different "closure rules" (choices of which variables are exogenous) produce different models from the same SAM. In this book's terms, the SAM supplies the coefficients from which W^P and the multipliers are built. The closure determines the causal story the model tells about them.

Part 1: The SAM as the foundation. Taylor's practice was to begin from a balanced SAM before writing behavioral equations. The SAM disciplines the modeler: every flow must come from somewhere and go somewhere. In this chapter's notation, the SAM delivers both the receipt-share matrix of Definition 20.3 and the expenditure coefficients A_n of Definition 20.10. A modeler who starts from aggregate equations may violate the flow structure without noticing.

Part 2: Closure rules as causal assumptions. Taylor identified the closure as the modeler's most consequential choice, because the same accounting identities admit different causal stories:

- **savings-driven (neoclassical)**: full employment, with investment adjusting to available saving;
- **investment-driven (Keynes–Kalecki)**: output and employment adjust to exogenous investment;

- **Kaldorian**: the income distribution adjusts;
- **Johansen-style**: government absorption adjusts.

Whether the current account adjusts through the exchange rate or through income contraction is a further choice of the same kind. Closures are statements about which variables are exogenous, not properties of a weight matrix. The SAM multiplier of Definition 20.10 is the simplest demand-driven closure: fixed prices, exogenous government and RoW, quantities adjusting.

Part 3: Distribution in the SAM. Taylor's models disaggregate households (by income class, rural/urban status) and income types (wages, profits, rents), because in demand-driven models the distribution of income shapes the multiplier. This was shared with the SAM tradition of Pyatt, Round and Adelman–Robinson. It is this book's Chapter 18 at the macro level: the distributional structure of the SAM reveals which household types receive which factor payments, and therefore who benefits from growth. Between-block inequality σ_B itself still needs household microdata.

Taylor's contribution (1983)	**Formal object**	**This chapter**
SAM as modeling platform	Receipt shares $\operatorname{diag}(y)^{-1}T$; coefficients A_n	Defs. 20.3, 20.10
Circular flow	$M_{\text{SAM}} = (I - A_n)^{-1}$; complex eigenvalues of W	Def. 20.10, Ex. 20.8
Multiplier layers	$M = M_3 M_2 M_1$ (Pyatt–Round)	Prop. 20.11
Distribution shapes the multiplier	Thin capital-to-poor link in enclave sectors	Ex. 20.13
Closure rules	Choice of exogenous variables (not a property of W)	Part 2 above
Household disaggregation	Household accounts; σ_B needs microdata	Rem. 20.14

Taylor's books (1983; 1990; 2004) trained a generation of structuralist modelers in Latin America, Africa, and South Asia. His 1990 volume, *Socially Relevant Policy Analysis*, collected structuralist computable general equilibrium (CGE) models.

SAM-based CGE models, in the line of Dervis, de Melo, and Robinson (1982) and Lofgren, Harris, and Robinson (2002), remain a standard tool for evaluating policy in developing countries. This chapter places the SAM within the book's matrix framework. The multiplier decomposition locates where the circular flow is thin, and the counterfactual SAM evaluates interventions under an explicit closure.

Further Reading

SAM construction and methodology: Pyatt and Round (1979), "Accounting and Fixed Price Multipliers in a Social Accounting Matrix Framework." Round (2003), "Social Accounting Matrices and SAM-Based Multiplier Analysis." Robinson, Cattaneo, and El-Said (2001), "Updating and Estimating a Social Accounting Matrix Using Cross Entropy Methods."

Structuralist macroeconomics: Taylor (1983), *Structuralist Macroeconomics*. Taylor (1990), *Socially Relevant Policy Analysis: Structuralist Computable General Equilibrium Models for the Developing World*. Taylor (2004), *Reconstructing Macroeconomics*. Pyatt and Thorbecke (1976), *Planning Techniques for a Better Future*.

CGE modeling: Lofgren, Harris, and Robinson (2002), "A Standard Computable General Equilibrium (CGE) Model in GAMS." Dervis, de Melo, and Robinson (1982), *General Equilibrium Models for Development Policy*. For Bolivia-specific CGE models, see the work of Luis Carlos Jemio.

Bolivia SAM data: INE (`ine.gob.bo`): supply-use tables (base 2017) and input–output matrices (base 1990), National Accounts (Instituto Nacional de Estadística (INE), Bolivia, 2026). UDAPE: social indicators and public expenditure. MEFP (Ministerio de Economía y Finanzas Públicas): fiscal data.

Chapter 21

Measuring the Information Network

Where W^I Comes From

Chapter 20 showed how to measure W^P from the SAM. This chapter tackles the harder problem: measuring W^I, the information network that determines how agents form beliefs and preferences.

W^I is harder to measure than W^P for three reasons. First, information flows leave fewer paper trails than trade flows — there is no customs form for ideas. Second, the relevant signals are not prices but *narratives*: what people see, whom they listen to, which outcomes they observe. Third, W^I varies across subpopulations in ways that W^P does not: the gas executive and the quinoa farmer in Bolivia face the same W^P (they are in the same economy) but radically different W^I's (they consume different media, attend different schools, talk to different people).

Despite these difficulties, W^I can be estimated from four data sources: media audience data, social network data, survey-elicited reference groups, and digital trace data. Each captures a different layer of the information network. Together they approximate the matrix that Chapters 8–10 assumed as given. Two cautions run through the chapter. First, each source defines its own node set (groups, outlets, individuals), and layers can be combined only on a common one. Second, networks are almost never observed completely, so measurement error and identification (Section 21.4) are part of the estimate, not an afterthought.

21.1 Four Data Sources for W^I

21.1.1 Source 1: Media Audience Data

Definition 21.1 (Media-Derived W^I)**.** *Let there be K media outlets and G demographic groups with population shares p_g. Group g splits its attention across outlets with shares*

s_{gk} ($\sum_k s_{gk} = 1$), *and outlet* k *has reach* $m_k = \sum_g p_g s_{gk}$. *The* ***raw media overlap*** *is*

$$O_{gg'} = \sum_{k=1}^{K} s_{gk}\, s_{g'k}, \tag{21.1}$$

the probability that a member of g *and a member of* g'*, each sampling one outlet, sample the same one. High overlap means both groups see the same narratives; low overlap means each group lives in a separate information bubble. The* ***population-weighted media matrix*** *on groups is*

$$W^{I}_{media,gg'} = \sum_k s_{gk}\, \frac{p_{g'} s_{g'k}}{m_k}. \tag{21.2}$$

This is the probability that a member of g*, having picked an outlet, shares it with a member of* g' *drawn from that outlet's audience. It is row-stochastic by construction.*

Both media matrices are reversible random walks on the bipartite group–outlet graph. Their Perron vectors are therefore mechanical: population shares p for the group matrix (21.2), and reach m for the outlet projection $\mathrm{diag}(m)^{-1} S' \,\mathrm{diag}(p) S$. The two projections also share their nonzero eigenvalues, and hence Γ: with $S = (s_{gk})$, they are XY and YX for $X = S$ and $Y = \mathrm{diag}(m)^{-1} S' \,\mathrm{diag}(p)$. Media audience data thus measure *integration* (overlap, Γ, λ_k) and the *concentration of reach*. They do not measure who is believed: epistemic influence requires network or reference-group data.

Example 21.2 (Bolivia's Media W^I [ILLUSTRATIVE])**.** *Six demographic groups: (1) Urban non-indigenous men, (2) Urban non-indigenous women, (3) Urban indigenous, (4) Rural non-indigenous, (5) Rural indigenous men, (6) Rural indigenous women. Five media clusters: (A) Private TV/press, (B) State media, (C) Community radio (indigenous language), (D) Social media (Facebook, WhatsApp), (E) Word of mouth (local markets, community meetings).*

Illustrative audience shares (to be replaced with survey data):

Group	*Private TV*	*State*	*Community*	*Social*	*Word/mouth*
Urban NI men	*0.45*	*0.15*	*0.02*	*0.30*	*0.08*
Urban NI women	*0.40*	*0.18*	*0.03*	*0.28*	*0.11*
Urban indigenous	*0.20*	*0.30*	*0.15*	*0.20*	*0.15*
Rural NI	*0.25*	*0.20*	*0.10*	*0.15*	*0.30*
Rural indig. men	*0.08*	*0.22*	*0.30*	*0.10*	*0.30*
Rural indig. women	*0.05*	*0.15*	*0.25*	*0.08*	*0.47*

Illustrative population shares: $p = (0.17, 0.18, 0.25, 0.10, 0.15, 0.15)$.

Computing the raw overlap (21.1)*:*

- $O_{11} = 0.45^2 + 0.15^2 + 0.02^2 + 0.30^2 + 0.08^2 = 0.322$ *(self-overlap: how concentrated group 1's media diet is).*

- $O_{16} = 0.45 \times 0.05 + 0.15 \times 0.15 + 0.02 \times 0.25 + 0.30 \times 0.08 + 0.08 \times 0.47 = 0.112$ *(urban non-indigenous men and rural indigenous women: the lowest overlap, 35% of the two groups' self-overlap).*

- $O_{12} = 0.45 \times 0.40 + 0.15 \times 0.18 + \cdots = 0.300$ *(urban non-indigenous men and women: as high as self-overlap).*

Results:

- ***Integration.*** *The population-weighted matrix* (21.2) *has* $|\lambda_2| = 0.26$*, so* $\Gamma = 0.74$*. Simply row-normalizing O gives the same values.*

- ***Concentration.*** *Over groups,* $\pi = p$ *and* $N_{eff} = 5.59$ *of a possible 6. Over outlets, reach is* 0.243 *(private TV),* 0.208 *(state),* 0.139 *(community radio),* 0.193 *(social media) and* 0.216 *(word of mouth), so* $N_{eff} = 4.85$ *of 5.*

Taken at face value, these audience shares do not *describe separate information bubbles. Exposure overlaps substantially across groups, and reach is spread across five channels. If Bolivia's information structure is segmented, the segmentation must lie in what is said within each channel, in how it is interpreted, or in interpersonal networks — which is why the other three sources matter. (Numbers:* `code/ch21.py`*.)*

21.1.2 Source 2: Survey-Elicited Reference Groups

Definition 21.3 (Reference-Group W^I)**.** *Household surveys can elicit* W^I *directly by asking: "Whose economic outcomes do you observe and consider when making your own decisions?" (or proxies such as: "Who do you compare yourself to?", "Whose advice do you follow?", "Who would you like your children to be like?").*

Let $R_{ig} = 1$ *if agent* i *names someone in group* g *as a reference. The reference-group information matrix:*

$$W^I_{ref,gg'} = \frac{1}{N_g} \sum_{i \in g} R_{ig'}. \tag{21.3}$$

Entry $W^I_{ref,gg'}$*: the fraction of group* g *members who name someone in group* g' *as a reference.*

Stylized Fact 21.4 (Reference Groups Are Local and Assortative)**.** *The evidence suggests that reference groups are local and assortative:*

1. ***By income**: relative income matters for well-being, and the effect is asymmetric. Comparisons with richer reference groups lower the well-being of the poorer, more than the reverse (Ferrer-i Carbonell, 2005).*
2. ***By geography**: reference groups are local. Higher earnings of neighbors in the same area lower self-reported happiness (Luttmer, 2005).*
3. ***By gender**: same-gender role models shift aspirations. Randomly assigned female village leaders raised girls' educational aspirations in India (Beaman et al., 2012).*
4. ***By ethnicity**: plausible, but we know of no Bolivia-specific estimate; it is a target for the survey module of Appendix 21B.*

The assortative mixing parameters λ_k of Chapter 18 can be estimated directly from reference-group data.

21.1.3 Source 3: Social Network Data

Definition 21.5 (Network-Derived W^I)**.** *When social network data are available (friendship networks, communication records, co-membership in organizations), the information matrix is estimated as:*

$$W^I_{net,ij} = \frac{c_{ij}}{\sum_k c_{ik}}, \tag{21.4}$$

where c_{ij} is the communication frequency between i and j (messages, calls, meetings). Row-normalize to obtain a row-stochastic matrix.

Example 21.6 (Village Networks in Banerjee et al. (2013))**.** *Banerjee, Chandrasekhar, Duflo, and Jackson (2013) collected household-level network data in 75 villages in rural Karnataka, India, covering several relationships: who visits whom, kinship, borrowing and lending money or goods, and advice. Villages average roughly 200 households. In the villages where a microfinance institution entered, the MFI first informed pre-selected "leaders" (teachers, shopkeepers, self-help-group leaders). Eventual take-up was higher where those first-informed households had higher* diffusion centrality*, a measure closely related to eigenvector centrality. In this book's terms, the result concerns the centrality of the* injection points *in W^I. It is not a village-level concentration statistic such as N_{eff}.*

Data of this kind are the gold standard for W^I_{net} at the individual scale. They are expensive, and even here some households were not surveyed. That is the sampling problem of Section 21.4.

21.1.4 Source 4: Digital Trace Data

Definition 21.7 (Digital W^I)**.** *Digital platforms generate trace data on information flows: who follows whom (Twitter/X), who shares whose content (Facebook), who watches what*

(YouTube), who reads what (news aggregators). The digital information matrix:

$$W^I_{digital,ij} = \frac{\textit{content shared from } j \textit{ to } i}{\sum_k \textit{content shared from } k \textit{ to } i}. \tag{21.5}$$

Remark 21.8 (Digital W^I Is Growing). Internet use in Bolivia has spread fast: 47.5% of the population used the internet in 2019, 77.3% in 2023 and 79.7% in 2024 (ITU estimates, World Development Indicators series IT.NET.USER.ZS; World Bank, 2026b). Use is concentrated among urban populations under 35, for whom digital channels compete with traditional media. The consequences:

1. W^I is becoming more measurable (digital traces are recorded).

2. W^I is becoming more manipulable (algorithmic recommendation systems shape what people see — Chapter 27).

3. W^I is changing structure: cross-group exposure may increase (access to global content) or decrease (algorithmic echo chambers). The net effect on N_{eff} is empirically ambiguous.

21.2 Combining Sources: The Composite W^I

Definition 21.9 (Composite Information Matrix)**.** *The composite W^I combines the four sources:*

$$W^I = \omega_m W^I_{media} + \omega_r W^I_{ref} + \omega_n W^I_{net} + \omega_d W^I_{digital}, \tag{21.6}$$

where $\omega_m + \omega_r + \omega_n + \omega_d = 1$ are the source weights. The composite is defined only when all four layers are row-stochastic matrices on the same node set*. Group-level media and reference-group matrices must first be mapped to individuals (each person inherits her group's row), or individual-level network and digital data aggregated to groups (Definition 19.6). The weights can in principle be estimated by matching the composite's predictions (for belief convergence, preference formation, technology adoption) to observed outcomes. They are identified only if the layers make different predictions — for example, if they differ in who is connected to whom.*

Example 21.10 (Bolivia's Composite W^I Weights [ILLUSTRATIVE])**.** *Hypothetical weights, to show how heterogeneous the composite can be:*

Source	*Urban under 35*	*Urban over 50*	*Rural*
Media (ω_m)	*0.15*	*0.45*	*0.25*
Reference group (ω_r)	*0.20*	*0.30*	*0.40*
Social network (ω_n)	*0.15*	*0.15*	*0.25*
Digital (ω_d)	*0.50*	*0.10*	*0.10*

In this illustration, digital W^I dominates for urban youth: half of all information formation comes through social media. For rural populations, reference groups and traditional media dominate. For older urban populations, traditional media still matters. The weights vary across subpopulations, which means the composite W^I is heterogeneous: different Bolivians live in different information environments, even within the same city.

21.3 Estimating N_{eff} and π^I from Data

Definition 21.11 (Information Structural Parameters)**.** *From an estimated row-stochastic W^I on a stated node set, the key structural parameters are:*

***1. The epistemic Perron vector π^I**, computed by power iteration. Under DeGroot updating, π^I_j is node j's weight in the long-run consensus (Chapter 8). For media-overlap matrices it reduces to population shares or reach (Definition 21.1).*

***2. The effective number of voices** $N_{eff} = 1/\sum_j (\pi^I_j)^2 \in [1, n]$: the concentration of epistemic power. Compare it across node sets only as N_{eff}/n.*

***3. The spectral gap** $\Gamma(W^I) = 1 - |\lambda_2(W^I)|$: information integration, i.e. the speed of consensus. Small Γ^I means weakly coupled clusters (echo chambers); large Γ^I means information circulates freely. Γ and N_{eff} are independent coordinates (Proposition 6.16).*

***4. The assortative mixing parameters** λ_k: the ratio of mean cross-group to mean within-group entries, $\hat{\lambda}_k = \bar{O}_{cross,k}/\bar{O}_{within,k}$, computed on the raw overlaps or intensities.*

Example 21.12 (Bolivia's Information Parameters [ILLUSTRATIVE])**.** *From the illustrative audience table of Example 21.2:*

Parameter	*Value*	*Method*
$\Gamma(W^I_{media})$	*0.74*	*Spectral gap of the* 6×6 *group matrix*
N_{eff} *over groups*	*5.59 of 6*	π = *population shares*
N_{eff} *over outlets*	*4.85 of 5*	π = *reach*
λ_{gender}	*0.99 / 0.93*	*Cross-gender overlap (urban NI / rural indigenous)*
$\lambda_{ethnicity}$	*0.72*	*Cross-ethnic overlap*
$\lambda_{urban\text{-}rural}$	*0.68*	*Cross-geography overlap*
Reach: private TV, state	*0.243, 0.208*	$m_k = \sum_g p_g s_{gk}$
Reach: word of mouth, social	*0.216, 0.193*	
Reach: community radio	*0.139*	

The illustrative audience data show high integration and only mild concentration of reach. Private TV and state media together reach 45% of attention. Community radio — the only channel with substantial indigenous-language content — reaches 14%. Its small reach is concentrated in rural indigenous groups. Two tempting comparisons are not valid:

- *"*$N^I_{eff} < N^P_{eff}$*" compares counts on different node sets (5 outlets or 6 groups against dozens of sectors). Only* N_{eff}/n *is comparable.*

- *Reach is not epistemic influence. Who is* believed *requires network or reference-group data, not audience shares.*

(Numbers: `code/ch21.py`*.)*

21.4 Sampling, Measurement Error, and Identification

Every source above observes W^I with error, and most observe only part of it. Five problems, and the tools the literature has developed for them:

1. **Sampled networks.** Surveys interview a sample of nodes, so links among unsampled nodes are missing. Network statistics computed on the sampled graph (degree, centrality, Γ, π) are biased, and so are regressions that use them. Chandrasekhar and Lewis (2016) show how the bias arises and propose corrections, including reconstructing the missing part of the graph.

2. **Aggregated relational data (ARD).** When full network elicitation is too costly, ask "How many people with trait k do you know?" for a few traits. Breza,

Chandrasekhar, McCormick, and Pan (2020) show how to estimate a latent-space network model from ARD and recover network statistics such as centrality. ARD are well suited to the group-level W^I of this chapter: traits can be defined by ethnicity, gender and locality.

3. **Censored name generators.** "Name up to three people ..." (Appendix 21B) top-codes degree and drops weak ties. Griffith (2022) shows that such censoring biases peer-effect estimates, typically towards zero. Record the number of people a respondent *would* name, or use uncensored formats.

4. **Misreported links.** When both ends are surveyed, reports often disagree: i names j but j does not name i. Comola and Fafchamps (2014) model reporting errors explicitly and use the discordant reports to test whether links are formed unilaterally or bilaterally.

5. **Identification of influence.** Even a perfectly measured W^I does not by itself identify influence:

   - Manski's (1993) reflection problem: in linear-in-means models, endogenous effects (group behavior affects the individual) cannot be separated from contextual effects (group characteristics affect the individual). Correlated effects (common shocks, self-selection into groups) are a further confounder.
   - Bramoullé, Djebbari, and Fortin (2009) show that network structure can break the reflection problem: friends of friends who are not friends supply exclusion restrictions.
   - Homophily and contagion are generically confounded in observational network data (Shalizi and Thomas, 2011). Experiments or quasi-random variation in links (as in Banerjee et al.'s leader seeding) are the cleanest route.

Sampling error in W^I propagates to π^I and to Γ. The perturbation lemma of Chapter 6 (Lemma 6.19) gives the first-order sensitivity of π to an error matrix E; bootstrapping over sampled nodes gives standard errors. π is most fragile when an eigenvalue is close to $+1$, i.e. precisely in the weakly integrated networks this book cares about.

21.5 Validating W^I: Does It Predict Formation?

Definition 21.13 (Validation Tests for W^I)**.** *An estimated W^I passes validation if it predicts the formation outcomes derived in Chapters 8–10:*

1. ***Belief convergence**: Groups with high media overlap (large $O_{gg'}$) should have more similar beliefs about economic outcomes. Test: correlate $O_{gg'}$ with $|\bar{\theta}_g - \bar{\theta}_{g'}|$. The correlation should be negative. Dyads are not independent, so use permutation (QAP) or dyadic-robust inference.*

2. ***Aspiration prediction**: The W^I-weighted average return observed by group g should predict group g's aspirations. Test: regress $\bar{\theta}_g$ on $\sum_{g'} W^I_{gg'} \bar{r}_{g'}$. This requires many groups (e.g. municipality $\times$ ethnicity $\times$ gender cells), not six.*

3. ***Ψ ratio**: The formation-to-enforcement ratio (Chapter 11) should be higher in populations with more assortative W^I (lower λ_k). Test: compare $\hat{\Psi}$ across subpopulations with different λ_k values.*

4. ***Diffusion**: New information injected at more central nodes should spread further (as in Banerjee et al., 2013), and communities with higher $\Gamma(W^I)$ should converge faster.*

These tests are informative only if W^I is measured independently of the outcomes it is meant to predict, and if the identification problems of Section 21.4 are addressed.

Example 21.14 (A Validation Design for Bolivia)**.** *Bolivia's standard household surveys (the EH) and regional opinion surveys (Latinobarómetro) do not contain the media and aspiration modules these tests require, so no results are reported here. A feasible design:*

1. *Field the Appendix 21B module, with ARD questions by ethnicity and locality, in a household survey.*

2. *Build W^I at the municipality $\times$ group level.*

3. *Run Tests 1–3 with dyadic-robust inference.*

4. *Compare with an experimental benchmark, such as randomized seeding of an information campaign.*

21.6 The W^I Operator Revisited: What the Data Tell the Designer

Remark 21.15 (Data-Informed Information Policy). A measured W^I tells the policy designer (Chapter 15) where the dark entries are. On the illustrative audience table:

Dark entry in W^I	**Operator**	**What to simulate**
Urban NI men ↔ rural indigenous women (O_{16} at 35% of self-overlap)	Bilingual media program	$\Delta\Gamma$, $\Delta\lambda_{\text{ethnicity}}$
Community radio (reach 0.14, concentrated in rural indigenous groups)	Community journalism; carrying community content on mainstream channels	Δ reach, $\Delta\lambda_{\text{urban-rural}}$
Low digital access in rural areas	Rural broadband + digital literacy	Δ overlap of rural groups with urban ones needs network data
Women ↔ men in economic information (not visible in media data)	Female entrepreneurship mentoring	

The expected effects must be computed from the measured matrix, as in Exercise 21.6. In the illustrative table, adding 0.10 to the darkest overlap O_{16} raises Γ from 0.740 to 0.791 and leaves N_{eff} essentially unchanged: an integration operator, not a deconcentration operator.

Key Structural Insight

Measuring W^I converts the abstract "information policy" of Chapter 15 into a concrete investment schedule: which groups need which information channels, with what expected effect on integration (Γ, λ_k) and on concentration (N_{eff}). The measurement makes the operator design problem solvable from data rather than assumed from theory. It can also overturn the theory's presumption. In this chapter's illustrative audience table, channel choice is far less segregated than the "information bubbles" narrative suggests. The policy-relevant segmentation, if any, has to be sought in interpersonal networks and in content.

21.7 Summary

1. W^I can be estimated from four sources, each on its own node set: **media audience data** (overlap between groups or outlets), **survey-elicited reference groups** (who do you compare yourself to?), **social network data** (communication frequency), and **digital trace data** (platform interactions). A **composite** exists only on a common node set.

2. Media-overlap matrices are reversible random walks. Their Perron vectors are population shares (over groups) or reach (over outlets). Audience data therefore measure **integration** and the concentration of **reach**, not who is believed.

3. **Bolivia's information parameters** (illustrative audience table): $\Gamma^I = 0.74$; $N_{\text{eff}} = 5.59$ of 6 over groups and 4.85 of 5 over outlets; $\lambda_{\text{ethnicity}} = 0.72$, $\lambda_{\text{urban-rural}} = 0.68$, $\lambda_{\text{gender}} \approx 0.93$–$0.99$. Exposure is far less segregated than the "bubbles" narrative suggests; community radio reaches 14% of attention.

4. **Measurement and identification**: networks are sampled (Chandrasekhar and Lewis, 2016), can be recovered from aggregated relational data (Breza et al., 2020), are censored by fixed-length name generators (Griffith, 2022) and misreported (Comola and Fafchamps, 2014). Influence is not identified by W^I alone (Manski, 1993; Bramoullé et al., 2009; Shalizi and Thomas, 2011).

5. **Validation and design**: a measured W^I must predict belief convergence, aspirations and diffusion, using dyadic-robust inference and data that the matrix was not built from. Its dark entries tell the designer where integration operators should act; their effects must be simulated on the measured matrix.

21.8 Exercises

21.1 Media overlap. Using Example 21.2: compute the full 6×6 raw overlap matrix O and the population-weighted matrix W^I_{media}. Compute eigenvalues, Γ, $\boldsymbol{\pi}^I$, and N_{eff}. Show that π equals the population shares, and that the outlet projection has π equal to reach and the same Γ. (Answers: `code/ch21.py`.)

21.2 Reference groups. Design a 5-question survey module to elicit W^I reference groups for a Bolivian household survey. What questions would you ask? How would you construct W^I_{ref} from the responses?

21.3 Digital W^I from Facebook data. If you had access to Bolivia's Facebook advertising API (audience overlap estimates by demographic): construct W^I_{digital} for 6 groups. How does it compare to W^I_{media}? Is N_{eff} higher or lower?

21.4 Validation. Using Latinobarómetro data for Bolivia: test whether groups with high $W^I_{gg'}$ (from media overlap) have more similar views on: (a) expected inflation, (b) trust in government, (c) support for redistribution. Report correlations, with significance assessed by a permutation (QAP) test that respects dyadic dependence.

21.5 λ_k estimation. Estimate the assortative mixing parameters for Bolivia's W^I:

(a) By gender: $\lambda_g = \bar{O}_{\text{cross-gender}}/\bar{O}_{\text{within-gender}}$.

(b) By ethnicity: λ_r.

(c) By urban/rural: λ_u.

(d) Which dimension shows the strongest assortative mixing? How does this compare with the W^P assortative mixing from Chapter 18?

21.6 The operator from data. Using your estimated W^I:

(a) Identify the darkest entry. This is where a bilingual media or digital access program should target.

(b) Simulate: add 0.10 to the darkest raw overlap (symmetrically), renormalize, and recompute N_{eff} and Γ^I. Which moves, and why?

(c) Using an illustrative $\Psi = 2$ and the formation model (Proposition 18.13): estimate the long-run effect on the aspiration gap.

21.7 Cross-country comparison. Using media audience data or data on the concentration of media ownership (where available) and internet penetration (ITU) for Bolivia, Chile, and Sweden: estimate the outlet-level N^I_{eff}/n for each. Rank them. Does the ranking predict the Ψ ratio ranking?

Mathematical Appendices

Appendix 21A: From Media Shares to W^I — The Overlap Formula

Derivation. Assume agent i in group g samples information from outlet k with probability s_{gk}. Two agents $i \in g$ and $j \in g'$ receive correlated signals when they share a media source. The probability that both receive the same signal from some outlet:

$$\Pr(\text{shared signal}) = \sum_k s_{gk} \cdot s_{g'k} = O_{gg'}. \tag{21.7}$$

This is the inner product of the media consumption vectors: $O = SS'$, where S is the $G \times K$ audience share matrix. Its rank is at most $\min(G, K)$. Rank does not bound N_{eff}, which depends on the Perron vector and lies in $[1, n]$ for whichever node set is used.

Normalization. The raw overlap SS' is not row-stochastic. Two row-stochastic versions are natural:

- $\hat{W}_{gg'} = O_{gg'} / \sum_{g''} O_{gg''}$, which ignores group sizes;
- the population-weighted $W = S\,\mathrm{diag}(m)^{-1} S'\,\mathrm{diag}(p)$ of (21.2).

Both are reversible, with Perron vectors proportional to the row sums of O and to p respectively. W and the outlet projection $\mathrm{diag}(m)^{-1} S'\,\mathrm{diag}(p) S$ are products XY and YX of the same two matrices. They therefore share their non-unit eigenvalues, and hence Γ.

Limitations. Media overlap is a proxy for information sharing, not information sharing itself. Two people who watch the same channel may attend to different content or interpret it differently. The overlap formula assumes passive reception; in practice, agents filter, interpret, and discuss media content through their social networks (captured by W^I_{net}).

Appendix 21B: Estimation of W^I from Household Surveys

The "who do you talk to" module. Add to the household survey:

1. "Name the 3 people outside your household whose economic situation you pay most attention to." Record: relationship, location, gender, ethnicity, occupation.

2. "Name the 3 information sources you use most for economic decisions." Record: media type, frequency, language.

3. "If your neighbor earns more than you, does that make you want to earn more?" (Aspiration reference intensity, μ proxy.)

Construction. From the named alters (question 1): assign each alter to a demographic group g'. The fraction of group g's members naming alters in g' gives $W^I_{\text{ref},gg'}$. Row-normalize.

The Manski problem. Manski's (1993) reflection problem: in a linear-in-means model, the effect of the group's average behavior (endogenous effect) cannot be separated from the effect of the group's average characteristics (contextual effect), because the former is a linear function of the latter. Correlated effects — common shocks, or self-selection into groups — are a separate confounder. The cohort-year decomposition of Chapter 11 addresses common year shocks, but not the reflection problem itself. Network structure helps: with intransitive triads, the characteristics of friends' friends who are not one's own friends are valid instruments (Bramoullé, Djebbari, and Fortin, 2009). Name-generator censoring ("3 people") must also be corrected for (Section 21.4). The estimated W^I is credible for formation predictions only if it also survives the validation tests of Definition 21.13.

Structuralist Case Study: Information Structures in the Global South

The structuralist tradition has long recognized that the information structure shapes economic outcomes — from Prebisch's critique of the "ideological diffusion" of free-trade doctrine from center to periphery, to Furtado's analysis of cultural dependency (Chapter 9), to Quijano's coloniality of epistemic power (Chapter 8). What the tradition lacked was measurement.

This chapter provides the measurement infrastructure. Three findings deserve emphasis for the structuralist research program.

First: concentration must be compared on comparable node sets. It is tempting to compare the effective number of outlets with the effective number of sectors and conclude that the information bottleneck is tighter than the production bottleneck, as Furtado's (1974) thesis of cultural dependency would predict. That comparison is not valid: N_{eff} is bounded by the number of nodes, and only N_{eff}/n can be compared across node sets. On this chapter's illustrative audience table, outlet reach is not very concentrated ($N_{\text{eff}} = 4.85$ of 5). Furtado's thesis concerns the *origin and content* of information (center-produced models of consumption and

development). Audience shares across domestic outlets cannot capture that; it requires content data and a global W^I (Chapter 16).

Second: assortative mixing in W^I may or may not align with production inequality. The groups with the lowest per-capita production power (rural indigenous women, Chapter 18) have the most distinct media diet, centered on word of mouth and community radio. But on the illustrative table, their exposure still overlaps substantially with other groups ($\lambda_{\text{ethnicity}} = 0.72$). Whether W^I reinforces the production hierarchy is therefore an empirical question for network and reference-group data, not something audience shares settle. If the alignment is strong, creating cross-group information edges that cut across the production structure is the natural high-Ψ intervention.

Third: digital W^I is a structural opportunity and a structural risk. Widespread internet use (about four Bolivians in five by 2024) creates the physical infrastructure for cross-group information edges. But the algorithmic architecture of social media platforms may create new forms of assortative mixing (echo chambers, content personalization) that replicate or intensify the existing W^I structure. Chapter 27 addresses this in detail. For now: the measurement infrastructure developed in this chapter is the prerequisite for monitoring whether digital W^I is raising or lowering integration and concentration over time.

Structuralist insight	**This chapter's measurement**	**Reference**
Cultural dependency (Furtado)	Origin and content of information; needs content data and global W^I (not outlet N_{eff})	Ch. 9
Coloniality of knowledge (Quijano)	Reach of indigenous-language channels (community radio: 0.14) and whether it is believed	Ch. 8
Formation as deeper constraint	Ψ higher where W^I more assortative	Ch. 11
Information as policy lever	Dark entries $\rightarrow$ targeted operators	Ch. 15

Further Reading

Social networks and information: Banerjee et al. (2013), "The Diffusion of Microfinance." Jackson (2008), *Social and Economic Networks*, Chapters 7–8. Banerjee et al. (2019), "Using Gossips to Spread Information: Theory and Evidence from Two Randomized Controlled Trials."

Media and beliefs: DellaVigna and Kaplan (2007), "The Fox News Effect." Gentzkow and Shapiro (2011), "Ideological Segregation Online and Offline." Bursztyn and Cantoni (2016), "A Tear in the Iron Curtain: The Impact of Western Television on Consumption Behavior."

Reference groups and aspirations: Ferrer-i Carbonell (2005), "Income and Well-Being." Luttmer (2005), "Neighbors as Negatives." Genicot and Ray (2017), "Aspirations and Inequality."

Measuring networks and identifying influence: Manski (1993), "Identification of Endogenous Social Effects: The Reflection Problem." Bramoullé, Djebbari, and Fortin (2009), "Identification of Peer Effects through Social Networks." Shalizi and Thomas (2011), "Homophily and Contagion Are Generically Confounded in Observational Social Network Studies." Chandrasekhar and Lewis (2016), "Econometrics of Sampled Networks." Breza, Chandrasekhar, McCormick, and Pan (2020), "Using Aggregated Relational Data to Feasibly Identify Network Structure without Network Data." Comola and Fafchamps (2014), "Testing Unilateral and Bilateral Link Formation." Griffith (2022), "Name Your Friends, but Only Five? The Importance of Censoring in Peer Effects Estimates Using Social Network Data."

Media in Latin America: Boas (2005), "Television and Neopopulism in Latin America." Reporters Without Borders, *World Press Freedom Index* (annual; press freedom, not audience concentration).

Chapter 22

Measuring the Utility Network

Who Cares About Whom, and How Much

> *"[La solidaridad] no es, pues, un sentimiento superficial por los males de tantas personas, cercanas o lejanas. Al contrario, es la determinación firme y perseverante de empeñarse por el bien común."*
>
> ["[Solidarity] is not, then, a superficial feeling about the misfortunes of so many people, near or far. On the contrary, it is the firm and persevering determination to commit oneself to the common good."]
>
> — John Paul II, *Sollicitudo Rei Socialis*, §38 (1987)

Chapter 20 measured W^P from the SAM. Chapter 21 measured W^I from media, surveys, and digital data. This chapter completes the measurement trilogy by tackling W^U: the utility network that records who cares about whom.

W^U is the most elusive of the three networks. Trade flows (W^P) are recorded in customs data. Information flows (W^I) leave media and digital traces. But caring — the weight agent i places on agent j's welfare in her utility function — is an internal state. It manifests only through behavior: private transfers, reciprocal exchange, volunteering, collective action, voting patterns, and survey responses about solidarity.

The chapter develops five measurement strategies, applies them to Bolivia, and shows how W^U interacts with W^P and W^I to determine the composite power vector and the welfare loss decomposition.

22.1 What W^U Measures

Definition 22.1 (The Utility Network — Recall)**.** *From Chapter 5: agent i's utility is $U_i(\mathbf{x}) = u_i(x_i) + \sum_{j \neq i} W^U_{ij} u_j(x_j)$. Entry $W^U_{ij} \in [0, 1]$ is the weight agent i places on*

agent j's utility. When $W^U_{ij} = 1$ for all $j \neq i$: i weights everyone's utility as her own (utilitarian). Equal weights below one (e.g. $1/N$) are impartial but weaker concern. When $W^U_{ij} = 0$: i is indifferent to j's welfare. When W^U_{ij} is large for a small set of j's (family, community): i cares intensely but narrowly. Raw W^U need not be row-stochastic (Remark 5.3). Perron-based statistics (π^U, N_{eff}, Γ) use the row-normalized $D^{-1}W^U$, which records the direction *of concern; the row sums record its* intensity.

Remark 22.2 (W^U Determines Externality Centrality). Following Definition 2.2, agent j's externality centrality $c_j = \sum_i W^U_{ij}$ measures how much the rest of society cares about j's welfare. When c_j is high: j's welfare matters to many people. When c_j is low: j is socially invisible — her suffering does not register in others' utility.

The welfare loss decomposition (Theorem 7.8; Chapter 2, Proposition 2.9) includes $L_{\text{externality}} \approx \frac{N(1-\kappa m)^2}{2\kappa}\gamma_U^2 \operatorname{Var}(c_j)$, with $m = \Omega/N$, to second order in γ_U: when externality centrality is unequal (some people are cared about more than others), there is a welfare loss from the mismatch between who needs resources and who receives social support. (Chapter 18's $L_{\text{formation}}$ is a different object: the additional loss from formed preferences, not from unequal c_j.)

22.2 Five Measurement Strategies

22.2.1 Strategy 1: Private Transfers

Definition 22.3 (Transfer-Revealed W^U)**.** *Private transfers (remittances, gifts, informal loans, in-kind support) reveal who cares about whom. Let τ_{ij} be the transfer from agent i to agent j. The transfer-revealed utility weight:*

$$\hat{W}^U_{ij} \propto \frac{\tau_{ij}}{\sum_k \tau_{ik}}. \tag{22.1}$$

Under pure altruism, the transfer from i to j reflects i's concern for j, conditional on j's need and i's capacity. Transfer shares therefore confound concern with need. They also confound altruism with exchange and insurance motives (Appendix 22A).

Example 22.4 (A Transfer Network for Bolivia [ILLUSTRATIVE])**.** *Bolivia's Encuesta de Hogares records transfers* received*, by type of source (other households in the country, remittances from abroad). It does not record the recipient group of transfers sent, so the matrix below cannot be read off it: it is the kind of matrix a dyadic transfer module would produce. Rows give each group's domestic transfers by recipient group (shares).*

From ↓ / To →	*Urban NI*	*Urban Indig*	*Rural NI*	*Rural Indig*
Urban NI	*0.750*	*0.133*	*0.083*	*0.033*
Urban Indig	*0.125*	*0.500*	*0.062*	*0.312*
Rural NI	*0.113*	*0.042*	*0.704*	*0.141*
Rural Indig	*0.038*	*0.187*	*0.088*	*0.688*

Key patterns:

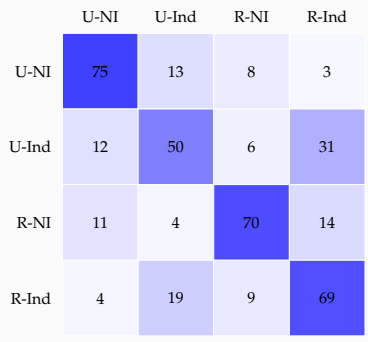
1. *Strong within-group transfers: own-group shares range from 0.50 to 0.75.*

2. *Cross-ethnic transfers are low. Urban NI → Rural Indigenous is 0.033. Overall,* $\lambda^{U}_{ethnicity} = 0.20$ *(mean cross-ethnic share over mean within-ethnic share), and* $\lambda^{U}_{geography} = 0.28$.

3. *Urban indigenous households send 31% of their transfers to rural indigenous households: kinship networks that span the urban–rural divide.*

The group-level matrix has $\Gamma = 0.30$ *and* $N_{eff} = 3.85$ *of 4. Transfers are strongly assortative, slowing integration, but the direction of concern is not concentrated on any one group. Remittances are a separate, exogenous inflow: Bolivia is a net* recipient *of remittances (US$1.34 billion, 2.7% of GDP, in 2019; World Bank, 2026b), so they add to, rather than drain, households' resources. (Numbers:* `code/ch22.py`.)

Visualization: Transfer shares among four groups



$\Gamma = 0.30$, $N_{\text{eff}} = 3.85$ of 4

The illustrative transfer-share matrix of Example 22.4, with stronger blue for larger shares; cell labels are shares in percent. The bright diagonal is within-group solidarity. The off-diagonal cell linking urban and rural indigenous households (31%) is the one strong cross-block link; cross-ethnic cells are faint.

22.2.2 Strategy 2: Dictator Games and Experimental Measures

Definition 22.5 (Experimentally-Elicited W^U)**.** *Dictator games, public goods games, and trust games elicit W^U directly:*

1. ***Dictator game**: Agent i receives an endowment E and can give any amount x to agent j. The share $s_{ij} = x/E$ is informative about W^U_{ij} but is not equal to it. With log utility, i maximizes $\log(E - x) + W^U_{ij} \log x$, so $s_{ij} = W^U_{ij}/(1 + W^U_{ij})$ and*

$$W^U_{ij} = \frac{s_{ij}}{1 - s_{ij}}.$$

   Other utility functions give other mappings, and giving in the lab also reflects norms and experimenter demand.

2. ***Modified dictator game with group variation**: Vary the identity of the recipient (same ethnicity vs. different; same community vs. different). The difference in giving reveals the assortative structure of W^U.*

Example 22.6 (A Directed Dictator Game [ILLUSTRATIVE])**.** *Hypothetical results of the modified dictator game, of the kind a Bolivian field experiment could produce. No existing study is the source. Cárdenas (2003) worked in rural Colombia. Henrich et al. (2010) included the Tsimane of lowland Bolivia but did not vary the recipient's ethnicity or community. For directed giving to identified network partners, see Leider et al. (2009) and Ligon and Schechter (2012).*

Recipient type	*Mean share given s*	$W^U = s/(1-s)$
Same community, same ethnicity	*0.38*	*0.61*
Same community, different ethnicity	*0.25*	*0.33*
Different community, same ethnicity	*0.22*	*0.28*
Different community, different ethnicity	*0.12*	*0.14*
Anonymous stranger	*0.18*	*0.22*

In this illustration, community membership matters more than ethnicity: giving to a different-ethnicity member of one's own community (0.25) exceeds giving to a same-ethnicity member of another community (0.22). On the W^U scale, the implied mixing parameters are $\lambda^U_{community} \approx 0.41$–$0.46$ and $\lambda^U_{ethnicity} \approx 0.48$–$0.54$; the range comes from which cell is held fixed. (The anonymous stranger may come from the respondent's own community, so that baseline need not be the lowest.) (Numbers: `code/ch22.py`.)

22.2.3 Strategy 3: Voting and Collective Action

Definition 22.7 (Revealed Solidarity from Collective Action)**.** *Voting for redistributive policies is informative about cross-group solidarity: a voter who supports transfers from her group to another group, against her material interest, reveals $W^U_{ij} > 0$ for members of the recipient group. Votes, however, bundle many motives (material interest, partisanship, regional identity, views on the other provisions on the ballot), and group-level vote shares must be inferred from aggregate returns by ecological inference. Let v_{gp} be the vote share of group g for policy p that transfers from g to g':*

$$\hat{W}^U_{gg'} \propto v_{gp(\text{transfer to } g')}. \tag{22.2}$$

Example 22.8 (Bolivia's 2009 Constitutional Referendum)**.** *The January 2009 referendum approved the new constitution, which includes provisions for indigenous autonomy, with about 61% of the national vote. A separate question set a maximum size for landholdings. The constitution was rejected in the eastern lowland departments, Santa Cruz among them.*

Official results are reported by territory, not by ethnicity or class. Group-level approval rates would have to be estimated by ecological inference from municipal returns and census composition. Even then, they would not isolate W^U. Low approval in a group mixes material interest, opposition to other provisions (notably on regional autonomy), and partisanship. High approval among indigenous voters mixes self-interest and solidarity. A credible voting-based measure of W^U needs variation in which group benefits from a measure, holding the voter's own stake fixed.

22.2.4 Strategy 4: Survey-Elicited Social Distance

Definition 22.9 (Social Distance Scale)**.** *The Bogardus Social Distance Scale and its variants ask respondents about their willingness to interact with members of other groups: "Would you accept [group X] as: a neighbor / a coworker / a friend / a family member?" Each acceptance level implies a higher W^U_{ig}.*

More specific surveys ask: "How concerned are you about the welfare of [group X]?" on a Likert scale. The response directly estimates W^U_{ig}.

Example 22.10 (Group-Specific Trust [ILLUSTRATIVE])**.** *Hypothetical responses to a group-specific trust module, "How much do you trust people from [group]?" (1–4 scale). Standard regional surveys such as Latinobarómetro ask about generalized interpersonal trust, so a module like this would have to be added.*

Respondent group	*Own group*	*Other ethnic*	*Other class*	*Foreigners*
Indigenous	*3.2*	*2.4*	*1.8*	*1.5*
Mestizo	*3.0*	*2.6*	*2.2*	*2.0*
Upper class	*3.4*	*2.1*	*1.5*	*2.5*

The illustrative pattern: high within-group trust (diagonal), moderate cross-ethnic trust (off-diagonal), low cross-class trust. In this hypothetical table, upper-class respondents trust foreigners more than they trust other classes at home. If real data showed this, it would be Sunkel's transnational integration with national disintegration operating on W^U: social bonds that connect outward more than across the domestic class structure.

22.2.5 Strategy 5: Fiscal Revealed Preference

Definition 22.11 (Fiscal W^U)**.** *The tax-and-transfer system reveals the polity's aggregate allocation rule. Taxes paid by group g that finance transfers to group g' are, at most, a revealed social-welfare weight in the inverse-optimal-tax sense (e.g., Bourguignon and Spadaro, 2012). They are not any individual's W^U: they reflect political bargaining, institutions and who controls the fiscal process. With that caveat, define:*

$$W^U_{fiscal,gg'} = \frac{\textit{net transfer from } g \textit{ to } g'}{\textit{total fiscal capacity of } g}. \tag{22.3}$$

Example 22.12 (Bolivia's Fiscal Channel [ILLUSTRATIVE])**.** *Orders of magnitude, to be checked against MEFP data.*

- *The direct hydrocarbons tax (IDH) raises on the order of 8 billion Bs, paid by the gas sector (and ultimately by its export customers).*
- *The Bono Juancito Pinto (a school-attendance transfer per pupil) distributes about 0.5 billion Bs.*
- *The Renta Dignidad (a universal non-contributory pension for residents aged 60 and over) distributes about 3.5 billion Bs.*

Illustrative fiscal incidence of gas revenue: about 12% reaches poor households through transfers, about 8% reaches non-poor households (public salaries, urban infrastructure), and about 25% is reinvested in the sector. These are shares of a sector's tax revenue, not utility weights of the gas sector: a sector has no utility.

The fiscal channel is less assortative than the private channel: the state redistributes across groups more than households do. It is probably also larger in aggregate. Remittances received from abroad, the bulk of measured private transfers, were US$1.336 billion

in 2019, about 9.3 billion Bs at 6.96 Bs per dollar, or 2.7% of GDP, down from a peak of 8.0% in 2007 (WDI series BX.TRF.PWKR.CD.DT and BX.TRF.PWKR.DT.GD.ZS; World Bank, 2026b). Government transfers are of the order of 15 billion Bs, a figure not yet checked against MEFP data. If that order of magnitude holds, the more redistributive channel is the bigger one, but the private channel is where cross-group concern, or its absence, shows directly.

22.3 Assembling the Composite W^U

Definition 22.13 (Composite Utility Network)**.**

$$W^U = \omega_\tau W^U_{transfer} + \omega_v W^U_{voting} + \omega_s W^U_{survey} + \omega_f W^U_{fiscal}, \tag{22.4}$$

where the weights reflect the relative importance of each channel in determining who actually receives social support. As for W^I (Chapter 21), the composite exists only if every layer is a row-stochastic matrix on the same *node set. The five strategies above produce matrices on different node sets: four groups for transfers, territorial units for votes, survey groups, fiscal categories. They must first be mapped to a common set of groups.*

Remark 22.14 (Comparing networks). It is tempting to rank the three networks by N_{eff} and conclude that "caring is the most concentrated network". But N_{eff} lies in $[1, n]$, so counts on node sets of different size (four groups, five outlets, dozens of sectors) cannot be ranked. The comparable statistic is N_{eff}/n on each layer's own node set, or N_{eff} on a common node set. On this chapter's illustrative transfer matrix, $N_{\text{eff}}/n = 0.96$. The utility layer is strongly *assortative* ($\lambda^U_{\text{ethnicity}} = 0.20$, $\Gamma = 0.30$), but the direction of concern is not concentrated: integration and concentration are different coordinates (Proposition 6.16).

22.4 The Alignment Problem

When the same groups are marginal in production, information and solidarity, disadvantage compounds. How much does alignment concentrate the composite Perron vector? The answer is exact in one important case.

Proposition 22.15 (Alignment and Composite Concentration)**.** *Let $W = \sum_k \alpha_k W^k$ be a composite of row-stochastic layers on a common node set, with $\alpha_k \geq 0$ and $\sum_k \alpha_k = 1$.*

(a) ***Shared Perron vector.*** *If all layers have the same stationary vector π and W is irreducible (for instance, some layer with $\alpha_k > 0$ is irreducible), the composite has Perron vector π.*

(b) ***Rank-one layers.*** *If* $W^k = \mathbf{1}\pi^{k\prime}$*, then* $\pi = \sum_k \alpha_k \pi^k$*, and*

$$\mathrm{Var}(\pi) = \sum_{k,l} \alpha_k \alpha_l \,\mathrm{Cov}(\pi^k, \pi^l) \;\le\; \Big(\sum_k \alpha_k \sqrt{\mathrm{Var}(\pi^k)} \Big)^2 \;\le\; \max_k \mathrm{Var}(\pi^k).$$

Here Var *and* Cov *are taken across the* N *nodes. For given layer variances,* $\mathrm{Var}(\pi)$ *is nondecreasing in every pairwise covariance (strictly when* $\alpha_k \alpha_l > 0$*). It is largest under perfect alignment (all correlations one), where the first bound holds with equality, and it falls as correlations become negative; with two layers it is smallest at correlation* -1*, where* $\mathrm{Var}(\pi) = (\alpha_1 \sigma_1 - \alpha_2 \sigma_2)^2$ *with* $\sigma_k^2 = \mathrm{Var}(\pi^k)$*. It never exceeds the most concentrated layer's variance.*

Proof. (a) $\pi' W = \sum_k \alpha_k \pi' W^k = \pi'$, and the stationary vector of an irreducible W is unique (Seneta, 2006, Thm. 1.5). (b) Let $\bar{\pi} = \sum_k \alpha_k \pi^k$, which sums to one. Then $W = \mathbf{1}\bar{\pi}'$, and for each l, $\bar{\pi}' \mathbf{1} \pi^{l\prime} = \pi^{l\prime}$, so

$$\bar{\pi}' W = \sum_l \alpha_l \pi^{l\prime} = \bar{\pi}'.$$

It is the only stationary vector: $x' \mathbf{1} \bar{\pi}' = x'$ with $x' \mathbf{1} = 1$ forces $x = \bar{\pi}$. Every π^k has mean $1/N$, so $\bar{\pi} - \mathbf{1}/N = \sum_k \alpha_k (\pi^k - \mathbf{1}/N)$, and expanding the square gives the variance identity. The first inequality is Cauchy–Schwarz, $\mathrm{Cov}(\pi^k, \pi^l) \le \sigma_k \sigma_l$, applied term by term with $\alpha_k \alpha_l \ge 0$; it gives $\sum_{k,l} \alpha_k \alpha_l \sigma_k \sigma_l = (\sum_k \alpha_k \sigma_k)^2$. The second holds because a weighted mean of standard deviations is at most their maximum. The monotonicity statements follow because each covariance enters with coefficient $\alpha_k \alpha_l \ge 0$; with two layers, $\mathrm{Cov} = \sigma_1 \sigma_2 \,\mathrm{corr}$ and $\mathrm{corr} \ge -1$. □

With rank-one layers, alignment therefore makes compound disadvantage as large as it can be, but no larger than the worst single layer. In particular, an aligned rank-one composite is *not* more concentrated than every layer. For layers that are not rank-one there is no closed form, and the bound can fail. In `code/ch22.py`, two irreducible 3×3 layers with Perron vectors $(0.45, 0.10, 0.45)$ and $(0.49, 0.36, 0.16)$ have an equal-weight composite with Perron vector $(0.64, 0.21, 0.15)$, whose variance is 1.7 times the larger layer variance. In other examples the bound holds: three aligned layers with Perron vectors $(0.4, 0.3, 0.2, 0.1)$, $(0.30, 0.27, 0.23, 0.20)$ and $(0.28, 0.26, 0.24, 0.22)$ give a composite with $\mathrm{Var}(\pi) = 0.0028$, against 0.0125 for the most concentrated layer. The perturbation lemma (Lemma 6.19) gives the local effect of adding one layer to another.

Example 22.16 (Bolivia's Network Alignment [ILLUSTRATIVE])**.** *Four groups (urban non-indigenous, urban indigenous, rural non-indigenous, rural indigenous), with*

population shares $(0.35, 0.25, 0.15, 0.25)$. *Rank-one layers with equal weights. Per-capita positions* $r_g = \pi_g / p_g$ *(illustrative values; 1 is the population average) for rural indigenous households:*

Network	*Relative position*	$r_{\text{rural indig.}}$
W^P	*Low-linkage sectors*	*0.46*
W^I	*Assortative, word-of-mouth*	*0.35*
W^U	*Within-community only*	*0.30*
Composite	*All aligned*	*0.37*

By part (b) of Proposition 22.15, the composite position is the weighted average of the layers' positions: it lies between *them, not below all of them. Alignment shows up in the correlations of the group-level Perron vectors:* 0.96 *(P, U),* 0.98 *(P, I) and* 0.99 *(I, U).*

Now suppose the utility layer were anti-aligned on this group, with rural indigenous households at $r^U = 1.00$ *because their strong solidarity networks make them central in* W^U*. Their composite position would rise from* 0.37 *to* 0.60*. Cross-network compensation of this kind is the mathematical form of "compensating institutions". Whether Bolivia's networks are aligned is an empirical question; the colonial racial hierarchy (Quijano, Chapter 18) gives reasons to expect alignment, and the ayllu (below) reasons to expect compensation. (Numbers:* `code/ch22.py`*.)*

Key Structural Insight

The three networks are analytically separable (W^P, W^I, W^U) and plausibly correlated in practice: the same historical processes (colonialism, racial classification, patriarchy, geographic isolation) shaped all three. The structuralist hypothesis is not "this economy has a bad W^P" or "a bad W^I" or "a bad W^U", but "the same structural forces produced block-diagonal structure in all three, aligned on the same axis". If that is right, interventions must address the alignment, not just one network in isolation.

This is why Chapters 20–22 are a trilogy: measuring each network separately is necessary but not sufficient. The structural diagnosis requires measuring all three *on a common node set* and computing the composite — because the welfare loss (Chapter 2) depends on the composite W, not on any single layer.

22.5 Summary

1. W^U measures who cares about whom. Five sources are informative about it: **private transfers**, **experiments** (directed dictator games, with $W = s/(1-s)$ under log altruism), **voting**, **surveys** (trust, social distance) and **fiscal data**. Each confounds concern with something else: need, exchange and insurance motives, norms, partisanship, or political bargaining.

2. From transfer data, the altruism FOC identifies $W^U_{ij} = (c_j/c_i)^\sigma$ for pairs with positive transfers, and only a bound for pairs without. Separating altruism from exchange and insurance needs dyadic data and exclusion restrictions.

3. Private W^U is **highly assortative** in the chapter's illustrative transfer matrix ($\lambda^U_{\text{ethnicity}} = 0.20$, $\Gamma = 0.30$) without being concentrated ($N_{\text{eff}}/n = 0.96$). N_{eff} can be compared across networks only on a common node set or as N_{eff}/n. The fiscal channel is less assortative and probably larger in aggregate (remittances were 9.3 billion Bs in 2019; government transfers are not yet sourced).

4. **Alignment**: with rank-one layers, the composite Perron vector is the weighted average of the layers', so its variance rises with the layers' covariances but never exceeds the most concentrated layer's. Alignment maximizes compound disadvantage, and anti-alignment ("compensating institutions") reduces it.

5. The welfare loss depends on the **composite** W, which exists only when all layers are measured on a common node set. Andean institutions such as the ayllu show that solidarity networks can be both dense within communities and federated across them, so their place in the composite is an empirical question.

22.6 Exercises

22.1 Transfer-revealed W^U. Using Bolivia's EH: identify all reported transfers *received* (from other households, remittances, in-kind) by recipient group (urban NI, urban indigenous, rural NI, rural indigenous). Explain why these data do not identify the sender group, and what a dyadic module would add. Then, for the illustrative matrix of Example 22.4, compute Γ, $\boldsymbol{\pi}$, N_{eff} and $\lambda^U_{\text{ethnicity}}$. (Answers: `code/ch22.py`.)

22.2 Experimental design. Design a modified dictator game for a Bolivian community that estimates W_{ij}^U varying by: (a) ethnicity, (b) geography, (c) gender. How many treatments do you need? What sample size?

22.3 Fiscal W^U. Using MEFP budget data: trace the flow of the IDH (gas tax) through the fiscal system. What fraction reaches each household group? Construct W_{fiscal}^U, and explain why it is a revealed allocation rule of the polity rather than anyone's utility weights.

22.4 The alignment test. For Bolivia: map W^P (Ch. 20 SAM), W^I (Ch. 21) and W^U (this chapter) to a common set of four household groups. Compute $\boldsymbol{\pi}^P$, $\boldsymbol{\pi}^I$ and $\boldsymbol{\pi}^U$ and the correlations between them. Is the correlation positive (aligned) or negative (compensating)?

22.5 Misalignment as policy. Suppose a policy strengthens W^U for indigenous communities (solidarity programs, mutual aid networks) without changing W^P or W^I. This creates misalignment. Simulate: compute the composite $\boldsymbol{\pi}$ before and after. Does compound disadvantage decrease?

22.6 Cross-country comparison. Using Latinobarómetro trust data for Bolivia, Chile, and Uruguay: estimate a survey-based W^U for each, on the same set of groups. Which country has the highest N_{eff}^U/n, and the highest Γ? Do they correlate with income inequality?

22.7 The complete measurement. Combine Chapters 20–22: for Bolivia, you now have W^P (SAM), W^I (media + surveys), W^U (transfers + voting + surveys). Map all three to a common node set (household groups), and compute the composite $W = \alpha_U W^U + \alpha_P W^P + \alpha_I W^I$ with weights from Chapter 5. Compute the composite Γ, $\boldsymbol{\pi}$, and the welfare loss decomposition $L = L_{\text{power}} + L_{\text{externality}} + L_{\text{interaction}}$ (Theorem 7.8). What fraction of Bolivia's welfare loss comes from each component?

Mathematical Appendices

Appendix 22A: Inferring W^U from Transfer Data

The structural model of transfers. Agent i with income y_i maximizes $u(c_i) + \sum_j W^U_{ij} u(c_j)$ subject to $c_i + \sum_j \tau_{ij} = y_i$ and $c_j = y_j + \sum_k \tau_{kj} - \sum_k \tau_{jk}$. The first-order condition for the transfer $\tau^*_{ij} \geq 0$:

$$u'(c^*_i) \geq W^U_{ij}\, u'(c^*_j), \quad \text{with equality if } \tau^*_{ij} > 0. \tag{22.5}$$

With CRRA utility $u(c) = c^{1-\sigma}/(1-\sigma)$, $u'(c) = c^{-\sigma}$, so for pairs with positive transfers

$$W^U_{ij} = \left(\frac{c_j}{c_i}\right)^{\sigma},$$

which is below one when the giver is better off ($c_i > c_j$). For example, $c_i = 3$, $c_j = 1$, $\sigma = 2$ gives $W^U_{ij} = 1/9$; the inverted formula $(c_i/c_j)^\sigma$ would give 9. The condition identifies W^U_{ij} from consumption levels alone, given σ. For pairs with $\tau_{ij} = 0$ it gives only the bound $W^U_{ij} \leq (c_j/c_i)^\sigma$.

Identification. Three problems.

1. **Corner solutions.** $\tau_{ij} = 0$ does not mean $W^U_{ij} = 0$: the transfer cost may exceed the gain. A Tobit or Heckman selection model separates the extensive margin (whether to transfer) from the intensive margin (how much), but it needs an exclusion restriction, i.e. a variable that shifts the cost of transferring without shifting concern.

2. **Other motives.** Transfers are also driven by exchange (payment for services; Cox, 1987), by informal insurance under limited commitment (Ligon et al., 2002), and by mixtures of altruism and imperfect commitment within families (Foster and Rosenzweig, 2001). These motives predict different responses of transfers to the recipient's income, which is how the literature tells them apart.

3. **Dyadic data.** Transfer networks are dyadic. Estimation should use dyadic regressions with dyad-robust standard errors (Fafchamps and Gubert, 2007) and, where possible, directed games with real network partners (Leider et al., 2009; Ligon and Schechter, 2012).

Appendix 22B: Network Alignment and Composite Eigenvalues

The composite weight matrix. $W = \alpha_U W^U + \alpha_P W^P + \alpha_I W^I$ with $\alpha_U + \alpha_P + \alpha_I = 1$. All layers must be row-stochastic on a common node set. The map $W \mapsto \pi(W)$ is nonlinear, so the Perron vector of W is not generally the weighted average of the layers' Perron vectors. Two exceptions are proved in Proposition 22.15: a shared Perron vector, and rank-one layers.

Alignment as eigenvector correlation. Define the alignment coefficient: $A = \mathrm{Corr}(\boldsymbol{\pi}^P, \boldsymbol{\pi}^U)$. When $A \approx 1$ (aligned): the same agents are central in both networks. For rank-one layers (taking two of them),

$$\mathrm{Var}(\pi) = \alpha_P^2 \,\mathrm{Var}(\pi^P) + \alpha_U^2 \,\mathrm{Var}(\pi^U) + 2\alpha_P \alpha_U \,\mathrm{Cov}(\pi^P, \pi^U)$$

exactly. For given layer variances it is largest under alignment ($A = 1$), and it never exceeds $\max(\mathrm{Var}(\pi^P), \mathrm{Var}(\pi^U))$. Both statements are for rank-one layers; for general layers the composite can be more concentrated than either layer (Section 22.4).

When $A \approx -1$ (anti-aligned): the composite $\mathrm{Var}(\pi)$ is reduced by the negative covariance. Anti-alignment is the mathematical structure of "compensating institutions": if the economically powerful (W^P) are the socially isolated (W^U), the composite disadvantage is smaller than either alone.

Bolivia's alignment coefficients. They can be computed only once the three layers are measured on a common node set; no such estimate exists yet. In the illustrative four-group example (Example 22.16), the correlations are $A(\pi^P, \pi^U) = 0.96$, $A(\pi^P, \pi^I) = 0.98$ and $A(\pi^I, \pi^U) = 0.99$.

Structuralist Case Study: Solidarity, Reciprocity, and the Andean Ayllu

The measurement of W^U in Bolivia cannot be divorced from the Andean institution of the *ayllu*: the community-based organization that has governed land, labor and social relations in the highlands since before the Spanish conquest and continues to do so, in changed forms, today. The ayllu is a living institution of structured solidarity, and it tests the measurement tools of this chapter. We follow John Murra's reconstruction of Andean economic organization, and the historians and anthropologists who extended it.

Dense within, federated across. Several principles organize the ayllu:

- reciprocity (*ayni*: labor or goods given now, to be returned in kind later);

- *mink'a*: labor provided at a household's or the community's request, typically compensated with food, drink or produce (Alberti and Mayer, 1974);
- communal work obligations;
- Murra's "vertical archipelago" (1972): a community's control of multiple ecological floors, often far apart.

These generate dense within-community solidarity: in this chapter's notation, a bright diagonal block of W^U. But the ayllu was never simply a closed block. Verticality linked distant ecological zones. Ayllus were nested in larger units (*markas* and the moieties above them), and today they are federated: the Consejo Nacional de Ayllus y Markas del Qullasuyu (CONAMAQ) and the Katarista-influenced peasant confederation CSUTCB organize across communities and regions. The mobilizations of 2000–2005 drew heavily on such community and union federations. In matrix terms, the ayllu combines strong within-block weights with *institutionalized* cross-block links. It is a counterexample to the presumption that strong local solidarity must mean fragmentation.

Colonial disruption. Historians document how the colonial order reorganized this structure. The Toledan *reducciones* resettled dispersed populations into new towns. The *mita* and tribute were levied through community authorities, mobilizing within-community obligations for extraction (Stern, 1982; Platt, 1982). In network terms, colonial rule cut some cross-block links (the vertical archipelagos) and redirected within-block obligations upward to a colonial hierarchy. How far this made W^U more block-diagonal than before is a historical question we flag rather than settle.

Contemporary measurement. The tools of this chapter could measure the ayllu's solidarity directly: directed dictator games with recipients from inside and outside the community (Example 22.6 shows the design, with illustrative numbers), dyadic transfer modules, and records of communal labor. Their cross-community entries matter as much as the diagonal: a federated solidarity network can have both a bright diagonal and a high $\Gamma(W^U)$. The formation model offers one account of persistence: solidarity norms ($\theta_{\text{solidarity}}$) are reproduced each generation through community information networks, with half-life $\ln 2 / \ln(1/\rho)$ generations, about 4 at $\rho = 0.85$.

The policy implication follows. Within-community solidarity is valuable for insurance and resilience. Cross-community links already exist in the ayllu federations, and policy can build on them (inter-community exchange, regional federations, shared infrastructure) rather than treat community solidarity as an obstacle.

And if indigenous communities are central in W^U while marginal in W^P, the networks are partly *anti-aligned* — the compensating case of Proposition 22.15.

Andean concept	**Formal object**	**Measurement**
Ayni (reciprocity)	High W^U_{within}	Dyadic transfer data, directed dictator games
Mink'a (requested labor)	Within-community exchange of labor	Records of labor exchange
Verticality and federation (Murra; CONAMAQ)	Institutionalized cross-block links, higher $\Gamma(W^U)$	Cross-community transfer and labor rates
Colonial reorganization	Cross-block links cut; obligations redirected upward	Historical analysis
Contemporary persistence	$\rho \approx 0.85$, half-life ≈ 4 gen.	Cohort comparison
Compensation	Anti-alignment of W^U with W^P	Prop. 22.15

Further Reading

Social preferences and networks: Fehr and Schmidt (1999), "A Theory of Fairness, Competition, and Cooperation." Charness and Rabin (2002), "Understanding Social Preferences with Simple Tests." Leider et al. (2009), "Directed Altruism and Enforced Reciprocity in Social Networks."

Transfers and informal insurance: Townsend (1994), "Risk and Insurance in Village India." Ligon et al. (2002), "Informal Insurance Arrangements with Limited Commitment." Cox (1987), "Motives for Private Income Transfers." Foster and Rosenzweig (2001), "Imperfect Commitment, Altruism, and the Family: Evidence from Transfer Behavior in Low-Income Rural Areas." Fafchamps and Gubert (2007), "The Formation of Risk Sharing Networks." Ligon and Schechter (2012), "Motives for Sharing in Social Networks." Bourguignon and Spadaro (2012), "Tax–Benefit Revealed Social Preferences."

Andean institutions: Murra (1972), "El control vertical de un máximo de pisos ecológicos en la economía de las sociedades andinas." Alberti and Mayer (1974), *Reciprocidad e intercambio en los Andes peruanos*. Patzi Paco (2004), *Sistema comunal: una propuesta alternativa al sistema liberal*. Platt (1982), *Estado boliviano y ayllu andino*. Stern (1982), *Peru's Indian Peoples and the Challenge of Spanish Conquest*.

Experimental economics in developing countries: Henrich et al. (2010), "Markets, Religion, Community Size, and the Evolution of Fairness and Punishment." Cárdenas (2003), "Real Wealth and Experimental Cooperation."

Chapter 23

Tracking Structural Change Over Time

The Moving Heatmap

> *"El cambio estructural no es un evento sino un proceso: una secuencia de transformaciones en las relaciones entre los componentes del sistema económico."*
>
> ["Structural change is not an event but a process: a sequence of transformations in the relations between the components of the economic system."]
> — CEPAL, *Cambio estructural para la igualdad: una visión integrada del desarrollo* (2012)

Chapters 20–22 showed how to measure W^P, W^I, and W^U at a single point in time. This chapter adds the time dimension: how to track W_t across years, detect structural breaks, measure the speed of change, and distinguish structural change from cyclical fluctuation.

The central object is the **structural trajectory**: the time series of Γ_t, $\sigma_{B,t}$, $G(\boldsymbol{\pi}_t)$ and $N_{\text{eff},t}$ computed from a sequence of SAMs or equivalent data. The trajectory tells you whether the economy is integrating or fragmenting, at what speed, and whether specific events (a nationalization, a trade reform, a crisis) produced detectable structural effects.

Chapter 17 previewed Bolivia's trajectory. This chapter develops the methods and applies them to data. Two conventions are in force throughout (Chapter 1, Section 1.7): Γ measures *integration* (mixing speed) only, and the concentration of power is measured by N_{eff}, $\text{Var}(\boldsymbol{\pi})$ or $G(\boldsymbol{\pi})$, never by Γ (Proposition 6.16).

The production-network trajectories are sourced. For Bolivia they come from INE's symmetric input–output matrices (base 1990, 1988–2014) and supply-use tables (base 2017, 2017–2024) (Instituto Nacional de Estadística (INE), Bolivia, 2026). The two bases differ in classification and national-accounts vintage, so they are kept as two separate series and no change is ever computed across the break. For Korea and the comparators they come from the domestic blocks of the OECD Inter-Country Input-Output tables, 2021 edition for 1995–2014 and 2023 edition for 2011–2020 (OECD, 2021, 2023). Complexity is the Growth Lab's ECI (The Growth Lab at Harvard University, 2026). Throughout, Γ is that of the 12-sector domestic input-share matrix $W^P = D^{-1}A$ with the diagonal kept: the book's 12-sector classification for Bolivia's own series, and an ISIC-based 12-sector classification for comparisons across countries (Bolivia 2019: $\Gamma = 0.467$ and 0.475 respectively). Γ depends on aggregation and normalization. For Bolivia in 2019 it ranges from 0.33 to 0.69 across input and sales shares, with and without the diagonal, at 12 and 39 sectors, so only like-for-like comparisons are meaningful. Between-block inequality σ_B and information concentration $N_{\text{eff}}(W^I)$ need household and audience microdata that were not available: their values remain [ILLUSTRATIVE]. Every sourced number is read from `data/processed` and checked by `code/ch23.py`.

The data do not tell the story usually told about Bolivia, and the chapter follows the data. Bolivia's Γ is not low by comparison; it drifted *down*, and so did Korea's. What separates the two economies is complexity, not aggregate integration.

23.1 The Time-Varying Weight Matrix

23.1.1 Construction

Definition 23.1 (Time-Indexed Weight Matrix)**.** *For each year t, construct W_t from the available data:*

1. *W_t^P: from the I-O table (if available for year t) or interpolated from benchmark I-O tables using annual National Accounts.*
2. *W_t^I: from media audience data (annual surveys) or digital trace data (continuous).*
3. *W_t^U: from household surveys (typically annual or biennial).*

The structural indicators are computed annually: $\Gamma_t = \Gamma(W_t)$, $\boldsymbol{\pi}_t = \boldsymbol{\pi}(W_t)$, etc.

Definition 23.2 (Interpolation of W^P Between Benchmark I-O Tables)**.** *Most countries produce I-O tables only every 5–10 years. Between benchmarks, W^P_t can be approximated by:*

$$W^P_t \approx W^P_{\tau_0} + \frac{t-\tau_0}{\tau_1-\tau_0}(W^P_{\tau_1} - W^P_{\tau_0}), \tag{23.1}$$

where τ_0 and τ_1 are the benchmark years. Because the right-hand side is a convex combination of two row-stochastic matrices, it is itself row-stochastic. Alternatively, use the RAS method on the intermediate-transactions matrix Z_{τ_0} (not on W^P): given the year-t intermediate sales $\mathbf{u}_t$ (row totals of Z_t) and intermediate purchases $\mathbf{v}_t$ (column totals) from annual National Accounts, with $\mathbf{1}'\mathbf{u}_t = \mathbf{1}'\mathbf{v}_t$, rescale Z_{τ_0} biproportionally to these margins and then row-normalize:

$$Z_t = RAS(Z_{\tau_0}; \mathbf{u}_t, \mathbf{v}_t), \qquad W^P_t = D_t^{-1} Z_t, \quad D_t = \operatorname{diag}(Z_t \mathbf{1}). \tag{23.2}$$

RAS preserves the zero pattern (hence the block structure) of Z_{τ_0} while adjusting for changes in sectoral composition. A solution exists only if the margins are consistent and compatible with the zero pattern (Appendix 23.8).

23.1.2 The Structural Trajectory

Definition 23.3 (Structural Trajectory)**.** *The **structural trajectory** of an economy is the time series:*

$$\mathcal{T} = \{(\Gamma_t, \sigma_{B,t}, G(\boldsymbol{\pi}_t), N_{eff,t}, ECI_t) : t = t_0, \ldots, T\}. \tag{23.3}$$

The trajectory is plotted in the five-dimensional structural health space (Chapter 17). Projection onto the $(\Gamma_t, \sigma_{B,t})$ plane gives a useful two-dimensional visualization: integration on one axis, between-block inequality on the other. Power concentration ($G(\boldsymbol{\pi}_t)$, $N_{eff,t}$) is a separate coordinate and must be read separately.

23.1.3 Products of Time-Varying Matrices

The per-period indicators $\Gamma(W_t)$ and $\boldsymbol{\pi}(W_t)$ describe a frozen snapshot. The economy, however, is driven by the *sequence*. In the information reading, beliefs evolve as $\mathbf{b}_{t+1} = W_t \mathbf{b}_t$, so after several periods

$$\mathbf{b}_t = P_{t,s}\, \mathbf{b}_s, \qquad P_{t,s} := W_{t-1} W_{t-2} \cdots W_s \quad (t > s), \tag{23.4}$$

a *backward product* of row-stochastic matrices. What matters for integration over a decade is the mixing of $P_{t,s}$, not of each factor.

Remark 23.4 (Per-period Γ does not control the product). $|\lambda_2|$ is not submultiplicative. Take

$$X = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0.85 & 0 & 0.15 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & 0.95 & 0.05 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Then $\Gamma(X) = 0.850$ and $\Gamma(Y) = 0.776$, but $\Gamma(XY) = 0.042$. A trajectory that alternates X and Y shows a "healthy" per-period Γ_t while the realized two-period dynamics barely mix (realized gap per period $1 - |\lambda_2(XY)|^{1/2} = 0.021$). Numbers from `code/ch23.py`.

The right instrument is the **Dobrushin ergodicity coefficient**

$$\tau_1(W) := \tfrac{1}{2} \max_{i,j} \sum_k |W_{ik} - W_{jk}| \ \in [0, 1], \tag{23.5}$$

the largest total-variation distance between two rows. $\tau_1(W) < 1$ iff every two rows share some column with positive weight ("W is scrambling").

Proposition 23.5 (Properties of τ_1)**.** *For row-stochastic A, B: (i) $\tau_1(AB) \le \tau_1(A)\,\tau_1(B)$; (ii) every eigenvalue $\lambda \ne 1$ of A satisfies $|\lambda| \le \tau_1(A)$, so $\Gamma(A) \ge 1 - \tau_1(A)$.*

Proof. Step 1: τ_1 is an operator norm. Let $\mathcal{Z} := \{\delta \in \mathbb{R}^N : \delta'\mathbf{1} = 0\}$. We show $\|\delta' A\|_1 \le \tau_1(A)\|\delta\|_1$ for all $\delta \in \mathcal{Z}$, with equality for $\delta = \mathbf{e}_i - \mathbf{e}_j$ at the maximizing pair (i, j) in (23.5). If $\delta \in \mathcal{Z} \setminus \{0\}$, write $\delta = c(\mathbf{p} - \mathbf{q})$ with $c = \|\delta\|_1/2$ and $\mathbf{p} = \delta^+/c$, $\mathbf{q} = \delta^-/c$ probability vectors (positive and negative parts have equal mass because $\delta'\mathbf{1} = 0$). Then $\mathbf{p}'A - \mathbf{q}'A = \sum_{i,j} p_i q_j (A_{i\cdot} - A_{j\cdot})$, so $\|\delta' A\|_1 \le c \max_{i,j} \|A_{i\cdot} - A_{j\cdot}\|_1 = 2c\,\tau_1(A) = \tau_1(A)\|\delta\|_1$. Since $A\mathbf{1} = \mathbf{1}$, $\delta' A\mathbf{1} = 0$: $\mathcal{Z}$ is invariant under $\delta' \mapsto \delta' A$.

(i) For $\delta \in \mathcal{Z}$, $\delta' A \in \mathcal{Z}$, so $\|\delta' AB\|_1 \le \tau_1(B)\|\delta' A\|_1 \le \tau_1(B)\tau_1(A)\|\delta\|_1$. Taking $\delta = \mathbf{e}_i - \mathbf{e}_j$ ($\|\delta\|_1 = 2$) gives $\frac{1}{2}\|(AB)_{i\cdot} - (AB)_{j\cdot}\|_1 \le \tau_1(A)\tau_1(B)$ for every pair (i, j).

(ii) Let $\mathbf{u} \in \mathbb{C}^N \setminus \{0\}$ satisfy $\mathbf{u}' A = \lambda \mathbf{u}'$ with $\lambda \ne 1$. Then $\lambda\,\mathbf{u}'\mathbf{1} = \mathbf{u}' A\mathbf{1} = \mathbf{u}'\mathbf{1}$, so $\mathbf{u}'\mathbf{1} = 0$. Write $\mathbf{u} = \mathbf{x} + i\mathbf{y}$ with $\mathbf{x}, \mathbf{y} \in \mathcal{Z}$ real. Since A is real, $\mathbf{x}' A^n = \mathrm{Re}(\lambda^n \mathbf{u}')$ and $\mathbf{y}' A^n = \mathrm{Im}(\lambda^n \mathbf{u}')$, and by (i) applied n times, $\|\mathbf{x}' A^n\|_1 \le \tau_1(A)^n \|\mathbf{x}\|_1$, and likewise for $\mathbf{y}$. Hence

$$|\lambda|^n \|\mathbf{u}\|_1 = \|\lambda^n \mathbf{u}'\|_1 \le \|\mathbf{x}' A^n\|_1 + \|\mathbf{y}' A^n\|_1 \le \tau_1(A)^n (\|\mathbf{x}\|_1 + \|\mathbf{y}\|_1) \le 2\,\tau_1(A)^n \|\mathbf{u}\|_1,$$

so $|\lambda| \le 2^{1/n} \tau_1(A)$ for every n; let $n \to \infty$. (For real λ the case $n = 1$ already suffices.) For the bound on Γ: if 1 is a simple eigenvalue, $|\lambda_2| \le \tau_1(A)$ by what was just shown. If 1 is a repeated eigenvalue then $\Gamma(A) = 0$; since A^n is bounded,

1 is semisimple, so there are two independent left eigenvectors for 1, and a combination $\mathbf{u} \in \mathcal{Z} \setminus \{0\}$ of them satisfies $\mathbf{u}'A = \mathbf{u}'$; Step 1 then forces $\tau_1(A) \geq 1$, i.e. $\tau_1(A) = 1$, and the bound holds with equality. These are the standard properties of the coefficient of ergodicity (Seneta, 2006, Sect. 4.3); `code/ch23.py` checks (i), (ii) and the norm identity of Step 1 on random matrices. □

In the counterexample, $\tau_1(X) = \tau_1(Y) = \tau_1(XY) = 1$: neither matrix is scrambling, which flags the problem that Γ_t hides.

Definition 23.6 (Weak and Strong Ergodicity)**.** *The sequence (W_t) is **weakly ergodic** if $\tau_1(P_{t,s}) \to 0$ as $t \to \infty$ for every s: the rows of the product become equal, so the system forgets its initial condition and reaches consensus. It is **strongly ergodic** if, in addition, $P_{t,s} \to \mathbf{1}\mathbf{v}_s'$ for some probability vector $\mathbf{v}_s$.*

Proposition 23.7 (A Sufficient Condition for Weak Ergodicity)**.** *If $\sum_t \big(1 - \tau_1(W_t)\big) = \infty$, then (W_t) is weakly ergodic.*

Proof. Fix s. By Proposition 23.5(i), applied $t - s - 1$ times, $\tau_1(P_{t,s}) \leq \prod_{r=s}^{t-1} \tau_1(W_r) \leq \exp\big(-\sum_{r=s}^{t-1}(1 - \tau_1(W_r))\big)$, using $x \leq e^{-(1-x)}$ for $x \in [0,1]$ (convexity of the exponential: $e^z \geq 1 + z$ with $z = x - 1$). Removing the finitely many terms $r < s$ from a divergent series of nonnegative terms leaves it divergent, so the exponent tends to $-\infty$ and $\tau_1(P_{t,s}) \to 0$. This is Hajnal's condition (Hajnal, 1958); see Seneta (2006, Sect. 4.4). □

For backward products such as (23.4), weak and strong ergodicity coincide (Chatterjee and Seneta, 1977; Seneta, 2006, ch. 4), and the limit vector has a direct power interpretation. An **absolute probability sequence** is a sequence of probability vectors $(\boldsymbol{\pi}_t)$ with $\boldsymbol{\pi}_{t+1}'W_t = \boldsymbol{\pi}_t'$ (Kolmogoroff, 1936; Touri and Nedić, 2014). Such a sequence always exists, and it makes $\boldsymbol{\pi}_t'\mathbf{b}_t$ invariant, since $\boldsymbol{\pi}_{t+1}'\mathbf{b}_{t+1} = \boldsymbol{\pi}_{t+1}'W_t\mathbf{b}_t = \boldsymbol{\pi}_t'\mathbf{b}_t$. Under weak ergodicity the eventual consensus is therefore $\boldsymbol{\pi}_s'\mathbf{b}_s$: $\boldsymbol{\pi}_s$, *not* $\boldsymbol{\pi}(W_s)$, *is the time-varying vector of epistemic power.* When W_t changes slowly, $\boldsymbol{\pi}_t \approx \boldsymbol{\pi}(W_t)$, with an error controlled by the perturbation lemma (Lemma 6.19).

Key Structural Insight

Report three integration diagnostics, not one: the per-period gap $\Gamma(W_t)$; the Dobrushin coefficient $\tau_1(W_t)$, which composes over time; and the *realized-product gap* over a window of k years, $\Gamma_t^{(k)} := 1 - |\lambda_2(W_{t+k-1} \cdots W_t)|^{1/k}$. When they disagree, the realized-product gap is the one that describes what actually happened.

23.2 Detecting Structural Breaks

Definition 23.8 (Structural Break)**.** *A **structural break** at time t^* is a discontinuity in the trajectory $\mathcal{T}$: a change in Γ, σ_B, or $\boldsymbol{\pi}$ that is larger than can be explained by normal cyclical variation. Formally, a break at t^* if:*

$$|\Gamma_{t^*+1} - \Gamma_{t^*}| > k \cdot \hat{\sigma}_{\Delta\Gamma}, \tag{23.6}$$

where $\hat{\sigma}_{\Delta\Gamma}$ is the standard deviation of year-to-year changes in Γ during a stable period, and k is the detection threshold (typically $k = 2$ or $k = 3$).

Remark 23.9 (Breaks in $\boldsymbol{\pi}$ need error bands). The Perron vector is badly conditioned exactly when Γ is small. By the perturbation lemma (Lemma 6.19), $d\boldsymbol{\pi}' = \boldsymbol{\pi}' E\, Z$ with $Z = (I - W + \mathbf{1}\boldsymbol{\pi}')^{-1}$, and Z is large when λ_2 is close to $+1$. In the two-block matrix of `code/ch23.py`, moving weight δ from a within-block to a cross-block entry of one row changes block 1's power at rate -1.28 when $\Gamma = 0.13$, against -0.37 when $\Gamma = 0.45$: about 3.5 times more sensitive. In a weakly coupled dual economy with $\Gamma = 0.13$, measurement error in W_t is strongly amplified in $\boldsymbol{\pi}_t$. Bolivia's measured Γ (0.47–0.57 at 12 sectors) is in the second regime, so amplification through Z is moderate there. The larger source of spurious breaks in Bolivia's series is construction: the change of base, years without a published make matrix, and inconsistencies in the published tables (Example 23.11). Before declaring a break in Γ or $\boldsymbol{\pi}$, bootstrap the underlying data (resample survey households, perturb I-O cells within their revision ranges), recompute Γ_t and $\boldsymbol{\pi}_t$, and require the change to exceed the bootstrap band as well as $k\hat{\sigma}$.

Definition 23.10 (Types of Structural Breaks)**.** *Three types of structural breaks correspond to three types of ΔW:*

1. ***Edge creation break**: new edges appear (a trade agreement, infrastructure project, market opening). New* cross-block *edges in a weakly coupled structure raise Γ (they raise the conductance, which bounds Γ from below for reversible W); new* within-block *edges can lower it. Example: South Korea's export push (1960s).*

2. ***Edge destruction break**: edges are severed (a crisis, sanctions, nationalization). Γ may decrease or increase depending on which edges are destroyed (cross-block destruction lowers Γ; within-block destruction may increase it). Example: Soviet collapse (1991).*

3. ***Reweighting break**: existing edge weights shift (a terms-of-trade shock, a technological change that alters input shares). Γ changes smoothly rather than discretely.*

Example: the commodity supercycle (2003–2014).

Whether a break is structural *or* cyclical *is a separate question (Definition 23.15): any of the three types is structural if it persists and cyclical if it reverses.*

23.3 Bolivia's Structural Trajectory, 1988–2024

Example 23.11 (Bolivia: Five Structural Epochs)**.** *INE publishes symmetric input–output matrices for every year 1988–2009 and 2011–2014 on base 1990, and supply-use tables for 2017–2024 on base 2017. There are no tables for 2010, 2015 or 2016.* Γ *below is that of the 12-sector domestic input-share matrix; complexity is the ECI (HS92, available from 1995). Household microdata were not available, so the* σ_B *values are* [ILLUSTRATIVE].

Epoch 1: Structural Adjustment (1985–1993). *The NEP (New Economic Policy) liberalized trade, closed state mines, and privatized public enterprises. Expected effect: edge destruction (state mining edges severed) and edge creation (new private-sector trade edges).*

Trajectory: the tables begin in 1988, after the 1985–87 mine closures, so the shock itself is not observed. From 1988 to 1993, Γ *was flat (0.563 to 0.573), and the hydrocarbons sector's Perron weight was small (0.025 in 1988). Illustratively,* σ_B *rose from* ≈ 0.65 *to* ≈ 0.68 *as mining workers were expelled into the informal sector.*

Epoch 2: Capitalization and Gas Discovery (1994–2005). *The "capitalization" of state enterprises brought FDI into gas, telecoms, and transport. Gas production began at Sabálo and San Alberto fields.*

Trajectory: Γ fell*, from 0.567 in 1994 to 0.502 in 2005, most of it in a level shift around 1999–2000 (see the break test below). The gas sector connected primarily to the rest of the world: the share of its output exported rose from 22% in 2000 to 69% in 2006. Its Perron weight in the domestic* W^P *rose from 0.022 (1994) to 0.074 (2005), still below the uniform share* $1/N$*. Bolivia's weight in the global trade network rose from 0.00018 (1995) to 0.00047 (2006) (Chapter 16). ECI fell by 0.24 (1995 to 2005). Illustratively,* σ_B *fell from 0.68 to 0.63 as gas revenue created government jobs, moving some workers from the informal to the formal block.*

Epoch 3: Gas Nationalization (2006–2009). *The Hydrocarbons Law 3058 (2005) had already raised the state's take to 50% (18% royalties plus the 32% direct hydrocarbons tax, IDH). Supreme Decree 28701 (May 2006) imposed a transitional 82% take on the two largest fields (San Alberto and Sabálo) while all contracts were renegotiated.*

Trajectory: no detectable structural break in Γ*: 0.518 in 2006 and 0.504 in 2009. The nationalization changed the distribution of existing flows (gas* $\to$ *MNC profit repatriation*

shifted to gas → treasury), a persistent reweighting of ownership and fiscal flows that the inter-industry W^P does not record. It did not create new domestic production edges: hydrocarbons bought 0.381 of domestic intermediate inputs per unit of output in 2006, and their Perron weight moved within 0.064–0.087. The government's and foreign owners' shares of the circular flow shifted, but measuring that needs a SAM (Chapter 20), not W^P. ECI fell by 0.14. Illustratively, σ_B fell modestly (0.63 to 0.61) as fiscal transfers reached poor households.

Epoch 4: Redistribution Without Structural Change (2010–2019). *A decade of high gas revenue, large fiscal transfers (Bono Juancito Pinto, Renta Dignidad, subsidies), and poverty reduction.*

Trajectory: no single base spans the decade. On base 1990, Γ was 0.478 in 2011 and 0.484 in 2013; on base 2017, 0.506 in 2017 and 0.467 in 2019. ECI stayed at its low level: -1.28 in 2010, -1.30 in 2019, with Bolivia's rank falling from 127th to 131st. Illustratively, σ_B held at 0.61–0.62 and $N_{eff}(W^I)$ rose slowly (3.0 → 3.4, driven by social media; 3.4 is the end-of-period value on Chapter 17's four media ecosystems, Example 17.9). Meanwhile the distributional record was remarkable. The Gini of per-capita household income fell from 0.600 (2005) to 0.426 (2019), and extreme poverty from 38.2% (2005) to 15.2% (2018). Ten years of redistribution produced no rise in Γ and no gain in complexity, and large equity gains anyway.

Epoch 5: Crisis and Stagnation (2019–present). *Political crisis (2019), pandemic (2020–2021), declining gas reserves.*

Trajectory: on base 2017, Γ fell from 0.467 (2019) to 0.381 (2023) and was 0.419 in 2024. The hydrocarbons sector shrank from 6.1% of gross value added (2019) to 3.3% (2024), and its Perron weight nearly halved (0.039 to 0.021). ECI rose, *from -1.30 to -1.10, as gas left the export basket: fuels were 32.1% of merchandise exports in 2019 and 18.9% in 2024. The base-2017 series is eight points long and includes the pandemic, so it is too early to call the fall in Γ structural.*

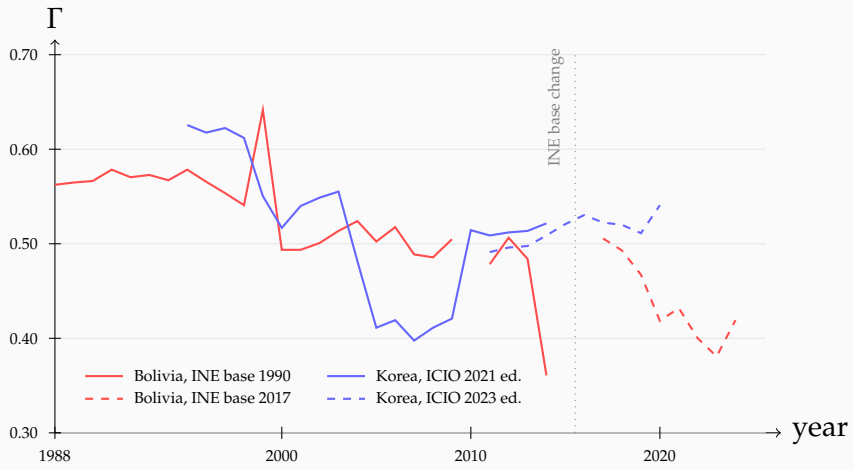
Epoch	*Period*	Γ *(start → end)*	$\Delta\Gamma$	ΔECI	*Structural break?*
1. Adjustment	*1988–93*	*0.563 → 0.573*	$+0.01$	—	*Not observed*
2. Gas discovery	*1994–05*	*0.567 → 0.502*	-0.07	-0.24	*1999–2000 (artefact?)*
3. Nationalization	*2006–09*	*0.518 → 0.504*	-0.01	-0.14	*No*
4. Redistribution	*2011–13*	*0.478 → 0.484*	$+0.01$	-0.02	*No*
	2017–19	*0.506 → 0.467*	-0.04		
5. Crisis	*2019–24*	*0.467 → 0.419*	-0.05	$+0.20$	*Too short*

Source: Γ, INE input–output matrices (base 1990, epochs 1–4, first row) and supply-use tables (base 2017, epoch 4 second row and epoch 5), 12 sectors, domestic flows, input shares (Instituto Nacional de Estadística (INE), Bolivia, 2026); author's calculations (`data/compute/compute_bolivia_io.py`, `data/`

processed/bolivia_io_summary.csv). The two bases are not comparable, so epoch 4 is split. ECI: Growth Lab, HS92 (The Growth Lab at Harvard University, 2026), changes 1995→2005, 2006→2009, 2010→2019 and 2019→2024 (data/processed/eci_epochs_bolivia.csv). Inequality and poverty: INE, Encuestas de Hogares (old poverty lines for 2005–2018). The illustrative $\Delta\sigma_B$ *are* $+0.03$, -0.05, -0.02, ≈ 0 *and unknown.*

Break test. *Apply Definition 23.8 to the base-1990 series, with* $\hat{\sigma}_{\Delta\Gamma}$ *estimated on the quiet years 1989–1997 (year-to-year changes of about 0.01). Three moves are an order of magnitude larger: 1999 (*$+0.10$*), 2000 (*-0.15*) and 2014 (*-0.12*). At* $k = 3$*, 2007 (*-0.03*) and 2012 (*$+0.03$*) are flagged as well. None of the large moves matches a policy event. INE published make matrices only for 2002, 2003, 2006 and 2011–2014. The 1999 spike and the 2000 drop fall in a run of years built on a diagonal-make assumption; 2014 is the last year of the base, drops 0.12 below 2013, and has no successor to confirm it. The 2007 move coincides with a change in the number of sectors with a defined row (public administration drops out of* W^P *in years without a make matrix), and the 2012 move with a known inconsistency in that year's published table (one industry's output does not equal its intermediate plus final sales). These are, most likely, construction effects. That is exactly the situation Remark 23.9 warns about: the test finds breaks, and the bootstrap over the table's construction is what decides whether they are structural.*

Visualization: Bolivia's and Korea's Γ trajectories, 1988–2024



Γ of the 12-sector domestic input-share matrix, by year. Bolivia (red): INE base 1990, 1988–2014 (there is no 2010 table), and base 2017, 2017–2024; the two series are not comparable and are not joined. Korea (blue): OECD ICIO 2021 edition, 1995–2014, and 2023 edition, 2011–2020 (the editions differ by 0.017 in the overlap year 2011). Bolivia is on the book's 12-sector classification and Korea on the ISIC-based one; in 2019 the two classifications give Bolivia 0.467 and 0.475. Both economies' Γ fell: Bolivia's from 0.567 (1990) to 0.484 (2013), Korea's from 0.626 (1995) to a trough of 0.397 (2007) and 0.511 in 2019. Over 1995–2019, Korea's ECI rose from 1.10 to 1.81 and Bolivia's fell from -0.85 to -1.30. Bolivia's 1999 spike and 2014 drop are the break-test flags of Example 23.11. σ_B is not plotted: there is no sourced series. Sources: Instituto Nacional de Estadística (INE), Bolivia (2026); OECD (2021, 2023); The Growth Lab at Harvard University (2026); author's calculations (`data/compute/compute_bolivia_io.py`, `data/compute/compute_oecd_io.py`); coordinates from `code/ch23.py`.

Key Structural Insight

The trajectory panel does not show what the usual story leads one to expect. On a like-for-like 12-sector basis, Bolivia's production network is not a fragmented outlier: its Γ (0.475 in 2019, ISIC basis) lies inside the range of the comparators, between the United States (0.463) and Korea (0.511), below Germany (0.544) and Mexico (0.649). And Γ fell in both economies: in Bolivia from 0.567 (1990) to 0.484 (2013), in Korea from 0.626 (1995) to 0.511 (2019). Aggregate Γ does not separate the two development paths.

What separates them is complexity and where power sits. Korea's ECI rose from 1.10 to 1.81 over 1995–2019, and the Perron vector of its production network concentrated on manufacturing ($G(\boldsymbol{\pi})$ rose from 0.618 to 0.658). That concentration is industrial depth, not fragmentation, and not oppression: in an input-share matrix, a high π marks a hub that many sectors buy from. Bolivia's ECI fell from -0.85 to -1.30. Its enclave shows up not in Γ but in $\boldsymbol{\pi}$ and export orientation: in 2019 hydrocarbons had a Perron weight of 0.039 and exported 70% of their output (Chapter 20).

Nor did distributional progress wait for Γ. Bolivia grew at 4.0% a year over 1990–2019, its Gini fell from 0.600 (2005) to 0.426 (2019), and extreme poverty from 38.2% to 15.2% (2005–2018), while Γ drifted down and ECI fell. On this chapter's reading, the gains came mostly through transfers and labour-market tightening within the existing structure ($\Delta\boldsymbol{\omega}$) rather than through ΔW. Whether they can be sustained without structural change is the open question of Chapter 26. That they happened at all is evidence against any claim that equity gains *require* rising Γ or positive ECI.

Korea's transformation (land reform, mass education, export discipline, industrial policy) began decades before 1995, and its Γ in those decades is not measured here: pre-1995 Korean tables were not available. Whatever that path was, Korea entered 1995 with Γ above 0.6, so its development cannot be read as a rise in aggregate integration over the years the data cover.

A structural report card needs several coordinates. Γ alone grades Korea and Bolivia alike.

23.4 Speed of Structural Change

Definition 23.12 (Structural Velocity)**.** *The **structural velocity** is the rate of change of the spectral gap:*

$$v_\Gamma(t) = \frac{\Gamma_{t+1} - \Gamma_t}{\Delta t}. \tag{23.7}$$

*The **structural acceleration** is* $a_\Gamma(t) = v_\Gamma(t+1) - v_\Gamma(t)$.

Example 23.13 (Structural Velocities, Selected Countries). *Endpoint velocities* $v_\Gamma = (\Gamma_T - \Gamma_{t_0})/(T - t_0)$*, per decade, computed within each data vintage:*

Country	*1995–2014 (ICIO 2021)*	*2011–2020 (ICIO 2023)*
South Korea	-0.055	$+0.055$
Chile	$+0.091$	$+0.107$
Argentina	$+0.037$	-0.050
Germany	-0.002	$+0.043$
United States	-0.010	-0.016
China	*n/a*	-0.038
	1990–2013 (INE base 1990)	*2017–2024 (INE base 2017)*
Bolivia	-0.036	-0.12

Source: OECD ICIO, 2021 and 2023 editions (OECD, 2021, 2023), ISIC-based 12 sectors; INE input–output matrices and supply-use tables (Instituto Nacional de Estadística (INE), Bolivia, 2026), book's 12 sectors; domestic flows, input shares. Author's calculations (`data/compute/compute_velocity.py`, `data/processed/gamma_velocity.csv`). China is split into two blocks in the 2021 edition and could not be extracted for 1995–2010. Venezuela is not a separate economy in ICIO. OLS slopes for Bolivia: -0.046 *and* -0.16.

Three things stand out. First, the signs flip. Korea's Γ *fell at 0.055 per decade over 1995–2014 and rose at the same speed over 2011–2020; Argentina's did the reverse. At the 12-sector level,* v_Γ *is not a stable property of a country, and over a decade it is of the same order as the year-to-year noise in* Γ_t*. Second,* v_Γ *does not track development. The fastest "integrator" in the table is Chile, whose ECI barely moved (about 0.1 in 1995,* -0.06 *in 2019). Among the most negative entries are Korea over 1995–2014, when its ECI rose from 1.10 to 1.70, and China, whose ECI also rose (1.15 in 2010, 1.30 in 2019). Third, Bolivia's* Γ *fell in both of its series (*-0.036 *per decade over 1990–2013;* -0.12 *over 2017–2024, a window that includes the pandemic). There is no "distance to Korea" to close in* Γ*: on the cross-country basis, Bolivia's 2019 value (0.475) is within 0.04 of Korea's (0.511).*

Remark 23.14 (What Determines Structural Velocity?). The co-evolutionary dynamics of Chapter 13 suggest that structural velocity depends on the following (a list of channels, not a derived result):

1. **Cross-block surplus** s_{cross}: higher surplus makes new cross-block edges profitable. Industrial policy that creates cross-block linkages increases v_Γ.

2. **Edge destruction rate** δ_{cross}: crises, policy reversals, and political instability destroy cross-block edges, reducing v_Γ or making it negative.

3. **Proximity bias** p_1: strong path dependence (high p_1) slows diversification into structurally distant activities, capping v_Γ.

4. **Formation velocity**: the speed at which preferences adjust $(1 - \rho)$. Fast preference adjustment (ρ small) allows rapid norm change, which supports structural change. Slow adjustment (ρ large; the book's illustrative Bolivian value is $\rho \approx 0.85$) means preferences lag behind structure, creating resistance to change.

23.5 Distinguishing Structural Change from Cyclical Fluctuation

Definition 23.15 (Structural-Cyclical Decomposition)**.** *Any change in the Perron eigenvector can be written as*

$$\Delta\pi_{i,t} = \underbrace{\Delta\pi^S_{i,t}}_{\textit{structural}} + \underbrace{\Delta\pi^C_{i,t}}_{\textit{cyclical}}, \tag{23.8}$$

where:

1. *The **structural component** $\Delta\pi^S$ is the part of the change that* persists*: the change in the trend of W_t, whatever its origin (new edges, destroyed edges, or a lasting reweighting such as a new fiscal regime).*

2. *The **cyclical component** $\Delta\pi^C$ is the part that* reverses*: transitory deviations of W_t from its trend due to price fluctuations, demand shocks, or temporary policy.*

The criterion is persistence, not the type of break (Definition 23.10). Identification: filter each series $\pi_{i,t}$ separately (HP or band-pass, Appendix 23.8); its trend is $\pi^S_{i,t}$ and the residual is $\pi^C_{i,t}$. The same filter applied to Γ_t gives the structural and cyclical parts of integration. This is a definition, not a theorem: the split depends on the filter and its smoothing parameter.

Example 23.16 (Bolivia: The Gas Boom in $\boldsymbol{\pi}$, 2003–2024)**.** *During the gas boom, the hydrocarbons sector's Perron weight in the domestic W^P rose from 0.047 (2003) to 0.085 (2013), and to 0.117 in 2014, a year whose Γ is an outlier (Example 23.11). On base 2017 it was 0.045 in 2017, 0.039 in 2019 and 0.021 in 2024. The sector's share of gross value added followed the same arc: 3.5% (2000), 7.9% (2006) and 10.0% (2013) on base 1990; 6.3% (2017) and 3.3% (2024) on base 2017.*

Source: INE input–output matrices (base 1990) and supply-use tables (base 2017), 12 sectors, domestic flows, input shares (Instituto Nacional de Estadística (INE), Bolivia, 2026); author's calculations (`data/compute/compute_bolivia_io.py`, `data/compute/compute_bolivia_facts.py`).

The change of base blocks the decomposition that Definition 23.15 calls for. No single series runs from before the boom to after it, so the 2019 level cannot be compared with the

2003 level, and a filter run across the break would read the change of base as structural. Within each base the pattern is clear. The rise in π_{gas} tracked the price and volume boom, and on the new base the weight kept falling as production declined. The data show no sign of a persistent increase in the sector's place in the domestic network. Whatever the nationalization changed permanently, it was the fiscal capture of rents (Epoch 3), which W^P does not record.

Note also the scale. Even at its peak, π_{gas} was never far above the uniform share $1/N \approx 0.09$: the sector was never a hub of the domestic network. Its weight lay in exports and fiscal revenue, not in domestic linkages. An analysis that counted Bolivia's boom as "structural transformation" was reading a cyclical rise in a peripheral sector's weight.

Prebisch Connection: Commodity booms as cyclical mirages

Latin American economic history is a sequence of commodity booms mistaken for structural change. Guano in Peru (1840–1870), nitrate in Chile (1880–1920), coffee in Brazil (1890–1930), oil in Venezuela (1920–2015), gas in Bolivia (2003–2014). Each boom raised incomes, reduced poverty, and generated optimism about "transformation." Each bust revealed that the production structure had not changed: the economy remained an enclave connected to the world through a single commodity edge. The enclave is not a matter of aggregate Γ, which in Bolivia is in the range of industrial economies. It is a matter of where the commodity sector sits: at the periphery of the domestic network (a small π), exporting most of its output, buying few domestic inputs, and paying a small share of its value added as wages.

The structural-cyclical decomposition is the instrument that detects the mirage. During the boom: $\Delta\pi^C$ is large (the commodity sector's weight inflates). After the bust: $\Delta\pi^C$ reverses. The structural component $\Delta\pi^S$ is what remains — and it is typically near zero for commodity booms, because commodity booms do not create cross-block domestic linkages.

The exception: commodity booms CAN produce structural change if the revenue is invested in the operators of Chapter 15 (information, education, infrastructure, industrial policy). Norway is the canonical example, though not through its oil fund: the Government Pension Fund Global invests only abroad, precisely to keep oil revenue out of the domestic economy. Norway's structural change came from building domestic supplier, engineering and technology capabilities around the petroleum sector. Neither aggregate used in this chapter shows it, however. On the 12-sector ICIO basis Norway's Γ fell from 0.644 in 1995 to 0.313 in 2014, as the economy reorganized around petroleum and its supply industries, and its ECI fell from 1.24 to 1.01 (OECD, 2021; The Growth Lab at Harvard University, 2026; `data/processed/gamma_velocity.csv`, `eci_facts.csv`). If the petroleum cluster is structural change, it is visible only at a finer grain: in the domestic input intensity of the petroleum sector and the exports of its suppliers. That is where a test would have to look. Bolivia's gas revenue went largely into transfers ($\Delta\omega$): poverty fell sharply, Γ drifted down, and ECI fell.

23.6 The Structural Change Index

Definition 23.17 (Structural Change Index (SCI))**.** *The **Structural Change Index** summarizes the entire trajectory in a single number:*

$$SCI(t_0, T) = \frac{\Gamma_T - \Gamma_{t_0}}{\Gamma^{ref} - \Gamma_{t_0}}, \tag{23.9}$$

where $\Gamma^{ref} > \Gamma_{t_0}$ is a reference level of integration chosen by the analyst (here $\Gamma^{ref} = 0.50$, a round number near Germany's 0.544 in 2019 on the 12-sector ICIO basis). It is a normalization, not the estimated threshold Γ^* of Chapter 26. SCI is the fraction of the distance to the reference closed over the period: it equals 0 if nothing changed and 1 if the reference was reached, is negative when integration fell (the United States in Example 23.18), can exceed 1 on overshoot, and is undefined when $\Gamma_{t_0} \geq \Gamma^{ref}$.

Example 23.18 (SCI for Selected Countries)**.** *Using $\Gamma^{ref} = 0.50$ and the 1995–2014 window of the ICIO 2021 edition (Bolivia: INE base 1990, 1990–2013):*

Country	Γ_{t_0}	Γ_T	*Period*	*SCI*
Chile	*0.467*	*0.640*	*1995–2014*	5.30
Argentina	*0.407*	*0.476*	*1995–2014*	0.75
United States	*0.475*	*0.457*	*1995–2014*	-0.72
South Korea	*0.626*	*0.522*	*1995–2014*	*undefined*
Germany	*0.554*	*0.549*	*1995–2014*	*undefined*
Bolivia	*0.567*	*0.484*	*1990–2013*	*undefined*

Source: as for Example 23.13. The SCI is recomputed by `code/ch23.py` *from the file* `data/processed/gamma_velocity.csv`.

With the real trajectories the index is mostly undefined. Korea, Germany and Bolivia all start above Γ^{ref}, so on this basis they have "no gap to close". Where the index is defined, the denominator dominates it: Chile's 5.30 comes from a gap of only 0.033, overshot fivefold. The SCI presumes that development means moving Γ from far below a reference to the reference. At the 12-sector level the data supply no such gradient: in 2019, Γ for economies as different as Bolivia, Korea, Germany and the United States lies between 0.46 and 0.54. An index of this form is informative only for an indicator with a clear developmental gradient and starting points far from the reference. On this evidence that points to ECI, or to finer-grained network measures, rather than to aggregate Γ.

23.7 Summary

1. The **structural trajectory** $\{(\Gamma_t, \sigma_{B,t}, G(\boldsymbol{\pi}_t), N_{\text{eff},t}, \text{ECI}_t)\}$ is the time series of structural health. In the data (INE base 1990 and base 2017, kept separate; OECD ICIO 1995–2019), Γ *fell* in both Bolivia (0.567 in 1990, 0.484 in 2013) and Korea (0.626 in 1995, 0.511 in 2019). Complexity separates them: Korea's ECI rose from 1.10 to 1.81, Bolivia's fell from -0.85 to -1.30.

2. **Structural breaks** are detectable as discontinuities in Γ_t, once bootstrap error bands are respected. Per-period Γ_t does not control the product $W_t \cdots W_s$: the Dobrushin coefficient τ_1 and the realized-product gap do, and the absolute probability sequence is the time-varying power vector. Bolivia's series shows no break that matches a policy event; the flagged years (1999–2000, 2014) look like effects of how the tables were built.

3. **Structural velocity** v_Γ measures the speed of integration. Bolivia: -0.036 per decade (1990–2013) and -0.12 (2017–2024). Korea: -0.055 (1995–2014) and $+0.055$ (2011–2020). At the 12-sector level, velocities flip sign between windows and do not track development.

4. The **structural-cyclical decomposition** (a definition, based on persistence) separates permanent changes in W from temporary price effects. Commodity booms are cyclical: $\Delta\pi^{\text{C}}$ reverses after the boom. Bolivia's π_{gas} rose from 0.047 (2003) to 0.085 (2013) and, on the new base, fell to 0.021 (2024); the change of base prevents a clean decomposition. The sector stayed peripheral in the domestic network throughout.

5. The **Structural Change Index** (SCI): fraction of the distance to a reference Γ closed (negative if integration falls). With $\Gamma^{\text{ref}} = 0.50$ it is undefined for Korea, Germany and Bolivia, which start above the reference: aggregate Γ supplies no developmental gradient for it to measure.

23.8 Exercises

23.1 Bolivia's trajectory. Using INE's input–output matrices (base 1990) and supply-use tables (base 2017), and EH microdata: construct Bolivia's structural trajectory for 2000–2019, keeping the two bases separate. Plot Γ_t against $G(\boldsymbol{\pi}_t)$ and, if you can compute it from the EH, against $\sigma_{B,t}$. Which coordinates move? Repeat with sales shares and with the diagonal removed: does the direction of change survive?

23.2 Structural break detection. Compute $\Delta\Gamma_t$ for each year. Apply the break detection criterion with $k = 2$, and a bootstrap band (Remark 23.9). Are there any structural breaks? At what years? Do they correspond to policy events? Compute $\tau_1(W_t)$ and the realized-product gap $\Gamma_t^{(5)}$ as well: do they tell the same story as Γ_t?

23.3 Structural velocity. Compute v_Γ for Bolivia (INE), Chile and Peru (OECD ICIO), by endpoints and by OLS. Does each country's velocity keep its sign across the 1995–2014 and 2011–2020 windows? Compare it with each country's change in ECI over the same years. What does the comparison say about v_Γ as a measure of development?

23.4 Structural-cyclical decomposition. For Bolivia, decompose $\pi_{\text{gas},t}$ into structural and cyclical components with an HP filter, on the base-1990 series only (1988–2014). How much of the 2003–2013 rise was structural? How sensitive is the answer to λ, and to dropping the 2014 outlier? Why must the base-2017 years not be appended to the series?

23.5 Commodity boom counterfactual. Suppose Bolivia had invested 30% of gas revenue in cross-block operators (infrastructure connecting gas to traditional sector, technical education for rural workers). Simulate the effect on the hydrocarbons sector's domestic input intensity (0.225 per unit of output in 2019), on its Perron weight, and on Γ. Which of the three indicators would detect the change, and why might Γ not?

23.6 SCI computation. For the Latin American countries covered by the OECD Inter-Country Input-Output tables, compute Γ of the row-normalized domestic input-share matrix $W^P = D^{-1}A$ for the first and last available years, using the same sector classification for every country. Compute SCI with $\Gamma^{\text{ref}} = 0.50$ and rank the countries. For how many is it undefined? Now rank them by the change in ECI. Why is ECI *not* a proxy for Γ? (Hint: ECI is computed from the country–product export matrix and is standardized across countries each year.)

23.7 Trajectory projection. Bolivia's ECI changed by -0.44 over 1995–2019 and by $+0.20$ over 2019–2024; Korea's rose by $+0.70$ over 1995–2019 (`data/processed/eci_epochs_bolivia.csv`, `eci_facts.csv`). Project Bolivia's ECI for 2025–2060 under three scenarios: (a) the 1995–2019 trend, (b) the 2019–2024 trend, (c) Korea's 1995–2019 pace. Plot all three. When, if ever, does

each reach ECI $= 0$? ECI is standardized across countries each year: what does a "trend" in it mean?

Mathematical Appendices

Appendix 23A: The RAS Method for I-O Table Updating

Problem. Given a benchmark I-O coefficient matrix A_0 and new row totals $\mathbf{r}$ and column totals $\mathbf{c}$, find the updated matrix A^* that matches the new margins while staying "close" to A_0.

Algorithm. The RAS (or biproportional) method alternates row and column scaling:

1. Initialize $A^{(0)} = A_0$.
2. Scale rows: $A_{ij}^{(2k+1)} = A_{ij}^{(2k)} \cdot r_i / \sum_j A_{ij}^{(2k)}$.
3. Scale columns: $A_{ij}^{(2k+2)} = A_{ij}^{(2k+1)} \cdot c_j / \sum_i A_{ij}^{(2k+1)}$.
4. Repeat until convergence.

When it exists, the RAS solution minimizes the KL-divergence

$$\sum_{ij} A_{ij}^* \ln(A_{ij}^* / A_{0,ij})$$

subject to the margin constraints. It preserves zeros (if $A_{0,ij} = 0$ then $A_{ij}^* = 0$) and the block structure of A_0. Existence requires $\sum_i r_i = \sum_j c_j$ and margins compatible with the zero pattern of A_0 (Bacharach, 1970); in practice the iteration is applied to intermediate transactions, whose margins are intermediate sales and purchases, and the result is row-normalized afterwards to obtain W_t^P.

Structural interpretation. RAS updating assumes the block structure of W^P is unchanged between benchmarks — only the relative sizes of sectors change. This is conservative: it cannot detect edge creation or destruction between benchmark I-O tables. Between benchmarks, Γ_t moves only through the margins. Real structural change therefore shows up as a jump *at the next benchmark year*, which then bundles several years of accumulated change; do not read the jump as a one-year break.

Appendix 23B: Structural vs. Cyclical Decomposition

HP filter. The Hodrick-Prescott filter decomposes $\Gamma_t = \Gamma_t^S + \Gamma_t^C$ by minimizing $\sum_t (\Gamma_t - \Gamma_t^S)^2 + \lambda \sum_t [(\Gamma_{t+1}^S - \Gamma_t^S) - (\Gamma_t^S - \Gamma_{t-1}^S)]^2$. The smoothing parameter λ controls the trade-off between fit and smoothness. For annual data: $\lambda = 6.25$ (Ravn

and Uhlig, 2002) or $\lambda = 100$ (traditional). These choices matter: the gain-$\frac{1}{2}$ cutoff period of the annual HP filter is about 10 years for $\lambda = 6.25$, 20 years for $\lambda = 100$, 28 years for $\lambda = 400$ and 40 years for $\lambda = 1600$ (`code/ch23.py`). To separate a 10–20 year cycle from structural trends use $\lambda \approx 6.25$–100; $\lambda = 400$–1600 treats everything shorter than 28–40 years as cyclical.

Band-pass filter. The Baxter-King or Christiano-Fitzgerald filter isolates fluctuations at specific frequencies. Structural change: periods > 20 years. Business cycle: 2–8 years. The band-pass filter cleanly separates the two.

Structural breaks. The HP and band-pass filters assume smooth trends. For detecting abrupt structural breaks (nationalization, crisis, regime change): use the Bai and Perron (1998) multiple structural break test applied to Γ_t. The null hypothesis is no break; the alternative is one or more breaks at unknown dates.

Structuralist Case Study: CEPAL's Structural Change Program

CEPAL's (2012b) report *Structural Change for Equality: An Integrated Approach to Development* represents the institutional synthesis of the structuralist tradition's diagnosis and prescription. Published 63 years after Prebisch's 1949 manifesto, it argues that Latin America's central challenge remains structural: the region has achieved macroeconomic stability and poverty reduction but has not changed its production structure.

The report maps onto this chapter's framework precisely.

Part 1: Structural change is slow. CEPAL documents that Latin America's export composition changed little over 1990–2010: the region remained specialized in natural resources and low-technology manufactures. In this chapter's terms, the complexity trajectory was flat or falling: Chile's ECI was about 0.1 in 1995 and -0.06 in 2019, and Bolivia's fell from -0.85 to -1.30. The Γ velocities are small and of mixed sign (1995–2014, per decade: Chile $+0.091$, Argentina $+0.037$, Brazil $+0.020$, Colombia -0.036, Peru -0.060; Example 23.13), and, as for Korea, they do not track development.

Part 2: Redistribution without structural change is unsustainable. CEPAL notes that the region's poverty reduction (2000–2012) was driven by commodity booms and fiscal transfers, not by structural transformation. In this chapter's terms: $\Delta\omega \gg \Delta W$. The improvement is cyclical ($\Delta\pi^C > 0$ during the boom) not structural ($\Delta\pi^S \approx 0$). When the commodity cycle reversed (post-2014), poverty stagnated and in some countries reversed.

Part 3: The prescription is "change for equality." CEPAL argues for industrial policy, technological upgrading, and institutional reform — the operators of Chapter 15, stated in institutional language rather than matrix notation. The report's title states a claim: structural change ($\Delta\Gamma > 0$) is necessary for sustained equality ($\sigma_B \downarrow$). In this book that claim is not a theorem; it is the testable conjecture of Chapter 26 (Conjecture 26.1). Bolivia's record is evidence against it in its Γ form: between 2005 and 2019 the Gini fell from 0.600 to 0.426 while Γ drifted down and ECI fell. The claim survives only as one about *sustaining* equity gains, not achieving them.

CEPAL (2012b)	This chapter	Cross-reference
Structural change is slow	ECI flat or falling; v_Γ of mixed sign	Ex. 23.13
Redistribution without structural change	$\Delta\boldsymbol{\omega} \gg \Delta W$	Sec. 23.3, Epoch 4
Commodity boom as mirage	$\Delta\pi^C$ reverses	Prebisch Connection
Change for equality	$\Delta\Gamma > 0$ needed? Not in Bolivia 2005–19	Conjecture (Ch. 26)
Policy: industrial upgrading	Operators from Ch. 15	Ch. 15
SCI as metric	Fraction of gap closed	Def. 23.17

CEPAL wrote in the language of institutions and policy. This chapter translates into eigenvalues and trajectories. The message is largely the same: Latin America has been redistributing within a structure that, measured by what it can produce and export, has not changed. The data add a warning CEPAL did not need to make. At this level of aggregation, the integration of the input–output network is not where the missing structural change shows up: Bolivia's Γ is like Korea's, and both fell.

Further Reading

Structural change measurement: CEPAL (2012b), *Structural Change for Equality*. McMillan et al. (2014), "Globalization, Structural Change, and Productivity Growth." de Vries et al. (2012), "Deconstructing the BRICs."

I-O table updating: Miller and Blair (2009), *Input-Output Analysis: Foundations and Extensions*, Chapter 8 (RAS and cross-entropy methods). Temursho (2018), "Entropy-Based Benchmarking Methods."

Structural breaks: Bai and Perron (1998), "Estimating and Testing Linear Models with Multiple Structural Changes." For application to development: Pritchett (2000), "Understanding Patterns of Economic Growth."

Data used in this chapter: INE input–output matrices and supply-use tables (Instituto Nacional de Estadística (INE), Bolivia, 2026); OECD ICIO (OECD, 2021, 2023); Growth Lab complexity data (The Growth Lab at Harvard University, 2026); World Development Indicators (World Bank, 2026b). Scripts and processed files are in `data/`.

Bolivia's economic history: Morales and Sachs (1990), "Bolivia's Economic Crisis." Weisbrot et al. (2009), "Bolivia: The Economy During the Morales Administration." Andersen et al. (2020), "Bolivia's Twenty-First Century Commodity Boom," INESAD Working Paper.

Part VII

Political Economy and Conclusion

Chapter 24

Hegemony and Counter-Hegemony

When the Dominated Accept the Matrix, and When They Don't

> *"La crisi consiste appunto nel fatto che il vecchio muore e il nuovo non può nascere: in questo interregno si verificano i fenomeni morbosi più svariati."*
>
> ["The crisis consists precisely in the fact that the old is dying and the new cannot be born; in this interregnum a great variety of morbid symptoms appear."]
>
> — Antonio Gramsci, *Quaderni del carcere*, Q3 §34 (1930)

Part VII opens where Parts I–VI left a question unanswered: if the current W is far from W^{CE}, and if structural reform would make nearly everyone better off (the Structural Welfare Theorem, Chapter 12), why does the structure persist?

The answer from Chapter 14 is political: coalitions block structural change that threatens their power. But this answer is incomplete. It explains why the powerful resist change. It does not explain why the powerless *accept* the structure — why the indigenous woman in Potosí does not demand the operators that would raise her π_I from 10^{-5} to something closer to $1/N$.

The answer is hegemony: the process by which the dominant coalition's preferred W becomes the *accepted* W — not through coercion but through the formation of preferences that regard the structure as natural, inevitable, or even desirable. In this book's notation: hegemony is the steady state of the formation model (Chapter 10) in which the W^I-shaped preferences of the dominated classes are calibrated to accept the W^P-determined allocation.

Counter-hegemony is the disruption: a change in W^I that produces new preferences, new demands, new coalitions, and ultimately a new W. This chapter formalizes both.

24.1 Hegemony as a Structural Equilibrium

24.1.1 Gramsci's Concept, Formalized

Definition 24.1 (Hegemony)**.** *A weight matrix W is **hegemonic** if it satisfies three conditions simultaneously:*

1. ***Power concentration**: $N_{eff}(\boldsymbol{\pi}^P)$ is well below N (equivalently, $\mathrm{Var}(\boldsymbol{\pi}^P)$ is well above 0; Chapter 6): the structure benefits some agents substantially more than others. ($\mathrm{Var} > 0$ alone holds for almost every W and would not discriminate.)*

2. ***Consent**: the steady-state preferences $\boldsymbol{\theta}^*(W^I)$ evaluate the allocation $\mathbf{x}^*(W, \boldsymbol{\theta}^*)$ favorably: $U_i(x_i^*, \theta_i^*) \geq U_i(x_i', \theta_i^*)$ for all feasible x_i' (each agent is satisfied given her preferences).*

3. ***Formation alignment**: the information network W^I is controlled (directly or indirectly) by the coalition that benefits from W: on a common node set of agents or groups, the agents with high π_i^P also have high π_i^I (a positive rank correlation between $\boldsymbol{\pi}^I$ and $\boldsymbol{\pi}^P$; exact proportionality would be a knife-edge case).*

Condition 2 formalizes consent only. Gramsci's hegemony is "consent armoured by coercion": the state combines political society (coercion) with civil society (consent). This chapter models the consent half; coercion enters only as the outside option that the dominant coalition holds in reserve.

Definition 24.2 (Hegemonic Equilibrium)**.** *A **hegemonic equilibrium** is a structural equilibrium (Definition 13.7) with the additional property that the formation channel (W^I) stabilizes the power structure (W^P) by producing preferences that consent to it.*

Formally: a hegemonic equilibrium is a fixed point $(W^*, \boldsymbol{\theta}^*, \mathbf{x}^*)$ of the co-evolutionary system where:

$$\boldsymbol{\theta}^* = T(\boldsymbol{\theta}^*; W^*), \quad \mathbf{x}^* = h(\boldsymbol{\pi}(W^*), \boldsymbol{\theta}^*), \quad W^* = \mathcal{E}(W^*; \mathbf{x}^*), \tag{24.1}$$

AND the three conditions of Definition 24.1 hold at $(W^*, \boldsymbol{\theta}^*, \mathbf{x}^*)$; in particular, no agent, given her preferences θ_i^*, prefers a feasible alternative allocation, so none demands a change in W.

This definition asserts no existence result; whether such fixed points exist depends on the formation map T, the bargaining map h and the network evolution map $\mathcal{E}$ (Chapter 13, Definition 13.6). It is the Satisfaction Paradox (Proposition 12.2) with a political interpretation: the dominated classes do not demand structural change because their W^I-formed preferences are calibrated to the W^P-determined returns. The hegemony is self-reproducing.

Intuition: Why hegemony is stable

Three reinforcing loops:

Loop 1 (Production): W^P determines the allocation $\mathbf{x}^*$ through the bargaining equilibrium. The dominated receive less, but their sectors are structurally peripheral (low π_i), so there is no economic lever for change.

Loop 2 (Information): W^I shows the dominated their own group's returns (assortative mixing, Chapter 18). They observe low returns and form low aspirations ($\theta^*_{\text{disadv}} < \theta^*_{\text{adv}}$, Proposition 18.13). Low aspirations produce low investment, low returns, confirming the signal.

Loop 3 (Solidarity): W^U is assortative by the same groups. The dominated care about their own group, not about the advantaged group. Cross-group solidarity is low ($\lambda^U \approx 0.2$, Chapter 22). The dominated do not form cross-class coalitions because they do not identify with the advantaged class's interests.

Hegemony is the triple alignment of loops 1, 2, and 3: the same groups are peripheral in W^P, informationally isolated in W^I, and solidaristically self-contained in W^U. The alignment IS the hegemony. Breaking it requires disrupting at least one loop.

24.1.2 The Cost of Maintaining Hegemony

Definition 24.3 (Hegemony Maintenance Cost)**.** *Maintaining hegemony requires the dominant coalition to keep the consensus $b^* = (\boldsymbol{\pi}^I)'\mathbf{b}^{(0)}$ at the target belief b_T ("the structure is natural/efficient") against the drift created by independent outlets. Model the per-period cost as*

$$C_H = \mu_H \cdot \sum_{j \notin S_H} \pi_j^I \, |b_T - b_j^{(0)}| \cdot \big(1 - \lambda_2(W^I)\big), \tag{24.2}$$

where S_H is the set of outlets the coalition already controls, $b_j^{(0)}$ is outlet j's belief absent manipulation, and $\mu_H > 0$ is a cost scale. The factor $1 - \lambda_2(W^I)$ comes from the corrected manipulation cost of Chapter 8: by the perturbation lemma (Lemma 6.19), $\boldsymbol{\pi}^I$ is cheapest to move when λ_2 is close to $+1$, i.e. when W^I is weakly coupled.

Two different properties of W^I matter, and they must not be confused:

- **Concentration** (N^I_{eff} small). When a few outlets the coalition controls hold most of $\boldsymbol{\pi}^I$, the sum in (24.2) is small: the coalition owns the consensus and barely needs to manipulate. As N^I_{eff} rises, more independent weight must be

countered and C_H rises. A hub that everyone listens to has a *large* spectral gap (Proposition 6.16), so concentration is measured by N^I_{eff}, not by Γ^I.

- **Weak coupling** ($1 - \lambda_2$ small). When W^I splits into weakly connected blocks, the consensus is fragile: it is cheap for *anyone*, the coalition or its challengers, to move it.

Example 24.4 (Bolivia's Hegemonic Transitions [ILLUSTRATIVE])**.**

Period	*Hegemonic structure*	N^I_{eff}	*Stability*
Pre-1952	*Tin-mining oligarchy*	≈ 1.5	*Stable (few media)*
1952–1964	*MNR revolutionary state*	≈ 2.0	*Moderately stable*
1964–1982	*Military governments*	≈ 1.8	*Coercion-dependent*
1982–1985	*UDP government, hyperinflation*	—	*No stable hegemony*
1985–2003	*Neoliberal consensus*	≈ 2.5	*Stable, then collapsed*
2006–2019	*MAS/indigenous hegemony*	≈ 3.0–3.4	*Fragile (social media)*
2019–	*Contested*	≈ 3.5	*No hegemony*

The trend: N^I_{eff} has been increasing (from 1.5 to 3.5 over 70 years). On the reading of Definition 24.3, each increase made the existing hegemony more expensive to maintain. The neoliberal consensus (1985–2003) collapsed after social movements (the Water War 2000, Gas War 2003) generated alternative W^I channels that broke the information monopoly. The MAS hegemony (2006–2019) was already fragile at $N^I_{eff} \approx 3.0$–3.4 (the end-of-period value 3.4 is Chapter 17's, on four media ecosystems; Example 17.9); new digital channels contributed to destabilizing it in 2019. Whether digital media raise or lower N^I_{eff} in the long run depends on the balance between new channels and algorithmic concentration within platforms (Chapter 27).

24.2 Counter-Hegemony as ΔW^I

Definition 24.5 (Counter-Hegemonic Movement)**.** *A **counter-hegemonic movement** is a coalition S of dominated agents that implements ΔW^I — a change in the information network that disrupts the formation channel supporting the current hegemony. The movement creates alternative W^I edges: new media, new narratives, new reference groups, new signals that show the dominated a different set of returns.*

With the linear formation rule of Chapter 10, $\theta_{t+1} = \rho\theta_t + \mu(\cdot)$, the steady state is unique, so there is no basin boundary to cross and no threshold. A threshold needs a nonlinearity. The simplest one is *conformity*: the return an agent observes from acting on the alternative view rises with the share of her reference group already acting on it. Let $\theta_t \in \mathbb{R}$ index the intensity of counter-hegemonic preference

in the dominated group ($\theta \approx 0$: consent; $\theta \approx 1$: the alternative view), and let

$$\theta_{t+1} = \rho\,\theta_t + (1-\rho)\big[g(\theta_t) + s\big], \qquad g(\theta) = \frac{1}{1+e^{-\beta(\theta-1/2)}}, \tag{24.3}$$

where g is the conformity response and $s \geq 0$ is the persistent counter-hegemonic signal delivered by the new W^I edges, scaled so that $(1-\rho)s = \mu\,\|\Delta(W^I\mathbf{r})\|$.

Proposition 24.6 (Counter-Hegemony Threshold and Saddle-Node)**.** *In model* (24.3) *with $\beta > 4$ and $0 < \rho < 1$:*

1. *For $s = 0$ there are three fixed points $\theta_h < \frac{1}{2} < \theta_c$ with $\theta_c = 1 - \theta_h$. The outer two (hegemonic and counter-hegemonic) are stable; the middle one, $\theta_u = \frac{1}{2}$, is unstable and is the separatrix. A one-time shock moves the system to the counter-hegemonic equilibrium iff it pushes θ above θ_u.*

2. *As the persistent signal s rises, θ_h and the separatrix move toward each other and merge at s^*, defined by $g'(\theta) = 1$, $\theta < \frac{1}{2}$ and $\theta = g(\theta) + s^*$ (a **saddle-node bifurcation**). For $s > s^*$ the hegemonic equilibrium no longer exists and θ converges to the counter-hegemonic one from every initial condition. The collapse condition is therefore*

$$\mu\,\|\Delta(W^I\mathbf{r})\| > (1-\rho)\,s^*. \tag{24.4}$$

For $\beta = 10$, $\rho = 0.85$: $\theta_h = 0.007$, $\theta_u = 0.5$, $\theta_c = 0.993$ and $s^* = 0.181$; an impulse to $\theta = 0.457$ returns to θ_h, an impulse to 0.557 goes to θ_c (`code/ch24.py`).

Proof. Write $F(\theta) = \rho\theta + (1-\rho)[g(\theta) + s]$. Fixed points of F solve $h(\theta) = -s$, where $h(\theta) := g(\theta) - \theta$. Since $g' > 0$, $F'(\theta) = \rho + (1-\rho)g'(\theta) > 0$, so F is strictly increasing, and a fixed point is stable iff $F' < 1$ iff $g' < 1$ (no oscillatory instability is possible).

Shape of h. $g'(\theta) = \beta g(1-g)$ increases on $(-\infty, \frac{1}{2})$, decreases on $(\frac{1}{2}, \infty)$, has maximum $\beta/4 > 1$ at $\frac{1}{2}$ and tends to 0 at $\pm\infty$. Hence $g' = 1$ at exactly two points $a < \frac{1}{2} < b$, and $h' = g' - 1$ is negative on $(-\infty, a)$, positive on (a, b), negative on (b, ∞). Symmetry of the logistic, $g(1-\theta) = 1 - g(\theta)$, gives $h(1-\theta) = -h(\theta)$, so $b = 1 - a$, $h(\frac{1}{2}) = 0$ and $h(b) = -h(a)$. Since h is increasing on $(a, b) \ni \frac{1}{2}$, $h(a) < 0 < h(b)$. Finally $h(\theta) \to +\infty$ as $\theta \to -\infty$ and $h(\theta) \to -\infty$ as $\theta \to +\infty$.

Part 1 ($s = 0$). By strict monotonicity on each of the three intervals, $h = 0$ has exactly one root in each: $\theta_h \in (-\infty, a)$, $\theta_u = \frac{1}{2}$, $\theta_c \in (b, \infty)$, with $\theta_c = 1 - \theta_h$ by the antisymmetry of h. As $g \in (0, 1)$, every fixed point lies in $(0, 1)$. At θ_h and θ_c, $g' < 1$ (stable); at $\frac{1}{2}$, $g' = \beta/4 > 1$ (unstable). For the basins: F is increasing, so $\theta_0 > \frac{1}{2}$ implies $\theta_t > F(\frac{1}{2}) = \frac{1}{2}$ for all t, and (θ_t) is monotone; a bounded monotone

sequence converges, and its limit is a fixed point of the continuous map F, hence θ_c (the only fixed point above $\frac{1}{2}$). Symmetrically $\theta_0 < \frac{1}{2}$ gives $\theta_t \to \theta_h$.

Part 2 ($s > 0$). On $(-\infty, a]$, h falls continuously from $+\infty$ to $h(a)$; on $[a, b]$ it rises to $h(b) > 0$; on $[b, \infty)$ it falls to $-\infty$. So for $0 \leq s < s^* := -h(a) = a - g(a)$ the equation $h = -s$ has three roots, $\theta_h(s) < a < \theta_u(s) < b < \theta_c(s)$; as s rises, $\theta_h(s)$ increases and $\theta_u(s)$ decreases (because h is decreasing, respectively increasing, on their intervals), and they meet at a when $s = s^*$; there $g'(a) = 1$, $a < \frac{1}{2}$ and $a = g(a) + s^*$. For $s > s^*$, $-s < h(a) < h(b)$, so the only root is $\theta_c(s) > b$. Then $F(\theta) > \theta$ for $\theta < \theta_c(s)$ and $F(\theta) < \theta$ above it, and the monotone-sequence argument of Part 1 gives $\theta_t \to \theta_c(s)$ from every θ_0. Since $(1 - \rho)s = \mu \|\Delta(W^I \mathbf{r})\|$, $s > s^*$ is (24.4). □

The mechanism is the standard one of threshold and social-interaction models: Granovetter's thresholds (Granovetter, 1978), binary choice with conformity effects (Brock and Durlauf, 2001), and preference falsification (Kuran, 1989). What this chapter adds is only the reading of the signal s as a ΔW^I.

In words: the counter-hegemonic signal must either deliver a shock large enough to carry preferences past the separatrix, or be persistent and strong enough to remove the hegemonic equilibrium altogether. Once preferences cross, the formation dynamics carry them to the new steady state — and the demand for structural change becomes self-sustaining.

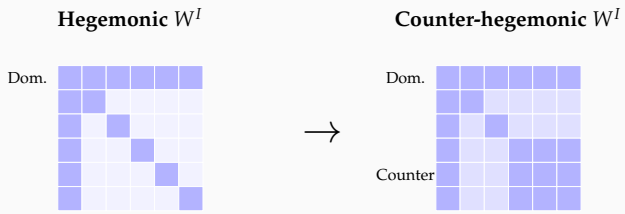
Example 24.7 (Three Counter-Hegemonic Episodes in Bolivia)**.** ***1. The 1952 Revolution.*** *Alternative W^I channels grew around the MNR: the miners' radio stations founded and run by the mining unions (e.g., Radio 21 de Diciembre), campesino unions, and the COB (Central Obrera Boliviana, founded in 1952).*

2. The Water War (2000) and Gas War (2003). *The neoliberal hegemony (1985–2003) was sustained by a W^I dominated by private media that framed privatization as modernization. The counter-hegemonic movement created alternative channels: community radio, neighborhood assemblies (juntas vecinales), and in Cochabamba the Coordinadora de Defensa del Agua y de la Vida (factory workers, irrigators' associations and neighbourhood groups); in 2003 the El Alto juntas were central. These channels showed alternative returns: the cost of water privatization (sharp tariff increases), the destination of gas revenue (foreign companies). On the reading of Proposition 24.6, the signal was strong and persistent enough to pass the threshold (24.4). Preferences shifted: privatization went from "modern" to "theft." Coalition shift: $\{M, W\}$ formed and eventually became the MAS coalition.*

3. The 2019 Crisis. *The MAS hegemony (2006–2019) was sustained by W^I dominated by state media and indigenous-movement networks. Counter-hegemonic chan-*

nels: social media (Facebook, WhatsApp), urban civic committees, the "pititas" movement. These channels showed alternative signals: allegations of electoral irregularities, environmental devastation (Chiquitano fires). The formation signal was amplified by the digital W^I (Chapter 21: $\omega_d = 0.50$ for urban youth). The MAS hegemony fractured: preferences in urban areas shifted away from the "process of change" narrative.

Visualization: Counter-hegemony as W^I rewiring



Left: hegemonic W^I — hub-spoke, centered on the dominant coalition's media (column 1, attention paid to the dominant outlet, is bright). Right: counter-hegemonic W^I — the dominated agents (lower-right block) have formed their own information network. Row-normalizing the cell values (`code/ch24.py`): hegemonic $\pi_{\text{dom}} = 0.37$, $N_{\text{eff}} = 4.63$, $\Gamma = 0.72$; counter-hegemonic $\pi_{\text{dom}} = 0.15$, $N_{\text{eff}} = 5.88$, $\Gamma = 0.67$. Power concentration falls, as N_{eff} shows. Γ falls slightly too: a hub everyone listens to mixes *fast*, so the hegemonic network is the better integrated one. Γ measures integration, not concentration.

24.3 State Crisis as Hegemonic Collapse

Definition 24.8 (State Crisis)**.** *A **state crisis** occurs when the hegemonic equilibrium loses stability or ceases to exist: the three loops (production, information, solidarity) decouple, and the dominant coalition can no longer maintain consent. In the co-evolutionary dynamics (Chapter 13), this corresponds to the Jacobian $\mathcal{J}$ of the hegemonic fixed point acquiring an eigenvalue of modulus ≥ 1; in the saddle-node of Proposition 24.6 the eigenvalue reaches $+1$ and the equilibrium disappears.*

This definition is the book's, not Skocpol's. Skocpol (1979) is explicitly structural and non-voluntarist: social revolutions are not made by movements with new ideas but happen when international (military and fiscal) pressure breaks the administrative and coercive apparatus of agrarian-bureaucratic states, often because dominant classes block the reforms that would meet the pressure, and when agrarian structures give peasant communities the autonomy to revolt. Her theory concerns social revolutions within these scope conditions. Bolivia 1952 fits it far better than 2003, which was a crisis of government in a democracy.

Conjecture 24.9 (Conditions for Hegemonic Collapse)**.** *The hegemonic equilibrium loses stability when one or more of the following holds (the first two are Gramscian channels; the last two are Skocpol's):*

1. ***The counter-signal crosses the threshold****: the information network becomes too diverse (N^I_{eff} rises) for the dominant coalition to control, and counter-hegemonic channels accumulate until $\mu\|\Delta(W^I\mathbf{r})\| > (1-\rho)s^*$ (Proposition 24.6).*

2. ***An exogenous shock exposes the gap****: a crisis (financial, environmental, health) reveals the allocation inequality that the hegemonic W^I had normalized. The shock changes the returns $\mathbf{r}_t$ dramatically, and the W^I-weighted signal shifts even under the existing information structure.*

3. ***State breakdown under external pressure*** *(Skocpol): international military or fiscal pressure strains the state, and dominant-class resistance to reform splits the elite and paralyzes the state's coercive and administrative capacity. In Chapter 14's terms: the $\{F, B\}$ coalition's feasibility set $\mathcal{F}(S)$ shrinks and some B agents defect.*

4. ***Revolt from below enabled by structure*** *(Skocpol): agrarian or communal structures that give the dominated autonomous organization (peasant communities, unions) allow mass revolt once the state is weakened. In Chapter 14's terms: the alliance $\{M, W\}$ or $\{B, M, W\}$ becomes feasible.*

Example 24.10 (Bolivia 2000–2006: From Crisis to Counter-Hegemony)**.** *The neoliberal hegemony collapsed through a sequence in which all four channels were present (N^I_{eff} values are* [ILLUSTRATIVE]*):*

Condition 1*: Community radio and social movements created alternative W^I channels throughout the 1990s. N^I_{eff} rose from ≈ 2.0 to ≈ 2.8.*

Condition 2*: The Water War (2000) exposed the cost of privatization (sharp tariff increases). The shock changed $\mathbf{r}_t$: the "returns to privatization" signal turned negative for poor households.*

Condition 3*: Fiscal stress weakened the state: the 1999–2002 slowdown and a large fiscal deficit led to the IMF-backed income-tax proposal of February 2003, which provoked riots and a police mutiny in La Paz. The governing coalition split.*

Condition 4*: The Gas War (2003) formed a cross-class coalition from autonomous organizations: El Alto's juntas vecinales (working class), rural cocaleros and campesino unions, and middle-class professionals united against gas export plans. The alliance $\{M, W\}$ forced President Sánchez de Lozada's resignation in October 2003; renewed protests forced President Mesa's resignation in June 2005.*

The hegemonic equilibrium was unstable by 2003. The transition to the MAS hegemony (2006) was the co-evolutionary system settling into a new structural equilibrium —

with different ownership of and fiscal claims on W^P (nationalized gas), different W^I (state media + indigenous networks), and different $\boldsymbol{\theta}^*$ (indigenous identity, state-led development, plurinationalism). The change in W^P was one of ownership and rent capture, not of production linkages: the integration of the domestic input-share network barely moved ($\Gamma = 0.518$ in 2006 and 0.504 in 2009), and the hydrocarbons sector stayed at its periphery, with a Perron weight below 0.09 in every year of 2006–2009 (INE input–output matrices, base 1990, 12 sectors; author's calculations, `data/compute/compute_bolivia_io.py`; Instituto Nacional de Estadística (INE), Bolivia, 2026; Chapter 23).

24.4 The Hegemonic Cycle

Conjecture 24.11 (The Hegemonic Cycle)**.** *In the co-evolutionary system with formation and coalition dynamics, the long-run trajectory exhibits a hegemonic cycle:*

1. ***Establishment**: A coalition wins and implements ΔW (production restructuring) + ΔW^I (narrative control). The new structure becomes a structural equilibrium. Consent is formed through W^I.*

2. ***Consolidation**: The hegemony stabilizes. Preferences converge to $\boldsymbol{\theta}^*$. Growth occurs within the structure. Inequality is accepted.*

3. ***Erosion**: N^I_{eff} increases (new media, new movements). The formation signal diversifies. Peripheral agents begin to see alternative returns. $\Delta\theta$ begins.*

4. ***Crisis**: An exogenous shock or accumulated ΔW^I pushes preferences past the basin boundary. The hegemonic equilibrium becomes unstable. The dominant coalition fragments.*

5. ***Transition**: A new coalition forms and implements a new ΔW. Return to step 1 with a different structure.*

The cycle period depends on: (a) the speed of N^I_{eff} growth (faster media diversification = shorter cycle), (b) the persistence parameter ρ (stickier preferences = longer consolidation phase), (c) the frequency of exogenous shocks (more shocks = more crisis triggers).

Remark 24.12 (What a cycle requires). This is a conjecture, not a theorem. A system whose fixed points are stable, driven by a monotonically rising N^I_{eff}, does not cycle: it moves once from one equilibrium to another and stays. A genuine cycle needs either recurring exogenous shocks or a slow feedback that reverses the drift. In model (24.3), a hysteresis loop arises if the counter-signal s drifts up slowly while the old hegemony rules (erosion), the system jumps at s^* (crisis), and

the new hegemony then uses its control of W^I to push s back down past the lower fold of the reversed model (consolidation of the new order). Writing down and estimating that slow equation is an open problem.

Intuition: The Latin American hegemonic cycle

The Latin American political cycle (Chapter 14's pendulum) is the hegemonic cycle operating at the regional scale:
Oligarchic hegemony (pre-1930): export agriculture and mining, $N^I_{\text{eff}} \approx 1$–2. Collapsed with the Great Depression (exogenous shock to W^P_{global}).
National-populist hegemony (1930–1960s): ISI, mass media controlled by the state, labor incorporation. Collapsed as the bourgeoisie defected (ISI success raised π_B, triggering alliance switch per Chapter 14).
Bureaucratic-authoritarian hegemony (1960s–1980s): military rule, controlled media (N^I_{eff} suppressed by censorship), alliance $\{F, B, M\}$. Collapsed with the debt crisis (exogenous shock) and democratization.
Neoliberal hegemony (1985–2003): Washington Consensus, private media, $N^I_{\text{eff}} \approx 2$–3. Collapsed as social movements created alternative W^I.
Left-populist hegemony (2006–2019 in Bolivia): resource nationalism, state + indigenous media, $N^I_{\text{eff}} \approx 3$. Eroding as social media diversifies W^I.
The hegemonies lasted between about 13 and 40 years. The first three collapsed after exogenous shocks (the Depression, the debt crisis) or elite defection; the information channel was decisive mainly in the last two. If digital media keep adding channels faster than any coalition can control them, the cycle may be accelerating — a hypothesis that five episodes cannot test (Exercise 24.6), and one that depends on whether platforms raise or lower N^I_{eff} (Chapter 27).

24.5 Summary

1. **Hegemony** is a structural equilibrium where the dominant coalition's preferred W is sustained by W^I-formed preferences that consent to it (the consent half of Gramsci's "consent armoured by coercion"). Three conditions: power concentration ($N^P_{\text{eff}} \ll N$), consent, formation alignment. Three reinforcing loops (W^P, W^I, W^U) make it stable.

2. The **maintenance cost** of hegemony rises with N^I_{eff} (more independent weight to counter) and falls when W^I is weakly coupled ($1 - \lambda_2$ small: the consensus is cheap for anyone to move). Concentration is measured by N^I_{eff}, never

by Γ^I: a hub everyone listens to has a large Γ.

3. **Counter-hegemony** is ΔW^I. Under conformity, it succeeds by a shock that carries preferences past the separatrix, or by a persistent signal above the saddle-node threshold, $\mu\|\Delta(W^I\mathbf{r})\| > (1-\rho)s^*$.

4. **State crisis** combines Gramscian channels (counter-signals, shocks that expose inequality) with Skocpol's structural ones (state breakdown under external pressure, revolt enabled by autonomous organization). Bolivia 1952 fits Skocpol's scope; 2000–2005 was a crisis of government.

5. The **hegemonic cycle** (establishment $\to$ consolidation $\to$ erosion $\to$ crisis $\to$ transition) is a conjecture: it requires recurring shocks or a slow reversing feedback, which remains to be modelled and estimated.

24.6 Exercises

24.1 Hegemony as fixed point. For a 2-group economy (Elite E, Masses M): write the co-evolutionary system (Chapter 13) with W^I controlled by E. Find parameter values under which a fixed point of this system satisfies all three conditions of Definition 24.1, and is therefore a hegemonic equilibrium (Definition 24.2; that definition holds by construction, so the work is in finding the fixed point and checking the conditions). Under what parameter values does a counter-hegemonic equilibrium also exist?

24.2 Maintenance cost. For Bolivia's MAS hegemony (2006–2019): estimate the cost of W^I control (state media budget, community communication spending). How does it compare with the government's total budget? At what N^I_{eff} does the cost exceed the coalition's willingness to pay?

24.3 Counter-hegemony simulation. Start from a hegemonic equilibrium. Add counter-hegemonic ΔW^I edges incrementally ($\Delta\lambda^I = +0.02$ per period). Simulate 50 periods. At what period does the hegemonic equilibrium collapse? How does this depend on ρ?

24.4 Bolivia's hegemonic transitions. Map all five hegemonies from the Prebisch Connection box to the structural power game (Chapter 14): for each, identify the coalition, the W^I control mechanism, the N^I_{eff}, and the trigger for collapse.

24.5 Skocpol's conditions. For the 2000–2006 Bolivian transition: verify each of the four conditions in Conjecture 24.9. Which was the binding condition? Could the neoliberal hegemony have survived if any one condition had not been met?

24.6 The acceleration of the cycle. Plot the duration of each Latin American hegemonic epoch (from the Prebisch Connection box) against N_{eff}^{I}. Is there a negative correlation (more information diversity = shorter hegemonic periods)? What does this predict for the current period?

24.7 Gramsci vs. Skocpol. Gramsci emphasized consent (the formation of preferences). Skocpol emphasized structural conditions (fiscal crisis, military defeat, elite splits). In the model: Gramsci's hegemony is the W^I loop (preference formation); Skocpol's crisis is the W^P loop (economic disruption) + the coalition loop (elite defection). Are these complementary or competing? Under what parameter values does each mechanism dominate?

Mathematical Appendices

Appendix 24A: Stability of the Hegemonic Equilibrium

The hegemonic fixed point. Let $(W^*, \boldsymbol{\theta}^*, \mathbf{x}^*)$ be a hegemonic equilibrium. The stability is governed by the eigenvalues of the co-evolutionary Jacobian (Chapter 13, Appendix 13B):

$$\mathcal{J} = \begin{pmatrix} \partial T/\partial \boldsymbol{\theta} & \partial T/\partial W \\ \partial \mathcal{E}/\partial \boldsymbol{\theta} & \partial \mathcal{E}/\partial W \end{pmatrix}. \tag{24.5}$$

The hegemony is stable if $\rho(\mathcal{J}) < 1$. It loses stability when:

1. The formation feedback is too strong: $\rho(J) \to 1$, where $J = \rho I + \mu\phi W^I CP$ is the formation Jacobian (Proposition 10.2). This may occur as counter-hegemonic channels strengthen the conformity feedback; in the scalar model (24.3) the relevant eigenvalue is $\rho + (1-\rho)g'(\theta_h)$, which reaches 1 exactly at the saddle-node.

2. An exogenous shock changes the returns $\mathbf{r}$ suddenly, shifting the formation map T and moving the system outside the basin of attraction.

3. The coalition dynamics (Chapter 14) produce an alliance switch that changes W discretely.

Bifurcation. The relevant bifurcation is a **saddle-node** (fold), not a pitchfork. In model (24.3), as the counter-signal s rises the stable hegemonic fixed point and the unstable separatrix approach each other, merge at s^* ($s^* = 0.181$ for $\beta = 10$; `code/ch24.py`) and vanish. Below s^*: two stable equilibria (hegemony and counter-hegemony) separated by the separatrix, so a large enough shock can push the system across. Above s^*: only the counter-hegemonic equilibrium survives. Running s back down does not restore the old hegemony until a second fold is reached, at $s = -s^*$ (a pro-hegemonic signal; by the symmetry of g): hysteresis, which is why hegemonic transitions are not reversed by small reversals of the signal.

Appendix 24B: Gramsci's Categories in Matrix Notation

Hegemony. A structural equilibrium where the dominant group's power ($\boldsymbol{\pi}$) is reproduced by W^I-formed consent ($\boldsymbol{\theta}^*$). For Gramsci consent never stands alone: it is "armoured by coercion", and the state is political society plus civil society. The model captures the consent half; coercion is the coalition's reserve option.

Historic bloc. The alignment of a social coalition, its economic base and its worldview into a durable unit. In this chapter: the triple alignment of loops 1–3 (production, information, solidarity) around one coalition. The formalization is ours; Gramsci's concept is broader.

Passive revolution. Transformation directed from above that absorbs the leaders and demands of subaltern groups without their autonomous mobilization: in Gramsci's phrase, a "revolution-restoration". It is arguably the Gramscian concept most used for Latin America. In this chapter: a ΔW implemented by the dominant coalition that co-opts the counter-hegemonic channels, moving $\boldsymbol{\theta}$ toward consent before s reaches s^*.

Organic intellectuals. For Gramsci, intellectuals tied organically to a class — including subaltern classes — who give it homogeneity and awareness of its function; they are defined by that tie, not by audience size. In this chapter: the agents, on either side, whose position in W^I lets them shape a class's formation signal. When they belong to the dominant coalition they hold high π_j^I, and in Chapter 8's terms they are the DeGroot influencers whose beliefs weigh most in the consensus.

War of position. The slow process of building alternative W^I channels (counter-hegemonic institutions: schools, media, unions, parties). In this chapter: the gradual increase of N_{eff}^I through new information edges. The "war of position" IS the accumulation of ΔW^I over time.

War of maneuver. The rapid seizure of state power once the counter-hegemonic movement has sufficient support. In this chapter: the moment when preferences cross the basin boundary and the demand for structural change becomes politically irresistible. The "war of maneuver" IS the coalition switch (Chapter 14) triggered by the preference shift.

Interregnum. The period between the collapse of one hegemony and the establishment of the next: "the old is dying and the new cannot be born" (Q3 §34). In this chapter: the co-evolutionary system is between structural equilibria. W^I is contested; $\boldsymbol{\theta}$ is in flux; W^P is unstable. This is the period of maximum vulnerability — of Gramsci's "morbid symptoms" (political instability, democratic backsliding, authoritarianism).

Structuralist Case Study: Gramsci and Skocpol on Structural Crisis

Antonio Gramsci (1891–1937) and Theda Skocpol (b. 1947) represent two approaches to the same question: why do structures of domination persist, and how do they collapse? Gramsci emphasized consent and cultural production. Skocpol emphasized structural conditions and state capacity. This chapter argues that both are correct about different loops of the same co-evolutionary system.

Gramsci: hegemony as formation. Gramsci, writing from Mussolini's prison, asked why the Italian working class accepted capitalist domination even when it was materially disadvantaged. His answer: hegemony — the process by which the dominant class's worldview becomes the "common sense" of society. In this book's notation: hegemony is the steady state of the formation model where W^I produces preferences that consent to the W^P-determined allocation. Gramsci's "common sense" is $\boldsymbol{\theta}^*$; the formative position of his organic intellectuals is a position in W^I; his "war of position" is the accumulation of counter-hegemonic ΔW^I.

This is a formal reinterpretation, and Gramsci would resist parts of it. The mechanism of Chapters 8–10 — ideas produced by specific agents, disseminated through specific channels (media, schools, churches = W^I edges), and internalized as preferences — captures the consent side of his concept. It leaves out coercion (hegemony is "armoured by coercion"), the historic bloc that welds an economic base to a worldview, and passive revolution, the reorganization from above that absorbs subaltern demands (Appendix 24B). The information structure is the part of hegemony the model can see.

Skocpol: crisis as structural breakdown. Skocpol, analyzing the French, Russian, and Chinese revolutions, argued against voluntarist explanations, borrowing Wendell Phillips's line that "revolutions are not made; they come." Social revolutions occurred when international military and fiscal competition broke the administrative and coercive apparatus of agrarian-bureaucratic old regimes — often because landed dominant classes blocked the reforms that would have met the pressure — and when agrarian structures gave peasant communities the solidarity and autonomy to revolt. Ideas and movements are, in her account, not the cause. In this chapter's notation: state crisis is the co-evolutionary system losing stability because the dominant coalition's resource base (W^P position, $\mathcal{F}(S)$) collapses under external pressure and its members split, while the structure of W^U among the dominated (communal solidarity) makes revolt feasible.

The mapping: international pressure = an exogenous shock to the global W^P and to the fiscal capacity of the state (shrinking $\mathcal{F}(S)$). Dominant-class obstruction of reform = B and F refusing the ΔW that would meet the shock, and splitting

over it (Chapter 14). Peasant revolt = autonomous organization of the dominated (high within-group W^U) once the state is weakened. The information channel, central for Gramsci, plays no causal role in Skocpol; this chapter's Conjecture 24.9 combines the two, which is our synthesis, not hers. Her scope conditions fit Bolivia 1952 far better than the 2003–2005 crisis of government.

The synthesis. Gramsci explains why structures persist (the formation loop produces consent). Skocpol explains when they collapse (structural conditions destabilize the fixed point). Both are needed: without Gramsci, we cannot explain why the dominated accept the structure during "normal" times. Without Skocpol, we cannot explain the timing of crises. The co-evolutionary system offers a way to combine them: hegemony as a stable fixed point (Gramsci's consent), crisis as its loss of stability (driven, on Skocpol's account, by external pressure on the state rather than by ideas).

Concept	**Gramsci**	**Skocpol**
Central question	Why do the dominated consent?	When does the state collapse?
Mechanism	W^I forms preferences (consent, armoured by coercion)	External pressure breaks the state; agrarian structure enables revolt
Formal object	Hegemonic equilibrium (Def. 24.2)	Stability loss (Conj. 24.9)
Time horizon	Long-run (generations)	Medium-run (decades)
Policy variable	ΔW^I (counter-hegemony)	Exogenous shock + coalition
Bolivia application	MAS narrative control	1952 revolution (best fit)

Gramsci wrote from prison. Skocpol wrote from Harvard. Between them, they describe the two phases of structural politics: the long war of position (building alternative W^I) and the sudden war of maneuver (the crisis that topples the structure). The co-evolutionary system of this book is one way to put both in a single dynamical model.

Further Reading

Gramsci: Gramsci (1971), *Selections from the Prison Notebooks*, edited by Q. Hoare and G. N. Smith. For the Latin American reception: Portantiero (1981), *Los usos de Gramsci*. For the application to media: Hall (1986), "Gramsci's Relevance for the Study of Race and Ethnicity."

Skocpol: Skocpol (1979), *States and Social Revolutions: A Comparative Analysis of France, Russia, and China*. For the extension to the "third wave": Goodwin (2001), *No Other Way Out: States and Revolutionary Movements*.

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Chapter 25

Truncated Industrialization

The Missing Dense Subgraph

> *"La industrialización latinoamericana se caracteriza por ser una industrialización trunca: carece del núcleo endógeno de dinamización del progreso técnico — el sector de bienes de capital."*
>
> ["Latin American industrialization is characterized as truncated industrialization: it lacks the endogenous nucleus for the dynamization of technical progress — the capital goods sector."]
>
> — Fernando Fajnzylber, *La industrialización trunca de América Latina* (1983)

Fajnzylber (1983) identified a specific structural absence at the heart of Latin American economies: the capital goods sector. Not just any sector — the sector that produces the machines that produce other goods. The sector where learning-by-doing, scale economies, and technological spillovers are concentrated. The sector that, in East Asia, served as the engine of structural transformation. The sector that Latin America never developed.

In this book's notation: the capital goods sector is identified with the **dense core** of the product space — the cluster of products (machinery, metal products, electronics, chemicals) where the proximity matrix $\phi_{pp'}$ is densest. It is the bright core of the heatmap. And Latin America's heatmap has a hole where that core should be. (The identification is ours: Fajnzylber's nucleus is a sector *plus* the domestic engineering and innovation capacity around it, not a set of export products.)

This chapter formalizes Fajnzylber's diagnosis in terms of missing subgraphs. One distinction runs through it: a *dense module* (products tightly related to each

other) and a *bridge* (products that connect otherwise separate clusters) have opposite effects on integration Γ. Removing a bridge provably lowers Γ; removing a dense module can raise it. Whether the capital-goods core is a bridge is therefore an empirical question, stated below as a conjecture. The product-space quantities (the truncation index τ, the size and density of $\mathcal{D}$, relatedness ω) are [ILLUSTRATIVE]. Complexity (ECI, PCI), GDP per capita and the domestic Γ of Bolivia and its comparators are sourced, and `code/ch25.py` reads them from `data/processed`. At the 12-sector level those sourced values give the chapter's central conjecture no support (Section 25.1.3).

25.1 The Dense Subgraph

25.1.1 Definition

Definition 25.1 (Dense Subgraph of the Product Space)**.** *Let ϕ be the symmetric product-proximity matrix ($\phi_{pp'} \in [0,1]$, diagonal excluded). For a set of products S, let $e(S) = \sum_{p<p',\, p,p' \in S} \phi_{pp'}$ be its internal weight and*

$$\bar{\phi}_S = \frac{2\, e(S)}{|S|(|S|-1)}$$

*its average internal proximity. The **dense subgraph** is the maximizer of internal weight per product,*

$$\mathcal{D} = \arg\max_{S \subseteq \{1,\dots,P\}} \frac{e(S)}{|S|}, \tag{25.1}$$

*(Goldberg's densest-subgraph problem, solvable exactly by max-flow; Charikar's greedy peeling gives a 2-approximation; Appendix 25.6). Report alongside it the **density ratio** $\bar{\phi}_{\mathcal{D}} / \bar{\phi}_{all}$ and $|\mathcal{D}|$, and check robustness to the product classification.*

Excluding the diagonal matters: if $\phi_{pp} = 1$ is included in an average over $|S|^2$ pairs, every single product has density 1 and the "densest" set is a single product. Note also what the definition does *not* say: a dense set is a set whose members are related to each other; it need not be related to anything else.

Stylized Fact 25.2 (The Product-Space Core)**.** *Hidalgo et al. (2007) built the product space from about 775 products at the SITC (rev. 2) 4-digit level and found a densely connected core of metal products, machinery and chemicals, with a sparse periphery of primary commodities. The quantitative statements below are* [ILLUSTRATIVE] *placeholders for that structure, to be replaced by the output of Exercise 25.1:*

1. *The densest subgraph contains machinery (non-electrical and electrical), precision and optical instruments, metal products, organic chemicals, and plastics. Average internal proximity:* $\bar{\phi}_{\mathcal{D}} \approx 0.35$.

2. *The average proximity in the full product space:* $\bar{\phi}_{all} \approx 0.08$.

3. *The density ratio:* $\bar{\phi}_{\mathcal{D}} / \bar{\phi}_{all} \approx 4.4$. *The capital goods cluster is 4.4 times denser than the product space average.*

4. *Products in* $\mathcal{D}$ *have the highest average PCI (Product Complexity Index):* $\overline{PCI}_{\mathcal{D}} \approx 1.8$, *compared with* $\overline{PCI}_{all} = 0$ *by construction (PCI is standardized across products).*

25.1.2 The Missing Subgraph

Definition 25.3 (Truncated Product Space)**.** *A country c's product space is* ***truncated*** *if it lacks the dense subgraph:* $M_{cp} = 0$ *for most* $p \in \mathcal{D}$. *The* ***truncation index****:*

$$\tau_c = 1 - \frac{|\{p \in \mathcal{D} : M_{cp} = 1\}|}{|\mathcal{D}|} \in [0, 1]. \tag{25.2}$$

$\tau_c = 0$*: the country has the complete dense subgraph.* $\tau_c = 1$*: the country has none of it. Illustrative values: Latin America* $\tau \approx 0.70$–0.95*; East Asia* $\tau \approx 0.10$–0.40*; Bolivia* $\tau \approx 0.98$.

To connect the product space to integration we need an assumption about how the domestic production network relates to it.

Assumption (product-space proxy). The domestic network of country c is proxied by the proximity among the products c produces: $W_c = D_c^{-1}\phi_c$, where ϕ_c is ϕ restricted to c's products and $D_c = \mathrm{diag}(\phi_c \mathbf{1})$. This W_c is row-stochastic and reversible, so its eigenvalues are real (we assume c produces at least two products and every product has positive proximity to some other product, so that D_c is invertible). Order them $1 = \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N$: in this section $\lambda_2(W_c)$ is the second-*largest* eigenvalue, which differs from the eigenvalue of second-largest modulus that defines Γ when $|\lambda_N| > |\lambda_2|$. (It is a proxy: the input-share matrix $W^P = D^{-1}A$ of an I-O table is a different object, and Exercise 25.3 asks how well the proxy tracks it.)

For a set S of c's products, the **conductance** of the cut (S, S^c) is $\Phi_c(S) = \phi_c(S, S^c) / \min\big(\mathrm{vol}(S), \mathrm{vol}(S^c)\big)$, where $\phi_c(S, S^c)$ is the total proximity across the cut and $\mathrm{vol}(S) = \sum_{p \in S} (\phi_c \mathbf{1})_p$.

Proposition 25.4 (Removing a Bridge Bounds Γ)**.** *Under the proxy assumption, for every cut (S, S^c) of c's products,*

$$\Gamma(W_c) \;\le\; 1 - \lambda_2(W_c) \;\le\; 2\,\Phi_c(S). \tag{25.3}$$

Hence if c lacks a set $\mathcal{B}$ of products that bridges *two clusters S and S^c — so that, without $\mathcal{B}$, the direct proximity across the cut is small — then c's integration is at most twice that small conductance, however large the conductance of the same cut would be with $\mathcal{B}$ present.*

Proof. First inequality. The eigenvalue of second-largest modulus has modulus $\max(|\lambda_2|, |\lambda_N|) \ge \lambda_2$, so $\Gamma(W_c) \le 1 - \lambda_2(W_c)$.

Second inequality (the easy half of Cheeger's inequality for reversible chains; Lawler and Sokal, 1988; Sinclair and Jerrum, 1989). Write $\phi = \phi_c$, $D = D_c$, $\mathbf{d} = \phi\mathbf{1}$. $W_c = D^{-1}\phi$ is similar to the symmetric matrix $M = D^{-1/2}\phi D^{-1/2}$, whose top eigenvector is $D^{1/2}\mathbf{1}$. By Courant–Fischer, with $\mathbf{g} = D^{1/2}\mathbf{f}$,

$$\lambda_2 = \max_{\mathbf{g}\perp D^{1/2}\mathbf{1},\, \mathbf{g}\neq 0} \frac{\mathbf{g}'M\mathbf{g}}{\mathbf{g}'\mathbf{g}} \quad\Longrightarrow\quad 1-\lambda_2 = \min_{\mathbf{f}'\mathbf{d}=0,\, \mathbf{f}\neq 0} \frac{\mathbf{f}'D\mathbf{f} - \mathbf{f}'\phi\mathbf{f}}{\mathbf{f}'D\mathbf{f}} = \min_{\mathbf{f}'\mathbf{d}=0,\, \mathbf{f}\neq 0} \frac{\tfrac{1}{2}\sum_{p,q}\phi_{pq}(f_p - f_q)^2}{\sum_p d_p f_p^2},$$

where the last step uses $d_p = \sum_q \phi_{pq}$ and the symmetry of ϕ. Take a cut with S and S^c nonempty (so $\mathrm{vol}(S), \mathrm{vol}(S^c) > 0$) and the test vector $\mathbf{f} = \mathbf{1}_S/\mathrm{vol}(S) - \mathbf{1}_{S^c}/\mathrm{vol}(S^c)$, for which $\mathbf{f}'\mathbf{d} = 1 - 1 = 0$. Only pairs across the cut contribute to the numerator, each with $(f_p - f_q)^2 = \kappa^2$, $\kappa := 1/\mathrm{vol}(S) + 1/\mathrm{vol}(S^c)$, so the numerator equals $\phi(S, S^c)\,\kappa^2$ (each unordered cross pair is counted twice in the double sum, cancelling the $\frac{1}{2}$). The denominator is $\mathrm{vol}(S)/\mathrm{vol}(S)^2 + \mathrm{vol}(S^c)/\mathrm{vol}(S^c)^2 = \kappa$. Hence

$$1 - \lambda_2 \le \phi(S, S^c)\Big(\frac{1}{\mathrm{vol}(S)} + \frac{1}{\mathrm{vol}(S^c)}\Big) \le \frac{2\,\phi(S, S^c)}{\min(\mathrm{vol}(S), \mathrm{vol}(S^c))} = 2\,\Phi_c(S).$$

The "hence" clause applies this to c's product set, from which $\mathcal{B}$ is absent, so $\Phi_c(S)$ is the conductance without $\mathcal{B}$. `code/ch25.py` checks both inequalities on random symmetric proximity matrices and all cuts. □

Remark 25.5 (A dense module is not a bridge). The proposition is about bridges, not about dense sets. In `code/ch25.py`, two ten- and thirteen-product examples have a dense core $\mathcal{D}$ of four products:

- *Bridge*: the core connects three periphery clusters. $\Gamma = 0.548$ with $\mathcal{D}$, 0.009 without it; without $\mathcal{D}$ the block cut has conductance 0.006, so (25.3) gives $\Gamma \le 0.012$.

- *Module*: the core is dense (density ratio 3.65) but weakly tied to a well-mixed periphery. $\Gamma = 0.218$ with $\mathcal{D}$, 0.800 without it: removing the dense set *raises* integration, because the module itself was the bottleneck.

The densest-subgraph algorithm finds $\mathcal{D}$ in both cases; it cannot tell them apart. The old "truncation theorem" (removing $\mathcal{D}$ cannot raise Γ, by interlacing) is false: interlacing applies to principal submatrices of symmetric matrices, not to renormalized row-stochastic ones.

Conjecture 25.6 (Truncation Lowers Integration)**.** *In the observed product space the capital-goods core is a bridge set: without it, the clusters that make up a commodity exporter's production (mining, agriculture, services) are connected only through low-conductance cuts. If so, by Proposition 25.4, domestic integration is bounded above by a decreasing function of the truncation index,*

$$\Gamma(W_c) \leq f(\tau_c), \qquad f'(\tau) < 0.$$

This is testable (Exercise 25.3): measure the participation of $\mathcal{D}$'s products across clusters (betweenness, or the Guimerà–Amaral participation coefficient), compute the conductance of the commodity clusters with and without $\mathcal{D}$, and regress Γ_c on τ_c.

Example 25.7 (The Missing Core: Bolivia vs. South Korea)**.**

	Bolivia	*South Korea*
Truncation index $\tau^{\dagger}$	*0.98*	*0.18*
Products in $\mathcal{D}$ *with RCA*†	*2 of* $\approx$*100*	*82 of* $\approx$*100*
ECI (2019, HS92)	-1.30	$+1.81$
$\Gamma(W^P)$ *(2019, 12 sectors, like-for-like)*	*0.475*	*0.511*
${\sigma_B}^{\dagger}$	*0.62*	*0.20*
GDP per capita (PPP, current intl. \$, 2019)	*\$11,005*	*\$46,511*

†[ILLUSTRATIVE]: *τ and the counts in $\mathcal{D}$ need country $\times$ product RCA data, not computed here; σ_B needs household microdata. Source for the other rows: ECI, Growth Lab, Atlas of Economic Complexity, HS92 country-year data (The Growth Lab at Harvard University, 2026); Γ, domestic input-share W^P on the ISIC-based 12-sector aggregation, basic prices, diagonal kept, Bolivia from the INE supply-use tables, base 2017 (Instituto Nacional de Estadística (INE), Bolivia, 2026), Korea from OECD ICIO 2023 (OECD, 2023), author's calculations (data/compute/compute_bolivia_io.py, data/compute/compute_oecd_io.py); GDP per capita, WDI series NY.GDP.PCAP.PP.CD (World Bank, 2026b).*

On the illustrative counts, Bolivia has 2 products in the dense subgraph (tin processing, some basic chemicals) and South Korea has 82. The sourced rows say something sharper. The complexity gap is large: Bolivia ranks 131st on ECI in 2019, Korea 2nd. The integration gap is not: at the 12-sector level Bolivia's domestic W^P is about as integrated

as Korea's, and more integrated than that of the United States ($\Gamma = 0.463$). An earlier draft of this example described Bolivia's W^P as three weakly connected blocks (gas/mining, urban services, rural agriculture); the INE tables do not show that. What they show is an enclave. Hydrocarbons, the leading export sector, have a domestic power share $\pi = 0.039$, export 70% of their output, and buy 0.225 of domestic intermediate inputs per unit of output (manufacturing: 0.558) (Instituto Nacional de Estadística (INE), Bolivia, 2026). The enclave is measured by π and by export orientation, not by Γ.

25.1.3 What the 12-Sector Data Say

Empirical Result 25.8 (No Support for the Conjecture at 12 Sectors)**.** *In 2019, on a common 12-sector classification (domestic flows, basic prices, input shares, diagonal kept):*

	BOL	*KOR*	*DEU*	*USA*	*CHL*	*CHN*	*MEX*
$\Gamma(W^P)$	*0.475*	*0.511*	*0.544*	*0.463*	*0.557*	*0.582*	*0.649*
ECI (HS92)	-1.30	$+1.81$	$+1.54$	$+1.29$	-0.06	$+1.30$	$+0.95$

Source: as Example 25.7; comparators' Γ from OECD ICIO 2023 (OECD, 2023), author's calculations (data/compute/compute_oecd_io.py).

1. Cross-section. *If truncation lowered integration, Bolivia, the least complex economy in the table, should have the lowest Γ. It sits inside the comparator range, above the United States.*

2. Time series. *Bolivia's Γ fell from 0.567 (1990) to 0.484 (2013) (INE base 1990; the base-2017 value of 0.467 for 2019 is not strictly comparable). Korea's fell from 0.626 (1995) to 0.511 (2019) (OECD ICIO, 2021 and 2023 editions), while its ECI rose from $+1.10$ to $+1.81$. Aggregate Γ does not separate the two development paths.*

3. Measurement. *The Bolivia–Korea gap in Γ (0.04) is an order of magnitude smaller than the range of Bolivia's own 2019 Γ across normalizations (0.33–0.69: input or sales shares, with or without the diagonal, 12 or 39 sectors).*

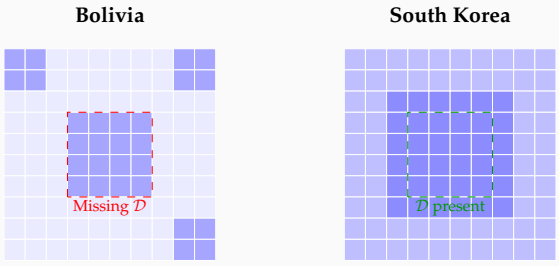
The result does not refute Conjecture 25.6, but it removes the evidence this chapter used to cite for it, and the chapter's quantitative claims about Bolivia's "integration deficit" are withdrawn. The conjecture is about bridges in the product space, and a 12-sector table cannot see them: the capital-goods core sits inside "Manufacturing", and aggregation can raise Γ (Chapter 19). Two finer tests are needed.

1. *Industry level.* INE's 39-industry detail against comparator tables at the same detail, with the capital-goods industries as separate nodes: compute the conductance of the commodity clusters with and without them. The detail values available now are not comparable (Bolivia 0.410 at INE's 39 industries, Korea 0.455 at ICIO's 45-industry detail); a concordance is needed first.

2. *Product level.* τ_c from Growth Lab country × product RCA data, the proxy W_c, the betweenness and participation of $\mathcal{D}$'s products, and the regression of Γ_c on τ_c (Exercise 25.3).

Until one of them is done, Conjecture 25.6 is a conjecture without empirical support. The complexity gap is real; whether it is also an integration gap is open.

Visualization: The product space: Bolivia vs. South Korea



Bolivia (left): the periphery is lit (commodities), but the dense core (dashed red box) is dark. This is the "truncated" product space — the missing subgraph. South Korea (right): the core is the brightest region, with moderate brightness extending outward. This is the "complete" product space. The panels are schematic (shading marks product presence, not matrix entries). The measured domestic Γ is 0.475 for Bolivia and 0.511 for Korea (Result 25.8): the difference the panels draw is a difference in complexity, and it does not show up in 12-sector integration.

25.2 Why the Dense Subgraph Matters for Γ

Remark 25.9 (The Capital Goods Sector as Structural Bridge). The capital goods sector serves three structural functions that no other sector performs:

1. **Backward linkages to all sectors**: Capital goods require inputs from metals, chemicals, plastics, electronics, and precision engineering. The sector's demand creates edges to upstream blocks that would otherwise be disconnected.

2. **Forward linkages to all sectors**: Every sector uses capital goods. Agriculture needs tractors; services need computers; mining needs drilling equipment.

The sector's output creates edges to downstream blocks.

3. **Knowledge spillovers**: The capital goods sector is where process innovation occurs — improvements in how things are made, not just what is made. These spillovers propagate through W^I (engineers and technicians who move between sectors carry knowledge) and through W^P (better machines improve productivity in the sectors that use them).

In graph-theoretic terms, functions 1–2 are a claim that the capital goods sector has high **betweenness centrality** (it sits on the shortest paths between many pairs of products) and a high participation coefficient (its links are spread across clusters). If so, removing it fragments the product space into weakly connected clusters, and Proposition 25.4 applies. These are empirical claims about ϕ, distinct from the density of $\mathcal{D}$ (Remark 25.5), and they are what Conjecture 25.6 needs.

How large is the cost of truncation? An earlier draft measured it as a welfare loss proportional to $(1-\Gamma)/\Gamma$. That rested on a claimed bound of $\mathrm{Var}(\boldsymbol{\pi})$ by Γ, which is false: the distributional loss is $(\kappa N/2)\Phi^2 \mathrm{Var}(\boldsymbol{\pi})$ and is unrelated to Γ (Proposition 6.16). What $(1-\Gamma)/\Gamma$ does measure is *integration*: $1/\Gamma$ is the relaxation time of W (the number of periods over which a shock's footprint decays by a factor e, to first order), so $(1-\Gamma)/\Gamma = 1/\Gamma - 1$ is the *excess relaxation time* over a fully integrated network.

Example 25.10 (The Truncation Penalty in Relaxation Time: A Stylized Economy [ILLUSTRATIVE])**.** *Define the truncation penalty as the part of the excess relaxation time removed by acquiring $\mathcal{D}$:*

$$P_{trunc} = \frac{1-\Gamma(\tau_c)}{\Gamma(\tau_c)} - \frac{1-\Gamma(0)}{\Gamma(0)}. \tag{25.4}$$

For a stylized truncated economy ($\tau = 0.98$, $\Gamma = 0.13$, the value an earlier draft attributed to Bolivia) against a counterfactual with the full dense subgraph ($\Gamma(0) \approx 0.35$):

$$P_{trunc} = \frac{0.87}{0.13} - \frac{0.65}{0.35} = 6.69 - 1.86 = 4.83, \tag{25.5}$$

which is $4.83/6.69 = 72\%$ of its excess relaxation time. The share is driven by the assumed counterfactual $\Gamma(0)$, which nobody observes (`code/ch25.py`):

$\Gamma(0)$	*0.25*	*0.30*	*0.35*	*0.45*	*0.52*
Share of excess relaxation time	*55%*	*65%*	*72%*	*82%*	*86%*

The range is wide: truncation would account for between about half and nine-tenths of this economy's integration deficit, if Conjecture 25.6 holds.

Bolivia is not this economy. Its measured Γ *(0.475, Result 25.8) gives an excess relaxation time of* $1/0.475 - 1 = 1.11$*, against Korea's* $1/0.511 - 1 = 0.96$*. With Korea's* Γ *as the counterfactual, the penalty would be 13% of Bolivia's excess relaxation time, a smaller effect than a change of normalization produces. At the 12-sector level there is no integration deficit for truncation to explain.*

Key Structural Insight

If Conjecture 25.6 holds, the missing capital goods sector is not one gap among many but the central one. Its absence would explain:

1. Why Γ is low (the sector that bridges blocks is missing). For Bolivia this prediction fails at the 12-sector level: its Γ is not low (Result 25.8).
2. Why σ_B is high (without the bridge, the blocks are independent, and between-block inequality dominates). Untested here: σ_B needs household microdata.
3. Why ECI is negative (the missing products are precisely the highest-PCI products). This prediction the data bear out: Bolivia's ECI is -1.30 in 2019, 131st.
4. Why the export sector's multiplier is small (the sector with the highest backward linkages is absent). In INE's 2019 tables the output multiplier (Rasmussen backward linkage, domestic flows) is 1.34 for hydrocarbons and 1.86 for manufacturing (Instituto Nacional de Estadística (INE), Bolivia, 2026).
5. Why the product space offers few diversification paths (the dense core, which is the hub of the product space, is missing — relatedness density to core products is near zero).

Fajnzylber saw this in 1983. The product space literature rediscovered it in 2007. This chapter proves the part that is mathematics (removing a bridge bounds Γ by twice the remaining conductance) and states the part that is empirical as a conjecture. The first test the data allow, 12-sector Γ, gives that conjecture no support. The complexity gap is real; whether it is also an integration gap must be tested at finer resolution.

25.3 Why Latin America's Industrialization Was Truncated

Remark 25.11 (Reasons for Truncation).

1. **ISI stalled before the capital-goods core** (Chapter 15, Tavares). Import substitution began with consumer goods and moved toward intermediates and capital goods, but in most countries it stalled before a domestic capital-goods sector with its own technological capacity emerged. Tavares's (1964) analysis of Brazil attributed the stalling to the external constraint and to a market narrowed by income concentration. In operator terms, $\mathcal{L}_{\text{ISI}}$ densified the consumer block internally more than it created cross-block entries (intermediates $\to$ capital goods).

2. **The coalition blocked it** (Chapter 14). Capital-goods production requires massive investment, long time horizons, and state support for infant industries. The industrial bourgeoisie (B) preferred to import capital goods (cheaper in the short run) than to invest in producing them domestically. The coalition $\{B, M, W\}$ chose consumer ISI because it generated immediate consumer-goods availability (politically popular) rather than capital-goods production (politically costly, economically transformative).

3. **The formation channel maintained the status quo** (Chapter 10). Without a domestic capital-goods sector, the information signals in W^I did not include "returns to precision engineering" or "opportunities in advanced manufacturing." Aspirations and educational investments were calibrated to the existing product space: commodities, basic manufactures, services. The formation channel produced preferences consistent with the truncated structure, preventing the demand for capital-goods capabilities from emerging endogenously.

4. **Fajnzylber's own mechanisms.** Fajnzylber (1983) emphasized two further causes: an *imitative consumption pattern*, in which high-income groups copied the durable-goods consumption of the United States, steering industry toward assembling those goods rather than toward the machines that make them; and the *leadership of transnational corporations* in the dynamic industries, which imported their technology from headquarters and had little reason to build a local capital-goods and engineering base, while national industrial leadership remained weak.

Example 25.12 (The Divergence: Latin America vs. East Asia [ILLUSTRATIVE])**.** *In 1960 the large Latin American economies (Brazil, Mexico, Argentina) were richer and*

more industrialized than South Korea and Taiwan. Both regions pursued ISI. The divergence:

***East Asia**: Targeted the dense subgraph deliberately. South Korea's Heavy and Chemical Industrialization (HCI) drive (1973–1979) created steel, shipbuilding, petrochemicals, machinery, and electronics from scratch. Taiwan's state research institute ITRI spun off UMC (1980) and TSMC (1987), anchored in the Hsinchu Science Park (1980). The strategy was politically costly (the Korean government bet the economy on heavy industry, with enormous fiscal risk) but structurally transformative: τ fell from ≈ 0.70 to ≈ 0.20 in 20 years. Whether Korea's domestic Γ rose during the HCI drive cannot be checked here: comparable input-output tables for Korea start in 1995 (Chapter 23), and from 1995 to 2019 Korea's Γ* fell *(0.626 to 0.511) while its ECI rose.. From 1995, when the OECD ICIO data begin, Korea's 12-sector Γ* fell*, from 0.626 to 0.511 in 2019, while its ECI rose from $+1.10$ to $+1.81$. On the data available, deepening the core did not show up as rising aggregate Γ.*

***Latin America**: Mostly stalled at consumer and intermediate ISI. Brazil came closest: its Second National Development Plan (II PND, 1974–1979) was an explicit push into capital goods and basic inputs, and it built Embraer and Petrobras's engineering capacity. But the push was debt-financed and was cut short by the debt crisis of the 1980s before the core reached critical mass. Mexico built maquiladoras (assembly of imported components) — a within-block operator that imported the dense subgraph as a service rather than building it domestically. Argentina deindustrialized from the 1970s onward. Bolivia never even attempted it.*

The result: East Asia entered the dense subgraph and achieved "genuine competitiveness" (Fajnzylber's term). Latin America remained truncated, with consumer manufacturing but no capital-goods core. The casillero vacío (Chapter 16) — the empty cell of growth and *equity — is, on this chapter's conjecture and the conjecture of Chapter 26, the consequence: without the dense subgraph, ECI stays low, as it has (2019: Chile -0.06, Bolivia -1.30, against Korea $+1.81$). The conjecture adds that Γ stays low too. The 12-sector data do not show this: Argentina's Γ is 0.522, Chile's 0.557 and Mexico's 0.649, all above Korea's 0.511.*

25.4 The Path to the Dense Subgraph

Remark 25.13 (Three Strategies for Building $\mathcal{D}$). 1. **The Korean strategy: brute-force creation.** The state invests directly in capital-goods production (steel mills, semiconductor fabs, machinery plants). High cost, high risk, high reward. Requires: a strong developmental state, fiscal capacity ($\pi_{\text{government}} > 0.2$), political insulation from short-run pressures (authoritarian or consensus-

based governance). Applicable to: large economies with pre-existing manufacturing bases (Brazil, Mexico, Indonesia — not Bolivia).

2. **The Singapore strategy: attract the subgraph.** Foreign direct investment brings the dense subgraph to the country. Multinational corporations build plants that produce capital goods using domestic labor. Over time, capabilities transfer to domestic firms. Requires: human capital (educated workforce), institutional quality (property rights, contract enforcement), geographic advantage (proximity to markets). Applicable to: small open economies with high human capital (Costa Rica, Uruguay — possibly Bolivia for specific niches).

3. **The incremental strategy: grow toward $\mathcal{D}$ through the product space.** Start from current exports and diversify step by step through products with high relatedness density to the dense subgraph. Each step creates new capabilities that make the next step feasible. Requires: patience (decades), sequenced operators (Chapter 15), and political sustainability across electoral cycles. Applicable to: all countries, but slow for those far from $\mathcal{D}$ (Bolivia's illustrative average relatedness to $\mathcal{D}$ is 0.06 — the incremental path would take about 4 product-space steps (Example 25.14), roughly 20–30 years per step at the current pace of diversification, an illustrative figure).

Example 25.14 (Bolivia's Incremental Path [ILLUSTRATIVE])**.** *Starting from Bolivia's current exports (gas, zinc, gold, soy, quinoa):*

Step 1 *(adjacent, $\omega \approx 0.15$): Gas processing, zinc smelting, lithium extraction (given Bolivia's lithium resources in the Salar de Uyuni, the largest identified resources in the world, though not the largest economically proven reserves). These add within-periphery value but begin to create basic chemical/metallurgical capabilities.*

Step 2 *(moderate distance, $\omega \approx 0.10$): Basic chemicals, fertilizers, processed metals. These products require capabilities developed in Step 1 and connect to agriculture (fertilizers) and construction (metals).*

Step 3 *(approaching $\mathcal{D}$, $\omega \approx 0.06$): Specialized plastics, basic electrical components, simple machinery. This is the edge of the dense subgraph. Reaching it requires the human-capital and institutional operators from Chapter 15.*

Step 4 *(entering $\mathcal{D}$, $\omega \approx 0.04$): Precision instruments, advanced chemicals, electronic components. This IS the dense subgraph. Reaching it from Bolivia's current position requires the full multi-scale structural plan from Exercise 19.7.*

At an illustrative 20–30 years per step, the path to Step 4 takes approximately 80–120 years. An earlier draft tied the step time to the structural velocity of Γ and scaled it by

Korea's. That cannot work. Bolivia's measured velocity is negative (−0.036 per decade, 1990–2013, INE base 1990), and so was Korea's over 1995–2014 (−0.055 per decade, OECD ICIO 2021) while it deepened its core. The clock for this path is the rate of entry into new products, measured in the product space, not v_Γ. If an active strategy made each step 7.5 times faster (an illustrative ratio), the path would take approximately 11–16 years. The gap between 120 years and 16 is the gap between passive diversification and active operator design.

Prebisch Connection: From truncated to genuine competitiveness

In later work (Fajnzylber, 1988), Fajnzylber distinguished between "spurious competitiveness" (based on low wages, resource extraction, currency depreciation) and "genuine competitiveness" (based on knowledge, technology, and institutional quality). In this chapter's notation:

Spurious competitiveness: high π_c in W^P_{global} through commodity exports. High income during booms ($\Delta\omega$), but no structural change ($\Delta\Gamma = 0$). Bolivia's gas revenue era fits: GDP grew 5.06% a year on average in 2006–2014 (World Bank, 2026b), Γ was roughly flat (0.518 in 2006, 0.484 in 2013, INE base 1990), and ECI fell (-1.21 to -1.34).

Genuine competitiveness: high π_c through capital-goods and high-tech exports. The dense subgraph IS the source of genuine competitiveness: it generates learning, spillovers, and diversification opportunities that sustain growth without commodity dependence.

The transition from spurious to genuine is the transition from the periphery of the product space to the dense core. On Conjecture 25.6, it also raises Γ, as Chapters 15–17 prescribe. And it requires building the missing subgraph — the capital goods sector that Fajnzylber identified 40 years ago as the key to Latin American development.

25.5 Summary

1. The **dense subgraph** is the maximizer of internal proximity per product, $e(S)/|S|$ (Goldberg; Charikar's peeling approximates it). In the product space it is the core of machinery, metal products and chemicals (Hidalgo et al., 2007), which holds the highest-PCI products.

2. **Truncated industrialization**: Latin America lacks the dense subgraph. Bolivia's truncation index: 0.98 (2 of ≈100 products in $\mathcal{D}$); Korea's: 0.18 (both illustrative). The complexity gap is sourced: ECI -1.30 against $+1.81$ (2019).

3. **Bridges, not density, drive integration.** Removing a bridge set bounds Γ by twice the conductance that remains (Proposition 25.4); removing a dense module can *raise* Γ. That the capital-goods core is a bridge, so that truncation lowers Γ, is a testable conjecture.

4. The **truncation penalty**, measured in excess relaxation time $1/\Gamma - 1$ (not in welfare), can be large in a stylized truncated economy (55–86% across counterfactuals). For Bolivia the conjecture gets **no support at the 12-sector level**: its Γ (0.475) is inside the comparator range (Korea 0.511, United States 0.463), and aggregate Γ fell in both Bolivia and Korea. Testing it needs 39-industry or product-level data.

5. **Causes and strategies.** ISI stalled before the capital-goods core (Tavares), coalitions and the formation channel kept it there, and Fajnzylber added imitative consumption and transnational leadership. Strategies: build (Korea), attract (Singapore), or grow toward $\mathcal{D}$ (Bolivia: 80–120 years passively, 11–16 at a 7.5-fold faster pace; both illustrative).

25.6 Exercises

25.1 Identify the dense subgraph. Using the Atlas of Economic Complexity product-space data (SITC rev. 2, 4-digit): compute the average internal proximity $\bar{\phi}$ for each SITC 2-digit group, then run Charikar's peeling algorithm (Appendix 25.6) on the whole space. Which products form $\mathcal{D}$? Do they match the capital goods cluster? How large is the density ratio $\bar{\phi}_{\mathcal{D}}/\bar{\phi}_{\text{all}}$?

25.2 Truncation index. Compute τ_c for Bolivia, Brazil, Chile, South Korea, and Germany. Rank them. Plot τ_c vs. ECI. Is the correlation strong?

25.3 The truncation-Γ relationship. For 30 countries: estimate τ_c (from product-space data) and Γ_c, both from the product-space proxy $W_c = D_c^{-1}\phi_c$ and from OECD ICIO input shares. Is $\mathcal{D}$ a bridge (compute its betweenness and the conductance of the commodity clusters with and without it)? Plot Γ_c vs. τ_c. Estimate $f(\tau)$ by regression. Is $f'(\tau) < 0$ as predicted? Start from the 12-sector values of Result 25.8, then repeat with INE's 39 industries and a concordance to the ICIO industries. Does the answer change with the resolution?

25.4 Counterfactual: Bolivia with the dense subgraph. Suppose Bolivia magically acquired RCA in the 20 highest-PCI products in $\mathcal{D}$. Recompute τ, ECI, and (approximately) Γ. How much would Γ increase? How much would σ_B decrease?

25.5 The divergence. Using trade data for 1960, 1980, and 2000: compute τ for Brazil, Mexico, South Korea, and Taiwan at each date. Plot the trajectories. At what date did the paths diverge? Does it correspond to a specific policy (Korea's HCI drive, Mexico's maquiladora program)?

25.6 Bolivia's incremental path. Using OEC relatedness data:

(a) Identify the 5 products with the highest relatedness density from Bolivia's current exports to the dense subgraph.

(b) For each: what capabilities are needed? What operators (Chapter 15)?

(c) Estimate the time required at Bolivia's observed rate of entry into new products (with RCA) vs. Korea's over 1995–2019. Why is the structural velocity v_Γ the wrong clock here?

25.7 The lithium question. Bolivia has the world's largest identified lithium resources (though not the largest proven reserves). Suppose lithium batteries are in the dense subgraph. In the Atlas (HS92, 2019), electric accumulators (HS 8507, which include lithium-ion batteries) have PCI 0.549 (0.811 in 2023); carbonates including lithium carbonate (HS 2836) have -0.573, and natural gas (HS 2711) -2.226 (The Growth Lab at Harvard University, 2026).

(a) Compute Bolivia's relatedness density to lithium batteries ($\omega_{c,\text{LiB}}$).

(b) What intermediate products connect Bolivia's current exports to lithium batteries?

(c) What is the operator sequence needed? (Hint: lithium extraction $\to$ lithium carbonate $\to$ cathode materials $\to$ battery cells. Each step requires different capabilities.)

(d) Is this a realistic entry point to $\mathcal{D}$? What are the risks?

(e) A PCI of 0.55 is above the product mean of zero but well below that of most machinery. Is HS 8507 in the $\mathcal{D}$ you found in Exercise 25.1?

Mathematical Appendices

Appendix 25A: Dense Subgraphs and Spectral Gaps

Computing $\mathcal{D}$. Goldberg's problem $\max_S e(S)/|S|$ is solvable exactly in polynomial time by a sequence of minimum-cut computations. For large product spaces, Charikar's (2000) greedy peeling is simpler and guarantees at least half the optimal value: repeatedly delete the product with the smallest remaining weighted degree, and keep the intermediate set with the highest $e(S)/|S|$. On the ten-product example of `code/ch25.py`, peeling returns the four-product core, which coincides with the brute-force optimum ($e/|S| = 1.35$). A lower size bound $|S| \geq k$ makes the problem NP-hard; if one is needed, report peeling with the bound and state that it is a heuristic. Modularity-based community detection is an alternative, but it partitions the whole space into communities and needs a resolution parameter γ and a density threshold to single out $\mathcal{D}$; whichever method is used, state the parameters.

Integration after removing a set. For a symmetric ϕ and $W = D^{-1}\phi$ (reversible), the eigenvalues are real and, with λ_2 the second-largest eigenvalue, Cheeger's inequality holds (Lawler and Sokal, 1988; Sinclair and Jerrum, 1989):

$$\frac{\Phi(W)^2}{2} \;\leq\; 1 - \lambda_2(W) \;\leq\; 2\,\Phi(W), \qquad \Phi(W) = \min_{S:\ \mathrm{vol}(S) \leq \mathrm{vol}/2} \frac{\phi(S, S^c)}{\mathrm{vol}(S)}.$$

Removing a set of products and renormalizing changes every row that touched it, so neither eigenvalue interlacing (which needs principal submatrices of a symmetric matrix) nor any monotonicity argument applies. What does apply is the upper bound: after removal, $\Gamma \leq 2\Phi$ for every cut of the remaining products (Proposition 25.4). If the removed set was the only strong connection between two clusters, the remaining conductance of that cut is small and so is Γ. If it was a dense module with few outside connections, its removal can raise Γ (Remark 25.5).

Appendix 25B: The Product Complexity Index and the Dense Subgraph

PCI and density. The PCI is the product-level counterpart of the ECI: the second eigenvector of the product-product similarity matrix $\tilde{M}_{pp'}$ (Chapter 16). Products with high PCI are those that are produced mainly by complex (high-ECI) countries. High-PCI products tend to cluster in the dense subgraph because: (i) they require diverse capabilities (which means they co-occur with many other complex

products), and (ii) they are produced only by diversified countries (which means their presence is correlated with the presence of many other products).

The core-periphery structure of PCI. Plotting PCI against product density (number of neighbors in the product space) reveals a strong positive correlation: the densest products are the most complex. The dense subgraph therefore contains many of the high-PCI products, and the periphery (low density) corresponds to low-PCI products (commodities). The truncation index τ_c should then correlate negatively with ECI, roughly $\text{ECI}_c \approx -a\tau_c + b$ with $a > 0$; Exercise 25.2 tests this.

Structuralist Case Study: Fajnzylber's Industrialización Trunca

Fernando Fajnzylber (1940–1991) was a Chilean economist who worked for many years in Mexico and directed the joint CEPAL/UNIDO Industrial Development Division. His 1983 book *La industrialización trunca de América Latina* diagnosed the region's structural failure with a precision that anticipated the product space literature by a quarter of a century.

Fajnzylber's argument has three parts. The mapping below is our formal reinterpretation; where he would disagree, we say so.

Part 1: The missing "endogenous nucleus of technical dynamism." Fajnzylber identified the capital goods sector, together with the engineering and innovation capacity around it, as the missing piece. Comparing Latin America with the advanced economies, and above all with Japan and the East Asian newly industrializing countries, he argued that the capital goods sector was the engine of technical progress. It was the sector where learning-by-doing was fastest, where scale economies were largest, and where spillovers to other sectors were strongest. Latin America had built consumer-goods industries (textiles, food processing, household appliances) but had never built the machines that make those goods. The "nucleus" was imported. Two mechanisms kept it so: an imitative consumption pattern among high-income groups, and the leadership of transnational corporations in the dynamic industries (Remark 25.11). In this chapter's notation: the dense subgraph $\mathcal{D}$ was missing (illustratively, $\tau \approx 0.80$). The notation captures the absence, not the consumption and ownership mechanisms that produced it.

Part 2: Truncation explains the casillero vacío. Seven years before his 1990 paper on the empty box, Fajnzylber had already identified the mechanism: without the capital goods sector, industrialization could not generate enough inter-sectoral linkages to spread growth broadly. Consumer ISI created modern-sector jobs but did not transform the traditional sector (Tavares's diagnosis, Chapter 15). The

dual economy persisted because the structural bridge was never built. Growth without equity was the consequence: the modern sector grew (increasing average income) while the traditional sector stagnated (maintaining inequality). In 1990 he showed that no Latin American country occupied the cell of high growth *and* high equity — the casillero vacío. In this chapter's terms, truncation is a candidate cause of that empty cell (Conjecture 25.6), not an identity.

Part 3: The prescription is "genuine competitiveness." In his later work (1988, 1990) Fajnzylber argued that the path forward was not more ISI (which had targeted the wrong products) nor liberalization (which would expose the truncated structure to global competition it could not survive) but "genuine competitiveness": building the knowledge base, institutional infrastructure, and industrial capabilities needed to produce complex goods. In this chapter's notation: the prescription is to build $\mathcal{D}$ through the sequenced operators of Chapter 15, starting with information and education (which create the human-capital preconditions) and culminating in industrial policy targeting the dense subgraph.

Fajnzylber (1983, 1990)	**Formal object**	**This chapter**
Missing capital goods sector	Truncation $\tau_c \to 1$	Def. 25.3
Endogenous nucleus	Dense subgraph $\mathcal{D}$ (plus capabilities not in the notation)	Def. 25.1
Truncation $\to$ low linkages	Bridge removal bounds Γ; $\tau \uparrow \Rightarrow \Gamma \downarrow$ if $\mathcal{D}$ is a bridge	Prop. 25.4, Conj. 25.6
Size of the gap	Stylized: 55–86% of excess relaxation time; Bolivia: no 12-sector integration gap	Ex. 25.10, Result 25.8
Imitative consumption, TNC leadership	Causes of truncation	Remark 25.11
Spurious vs. genuine (1988)	Commodity π vs. $\mathcal{D}$-based π	Prebisch Connection
Three strategies	Build, attract, or grow toward $\mathcal{D}$	Remark 25.13

Fajnzylber died in 1991 at age 51, before seeing either the product space or the economic complexity literature that would echo his diagnosis. The casillero vacío

became the defining image of the structuralist diagnosis. This chapter gives his truncated industrialization a truncation index, a bridge bound, a testable conjecture, and a measure of its cost in integration. For Bolivia, the 12-sector data show his missing nucleus in complexity, not in aggregate integration.

Further Reading

Fajnzylber: Fajnzylber (1983), *La industrialización trunca de América Latina*. Fajnzylber (1988), "Competitividad internacional: evolución y lecciones." Fajnzylber (1990), "Industrialización en América Latina: de la 'caja negra' al 'casillero vacío' ." CEPAL (2006), *Fernando Fajnzylber: Una visión renovadora del desarrollo en América Latina*.

Product space and diversification: Hidalgo et al. (2007), "The Product Space Conditions the Development of Nations." Hausmann et al. (2007), "What You Export Matters." Hidalgo (2021), "Economic Complexity Theory and Applications."

Industrial policy and the capital goods sector: Amsden (1989), *Asia's Next Giant*. Chang (2002), *Kicking Away the Ladder*. Cimoli et al. (2009), *Industrial Policy and Development*.

Bolivia's industrialization prospects: Fundación Milenio (various years), *Informe Económico*. For the lithium question: Obaya (2019), "Estudio de caso sobre la gobernanza del litio en el Estado Plurinacional de Bolivia."

Chapter 26

The Neo-Structuralist Synthesis

Growth with Equity as a Spectral Condition

> *"El desafío fundamental del desarrollo latinoamericano es el de transformar las estructuras productivas en un contexto de creciente equidad social."*
>
> ["The fundamental challenge of Latin American development is to transform productive structures in a context of growing social equity."]
> — José Antonio Ocampo, "La macroeconomía de la bonanza económica latinoamericana," *Revista de la CEPAL* 93 (2007)

This chapter states the book's policy conclusion. Twenty-five chapters have built the framework: the weight matrix, the Perron eigenvector, the spectral gap, the formation model, the co-evolutionary dynamics, the measurement infrastructure, the political economy. What does it all add up to?

One conjecture: *sustained growth with equity requires high* Γ *and high ECI simultaneously*, together with a diverse information network. Neither alone is sufficient. High ECI without equity: Brazil (complex but unequal). High Γ without high ECI: Cuba (equal but stagnant). Both together: South Korea, Taiwan, the Nordic countries. The intersection is small but achievable. The data in this chapter strain the conjecture in two places. At the 12-sector level, aggregate Γ hardly varies across very different economies (Brazil's is close to Korea's), so condition (1) does not discriminate. And Bolivia cut inequality sharply over 2005–2019 with a negative and falling ECI (Remark 26.4).

Ocampo's neo-structuralism — the synthesis of Prebisch's structuralism, CEPAL's institutional wisdom, and modern open-economy macroeconomics — provides the policy framework. This chapter translates it into the language of this book — with the caveat that the translation is ours, and that Ocampo's own framework

is not built around these three variables (see the case study) — and asks: what would it take for Bolivia to enter the intersection? Country values of Γ, ECI, GDP and distribution are sourced (INE, OECD ICIO, Growth Lab, WDI); thresholds, the plan and N_{eff}^{I} remain [ILLUSTRATIVE]. `code/ch26.py` reads the sourced values from `data/processed` and checks them.

26.1 The Growth-Equity Conjecture

The conjecture involves three objects that live on *different* matrices. Keeping them apart is the first requirement of a testable statement.

1. **Integration** $\Gamma_c := \Gamma(W_c^P)$, the spectral gap of country c's domestic input-share matrix $W_c^P = D^{-1}A_c$, computed on a common sector classification for all countries (e.g., OECD ICIO). It measures how fast shocks and demand propagate across sectors (Chapter 6), not how power is distributed.

2. **Complexity** ECI_c, computed from the *global* country × product matrix M ($M_{cp} = 1$ if c exports p with revealed comparative advantage): ECI is the standardized eigenvector for the second-largest eigenvalue of

$$\tilde{M} = D^{-1}MU^{-1}M',$$

   where D and U are the diagonal matrices of country diversity and product ubiquity (Chapter 16). ECI is *not* a function of W_c^P, and it is z-scored across countries each year: $\text{ECI}_c > 0$ means "above the world mean", a positional condition that roughly half of all countries fail by construction.

3. **Information diversity** $N_{\text{eff}}(W_c^I) = 1/\sum_j (\pi_j^I)^2$ for the domestic information network (Chapter 21). Compare it across countries only on a common node set, or as N_{eff}/N.

None of the three measures the concentration of production power $\boldsymbol{\pi}^P$; that is a fourth coordinate, $G(\boldsymbol{\pi}^P)$ or $N_{\text{eff}}(\boldsymbol{\pi}^P)$, independent of Γ (Proposition 6.16).

Conjecture 26.1 (Growth with Equity)**.** *Sustained growth with declining inequality requires:*

1. $\Gamma_c > \Gamma^*$ ***(structural integration)****: growth propagates across blocks instead of staying in an enclave. Illustrative threshold* $\Gamma^* \approx 0.30$*. (Whether aggregate* Γ *measures the enclave at all is questioned in Remark 26.4.)*

2. $ECI_c > 0$ ***(productive complexity)***: *the economy is in the dense core of the global product space (Chapter 16), giving diversification options, rising terms of trade and resilience to commodity shocks.*

3. $N_{eff}(W_c^I) > N_{eff}^*$ ***(information democracy)***: *the information network is diverse enough that the formation channel produces aspirations calibrated to the full range of economic opportunities, not just the local block. Illustrative threshold* $N_{eff}^* \approx 8$.

The thresholds are placeholders, not estimates: estimating them requires the cross-country data of Exercise 26.1. Violating any one condition is conjectured to produce a specific pathology (the examples are the conventional readings; for the measured values, see the note below the table):

Condition violated	Γ	*ECI*	N_{eff}	*Pathology (illustrative example)*
(1) only	*Low*	*High*	*High*	*Growth without equity (Brazil, Mexico)*
(2) only	*High*	*Low*	*High*	*Equity without growth (—)*
(3) only	*High*	*High*	*Low*	*Fragile growth (China; Korea before 1987)*
(1) and (2)	*Low*	*Low*	*High*	*Stagnation (—)*
(1) and (3)	*Low*	*High*	*Low*	*Enclave industrialization (—)*
(2) and (3)	*High*	*Low*	*Low*	*Trapped equality (Cuba)*
All three	*Low*	*Low*	*Low*	*Structural collapse (Venezuela, 2010s)*
None	*High*	*High*	*High*	*Growth with equity (Korea today, Nordics)*

Note: the "Low Γ*" labels for Brazil and Mexico are not borne out at the 12-sector level, where their 2019* Γ *is 0.496 and 0.649 against Korea's 0.511 (OECD ICIO 2023 (OECD, 2023); author's calculations, data/compute/compute_oecd_io.py). An earlier draft put Bolivia in the "Stagnation" row; its measured* Γ *is not low and its distribution improved strongly (Remark 26.4).*

Remark 26.2 (Status of the conjecture). An earlier draft presented this as a theorem, with a proof of condition (1) from the claim that Γ bounds the distributional loss $L_{\text{power}} = (\kappa N/2)\Phi^2 \operatorname{Var}(\boldsymbol{\pi})$. That claim is false in both directions (`code/ch26.py`): the rank-one matrix $\mathbf{1}\boldsymbol{\pi}'$ with $\boldsymbol{\pi} = (0.97, 0.01, 0.01, 0.01)$ has $\Gamma = 1$ and $\operatorname{Var}(\boldsymbol{\pi}) = 0.173$, close to the maximum $3/16$; a doubly stochastic, weakly coupled block matrix has $\Gamma = 0.02$ and perfectly uniform $\boldsymbol{\pi}$. High Γ is neither necessary nor sufficient for equal power. What remains is weaker and empirical: integration plausibly matters for equity through *between-block* inequality σ_B (growth reaches the traditional blocks), complexity through the growth path, and information diversity through formation (Chapters 8–11, 24). Each link is a hypothesis to be tested, not a derived result.

Remark 26.3 (Confronting Korea). The conjecture's own leading example contradicts it as a statement about *all* episodes of growth with equity. Korea grew fast

with low and stable inequality from the 1960s, yet it was ruled by authoritarian governments with controlled media until 1987, violating condition (3); on Chapter 23's illustrative path its Γ stayed below $\Gamma^* = 0.30$ until about 1980, violating condition (1) (pre-1995 tables are not used here; from 1995 the OECD ICIO data show Korea's 12-sector Γ *falling*, from 0.626 to 0.511 in 2019, while its ECI rose from $+1.10$ to $+1.81$); and its ECI in the 1960s, when it exported wigs, plywood and textiles, was probably not above the world mean (check with the Atlas SITC series from 1962). Taiwan is similar. Two readings are possible. Either the conjecture is false as a necessity claim, or it concerns *sustaining* growth with equity beyond middle income, where early land reform and mass education (which Korea and Taiwan had) can substitute for integration and information diversity for a time. The second reading is narrower and still testable: it predicts that countries in Korea's 1970s position which did not later raise Γ and N^I_{eff} stalled.

Remark 26.4 (Confronting Bolivia). Bolivia's 2005–2019 record is the sharpest test the book's own case offers, and the conjecture does not pass it as a claim about equity gains.

- *Distribution.* The Gini of per-capita household income fell from 0.600 (2005) to 0.426 (2019). Extreme poverty fell from 38.2% (2005) to 15.2% (2018) on the old poverty lines, and stood at 13.2% in 2019 on the updated lines (INE, *Pobreza y desigualdad*, Encuesta de Hogares, cuadros 3.06.01.02 and 3.06.01.03, 2025 revision). The World Bank's income Gini fell from 0.585 to 0.416 over the same years (World Bank, 2026a).

- *Growth.* GDP grew 4.52% a year on average over 2006–2019 (World Bank, 2026b).

- *Integration.* Γ stayed in the comparator range and, if anything, fell: 0.502 (2005), 0.518 (2006), 0.484 (2013) on INE's base 1990, and 0.467 in 2019 on base 2017 (not strictly comparable). Like-for-like in 2019 Bolivia's 0.475 sits between the United States (0.463) and Korea (0.511), below Germany (0.544), Chile (0.557), China (0.582) and Mexico (0.649). All of them clear the placeholder $\Gamma^* = 0.30$.

- *Complexity.* ECI was negative and falling: -1.10 (2005), -1.21 (2006), -1.46 (2018), -1.30 (2019); Bolivia's rank fell from 120th to 131st (The Growth Lab at Harvard University, 2026).

So inequality and poverty fell steeply for fourteen years, with sustained growth, while condition (2) was violated throughout and condition (1) held only in the

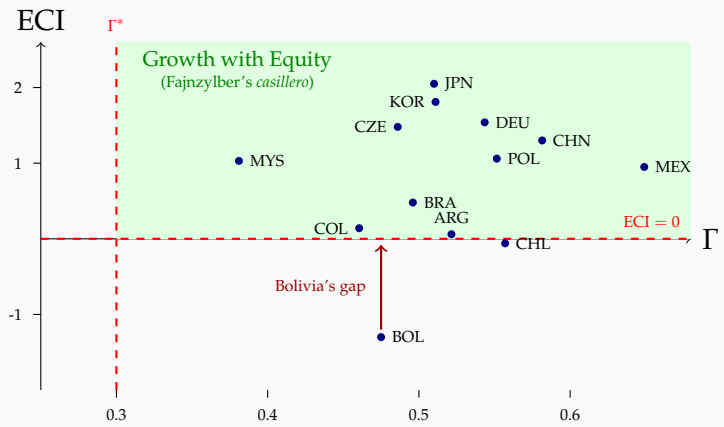
weak sense that every comparator also meets it. This is evidence against any claim that equity gains *require* high Γ or ECI > 0. Three readings remain, and they should be stated plainly:

1. *The conjecture is false as a necessity claim about equity gains.* Redistribution financed by an export enclave can lower inequality for well over a decade without structural change.

2. *It is a claim about sustaining them.* The gains rested on commodity revenue: fuels were 48.8% of merchandise exports in 2006 and 54.5% in 2013, and 32.1% in 2019 (World Bank, 2026b). Growth averaged 3.46% a year over 2006–2024, against 4.52% to 2019. The Gini was 0.418 in 2024 and 0.426 in 2025; extreme poverty was 13.3% in 2024 and 16.7% in 2025 on the updated lines (INE, same cuadros). Whether this is the beginning of a reversal is what the narrow reading predicts and what the next years will show.

3. *Condition (1) is stated on the wrong object.* Aggregate Γ does not separate development paths (Chapter 25, Result 25.8). What distinguishes Bolivia's production structure is its export enclave: hydrocarbons have a domestic power share $\pi = 0.039$, export 70% of their output, buy 0.225 of domestic inputs per unit of output and pay 6% of their value added as compensation of employees (Instituto Nacional de Estadística (INE), Bolivia, 2026). The rent was captured fiscally (through the IDH tax), not through linkages. If condition (1) is to carry content, it must be restated on the enclave's linkages (π and domestic input intensity of the export sectors) or on Γ at a resolution where the capital-goods core is a node.

Visualization: The growth-equity conjecture in (Γ, ECI) space, 2019



The green zone is the region of conditions (1)–(2): $\Gamma > \Gamma^*$ and ECI > 0, with the placeholder $\Gamma^* = 0.30$ ([ILLUSTRATIVE]). It is the cell that Fajnzylber's (1990) *casillero vacío* refers to: the combination of growth *and* equity that no Latin American country occupied in 1965–84. With the measured values, every country plotted lies to the right of Γ^*, between 0.38 (Malaysia) and 0.65 (Mexico), so the panel separates them by ECI alone. Bolivia's Γ (0.475) is in the middle of the range; its gap is vertical, in complexity (ECI -1.30). Chile (-0.06) and Argentina ($+0.06$) sit on the ECI line. Brazil and Mexico are not low-Γ economies at this resolution, so "complexity without integration" is not visible in aggregate Γ. China is inside but fails condition (3) (low N^I_{eff}), a reminder that the panel shows only two of the three conditions. Venezuela (ECI -1.02) is not plotted: it is not a separate economy in OECD ICIO.

Source: Γ, domestic input-share W^P, ISIC-based 12 sectors, basic prices, diagonal kept; Bolivia from INE supply-use tables, base 2017 (Instituto Nacional de Estadística (INE), Bolivia, 2026), others from OECD ICIO 2023 (OECD, 2023); author's calculations (data/compute/compute_bolivia_io.py, data/compute/compute_oecd_io.py). ECI: Growth Lab, HS92 (The Growth Lab at Harvard University, 2026). Bolivia's Γ ranges over 0.33–0.69 across normalizations.

26.2 The Neo-Structuralist Policy Framework

Definition 26.5 (The Neo-Structuralist Policy Package)**.** *This chapter's policy package, inspired by Ocampo's neo-structuralist synthesis and written in the operator language of Chapter 15, consists of eight interventions grouped in three* policy pillars *targeting the three conditions of Conjecture 26.1. (These pillars are the chapter's, not Ocampo's; his are discussed in the case study.)*

Pillar 1: Productive transformation *(raise ECI).*

1. *Industrial policy targeting the dense subgraph (Chapter 25): build the capital goods*

sector through sequenced operators ($\mathcal{L}_{info} \to \mathcal{L}_{HC} \to \mathcal{L}_{infra} \to \mathcal{L}_{ISI\text{-}B}$).

2. *Technology policy: invest in R&D, create technology transfer institutions, build the human capital base for the dense subgraph.*

3. *Trade policy: strategic integration into global value chains that connect to the dense core of the product space, not just commodity markets.*

Pillar 2: Structural integration *(raise Γ).*

4. *Domestic linkage policy: the operator $\mathcal{L}_{ISI\text{-}B}$ (cross-block edges, Chapter 15). Build the connections between the gas sector and the traditional economy, between urban manufacturing and rural agriculture, between the formal and informal sectors. For an enclave economy the target is the export sector's domestic linkages (its π and its domestic input intensity), not aggregate Γ, which may already be in the comparator range (Remark 26.4).*

5. *Infrastructure: roads, telecommunications, energy connections that create physical W^P edges between disconnected blocks.*

Pillar 3: Social cohesion *(raise N_{eff}, reduce attribute constraints).*

6. *Education: the dual operator on $W^P + W^I$ (Chapter 15). Universal, high-quality, and cross-group (reducing assortative mixing in W^I).*

7. *Information democracy: media pluralism, digital access, community communication strengthening (Chapter 21).*

8. *Anti-discrimination: reduce λ_k constraints (Chapter 18). Gender equality, indigenous inclusion, anti-racism.*

Conjecture 26.6 (The Three Pillars Are Complementary)**.** *The three pillars reinforce each other:*

1. *Productive transformation (ECI $\uparrow$) creates cross-block edges that raise Γ: new industries connect sectors that were previously disconnected.*

2. *Structural integration ($\Gamma \uparrow$) makes productive transformation feasible: cross-block linkages provide the intermediate inputs, labor pools, and demand that new industries need.*

3. *Social cohesion ($N_{eff} \uparrow$, $\lambda_k \uparrow$) supports both: an educated, informed population with high aspirations generates the demand for structural change and provides the human capital for productive transformation.*

The complementarity means that the return to each pillar is increasing in the level of the others. Investing in productive transformation without social cohesion (Brazil's path) produces growth without equity. Investing in social cohesion without productive transformation (Cuba's path) produces equity without growth. Only the joint investment produces both.

26.3 Bolivia's Plan: A Worked Application

Example 26.7 (A 30-Year Structural Plan for Bolivia [ILLUSTRATIVE]). ***Current position*** *(2019):* $\Gamma = 0.467$ *on the book's 12 sectors (0.475 like-for-like; 0.33–0.69 across normalizations),* $ECI = -1.30$*, hydrocarbons' domestic* $\pi = 0.039$ *with 70% of output exported (all sourced: Chapter 17, Remark 26.4);* $N_{eff} = 3.4$ *(Chapter 17, four media ecosystems) and* $\sigma_B = 0.62$ *(both illustrative).*

Target *(2055):* $ECI \geq 0$*,* $N_{eff} \geq 8$*,* $\sigma_B \leq 0.30$*, and export sectors whose domestic linkages (their* π *and domestic input intensity) rise. Aggregate* Γ *is not a target: it is already inside the comparator range. The ECI target is demanding. It requires* $+1.30$*, while Bolivia's ECI* fell *by 0.44 over 1995–2019 and Korea's rose by 0.70 (The Growth Lab at Harvard University, 2026).*

Phase 1 (2025–2035): Foundation.

Pillar 3 first (high-Ψ *operators):*

1. *Universal broadband to all municipalities (*$\Delta N_{eff} \approx +2$*, cost: \$1.5B over 10 years).*
2. *Bilingual education reform: curriculum in Quechua/Aymara and Spanish, with cross-ethnic school integration (*$\Delta\lambda_r^I \approx +0.15$*).*
3. *Female entrepreneurship mentoring program (Chapter 18 operator:* $\Delta\lambda_g^I \approx +0.10$*).*
4. *Community journalism strengthening (*$\Delta\pi^I_{community} = 0.05 \to 0.12$*).*

Expected effect by 2035: $N_{eff} \approx 5.5$*,* Ψ *reduced from 1.56 to* ≈ 1.1 *(formation channel weakens as information diversifies). Aspirations in the traditional sector begin rising.*

Phase 2 (2030–2045): Building Blocks.

Pillar 1 + 2 (medium-Ψ *operators):*

5. *Technical education: 10 vocational polytechnics for metallurgy, chemicals, and electronics (the Step 1–2 products from Chapter 25). Cost: \$2B over 15 years.*
6. *Infrastructure: 3 cross-block corridors connecting altiplano mining to lowland agriculture to urban manufacturing. Cost: \$5B over 15 years.*

7. *Lithium industrialization: from extraction (current) to carbonate (Step 1) to cathode materials (Step 2). Public-private partnership targeting Step 3 by 2040. Cost: $3B over 15 years.*

8. *Agricultural value-chain integration: connect quinoa, soy, and cattle to food processing and packaging (cross-block W^P edges). Cost: $1B.*

Expected effect by 2045: ECI ≈ -0.30, $\sigma_B \approx 0.40$, and the first domestic supply chains into and out of the extractive sectors.

Phase 3 (2040–2055): The Dense Subgraph.

Pillar 1 (low-Ψ operators, now feasible because Phases 1–2 created the preconditions):

9. *Industrial policy targeting the dense subgraph: specialized machinery, electronic components, pharmaceutical intermediates. Public procurement as demand anchor. Cost: $4B over 15 years.*

10. *FDI attraction for assembly and manufacturing of capital goods (the Singapore strategy adapted to Bolivia's scale).*

11. *R&D institutes linked to the polytechnics from Phase 2.*

Expected effect by 2055: ECI ≈ 0, $N_{eff} \approx 8$, $\sigma_B \approx 0.30$.

***Total cost**: approximately $17B over 30 years ($570M/year). For context: Bolivia's GDP was $49.1B in 2019 and $54.9B in 2024, in current dollars (World Bank, 2026b). The plan requires sustained investment of $\approx 1.2\%$ of 2019 GDP per year (1.0% of 2024 GDP) in structural transformation. This is far smaller than the resources South Korea mobilized for its Heavy and Chemical Industrialization drive, which absorbed most manufacturing investment in the late 1970s, and well below Bolivia's natural gas rents during the 2006–2014 boom, which peaked at 5.56% of GDP in 2013 (World Bank, 2026b).*

***Political feasibility** (Chapter 14): Phase 1 is politically easy (education, broadband, and mentoring have broad support — no coalition's π is directly threatened). Phase 2 is harder (infrastructure and industrial investment require fiscal commitment and threaten import-dependent interests). Phase 3 is hardest (industrial policy targeting the dense subgraph requires a developmental-state coalition that does not yet exist in Bolivia).*

The sequencing exploits the formation amplifier: Phase 1 changes W^I and preferences, which shifts coalition incentives (Chapter 14), which makes Phases 2 and 3 politically feasible. The information operator makes the structural operator possible. This is the sequencing conjecture of Chapter 15 (Conjecture 15.14) in action, and is only as good as that conjecture.

Key Structural Insight

The neo-structuralist synthesis, in this book's terms, is one conjecture and one plan:

The conjecture: sustained growth with equity requires $\Gamma > \Gamma^*$, ECI > 0, and $N_{\text{eff}} > N_{\text{eff}}^*$ simultaneously. It is a precise, testable claim built on three measurable objects — not a theorem, and not yet an estimate: the thresholds are placeholders, Korea's early decades already strain it (Remark 26.3), and Bolivia's 2005–2019 fall in inequality, with a negative and falling ECI and Γ in the comparator range, contradicts it as a claim about equity gains (Remark 26.4).

The plan: three policy pillars (productive transformation, structural integration, social cohesion), sequenced by Ψ (information first, infrastructure second, industrial policy third), financed by sustained investment of $\approx 1.2\%$ of GDP.

Every element of the plan corresponds to a specific operator from Chapter 15 targeting a specific entry of W. The expected effects are illustrative magnitudes, not derived results; turning them into estimates is the work of Exercises 26.4–26.6. That is what separates an engineering specification from aspirational prose: every number has a place where it can be checked.

26.4 Why Neither Neoliberalism Nor Populism Suffices

Remark 26.8 (The Insufficiency of Both Alternatives). **The neoliberal prescription** (Washington Consensus): liberalize trade, privatize, deregulate. In operator terms: remove the barriers (tariffs, controls) that shaped W^P and let the market create new edges. The assumption: the market will create the "right" W^P (the one that maximizes efficiency). The result: the market creates W^P edges dominated by proximity (Chapter 13, the principle of relatedness), which means the existing structure reproduces itself. Commodity exporters liberalize into more commodity export. Γ does not change. ECI does not change. The prescription fails condition (1) and (2) of Conjecture 26.1.

The populist prescription: nationalize resources, redistribute revenue, expand social programs. In operator terms: change ω (endowment redistribution) without changing W. The result: poverty falls and inequality falls — in Bolivia substantially (Gini 0.600 in 2005, 0.426 in 2019) — while the structure does not change: Γ stayed roughly flat (Chapter 17) and ECI fell (Remark 26.4). The chapter's claim is that gains financed by commodity revenue are fiscally fragile when the cycle

turns. The post-2014 data are mixed so far: the Gini was 0.418 in 2024, but extreme poverty rose from 13.3% (2024) to 16.7% (2025). The prescription fails condition (2) because it does not raise ECI; on condition (1), aggregate Γ cannot tell.

The neo-structuralist synthesis: change W itself — the production structure (W^P), the information structure (W^I), and the solidarity structure (W^U) — through deliberate, sequenced operator design. It aims at all three conditions simultaneously, seeking permanent structural change rather than redistribution alone.

Prebisch Connection: From Prebisch to Ocampo: one argument across generations

The structuralist argument has been consistent across six decades:

Prebisch (1949): The global economy is asymmetric (W^P_{global} is not doubly stochastic). The periphery must industrialize to change its structural position.

Furtado (1961): Underdevelopment is a stable structural equilibrium, not a stage. The periphery's structure reproduces itself through the formation channel.

Pinto (1970): The domestic economy is heterogeneous. Three productivity strata coexist without structural integration.

Fajnzylber (1983, 1990): The industrialization was truncated — the capital goods sector is missing. The casillero vacío is the consequence.

Ocampo et al. (2009): The synthesis — productive transformation plus social equity, achieved through deliberate industrial policy, macroeconomic management, and institutional strengthening.

CEPAL (2012b): Structural change for equality — redistribution without structural change is unsustainable.

This book (2027): The benchmark model abstracts from structure: anonymous interaction through markets ($W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$, with $W^{\text{CE}} = \frac{1}{N}\mathbf{1}\mathbf{1}'$ as its equal-endowment case) and no externalities. Structural economics studies non-anonymous W — its integration Γ and its distribution of power $\boldsymbol{\pi}$, two independent coordinates — and how both are produced and changed. Many structuralist insights from Prebisch to Ocampo can be restated as claims about these coordinates: a few as theorems, more as testable conjectures.

The argument is one argument, stated with increasing precision over 78 years: *structure matters, structure can be changed, and changing it requires different tools than the ones the standard model recommends.*

26.5 Summary

1. The **growth-equity conjecture**: sustained growth with equity requires $\Gamma > \Gamma^*$ (integration of the domestic input-share matrix), ECI > 0 (complexity, computed from the global export matrix and positional by construction) and $N_{\text{eff}} > N^*_{\text{eff}}$ (information diversity). It is testable, not proved; the old proof rested on a false Γ–Var($\boldsymbol{\pi}$) bound, Korea's early decades violate conditions (1) and (3), and Bolivia's 2005–2019 record (Gini 0.600 $\rightarrow$ 0.426 with ECI $-1.10 \rightarrow -1.30$ and Γ in the comparator range) contradicts it as a claim about equity gains. Aggregate Γ does not discriminate among countries at the 12-sector level.

2. The **neo-structuralist policy package** has three pillars: productive transformation (raise ECI), structural integration (raise Γ), social cohesion (raise N_{eff}, reduce attribute constraints). The pillars are complementary.

3. **Bolivia's 30-year plan** (illustrative): \$17B, three phases (W^I first, infrastructure second, industrial policy third), aiming at ECI ≈ 0 (from -1.30), $N_{\text{eff}} \approx 8$ and stronger domestic linkages of the export enclave by 2055. Cost: about 1.2% of 2019 GDP a year.

4. **Neither neoliberalism nor populism** addresses all three conditions. Neoliberalism changes prices but not structure. Populism changes endowments but not structure. The neo-structuralist synthesis changes W itself.

5. The structuralist argument from **Prebisch (1949) to Ocampo et al. (2009)** is one argument with increasing precision: structure matters, and it can be changed by deliberate operator design.

26.6 Exercises

26.1 The three conditions. For 30 countries: estimate Γ from the row-normalized domestic input-share matrices of the OECD ICIO tables (same sector classification for all), ECI from the Atlas of Economic Complexity, and N_{eff} from media concentration data. Why must Γ *not* be proxied by ECI? Plot each country in the (Γ, ECI) plane. Identify which countries satisfy all three conditions and which violate one or more.

26.2 Pathology classification. Classify the following countries using the pathology table from Conjecture 26.1: (a) Brazil (positive ECI and high inequality;

its measured 12-sector Γ is 0.496, close to Korea's), (b) Cuba (high Γ, low ECI; what data would you need to check either?), (c) Venezuela (low Γ, low ECI, low N_{eff}), (d) Costa Rica (moderate Γ, moderate ECI, high N_{eff}), (e) Bolivia, using the measured values of Remark 26.4. Which row of the table fits Bolivia over 2005–2019, and what does that imply for the table?

26.3 Complementarity. Simulate a 3-variable system $(\Gamma_t, \text{ECI}_t, N_{\text{eff},t})$ with complementary dynamics: $\Delta\Gamma_{t+1} = f(\Gamma_t, \text{ECI}_t, N_{\text{eff},t})$ increasing in all three. Compare: (a) investing in all three simultaneously, (b) investing in one at a time. Which reaches the target zone faster?

26.4 Bolivia's plan: cost-benefit. Using the cost estimates from Example 26.7 and the expected $\Delta\Gamma$, ΔECI:

(a) Bolivia's Γ (0.475) is already in the comparator range. Is there a cross-country Γ–GDP relationship at the 12-sector level at all? Estimate instead the GDP gain from moving ECI from -1.30 to 0 (use the cross-country ECI–growth relationship).

(b) Compute the net present value of the plan at a 5% discount rate.

(c) At what discount rate does the plan break even?

26.5 The neoliberal counterfactual. Suppose Bolivia had continued the 1985 neoliberal policy without the 2006 nationalization. Using the structural velocity from the 1985–2005 period: project Γ, ECI, and σ_B to 2020. Would Bolivia have entered the target zone? Compare with the actual trajectory.

26.6 The populist counterfactual. Suppose Bolivia had invested all gas revenue (2006–2019, total ≈\$50B, illustrative) in the structural plan instead of transfers. Using the formation model: what Γ and ECI would Bolivia have achieved by 2019? Would it be in the target zone?

26.7 The 78-year argument. Read one page from each of: Prebisch (1949), Furtado (1961), Pinto (1970), Fajnzylber (1983), CEPAL (2012b), and Ocampo et al. (2009). For each: identify the central claim and map it to a result or conjecture in this book. Write a one-paragraph synthesis: what is the one argument stated six times?

Mathematical Appendices

Appendix 26A: The Growth-Equity Frontier

Definition. The growth-equity frontier is the lower envelope of observed (Γ, Gini) pairs: the lowest Gini observed at each level of integration. There is no theorem that bounds Gini by a function of Γ (the claimed spectral bound of Chapter 2 is false; see Remark 26.2). The hypothesis is empirical: that the envelope $g(\Gamma)$ is decreasing.

Estimation. The envelope can be estimated by quantile regression, e.g. $Q_{0.05}(\text{Gini}_c \mid \Gamma_c) = a + b\Gamma_c + c\Gamma_c^2$. This is a conditional quantile, not a bound: about 5% of observations lie below it by construction, and a quadratic need not be monotone, so check $g' < 0$ over the observed range rather than assume it.

Distance to frontier. Country c's distance to the frontier: $d_c = \text{Gini}_c - g(\Gamma_c)$. When $d_c \approx 0$, the country's inequality is as low as that of the most equal countries with similar integration; further equality would require more integration ($\Delta\Gamma$). When $d_c > 0$, the country has "excess inequality" relative to its peers, and redistribution within the existing structure ($\Delta\omega$) has room to work; the larger d_c, the more room.

Appendix 26B: The Complementarity Condition

Formal statement. Let $V(\Gamma, \text{ECI}, N_{\text{eff}})$ be a reduced-form social welfare function (V, not W, to avoid a clash with the weight matrix). The three conditions are complementary if V is supermodular:

$$\frac{\partial^2 V}{\partial \Gamma \, \partial \text{ECI}} > 0, \quad \frac{\partial^2 V}{\partial \Gamma \, \partial N_{\text{eff}}} > 0, \quad \frac{\partial^2 V}{\partial \text{ECI} \, \partial N_{\text{eff}}} > 0. \tag{26.1}$$

This is an assumption (Conjecture 26.6), not a derived property. The arguments usually offered for it — (i) high ECI creates cross-block edges that raise Γ (if the dense core is a bridge; Chapter 25), (ii) high N_{eff} supports the aspiration formation that makes structural investment politically feasible, (iii) high Γ creates the domestic market and supply chains that support complex production — are dynamic causal links between the three variables. They motivate supermodularity of the payoff but are a different claim, and each needs its own evidence.

Implication for investment. Supermodularity means the marginal return to investing in any one pillar rises with the level of the other two, so piecemeal reforms understate the return to joint reform. It does *not* by itself say where to start. The sequence used in this chapter (information first) comes from a separate argument — Chapter 15's ordering by Ψ and by cost — and is only as good as that argument.

Structuralist Case Study: Ocampo's Neo-Structuralist Synthesis

José Antonio Ocampo (b. 1952), Colombia's former Finance Minister and CEPAL Executive Secretary, has been the leading architect of neo-structuralism — the synthesis of classical structuralism (Prebisch, Furtado, Pinto) with modern macroeconomics that retains the structuralist emphasis on productive transformation while incorporating lessons from the debt crises of the 1980s, the Washington Consensus era, and the commodity boom of the 2000s.

Ocampo's synthesis, developed across numerous publications and most systematically in Ocampo et al. (2009), *Growth and Policy in Developing Countries: A Structuralist Approach*, has three pillars. They overlap with this chapter's three conditions only in part, and the differences matter.

Pillar 1: Productive structure dynamics. Growth is driven by changes in the production structure: innovation (new activities and the upgrading of existing ones) and the complementarities, or linkages, among sectors that let those changes spread. The related claim that "what you export matters" is Hausmann et al.'s (2007); ECI is its later measurement. In this book's terms, Ocampo's innovation corresponds most closely to condition (2) (ECI) and his complementarities to condition (1) (integration Γ). The policy implication: industrial policy is not a relic of the ISI era but a necessity for structural transformation.

Pillar 2: Macroeconomic management for structural change. Ocampo's distinctive contribution: macroeconomic policy (fiscal, monetary, exchange rate) should be designed not just for stability but for structural transformation. Counter-cyclical fiscal policy prevents the commodity booms from being wasted on consumption (Chapter 23's structural-cyclical decomposition). Real exchange rate management prevents "Dutch disease" (commodity exports appreciating the currency and destroying manufacturing competitiveness), and the balance-of-payments constraint on growth must be managed rather than ignored. This pillar has *no* counterpart among the conjecture's three conditions: in the language of this book it concerns the stability of the trajectory (Chapter 23), not the structure of W.

Pillar 3: Social equity as a condition for growth. Ocampo reverses the conventional sequence ("grow first, redistribute later"): equity is not a consequence

of growth but a *precondition* for it. Ocampo's equity concerns income distribution, social protection and rights. Condition (3) of the conjecture — information diversity N_{eff} — is this book's addition, not his; the closest link is our claim that low attribute constraints (λ_k) and diverse information are needed for the human capital base, domestic demand and political stability that structural transformation requires.

Ocampo's synthesis	**This book's object**	**Book reference**
Innovation, productive transformation	ECI > 0; dense subgraph $\mathcal{D}$	Ch. 16, 25
Complementarities (linkages)	Integration $\Gamma > \Gamma^*$	Ch. 6, 17, 25
Counter-cyclical policy	$\Delta\pi^S$ vs. $\Delta\pi^C$	Ch. 23
Macro for structural change, BoP constraint	No counterpart in the conjecture; $v_\Gamma > 0$ sustained	Ch. 23
Social equity as precondition	Partly: $\lambda_k \uparrow$; $N_{\text{eff}} > N^*_{\text{eff}}$ is our addition	Ch. 18, 21
Complementarity of policies	Supermodular V (assumed)	App. 26B
Growth with equity	$\Gamma > \Gamma^*$ + ECI > 0 + $N_{\text{eff}} > N^*$	Conj. 26.1

Ocampo's work completes the structuralist arc: Prebisch diagnosed the asymmetry; Furtado explained the reproduction; Pinto measured the heterogeneity; Fajnzylber identified the missing piece; CEPAL documented the stagnation; Ocampo synthesized the prescription. This book offers the synthesis a formal language. The weight matrix represents Prebisch's structure; the spectral gap measures integration, though in Bolivia's data Pinto's structural heterogeneity shows up in the export enclave's π and export orientation rather than in aggregate Γ; ECI is closely related to, but distinct from, Fajnzylber's truncation (Chapter 25); the formation model represents Furtado's cultural dependency. Conjecture 26.1 borrows two of Ocampo's pillars, adds one of its own, and leaves his macroeconomics aside — a precise claim that the next generation of data can confirm or refute.

Further Reading

Ocampo: Ocampo (2004), "Latin America's Growth and Equity Frustrations During Structural Reforms." Ocampo (2007), "La macroeconomía de la bonanza económica

latinoamericana." Ocampo et al. (2009), *Growth and Policy in Developing Countries: A Structuralist Approach*. Ocampo (2009), "Latin America and the Global Financial Crisis." Ocampo and Ros (2011), *The Oxford Handbook of Latin American Economics*. Ocampo (2014), "The Latin American Debt Crisis in Historical Perspective," in Stiglitz and Heymann (eds.), *Life After Debt*.

Neo-structuralism: CEPAL (2012b), *Structural Change for Equality*. CEPAL (2016), *Horizons 2030: Equality at the Centre of Sustainable Development*. Bielschowsky (2009), "Sixty Years of ECLAC."

Growth with equity: Hausmann et al. (2007), "What You Export Matters." Rodrik (2007), *One Economics, Many Recipes*. Stiglitz (2002), *Globalization and Its Discontents*. Lin and Chang (2009), "Should Industrial Policy in Developing Countries Conform to Comparative Advantage?" (the debate between the two approaches).

Chapter 27

Algorithmic W^I

When the Machine Rewires Who Sees What

> *"The algorithms that determine what you see are not neutral. They are structural interventions on the information network — operators on W^I with objectives that may have nothing to do with your welfare or your society's development."*
> — This book

Every chapter until now has treated W^I as shaped by human institutions: media organizations, schools, social networks, community structures. This chapter confronts the twenty-first century's structural novelty: W^I is now substantially designed by algorithms.

Recommendation systems on Facebook, Instagram, TikTok, YouTube, and Google determine what billions of people see, whom they hear, and which signals enter their formation equation. (WhatsApp, the dominant platform in much of Latin America, is different: messages arrive in chronological order from contacts and groups, with no ranking algorithm. Its W^I is shaped by group membership and forwarding, not by a recommender, so the results of this chapter apply to it only indirectly.) These algorithms are operators on W^I — they add, remove, and reweight information edges in real time. They operate at a scale and speed that no human institution has achieved.

The question for structural economics: do these algorithms increase or decrease N_{eff}? Do they break assortative mixing or reinforce it? Do they connect the quinoa farmer in Potosí to the engineer in Santa Cruz, or do they lock each into a personalized information silo?

The answer: it depends on the algorithm's objective function — and on what users themselves choose, which the evidence below shows matters at least as

much. And that objective function is not set by the farmer, the engineer, or the Bolivian state. It is set by firms in Menlo Park, Beijing, and Mountain View.

27.1 The Algorithm as an Operator on W^I

27.1.1 Formalization

Definition 27.1 (Algorithmic W^I Operator)**.** *A **recommendation algorithm** is a function $\mathcal{A} : (W^I, \mathbf{h}) \to W^{I'}$ that takes the current information network W^I and a vector of engagement histories $\mathbf{h} = (h_1, \ldots, h_N)$ and outputs a modified information network $W^{I'}$:*

$$W^{I'}_{ij} = \mathcal{A}_{ij}(W^I, h_i, h_j, \boldsymbol{\eta}), \tag{27.1}$$

where $\boldsymbol{\eta}$ are the algorithm's parameters (learned from data to maximize an objective).

The algorithm modifies W^I by choosing which content from source j to show user i. If the algorithm shows i more content from j: $W^{I'}_{ij} > W^I_{ij}$ (the edge strengthens). If it suppresses content from j: $W^{I'}_{ij} < W^I_{ij}$ (the edge weakens).

Conjecture 27.2 (The Algorithm's Objective Shapes W^I's Structure)**.** *Three objective functions produce qualitatively different W^I structures:*

Objective 1: Maximize engagement *(the standard commercial objective). The algorithm shows i content that maximizes i's time on platform. Engagement is maximized by content that is emotionally arousing, confirms existing beliefs, and comes from sources similar to i. This produces **assortative amplification**: $W^{I'}_{ij}$ is large when i and j are similar (same demographic, same beliefs, same consumption pattern). The algorithm reinforces the existing block structure of W^I, lowering cross-group mixing λ_k and integration Γ (Proposition 27.3); by itself, block reinforcement need not change N_{eff} (Remark 27.6).*

Objective 2: Maximize information accuracy *(a hypothetical public-interest objective). The algorithm shows i content from the most accurate and diverse sources. This produces **bridge-building**: $W^{I'}_{ij}$ is large when j has information that i lacks (cross-block edges). The algorithm breaks assortative mixing and raises λ_k and Γ.*

Objective 3: Maximize advertiser revenue *(a variant of engagement). Similar to Objective 1 but weighted by the commercial value of the user's attention. This produces **stratified exposure**: high-value users (urban, educated, high-income) see diverse, high-quality content; low-value users (rural, less-educated, low-income) see lower-quality, more assortative content. The algorithm raises the diversity of the elite's feeds and lowers that of the periphery's. (Diversity of a single user's feed is the row entropy $H_i = -\sum_j W^I_{ij} \ln W^I_{ij}$, or its exponential e^{H_i}, the effective number of sources user i sees. N_{eff} is a property of the whole network and has no per-user version.)*

Intuition: The algorithm IS the new organic intellectual

In Gramsci's framework (Chapter 24), the "organic intellectual" is the agent with high π^I who produces the narratives that sustain the hegemony. In the algorithmic era, the organic intellectual is the recommendation system: it determines which narratives reach which audiences, with what frequency, and in what context. The algorithm has π^I that dwarfs any human journalist, professor, or politician — it reaches billions of users simultaneously and personalizes the message for each.

The difference: Gramsci's organic intellectual had an ideology (a preferred W^I structure). The algorithm has an objective function (maximize engagement, revenue, or some other metric). The objective function may or may not align with any political project, any development goal, or any welfare criterion from Chapter 12. It is structural power without structural intent.

27.2 Three Effects on Structural Economics

27.2.1 Effect 1: Echo Chambers and the Formation Trap

A precise version needs a model of what the algorithm does. The simplest is **multiplicative engagement ranking**: the algorithm reweights the organic feed by expected engagement and renormalizes,

$$W_{ij}^{I'} = \frac{W_{ij}^I \, e_{ij}}{\sum_l W_{il}^I \, e_{il}}, \tag{27.2}$$

where $e_{ij} > 0$ is the expected engagement of user i with content from source j. Belief confirmation means $e_{ij} = e_s$ when i and j share attribute k (same group) and $e_{ij} = e_c < e_s$ otherwise.

Proposition 27.3 (Engagement Ranking Lowers Mixing on the Sorted Attribute)**.** *Let W^I be row-stochastic and let (27.2) hold with group-level engagement $e_s > e_c > 0$ on attribute k. (i) A user whose organic cross-group share is $\lambda_i = \sum_{j \notin g(i)} W_{ij}^I$ (where $g(i)$ is i's group, i included) ends with*

$$\lambda_i' = \frac{\lambda_i \, e_c}{\lambda_i \, e_c + (1 - \lambda_i) \, e_s} < \lambda_i \qquad (0 < \lambda_i < 1). \tag{27.3}$$

(ii) For two equal groups of size m with uniform weights inside, $(1-\lambda)/m$, and across, λ/m, the matrix is doubly stochastic with $\Gamma = 1 - |1 - 2\lambda|$, which equals 2λ for $\lambda \leq \frac{1}{2}$. Reweighting maps it to the matrix of the same form with λ' from (27.3). Hence, for

$0 < \lambda \leq \frac{1}{2}$, *integration falls,* $\Gamma' = 2\lambda' < 2\lambda = \Gamma$, *while* $\boldsymbol{\pi}^I$ *stays uniform and* $N_{eff} = N = 2m$ *before and after.*

Proof. (i) Row i's cross-group mass after reweighting (before normalization) is $\sum_{j \notin g(i)} W^I_{ij} e_c = \lambda_i e_c$, and its own-group mass is $(1 - \lambda_i) e_s$; dividing by their sum gives (27.3). Since $\lambda_i > 0$, the inequality $\lambda_i' < \lambda_i$ is equivalent to $e_c < \lambda_i e_c + (1 - \lambda_i) e_s$, i.e. $(1 - \lambda_i)(e_s - e_c) > 0$, true for $\lambda_i < 1$.

(ii) Write the matrix as $B \otimes \frac{1}{m} \mathbf{1}_m \mathbf{1}_m'$ with $B = \left(\begin{smallmatrix} 1-\lambda & \lambda \\ \lambda & 1-\lambda \end{smallmatrix}\right)$. The eigenvalues of a Kronecker product are the products of the factors' eigenvalues: B has eigenvalues 1 and $1 - 2\lambda$, and $\frac{1}{m} \mathbf{1}\mathbf{1}'$ has 1 and 0 (multiplicity $m - 1$), so the spectrum is $\{1, 1 - 2\lambda, 0\}$ and $\Gamma = 1 - |1 - 2\lambda|$ (for $m = 1$ the zero eigenvalue is absent, which does not change Γ). The matrix is symmetric, hence doubly stochastic, so $\boldsymbol{\pi} = \mathbf{1}/N$ and $N_{\text{eff}} = N$. Under (27.2) every within-group entry of a row is multiplied by e_s and every cross-group entry by e_c, and every row has the same $\lambda_i = \lambda$; so after normalization the within entries are all $(1 - \lambda')/m$ and the cross entries all λ'/m. The reweighted matrix is therefore of the same form with λ', and by (i) $0 < \lambda' < \lambda \leq \frac{1}{2}$, so $\Gamma' = 2\lambda' < 2\lambda = \Gamma$. (For $\lambda > \frac{1}{2}$ the conclusion can fail: $\Gamma = 2(1 - \lambda)$ rises as λ falls toward $\frac{1}{2}$.) □

The result holds for the attribute the algorithm sorts on. It does *not* extend to "all attributes k", as an earlier draft claimed. In `code/ch27.py`, the organic feed is homophilous by geography only, urban users mostly hold belief L and rural users mostly R, and the algorithm triples the weight of same-belief content. Cross-belief exposure falls from 0.426 to 0.218, but rural users who hold the urban majority's belief (and urban users with the rural majority's belief) see their cross-geography exposure *rise* from 0.217 to 0.371: the algorithm sends them to like-minded sources in the other region.

Consequence for the formation model, on the sorted attribute: the self-confirming trap of Chapter 10 becomes deeper. The aspiration gap between groups widens because each group sees a more biased sample of returns. In the formation model, the Ψ ratio increases (the formation channel becomes more dominant) and the welfare cost of discrimination (Chapter 18) rises.

Example 27.4 (Bolivia: Facebook and the Urban-Rural Information Divide [ILLUSTRATIVE])**.** *Bolivia has on the order of 7 million Facebook users (early 2020s; source to be added). The recommendation algorithm shows users content similar to what they have previously engaged with. Urban Bolivians engage with commercial media, political commentary, and global content. Rural Bolivians engage with local community posts, agricultural content, and regional news.*

The algorithm amplifies this difference: urban users see progressively more urban content; rural users see progressively more rural content. The cross-group mixing parameter between rural indigenous women and urban audiences, already low (illustratively $\lambda \approx 0.22$; in Chapter 21's audience table this pair has the lowest overlap, 35% of self-overlap), may decrease further under algorithmic filtering.

Illustrative effect: by (27.3), an engagement premium of same-group content e_s/e_c between 1.38 and 2.07 lowers λ from 0.22 to between 0.17 and 0.12, i.e. $\Delta\lambda \approx -0.05$ to -0.10 (a 23–45% cut in cross-group exposure relative to the organic feed; `code/ch27.py`). In the formation model, the steady-state aspiration gap between urban and rural populations widens by $\frac{\mu}{1-\rho}|\Delta\lambda|\,\Delta\bar{r}$, which is 0.03–0.06 preference units for $\rho = 0.85$ and an illustrative $\mu\Delta\bar{r} = 0.09$.

27.2.2 Effect 2: Epistemic Power Concentration

Conjecture 27.5 (Algorithmic Concentration of $\boldsymbol{\pi}^I$)**.** *Engagement-maximizing algorithms produce **winner-take-all dynamics** in W^I: a small number of viral content producers capture a large share of attention, while the vast majority of voices are suppressed. The Perron eigenvector $\boldsymbol{\pi}^I$ becomes more concentrated:*

$$N_{eff}(W^{I'}) \leq N_{eff}(W^I). \tag{27.4}$$

The mechanism: the algorithm amplifies content with high initial engagement (likes, shares, comments). Content from well-connected, emotionally compelling sources gets shown to more users, generating more engagement, generating more amplification. Content from less-connected sources (community radio, local journalists, indigenous-language media) gets suppressed because it generates less engagement.

The consequence: Bolivia's $\pi^I_{community}$ (already small: 0.12 on Chapter 17's four ecosystems; community radio reaches 14% of attention in Chapter 21) may decline further as algorithmic curation redirects attention to high-engagement commercial content, and N^I_{eff} could drop from 3.4 (Chapter 17) toward 2.5–3.0 (illustrative).

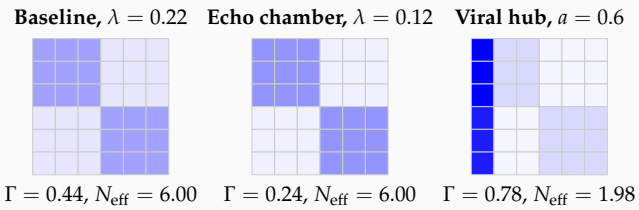
Remark 27.6 (Echo chambers and viral hubs move Γ and N_{eff} differently). The two effects of engagement ranking are different matrix operations and register differently on the two diagnostics (Proposition 6.16). In the six-user, two-group network of `code/ch27.py` ($\lambda = 0.22$: $\Gamma = 0.44$, $N_{\text{eff}} = 6$):

- an *echo chamber* that lowers λ to 0.12 halves integration ($\Gamma = 0.24$) but leaves power untouched ($N_{\text{eff}} = 6$);
- a *viral hub* that takes a share a of everyone's attention concentrates power

($N_{\text{eff}} = 3.74$ at $a = 0.3$, 1.98 at $a = 0.6$) while *raising* integration ($\Gamma = 0.61$, 0.78): a hub everyone listens to mixes fast.

A dashboard that read Γ as "structural health" would score the winner-take-all platform as an improvement. Monitoring must track both: Γ (or λ_k) for fragmentation, N_{eff} for concentration. Note also that a hub can *raise* cross-group exposure for some groups (rural users who all follow one urban viral source see more urban content), another reason the "for all k" claim fails.

Visualization: Echo chamber vs. viral hub: two different matrices [ILLUSTRATIVE]



Rows are users, columns are the sources they attend to; darker blue is more weight. Users 1–3 and 4–6 are two groups. The echo chamber darkens the diagonal blocks (less mixing, lower Γ) but keeps every column equally important ($N_{\text{eff}} = 6$). The viral hub darkens one column: everyone listens to source 1, power concentrates ($N_{\text{eff}} \approx 2$), and Γ rises. Γ measures integration; N_{eff} measures concentration; they are independent.

27.2.3 Effect 3: Cross-Border Information Integration

Conjecture 27.7 (The Paradox of Algorithmic Globalization)**.** *Algorithms also create cross-border information edges that did not previously exist. A TikTok user in El Alto can watch a Korean entrepreneur, a Brazilian farmer, or an Indian programmer. The algorithm sometimes shows content from radically different contexts, creating W^I edges that span the global product space.*

This is Sunkel's integration/disintegration (Chapter 13) applied to W^I: the algorithm simultaneously integrates globally (showing diverse international content to some users) and fragments domestically (personalizing feeds within national populations). The net effect on the domestic network is ambiguous. New cross-border sources spread attention over more outlets (raising N^I_{eff}); viral concentration lowers it (Conjecture 27.5); personalization lowers cross-group mixing (Proposition 27.3). N_{eff} is not additive across these channels, so the net effect must be computed on the combined matrix, not summed.

For urban educated Bolivians, the global effect may dominate the diversity of their feeds (e^{H_i} rises: exposure to international entrepreneurship, technology, ideas). For rural

indigenous Bolivians, the domestic effect may dominate (locked into algorithmically reinforced local content; e^{H_i} falls). Feed diversity would then rise for the elite and fall for the periphery — amplifying the compound disadvantage of Chapter 19.

Remark 27.8 (Reconciling with Chapter 24). Chapter 24 argued that new digital channels helped break the information monopoly of Bolivia's MAS hegemony in 2019, i.e. that they *raised* N_{eff}^{I}. This chapter argues that engagement ranking can *lower* it. Both can be true, because they operate at different levels: *across* channels, digital media add outlets that no coalition controls (state media lose weight, so N_{eff}^{I} rises); *within* a ranked platform, attention concentrates on a few viral sources (so N_{eff}^{I} can fall). Which dominates is an empirical question for each country and period, and the answer can change as platforms mature.

27.2.4 What the Evidence Says

The empirical literature is more mixed than the echo-chamber narrative suggests, and its best evidence comes from the United States:

- On Facebook in 2014, users' own choices reduced their exposure to cross-cutting news more than the ranking algorithm did (Bakshy et al., 2015).

- In the 2020 U.S. election experiments run with Meta, replacing the ranked feed with a chronological one, or reducing exposure to like-minded sources, did not measurably reduce polarization over three months (Guess et al., 2023; Nyhan et al., 2023), although ideological segregation in news exposure was high, especially in Pages and Groups (González-Bailón et al., 2023).

- Twitter's ranking algorithm amplified the mainstream political right more than the left in six of seven countries studied (Huszár et al., 2022); Facebook's algorithm limited exposure to counter-attitudinal news (Levy, 2021).

In the Global South the dominant channel is often not a ranked feed at all. Misinformation in Brazil's 2018 election and in India circulated largely through WhatsApp groups (Machado et al., 2019; Resende et al., 2019; Badrinathan, 2021); the UN fact-finding mission on Myanmar (2018) identified Facebook as a major vector of hate speech; and in Bolivia, Facebook removed in 2020 a network of inauthentic accounts targeting Bolivian politics that it attributed to the U.S. firm CLS Strategies. Proposition 27.3 is therefore a statement about one mechanism, whose size is an empirical question.

27.3 The Regulatory Response: W^I as a Public Good

Definition 27.9 (Information Network as a Public Good)**.** *The structure of W^I — who sees what — is a **public good** in the structural model: it determines the formation of preferences for the entire population, which determines the welfare loss (Chapter 2), the formation multiplier (Chapter 10), and the political equilibrium (Chapter 24). No individual user can choose the aggregate W^I structure; it is the emergent outcome of billions of algorithmic decisions.*

Remark 27.10 (Four Regulatory Instruments). Four instruments are available:

1. **Transparency requirements**: Mandate disclosure of the algorithmic objective function and its effect on W^I structure ($\Delta\lambda_k$, ΔN_{eff}). If users and regulators can observe what the algorithm does to W^I, they can demand changes.

2. **Diversity mandates**: Require recommendation algorithms to maintain a minimum cross-group exposure ratio: $\lambda_k^{I'} \geq \lambda_k^{\min}$ for all attribute groups k. This prevents the algorithm from amplifying assortative mixing below a threshold.

3. **Public-interest algorithms**: Develop and deploy alternative recommendation systems with Objective 2 (maximize information accuracy/diversity) rather than Objective 1 (maximize engagement). These could be operated by public broadcasters, educational institutions, or platform cooperatives.

4. N_{eff} **monitoring**: Track $N_{\text{eff}}(W^I)$ over time (Chapter 21) and treat declines as a regulatory trigger. If the algorithmically-mediated N_{eff} falls below a threshold, regulatory intervention is warranted. Track Γ or λ_k alongside it (Remark 27.6): they detect fragmentation, which N_{eff} misses.

Key Structural Insight

The algorithmic W^I is one of the major structural developments of the twenty-first century. A large part of the information network that shapes preference formation is now designed by a small number of firms, operating at global scale, with objectives that are not aligned with any development goal.

The structural economics framework provides the language: the algorithm is an operator on W^I. Its effect is measurable ($\Delta\lambda_k$, $\Delta\Gamma$, ΔN_{eff}, $\Delta\Psi$) — and the two dimensions, fragmentation and concentration, must be measured separately. Its welfare cost is computable within the model (Chapter 18's welfare loss formula, amplified by $1 + \Psi$). And its regulation is an operator design problem (Chapter 15): choose the algorithmic parameters $\boldsymbol{\eta}$ that keep both λ_k and N_{eff} above floors, subject to commercial viability.

The framework transforms the debate from "social media is good/bad" to: "what is the algorithm's effect on λ_k, N_{eff}, and Ψ, and does that effect move the structural trajectory toward or away from the growth-with-equity zone of Chapter 26?"

27.4 AI and the Future of Structural Analysis

27.4.1 AI as Measurement Tool

Remark 27.11 (AI for Structural Diagnostics). Machine learning techniques can improve the measurement of all three weight matrices:

1. W^P: Satellite imagery + NLP on firm registries can estimate production linkages in real time, filling gaps between benchmark I-O tables (Chapter 20).

2. W^I: Natural language processing of social media content can estimate media overlap (W^I_{media}, Chapter 21) at daily frequency, tracking N_{eff} in real time.

3. W^U: Sentiment analysis of social media posts can estimate solidarity patterns (W^U_{digital}, Chapter 22), supplementing survey-based measures.

4. The structural trajectory (Chapter 23) can be computed in near-real-time rather than at annual frequency.

27.4.2 AI as Structural Operator

Remark 27.12 (AI-Powered Structural Intervention). AI can also serve as a deliberate operator on W^I for development purposes:

1. **Mentoring at scale**: AI systems that connect would-be entrepreneurs in the traditional sector with role models, market information, and technical knowledge from the modern sector. This is the $\mathcal{L}_{\text{info}}$ operator (Chapter 15) deployed through an AI system rather than a human program.

2. **Adaptive education**: AI tutoring systems that adjust content to the learner's context while exposing them to cross-group knowledge ($\Delta\lambda^I > 0$).

3. **Market information**: AI systems that provide real-time price and demand information to rural producers, creating W^I edges from the traditional sector to the market that the assortative structure suppresses.

Prebisch Connection: The new center-periphery

The algorithmic W^I creates a new center-periphery structure that operates alongside the trade-based W^P:

The algorithmic center: Firms in Silicon Valley, Shenzhen, and a few other nodes control the parameters $\boldsymbol{\eta}$ of the global W^I. Their π^I is enormous: a single change to Facebook's algorithm can shift the formation signals received by 3 billion people.

The algorithmic periphery: Every other country is a consumer of algorithmic W^I designed elsewhere. Bolivia's W^I is shaped by algorithms designed, tuned and optimized for advertising markets elsewhere, and deployed without regard to Bolivian development needs. (How much local data and local adaptation enter these systems is not public, which is itself the point of the transparency instrument above.)

This is Prebisch's center-periphery thesis applied to W^I: the center produces the information infrastructure; the periphery consumes it. The "terms of trade" in W^I are the exchange of attention (from the periphery) for algorithmic curation (from the center). The formation signals that Bolivian youth receive are filtered through an objective function designed in Menlo Park.

The structuralist response: just as Prebisch advocated for ISI to reduce W^P dependency, the algorithmic era requires "information substitution" — the development of domestic or regional algorithmic capacities that can design W^I operators aligned with national development objectives. This is the information-age equivalent of building the capital goods sector: building the algorithmic infrastructure that shapes your own W^I.

27.5 Summary

1. Recommendation algorithms are **operators on** W^I: they determine who sees what, with effects measurable as $\Delta\lambda_k$, $\Delta\Gamma$, ΔN_{eff}, and $\Delta\Psi$. WhatsApp, central in Latin America, is not a recommender; its W^I is shaped by groups and forwarding.

2. Under multiplicative **engagement ranking**, cross-group mixing falls on the attribute the algorithm sorts on, $\lambda' = \lambda e_c/(\lambda e_c + (1-\lambda)e_s)$, and so does Γ; mixing on other attributes can rise. Echo chambers lower Γ but need not change N_{eff}; viral hubs lower N_{eff} but *raise* Γ. Monitor both.

3. The **evidence** is mixed: in the U.S., user choice matters as much as ranking,

and chronological feeds did not reduce polarization in the 2020 experiments; in the Global South, messaging groups and coordinated campaigns matter as much as feeds. Across channels digital media can raise N_{eff}^I (Chapter 24) while ranking concentrates attention within platforms.

4. With belief-dependent $W^I(\mathbf{b})$, a primitive $W^I(\mathbf{b}^*)$ forces consensus; persistent disagreement requires a reducible $W^I(\mathbf{b}^*)$, i.e. clusters, as in bounded-confidence models. **Regulation**: transparency, diversity mandates, public-interest algorithms, and monitoring of both λ_k and N_{eff}.

5. The **algorithmic center-periphery**: a few firms design the global W^I; the periphery consumes it. AI can also measure all three W's and serve as a development operator; information sovereignty requires building domestic algorithmic capacity — the information-age capital goods sector.

27.6 Exercises

27.1 Algorithmic $\Delta\lambda$. For a 4×4 W^I (urban men, urban women, rural men, rural women): simulate the effect of an engagement-maximizing algorithm that increases within-group exposure by 20% and decreases cross-group exposure by 15% (then renormalize rows). Compute $\Delta\lambda_k$ for gender and geography, $\Delta\Gamma$, and ΔN_{eff}. Why is $\Delta N_{\text{eff}} = 0$ if the four groups are equal-sized and the matrix is symmetric?

27.2 The echo chamber result. (a) Reproduce the proof of Proposition 27.3 for user-specific engagement e_{ij} that is higher for same-group sources ($\min_{j \in g(i)} e_{ij} > \max_{j \notin g(i)} e_{ij}$ for each user i); then show that the weaker condition "the W^I-weighted mean engagement of i's same-group sources exceeds that of her cross-group sources" is necessary and sufficient for $\lambda_i' < \lambda_i$. (b) Construct a two-attribute example (geography and belief) in which sorting on belief raises some group's cross-geography exposure, and check it against `code/ch27.py`. (c) Replace multiplicative reweighting by "show only the top-K sources by engagement". Does the result still hold?

27.3 Bolivia's digital divide. Using internet penetration data by department and demographic group: estimate the fraction of Bolivia's population whose W^I is algorithmically mediated (vs. traditional media/word-of-mouth). How does this fraction correlate with N_{eff} estimates from Chapter 21?

27.4 Objective function design. Propose an alternative algorithmic objective function for Bolivia that maximizes N_{eff} subject to a minimum engagement constraint. Write the optimization problem formally. What trade-offs does it involve?

27.5 AI mentoring simulation. Design an AI-powered mentoring system for rural Bolivian entrepreneurs. Specify: (a) the ΔW^I edges it creates, (b) the expected $\Delta\lambda^I$ and ΔN_{eff}, (c) the formation model prediction for $\Delta\theta^*$ after one generation.

27.6 Algorithmic center-periphery. Using SimilarWeb or a comparable web-traffic panel (Alexa was retired in 2022): estimate the fraction of Bolivia's internet traffic that goes to US-based platforms (Google, Facebook, YouTube, TikTok) vs. domestic content. What fraction of Bolivia's W^I is algorithmically designed abroad?

27.7 The policy frontier. Compare four regulatory regimes: (a) no regulation (current), (b) transparency only, (c) diversity mandates ($\lambda_k^{\min} = 0.3$), (d) public-interest algorithm. For each: estimate the effect on N_{eff}, the political feasibility, and the cost. Which regime best advances Bolivia's structural trajectory?

Mathematical Appendices

Appendix 27A: Recommendation Algorithms as Matrix Operators

Collaborative filtering. The standard recommendation algorithm estimates a user-content affinity matrix $R \approx UV'$, where U is a user-factor matrix and V is a content-factor matrix (matrix factorization, analogous to the SVD of the specialization matrix in Chapter 16). Content shown to user i is drawn from the rows of V with highest inner product $\mathbf{u}_i'\mathbf{v}_j$.

Effect on W^I. The user-factor matrix U clusters users by "type" (implicit demographic, behavioral, or preference clusters), and the content factor $\mathbf{v}_j$ of source j comes to resemble the factors of j's existing audience. Content from j is therefore ranked high for user i when $\mathbf{u}_i'\mathbf{v}_j$ is high, which tends to happen when i resembles j's audience. This produces assortative W^I. A convenient (assumed, not derived) parametrization of the result is $\lambda_k^{I'} \approx \exp(-d(\bar{\mathbf{u}}_g, \bar{\mathbf{u}}_{g'}))$, where d is the distance between group-average user factors: when groups are distinct in the factor space (different demographics, languages, consumption patterns), d is large and $\lambda^{I'}$ is small.

From ranking scores to W^I. Collaborative filtering produces predicted engagement $\hat{e}_{ij} = \mathbf{u}_i'\mathbf{v}_j$; a ranked feed then allocates attention increasingly in $\hat{e}_{ij}$. Equation (27.2) is the case where attention is proportional to organic weight times engagement; Proposition 27.3 is proved for it. Maximizing a *linear* objective $\sum_j R_{ij}\hat{e}_{ij}$ over a row-stochastic R is not a useful model: its solution puts all of i's attention on the single highest-$\hat{e}$ source, which may lie in the other group, so no monotone statement about λ follows from it.

Appendix 27B: The DeGroot Model with Algorithmic Intermediation

Modified DeGroot. In Chapter 8, the DeGroot model had beliefs converging through a fixed W^I. With algorithmic intermediation, W^I itself depends on beliefs:

$$\mathbf{b}_{t+1} = W^I(\mathbf{b}_t) \cdot \mathbf{b}_t, \tag{27.5}$$

where $W^I(\mathbf{b}_t)$ is the algorithmically modified information matrix (the algorithm adjusts the feed based on current beliefs/engagement patterns). This is a *nonlinear* dynamical system: the matrix changes with the state.

Fixed points. A fixed point $\mathbf{b}^*$ satisfies $\mathbf{b}^* = W^I(\mathbf{b}^*)\mathbf{b}^*$, i.e. $\mathbf{b}^*$ is a right eigenvector of $W^I(\mathbf{b}^*)$ for the eigenvalue 1. Two facts follow from Perron–Frobenius. (i) Every consensus vector $\mathbf{b}^* = c\mathbf{1}$ is a fixed point, for any W^I. (ii) If $W^I(\mathbf{b}^*)$ is irreducible, the eigenvalue 1 is simple with eigenvector $\mathbf{1}$, so $\mathbf{b}^*$ *must* be a consensus. Persistent disagreement is possible only if $W^I(\mathbf{b}^*)$ is *reducible*: the population splits into closed groups that put no weight on each other. This is a deeper form of the self-confirming trap: the algorithm creates the W^I that produces the beliefs that the algorithm was designed to serve.

Multiple equilibria: bounded confidence. An earlier draft described a "diverse" equilibrium in which beliefs stay varied while $W^I(\mathbf{b}^*)$ remains well connected. By (ii) this is impossible. The correct dichotomy is *consensus* versus *clusters*. The canonical model is bounded confidence (Hegselmann and Krause, 2002; Deffuant et al., 2000): $W^I_{ij}(\mathbf{b}) > 0$ iff $|b_i - b_j| \leq \varepsilon$, with equal weights on all such j. Beliefs converge in finite time to clusters more than ε apart, and the clusters are absorbing: once separated, they never interact again. In `code/ch27.py`, 60 agents with uniform initial beliefs end in 9 clusters for $\varepsilon = 0.05$, 2 for $\varepsilon = 0.15$ and full consensus for $\varepsilon = 0.30$. Engagement ranking acts like a smaller ε: it cuts the weight on distant sources, and with it the bridges that would have produced consensus. Note what "diversity" means here: a high-N_{eff}, well-connected $W^I(\mathbf{b}^*)$ goes with *agreement*; persistent variety of beliefs goes with a fragmented, reducible network.

Structuralist Case Study: The Algorithmic W^I in the Global South

This chapter's case study departs from the historical structuralist tradition to address a contemporary structural phenomenon. The algorithmic W^I invites a comparison with the colonial imposition of the racial classification system (Quijano, Chapter 18): both are structures designed outside the society that shape W^I for entire populations without their consent. The comparison is about structure, not about the scale of harm.

Three observations ground the structural analysis.

First: the speed of W^I transformation is unusual. Bolivia went from a small online minority around 2010 to roughly 7 million users of a single platform in the early 2020s (order of magnitude; source to be added). The share of the population using the internet rose from 22.4% in 2010 to 47.5% in 2019 and 77.3% in 2023 (ITU, via WDI series IT.NET.USER.ZS (World Bank, 2026b)). In Chapter 23's terms: W^I underwent a structural break — not gradual evolution but a discrete regime change. The digital W^I did not simply grow out of the existing information struc-

ture; for many users it displaced it. For urban Bolivians under 35, digital channels carry about half of W^I ($\omega_d = 0.50$, Chapter 21), much of it through WhatsApp groups rather than ranked feeds. This happened faster than the other structural changes in the book: faster than ISI or nationalization.

Second: the structural effects are unmeasured. Despite the speed and scale of the W^I transformation, we know of no developing country that systematically monitors N_{eff}, λ_k, or Ψ in the algorithmic W^I. The measurement infrastructure of Chapter 21 exists in principle but has not been deployed, and the platforms' own data are largely closed to outside researchers. A very powerful W^I operator is operating without structural monitoring.

Third: the regulatory capacity is asymmetric. The EU regulates algorithmic platforms (the Digital Services Act, 2022). The US debates regulation. In the Global South, where the structural effects may be largest (because N_{eff} was already low and attribute constraints were already severe), capacity is uneven: Brazil's electoral court has negotiated agreements with platforms and its Supreme Court has acted against them, and India regulates intermediaries under its 2021 IT Rules, but most smaller countries have little capacity. To our knowledge, Bolivia's telecommunications regulator (ATT) has no mandate, tools, or data to assess the structural effects of platform algorithms on Bolivia's W^I.

The structuralist response is clear: treat algorithmic W^I as a structural variable that requires measurement (Chapter 21), monitoring (Chapter 23), and regulation (this chapter's Remark 27.10). The alternative is to cede an important part of the formation of Bolivian preferences to algorithms designed elsewhere for other purposes.

Structural parallel	**Historical**	**Algorithmic**
External control of W	Colonial administration	Platform algorithms
Center-periphery	Trade asymmetry (W^P)	Information asymmetry (W^I)
Hegemonic formation	State media, schools	Engagement-maximizing feeds
Counter-hegemony	Community radio, movements	Public-interest algorithms?
Regulatory response	ISI, nationalization	Diversity mandates, λ_k and N_{eff} monitoring

Further Reading

Algorithmic curation and society: Pariser (2011), *The Filter Bubble*. Sunstein (2017), *#Republic: Divided Democracy in the Age of Social Media*. Zuboff (2019), *The Age of Surveillance Capitalism*.

Echo chambers and polarization: Levy (2021), "Social Media, News Consumption, and Polarization." Allcott et al. (2020), "The Welfare Effects of Social Media." Bail et al. (2018), "Exposure to Opposing Views on Social Media Can Increase Political Polarization." Bakshy et al. (2015), "Exposure to Ideologically Diverse News and Opinion on Facebook." Guess et al. (2023), "How Do Social Media Feed Algorithms Affect Attitudes and Behavior in an Election Campaign?" Nyhan et al. (2023), "Like-Minded Sources on Facebook Are Prevalent but Not Polarizing." González-Bailón et al. (2023), "Asymmetric Ideological Segregation in Exposure to Political News on Facebook." Huszár et al. (2022), "Algorithmic Amplification of Politics on Twitter." Golub and Jackson (2012), "How Homophily Affects the Speed of Learning and Best-Response Dynamics."

Opinion dynamics: Hegselmann and Krause (2002), "Opinion Dynamics and Bounded Confidence." Deffuant et al. (2000), "Mixing Beliefs among Interacting Agents." Dandekar et al. (2013), "Biased Assimilation, Homophily, and the Dynamics of Polarization."

Global South: Machado et al. (2019), "A Study of Misinformation in WhatsApp Groups with a Focus on the Brazilian Presidential Elections"; Resende et al. (2019), "(Mis)Information Dissemination in WhatsApp"; Badrinathan (2021), "Educative Interventions to Combat Misinformation: Evidence from a Field Experiment in India."

AI and development: Acemoglu and Johnson (2023), *Power and Progress: Our Thousand-Year Struggle over Technology and Prosperity*. World Bank (2019), *World Development Report: The Changing Nature of Work*. For the Global South specifically: Arora (2019), *The Next Billion Users: Digital Life Beyond the West*.

Platform regulation: EU Digital Services Act (European Union, 2022). Stigler Committee on Digital Platforms (2019), *Final Report*. For developing countries: Gurumurthy et al. (2019), "Platform Planet: Development in the Intelligence Economy."

Chapter 28

The General Case

Where the Standard Model Is the Special Case, and Structure Is Everything Else

> *"O subdesenvolvimento não constitui uma etapa necessária do processo de formação das economias capitalistas modernas. É, em si, um processo particular, resultante da penetração de empresas capitalistas modernas em estruturas arcaicas."*
>
> ["Underdevelopment does not constitute a necessary stage in the formation of modern capitalist economies. It is, in itself, a particular process, resulting from the penetration of modern capitalist enterprises into archaic structures."]
>
> — Celso Furtado, *Desenvolvimento e Subdesenvolvimento* (1961)

This is the last chapter. It has three purposes: to state the book's thesis in its final, corrected form; to restate each chapter's result in terms of W and the primitives it adds; and to turn what the book has not done into a research programme.

28.1 The Thesis

Remark 28.1 (Framing Principle: The General Case). The benchmark competitive model abstracts from social structure. Agents interact anonymously, through prices: aggregate demand weights each agent by her expenditure share s_i, so the implied interaction matrix is rank one, $W^{\mathrm{mkt}} = \mathbf{1}\mathbf{s}'$, with $\Gamma = 1$ and $\boldsymbol{\pi} = \mathbf{s}$. With equal endowments it is $W^{\mathrm{CE}} = \frac{1}{N}\mathbf{1}\mathbf{1}'$. Utility interdependence is absent ($W^{U} = 0$, no externalities), which is what the First Welfare Theorem needs; it holds for *any* wealth distribution $\mathbf{s}$ (Chapter 1, Section 1.7).

Structural economics studies the general case: non-anonymous, row-stochastic interaction matrices W — production, information and utility networks — and how they are produced and changed. Every W can be written as

$$W = W^{\mathrm{CE}} + \Delta(W), \qquad \Delta(W) := W - \tfrac{1}{N}\mathbf{1}\mathbf{1}', \tag{28.1}$$

and the departure is summarized by two coordinates that are independent of each other (Proposition 6.16):

- **power distance** $d_\pi = \frac{1}{2}\|\boldsymbol{\pi} - \frac{1}{N}\mathbf{1}\|_1$, or equivalently N_{eff}/N, which measures how unequally the structure distributes power;
- **integration deficit** $1 - \Gamma$, which measures how slowly shocks, beliefs and demand propagate across the network.

Neither determines outcomes alone. Allocations, beliefs, preferences and welfare depend on W *and* on primitives that are not functions of $\Delta(W)$: the bargaining surplus Φ and curvature κ, the formation parameters ρ and μ, endowments, the technical coefficients A, the coalition game, the product space.

This is a framing principle, not a theorem: the decomposition (28.1) holds for every matrix, so its content lies entirely in the maps from W and the primitives to outcomes. Those maps are what the previous 27 chapters study — some with theorems, more with conjectures and illustrative calculations.

28.2 The Book in One Page

The table restates each chapter's main result in terms of W and lists the primitives, beyond W, on which it depends. Integration is Γ; power is $\boldsymbol{\pi}$, with concentration measured by N_{eff}, $\mathrm{Var}(\boldsymbol{\pi})$ or $G(\boldsymbol{\pi})$. Bolivian production-network, complexity and distribution values are sourced (Section 28.4); σ_B and the W^I and W^U values remain illustrative.

Ch.	Topic	Result in terms of W	Primitives X
1	Matrix notation	Allocation, production and interaction as matrices. The benchmark is anonymous market interaction $W^{\mathrm{mkt}} = \mathbf{1}\mathbf{s}'$; W^{CE} is its equal-endowment case.	Endowments, preferences

Ch.	Topic	Result in terms of W	Primitives X
2	Welfare theorems	FWT holds for any $\mathbf{s}$ given complete markets, price-taking and no externalities ($W^U = 0$). Unequal $\boldsymbol{\pi}$ is distributional: $L_{\text{power}} = (\kappa N/2)\Phi^2 \operatorname{Var}(\boldsymbol{\pi})$.	κ, Φ, U
3	Growth & distribution	Growth spreads across blocks at a speed set by the integration of W^P; multipliers come from A, not from W.	A, saving, technology
4	Benchmark assumptions	Six assumptions restated as restrictions on W: $W^U = 0$, W^P exogenous, no W^I, anonymity, $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$ (equal endowments), fixed θ.	—
5	Three networks	W^P, W^I, W^U; a composite $\sum_k \alpha_k W^k$ exists only on a common node set.	α_k
6	Power	$\boldsymbol{\pi}'W = \boldsymbol{\pi}'$; $\boldsymbol{\pi}$ uniform iff W doubly stochastic; Γ and $\boldsymbol{\pi}$ independent.	Reading of $\boldsymbol{\pi}$
7	Bargaining	With $\boldsymbol{\pi}$ as Nash weights (an assumption): $x_i = \Omega/N + (\pi_i - 1/N)\Phi$.	d_i, S, Φ
8	Beliefs	DeGroot consensus $b^* = \boldsymbol{\pi}'\mathbf{b}^{(0)}$, reached at a speed set by $\Gamma(W^I)$; $N_{\text{eff}} = 1/\sum(\pi_j^I)^2$.	$\mathbf{b}^{(0)}$
9	Preferences	$\theta_{i,t+1} = \rho\theta_{i,t} + (1-\rho)\theta^\circ + \mu(\sum_j W_{ij}^I r_j - r^*)$: preferences formed through W^I. Definition: $\Psi_i(\Delta W, T) = F_i/E_i$, for a stated perturbation and horizon.	ρ, μ, θ°, r^*
10	Formation model	The loop $\boldsymbol{\theta} \to \mathbf{x}^* \to \mathbf{r} \to \boldsymbol{\theta}$ is a map T with Jacobian $J = \rho I + \mu\phi W^I CP$; unique stable steady state if $\rho(J) < 1$ (affine case); traps need nonlinear returns.	ρ, μ, returns
11	Measuring Ψ	The cohort–year ratio $\hat{\Psi}_{\text{var}}$ is descriptive, not Ψ; identifying Ψ needs strong assumptions or randomized W^I.	Identification design
12	Welfare	W^I shapes the preferences that evaluate W^P ("the ruler is rubber"). Under fixed aspirations, structures rank by $\operatorname{Var}(\boldsymbol{\pi})$, not by Γ.	Welfare criterion
13	Network evolution	$W_{t+1} = \mathcal{E}(W_t; \mathbf{x}_t^*)$: proximity-driven edge creation and destruction; structural equilibria exist and can be multiple (illustratively $\Gamma = 0.072$ and 0.643, same $\boldsymbol{\pi}$).	Creation, destruction rates

Ch.	Topic	Result in terms of W	Primitives X
14	Coalitions	Payoff $v_i \approx (1+\Psi)(\Delta\Omega/N + \Phi\,\Delta\pi_i)$: the power part is constant-sum; alliance cycling is a conjecture.	Game form
15	Operator design	Choose ΔW to raise integration and to lower concentration; because $\Gamma \perp \boldsymbol{\pi}$, raising Γ need cost no one power.	Budget, costs
16	Global W^P	Global input and trade shares; ECI comes from the country $\times$ product matrix M, not from W.	M, RCA rule
17	Domestic diagnostics	Γ, σ_B, $G(\boldsymbol{\pi})$, N_{eff}, Ψ, read separately. Bolivia 2019: $\Gamma = 0.467$ (0.475 like-for-like, inside the comparator range 0.463–0.649; 0.33–0.69 across normalizations), $G(\boldsymbol{\pi}^P) = 0.445$ (Korea 0.658). The enclave shows in $\pi_{\text{hydrocarbons}} = 0.039$ with 70% of output exported, not in Γ. Illustrative: $\sigma_B = 0.62$, $N^I_{\text{eff}} = 3.4$.	Data, aggregation, normalization
18	Gender & race	Node attributes constrain W through assortative mixing λ_k, which lowers Γ when within-group links dominate; power gaps need unequal group size or opportunity structure. Intersectionality is the interaction term.	λ_k
19	Scale invariance	$\pi_I = \pi_G \cdot \pi_{N\|c} \cdot \pi_{\text{within}}$ exactly (stochastic complementation); under exact lumpability aggregation raises Γ; isomorphism is a conjecture.	Nesting of scales
20	SAMs	A row-normalized SAM is a W whose Perron vector is account size; multipliers come from the SAM coefficient matrix.	SAM accounts
21	W^I measurement	Media overlap, reference groups, social networks, digital traces; audience data measure integration and reach, not who is believed. Bolivia: $\Gamma^I = 0.74$, $N_{\text{eff}} = 5.59$ of 6 groups; no information bubbles in audience shares.	Survey design

Ch.	Topic	Result in terms of W	Primitives X
22	W^U measurement	Transfers, experiments, voting, surveys. Bolivia: assortative ($\Gamma = 0.30$) but not concentrated ($N_{\text{eff}}^U = 3.85$ of 4 groups).	Survey design
23	Time series	Per-period Γ_t does not control $W_t \cdots W_s$: use τ_1, the realized-product gap and the absolute probability sequence. Bolivia's Γ fell from 0.567 (1990) to 0.484 (2013), $v_\Gamma = -0.036$ per decade; Korea's fell from 0.626 (1995) to 0.511 (2019). Aggregate Γ does not separate development paths, and the SCI is undefined when Γ starts above Γ^{ref}, as both do.	Filter, Γ^{ref}
24	Hegemony	Hegemonic equilibrium: W^I-formed consent for W^P. Under conformity, counter-hegemony succeeds past a saddle-node threshold; the cycle is a conjecture.	ρ, β, s
25	Truncation	Removing a bridge set bounds Γ by twice the remaining conductance; that the capital-goods core is a bridge is a conjecture. It gets no support at 12 sectors: Bolivia's Γ (0.475) is close to Korea's (0.511) while its ECI (-1.30) is far below ($+1.81$). Testing it needs 39-industry or product-level data.	ϕ, resolution
26	Synthesis	Conjecture: sustained growth with equity needs $\Gamma > \Gamma^*$, ECI > 0 and $N_{\text{eff}} > N_{\text{eff}}^*$. Bolivia's 2005–2019 fall in inequality (Gini 0.600 to 0.426) with a negative, falling ECI contradicts it as a claim about equity gains.	Thresholds
27	Algorithms	Engagement ranking lowers λ_k and Γ on the sorted attribute; viral hubs lower N_{eff} but raise Γ.	Engagement e_{ij}

28.3 What Furtado Knew

Celso Furtado's 1961 thesis — that underdevelopment is not a stage but a structure — is the sentence that generated this book. Every chapter is an elaboration.

Remark 28.2 (Furtado's Thesis in the Language of W). Furtado (1961) made three claims. Their translation is ours, and it is a reinterpretation rather than a proof:

1. **Underdevelopment is not the absence of development.** It is a specific structural configuration, the product of the *interaction* of a modern sector, tied to the world economy, with an archaic one. In W terms it has *two* features that the book's two coordinates keep apart. The first is an *enclave*: a modern export sector weakly tied to the domestic production network. It shows in the sector's small Perron weight in W^P and its high export share, not necessarily in a small aggregate $\Gamma(W^P)$. The second is an extraction channel through which surplus flows from the archaic to the modern sector, which concentrates bargaining power ($\boldsymbol{\pi}$ far from uniform under Assumption 7.A, Chapter 7). Neither implies the other. In Bolivia's measured W^P the enclave is there (hydrocarbons: $\pi = 0.039$, 70% of output exported), while aggregate Γ is in the comparator range and production power is less concentrated than Korea's.

2. **The structure reproduces itself.** Underdevelopment is a stable fixed point, not a transient state. The mechanisms of reproduction are: proximity-driven edge formation (Chapter 13), self-confirming preferences (Chapter 10), coalition dynamics that protect the incumbent W (Chapter 14), and hegemonic formation that produces consent (Chapter 24).

3. **The structure can be changed, but not by the standard tools.** Price adjustments, trade liberalization, and transfers do not change W. They change $\boldsymbol{\omega}$ (endowments) and relative prices within the existing structure. Changing W requires the operators of Chapter 15: information policy, institutional reform, education, infrastructure, and industrial policy, sequenced by Ψ and constrained by the political economy of Chapter 14.

Every chapter in this book develops one of these three claims:

Furtado's claim	**Formal content**	**Chapters**
Not a stage but a structure	Non-anonymous W: an export enclave (low π, high export share) and concentrated bargaining power	1–4
Self-reproducing	Stable fixed point of the co-evolutionary system	5–12, 24
Changeable but not by standard tools	Operators on W, not on $\boldsymbol{\omega}$	13–15, 25–26
Measurement	SAMs, W^I, W^U, trajectories	16–23
Scale invariance	Same at all scales	19
Contemporary	Algorithmic W^I	27

Intuition: Why "the general case"

The benchmark Arrow–Debreu model abstracts from social structure: agents interact anonymously through prices ($W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$), there are no externalities ($W^U = 0$), and preferences are taken as given rather than formed through W^I. This is a deliberate abstraction, not a mistake, and modern economics has relaxed each part of it — production-network macroeconomics (Long and Plosser, 1983; Acemoglu et al., 2012a; Baqaee and Farhi, 2019), network learning (DeGroot, 1974; Golub and Jackson, 2010, 2012) and endogenous preferences already work with general W. Those literatures are this book's allies.

What structural economics adds is a single language for the three networks at once, and a structuralist agenda for it: who holds power in W, how W forms the preferences that judge it, and how W can be changed. In that sense it *removes a restriction*: the anonymous benchmark restricts W to rank-one matrices $\mathbf{1}\mathbf{s}'$, a set of measure zero among row-stochastic matrices, and structural economics allows W to be anywhere.

The geometric analogy (Euclid as the flat special case of Riemann) holds only in part. The benchmark is not identified by a single number: $\Gamma = 1$ holds for every rank-one $W = \mathbf{1}\boldsymbol{\pi}'$, including the dictatorship $\mathbf{1}\mathbf{e}_1'$. The "flat" case needs two coordinates: full integration ($\Gamma = 1$) *and* equal power ($\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$). Structure is a departure in either coordinate, and the two can move independently.

28.4 What Bolivia Taught Us

Bolivia appeared in every chapter — not because Bolivia is the only developing country, but because it is a demanding test case. The data confirm some of the pathologies the book expected (an export enclave, low and falling complexity), contradict others (low integration, concentrated production power), and show equity gains that the growth-equity conjecture did not allow for.

Example 28.3 (Bolivia: The Structural Portrait, Complete)**.**

Instrument	*Bolivia*	*Benchmark*	*Chapter*
Sourced			
$\Gamma(W^P)$, *12 sectors, 2019*	*0.467*	*1.00*	*17*
like-for-like (others: 0.463–0.649)	*0.475*		*17, 25*
$G(\boldsymbol{\pi}^P)$ *(KOR 0.658, DEU 0.611)*	*0.445*	*0.00*	*17*
π *of hydrocarbons in* W^P	*0.039*		*17*
Hydrocarbons' output exported	*70%*		*17, 20*
ECI, 2006 $\rightarrow$ *2019*	$-1.21 \rightarrow -1.30$	> 0	*16*
π_G *(global trade* W^P*)*	*0.00053*	$1/N_G$	*16*
v_Γ *(1990–2013)*	-0.036*/decade*		*23*
Gini, 2005 $\rightarrow$ *2019*	$0.600 \rightarrow 0.426$		*17, 26*
Extreme poverty, 2005 $\rightarrow$ *2018*	$38.2\% \rightarrow 15.2\%$		*17*
Illustrative [ILLUSTRATIVE]			
σ_B	*0.62*	*0.00*	*17*
$N_{eff}(W^I)$*, four ecosystems*	*3.4*	*4*	*17*
$N_{eff}(W^U)$*, four groups*	*3.85*	*4*	*22*
$\hat{\Psi}$ *(self-employment)*	*1.56*	—	*17*
Truncation τ	*0.98*	*0.00*	*25*

Power ratio per capita: indigenous woman (Potosí) to German engineer $\approx 1 : 467$ *(Chapter 19, illustrative)*

SCI (1990–2019): undefined, because Bolivia's Γ *starts above* $\Gamma^{ref} = 0.50$ *(Chapter 23)*

Truncation share of the integration deficit: not identified at 12 sectors (Chapter 25)

2005–2019: large falls in poverty and inequality; aggregate Γ *flat to falling; ECI falling*

Sources: Γ, $G(\boldsymbol{\pi}^P)$ *and* π*, domestic input-share* W^P*, basic prices, diagonal kept, from the INE supply-use tables (base 2017 for 2019; base 1990 for* v_Γ*) (Instituto Nacional de Estadística (INE), Bolivia, 2026) and, for comparators, OECD ICIO 2021 and 2023 (OECD, 2021, 2023), author's calculations (data/compute/compute_bolivia_io.py, compute_oecd_io.py, compute_velocity.py); the like-for-like row uses the ISIC-based 12-sector aggregation, as do the comparators (Korea, Germany, the United States, Chile, China, Mexico). ECI: Growth Lab, HS92 (The Growth Lab at Harvard University, 2026).* π_G*: Growth Lab bilateral trade, 230 economies, author's calculations (data/compute/compute_global_wp.py). Gini of per-capita household income and extreme poverty (old lines; 13.2% in 2019 on the updated lines): INE, Encuesta de Hogares, cuadros 3.06.01.02–03.*

This table is the structural health record of an economy. In this edition its rows on the production network, complexity, global trade and distribution are sourced, and the chapter

scripts in `code/` *read them from* `data/processed`*; the between-block, information, utility and formation rows remain illustrative placeholders. Read together, the sourced rows tell a different story from the one this book set out to tell. Bolivia's domestic production network is not a fragmented outlier: its* Γ *is inside the comparator range, and* Γ *fell in Bolivia (1990–2013) as it did in Korea (1995–2019), so aggregate* Γ *does not separate development paths. Its production power is* less *concentrated than Korea's or Germany's; in industrial economies* $\pi(W^P)$ *concentrates on manufacturing and business-service hubs, which is industrial depth, not oppression. What is distinctive is the enclave, measured by* π *and export orientation; complexity, low and falling; and, against the book's expectations, a large fall in inequality and poverty.*

The numbers are not the ones an earlier draft expected, and that is the point of computing them. They are *specific*. The structural diagnosis does not say "Bolivia is poor." It says: "Bolivia's domestic production network is about as integrated as Korea's at the 12-sector level ($\Gamma = 0.475$ against 0.511), so aggregate integration is not its binding constraint, and the truncation conjecture gets no support at that level (Chapter 25). Its export sector is an enclave: hydrocarbons have $\pi = 0.039$ in the domestic network, export 70% of their output, buy 0.225 of domestic inputs per unit of output and pay 6% of their value added to employees; the rent is captured fiscally (IDH), not through linkages. Its complexity is low and falling (ECI -1.30, 131st). Its inequality nonetheless fell, from a Gini of 0.600 to 0.426 over 2005–2019, which Chapter 26's conjecture does not allow for. The information network has 3.4 effective voices and the formation channel produces self-confirming preferences ($\Psi = 1.56$), both illustrative until measured. The proposed intervention sequence is information $\rightarrow$ education $\rightarrow$ infrastructure $\rightarrow$ industrial policy, aimed at complexity and at the enclave's linkages, at an illustrative cost of \$17B over 30 years (about 1.2% of 2019 GDP a year). The political precondition is a coalition shift enabled by Phase 1's W^I reforms."

Precision is the gift structural economics offers. The standard model says: "get the prices right." The structuralist tradition says: "change the structure." This book says: "here is the structure, here are its eigenvalues and its distribution of power, here is what we conjecture is wrong, here is where the data have already shown us wrong, and here is a specification for fixing it that the data can check."

28.5 What the Book Does Not Do

Intellectual honesty requires stating the limits.

1. The model is static in important ways. The co-evolutionary system (Chap-

ter 13) is a difference equation, not a continuous-time model with stochastic shocks, financial frictions, or expectations. The formation model (Chapter 10) has a single preference parameter per agent. Reality has many. The model captures the structural skeleton; it does not capture the flesh.

2. The empirical estimates are rough. Γ, N_{eff}, and Ψ are estimated from imperfect proxies (I-O tables that are updated every 5–10 years, media surveys with small samples, cohort-year decompositions that rely on strong identifying assumptions). The structural health dashboard is a first approximation. The values for the production network, complexity, trade and distribution of Bolivia and its comparators are now sourced (INE, OECD ICIO, the Growth Lab, the World Bank, ILOSTAT; `data/README.md`), and Γ itself depends on aggregation and normalization (Bolivia's 2019 value ranges over 0.33–0.69). The between-block, information and utility values remain illustrative placeholders, internally consistent but not yet estimated. The quantitative claims (thresholds such as Γ^* and N^*_{eff}, shares such as the truncation penalty) stand or fall with the estimates, and the first estimates have already overturned one premise: Bolivia is not a low-Γ outlier.

3. The political economy is underdeveloped. Chapter 14's coalition model is a one-shot game. Real political dynamics involve repeated interaction, institutions, ideology, identity, and contingency. The hegemonic cycle (Chapter 24) is a conjecture, not a fully specified model. Reading $\boldsymbol{\pi}$ as bargaining power is an assumption; network-bargaining theory (Manea, 2011; Abreu and Manea, 2012a) shows that bargaining payoffs need not equal eigenvector centrality. The book identifies the political constraint but does not solve it.

4. Culture is underspecified. The formation model reduces culture to a preference parameter θ formed through W^I. Real culture involves narratives, identities, rituals, languages, cosmologies — things that do not reduce to a scalar. The model captures the *mechanism* of cultural transmission (network-mediated signal processing) but not the *content*.

5. The book is about structure, not about people. Every example features an "indigenous woman in Potosí" or a "gas executive in Santa Cruz" as a node in a matrix. Real people are not nodes. They have names, histories, hopes, contradictions, and dignity that no eigenvalue can capture. The structural portrait is necessary but not sufficient. The engineer who designs the bridge still needs to talk to the people who will cross it.

6. Many central claims are conjectures. The book proves what can be proved: the properties of $\boldsymbol{\pi}$ and Γ, the independence of the two, the Dobrushin and Hajnal results for time-varying W, the saddle-node of counter-hegemony, the bridge bound on Γ, the echo-chamber result under engagement ranking. Several of its

most important claims — that truncation lowers integration (Chapter 25), that growth with equity needs integration, complexity and information diversity together (Chapter 26), that hegemonies cycle (Chapter 24) — are conjectures. The first gets no support at the 12-sector level. Korea's early growth with equity under controlled media strains the second, and Bolivia's 2005–2019 fall in inequality with a negative, falling ECI contradicts it as a claim about equity gains. The mathematics says what would follow if they hold; the data must decide whether they do.

7. Structure has (at least) two dimensions. An earlier draft of this book used the spectral gap Γ as a single measure of "distance from the textbook." It is not: Γ measures integration, and power is measured by $\boldsymbol{\pi}$; any degree of concentration is compatible with any spectral gap. No single number summarizes W, and a dashboard that reports only Γ will score a winner-take-all network as healthy. Nor does aggregate Γ separate development paths: at 12 sectors Bolivia, Korea, Germany and the United States all have Γ between 0.46 and 0.55, and it fell in both Bolivia and Korea. Power concentration needs the same care: in a production network a high $G(\boldsymbol{\pi}^P)$ can mark industrial hubs, and Bolivia's (0.445) is lower than Korea's or Germany's. The distributional reading of $\boldsymbol{\pi}$ belongs to bargaining, under Assumption 7.A.

28.6 The Structuralist Research Program

This book opens a research program; it does not close one. Each limit above is also a question. Ten for the next generation:

1. **Estimation** (limit 2): Finish replacing illustrative numbers with sourced estimates computed by public code. The production, complexity and distribution values are done; σ_B (household microdata), W^I (audience data), W^U and the product-space quantities (τ, relatedness) are not. Then estimate W^P, W^I, W^U at high frequency and fine spatial resolution, using big data, satellite imagery, and digital trace data, with bootstrap error bands.

2. **Identification of** Ψ: Better designs for separating enforcement from formation. Randomized W^I interventions (following Bernard et al., 2014) in multiple countries.

3. **Testing the conjectures** (limit 6): Is the capital-goods core a bridge in the product space, so that truncation lowers integration? The 12-sector data cannot tell; the test needs INE's 39 industries against comparable detail, or

product-level RCA data. Does sustained growth with equity require $\Gamma > \Gamma^*$, ECI > 0 and $N_{\text{eff}} > N^*_{\text{eff}}$, and where are the thresholds? Bolivia's 2005–2019 record says equity gains do not; whether they last without structural change is the open question. What is the minimum viable capital-goods cluster for a small economy, and can it be shared regionally (a Mercosur or Andean Community program)?

4. **Algorithmic** W^I (limit 7): Real-time monitoring of both λ_k (fragmentation) and N_{eff} (concentration) under algorithmic curation, including messaging platforms. Experimental evaluation of diversity-mandated recommendation systems in the Global South.

5. **Climate and structure**: How does climate change rewire W^P (through agricultural disruption, migration, infrastructure damage)? What are the structural effects of green industrial policy?

6. **Gender formation**: Randomized evaluations of W^I interventions (role models, media campaigns, mentor programs) with long-run follow-up across generations.

7. **Multi-country structural trajectories**: Build the dashboard for every country, every year, and track the global structural trajectory. A "structural world development indicators" project.

8. **Political economy of operator sequencing** (limit 3): Under what political conditions is the Ψ-ordered sequence feasible? When does the information operator create the coalition for the structural operator? What microfoundation, if any, makes π a bargaining weight? What slow feedback, if any, makes hegemonies cycle?

9. **Endogenous institutions**: How do institutions (property rights, courts, regulations) emerge from the co-evolutionary dynamics? The institutional operator (Chapter 15) is exogenous in this book; it should be endogenous.

10. **The general equilibrium of structure** (limits 1, 4): Structuralist CGE models exist (Taylor, 1990, *Socially Relevant Policy Analysis: Structuralist Computable General Equilibrium Models for the Developing World*). What does not yet exist is one with all three networks, the formation channel, time-varying W and political equilibrium — and with culture represented by more than a scalar.

28.7 The Last Equation

The book began with one equation (Chapter 1), the equal-endowment benchmark:

$$W^{\mathrm{CE}} = \frac{1}{N}\mathbf{1}\mathbf{1}'. \tag{28.2}$$

It ends with its generalization:

$$W = W^{\mathrm{CE}} + \Delta(W). \tag{28.3}$$

Every economy is the benchmark plus a departure. The departure is the structure, and it has two independent coordinates: how integrated the network is (Γ) and how power is distributed in it ($\boldsymbol{\pi}$). Together with the primitives of each model, the structure shapes allocation ($\mathbf{x}^*$), beliefs ($\mathbf{b}^*$), preferences ($\boldsymbol{\theta}^*$), distributional welfare (L), dynamics (Γ_t, τ_1), politics (S^*), and the formation of the next generation's preferences ($\boldsymbol{\theta}^*_{t+1}$).

The benchmark model abstracts from the second term. This book studies it.

Key Structural Insight

Structural economics is not a school of thought. It is not a political position. It is not a regional tradition. It is the mathematics of social structure applied to economic life.

The weight matrix W is the object. The Perron eigenvector is the power. The spectral gap is the integration. The formation model is the dynamics. The operator design is the engineering. The coalition game is the politics. The dashboard is the diagnostic. The trajectory is the history.

One matrix, two coordinates, and the primitives that turn them into outcomes. What follows from them is partly proved, partly conjectured, and mostly still to be measured. That is the research programme.

Structuralist Case Study: Furtado's Return

Celso Furtado (1920–2004) appears three times in this book. In Chapter 2 (underdevelopment as a stable fixed point of W), in Chapter 9 (cultural dependency as endogenous preferences), and here, at the end.

Furtado was one of the greatest economic thinkers Latin America produced. He was trained in Paris (doctorate at the University of Paris, 1948), worked at CEPAL under Prebisch (1949–1957), led the working group on the Northeast (GTDN, 1958) and then directed SUDENE, the Northeast Brazil development agency, from its creation in 1959 until 1964, served as Brazil's Minister of Planning (1962–1963), and lost his political rights after the military coup of 1964. In exile he held a chair at the Sorbonne (1965–1985) and visiting posts at Yale and Cambridge; back in Brazil he was Minister of Culture (1986–1988). He wrote more than 30 books, across five decades. His intellectual range — from Keynesian macroeconomics to cultural theory to institutional analysis — anticipated almost every theme in this book.

The thesis of *Desenvolvimento e Subdesenvolvimento* (1961) can now be stated in its complete form.

Claim 1: Underdevelopment is a structure, not a stage. Rostow (1960) had argued that all economies pass through "stages of growth" from traditional to modern. Furtado rejected this: underdevelopment is not pre-development. It is a specific structural configuration produced by the *interaction* of modern and archaic elements within a single system. The modern sector does not gradually absorb the traditional sector; it coexists with it, extracting surplus from it, and reproducing the duality.

In this book: underdevelopment is a structural equilibrium (Definition 13.7) with a modern enclave weakly tied to the domestic network (in Bolivia's data, a small π and a high export share for hydrocarbons, rather than a small aggregate Γ), concentrated bargaining power $\boldsymbol{\pi}$ (the surplus-extraction channel), and self-reproducing dynamics. It is a fixed point, not a transient state. The fixed point exists because the co-evolutionary dynamics (proximity-dominated edge creation, self-confirming preferences, coalition protection) make it stable.

Claim 2: The structure produces the preferences that accept it. Furtado's deepest insight, developed in *O Mito do Desenvolvimento Econômico* (1974): underdevelopment persists not only because of economic structure but because of cultural structure — the patterns of consumption that the economic structure produces and that in turn reproduce it. The high-income minority of the periphery imitates the consumption patterns of the center; producing those goods requires capital-intensive technologies imported from the center; those technologies absorb

little labour and concentrate income further, which sustains the imitative minority. The "myth" is that the center's consumption standards could be generalized to the whole population of the periphery.

In this book: this is the formation model (Chapters 9–10). Preferences are endogenous to W^I. The formation equation $\theta_{i,t+1} = \rho\theta_{i,t} + (1-\rho)\theta^\circ + \mu(\sum W^I_{ij} r_j - r^*)$ captures part of Furtado's mechanism: agents form preferences by observing W^I-weighted signals, and when W^I is dominated by center signals (Chapter 8, epistemic Perron dominance), the formation produces center-mimicking preferences in the periphery — Furtado's "cultural dependency." What the formation equation does not capture is the link Furtado cared about most: from the composition of elite demand to technology choice and to the distribution of income. A faithful formalization would make the consumption basket, not a scalar aspiration, the formed object. The Ψ ratio (Definition 9.13; Chapter 11 asks what data can reveal about it) measures how much of a perturbation's effect comes through the formation channel.

Claim 3: Structural change requires changing the structure, not the prices. Furtado argued that neither market liberalization (which assumes the structure is efficient) nor simple redistribution (which operates within the structure) can solve underdevelopment. What is needed is structural transformation: changing the production structure (W^P), the information structure (W^I), and the institutional structure — simultaneously and deliberately.

In this book: this is Part IV (Chapters 13–15) and the neo-structuralist synthesis (Chapter 26). The operator design framework (Chapter 15) provides the engineering manual that Furtado's diagnosis called for. The Ψ-ordered sequence (information first, structure second, industrial policy third) is a proposed implementation protocol. The three conditions of the growth-equity conjecture ($\Gamma > \Gamma^*$, ECI > 0, $N_{\text{eff}} > N^*_{\text{eff}}$) are proposed success criteria, to be tested.

Furtado (1961, 1974)	**This book's formalization**	**Chapter**
Underdevelopment is a structure	Non-anonymous W: an enclave (low π, high export share), concentrated bargaining power	1–4
Not a stage	Stable fixed point of co-evolutionary system	13
Modern-archaic coexistence	Export enclave at the periphery of W^P, Pinto's heterogeneity	3, 17
Cultural dependency	W^I produces center-mimicking $\boldsymbol{\theta}^*$ (consumption link not modelled)	9, 10
Frustrated modernization	θ^*_{center} incompatible with peripheral W^P	10, 12
Structural change needed	Operators on W, not on $\boldsymbol{\omega}$	15
Simultaneity required	Three conditions (a conjecture), three pillars	26
The general case	W is any row-stochastic matrix; two coordinates	28

Furtado wrote much of his work in exile, in Portuguese and French, from a chair in Paris. He died in 2004, the year before Evo Morales was elected and five years before Hidalgo and Hausmann's 2009 paper on economic complexity. He did not see the product space, the ECI, or the formation experiments that would test his intuitions. He did not use the Perron eigenvector, the spectral gap, or the DeGroot model.

But he had the thesis: underdevelopment is a structure, not a stage. This book offers a formal language in which to state that thesis, to prove some of its parts, and to put the rest to the test.

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The structuralist tradition, complete: Prebisch (1949), the manifesto. Furtado (1961), the thesis. Pinto (1970), the diagnosis. Sunkel (1971), the integration/disintegration. Tavares (1972), the autopsy. Marini (1973), the exploitation. Fajnzylber (1983, 1990), the truncation and the empty box. Cardoso and Faletto (1969), the politics. Quijano (2000a), the coloniality. Sen (1999), the capabilities. CEPAL (2012b), the institutional synthesis. Ocampo et al. (2009), the policy synthesis. Hidalgo et al. (2007) and Hidalgo and Hausmann (2009), the empirical measurement. Taylor (1983, 1990), the structuralist macroeconomics and its CGE models.

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The textbook benchmark abstracts from social structure.

In its competitive form, agents interact anonymously through prices, information plays no structural role, and preferences are taken as given. With equal endowments, the interaction matrix is $W = \frac{1}{N}\mathbf{1}\mathbf{1}'$.

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Structural Economics: The General Case develops a complete mathematical framework in which social structure — production networks, information flows, and solidarity bonds — enters every equation. The Perron eigenvector measures power; the spectral gap measures how integrated the structure is — and the two are independent. The formation model explains how the structure produces the preferences that accept it. The operator design framework shows how to change it.

Drawing on 15 structuralist thinkers from Prebisch (1949) to Ocampo (2009), and using Bolivia as a running empirical case, the book restates central results of development economics — welfare theorems, growth and linkages, inequality dynamics, political economy — in one language: the structure of W and its departure $\Delta(W) = W - \frac{1}{N}\mathbf{1}\mathbf{1}'$ from the benchmark.

The competitive model is the special case.
This book is about the general case.

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$$W = \frac{1}{N}\mathbf{1}\mathbf{1}' + \Delta(W)$$